KEY

The New NATURAL SCIENCE EXAM BOOK

(Grade 9 - 12)

2009 E.C - 2013 E.C NGSEC & EHEEC Examinations With answers and detailed explanations

2014 & 2015 Ethiopian Secondary School Leaving Certificate Examination With answers and detailed explanations (ESSLCE)

English
Mathematics
Physics
Chemistry
Biology
Civic & Ethical Education

2016 Edition

Scolastic Apptitude Test

Compiled By:- G.P

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ETHIOPIAN GENERAL SECONDARY EDUCTION CERTIFICATE ENGLISH EXAMINATION GINBOT 2009/MAY 2017

BOOKLET CONDE: 015

Direction: For items 1-30, out of the choice A, B, C, D only one is correct. Choose the one that best completes each sentence

1. We practice speaking and writing English_____ six hours in a week.
   A. from B. for C. about D. in
2. The charity organization consists_____ different partners having a common interest of helping the poor.
   A. with B. up C. in D. of
3. _____! You've got the job you applied for.
   A. Take my regards C. Very good
   B. Congratualtions D. Many thanks
4. My sister graduated from a famous college in February last year but she hasn't got a job_____
   A. every since B. ever C. yet D. still
5. _____ to take a seat or are you in a hurry to go?
   A. Could you like C. Should you like
   B. Would you like D. Will you like
6. Tolosa promised_____ her after finding a good job.
   A. to marry C. marrying
   B. to marrying D. marry
7. The site for the building should be certified by_____exprt from_____Municipality of Addis Ababa.
   A. an/ a B. the/ the C. the/ a D. an/the
8. If you smile_____me once a day like this, I will give you anything you ask for.
   A. at B. to C. with D. for
9. As a matter of fact, there is plenty of space in this room. You can put your bag_____you like now.
   A. on where B. no where C. any where D. every where
10. The secretary is very slow. Do you know how many letters she has written_____we last met?
    A. since B. while C. when D. before
11. Children mostly depend on their mothers for food and protection. In this case, the love_____ a mother for her child is quite immeasurable.
    A. from B. by C. in D. of
12. I feel completely lonely today and I wish I_____ hear her lovely voice where she is from.

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A. could B. will C. may D. can

13. Beza won't be able to join a medical college_____she obtains the top most scores.
A. unless B. before C. as long as D. if not

14. To be frank with you, Samrawit is a very beautiful girl. She lives in Geneva and has got married to_____Europan man recently.
A. that B. an C. a D. the

15. _____ is an essential but difficult skill in learning a new language.
A. A writing B. Write C. Writing D. To write

16. I do not know how_____ this machine can operate. The technicians might help us with that.
A. quickest B. quick C. quickly D. quicker

17. Million is_____industrious worker_____he was awarded 100,000 Birr.
A. such/ that C. too an/ that
B. so an/ that D. such an/ that

18. Girmachew is an Ethiopian migrant residing in the USA. He has lived in Washington DC_____1974.
A. before B. after C. since D. for

19. Moe than eighty vernacular languages_____in Ethiopia,
A. were spoken C. will be spoken
B. are spoken D. have been spoken

20. Now, the group discussion is over. Do you have_____quesion for further explanation?
A. a few B. one C. any D. some

21. The demonstrators_____be allowed to destroy public property.
A. ought not to C. shouldn't have
B. should have not D. ought to not

22. The graduation ceremony_____when the photographers arrived.
A. will have been closed C. has been closed
B. had been closed D. have been closed

23. The referee ordered the players_____
A. "Don't take their' postions in the filed."
B. to take their postions in the field.
C. " Take your postions in the field."
D. to not take their postions in the field.

24. Which one of the following sentences shows possibility?
A. I can play tennis better than you.
B. He can definitely win the presidential election.
C. She can safely repair the pipe soon.

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D. Can you play the guitar, Sosina?

25. Now it is too late. The students_____their assignment before three days.
A. should have been submitting
B. should have been submitted
C. should have submitted
D. should submit

26. The writer_____book became the bestseller of the year is my grandfather.
A. whose B. whom C. where D. who is

27. This chemical is _____used in science laboratories because it is very expensive and unavailable.
A. usually B. rarely C. fairly D. commonly

28. Have you_____ visited the Awash and the Semen National Parks?
A. even B. hardly C. ever D. never

29. Sabur neither talks nor smiles at all when he is with me. As a matter of fact, he is one of the_____ boring friends I have ever made,
A. more B. most C. less D. least

30. We_____aromatic flowers during spring for pleasure.
A. have smelled
B. smell
C. were smelling
D. are smelling

II. COMPREHENSION

PART ONE: TEXT VOCABULARY

SECTION ONE

Directions: For items **31-35** choose the most appropriate word that can complete each sentence meaningfully.

31. The _____card was beautifully decorated with prismatic colours.
A Promoiton B. Poster C. request D. invitation

32. The_____ of our traditions due to foreign cultural influence needs some sort of protection.
A. renaissance B. revival C. erosion D. treatment

33. In many African countries, _____ of power has increasingly become a norm.
A. abuse B. corruption C. influence D. abnormality

34. Children should suitably be_____ with proper diet in order to improve their physical and mental growth.
A. underfed B. nourished C. furnished D. promoted

35. Although we had a number of appropriate demands at that time, none of them has been_____ addressed.
A. unfairly B. specially C. adequately D. poorly

SECTION TWO

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Directions: For items **36-40**, choose the most suitable meaning for the words written in bold in the sentences.

36. Before their departure, the passengers had ordered a **delicious** breakfast.
A. lovely B. costly C. tasty D. expensive

37. During our conversation, I realized that the director was **polite** and friendly.
A. cooperative B. forgetful C. kind D. respectful

38. If a dealer buys car for 1200 USD, he may **gain** 40 dollars from the resale.
A. lost B. balance C. profit D. increase

39. Many travelers keep a **diary** to record their experiences and observations.
A. milk product
B. record of one's impression
C. daily film script
D. memory of events

40. Unlike in the past, many rural people have these days learned to keep their **treasures** in bank vaults.
A. harvest B. property C. belongings D. valuables

PART TWO: READING PASSAGE I

Direction: Read the passage below carefully. Then, answer questions **41-50** according to the information in the passage.

1. Americans have a sense of space, not of place, Go to an American home, and almost the first thing you do is drift toward the window. How curious that the first compliment you pay your host inside the house is to say how lovely it is outside the house! He is pleased that you should admire his views. The distant horizon is not merely a line separating earth and sky. It is a symbol of future. The American is not rooted in his place. However lovely his place is, his eyes are drawn by the expanding space to a point on the horizon, and this, is his future.

2. By contrast, consider the traditional Chinese home. Blank walls enclose it. Step behind the spirit wall and you are in a courtyard with perhaps a miniature garden around the corner. Once inside the private compound, you are Wrapped in an atmosphere of calm beauty, an ordered world of buildings, pavement, rock and decorative vegetation. But you have no distant view. Nowhere does space open out before you. Raw nature in such a home is experienced only as weather, and the only open space is the sky above. The Chinese is rooted in his place. When he has to leave, it's not for the promised land on the terrestrial horizon, but for **another world** all together along the vertical, religious axis of his imagination.

3. The Chinese tie to place is deeply felt. Any desire to travel is alien sentiment. The Taoist classic Tao TeChung captures the ideal of

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rootedness in place with these words: "Though there may be another country in the neighborhood so close that they are with sight of each other and the crowing of cocks and barking of dogs in one place can be heard in the other, there is no traffic between them. Throughout their lives, the two peoples have noting to do with each other." In theory, if not practice, farmers have ranked high in Chinese society. The reason is not only that they are engaged in the "root" industry of producing food, but that, unlike merchants, they are tied to the land and do not **abandon** their country when it is in danger.

4. Nostalgia is a recurrent theme in Chinese poetry. An American reader of translated poems may well be taken aback- even put off- by the frequency as well as the sentimentality of the lament for home. To understand the strength of **this sentiment**, we need to know that the Chinese desire for stability and rootedness in place is prompted by the constant threat of war, exile, and the natural disasters of flood and drought. Forcible removal makes the Chinese keenly aware of their loss. By contrast, Americans move, for the most part, voluntarily. This nostalgia for home town is really longing for childhood to which they cannot return. In the mean time, the future beckons and the future is " *out there*". In open space. When we criticize American rootlessness we *tend to forget* that it is a result of ideals we admire, namely, social *mobility*, and optimism about the future. When we admire Chinese rootedness, we forget that the workplace means both location in space and position in society. To be tied to place is also to be bound to one's station in life, with little hope of betterment. Space symbolizes hope, place achievement and stability.

Source : Adapted from Real Writing (Bedford/ St Martin's: 204)

41. What sentiment does this refer to in the expression".... The strength of **this sentiment**...." (Paragraph 4)?
    A. Desire for stability
    B. Lament for home
    C. Chinese poetry
    D. Desire for rootedness in place
42. What does it refer to in the expression"..... we tend to forget that **It** is..." (Paragraph4)?
    A. Criticism
    B. Optimism
    C. Social mobility
    D. American rootlessness
43. Which of the following is true according to the passage?
    A. Practically, farmers and merchants share the same rank in Chinese society.
    B. Chinese farmers are less closely tied to the land than merchants are.
    C. Chinese farmers produce less food than is theoretically possible.
    D. The author used farmers to demonstrate rootedness in China.

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44. What does the word **abandon** mean as used in paragraph3?
A. stop B. leave C. end D. keep
45. What does another world imply in the expression"..... leve.... For another world altogether ....." (paragraph2)?
A. religion B. promised land C. death D. imagination
46. Which of the following is true about the Chinese traditional house?
A. It has plenty of open space.
B. It is enclosed by miniature gardens.
C. It has a calm yet beautiful compound.
D. It is surrounded by decorated walls.
47. Which one of the following is more important about the distant **horizon** seen from inside the American house(paragraph1)?
A. It is a line that separates earth and sky.
B. It is a line designed to serve as a symbol of the future of the American.
C. It is a line designed to show the beauty of the outside part of the American house.
D. It symbolizes the ever expanding feeling the American has for space.
48. What does **alien**, as used in ...an alien sentiment (paragraph 3), mean?
A. alone B. strange C. enemy D. familiar
49. Which of the sentences given below best reflects the main idea of the passage?
A. Americans' sense of space and place is much better than that of the Chinese.
B. There are cultural differences between the American and the Chinese concept of home.
C. American and Chinese homes are different in many ways.
D. The Chinese cultures are more nostalgic than the cultures of Americans.
50. Which word used in paragraph 2 has a fairly similar meaning to the word **small**?
A. spirit B. courtyard C. terrestrial D. miniature

Part Three:- Sentence Comprehension

Direction:- For items **51 and 52**, choose the answer which is best related in meaning to the head sentence.

51. Changes in life style have brought changes in the eating habits of many Ethiopians today.
A. Changes in Ethiopians' life style have improved their eating habit.
B. Life style has changed the eating styles of many Ethiopians.

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C. Life style and eating habits seem to be loosely connected in Ethiopia.

D. Ethiopians today do not eat as much as they did in the past.

52. The tourists from China could hardly understand English.

A. Their common of English is very high.

B. They speak and write English very well.

C. They understand little or no English.

D. They are fluent speakers of English.

PART FOUR: READING PASSAGE II

Direction:-For Items 53-65, read the passage carefully, And then, choose the best answer for each question according to the information given in the passage.

1. Development means improvement in a country's economic and social conditions. More specially, it refers to improvements in the management of the natural and human resources of a given country in order to create additional wealth and improve people's lives. Elaborating on the meaning of development, policy- makers suggest that while there can be value judgments on **what is** development and **what is not**, it should be a universally acceptable aim of development to make for conditions that lead to a realization of the potentials of human personality.

2. The people are held to be the principal actors in human scale development. Respecting the diversity of the people as well as the autonomy of the spaces in which they must act **converts** the present day object person to a subject person in the human scale development. Development of the variety that we have experienced so far has largely been one of the top-down approaches where there is little possibility of popular participation and decision making. Human scale development calls for a direct and participatory democracy where the state gives up its tradition **paternalistic** role in favour of a facilitator in enacting and consolidation people's solution flowing from the grassroots level.

3. Powerful economic interest groups set the national agenda of development which is unrepresentative of the heterogeneous and diverse nature of our civic society. This has always resulted in the **concentration** of power and resources in the hands of a few.

4. The slogans like" human centred development" and " integrated development" call for a more inclusive approach to socio-economic and socio-political changes such that all aspects of life of a people, **their** own history and collective consciousness make for a balanced advancement.

5. Social and Human Development, therefore, necessarily requires a unified approach, integrating the economic and social components in

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planning, policies and programmes for people's betterment. The challenge is to simultaneously integrate cross sectoral and regional developmental needs as well as to encourage participatory development. The issues of environment, pollution, women, habitat, hunger and employment have become the burning issues of the day to require public and institutional attention along with resources allocations. Two major **contemporary** concerns that require focus in any development initiative are that of human security and sustainability.

6. We need to ensure that development does not mean social dislocation, violence and war and that we meet" the needs of the present generation without compromising the ability of future generations to meet their own needs." The goal of development should be to develop man and not to end with developing things, Fulfillment of basic needs of mankind should be the true objective of development. Achievements that either do not contribute to this goal or even disrupt this basic requirement must not be pursued as a development goal.

Adapted from Sociology Discussion Communication Series, 44 (October 2016)

53. The contextual meaning of the word **concentration** in paragraph 3 relates to
A. focus of attention
B. strength
C. monopoly
D. large quantity

54. Which one of the following development approaches does the writer recommend in paragraph 4?
A. The paternalistic approach
B. The universal approach
C. The integrated approach
D. The basic needs approach

55. What does the word **Contemporary** n paragraph 5 suggest?
A. Chronological
B. current
C. retrospective
D. Prospective

56. According to the passage, development **does not** include
A. development economic and political decentralization
B. fulfillment of the basic needs of mankind
C. human security and sustainability
D. social dislocation ending

57. What does the pronoun **We** in paragraph 6 refer to?
A. The people
B. Development planners
C. Teachers
D. Students

58. What does the word **their** in paragraph 4 refer to?
A. National investors
B. Interest groups
C. People
D. Policy makers

59. The word **converts** in paragraph 2 means_____

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A. endorses
B. changes
C. Changes currency
D. adopts

60. Which one of the following phenomena is NOT a democratic aspect of development?
A. facilitator's role of government
B. top- down approach
C. empowerment of the people
D. popular participation

61. Which one of the following agenda is considered as development goal?
A. The improvement of the human living conditions.
B. The development of things and infra structures alone.
C. Dislocation of people and violence in different areas.
D. Concentration of power and resource in the hands of a few.

62. What does the word **this** in paragraph 6 refer to?
A. The people and their initiations to act
B. The state and its development steps
C. The fulfillment of the basic needs
D. policy- makers and their actions

63. According to the passage, development is understood as
A. the quantitative growth of material wealth in a given time.
B. the realization of human personality in terms of activities.
C. the proper management of human and natural resources.
D. the economic growth of a country in general terms.

64. In which paragraph is the concept of development well – defined?
A. paragraph four
B. paragraph five
C. paragraph two
D. Paragraph one

65. Which paragraph deals with the importance of participatory democracy in the human scale development?
A. Pargraph four
B.Paragraph two
C. paragraph one
D. Paragraph three

III. *Communicating Activities / Speaking*

Direction: *For items 66-71 choose the most appropriate alternatives to correctly complete the dialogues below.*

66. **Conductor:**_____
Passenger: *Here it is, sir,*
A. *I will check your ticker.*
B. *Ticket checking, Good morning.*
C. *You have a ticket, isn't it available?*
D. *May I check your ticket, please?*

67. **You:** Can I use your bicycle this afternoon?
Friend: Yes, of course, but_____
A. *I bought it just over a month ago.*

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B. we're close friends, in any case.
C. keep it safely with you forever.
D. don't ask me for it again.

68. **Customer:** The price of goods is increasing by the day, what do you think is the reason behind it?

Shopkeeper:_____

A. I really imagine that no one is to blame for that at all.
B. It is certainly related to the growing supply of goods.
C. That is temporary problem that is observed these days.
D. I think it is associated with the devaluation of the Birr.

69. **Hailu:** Isn't this item much easier than the rest of the questions?

Ahmed: _____

A. No, I imagine this isn't a moderate question.
B. Yes, I suppose you prefer the easiest one
C. No, I guess this is a fairly wrong question
D. Yes, I think it's the easiest item in the test

70. **Teacher:** Some student never attend classes regularly_____?

Director: You should report the case to the unit leader.

A. What will I do about them, sir
B. Which measure can I take, please
C. What should I do about this, sir
D. How should I do about that, please

71. **Tamiru:** My grandfather is suffering from loss of appetite. What do you adivise him, doctor?

Doctor: _____

A. It is risky to lose one's appetite.
B. I advise him to see a doctor soon.
C. Can you tell me what his name is?
D. How old is he at the present time?

IV: Writing

Part One: Reorganizing Jumble Words

Direction: For Items 72 and 73, reorganize the jumble words to make a meaningful sentence.

72. Language/ as/ status/ assured/ the/ not/ is /global/ English/ of/ future.

A. The status of the global future as English language is not assure.
B. The future English as the global status of language is not assured.
C. The future status of English as the global language is not assured.
D. The global status as English of the future language is not assured.

73. Was/ the/ gave/ that/ address/ me/ you/ incorrect.

A. The address was that incorrect you gave me.
B. .The address that you gave me was incorrect.

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C. That was incorrect you gave me the address.
D. That was you gave me the incorrect address.

PART TWO: SENTENCE COMBINATION

Direction: For Items 74-75, choose the best sentence combination for each of the pair of sentences below.

74. Many people aren't concerned about pollution. They aren't also worried about its impacts on life.
    A. Many people are neither concerned nor worried about pollution and its impacts on life.
    B. Many people aren't concerned about pollution but worried about its impacts on life.
    C. Many people are concerned and worried about both pollution and its impacts on life.
    D. Many people are either concerned about pollution or worried about its impacts on life.
75. There are many tourist attractions. Revenue is generated from them.
    A. There are many tourist attractions and revenue is generated.
    B. There are many tourist attractions for which revenue is generated.
    C. There are many tourist attractions when revenue is generated.
    D. There are many tourist attractions from which revenue is generated.

PART THREE: PARAGRAPH WRITING

Direction: For items 76-78, are based on three paragraphs given below, Choose the best answer which reflects the central of each paragraph.

We all know that monkeys are smart animals, but sometimes their intelligence is surprising. A psychologist once wanted to see just how smart a monkey was. He hung a banana high up in a monkey's cage. He put in several large boxes and a stick. He wanted to see if the monkey could use the boxes and the stick to get the banana. The monkey looked at the banana, the boxes and the stick. Then it took the psychologist's hand and led him to where the banana was hanging. It jumped up onto his shoulders and reached the banana from there.

76. What does the above paragraph imply?
    A. The boxes and the stick did not help the monkey to get the banana.
    B. The monkey made the psychologist smart and intelligent in the cage.
    C. Monkeys are more intelligent than people think they are.
    D. The psychologist made a monkey smart and intelligent in a cage.

For centuries, men who work as coal miners had many health problems. The worst of these problems is a disease called" Black Lung, "

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This disease is caused by the coal dust in the mines. In the United States, until very recently, about one in every five miners got this disease. For these men, it meant poor health and a shorter life. Then in 1969, a new law forced the mine companies to improve the working conditions for miners. Since then, fewer miners have got" Black Lung."

77. What is the main idea reflected in the paragraph?
    A. Compared to others, coal miners in the United States live longer and have better health.
    B. The 1969 miner company law in the United States improved miners "Black Lung"
    C. Coal dust causes" Black Lung", a disease that attacks the lung of coal miners.
    D. These days, no coal miner in the United States has got" Black Lung"

Even though Arizona and Rhode Island are both states of the U.S. they are strikingly different in many ways. For Example, the physical size of each state is different, Arizona is large, having an area of 114,000 square miles, whereas Rhode Island is only about a tenth the size, having an area of only 1,214 square miles. Climatic variability and population could also be considered. If we consider the size of the population of each state, Arizona has about four million people living in it, but Rhode Islan has less than one million.

78. What is the central theme of the above paragraph?
    A. The population of Arizona is smaller than that Rhodes Island.
    B. Arizona and Rhode Island differ in many ways.
    C. Rhode is the largest island in the USA.
    D. Arizona and Rhode are islands.

Part Four:- Punctuation

Direction: For item **79 and 80,** choose the sentence with the correct punctuation mark.

79. A Medical schools train doctors; dressers, nurses, and laboratory technicians.
    B. Medical schools train doctors: dressers, nurses and laboratory technicians.
    C. Medical schools train doctors- dressers, nurses and laboratory technicians.
    D. Medical schools train doctors, nurses, dressers and laboratory technicians.

80. A The teacher said, "There is an electric power failure in the school."
    B. "The teacher,"said There is an electric power failure" in the school.
    C. "The teacher said, There is an electric power failure in the school.
    D. The teacher said," There is an electric power failure in the school".

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ETHIOPIAN GENERAL SECONDARY EDUCATION CERTIFICATE ENGLISH EXAMINATION GINBOT 2010/JUN 2018

Instruction:- For items 1-30, out of the choices A, B, C D, only one is correct. Choose the one that best completes each sentence.

1. **Behailu:** _____ is the reason behind our poverty?
   Genet: One of the reasons could be the fact that our natural resources have not been mobilized sufficiently.
   A. Why B. How C. What D. Where
2. **Lemma:** They have tried to reach me several times. But they could not make it yesterday when I was idle.
   Receptionist: That is unfortunate! Can I call them now?
   Lemma: No, the moment I _____ on a patient.
   A. was operating C. will be operating
   B. am operating D. have been operating
3. Our curriculum _____ be reviewed at the interval of five years to cope with global trends.
   A. would have to be C. should have to be
   B. needs to D. shall have to be
4. Countries _____ foreign trade is dependent on a single export item are advised to revise their economic policy.
   A. who B. what C. whose D. whom
5. Agricultural transformation and industrialization are the leading economic sector of Ethiopia. The former has a long track record while the _____ is a recent phenomenon.
   A. last B. latter C. nearest D. latest
6. Many African countries are endowed with natural resources ranging from water bodies to precious minerals _____, they are still suffering from poverty.
   A. just B. and yet C. so D. already
7. We have _____seen a green horse on the planet except in movies.
   A. never B. always C. so D. more often
8. Patient: I am suffering from sleeplessness. What shall I do about it?
   Psychiatrist: This is a psychological problem with many people of your age. I think you _____.
   A. ought to reduce stress C. should try to sleep
   B. should ignore it D. ought to read the diagnosis
9. **Nardos:** To be frank with you, I am too late to reach the airport in time.
   Aster: If that is so, _____ you take a taxi?
   A. how do B. why aren't C. when do D. why don't

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10. Some citizens have already been dislocated from their villages. The government _____ resettle them in alternative quarters in the near future.
A. ought to B. is going to C. should be D. must be

11. African and Asian refugees _____ to Europe since the middle of 2015.
A. are flooding C. have been flooding
B. were flooding D. will be flooding

12. Breast milk _____ beneficial to children.
A. was B. is being C. was being D. is

13. It is evident that the ultimate goal of liberation movements taking place all over the world was to _____ history.
A. do B. make C. achieve D. work

14. **Mekbib:** Do students read school magazines to justify their weekly publication?
Hagos: No. They _____ do it.
A. ever hardly B. hardly ever C. merely D. mostly always

15. If you committed a crime, you _____ be punished in line with the law.
A. would B. ought to C. may D. shall

16. _____ is one of the deadliest health hazards in the developed world.
A. To smoke B. Smoking C. To smoking D. Smoke

17. Apollo 11 was the first Satellite _____ on the Moon in July 1969.
A. lands B. landing C. to land D. land

18. I have always advised low income groups _____.
A. don't spend less than they earn.
B. don't spend more than they earn.
C. not to spend more than they earn.
D. not earn more than they spend

19. Rishan: What did the inspector tell you about the accident?
Hanna: _____
A. He told me who was involved in it.
B. He asked me if the possible suspect could be .
C. He inquired whether or not I have witnessed the accident
D. He told me that the accident had been committed by a vagabond.

20. The reconstruction of ancient cities is desirable _____ the cost and human resource it incurs.
A. even though B. in spite of C. on behalf of D. although

21. The disaster of climate change is likely to increase _____ the G8 and G20 countries take a practical measure to control it.
A. unless B. even if C. if not D. till

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22. The refugee crisis of our time which has increased by threefold is actually said to be _____ the previous ones.
A. worse than C. as serious as
B. better than D. more worse than

23. Geographers tell us that he Dead Sea which is 400 meters below sea level is _____ point in Asia.
A. The least B. the lowest C. the lesser D. the highest

24. Many passengers were fatally injured during the plane accident. The doctors say that it is _____ to save their lives.
A. much late B. late enough C. much later D. too late

25. As a matter of fact, the guest speaker who was busy with other commitments had no _____ time to complete his speech.
A. most B. much C. many D. a little

26. Our school should be represented in the national conference on quality of education. I think _____ from the school administration, if not the director, is supposed to attend this conference.
A. every one B. not one C. someone D. no one

27. Geographical exploration is _____ fascinating experience that it exposes one to memorable discoveries.
A. so an B. such a C. such an D. so a

28. The migration of wild animals to neighboring countries would affect our tourism industry. _____, our National Parks should be carefully protected.
A. However C. Because of
B. On the contrary D. Therefore

29. The committee has exhaustively discussed all the agenda of the meeting. Is there _____ issue that has not been addressed?
A. nothing B. any C. some D. one

30. From among thousands of candidates who were invited for the interview, only _____ ones that produced genuine documents were accepted.
A. the B. any C. few D. an

II: Comprehension

Part one: Reading passage I

Instruction:- For items 31-45 read the passage below. And then, choose the best answer for each question according to the information given in the passage.

1. Poverty reduction has become one of the most pressing concerns of our time. Poverty cannot be equated merely with insufficient income. We are dealing with a complex multidimensional phenomenon closely linked to illness, illiteracy, infant mortality and the degradation of the environment, among many other

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factors. Up to now, tourism has not been accorded a substantial role in the majority of poverty reduction strategies drawn up by development aid organizations. The strong and sustained rise of tourism over the past fifty years is one of the most remarkable economic, social and cultural phenomena of our time. The events of September 11 dealt a sever blow to this major industry, but it is already back on the path of growth. The number of international tourist arrivals has grown from 25 million in 1950 to 693 million in 2001. This is equivalent to an average annual growth of 7 percent over a period of 50 years.

2. The revenues generated by these arrivals excluding airline ticket sales and revenues from domestic tourism have grown by 11 percent a year over the same span of time. This rate of growth far outstrips that of the world economy as a whole. International tourism receipts reached 462 billion dollars in 2001, making it one of the largest categories of international trade. Depending on the year, this trade volume equals or exceeds that of oil and petroleum product exports, or that of cars and transport equipment.
3. The number of international tourist arrivals is expected to surpass 1.5 billion in 2020 a threefold increase within a generation. The central question is whether we can harness this activity's potential to contribute to poverty **alleviation**. The initial signs indicate that this is indeed possible. Over the past decade, the annual growth of tourist arrivals in developing countries has been higher than the world average. In the 1990s, such countries experienced strong growth in their international tourism receipts, which demonstrates the existence of a competitive advantage in favor of them. Considering all the activities together, tourism accounts for more than a sixth of their non-petroleum exports far are surpassing their second and third largest export earners (raw cotton and textiles). Tourism has thus become one of the main components of the GDP of **these** countries.
4. These revealing figures show the important place tourism already occupies in countries that suffer from extreme poverty. Considering the speed with which the tourism industry is developing all over the world and the potential of developing countries in general, and that of LDCs in particular, it is possible to substantially improve on the results currently being obtained. Would not the situation of LDC s such as Egypt, Myanmar, Senegal, Tanzania, Sao Tome and Principe, and many others be much drier were it not for tourism? And where would the economic future of Cambodia, Nepal, Cape Verde, the Comoros, Haiti, Yemen, the

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Caribbean Islands and many others lie if not in tourism development?

5. Furthermore, in all of these countries, we can see the crucial contribution of tourism-generated foreign exchange receipts to the balance of payments. Such **revenues** reduce the foreign debt and their dependence on a single export sector, in most cases a raw material with low value and fluctuating price. Contrary to an oft-repeated misconception, tourism revenues in most developing countries are much larger than the induced imports or repatriation of benefits that it may generate. Due to the above reasons, tourism can play a major role in improving the Living standard of LDCs and help **them** lift themselves above the poverty line.

(Adapted from the of Statement of WTO, 29 August 2002, South Africa)

31. According to the passage poverty is linked to
    A. super natural forces
    B. sufficient income
    C. social and natural phenomena
    D. international aid
32. Which one of the paragraphs projects the rise in the number of international tourist arrivals?
    A. paragraph 2 B. paragraph 3 C. paragraph 4 D. paragraph 5
33. How are the past fifty years characterize?
    A. the decline of tourism industry
    B. the number of tourist arrivals has decreased
    C. the development of natural environment
    D. the adoption of tourism as poverty-reduction strategy.
34. The revenues generated by the international tourism over the past fifty years had grown by_____
    A. 7 per cent
    B. 11 per cent
    C. less than 7 percent
    D. more than 11 per cent
35. Which one of the following is not included in the manifestations of poverty?
    A. environmental degradation
    B. illiteracy and illness
    C. infant mortality rate
    D. food security and accessibility
36. _____ is one of the largest categories of international trade.
    A. Oil and petroleum export
    B. Agro-industry
    C. Construction industry
    D. Mining industry
37. By the year 2020, the number of international tourist arrivals will have grown to be
    A. 10 billion B. 1.5 billion C. 7 per cent D. 693 million
38. The writer appears to be one-sided because he _____
    A. discusses the multi advantages of tourism
    B. focuses on the economic benefits of tourism

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C. addresses the cultural impacts of tourism
D. compares ecotourism with irresponsible tourism

39. The most appropriate title of this passage could be _____
A. The Distinction of Tourism as Economic Sector.
B. Tourism and Poverty Reduction
C. The Cultural Pollution of Tourism
D. The Balance of Import and Export Trade

40. 'This' in paragraph 1 refers to _____
A. the number of international tourist arrivals
B. the event of September
C. an average growth of 7 percent over a period of fifty years
D. the remarkable economic, social and cultural progress

41. **These** in paragraph 3 stands for _____
A. Latin American countries C. African countries
B. Asian countries D. Developing countries

42. What does the pronoun **them** in paragraph 5 refer to?
A. The most developed countries C. International tourists
B. The Least Developed Countries D. The developing countries

43. The word **alleviation** (N) in paragraph 3 means _____
A. reduction B. control C. balance D. repatriation

44. "**Revenue**" in paragraph 5 denotes _____
A. capital B. income C. subsidy D. cost

45. What does the expression **such revenues** in paragraph **5** relate to?
A. revenue generated from industry
B. tax collected from agriculture
C. tourism generated foreign exchange
D. value added tax collected from tourists

Part Two: Vocabulary: - Section One

Instruction: For items 46-50, choose the most suitable meaning of the words in bold type.

46. The candidates' documents were rejected by the committee for **forgery**.
A. copy B. overwriting C. fraudulence D. authentication

47. The slow drivers attempted to **catch up with** the fast drivers in the national game.
A. stay back for a moment C. drop behind
B. stretch beyond D. reach the same level as

48. France **declared** a state of emergency in the wake of the terrorist attack.
A. officially announced C. claimed responsibility
B. ideally denounced D. repealed a law

49. The **dejected** players made their way back to the locker room, where they could mourn their defeat in private

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A. excited B. depressed C. exceptional D. emotional

50. **Domestic** flights are much cheaper than international flights.
A. private B. chartered C. temporary D. local

Section two

Instruction:- For items **51-55,** choose the most appropriate word which completes each sentence meaningfully.

51. It is obvious that one of the strategies which enhance interpersonal communication is _____.
A. shyness B. politeness C. punctuality D. superiority

52. They want to _____ their daughters before they sell their apartment.
A. disown B. marry off C. contract D. abandon

53. The Indian _____in Ethiopia is well-established since the 1940s.
A. community B. policy C. race D. aid

54. Some athletes have tested positive for drugs which _____ their performance.
A. disqualified B. held C. rose D. enhanced

55. The _____ of relief aid for drought-stricken people in some parts of the country is underway.
A. common B. distribution C. resettlement D. deportation

Part Three:- Reading Passage II

Instruction:- For items 56-65, read the passage carefully. Then choose the best answer for each question according to the information provided in the passage.

1) The terms 'tradition' and 'modernity' are expressions of values which help us observe the process of social and cultural transformation in societies as **they** pass from the 'primitive' to 'pre-industrial' to 'industrial' and 'post-industrial' phases of social development. Modernization is a conceptual tool which social scientists have widely used in analyzing the process as well as the quality of social change. All societies have tradition, but what we describe as 'traditional societies' refers to a specific historical phase of social and cultural development.

2) Traditional societies have substantial degree of differentiation of social strata, divisions between village, town and city, relatively higher level of technology that depends upon wide use of animal energy; have evolved written organized systems of polity with differentiation of political, military and religious offices of specialized elite, and a fairly advanced system of trade, commerce, finance and banking. The values, beliefs, ways of life, aesthetic and symbolic standards and forms of the society constitute its tradition which **maintains** continuity with the past. It is this element of

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continuity which characterizes a social or cultural attribute or value in a society as being traditional. The passage from traditional to modern stage of society initiates major social and cultural transformation.

3) As society passes from traditional to modern phase of development, the closure of opportunities to status mobility is rendered more and more open. New system more flexible and accommodative. These measures are both socio-cultural and structural. Science and technology play an important role in this process, which revolutionizes the outlook of people and also basically alters its production system and economy. These developments coincide, as they did in India, with basic changes in the political system. Traditional Indian society, which was anchored on the institutions of caste 'feudalistic and other worldly outlooks and values on life' underwent major changes through the rise of national movement for freedom.

4) The establishment of the Democratic Indian Republic cherishing the values of secularism, socialism and democracy challenged the traditional values of caste inequalities. The key to this process is the exposure of Indian society and its elite to the culture of science, technology and democracy in the West. The new institutions of education, law and justice, industry and commerce, health and medicine, transport and communication etc. were introduce. They ushered new processes of social and cultural changes in society. These dynamics came into contact with Western values of rationalism, science and technology which the British colonial administration had introduced in India mainly for the consolidation of colonization, but which paradoxically gave birth to the rise of cultural renaissance in India and the National Freedom Movement spearheaded by Mahatma Gandhi.

56. What is the most appropriate title for the passage?
   A. British Colonialism
   B. Cultural Revolution
   C. Traditional and Modernization
   D. The Indian Liberation Movement

57. One of the following assertions is **Not true** about traditional societies.
   A. Traditional societies are classless and amorphous.
   B. Traditional societies have a recognizable social structure.
   C. Traditional societies are eager to advanced technology.
   D. Traditional societies have evolved both oral and written literature.

58. According to the writer
   A. Tradition and change are self- contradictory.
   B. Institutional measures and social forces revolutionize social outlook
   C. Traditionalism is an all-time phenomenon.

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CamScanner

D. The Indian Caste system was not challenged by Western values like democracy.

59. Western values like secularism and rationalism were originally intended to
A. transform India to colonization.
B. support the national freedom Movement of Gandhi
C. consolidate the yoke of British colonialism
D. advance cultural freedom

60. The by-product of India's exposure to Western values is
A. the transformation of the traditional society
B. the transformation of the economic structure
C. the establishment of democratic India
D. The persistence of the Caste System

61. The pronoun **they** in paragraph 1 refers to _____
A. traditional values C. societies
B. modern values D. expressions

62. The word **maintains** (V) in paragraph 2 is synonymous with
A. recovers B. breaks with C. reverses D. keeps on

63. What does the word **West** in paragraph 4 refer to?
A. India C. Europe and the US
B. Ancient Greece D. Great Britain

64. The word consolidation in paragraph 4 means _____
A. communication B. movement C. assimilation D. negotiation

65. The whole passage could be summed up as
A. British colonialism has destroyed Indian traditional values.
B. India's exposure to Western Education has played a positive role towards its democratization process.
C. Tradition values in India and elsewhere should be preserved.
D. Social transformations and cultural renaissance are induced by only internal factors.

Part Four:- Sentence Comprehension

Instruction:- Item 66 and 67 are followed by four alternatives. Choose best sentence which has roughly the same meaning as the head sentence.

66. Realizing the current affairs of the country, the government postponed the national election for a year.
A. Before realizing the current situation of the country, the government postponed the national election.
B. Realizing the current situation of the country, the government rescheduled national election.
C. The government realized the current situation of the country when the election was over.

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D. There was coincidence between the national election and the current affairs of the country.

67. Some people read daily news papers and magazines more often than not.
A. Some people read periodicals regularly.
B. Some people never read periodicals at all.
C. Some people sometimes read periodicals almost every day.
D. Some people read periodicals very rarely.

III. Communicative Activities/ Speaking

Instruction:- For items 68-72, choose the best alternative to complete the dialogues below.

68. Driver:- I pray to the Almighty every morning just before breakfast, & you?
Mechanic:- _____
A. Neither do I B. I don't either C. So do I D. Nor Do I

69. Scientist:- Underdeveloped countries do not invest on space technology.
Assistant:- _____
A. Neither do the developing ones C. The developing ones can either
B. Either do the developing ones D.The developing ones neither do

70. Host:- How do you like the weather surrounding the Equator?
Traveler:- In fact, it is much warmer than the of Europe. But now_____.
A. I used to tropical weather
B. I am used to tropical weather
C. I am used to the cold temperature
D. I used to humid temperature

71.Visitor:- Who is responsible for saving endemic animals from extinction?
Agent:- It is indeed the great game parks all over the World. The giraffe, fortunately, _____ now lives peacefully in many of the Africa's game reserves protected from hunters and farmers.
A. however B. for instance C. on the contrary D. in addition

72. Journalist:- What is the mission of your Technical College?
Director: _____.
A. It is one of the best colleges in the country
B. To produce qualified technicians for emerging industries
C. Technical and vocational education colleges should be upgraded
D. We should review the curriculum

IV: Writing

Part One:- Punctuation and Capitalization

Instruction:-For items **73-75,** select the sentence with the correct punctuation mark and capitalization.

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73. A. Mr. Brown had been teaching English in Jamaica bermuda & Malaysia.
    B. MR. Brown had been teaching english in Jamaica, Bermuda, & Malaysia.
    C. Mr. Brown had been Teaching English in Jamaica, Bermuda & malaysia.
    D. Mr. Brown had been teaching English in Jamaica, Bermuda & Malaysia.
74. A. The coldest months in England are November, January, and February.
    B. The coldest months in England are November January and February.
    C. "An observer reported, The coldest months in England are November, January and February."
    D. An observer remarked, "The coldest months in England are November, January and February.
75. A. Vladimir Lenin said that, "war is the continuation of politics."
    B. Vladimir Lenin said, "war is the continuation of politics".
    C. Vladimir Lenin said, 'War is the continuation of politics."
    D. Vladimir Lenin said that war is the continuation of Politics.

Part Two:- Letter Writing

Instruction:- For items **76-77**, read the following letter and choose the best alternatives for each question.

P.O.Box 7690
Bahir Dar
Dear Mr Joseph,

When you were a guest at our restaurant in the Four Hills Plaza, you experienced an unfortunate mishap that resulted in a beverage being spilled on your coat. As we all know accidents happen but what matters is how the establishment responds to it as a customer unfortunately the staff on duty at the time did not reflect our customer service policy. I have investigated the incident talked to those involved and have scheduled re medical customer relations training for them. Please accept my sincere apology for the mishap and com again.

Yours faithfully,
Bmdd, General Manager

76. By classification, this letter could be described as _____
    A. unclassified letter
    B. a personal letter
    C. a formal letter
    D. an application
77. What is the letter addressed to?
    A. Bmdd
    B. the General Manager
    C. Mr Joseph
    D. Four Hills Plaza

Part Three:- Paragraph Writing

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Instruction:- Items 78-80 are based on the three paragraphs given below. Choose the most appropriate idea or topic sentence for each of them.

Language is a signaling system which operates with symbolic vocal sounds, and is used by a group of people for the purpose of communication. Language more than anything else, distinguishes mankind from the animal world. Even though animals communicate with one another by means of cries like the warning of birds at the imminence of danger, the ape's expression of anger, fear or pleasure. But the number of signals they produce is limited while in human language the number of utterances is indefinite.

78. What is the central idea of this paragraph?
   A. Defining the human language in connection with the apes.
   B. Defining the animal language in relation to birds.
   C. The distinction between the human and animal language.
   D. Verbal Vs non-verbal communication among animals.

Many years ago, mosquitoes were regarded as harmless insects with a nasty bite. Because of this, they were given little attention by scientists. Eventually they became the focus of scientific research. As a result, yellow fever has been eradicated. In contrast, malaria remains unconquered to date in many African countries. Hundreds of people in the low lands still die of malarias. In sum, scientists should redouble their effort to eradicate malaria from the surface of the Earth.

79. The central idea of this paragraph is _____
   A. Scientists should give more attention to yellow fever than malaria
   B. Malaria and yellow fever were eradicated simultaneously
   C. Scientists give more attention to malaria than today
   D. Both malaria and yellow fever are fatal diseases

Dictionaries and grammars are considered to be the supreme authority in meaning and usage. The task of writing a dictionary begins with the reading of vast amount of literature. As the editors read, they copy down on cards every interesting or rare word, every unusual usage and the sentences in which each of these words appear. In other words, the context of each word is collected together with the word itself. For a thorough job of dictionary writing such as the **Oxford English Dictionary**, millions of such cards are collected, alphabetized and sorted out. To define a word, then the lexicographer places before him, a stack of cards illustrating the contextual meaning of a given word.

80. What is the central idea highlighted in the above paragraph?
   A. The function of dictionaries
   B. The procedures of dictionary writing
   C. The types of dictionaries
   D. The difficulty of Oxford English Dictionary.

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ANSWER FOR GRADE 10 ENGISH EGSECE GINBOT 2009/MAY 2017

BOOKLET CODE: 015

1. **B** for + length of time(length of six hours in a week.
2. **D** consist is a verb followed b preposition of.
3. **B** That is expressing congratulation for his/her getting the job.
4. **C** 'Yes' is used in negative and question to mean up to now.
5. **B** That is polite asking if you want/ like to do any one of the two.
6. **A** promise is a verb followed by 'to inf'.
7. **B** Definite aticle refers to definite person or thing or name of organization.
8. **D** smile + for= to do something in favour of someone.
9. **C** Every where is free, so you can put it any where you like.
10. **A** since+ definite time(we last met) i.e beginning time.
11. **D** Love of ='of' shows belonging and it means the love belongs to a mother for her child.
12. **A** It shows present possible wish. To express wish the verb is changed to past. i.e can to could.
13. **A** unless + s+v_1= If + s+ not. Unless she obtains= If she does not obtain.
14. **C** Even if singular nouns like European, union and university start with vowel letter, they have'y' sound and use article 'a' before.
15. **C** Gerund is a word formed from a verb but uses as a noun to be subject' Writing' is a gerund used as a noun to be the subject of the sentence.
16. **C** How+ adv+ s+ v, uses to show manner. Means this machine can not operate quickly.
17. **D** such an + adjective+ singular noun+ that is used to express result.
18. **C** Since+ definit/ starting time
19. **B** Simple present passive to show truth i.e be+ v_3
20. **C** 'Any' is used in a question where negative answer 'no' is expected.
21. **A** Expresses advice/ suggestion not to do.
22. **B** Past perfect to show the first from two past events and the verb is passive.
23. **B** Reporting instruction or command with to+ inf.
24. **B** It is possible for him to win the presidential election.
25. **C** It shows past necessity. Means it was necessary for the students to submit their assignment before thee days.
26. **A** 'Whose' shows possession. The book belongs to the writer,
27. **B** The clue which says very expensive and unavailable indicates that it is rarely (almost no) used in the laboratory,

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28. C 'Ever' is used with present perfect(has/ have+ v_3) question to mean up to now.
29. B one of + the + superlative adjective means there are also others.
30. B Verbs of perception, such as smell, taste, feel, see and hear are used in simple present to show fact.
31. D invitation card
32. C Erosion means damage.
33. A Abuse means wrong use,
34. B Nourished means provided with something.
35. C Adequately means satisfactory or sufficient.
36. C Delicious means very tasty.
37. D Respectful means showing good manner.
38. C Gain means profit or increase in amount.
39. D Diary means daily record of events or thought.
40. D Treasures are highly valuable objects.
41. A Lament for home- expresses great sorrow or regret.
42. D paragraph 4 line 12 American rootlessness
43. D
44. B Go away from, leave
45. D imagination
46. C
47. B
48. B strange
49. B
50. D Miniature means small-scale copy or model of any object
51. B Eating habit or style depends on the change of life style.
52. C Adverb' hardly' means little or almost no.
53. C paragraph 3 line 4 power and resources in the hands of a few
54. C The integrated approach
55. B of the time or period-current point of focus
56. A development economic and political decentralization
57. A paragraph 6 line 1
58. C paragraph 4 line 4 people
59. B Changes
60. B Top- down approach where is little participation of people
61. A Paragraph 5 line 5-6
62. C paragraph 6 line 7 The fulfillment of the basic needs.
63. C paragraph 1 line 3
64. D paragraph one
65. B paragraph two line 7
66. D The answer' Here it is' shows that the passenger has been asked to show his ticket to the controller.

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67. C Allowing the borrower to use it with special care.
68. D Asking the shopkeeper his opinion about price increasing of goods these days.
69. D That is expressing agreement to what has been asked.
70. C The response of the director shows that the teacher has asked for advice.
71. B Advice forwarded by the doctor.
72. C subject+ passive negative verb
73. B The noun used as subject is qualified by relative clause (adjective clause)
74. A Two negative ideas are joined by neither..... nor, which means not this not that; none.
75. D The first sentence shows the source of revenue.
76. D This paragraph implies that the psychologist made the monkey smart to think how to reach the banana in the cage.
77. C The core idea of this paragraph is as coal dust couses a disease that attacks the lung of the miner.
78. B Arizona and Rhode islands are different in their size, number of population and climate.
79. D List of items/ nouns/ are separated by commas.
80. A It is direct speech, Here the reporting verb 'said' is separated by comma from the actual speech, the first letter of actual speech has to be capital and should be enclosed with in quotation marks/ inverted commas/.

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ANSWER FOR ENGLISH EXAMINATION GINBOT 2010/JUN 2018

1. C Wh-questions require some kind of information or complete answer rather than yes or no & 'what' asks about something or action.
2. B The adverb 'the moment' indicates real present and we use present continuous.
3. B Need-similar to must, have to, should indicates necessity; means it is necessary. It is necessary for the curriculum to be revised frequently.
   e.g He need think carefully about this before he takes action.
4. C The relative clause shows possession which belongs to 'countries' and is introduced by 'whose'.
   e.g Abera found the book. Its cover was torn.
   Abera found the book whose cover was torn.
   (Its-possessive adjective shows possession of thing)
5. B The former means the first (agricultural) and the latter refers to the second (industrialization).
6. D 'Already' here means previously; before now.
   e.g I have been there already, so I don't want to go again.
7. A 'Never' means at no time; on no occasion.
   e.g I have never seen a flying fish.
   i.e I have not seen a flying fish.
   He has never been in Antarctica.
   As he told me, my father has never smoked in his life.
8. A That is to express the person's feeling or suggestion, and ought to reduce stress can be the correct answer.
9. D The other possible means to reach there is taking a taxi if he is late.
10. A 'ought to' is a modal auxiliary verb that uses to express what is desirable or right at the time.
    e.g There ought to be more buses during rush hours.
11. C Present prefect continuous tense (has/have been $v_1$ing) is used to talk about an action which may not be finished or how long something has been going on.
    e.g She has been doing her assignment since last night.
    The man has been teaching at our school for over ten years.
12. D verb 'be' is used to show truth.
    e.g Man is a mortal creature.
13. B To <u>make</u> history, arrangement, a decision, an attempt, a mistake, a bed, a gesture etc.
    To <u>do</u> house works, exercise, business, favour, shopping etc.
14. B 'Hardly" means almost no and they do not regularly do it (read school magazines).

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15. A The second conditional /unlikely conditional sentence/ takes present conditional or would +inf in the main clause to express a condition which is less probable.
e.g If he knew that it is dangerous, he wouldn't come.
They would attend the ceremony if you invited them.

16. B Gerund (v+ing) is a word formed from a verb and uses as a noun to be subject or object.
e.g Smoking is bad for your health.
Eating nutritious food makes you healthy.

17. C Sequence indicators like the first, the second, the last, the only one etc are followed by infinitive with 'to'.
e.g He was the first candidate to finish his exam.

18. C It expresses negative purpose what for something is not done (what low income groups do not do).
e.g The boy walked quickly in order not to be late for class.
She turned her face not to see the accident.

19. D The question needs some kind of information, and the answer is he told me the accident had been committed by a vagabond. It was reported some time later and there is change of tense to its past.

20. B *Inspite of* / *despite* } + *noun, noun phrase, objective pronoun and present participle (ing form)*
are used to express contrasting ideas which state. Something that oppose main clause.
e.g Inspite of making laud noise, the speaker continued his speech.

21. A If --- not =unless
If the G8 and G20 countries do not take a practical measure, ---
Unless the G8 and G20 countries take a practical measure,---
e.g The plants will get dry if you don't water them regularly.

22. A The comparative degree (comparative + than) compares two things, persons of different quality. Here the recent and the previous situations are compared.
e.g This book is more interesting than the one you lent me.
Getaneh has played better than the other players in the team.

23. B The + superlative form of the adjective (the lowest) indicates it is the deepest place where compared with other places.

24. D s + v + so + adj/adv/+that+ statement of result.
If the result is negative, it can be replaced by too + adj/ adv + to inf.
It is too late to save their lives means it is so late that the doctors cannot save their lives.
e.g My boss is very busy. She cannot conduct the meeting.
My boss is so busy that she cannot conduct the meeting.
My boss is too busy to conduct the meeting.

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25. B Since the speaker was busy he had no much time to make a long speech. 'much' is usually used in questions and negative statements with uncountable noun.
e.g There hasn't been much good weather recently.

26. B Some and compound words formed from some are used in affirmative statement.
e.g There is someone outside your office and he wants to see you.

27. B Such a/ an+ adjective + singular noun + that→expresses result.
Such a fascinating experience that is the answer.
e.g It was such a dangerous job that I was afraid to do it.

28. D The discourse marker 'therefore' shows the effect what has to be done by national parks.

29. B 'Any' is used in a question to find out if anything is available.
e.g Are there any new comers in the class?
Do you need any of these old books?

30. A Definite article 'the' is used before a noun made known or specific by phrase or clause that follows it, and it defines those candidates who produce genuine or correct document.
e.g The teachers of this school are disciplined.
The athlete who made continuous training won the race.

31. C Paragraph 1.
32. B Paragraph 3-the number of international tourist arrivals is expected to surpass 1.5 billion in 2020.
33. D Paragraph 1 line 7.
34. A Paragraph 1 the last line.
35. D Paragraph 1.
36. A Paragraph 3.
37. B Paragraph 3 first line.
38. B The writer has more explained the economic benefits of tourism.
39. A 'The Distinction of Tourism as Economic Sector' is the most appropriate title to the passage.
40. A In paragraph 1 "**This**" refers to the number of international tourist 'arrivals.
41. D In paragraph 3 '**These**' stands for developing countries.
42. B In paragraph 5 '**them**' refers to the Least Developed Countries (LDC).
43. A In paragraph 3 the word **alleviation** means make lessor easier, reduce.
44. B In paragraph 5 the word **revenue** denotes income generated by tourism.
45. C In paragraph 5 **such revenue** relates to tourism generate foreign exchange.
46. C **Forgery** means fraudulence –deceitful.
47. D **catch up with** means reach the same level as.

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48. A **Declared** means announced, made known clearly.
49. B **Dejected** means made sad or gloomy (depressed).
50. D Domestic means of the home, family, household.
51. B Politeness- possession of good manners and consideration of other people.
52. B Marry off-give in marriage.
53. A Community-people living in one place as a whole.
54. D Enhance-add to the power.
55. B Distribution-giving or sending out.
56. C The most appropriate title to the passage is "Tradition and Modernization".
57. C The traditional societies maintain the continuity of values, believes, way of life etc in the past.
58. B Paragraph three
59. C Paragraph four line 9 and 10
60. B paragraph four
61. D The pronoun '**they**' in paragraph 1 refers to **expressions**.
62. D In paragraph 2 the word '**maintains**' is synonym to **keeps on**.
63. D In paragraph 4 the word '**west**' refers to **Great Britain**.
64. C **Consolidate** means unite or combine into one /assimilate/.
65. B Paragraph four.
66. B To postpone means to reschedule to change time table.
67. A More often is something that takes place regularly.
68. C That is positive addition to positive remark.
69. A It is negative addition to negative remark.
70. B The word 'but' shows that I am accustomed to the tropical weather.
71. B That is to mention examples.
72. B Purpose is expressed with to, in order to and so as to plus infinitive.
73. D Names of languages and places are written in capital but not common nouns and verbs.
74. D Names months in a year and places are all written in capital.
75. C That is how statement is punctuated in direct speech.
76. C It is a formal letter written from the general manager of the hotel to mr. Joseph.
77. C The letter is written to Mr. Joseph.
78. C The distinction between the human and animal language is the central idea.
79. C Scientists give more attention to malaria than today is the central idea.
80. B The paragraph talks about the procedures of dictionary writing.

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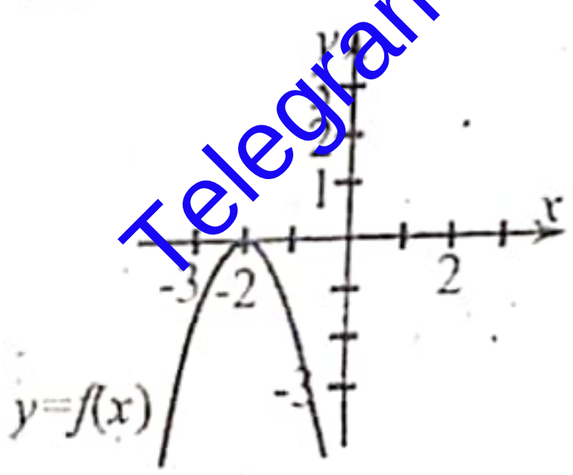
1. Let set M contains 25 elements. Then, which of the following sets is equivalent to M?
   A. $\{x: x \text{ is a natural number and } x < 25\}$
   B. $\{x: x \text{ is a natural number and } x < 26\}$
   C. $\{x: x \text{ is a natural number and } x = 25\}$
   D. $\{x: x \text{ is a natural number and } x > 25\}$
2. In a certain Ethiopian High School 100 students are studying English, 85 students are studying French, and 35 students are studying both languages. Then, how many students are studying any one of the foreign languages? A. A. 220 B. 15 C. 185 D. 150
3. Suppose F and G are two sets, where $F \cap G' = \{0, 2, 5\}$ and $F' \cap G = \{1, 4\}$. Which one of the following is equal to $F \Delta G$?
   A. $\emptyset$ B. $\{0, 2, 4\}$ C. $\{0, 1, 2, 4, 5\}$ D. $\{1, 4\}$
4. When $f(x) = x^5 + x - 1$ is divided by $x + 1$, the remainder is equal to:
   A. 1 B. -1 C. -2 D. -3
5. Which one of the following is considered as an exponential expression?
   A. $3x^2 + x + 1$ B. $\sqrt{x^2 + x}$ C. $3^x + 2x + 1$ D. $2^x + 3^x$
6. What is the interval notation of $\{x \in \mathbb{R}: -x < 9 \text{ and } x \neq 5\}$?
   A. $(-\infty, -9)$ B. $(-9, 5) \cup (5, \infty)$ C. $(9, \infty)$ D. $(-9, \infty)$
7. Let $f(x) = ax^2 + bx + c$ be a quadratic function whose graph is given as shown below.



   Which one of the following is the solution set of $ax^2 + bx + c < 0$?
   A. $(-\infty, -\infty)$ C. $(-2, \infty)$
   B. $(-\infty, -2)$ D. $(-\infty, \infty) \setminus \{-2\}$
8. Let $\theta = 420^o$. Which one of the following is true about θ?
   A. $\sin\theta = \frac{1}{2}$ B. $\cos\theta = \frac{1}{2}$ C. $\sin\theta = -\frac{1}{2}$ D. $\cos\theta = -\frac{1}{2}$
9. Which one of the following is the domain of $f(x) = 3x - \sqrt{2 - x}$?
   A. $\{x \in \mathbb{R}: x \leq 2\}$ C. $\{x \in \mathbb{R}: x \geq 0\}$
   B. $\{x \in \mathbb{R}: x \geq 2\}$ D. $\{x \in \mathbb{R}: x \leq 6\}$
10. A cylindrical tea cup of base radius 4 cmand height 10 cm has a hemispherical lid (top cover).
    What is the surface area of the lid part?

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A. $16cm^2$
B. $32\pi cm^2$
C. $40\pi cm^2$
D. $64\pi cm^2$

11. How many lines of symmetry does a regular hexagon have?
A. 6 B. 7 C. 5 D. 8

12. Which of the following is necessarily true about similar polygons?
A. They have congruent corresponding sides.
B. They have congruent corresponding angles.
C. They are always congruent.
D. Any two regular polygons are similar.

13. Consider an angle of 324°. Then which one of the following is the radian measure of this angle?
A. $\frac{5}{7}\pi$ B. $\frac{5}{9}\pi$ C. $\frac{7}{5}\pi$ D. $\frac{9}{5}\pi$

14. Let $u = (2, 5)$ & $v = (-3, 8)$ be vectors. Then the vector $u - v$ equals to:
A. (5, 13) B. (5, -3) C. (-1, 3) D. (-5, -3)

15. Which of the following describes the property of composite numbers?
A. All composite numbers are even except 9 and 15.
B. Each composite number has more than one prime factor.
C. Every composite number has at least two odd factors.
D. Every composite number has three or more different factors.

16. Let a be a rational number and b an irrational number. Then, which of the following is necessarily true?
A. $\sqrt{3}a + 2b$ is an irrational number.
B. $b^2 - 3a$ is a rational number.
C. $2b - 2(1 + b - a)$ is a rational number.
D. $3ab + 3a$ is a rational number.

17. The simplified form of $\sqrt[3]{-0.027} - (2^{-\frac{1}{2}})(1.28)^{0.5}$ is:
A. -1.1 B. -0.5 C. 0.5 D. 0.7

18. A farmer sold 57.36 kg of corn when measured corrected to two decimal places. Which of the following is a lower bound of this weight in kg?
A. 57.35 B. 57.355 C. 57.359 D. 57.36

19. Consider the polynomial function $p(x) = \frac{x^3-1}{3} - (x + 1)(x^2 - 2)$. Which of the following pairs is the leading coefficient and the constant term of $p(x)$ respectively?
A. $-\frac{2}{3}, \frac{5}{3}$ B. $-\frac{2}{3}, -\frac{4}{5}$ C. $\frac{1}{3}, \frac{5}{3}$ D. $-\frac{1}{3}, -\frac{7}{5}$

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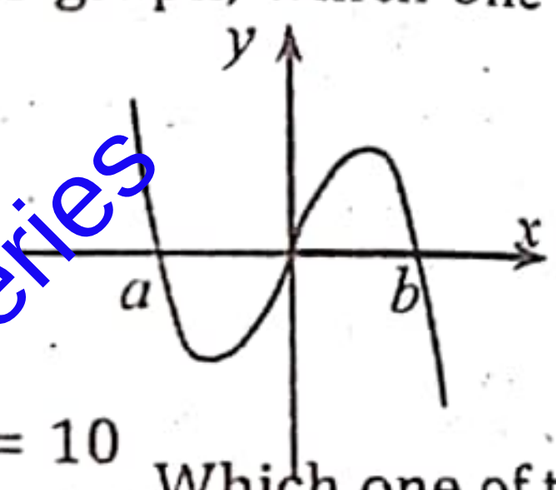
20. When dividing $x^3 + 2x^2 - 5x + 3$ by $x^2 + x - 1$, the quotient and the remainder respectively are:

A. $x + 1$ and $-3x + 4$
C. $x + 1$ and $-5x + 4$
B. $x + 1$ and $-5x + 2$
D. $x - 1$ and $-5x - 4$

21. Which of the following is NOT true?

A. $f(x) = x^4 - x - 1$ has a zero between -1 and 1.
B. $f(x) = x^3 + x - 3$ has a zero between 0 and 2.
C. $f(x) = (x^3 + 1)(x + 1)$ has a zero between 0 and 1.
D. $f(x) = x^3 - 2x + 1$ has a zero between -2 and -1.

22. The following figure is sketches of the graph of a polynomial function f, where a and b are real numbers.Referring to this graph, which one of the following is **NOT** true about $f(x)$?



A. $f(x) > 0$ only for $x \in (-\infty, a)$.
B. $f(x) < 0$on $(a, 0) \cup (b, \infty]$.
C. The degree of $f(x)$ is 3.
D. Zero is a simple root of $f(x)$.

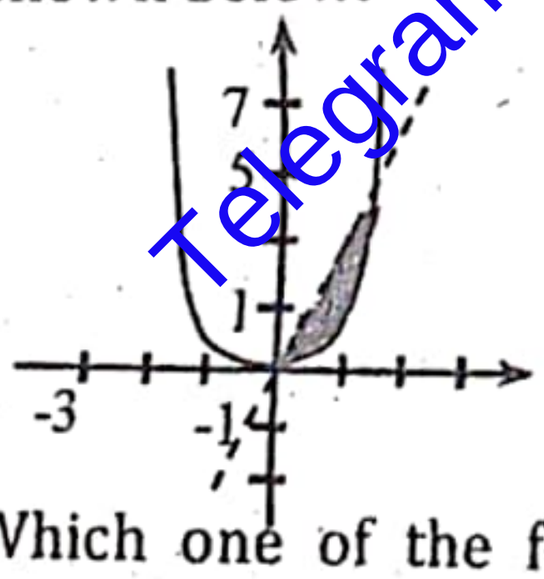
23. Consider the system of linear equations $\begin{cases} 3x - 2y = 10 \\ 2x + 2y = 5 \end{cases}$. Which one of the following is true about its solution?

A. It has infinitely many solutions.
C. It has unique solution.
B. It has no solution.
D. It is difficult to determine.

24. Consider the equation $2x^2 - a = 3x$. If -1 is its solution, then which one of the following is the other solution?

A. 5
B. 2.5
C. $\frac{1}{2}$
D. $-\frac{5}{2}$

25. Consider the graph of a relation $R=\{(x, y): y \geq x^2$ and $y < 2x\}$ that shown below.



Then which of the following is true of this relation?

A. The domain of R is $\{x \in \mathbb{R}: 0 \leq x \leq 2\}$
B. The range of R is $\{y \in \mathbb{R}: 0 < y < 4\}$
C. The domain of R is $\{x \in \mathbb{R}: x \leq 2\}$
D. The range of R is $\{y \in \mathbb{R}: y > 0\}$

26. Which one of the following is true about the graph of a quadratic function $f(x) = ax^2 + bx + c$?

A. It has a y-intercept.
C. It has exactly two zeros.
B. It is increasing.
D. It has exactly one x-intercept.

27. Let $A=\{x: |x - 1| = 1\}$. Then the power set of A equals to:

A. $\{0, 2\}$
C. $\{\{0\}, \{2\}, \{0, 2\}\}$
B. $\{\emptyset, \{0\}, \{2\}\}$
D. $\{\emptyset, \{0\}, \{2\}, \{0, 2\}\}$

28. For $a > 0 \& a \neq 1$, which one of the following is true about $f(x) = a^x$ and $g(x) = \log_a x$?

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A. They have the same domain.
B. They have the same range.
C. The domain of $f(x)$ is the same as the range of $g(x)$.
D. Their y-interceptes are the same.

29. The solution set of $3^{6(x-2)} = 9^{x+2}$ is equal to:
A. {4} B. {-4} C. {2} D. {-3}

30. The following table is the frequency distribution of certain set of data.

x	10	11	12	13	14	15
f	2	0	?	2	4	2

If the mean of this data is 12.8, then the frequency of 12 equals to:
A. 6 B. 5 C. 4 D. 3

31. Which one of the following is true about the Mode of a data set?
A. It can be the value out of the data set.
B. It is uniquely determined.
C. It is used for both qualitative and quantitative data.
D. It is the one which occurs exactly at the half way in the data set.

32. Which of the following is the property of the arithmetic mean of a data?
A. There can be more than one value for the mean.
B. It cannot be the value out of the data set.
C. We do not need all the data values to calculate it.
D. The arithmetic mean is affected by extreme values.

33. A fair coin will be tossed three times. What is the probability that at least two heads show up?
A. $\frac{1}{2}$ B. $\frac{3}{4}$ C. $\frac{3}{8}$ D. $\frac{5}{8}$

34. Which one of the following inequalities is true?
A. $\sin(-100^o) < 0$ C. $\tan(370^o) < 0$
B. $\cos(-180^o) > 0$ D. $\cos(200^o) > 0$

35. If $\sin\alpha = r$ and $\alpha + \theta = 90^o$, then which of the following is true?
A. $\cos\theta = \frac{1}{r}$ B. $\cos\theta = r$ C. $\sin\theta = r$ D. $\sec\theta = \frac{1}{r}$

36. Consider the function $g(\theta) = -\cos\theta$. Which of the following statements is correct about this function?
A. The period of g is 2π. C. The range of g is $(-\infty, \infty)$.
B. The period of g is -2π. D. The range of g is $(-\infty, 1]$.

37. Let $h(\theta) = \tan\theta$. Which of the following is NOT true?
A. $\tan 70^o > 1$ C. $\tan(-135^o) = 1$
B. $\tan(-20^o) < -1$ D. $\tan(270^o) = -1$

38. $\csc(330^o)$ is equal to:
A. $\frac{1}{2}$ B. $-\frac{1}{2}$ C. 2 D. -2

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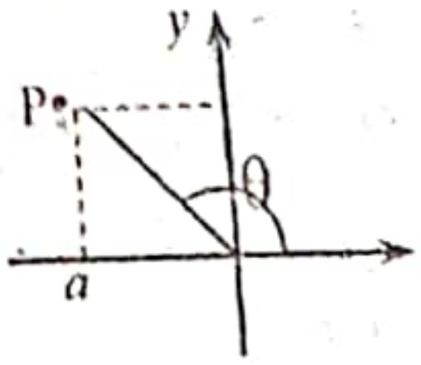
39. Let θ be an angle as shown in the figure below and d be the length of $\overline{OP}$.



Which of the following is equal to $\sec\theta$?

A. $\frac{d}{a}$ C. $\frac{d}{a}$

B. $-\frac{d}{b}$ D $\frac{d}{b}$

40. Which of the following is NOT true about a regular polygon of n sides?

A. Each interior angle measures $(1-\frac{2}{n})\pi$ radians.

B. The length of all its sides are equal,

C. The measure of each exterior angle $\frac{\pi}{n}$ radians.

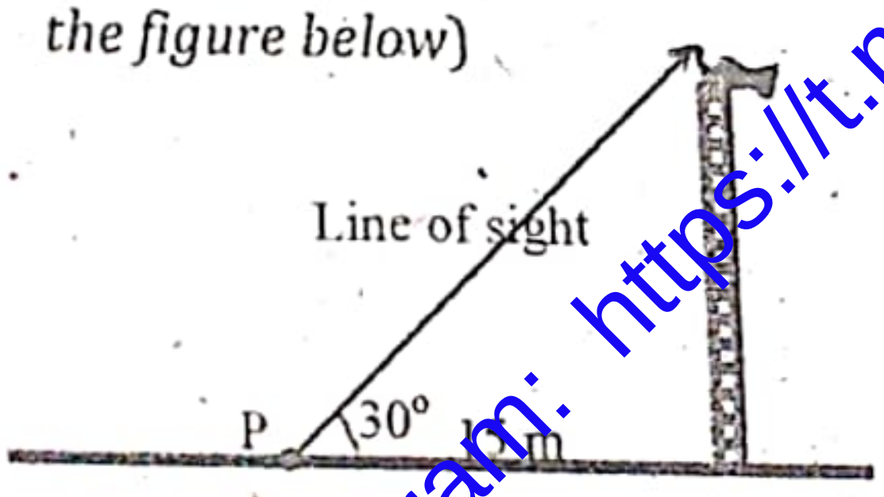
D. It is a convex polygon.

41. The sides of a given triangle are 6 units, 8 units and 10 units long. Which of the following is the area of the triangle?

A. 30 square units C. 40 square units

B. 24 square units D. 32 square units

42. A bird is at the top of a vertical pole posted on a level ground. At a point P on the level ground which is 15 meters away from the foot of the pole, the line of sight of the bird makes an angle of 30° with the ground. (*See the figure below*)



How far is the bird from point P?

A. $10\sqrt{3}$ meters

B. $\frac{15}{2}\sqrt{3}$ meters

C. 25.5 meters

D. 30 meters

43. If the point $A(b, 4)$ is 10 units away from B(0, -2), then the possible value(s) of b is(are):

A. 10 B. 8 and -8 C. $2\sqrt{3}$ D. 4 and 6

44. A right triangle has vertices P(-1, 1), Q(3, 4) and R(-1, 4). Which one of the following is the gradient of the hypotenuse of this triangle?

A. $\frac{3}{4}$ B. $\frac{4}{3}$ C. 4 D. $-\frac{3}{4}$

45. From the following pairs of equations of lines, which pair represents perpendicular lines?

A. $2x+5y+1=0$ and $4x+10y+2=0$

B. $2x+5y+1=0$ and $5x+2y+1=0$

C. $2x-5y+1=0$ and $5x-2y+1=0$

D. $2x-5y+1=0$ and $5x+2y+1=0$

46. The area of a regular 8 sided polygon with radius 4 units is:

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A. $8\sqrt{2}$ sq. units
C. $16\sqrt{2}$ sq. units
B. $64\sqrt{2}$ sq. units
D. $32\sqrt{2}$ sq. units

47. A regular pyramid has square base whose side is 8 cm long. If the length of the lateral edge is 12 cm, then which one of the following is the total surface area?
A. $64(1+2\sqrt{2})$ cm^2
C. $64\sqrt{5}$ cm^2
B. $128\sqrt{2}$ cm^2
D. $64(1+\sqrt{5})$ cm^2

48. A frustum of height 12 cm is formed from a right circular cone of height 16 cm and base radius 8 cm. The volume of the frustum is:
A. 330π cm^3 B. 104π cm^3 C. 336π cm^3 D. 312π cm^3

49. A right circular cylinder with base diameter 10 cm containsenough water that can immerse an object. If an iron ball of diameter 6 cm is immersed, how high will the water rise?
A. 12/25 cm B. 36/25 cm C.48/25 cm D. 72/25 cm

50.A factory has 30 male and 40 female workers. Out of these workers only 20 male workers have college degree. If 60% of the workers have college degree, then how many female workers do not have a college degree?
A. 10 B. 18 C. 22 D. 21

51. The simplified form of $\log_{100}(0.01\sqrt{0.01})$ is:
A. -3/2 B. -3/4 C. 3/4 D. 3/2

52. Let $f(x) = (\frac{1}{3})^x$ and $g(x) = (\frac{3}{4})^x$. Which of the following is true about these functions?
A. For $x < 0$, the graph of $f(x)$ lies above the graph of $g(x)$.
B. For $x \geq 0$, the graph of $f(x)$ lies above the graph of $g(x)$.
C. For $x > 0$, both the graphs of $f(x)$ and $g(x)$ liebelow the line $y = 1$.
D. For $x < 0$, both the graphs of $f(x)$ and $g(x)$ lie below the line $y = 1$.

53. Use the following table of logarithm to answer the question below it.

n	0	1	2	3	4	5	6	7	8	9
1.5	0.1761	0.1790	0.1818	0.1847	0.1875	0.1903	0.1931	0.1959	0.1987	0.2014
1.6	0.2041	0.2068	0.2095	0.2122	0.2148	0.2175	0.2201	0.2227	0.2253	0.2279
1.7	0.2304	0.2330	0.2355	0.2380	0.2405	0.2430	0.2455	0.2480	0.2504	0.2529
1.8	0.2553	0.2577	0.2601	0.2625	0.2648	0.2672	0.2695	0.2718	0.2742	0.2765
1.9	0.2788	0.2810	0.2833	0.2856	0.2878	0.2900	0.2923	0.2945	0.2967	0.2989
7.0	0.845	0.845	0.84	0.847	0.847	0.848	0.848	0.849	0.850	0.85

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	1	7	63	0	6		2	8	4	0	06
7.1	0.8513	0.8519	0.8525	0.8531	0.8537		0.8543	0.8549	0.8555	0.8561	0.8567
7.2	0.8573	0.8579	0.8585	0.8591	0.8597		0.8603	0.8609	0.8615	0.8621	0.8627
7.3	0.8633	0.8639	0.8645	0.8651	0.8657		0.8663	0.8669	0.8675	0.8681	0.8686
7.4	0.8692	0.8698	0.8704	0.8710	0.8716		0.8723	0.8727	0.8733	0.8739	0.8745

Which of the following is the approximate value of $\sqrt[4]{72300}$?

A. 15.4 B. 17.4 C. 14.7 D. 16.4

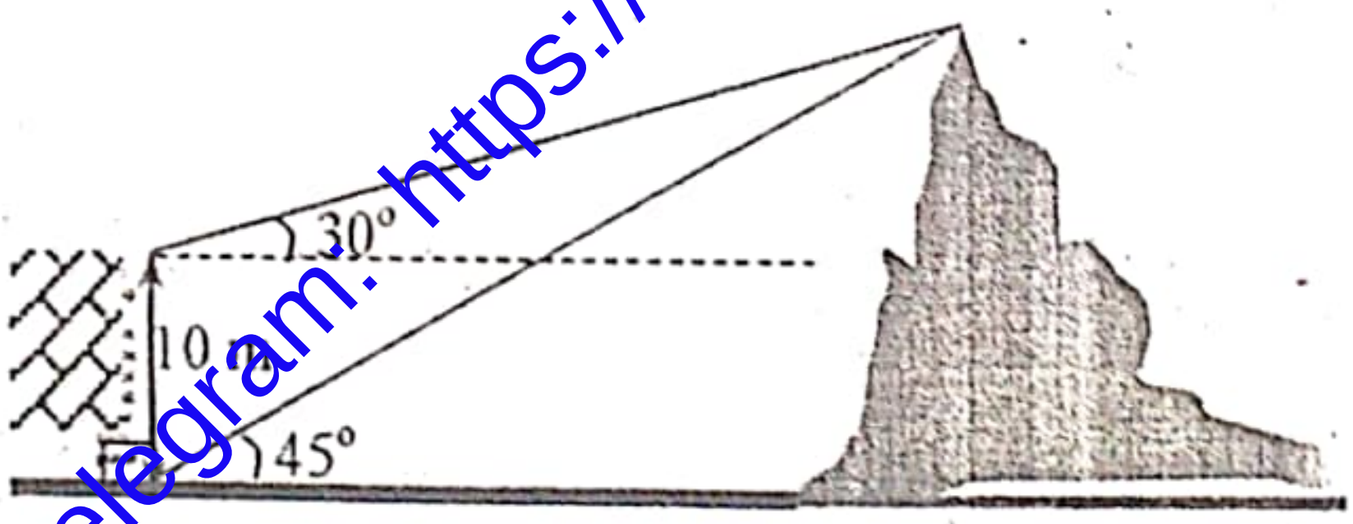
54. Let θ be a second quadrant angle in standard position whose terminal side lies on the line $y = -3x$. Then which of the following is equal to $\cos\theta$?

A. $-\frac{3}{\sqrt{10}}$ B. $\frac{1}{\sqrt{10}}$ C. $-\frac{1}{\sqrt{10}}$ D. $\frac{3}{\sqrt{10}}$

55. Let $f(\theta) = 1 + \sin\theta$. Which one of the following is NOT true about $f(\theta)$?

A. The domain of $f(\theta)$ is $(-\infty, \infty)$.
B. Range of $f(\theta)$ is $[0, 2]$.
C. $f(\theta) \geq 0$ for each values of θ.
D. The period of $f(\theta)$ is 2π.

56. As shown in the figure below, the angle of elevation of the top of the cliff when seen from the foot and top of a building are 45^o and 30^o respectively.



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If the height of the building is 10 m, then how high is the top of the cliff?

A 14.3 m B. 24.3 m C. 35.0 m D. 56.7 m

57. In a certain company, data of the workers' salary in Birr is given as shown below.

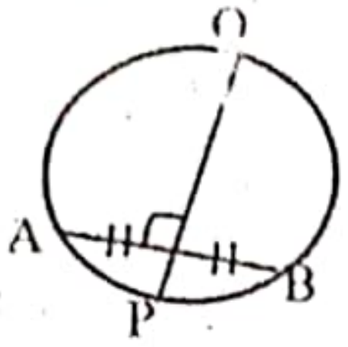
4700	5600	5000	4100	5300	4700	5000
4700	5300	5600	5300	4100	5300	4700
5000	5300	5300	5600	5000	5300	

Which one of the following is **NOT** true about this data?

A. Only two values have the same frequency.
B. The most frequent salary is 5300 Birr.
C. 20% of the workers receive the least salary.

D. &0% of the workers receive a salary more than 4900 Birr.

58. In the figure below, the circle has radius 5 cm and the length of the chord $\overline{AB}$ is 6 cm. Moreover $\overline{PQ}$ is perpendicular bisector of $\overline{AB}$.



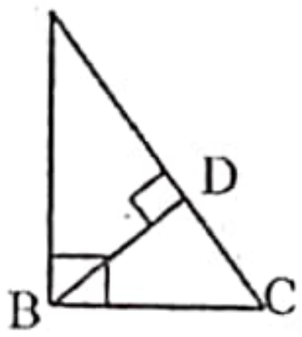
Which of the following equals the distance from Q to $\overline{AB}$?

A. $\sqrt{11}$ cm
B. 8 cm
C. $6\sqrt{2}$ cm
D. 9 cm

59. The base of a right prism is right angled triangle with legs 6 cm and 8 cm. If the altitude of the prism is 7 cm, then the total surface area is:

A. 216 cm² B. 182 cm² C. 158 cm² D.48 cm²

60. In right angled triangle ABC below, $\overline{BD}$ is altitude to hypotenuse $\overline{AC}$.



If AD=16 cm and DC=9 cm, then BD equals to:

A. 9 cm
B. 12 cm
C. 16 cm
D. 7 cm

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BOOKLET CODE: 019

1 Whencot(-340^o) is expressed in terms of an acute angle, it is equal to
A. $\cos 20^o$ B. $\sin 20^o$ C.$\cot 20^o$ D. $\tan 20^o$

2. Consider the function $f(\theta)= \cos\theta$, where $0\leq \theta \leq \pi$. Then, range of f is
A. $[\frac{1}{2}, 1]$ B. [0, 1] C. [-1, 0] D. [-1, 1]

3. If $\theta = -60^o$, then which of the following is true?
A. $\sin\theta = -\frac{1}{2}$ B. $\tan\theta = -\sqrt{3}$ C. $\cos\theta = -\frac{1}{2}$ D. $\sin\theta = \frac{\sqrt{3}}{2}$

4. Which one of the following is NOTa third quadrant angle?
A. 550^o B. 260^o C. -185^o D. -100^o

5. A regular octagon is inscribed in a circle of radius 12 cm as shown in the figure below.

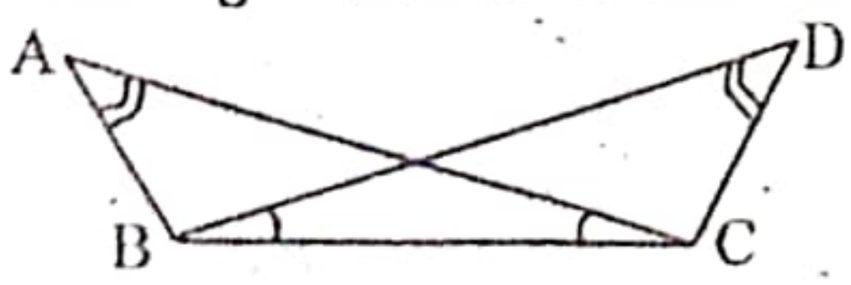
What is the area of the shaded region (the region between the circle and the octagon)?
A. $144(2\sqrt{3} - \pi)$ sq.cm C. $144(\pi - \sqrt{3})$ sq.cm
B. $144(4\sqrt{2} - \pi)$ sq.cm D. $144(\pi - 2\sqrt{2})$ sq.cm

6. On coordinate plane ΔPQR has vertices P(-1, 0), Q(3, 0) and R(2, 3). The distance from the point of concurrency of medians to vertex R is
A. $\frac{1}{3}\sqrt{10}$ units B. $\frac{1}{2}\sqrt{10}$ units C. $\frac{2}{3}\sqrt{10}$ units D. $\frac{3}{4}\sqrt{10}$ units

7. Let F_{30} ={1, 2, 3, 5, 6, 10, 15, 30} and F_{42} ={1, 2, 3, 6, 7, 14, 21, 42} be the set of all factors of 30 and 42 respectively. What is the greatest common factor of 30 and 42?
A. 42 B. 30 C. 6 D. 3

8. Which of the following is the radian measure of angle of 210^o?
A. $\frac{7}{3}\pi$ radians B. $\frac{6}{7}\pi$ radians C. $\frac{7}{6}\pi$ radians D. $-\frac{5}{6}\pi$ radians

9. In the figure below ∠BCA≡∠CBD and ∠BAC≡∠CDB.



Based on the given conditions in the figure above, which one of the following theorems can be used to prove that ΔABC≡ ΔDCB?
A. AA B. SSA C. SAS D. ASA

10. A regular polygon in which the measure of each interior angle is equal to the measure of each central angle is
A. regular hexagon C. regular pentagon
B. square D. equilateral triangle

11. The solution set of the inequality $2-x - x^2 < 0$ is given by
A. $(-\infty, 1) \cup (-2, \infty)$ C. $(-\infty, -2) \cup (1, \infty)$

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B. $(-2, 1)$ D. $(-\infty, 1) \cup (2, \infty)$

12. In which of the following intervals does a zero of the polynomial function $f(x) = x^3 + 2$ exist?
A. $(-1, 1)$ B. $(1, 2)$ C. $(-1, 0)$ D. $(-2, -1)$

13. A fair coin will be tossed three times. What is the probability of getting exactly two heads?
A. $\frac{5}{8}$ B. $\frac{3}{8}$ C. $\frac{1}{4}$ D. $\frac{1}{2}$

14. Let D $=\{0, 1, 2, 4\}$ and R $=\{(x,y): x, y \in D \text{ and } y < x\}$. From the following ordered pairs which one belongs to R?
A. $(4, 4)$ B. $(0, 4)$ C. $(1, 3)$ D. $(2, 0)$

15. What is the value of x for which $9^{x-1} + 6 = 3^{2x-1}$ is true?
A. $\frac{3}{2}$ B. 1 C. $\frac{1}{2}$ D. 0

16. $\csc(-405^o)$ is equal to
A. - 2 B. $-\sqrt{2}$ C. 2 D. $\sqrt{2}$

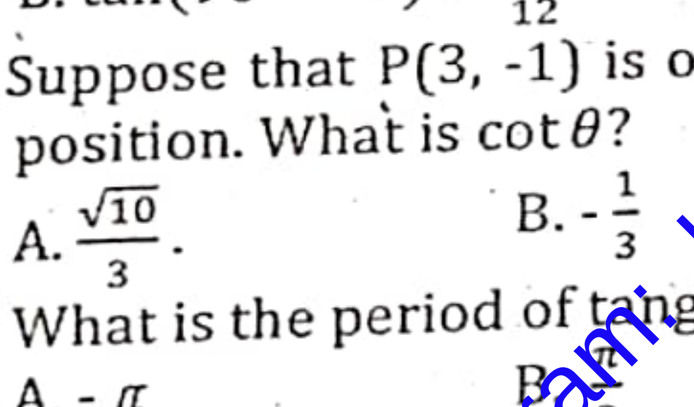
17. If θ is an angle in first quadrant such that $\sin\theta = \frac{5}{13}$, then which of the following is true?
A. $\cos(90^o - \theta) = -\frac{5}{13}$ C. $\sin(90^o - \theta) = \frac{5}{12}$
B. $\tan(90^o - \theta) = \frac{5}{12}$ D. $\cos(90^o - \theta) = \frac{5}{13}$

18. Suppose that P(3, -1) is on the terminal side of an angle θ in standard position. What is $\cot\theta$?
A. $\frac{\sqrt{10}}{3}$. B. $-\frac{1}{3}$ C. $-\frac{\sqrt{10}}{3}$ D. -3

19. What is the period of tangent function?
A. $-\pi$ B. $\frac{\pi}{2}$ C. π D. 2π

20. For an acute angle θ, the simplified form of $\cot\theta \cos(\frac{\pi}{2} - \theta)$ is
A. $\csc\theta$ B. $\sec\theta$ C. $\sin\theta$ D. $\cos\theta$

21. At a point P on a level ground, the angle of elevation of the top of a building is 60^o. At a point Q which is 10 meters away from P, the angle of elevation of the top of this building is found to be 30^o, as shown in the figure below.



What is the distance between point P and the foot of the building?
A. 15 meters C. 7 meters
B. $10\sqrt{3}$ meters D. 5 meters

22. Which of the following is the value of $\sin 270^o$?
A. 1 B. 0 C. - 0.5 D. - 1.

23. Let $f(x) = x^2 + 4x + 1$. The graph of f is obtained by shifting graph of
A. $g(x) = (x + 2)^2$ by 1 unit downward.

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B. $g(x)=(x+2)^2$ by 1 unit upward.
C. $g(x)=(x+2)^2$ by 3 units downward.
D. $g(x)=(x+2)^2$ by 3 units upward.

24. Let $R=\{(x, y): x, y \in \mathbb{R} \text{ and } y < 2x - x^2\}$ be a relation. What is the range of R?
A. $\{y \in \mathbb{R}: y < 2\}$
B. $\{y \in \mathbb{R}: y > 2\}$
C. $\{y \in \mathbb{R}: y < 1\}$
D. $\{y \in \mathbb{R}: y > 1\}$

25. Which one of the following lines is parallel to the line with equation $y + 2x = 1$?
A. A line having the equation, $\frac{1}{2}y - x = 0$
B. A line having the equation, $2y = 1 - x$
C. The line which contains points (2, -1) and (0, -3)
D. The line which contains points (2, -1) and (0, 3)

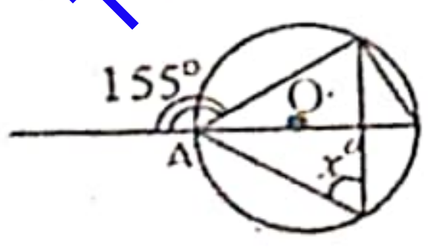
26. Coordinates of the vertices of a triangle are (-1, 1), (2, 1) and (-1, 3). What is the length of the longest side of this triangle?
A. $\sqrt{20}$ units
B. $\sqrt{13}$ units
C. 3 units
D. $\sqrt{8}$ units

27. Let A and B be two sets. If $A \setminus B = \emptyset$, then which of the following is always true?
A. $B \setminus A = A \setminus B$
B. $A \cup B = B$
C. $B \setminus A = \emptyset$
D. $A \cap B = B - A$

28. Let $a > 0$ and n be a natural number. Which one of the following is equal to $\sqrt{a}\sqrt[n]{a}$?
A. $\sqrt[n+2]{a^{2n}}$
B. $\sqrt[2n]{a^{n+2}}$
C. $\sqrt[2n]{a^2}$
D. $\sqrt[2n]{a}$

29. Which one of the following is a prime number?
A. 77
B. 71
C. 57
D. 51

30. For the circle shown below, O is its center and $\overline{AB}$ is its diameter.



If $m(\angle CAD)=155^\circ$ and $m(\angle AED)=x^\circ$, then which one of the following is equal to x?
A. 65
B. 45.5
C. 32.5
D. 25

31. Which of the following is NOT necessarily true about a regular polygon?
A. A cyclic quadrilateral is an example of a regular polygon.
B. It can be inscribed in circle.
C. The measures of all its interior angles are equal.
D. The lengths of all its sides are equal.

32. PQRS is a parallelogram and its perimeter is 32 units. If PQ=6 units long and $m(\angle PQR)=135^\circ$, then what is the area of the parallelogram?
A. 65 sq. units
B. $18\sqrt{2}$ sq. units
C. $30\sqrt{2}$ sq. units
D. 60 sq. units

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33. Let P={-1, 0, 1, 2} and S={0, 2, 3}. Then, the number of elements in (PΔS) equals

A. 5 B. 4 C. 3 D. 2

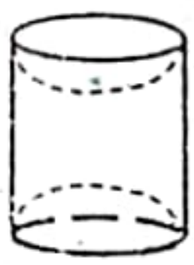
34. Which one of the following sets is equal to $S=\{x \in \mathbb{N}: -1 \leq x \leq 4\}$?

A. {-1, 0, 1, 2, 3}
B. {0, 1, 2, 3}
C. {1, 2, 3, 4}
D. {1, 2, 3}

35. Let $B=\{3, \{3, \sqrt{3}\}\}$ and P(B) be power set of B. Which one of the following is true?

A. $\{\{3, \sqrt{3}\}\} \in P(B)$
B. $\{\sqrt{3}\} \subseteq P(B)$
C. $\{3, \sqrt{3}\} \in P(B)$
D. $\{3\} \subseteq P(B)$

36. A solid is formed by a cylinder with radius 3 cm and height 5 cm and by two hemispheres as shown below.



Which one of the following is the volume of the solid object?

A. 16π cm^3
B. 15π cm^3
C. 13π cm^3
D. 9π cm^3

37. A right circular cone of base radius 4 cm and height 10 cm is cut off by a plane parallel to the base at a distance of 5 cm from the base. What is the volume of the resulting frustum?

A. $\frac{140}{3}\pi$ cubic cm
B. $\frac{28}{3}\pi$ cubic cm
C. $\frac{160}{3}\pi$ cubic cm
D. $16\sqrt{21}\pi$ cubic cm

38. Let a and b be positive real numbers and r and sbe real numbers. Which of the following is necessarily true?

A. $(\frac{b}{a^r})^r = a^{-r}b^r$
B. $\frac{a^r}{b^r} = (ab^{-1})^r$
C. $a^{-r} \times a^s = a^{-r \times s}$
D. $a^r + a^s = a^{r+s}$

39. Let $b > 1$ be real number, $f(x) = \log_b x$ and $g(x) = \log_{\frac{1}{b}} x$. Which of the following statements is true?

A. For $x > 1$, $0 < g(x) < 1$
B. For $x > 1$, $f(x) < 1$
C. For $0 < x < 1$, $g(x) > 0$
D. For $0 < x < 1$, $f(x) > 0$

40. In a laboratory a colony of 100 bacteria grows at a rate of 4% per minutes. Which one of the following expressions gives the time (in minutes) that takes for the colony of the bacteria to reach 10,000?

A. $\log_{1.04} 100$
B. $\log_{10} 1.4$
C. $\log_{100} 1.04$
D. $\log_{1.04} 10$

41. Which of the following is the solution set of the equation $\log_3 x - 2\log_x 9 = 0$?

A. {-9, 9} B. $\{3, \frac{1}{9}\}$ C. $\{9, \frac{1}{9}\}$ D. $\{-9, \frac{1}{9}\}$

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42. The solution set of the equation $(\frac{25}{9})^{x-1} = \sqrt{0.6}$ is

A. $\{\frac{3}{4}\}$ B. $\{\frac{4}{5}\}$ C. $\{\frac{4}{3}\}$ D. $\{\frac{5}{4}\}$

43. Which one of the following is true about the graph of the function $f(x)=1+(\frac{1}{5})^x$?

A. For $x > 0$, it lies between the lines $y = 1$ and $y = 2$

B. The line $y = 0$ is its asymptote

C. It intersects the y-axis at $(0, 1)$

D. It is increasing on $[0, \infty)$

44. Which of the following is the standard deviation of the data given by -2, -1, 1, 3, 4?

A. 5.4 B. 5.2 C. $\sqrt{5.4}$ D. $\sqrt{5.2}$

45. A teacher gave a test to 15 students and she calculated the average score as 7.6. Later on, she found out that the test paper of one student was missed. After she included the missed score, the average score of the 15 students became 8, while the original data for remaining 14 students was given as follows.

Score	5	7	8	9	10
No. of Students	1	3	5	2	3

What was the score on the missed test paper?

A. 10 B. 9 C. 8 D. 6

46. Which one of the following is true about the mean of a given data?

A. Subtracting a constant from each values of a data does not change the mean

B. The sum of the deviation of all values from the mean is zero.

C. It exists for any set of data values.

D. It is not affected by extreme values.

47. In the polynomial division theorem, if $f(x)$ and $d(x)$ are polynomials, then there exist polynomials $q(x)$ and $r(x)$ such that $f(x)= q(x)d(x)+r(x)$.

Which of the following is assumed?

A. The degree of $r(x)$>the degree of $f(x)$

B. The degree of $q(x)>0$

C. The degree of $r(x)$<the degree of $d(x)$

D. The degree of $r(x)$<the degree of $q(x)$

48. A polynomial function is given by: $y = 3x^2 - 3x^4$. Which of the following is NOT true about its graph?

A. As x takes large positive values, $|y|$ becomes large.

B. It touches the x-axs but doesn't cross it at $(0, 0)$.

C. It crosses x-axis two times.

D. It has exactly two turning points.

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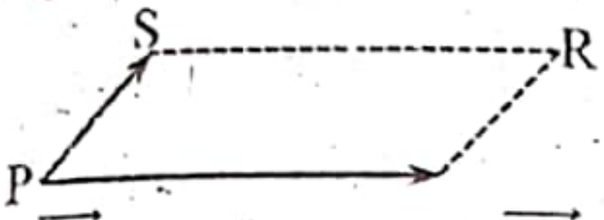
49. Which of the following is NOT true about $f(x)= x(1-x^2)^2 +2x^3$?
A. The constant term is 0
B. Coefficient of x^3 is 2
C. The leading coefficient is 1.
D. The degree of f is 5

50. When $f(x)= x^5 -6x +2$ is divided by $x - c$, the remainder is 7. Which of the following is a possible value of c?
A. 2 B. -2 C. 1 D. -1

51. Consider the parallelogram PQRS which is determined by vector $\overrightarrow{PS}$ and $\overrightarrow{PQ}$ as shown below.

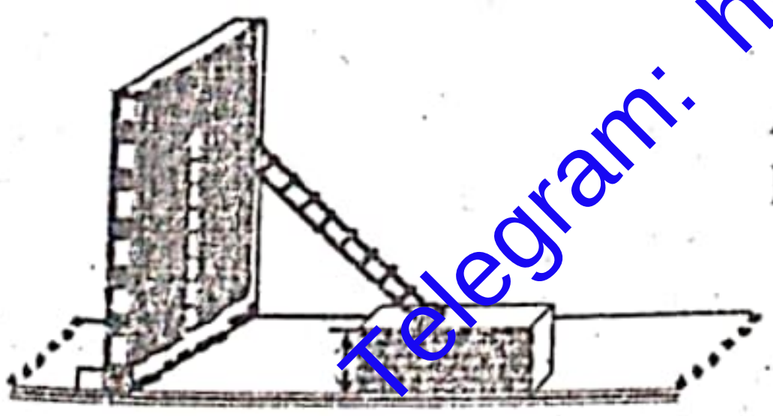


Which of the following is equal to the difference vector $\overrightarrow{PR} - \overrightarrow{PQ}$?
A. $\overrightarrow{SQ}$ B. $\overrightarrow{RQ}$ C. $\overrightarrow{SR}$ D. $\overrightarrow{PS}$

52. Which of the following is the volume of a sphere whose surface area is $144\pi cm^2$?
A. 288π cm^3 B. 268π cm^3 C. 162π cm^3 D. 122π cm^3

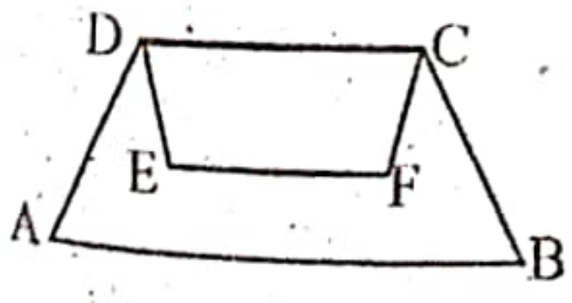
53. The base of a right triangular prism is an isosceles triangle of sides 7 cm, 6 cm, and 7 cm. If the height of the prism is 9 cm, what is the volume of the prism?
A. $54\sqrt{10}$ cm^3 B. $42\sqrt{10}$ cm^3 C. $27\sqrt{10}$ cm^3 D. $21\sqrt{10}$ cm^3

54. From a stand which is 2 meters high a 10 meters long ladder leans against a vertical wall and makes an angle of 60° with the wall as shown in the figure below.



How high (h) is the top of the ladder from the foot of the wall on the ground?
A. $(2+5\sqrt{3})$ meters C. $5\sqrt{3}$ meters
B. 7 meters D. 5 meters

55. In the figure below quadrilaterals ABCD and CDEF are similar and AB=10 cm, CD=8 cm, AC=12 cm and DE=6 cm.



The length CE is equal to
A. 6.4 cm
B. 7.2 cm
C. 7.5 cm
D. 9.6 cm

56. Which of the following is true about the standard deviation of the numbers $x_1, x_2, x_3,...,x_{10}$, with variance σ^2?
A. If each value is divided by $\frac{1}{2}$, the new standard deviation is 2σ.
B. If σ is subtracted from each value, the new standard deviation is 0.

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C. If each value is multiplied by $\frac{1}{4}$, the new standard deviation is $\frac{1}{2}\sigma$.

D. If 3 is added to each value, the new standard deviation is $\sqrt{\sigma}$.

57. Let r be a rational number and t be an irrational number. Which of the following is always an irrational number?

A. $\sqrt{1+r}$ B. $r \times t$ C. $r+t^2$ D. $r-t$

58. Let f be a function defined by $f(x)= \sqrt{x-1}$, and given by $f = \{(a, 1), (b, \sqrt{2}), (c, 2), (d, \sqrt{5})\}$. Then, what is the domain of f?

A. $\{x \in \mathbb{R} | x \geq 1\}$

B. $\{x \in \mathbb{R} | 1 \leq x \leq 10\}$

C. $\{1, 2, 4, 5\}$

D. $\{2, 3, 5, 6\}$

59. Given set $A=\{-1, \emptyset, \{0, -1\}\}$, how many subsets does A have?

A. 3 B. 4 C. 7 D. 8

60. Which of the following is the solution set of the equation $|1-3x| = 2|x-2|$?

A. $\{-3, 1\}$ B. $\{-1, 0\}$ C. $\{-1, 3\}$ D. $\{1, \frac{3}{5}\}$

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ETHIOPIAN GENERAL SECONDARY EDUCATION CERTIFICATE MATHEMATICS EXAMINATION
GINBOT 2009/ JUNE 2017 "Solution and Key"

BOOKLET CODE: 018

1. Let $n(\ )$ means "*the number of elements of...*".

Given: $n(\mathrm{M}) = 25$.

RECALL: "*Two sets* A *and* B *are said to be equivalent*; A≡B *if and only if there is a 1-1 correspondence between elements of A and B*, i.e. A≡B⇔ $n(\mathrm{A}) = n(\mathrm{B})$."

A. $n(\{x: x \text{ is a natural number and } x < 25\}) = 24 \neq 25$
⇔{$x: x$ is a natural number and $x < 25$}≢M

B. $n(\{x: x \text{ is a natural number and } x < 26\}) = 25 = 25$
⇔ {$x: x$ is a natural number and $x < 26$}≡M

C. $n(\{x: x \text{ is a natural number and } x = 25\}) = 1 \neq 25$
⇔ {$x: x$ is a natural number and $x = 25$}≢M

D. $n(\{x: x \text{ is a natural number and } x > 25\}) = \infty \neq 25$
⇔ {$x: x$ is a natural number and $x > 25$}≢M **Answer:-B**

2. Let $n(\ )$ be to say "*the number of elements of...*".

Givens:- $n(\mathrm{E}) = 100$, $n(\mathrm{F}) = 85$ and $n(\mathrm{E}\cap\mathrm{F}) = 35$

Required: Number of students studying any one of E and F, i.e. EΔF
(*Symmetric difference of* E and F)

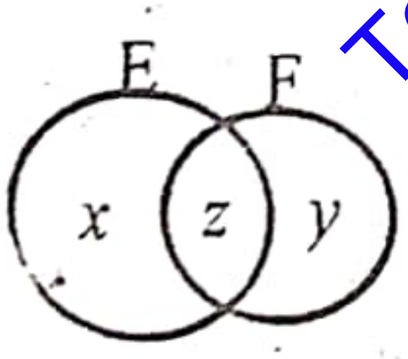
RECALL: i) $n(\mathrm{E}\Delta\mathrm{F}) = n(\mathrm{E}\cup\mathrm{F})\backslash(\mathrm{E}\cap\mathrm{F}) = n(\mathrm{E}\cup\mathrm{F}) - n(\mathrm{E}\cap\mathrm{F})$

ii) $n(\mathrm{E}\cup\mathrm{F}) = n(\mathrm{E}) + n(\mathrm{F}) - n(\mathrm{E}\cap\mathrm{F})$

Then, combining the above we get that

$$n(\mathrm{E}\Delta\mathrm{F}) = n(\mathrm{E}) + n(\mathrm{F}) - 2n(\mathrm{E}\cap\mathrm{F}) = 100 + 85 - 2(35) = 115$$

METHOD II: As shown in the Venn-diagram below:



x= Number of students studying E only
y= Number of students studying F only
z= Number of students studying both E and F
$x+y$= Number of students studying only E or F

Given $x+z=100$, $y+z=85$ and $z=35$. Thus

$\Rightarrow x+y=(100-z)+(85-z)=(100-35)+(85-35)=115$

Answer:- **No answer inthe alternative**

3. *Givens*: Sets F & G such that $\mathrm{F}\cap\mathrm{G}' = \{0, 2; 5\}$ & $\mathrm{F}'\cap\mathrm{G} = \{1, 4\}$

Required: FΔG, i.e. *Symmetric difference of* F & G)

RECALL: i) $\mathrm{F}\Delta\mathrm{G} = (\mathrm{F}\backslash\mathrm{G}) \cup (\mathrm{G}\backslash\mathrm{F})$ ii) $\mathrm{F}\backslash\mathrm{G} = \mathrm{F}\cap\mathrm{G}'$

Thus,by definition of Δ

$\mathrm{F}\Delta\mathrm{G} = (\mathrm{F}\backslash\mathrm{G}) \cup (\mathrm{G}\backslash\mathrm{F})$
$= (\mathrm{F}\cap\mathrm{G}') \cup (\mathrm{G}\cap\mathrm{F}')$; because $\mathrm{F}\backslash\mathrm{G} = \mathrm{F}\cap\mathrm{G}'$ & $\mathrm{G}\backslash\mathrm{F} = \mathrm{G}\cap\mathrm{F}'$
$= (\mathrm{F}\cap\mathrm{G}') \cup (\mathrm{F}'\cap\mathrm{G})$, because ∩ commutative

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$=\{0, 2, 5\} \cup \{1, 4\} = \{0, 1, 2, 4, 5\}$ **Answer:- C**

4. ***Givens***: Dividend polynomial $f(x) = x^5 + x - 1$& divisor $x + 1$

Required: Remainder

RECALL: *"Remainder theorem: If a polynomial $f(x)$is divided by $x - a$then the remainder is equal to $f(a)$."*

Thus, When $f(x) = x^5 + x - 1$ is divided by $x + 1 = x - (-1)$, the remainder is equal to $f(-1) = (-1)^5 + (-1) - 1$

$= -1 - 1 - 1 = -3$ **Answer:- D**

5.

RECALL: *"Exponential expression is an algebraic expression built only form the exponentialsa^x for $1 \neq a > 0$ and $x \in \mathbb{R}$."*

Thus,A. $3x^2 + x + 1$ is polynomial expression

B. $\sqrt{x^2 + 6}$ is square root of polynomial expression

C. $3^x + 2x + 1$ is the combination of exponential expression 3^x and polynomial expression $2x + 1$

D. $2^x + 3^x$ is the combination of exponential expressions 2^x and 3^x which is also exponential expression. **Answer:- D**

6. ***Given***: Set in *set builder* form as: $\{x \in \mathbb{R}: -x < 9$ and $x \neq 5\}$

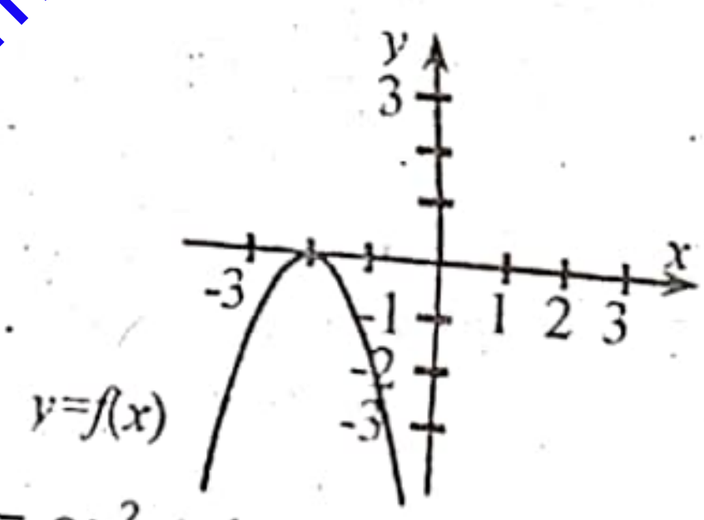
Required: Interval notation of the given set

RECALL: *"$x > a < b \neq x \Leftrightarrow x \in (a, \infty)$, then $x \in (a, b) \cup (b, \infty)$"*

Thus, $-x < 9 \Leftrightarrow x > -9$,; multiplying/dividing both sides by -1

$\Leftrightarrow x \in (-9, \infty)$, & if $x \neq 5 > a$then $x \in (-9, 5) \cup (5, \infty)$

Therefore, $\{x \in \mathbb{R}: -x < 9$ and $x \neq 5\} = (-9, 5) \cup (5, \infty)$ **Answer:- B**



7. ***Given***: Graph of $f(x) = ax^2 + bx + c$ as in figure below.

Required: Solution set of $ax^2 + bx + c < 0$, i.e. $f(x) < 0$

Then, $f(x) < 0$ in the interval of x where the graph of f lies below x-axis, i.e. on set of all real numbers except at -2 which is $(-\infty, \infty)\backslash\{-2\}$.

Answer:- D

8. ***Given***: $\theta = 420^\circ$

RECALL: If $\theta = \alpha + n360^\circ$ for $n \in \mathbb{Z}$; i.e θ and α are *co-terminal*, then θ and α have same trigonometric values.

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Recall that $\theta = 420^o$ is first quadrant angle because
$\theta = 420^o = 60^o + 1(360^o)$ and so 420^o co-terminal with 60^o.

Thus, $\sin\theta = \sin 420^o = \sin 60^o = \frac{\sqrt{3}}{2}$, and

$\cos\theta = \cos 420^o = \cos 60^o = \frac{1}{2}$ **Answer:-B**

9. *Given*: The function $f(x) = 3x - \sqrt{2-x}$

Required: Domain of f

RECALL: Domain of $g(x) - h(x)$ = Domain of $g(x) \cap$ Domain of $h(x)$.

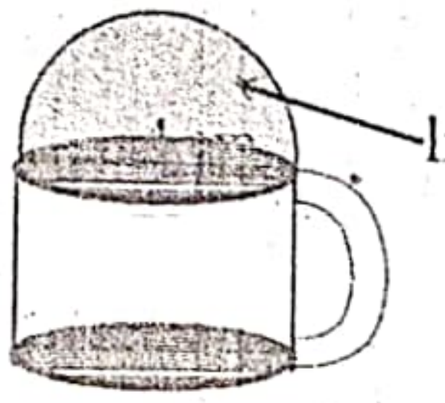
Thus, Domain of $f(x)$ = Domain of $(3x - \sqrt{2-x})$

= Domain of $(3x) \cap$ Domain of $(\sqrt{2-x})$

$= \mathbb{R} \cap \{x \in \mathbb{R}: 2 - x \geq 0\}$

$= \mathbb{R} \cap \{x \in \mathbb{R}: x \leq 2\} = \{x \in \mathbb{R}: x \leq 2\}$ **Answer:-A**

10. *Given*: A cylindrical tea cup of base radius 4 cm and height 10 cm has a hemispherical lid (top cover) as in the figure rigt.



Required: The surface area of the lid part.

RECALL: *Hemispherical* means *semi-sphere*. *Surface area* of a sphere with radius r is $4\pi r^2$ so that the surface area of its hemisphere will be $2\pi r^2$.

The hemispherical lid shares the same radius $r = 4$ cm with the cylindrical part of the tea cup.Thus, the surface area of the lid part is $2\pi r^2 = 2\pi(4\text{ cm})^2 = 32\pi\text{ cm}^2$. **Answer:- B**

11. *Given*: A regular hexagon

Required: Number of lines of symmetries

RECALL: *Lines of symmetry is a line that divides a geometrical figure in to two parts which are mirror image to each other. And an n-sided regular polygon has n lines of symmetry.*

Thus, a regularhexagon as shown in the figure left has 6 lines of symmetry.

Answer:-A

12. *Given*: Any two similar polygons

RECALL: *Two polygons are similar if and only if*

i) Corresponding angles are congruent, and

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ii)Corresponding sides are proportional.

A. False, because, corresponding sides' congruence is not necessary condition for similarity; onlyside proportionality is necessary.
B. True, because, by condition (i) above corresponding angles' congruence is necessary condition for similarity.
C. False, because, corresponding sides' may not be congruent; of course the converse is true, i.e. congruent polygons are similar.
D. False, for instance an equilateral triangle and a square are not similar.

Answer:-B

13. ***Given***: A degree measure 324^o

Required: Radian measure of this angle

RECALL: *The relation between a degree measure and radian measure t of an angle is* $t = \frac{\theta}{180^o}\pi$.

Thus, for $\theta = 324^o$ we have $t = \frac{\theta}{180^o}\pi = \frac{324}{180^o}\pi = \frac{9}{5}\pi$ **Answer:-D**

14. ***Given***: Vectors $\boldsymbol{u} = (2, 5)$ & $\boldsymbol{v} = (-3, 8)$

Required: $\boldsymbol{u} - \boldsymbol{v}$

RECALL: For vectors $\boldsymbol{u} = (a, b)$ & $\boldsymbol{v} = (c, d)$ we have $\boldsymbol{u} - \boldsymbol{v} = (a\text{-}c, b\text{-}d)$.

Thus, for $\boldsymbol{u} = (2, 5)$ & $\boldsymbol{v} = (-3, 8)$ we have $\boldsymbol{u} - \boldsymbol{v} = (2\text{-}(-3), 5\text{-}8) = (5, -3)$

Answer:-B

15. ***Given***: Composite numbers; i.e. 4, 6, 8, 9, 10, 12, 14, 15, 16, 18...

Required: Identifying the common property of all these numbers

RECALL: **Definition:** *"A natural number is composite if it is not prime or 1."*

A. False, for instance 21, 25, 27 ...are all composite but odd.
B. False, for instance 4 has only one prime factor; i.e. 2.
C. False, for instance $4 = 1 \times 2^2$ and so 1 is the only odd factor of 4.
D. True, for instance the smallest composite number is 4 and it has factors 1, 2 and 4.*In general, a composite number m at least has factors 1, m and multiple of any prime numbers.*

Answer:- D

16. ***Given***: A rational number a and an irrational number b

Required: Determining ir/rationality of expressions in a and b

RECALL: *Rational* × *Irrational*=*Rational* or *Irrational*
Irrational × *Irrational*= *Rational* or *Irrational*
Rational ± *Irrational* = *Irrational*
Irrational ± *Irrational* =*Rational* or *Irrational*

A. $\sqrt{3}a + 2b$ = (Irrational×Rational) + (Rational× Irrational)

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$=$(Rational or Irrational) + Irrational

$=$Rational or Irrational$\therefore$ A is not necessarily true.

For instance Let $a = 1$ and $b = -\sqrt{3}/2$. Then

$\sqrt{3}a + 2b = \sqrt{3}(1) + 2(-\sqrt{3}/2) = 0$ which is rational.

B. $b^2 - 3a$ = (Irrational)^2 − (Rational × Rational)

$=$(Rational or Irrational) − Rational

$=$Rational or Irrational $\therefore$ B is not necessarily true.

C. $2(1 + b - a)$ = Rational × (Rational + Irrational − Rational)

= Rational × (Rational + Irrational − Rational)

= Rational × Irrational = Irrational

Thus, $2b - 2(1 + b - a)$ = (Rational × Irrational) − Irrational

= Irrational − Irrational = Rational or Irrational

$\therefore$ C is not necessarily true.

D. $3ab$ is irrational or rational and $3a$ is rational so that

$3ab + 3a$ = (Irrational or Rational) + Rational = Irrational or Rational

$\therefore$ D is not necessarily true.

Answer:- No answer in the alternatives

17. Given: The expression $\sqrt[3]{-0.027} - (2^{-\frac{1}{2}})(1.28)^{0.5}$

Required: The simplified form

RECALL: i) $\sqrt[n]{a} = a^{1/n}$ for n is odd if $a \leq 0$ and even if $a \geq 0$

ii) $a^{-n} = (1/a)^n$ for $a \neq 0$

iii) $a^n b^n = (ab)^n$

First $\sqrt[3]{-0.027} = ((-0.3)^3)^{1/3} = -0.3$, and

$(2^{-\frac{1}{2}})(1.28)^{0.5} = (0.5)^{0.5}(1.28)^{0.5}$

$= (0.5 \times 1.28)^{0.5} = (0.64)^{0.5} = ((0.8)^2)^{0.5} = 0.8$

Thus, $\sqrt[3]{-0.027} - (2^{-\frac{1}{2}})(1.28)^{0.5} = -0.3 - 0.8 = -1.1$ Answer:- A

12. Given: 57.35 corrected to two decimal places

Required: A lower bound of 57.36

Corrected to two decimal place means

$57.36 - 0.005 \leq 57.36 < 57.36 + 0.005 \Rightarrow 57.355 \leq 57.36 < 57.365$

$\Rightarrow$ Lowerbound = 57.355

Answer:- B

13. Given: Polynomial function $p(x) = \frac{x^3-1}{3} - [(x+1)(x^2-2)]$

Required: Leading coefficient & constant term of p

RECALL: For n^{th} degree polynomial

$p(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0$

a_n is the leading coefficient & a_0 is the constant term of p.

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First Changing $p(x)$ in the standard form,

$$p(x) = \frac{x^3 - 1}{3} - [(x+1)(x^2 - 2)]$$

$$= \frac{1}{3}x^3 - \frac{1}{3} - [x^3 + x^2 - 2x - 2] = -\frac{2}{3}x^3 - x^2 + 2x + \frac{5}{3}$$

Thus, leading coefficient of $p = -\frac{2}{3}$& constant term of $p = \frac{5}{3}$ **Answer:- A**

20. *Given*: Dividend:$x^3 + 2x^2 - 5x + 3$ & Divisor:$x^2 + x - 1$

Required: Quotient and Remainder

RECALL: $\frac{\text{Dividend}}{\text{Divisor}} = \text{Quotient} + \frac{\text{Remainder}}{\text{Divisor}}$

Then, by long division $\frac{x^3+2x^2-5x+2}{x^2+x-1} = x + 1 + \frac{-5x+4}{x^2+x-1}$

⇒The quotient is $x + 1$&The remainder is $-5x + 4$ **Answer:- A**

21.*Given*: Polynomials

Required: Identify existence of zeros between two real numbers

RECALL: A polynomial $p(x)$ has at least one zero between a& b if $p(a)p(b) < 0$

A. $f(x) = x^4 - x - 1 \Rightarrow f(-1) = 1$& $f(1) = -1$ so that
$f(-1)f(1) = 1(-1) = -1 < 0 \Rightarrow f$ has zero between -1 and 1

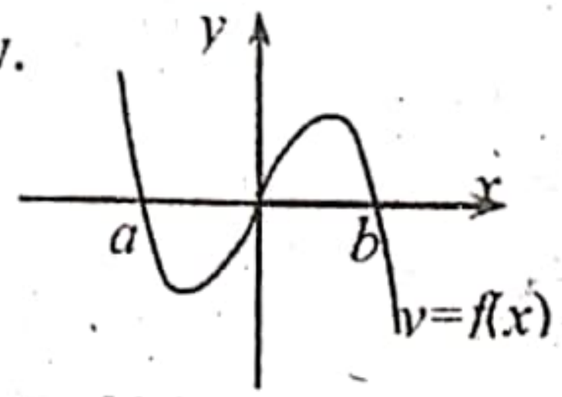
B. $f(x) = x^3 + x - 3 \Rightarrow f(0) = -3$& $f(2) = 7$ so that
$f(0)f(2) = -3(7) = -21 < 0 \Rightarrow f$ has a zero between 0 and 2

C. $f(x) = (x^3 + 1)(x + 1) \Rightarrow f(0) = 1$& $f(1) = 4$ so that
$f(0)f(1) = 1(4) = 4 > 0 \Rightarrow$ **Not sure!** f has a zero between 0 & 1

D. $f(x) = x^3 - 2x + 1 \Rightarrow f(-2) = -3$& $f(-1) = 2$ so that
$f(-2)f(-1) = -3(2) = -6 < 0 \Rightarrow f$ has a zero between -2 and -1

Answer:- C

22. *Given*: Graph of a polynomial as in figure below.



Required: Identify the **NOT** true statement about $f(x)$

A. The graph is above x-axis on $(-\infty, a) \cup (0, b)$
$\Rightarrow f(x) > 0$only for $x \in (-\infty, a) \cup (0, b) \Rightarrow$ A is **NOT** true.

B. The graph is below x-axis on $(a, 0) \cup (b, \infty)$
$\Rightarrow f(x) < 0$only for $x \in (a, 0) \cup (b, \infty) \Rightarrow$ B is **NOT** true.

C. Sincen^{th} degree polynomial may have no zeros or only one zero, the degree of $f(x)$ is not necessarily $3 \Rightarrow$ C is **NOT** true

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D. Since the graph of f crosses x-axis at $x = 0$, Zero is a simple root of $f(x) \Rightarrow$ D is true

Answer:- No answer in the alternatives

23. ***Given***: The system of linear equations $\begin{cases} 3x - 2y = 10 \\ 2x + 2y = 5 \end{cases}$

Required: Identifying the solution type.

RECALL: The system of linear equations $\begin{cases} ax + by = c \\ dx + ey = f \end{cases}$

i. has unique solution if $\frac{a}{d} \neq \frac{b}{e}$

ii. has infinitely many solutions if $\frac{a}{d} = \frac{b}{e} = \frac{c}{f}$

iii. has no solution if $\frac{a}{d} = \frac{b}{e} \neq \frac{c}{f}$

Thus, $\begin{cases} 3x - 2y = 10 \\ 2x + 2y = 5 \end{cases} \Rightarrow \frac{3}{2} \neq \frac{-2}{2} \Rightarrow$ It has unique solution **Answer:- C**

24. ***Given***: The equation $2x^2 - a = 3x$ with one of its solution $r_1 = -1$.

Required: The other solution, r_2.

RECALL: If r_1 and r_2 are the solution of $ax^2 + bx + c = 0$, then

i. $r_1 + r_2 = -\frac{b}{a}$ ii. $r_1 r_2 = \frac{c}{a}$

Then, $2x^2 - a = 3x$ with one of its solution $r_1 = -1$

$\Rightarrow 2(-1)^2 - a = 3(-1) \Rightarrow a = 5$

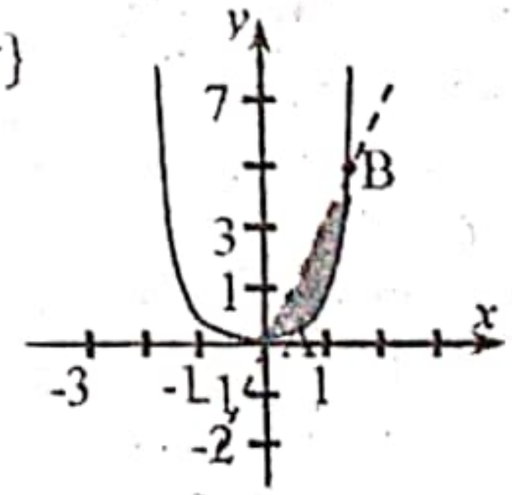
Now the equation will be $2x^2 - 5 = 3x$, i.e. $2x^2 - 3x - 5 = 0$

Thus, $r_1 = -1$, $a = 2, b = -3$ and $c = -5$. Then,

By $r_1 + r_2 = -\frac{b}{a}$ we get $r_2 = -\frac{b}{a} - r_1 = -\frac{-3}{2} - (-1) = \frac{5}{2} = 2.5$

Or by $r_1 r_2 = \frac{c}{a}$ we get $r_2 = \frac{c}{ar_1} = \frac{-5}{2(-1)} = \frac{5}{2} = 2.5$ **Answer:- B**

25. ***Given***: A relation $R = \{(x, y): y \geq x^2 \text{ and } y < 2x\}$ and its graph as



Required: Domain and Range of R.

RECALL: Domain of a relation is interval of x along its region horizontally.

Range of a relation is the interval of y along its region vertically

First we find points A and B as in the figure above solving $x^2 = 2x$

$\Rightarrow x^2 - 2x = 0 \Rightarrow x(x - 2) = 0 \Rightarrow x = 0 \text{ or } x = 2$

So $x = 0 \Rightarrow y = x^2 = 0^2 = 0 \& x = 2 \Rightarrow y = x^2 = 2^2 = 4$

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Thus, A=(0, 0) and B=(2, 4) and since one of the boundaries is broken curve A and B are not in R so that no "or equal to" sign and hence

Domain of R is $\{x \in \mathbb{R}: 0 < x < 2\}$ &

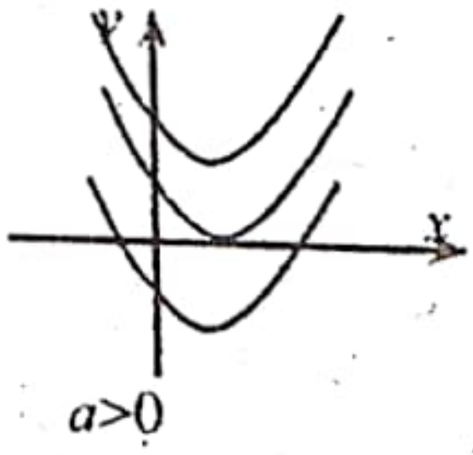
Range of R is $\{y \in \mathbb{R}: 0 < y < 4\}$ **Answer:- B**

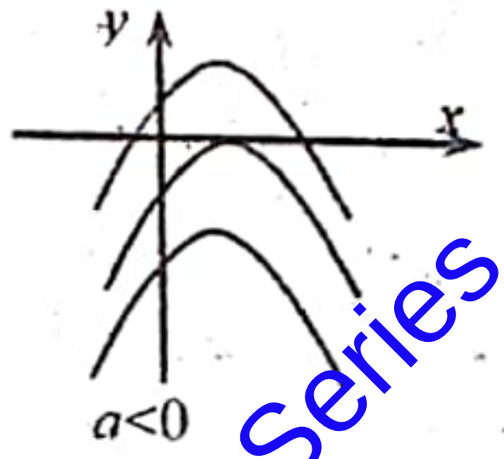
26. ***Given***: Graph of a quadratic function $f(x) = ax^2 + bx + c$

Required: Determining behavior of graph of f.

Consider the possible graphs of $f(x) = ax^2 + bx + c$ as in figure below.

RECALL:Possible graphs of $f(x) = ax^2 + bx + c; a \neq 0$





A. Since $x = 0 \in$ domain of f and $f(0) = c$, clearly f always has a y-intercept at a point $(0, c)$. (seein the figure above) ∴ A is True

"Actually every polynomial has a y-intercept."

B. As in figure above f has both increasing and decreasing intervals.

i.e. i) f decreases in $(-\infty, -b/2a)$ & increases in $(-b/2a, \infty)$ if $a > 0$.

ii) f increases in $(-\infty, -b/2a)$ & decreases in $(-b/2a, \infty)$ if $a < 0$.

∴ B is False

C. As in figure above f may touch x-axis twice or once or no time.

i.e. i) f crosses x-axis twice if $b^2 - 4ac > 0$.

ii) f touches x-axis once if $b^2 - 4ac = 0$.

iii) f doesn't touches x-axis if $b^2 - 4ac < 0$.

∴ C is False

D. Using C, f may havetwo x-intercepts or not at all.∴ D is False

Answer:- A

27. ***Given***: A set A=$\{x:|x - 1| = 1\}$

Required: The power set of A.

RECALL: Power set A denoted by P(A) or 2^A is the set of all subsets of A.

First $|x - 1| = 1 \Rightarrow x - 1 = \pm 1 \Rightarrow x = 0$ or $x = 2$ so that

A=$\{x:|x - 1| = 1\}$=$\{0, 2\}$

Then subsets of A are: $\emptyset$, $\{0\}$, $\{2\}$ and $\{0, 2\}$

Thus, P(A)= $\{\emptyset, \{0\}, \{2\}, \{0, 2\}\}$

28. ***Given***: For $a > 0$& $a \neq 1$, $f(x) = a^x$ and $g(x) = \log_a x$ **Answer:- D**

Required: Determining the relation between f and g.

RECALL: $f(x) = a^x$ and $g(x) = \log_a x$ are **inverses** to each other.

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Thus, Domain of f =Range of g& Domain of g =Range of f

Answer:- D

29. ***Given***: An equation $3^{6(x-2)} = 9^{x+2}$

Required: The solution set.

RECALL: Whenever defined i. $(a^m)^n = a^{mn}$ and ii. $a^n = a^m \Leftrightarrow n = m$

Both $3^{6(x-2)}$ and 9^{x+2} are defined for $x \in \mathbb{R}$.

So by i, $9^{x+2} = (3^2)^{x+2} = 3^{2x+4}$ so that by ii

$$3^{6(x-2)} = 9^{x+2} \Rightarrow 3^{6(x-2)} = 3^{2x+4} \Rightarrow 6(x-2) = 2x+4$$
$$\Rightarrow 6x - 12 = 2x + 4$$
$$\Rightarrow 4x = 16 \Rightarrow x = 4$$

Since $x \in \mathbb{R}$, solution set={4}

Answer:- A

30. ***Given***: A frequency distribution table

x	10	11	12	13	14	15
f	2	0	t	2	4	2

, and mean 12.8.

Required: Frequency of 12.

RECALL: Mean= $\frac{\sum_1^n f_k x_i}{\sum_1^n f_k} = \frac{f_1x_1+f_2x_2+\cdots+f_nx_n}{f_1+f_2+\cdots+f_n}$

Let Frequency of 12 be t as in the table above.

Then, $12.8 = \frac{2(10)+0(11)+t(12)+2(13)+4(14)+2(15)}{2+0+t+2+4+2}$

$$\Rightarrow 12.8 = \frac{12t+132}{t+10} \Rightarrow 12.8t + 128 = 12t + 132 \Rightarrow t = 5$$

∴ Frequency of 12 is 5

Answer:- B

31. ***Given***: Statements about mode of data set.

Required: Determining the true statement about the Mode.

RECALL: Definition: Mode is the most frequent value/item in qualitative or quantitative data set.

Thus, based on the above definition

A. It can't be the value out of the data set. ∴ A is False

B. A data set may have two or more equally most frequent values so that it isn't uniquely determined.

For instance: The data set 1, 2, 2, 4, 5, 5 has mod 2, 5. ∴ B is False

C. Mode measureslocation for both qualitative & quantitative data.

For instance:

The qualitative data set: Red, White, Black, Red ⇒ Mode= Red

The quantitative data set: 1, 2, 2, 4, 5 ⇒ Mode= 2. ∴ C is True

D. This is median's nature in arrayed data set. ∴ D is False **Answer:- C**

32. ***Given***: Statements about arithmetic mean of data set.

Required: Determining true statement about the arithmetic mean.

RECALL: Definition: Mean is the average value of quantitative

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data set.

Thus, based on the above definition

A. There is only one average value of a data set so that mean is uniquely determined. ∴ A is False

B. Mean can be the value out of the data set..

For instance: Consider the data set 1, 2, 4, 5. Then, the mean is $\bar{x} = \frac{1+2+4+5}{4} = 3$, but 3 is not in the data set.∴ B is False

C. Since it is the average value of the data set, we do need all the data values to calculate mean.∴ C is False

D. The arithmetic mean is affected by extreme values.

For instance: For the data set 1, 2, 4, 5 the mean is $\bar{x} = \frac{1+2+4+5}{4} = 3$,

For the data set 1, 2, 4, 57 the mean is $\bar{x} = \frac{1+2+4+57}{4} = 16$

∴ D is True

Answer:- D

33. *Given*: A fair coin will be tossed three times.

Required: The probability that at least two heads show up.

RECALL: $p(E) = \frac{n(E)}{n(S)}$ is probability of an event E from a sample space S.

In tossing a coin three times the sample space is

S={HHH, HHT,HTH, HTT, THH, THT, TTH, TTT} with $n(S) = 8$

The event "at least two heads show up" is

E={HHH, HHT,HTH, THH} with $n(E) = 4$

Thus, $p(E) = \frac{n(E)}{n(S)} = \frac{4}{8} = \frac{1}{2}$

Answer:- A

34. *Given*: Trigonometric inequalities.

Required: Identifying the true inequality.

RECALL: 1) $\sin\theta > 0$, $\cos\theta > 0$, $\tan\theta > 0$ if θ is in quadrant I

2) $\sin\theta < 0$, $\cos\theta < 0$, $\tan\theta > 0$ if θ is in quadrant III

3) $\cos(-\theta) = \cos\theta$

Thus,

A. -100^o is in quadrant III⇒ $\sin(-100^o) < 0$∴ A is True

B. $\cos(-180^o) = \cos 180^o = -1 < 0$∴ B is False

C. 370^o is in quadrant I⇒ $\tan 370^o > 0$∴ C is False

D. 200^o is in quadrant III⇒ $\cos 200^o < 0$∴ D is False **Answer:- A**

35. *Given*: $\sin\alpha = r$ and $\alpha + \theta = 90^o$.

Required: Identifying the true equation.

RECALL: 1) [*Co-function identity*] If $\alpha + \theta = 90^o$ for acutes $\alpha\&\theta$, then

$$\cos\theta = \sin\alpha \& \sin\theta = \cos\alpha$$

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2) [*Pythagorean identity*] $\sin^2\theta + \cos^2\theta = 1$

3) [*Reciprocal identity*] $\sec\theta = \frac{1}{\cos\theta}$

Thus,

A. $\cos\theta = \sin\alpha = r \therefore$ A is False

B. $\cos\theta = \sin\alpha = r \therefore$ B is True

C. $\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\sin^2\alpha} = \sqrt{1-r^2} \neq r \therefore$ C is False

D. $\sec\theta = \frac{1}{\cos\theta} = \frac{1}{\sin\alpha} = \frac{1}{r} \therefore$ D is True

Answer:- B & D

36. ***Given***: $g(\theta) = -\cos\theta$

Required: Identifying period and range of g

RECALL: 1) Period of g is the smallest positive number p if $g(\theta + p) = g(\theta)$

For instance: Period of $y = \sin\theta$ & $y = \cos\theta$ is $p = 2\pi$

2) Range of $\cos\theta$ is $[-1, 1]$; i.e. $-1 \leq \cos\theta \leq 1$

Thus, for $g(\theta) = -\cos\theta$

i) $g(\theta + 2\pi) = -\cos(\theta + 2\pi) = -\cos\theta$ Period of $\cos\theta$ is 2π

$= g(\theta)$ Thus, period of g is 2π

$\therefore$ A is correct and B is incorrect

ii) Multiplying by -1 in all sides;

$-1 \leq \cos\theta \leq 1 \Rightarrow 1 \geq -\cos\theta \geq -1$

$\Rightarrow -1 \leq -\cos\theta \leq 1 \Rightarrow -1 \leq g(\theta) \leq 1$

Thus, range of g is $[-1,1]$ $\therefore$ Both C & D are incorrect. **Answer:- A**

37. ***Given***: $h(\theta) = \tan\theta$.

Required: Identifying the sign and value of h.

RECALL: 1) $\tan 45^o = 1$, $\tan 90^o$ is undefined &$\tan 0^o = 0$

2) $\tan\theta$ is strictly increasing in its domain

, i.e. $\theta_1 > \theta_2 \Rightarrow h(\theta_1) > h(\theta_2)$

3) $\tan(-\theta) = -\tan\theta$

4) $\tan\theta = -\tan(180^o - \theta)$ if θ is in quadrant II

5) Period of $\tan\theta$ is π; i.e. $\tan(\theta + n\pi) = \tan\theta$

Then,

A. $70^o > 45^o \Rightarrow \tan 70^o > \tan 45^o = 1 \Rightarrow \tan 70^o > 1 \therefore$ A is True

B. $20^o < 45^o \Rightarrow -20^o > -45^o \Rightarrow \tan(-20^o) > \tan(-45^o)$

$\Rightarrow \tan(-20^o) > -\tan 45^o = -1$

$\Rightarrow \tan(-20^o) > -1 \therefore$ B is False

C. $\tan(-135^o) = -\tan 135^o = -(-\tan(180^o - 135^o))$

$= \tan 45^o = 1 \therefore$ C is True

D. $\tan(270^o) = \tan(270^o - 180^o) = \tan 90^o$ which is undefined

$\therefore$ D is False

Answer: A & C

38. ***Given***: $\csc(330^o)$.

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Required: Value of $\csc(330^o)$

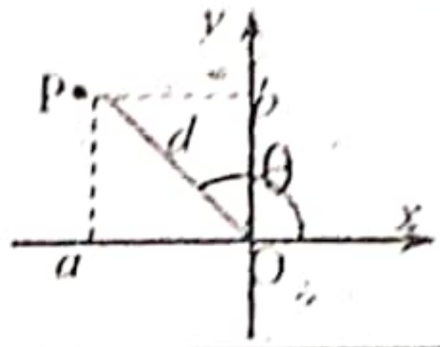
RECALL: 1) $\csc\theta = -\csc(360^o - \theta)$ if θ is in quadrant IV
2) $\sin 30^o = 0.5$

Then, since 330^o is in quadrant IV

$$\csc(330^o) = -\csc(360^o - 330^o) = -\csc(30^o) = \frac{-1}{\sin 30^o} = \frac{1}{0.5} = 2$$

Answer:- C

39. ***Given***: θ as in figure.



Required: Value of $\sec\theta$

RECALL: $\sec = \frac{Hypotenuse}{adjacent}$

Then, $\sec\theta = \frac{Hypotenuse}{adjacent} = \frac{d}{a}$, which is negative since $a < 0$ and $d > 0$

Answer:- C

40. ***Given***: A regular polygon of n sides.

Required: Identifying the wrong statement about the polygon

RECALL: Regular is a convex polygon with equal sides & interior angles.

Central angle $= \frac{2\pi}{n}$; Interior angle $= \frac{(n-2)\pi}{n}$; Exterior angle $= \frac{2\pi}{n}$

Answer:- C

41. ***Given***: Sides lengths of triangle 6, 8 and 10 units.

Required: Area of the triangle

RECALL: $A = \sqrt{(s-a)(s-b)(s-c)}$ for $s = \frac{a+b+c}{2}$

Thus $s = \frac{6+8+10}{2} = 12 \Rightarrow s - a = 6, s - a = 4$ and $s - c = 2$ so that

$$A = \sqrt{(s-a)(s-b)(s-c)} = \sqrt{12(6)(4)(2)} = 24$$

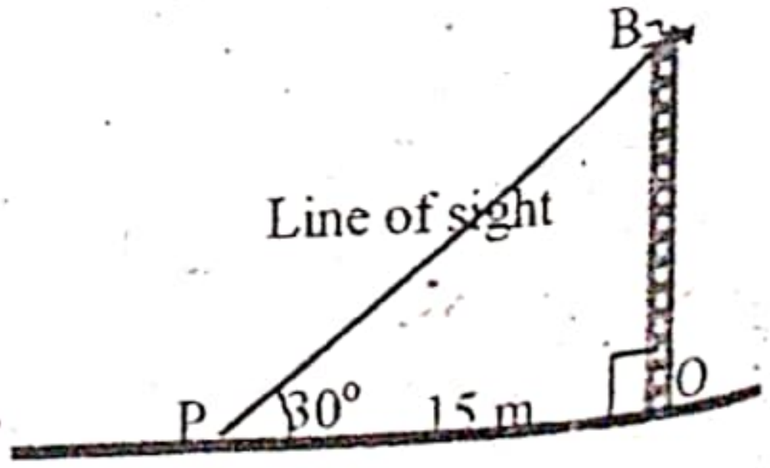
METHOD II: $6^2 + 8^2 = 10^2 \Rightarrow$ right triangle with legs 6 & units8

$$\Rightarrow A = \frac{1}{2}(6)(8) = 24$$

Answer:- B

42. ***Given***: A bird is at the top of a vertical pole posted on a level ground whose line of sight makes 30^o with the ground at point P as in figure below.

Required: Length of the line of sight



RECALL: $\cos = \frac{Adjacent}{Hypotenous}$

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As in figure above, assuming the bird as **point** B where O is the foot of the pole so that right triangleΔBOP, letLength of line of sight be L. Then$\cos 30^o = \frac{Adjacent}{Hypotenous} = \frac{15\text{ m}}{L} \Rightarrow L = \frac{15\text{ m}}{\cos 30^o} = \frac{15\text{ m}}{\sqrt{3}/2} = 10\sqrt{3}\text{ m}$

Answer:- A

43. ***Given***: Points A(b, 4) & B(0, -2) 10 units apart.

Required: Possible value of b

RECALL: Distance formula in a plane$D = \sqrt{(\Delta x)^2 + (\Delta y)^2}$

Thus$D = \sqrt{(\Delta x)^2 + (\Delta y)^2} \Rightarrow 10 = \sqrt{(0-b)^2 + (-2-4)^2}$.

$\Rightarrow 10 = \sqrt{b^2 + 36}$

$\Rightarrow 100 - 36 = b^2 \Rightarrow b = \pm 8$ **Answer:- B**

44.***Given***: A right triangle with vertices P(-1, 1), Q(3, 4) and R(-1, 4).

Required: Slope of the line containing the hypotenuse

RECALL: 1) Hypotenuse is the longest side & using$D = \sqrt{(\Delta x)^2 + (\Delta y)^2}$ 2) Slope in a plane is $m = \frac{\Delta y}{\Delta x}$

Then$PQ = \sqrt{(\Delta x)^2 + (\Delta y)^2} = \sqrt{(3-(-1))^2 + (4-1)^2} = \sqrt{25} = 5$

$PR = \sqrt{(\Delta x)^2 + (\Delta y)^2} = \sqrt{(-1-(-1))^2 + (4-1)^2} = \sqrt{9} = 3$

$QR = \sqrt{(\Delta x)^2 + (\Delta y)^2} = \sqrt{(-1-3)^2 + (4-4)^2} = \sqrt{16} = 4$

Thus$\overline{PQ}$ is the hypotenuse.

METHOD II: Dot product $\overrightarrow{PR} \cdot \overrightarrow{QR} = 0 \Rightarrow \overline{PQ}$ is the hypotenuse.

Therefore, that slope of the line containing $\overline{PQ}$ is $m = \frac{\Delta y}{\Delta x} = \frac{4-1}{3-(-1)} = \frac{3}{4}$

Answer:- A

45. ***Given***: Pairs of equations of lines.

Required: Identifying perpendicularity

RECALL: $Ax + By + C = 0 \perp Dx + Ey + F = 0$ $\Leftrightarrow (-\frac{A}{B})(-\frac{D}{E}) = -1$, i.e.$AD + BE = 0$

Then

A. $2x + 5y + 1 = 0$ and $4x + 10y + 2 = 0 \Rightarrow$A=2, B=5, D=4, E=10 so that
AD+BE=(2)(4)+(5)(10)=58≠0 ⇒Not perpendicular

B. $2x + 5y + 1 = 0$ and $5x + 2y + 1 = 0 \Rightarrow$A=2, B=5, D=5, E=2 so that
AD+BE=(2)(5)+(5)(2)=20≠0 ⇒Not perpendicular

C. $2x - 5y + 1 = 0$ and $5x - 2y + 1 = 0 \Rightarrow$A=2, B=-5, D=5, E=-2 so that
AD+BE=(2)(5)+(-5)(-2)=20≠0 ⇒Not perpendicular

D. $2x - 5y + 1 = 0$ and $5x + 2y + 1 = 0 \Rightarrow$A=2, B=-5, D=5, E=2 so that
AD+BE=(2)(5)+(-5)(2)=0 ⇒**Perpendicular** **Answer:- D**

46. ***Given***: A regular $n = 8$ sided polygon with radius $r = 4$.

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Required: Area

RECALL: Area of a regular n-sided polygon with radius r is $A = \frac{1}{2}nr^2 \sin\frac{360^o}{n}$

Thus, $A = \frac{1}{2}nr^2 \sin\frac{360^o}{n} = \frac{1}{2}8(4)^2 \sin\frac{360^o}{8} = 64\sin 45^o = 64(\frac{\sqrt{2}}{2}) = 32\sqrt{2}$

Answer:- D

47. ***Given***: A regular pyramid with square base of $s = 8$ cm side and the length of the lateral edge is $\ell = 12$ cm.

Required: Total surface area, A_{TS}

RECALL: Total surface area ofa regular pyramid with lateral edge ℓ &a base of regular n-sided polygon whose sides are s is given by:

$$A_{TS} = s^2 + \frac{ns}{4}\sqrt{4\ell^2 - s^2}\,.$$

Thus, for $n = 4$

$$A_{TS} = s^2 + \frac{ns}{4}\sqrt{4\ell^2 - s^2} = 8^2 + \frac{4(8)}{4}\sqrt{4(12)^2 - 8^2} = 64(1+2\sqrt{2})$$

Answer:- A

48. ***Given***: A frustum of height $h = 12$ cm is formed from a right circular cone of height $H = 16$ cm and base radius $R = 8$ cm.

Required: Volume, V

RECALL: Volume of frustum of height h from a right circular cone with height H and bases radii r and R is given by

$$V = \frac{\pi h}{3}(r^2 + rR + R^2) \text{ where } \frac{h}{H} + \frac{r}{R} = 1\,.$$

Then, for $h = 12$ cm, $H = 16$ cm and $R = 8$ cm, first

$$\frac{h}{H} + \frac{r}{R} = 1 \Rightarrow \frac{12}{16} + \frac{r}{8} = 1 \Rightarrow r = 8(1 - \frac{3}{4}) = 2 \Rightarrow r = 2 \text{ cm}$$

Thus $V = \frac{\pi h}{3}(r^2 + rR + R^2) = \frac{\pi(12)}{3}(2^2 + 2(8) + 8^2) = 336\pi \text{ cm}^3$

Answer:- C

49. ***Given***: An iron ball of diameter $d = 6$ cm is immersed in to a water in right circular cylinder with base diameter $D = 10$ cm.

Required: The height change of the water, Δh

RECALL: Volumesof displaced water & immersed object are equal.

Thus, Volume of displaced water in the cylinder= Volume of the ball

$$\Rightarrow \frac{1}{4}\pi D^2 \Delta h = \frac{1}{3}\pi d^2 \Rightarrow \Delta h = \frac{4d^2}{3D^2} = \frac{4(6)^2}{3(10)^2} = \frac{12}{25}$$ **Answer:- A**

50. ***Given***: 30 male & 40 female with total 70 and 60% of totalhave college degree and only 20 male workers have college degree.

Required: Number of female workers not having a college degree

Assign as follows: Male M, Female F, Total T, with degree D

Male with degree be M_D & with no degree be M_{ND}

Female with degree be F_D & with no degree be F_{ND}

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Then, we need F_{ND} so that

$$F_{ND} = F - F_D = F - (D - M_D) = F - (60\%T - M_D)$$
$$= F - (60\%(M + F) - M_D)$$
$$= 40 - (60\%(70) - 20)$$
$$= 40 - (0.6(70) - 20) = 18$$

Answer:- B

51. ***Given***: $\log_{100}(0.01\sqrt{0.01})$

Required: Simplified form

RECALL: Whenever defined

1) i) $\sqrt{x} = x^{\frac{1}{2}}$ ii) $x^a x^b = x^{a+b}$ iii) $(x^a)^b = x^{ab}$
2) i) $\log_a x^k = k \log_a x$ ii) $\log_{a^k} x = \frac{1}{k}\log_a x$ iii) $\log_a a = 1$

Then, $\log_{100}(0.01\sqrt{0.01}) = \log_{10^2}(0.01(0.01)^{\frac{1}{2}})$

$$= \log_{10^2} 0.01^{\frac{3}{2}}$$
$$= \frac{1}{2}\log_{10}(10^{-2})^{\frac{3}{2}}, \text{ by } 10^{-2} = 0.01$$
$$= \frac{1}{2}\log_{10} 10^{-2(\frac{3}{2})}$$
$$= \frac{1}{2}\log_{10} 10^{-3} = \frac{1}{2}(-3)\log_{10} 10 = -\frac{3}{2}$$

Answer:- A

52. ***Given***: $f(x) = (\frac{1}{3})^x$ and $g(x) = (\frac{3}{4})^x$

Required: Determining relative graph position.

RECALL: For $0 < a < b < 1$: we have

1) $a^x < b^x$ if $x > 0$ and $a^x > b^x$ if $x < 0$
2) $0 < a^x < 1$ if $x > 0$ and $a^x > 1$ if $x < 0$

A. For $x < 0, \frac{1}{3} < \frac{3}{4} \Rightarrow (\frac{1}{3})^x > (\frac{3}{4})^x \Rightarrow f(x) > g(x) \therefore$ A is True

For instance: $x = -1 \Rightarrow f(-1) = (\frac{1}{3})^{-1} = 3 \& g(-1) = (\frac{3}{4})^{-1} = \frac{4}{3}$

B. For $x \geq 0, \frac{1}{3} < \frac{3}{4} \Rightarrow (\frac{1}{3})^x \leq (\frac{3}{4})^x \Rightarrow f(x) \leq g(x)$

$\therefore$For $x \geq 0$, the graph of $f(x)$ lies above the graph of $g(x)$ but also concedes at $x = 0 \therefore$ B is False

For instance: $x = 1 \Rightarrow f(1) = (\frac{1}{3})^1 = \frac{1}{3} \& g(1) = (\frac{3}{4})^1 = \frac{3}{4}$ and

$x = 0 \Rightarrow f(0) = 1 \& g(0) = 1$ so that $f(x) = g(x)$

C. For $x > 0, 0 < \frac{1}{3} < 1 \Rightarrow 0 < (\frac{1}{3})^x < 1^x \Rightarrow 0 < f(x) < 1 = y$ and

$$0 < \frac{3}{4} < 1 \Rightarrow 0 < \left(\frac{3}{4}\right)^x < 1^x \Rightarrow 0 < g(x) < 1 = y$$

$\therefore$Graphs of $f(x) \& g(x)$ lie below the line $y = 1 \therefore$ C is True

D. For $x < 0, 0 < \frac{1}{3} < 1 \Rightarrow (\frac{1}{3})^x > 1^x \Rightarrow f(x) > 1 = y$ and

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$$0 < \frac{3}{4} < 1 \Rightarrow (\frac{3}{4})^x > 1^x \Rightarrow g(x) > 1 = y$$

$\therefore$Graphs of $f(x)$& $g(x)$ lie above the line $y = 1 \therefore$ D is False

Answer:- A & C

53.*Given*: Table of logarithm ***But Base is not indicated, Let it be 10!***

n	0	1	2	3	4	5	6	7	8	9
1.5	0.17 61	0.17 90	0.18 18	0.18 47	0.18 75	0.19 03	0.19 31	0.19 59	0.19 87	0.2 14
1.6	0.20 41	0.20 68	0.20 95	0.21 22	0.21 48	0.21 75	0.22 01	0.22 27	0.22 53	0.2 79
1.7	0.23 04	0.23 30	0.23 55	0.23 80	0.24 05	0.24 30	0.24 55	0.24 80	0.25 04	0.2 29
1.8	0.25 53	0.25 77	0.26 01	0.26 25	0.26 48	0.26 72	0.26 95	0.27 18	0.27 42	0.2 65
1.9	0.27 88	0.28 10	0.28 33	0.28 56	0.28 78	0.29 00	0.29 23	0.29 45	0.29 67	0.2 89
7.0	0.84 51	0.84 57	0.84 63	0.84 70	0.84 76	0.84 82	0.84 88	0.84 94	0.85 00	0.8 06
7.1	0.85 13	0.85 19	0.85 25	0.85 31	0.85 37	0.85 43	0.85 49	0.85 55	0.85 61	0.8 67
7.2	0.85 73	0.85 79	0.85 85	0.85 91	0.85 97	0.86 03	0.86 09	0.86 15	0.86 21	0.8 27
7.3	0.86 33	0.86 39	0.86 45	0.86 51	0.86 57	0.86 63	0.86 69	0.86 75	0.86 81	0.8 86
7.4	0.86 92	0.86 98	0.87 04	0.87 10	0.87 16	0.87 22	0.87 27	0.87 33	0.87 39	0.8 45

Required: Approximate value of $\sqrt[4]{72300}$.

RECALL: $\log x = \log_{10} x$
and $x = \text{antilog}(\log x)$

Let $x = \sqrt[4]{72300}$. Then, $\log x = \log \sqrt[4]{72300}$

$$= \frac{1}{4}\log(7.25\times10^4)$$
$$= \frac{1}{4}(\log 7.25 + \log 10^4)$$
$$\approx \frac{1}{4}(0.8591 + 4) \approx 1.2148, \text{ from the table}$$

Thus, $\log x \approx 1.2148 \Rightarrow x = 10^{1.2148} = 10^1 \times 10^{0.2148}$
$$= 10\times\text{antilog}(0.2148)$$
$$= 10\times1.74, \text{ from the table}$$
$$= 17.4$$

Answer:- B

54. *Given*: Second quadrant angle θ with terminal side on $y = -3x$

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Required: Value of $\cos\theta$.

RECALL: $\cos\theta = \frac{x}{r}$; $r = \sqrt{x^2 + y^2}$ and $x < 0$ in second quadrant

Then, $\cos\theta = \frac{x}{r} = \frac{x}{\sqrt{x^2+y^2}} = \frac{x}{\sqrt{x^2+(-3x)^2}}$

$= \frac{x}{\sqrt{10x^2}} = \frac{x}{|x|\sqrt{10}} = \frac{x}{(-x)\sqrt{10}}$ because $x < 0$

$= -\frac{1}{\sqrt{10}}$ **Answer:- C**

55. ***Given***: $f(\theta) = 1 + \sin\theta$

Required: Identifying False on f.

RECALL: 1) Domain of $y = \sin\theta$ is $(-\infty, \infty)$.

2) Range of $y = \sin\theta$ is $[-1, 1]$.

3) Period of $y = \sin\theta$ is 2π.

Then, for $f(\theta) = 1 + \sin\theta$

A. Domain of $y = \sin\theta$ is $(-\infty, \infty)$ $\Rightarrow$ Domain of $1 + \sin\theta$ is also $(-\infty, \infty)$.

$\Rightarrow$ Domain of $f(\theta)$ is $(-\infty, \infty)$

$\therefore$ A is True

B. Range of $y = \sin\theta$ is $[-1, 1] \Rightarrow -1 \leq \sin\theta \leq 1$

$\Rightarrow -1+1 \leq 1 + \sin\theta \leq 1+1$, adding 1

$\Rightarrow 0 \leq f(\theta) \leq 2$

$\Rightarrow$ Range of $f(\theta)$ is $[0, 2]$ $\therefore$ B is True

C. $0 \leq f(\theta) \leq 2 \Rightarrow f(\theta) \geq 0$ for each values of θ; Not! '>'.

For instance: $\theta = \square \Rightarrow f(\square) = 1 + \sin\square = 1+(-1) = 0$ $\therefore$ C is False

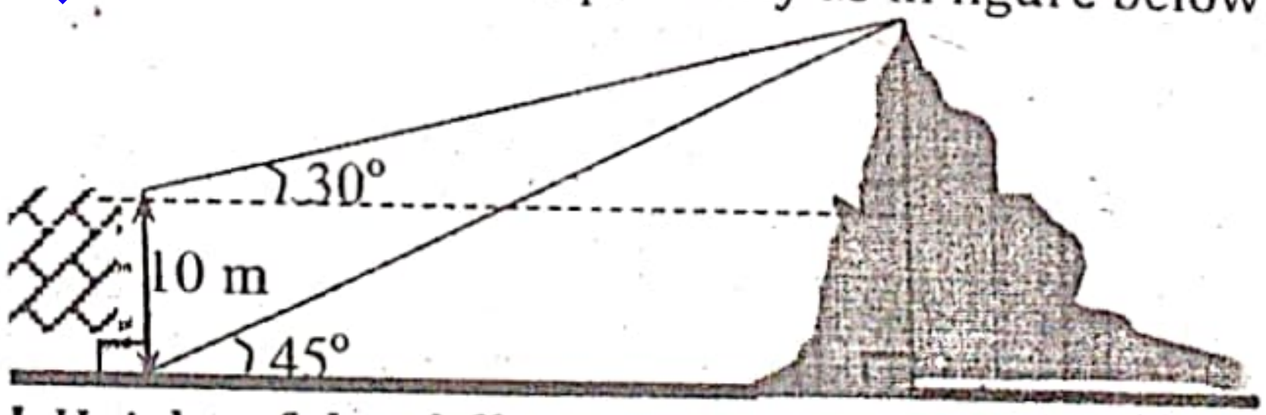
D. For $n \in \mathbb{Z}$; $f(\theta + n2\pi) = 1 + \sin(\theta + n2\pi)$

$= 1 + \sin\theta$, $\leftarrow$ Period of $y = \sin\theta$ is 2π

$= f(\theta) \Rightarrow$ Period of f is 2π $\therefore$ D is True

Answer:- C

56. ***Given***: Elevation angle of top of cliff from foot & top of a 10 m high building are 45^o and 30^o respectively as in figure below



Required: Height of the cliff.

RECALL: $\tan = \frac{opposite}{adjacent}$, $\tan 30^o = \frac{1}{\sqrt{3}}$, $\tan 45^o = 1$

In the given figure above assign points as:

A; Top of building B;Bottom of the building C; Top of cliff

D; Bottom of cliff E; on $\overline{CD}$ so that $\overline{CD} \perp \overline{AE}$

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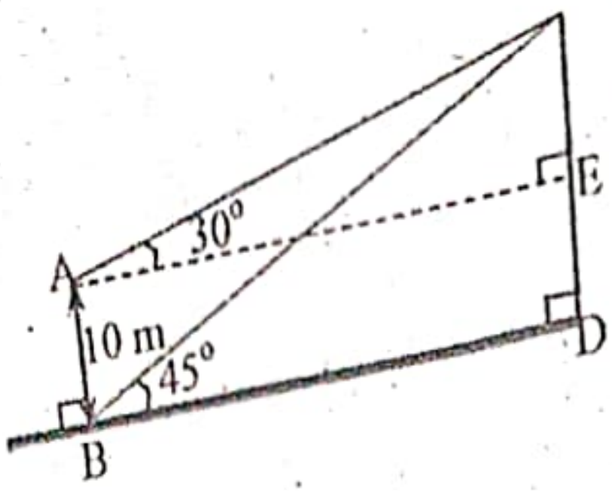

Then, we get the following free diagram. And we need CD.

Note that ED=AB, BD=AE

In right triangle ΔACE

$$\tan 30^o = \frac{CE}{AE} \Rightarrow AE = \frac{CE}{\tan 30^o}$$

$$\Rightarrow AE = \sqrt{3}CE \ldots (1)$$

Now, in right triangle ΔBCD

$$\tan 45^o = \frac{CD}{BD} \Rightarrow CD = BD = AE = \sqrt{3}CE$$

Thus, using $CE = CD - ED = CD - AB \Rightarrow CD = \sqrt{3}(CD - AB)$

$$\Rightarrow CD = \frac{\sqrt{3}AB}{\sqrt{3}-1} = \frac{10\sqrt{3}}{\sqrt{3}-1}$$

Rationalizing the denominator $CD = 15 + 5\sqrt{3}$ m ≈ 23.66 m

Answer:- No answer is given in the alternatives

57. ***Given***: The data set as follows

4700 5600 5000 4100 5300 4700 5000
4700 5300 5600 5300 4100 5300 4700
5000 5300 5300 5600 5000 5300

Required: Identifying False on the data set.

RECALL: Building frequency distribution table

Then, the frequency distribution table will be as follows.

x_i	4100	4700	5000	5300	5600
f_i	2	4	4	7	3

Then, observing $\sum_1^5 f_i = 20$,

A. Only two values have same frequency, i.e. 4700 & 5000. ∴ A is True

B. Highest frequency is 7, i.e. that of 5300 Birr. ∴ B is True

C. The least salary is 4100 Birr with frequency $f_1 = 2$

So, the percentage the workers receive the least salary is

$$\frac{f_1}{\sum_1^5 f_i} \times 100\% = \frac{2}{20} \times 100\% = 10\%$$ ∴ C is False

D. Salary more than 4900 are 5000, 5300 & 5600 with total frequency

$$f_3 + f_4 + f_5 = 4 + 7 + 3 = 14$$

So, the percentage the workers receive salary more than 4900 Birr is

$$\frac{f_3+f_4+f_5}{\sum_1^5 f_i} \times 100\% = \frac{14}{20} \times 100\% = 70\%$$ ∴D is True

Answer:- C

58. ***Given***: Circle with radius 5 cm & length of chord $\overline{AB}$ is 6 cm. $\overline{PQ}$ perpendicular bisector of $\overline{AB}$ as figure below.

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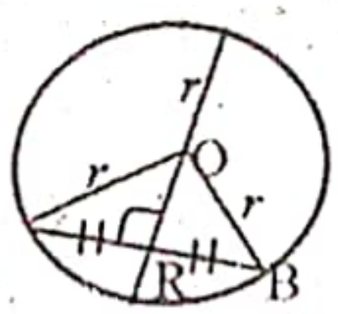


Required: distance fromQ to$\overline{AB}$.

RECALL: Perpendicular bisector of a chord of a circle passes through the center.

Let the center be at point O and $\overline{AB}$ meets $\overline{QP}$ at R. We need QR.

Then, AO=BO=QO=5 cm (Radius) and AR= $\frac{1}{2}$AB=3 cm so that in right triangle ΔOAP by OR= $\sqrt{OA^2 - AR^2} = \sqrt{5^2 - 3^2}$ =4 cm.

Thus, QR=QO+OR=5 cm+ 4 cm= 9 cm

9. ***Given***: Right prism with altitude 7 cmand right angled triangle base of legs 6 cm & 8 cm.

Answer: D

Required: Total surface area

RECALL: The total surface area A_t of right prism of altitude h with base area A_b and base perimeter p is given by$A_t = ph + 2A_b$

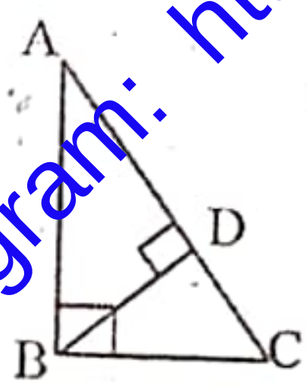
Right triangle base of legs 6 and 8⇒Hypotenuos= $\sqrt{6^2 + 8^2}$ =10

So base area is$A_b = \frac{1}{2}(6)(8)$ =24 cm² and

Lateral face area$A_l = ph$ =(6+8+10)7=168 cm² so that

$A_t = A_l + 2A_b$ =(168 cm²)+2(24 cm²)= 216 cm²

Answer:- A



. ***Given***: Right ΔABC as below,$\overline{BD}$ is altitude to hypotenuse$\overline{AC}$, AD=16 cm and DC=9 cm.

Required: BD

RECALL: In the above triangle ΔABC~ΔADB~ΔBDC by AA similarity.

Then, from ΔADB~ΔBDC by side proportionality and DB= BD

$\frac{AD}{BD} = \frac{DB}{DC} \Rightarrow \frac{16}{BD} = \frac{DB}{9} \Rightarrow BD^2 = 9(16) \Rightarrow BD^2 = \sqrt{9(16)}$ =12 cm

iswer:- B

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1. $y = \cot x$ has period $180^o \Rightarrow \cot x = \cot(x + n(180^o))$, for $n \in \mathbb{Z}$
$\Rightarrow \cot(-340^o) = \cot(-340^o + 2(180^o)) = \cot 20^o$ **Answer:-**
C

2. The function $f(\theta) = \cos\theta$ is strictly decreasing in $[0, \pi]$ so that
$0 \le \theta \le \pi \Rightarrow \cos 0 \ge \cos\theta \ge \cos\pi$
$\Rightarrow 1 \ge \cos\theta \ge -1 \Rightarrow 1 \ge f(\theta) \ge -1 \Rightarrow -1 \le f(\theta) \le 1$
$\therefore$ Range of $f(\theta) = \cos\theta$, where $0 \le \theta \le \pi$ is $[-1, 1]$ **Answer: D**

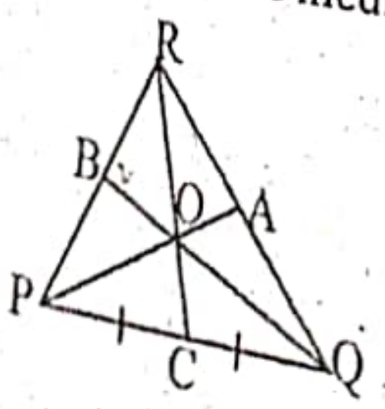
3. A & D. $\sin(-\theta) = -\sin\theta$ so that
$\sin\theta = \sin(-60^o) = -\sin 60^o = -\frac{\sqrt{3}}{2} \leftarrow$ A & D are False
B. $\tan(-\theta) = -\tan\theta$ so that
$\tan\theta = \tan(-60^o) = -\tan 60^o = -\sqrt{3} \leftarrow$ B is True
C. $\cos(-\theta) = \cos\theta$ so that
$\cos\theta = \cos(-60^o) = \cos 60^o = \frac{1}{2} \leftarrow$ C is False **Answer:-**
B

4. θ is quadrant III angle if $\theta + n360^o \in (180^o, 270^o)$ for $n \in \mathbb{Z}$. Then,
A. $n = -1 \Rightarrow 550^o - 360^o = 190^o \in (180^o, 270^o) \Rightarrow 550^o$ is in quadrant III
B. $n = 0 \Rightarrow 260^o \in (180^o, 270^o) \Rightarrow 260^o$ is in quadrant III
C. $n = 1 \Rightarrow -185^o + 360^o = 175^o \notin (180^o, 270^o) \Rightarrow -185^o$ is not in quadrant III
D. $n = 1 \Rightarrow -100^o + 360^o = 260^o \in (180^o, 270^o) \Rightarrow -100^o$ is in quadrant III
Answer:- C

5. Given: A regular octagon inscribed in a circle of radius, $r = 12$ cm as shown in the figure below. Then, for A=Area
A(Shaded region)
$= A(\text{Circle}) - A(\text{Octagon})$
$= \pi r^2 - \frac{1}{2} n r^2 \sin\frac{360^o}{n} = r^2(\pi - \frac{1}{2} n \sin\frac{360^o}{n})$
$= (12\text{cm})^2(\pi - \frac{1}{2}(8)\sin\frac{360^o}{8}) \leftarrow r = 12$ cm; $n = 8$
$= 144(\pi - 4\sin 45^o)$sq.cm
$= 144(\pi - 2\sqrt{2})$sq.cm $\leftarrow \sin 45^o = \frac{\sqrt{2}}{2}$

Answer:- D

6. Given: ΔPQR with vertices P(-1, 0), Q(3, 0) R(2, 3) as in figure below. Let $\overline{RC}$, $\overline{PA}$ & $\overline{QB}$ be medians concurrent at O. Then, $OR = \frac{2}{3}CR$



By mid-point formula
$C(\frac{x_P + x_Q}{2}, \frac{y_P + y_Q}{2}) = C(\frac{-1+3}{2}, \frac{0+0}{2}) = C(1, 0)$
Thus, by distance formula

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$OR = \frac{2}{3}CR = \frac{2}{3}\sqrt{(1-2)^2 + (0-3)^2} = \frac{2}{3}\sqrt{10}$ units

Answer:- C

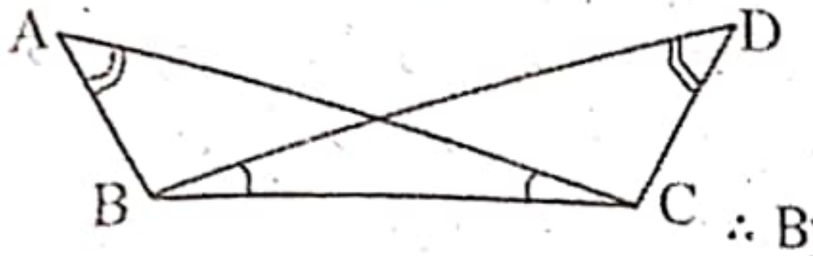
7. $30 = 2\times3\times5$ and $42 = 2\times3\times7$.
   Then, multiplying factors with higher exponent
   $\Rightarrow$ LCM $(30, 42) = 2\times3\times5\times7 = 210$ so that
   $GCF\,(30, 42) = \frac{30\times42}{LCM(30,42)} = \frac{1260}{210} = 6$

Answer:- C

8. Radian measure $= (\frac{\pi}{180^o} \times$ Degree measure) radians
   $= \frac{\pi}{180^o} \times 210^o$ radians $= \frac{7}{6}\pi$ radians

Answer:- C

9. Given figure below with $\angle BCA \equiv \angle CBD$ and $\angle BAC \equiv \angle CDB$.



i) $\angle BCA \equiv \angle CBD$, given
ii) $\angle BAC \equiv \angle CDB$, given
iii) $\overline{BC} \equiv \overline{CB}$, common side
$\therefore$ By ASA, $\Delta ABC \equiv \Delta DCB$

Answer:- D

10. For a regular n- sided polygon
    i) Measure of each interior angle $= \frac{180^o(n-2)}{n}$
    ii) Measure of each central angle $= \frac{360^o}{n}$
    Thus, $\frac{180^o(n-2)}{n} = \frac{360^o}{n} \Rightarrow 180^o(n-2) = 360^o$
    $\Rightarrow n = 4 \leftarrow$ no. of sides $\Rightarrow$ Square

Answer:- B

11. $2 - x - x^2 < 0 \Rightarrow (2+x)(1-x) < 0$
    $\Rightarrow 2+x < 0$ and $1-x > 0$ or $2+x > 0$ and $1-x < 0$
    $\Rightarrow x < -2$ and $x < 1$ or $x > -2$ and $x > 1$
    $\Rightarrow x < -2$ or $x > 1 = (-\infty, -2) \cup (1, \infty)$

Answer:- C

12. $f(x) = x^3 + 2 \Rightarrow f(-2) = (-2)^3 + 2 = -6$ & $f(-1) = (-1)^3 + 2 = 1$
    Since $f(-2) < 0$ and $f(-1) > 0$, by *location theorem*, f has at least one zero between $x = -2$ and $x = -1$, i.e. in the interval $(-2, -1)$.

Answer:- D

13. In tossing a coin three times the sample space is
    S = {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT} with $n(S) = 8$
    The event "*exactly two heads*" is
    E = {HHT, HTH, THH} with $n(E) = 3$
    Thus, $p(E) = \frac{n(E)}{n(S)} = \frac{3}{8}$

Answer:- B

14. Given: D = {0, 1, 2, 4} and R = {(x, y): $x, y \in$ D and $y < x$}.
    Since $0 < 2$ and $0, 2 \in$ D we have $(2, 0) \in$ R

Answer:- D

15. $9^{x-1} + 6 = 3^{2x-1} \Rightarrow 3^{2x-2} + 6 = 3^{2x-1} \Rightarrow \frac{3^{2x}}{3^2} + 6 = \frac{3^{2x}}{3} \leftarrow$ by $a^{x-y} = \frac{a^x}{a^y}$
    Let $3^{2x} = y$. Then, $\frac{3^{2x}}{3^2} + 6 = \frac{3^{2x}}{3} \Rightarrow \frac{y}{9} + 6 = \frac{y}{3} \Rightarrow y = 27$ so that
    $3^{2x} = y = 27 = 3^3 \Rightarrow 2x = 3 \Rightarrow x = \frac{3}{2}$

Answer:- A

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16. $\csc(-405^\circ) = -\csc 405^\circ \leftarrow \text{by} \csc(-x) = -\csc x = -\csc(405^\circ - 360^\circ)$
$= -\csc 45^\circ = -\frac{1}{\sin 45^\circ} \leftarrow \text{by } \csc x = \frac{1}{\sin x}$
$= -\frac{1}{1/\sqrt{2}} = -\sqrt{2}$, Answer:- B

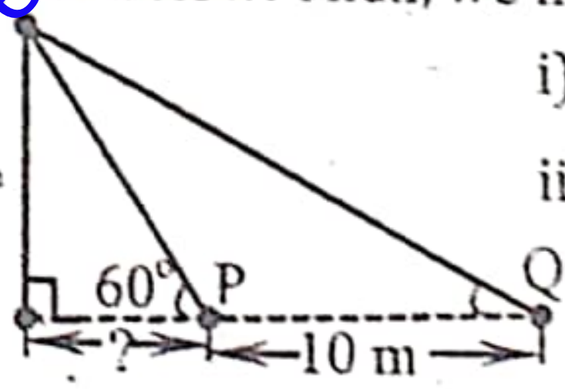
17. Since θ is first quadrant angle it is acute angle so that
1) [*Co-function identities*]
a) $\cos(90^\circ - \theta) = \sin\theta \& \sin(90^\circ - \theta) = \cos\theta$
b) $\cot(90^\circ - \theta) = \tan\theta \& \tan(90^\circ - \theta) = \cot\theta$
c) $\csc(90^\circ - \theta) = \sec\theta \& \sec(90^\circ - \theta) = \csc\theta$
2) [*Pythagorean identity*] $\sin^2\theta + \cos^2\theta = 1$
3) [*Ratio identity*] $\cot\theta = \frac{\cos\theta}{\sin\theta}$
4) Infirst quadrant,allsin θ,cos $\theta$and cot θ are positive.
A. $\cos(90^\circ - \theta) = \sin\theta = \frac{5}{13} \therefore$ A is false & D is true
B. $\tan(90^\circ - \theta) = \cot\theta = \frac{\cos\theta}{\sin\theta}$
$= \frac{\sqrt{1-\sin^2\theta}}{\sin\theta} = \frac{\sqrt{1-(\frac{5}{13})^2}}{\frac{5}{13}} = \frac{12}{5} \therefore$ B is false
C. $\sin(90^\circ - \theta) = \cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{1-(\frac{5}{13})^2} = \frac{12}{13} \therefore$ C is false
Answer:- D

18. $\cot\theta = \frac{x}{y} = \frac{3}{-1} = -3$ Answer:- D

19. $\tan(\theta + n\pi) = \tan\theta$ for all $n \in \mathbb{Z}$; and $n = 1$ is the smallest such positive integer so that the period of tangent function is π. Answer:- C

20. For acute θ, by co-function identity $\cos(\frac{\pi}{2} - \theta) = \sin\theta$.Thus,
$\cot\theta \cos(\frac{\pi}{2} - \theta) = \cot\theta \sin\theta = \frac{\cos\theta}{\sin\theta} \times \sin\theta = \cos\theta$ Answer:- D

21. Let top and bottom of the building be at points S and R respectively as in figure below. Then, we need RP so that



i) In rightΔSRP, $\tan 60^\circ = \frac{SR}{RP} \Rightarrow SR = \sqrt{3}RP$
ii) In right ΔSRQ, $\tan 30^\circ = \frac{SR}{RQ}$
$\Rightarrow \tan 30^\circ = \frac{SR}{RP + PQ}$
$\Rightarrow \frac{\sqrt{3}}{3} = \frac{\sqrt{3}RP}{RP+10} \leftarrow$given PQ=10 m& by (i) $SR = \sqrt{3}RP$
$\Rightarrow RP+10 = 3RP \Rightarrow RP = 5$ m Answer:- D

22. Consider the following unit circle in the figure below.
Terminal side of angle 270°intersects the unit circle at point $(x, y)=(0, -1)$ so that

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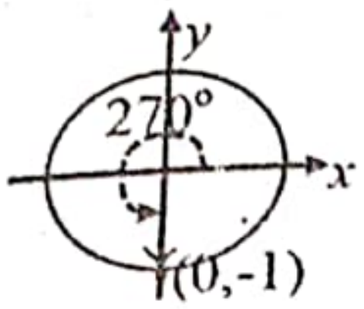


$x=0, y=-1$ and $r=1$ gives

$$\sin 270^o = \frac{y}{r} = \frac{-1}{1} = -1$$

Answer:- D

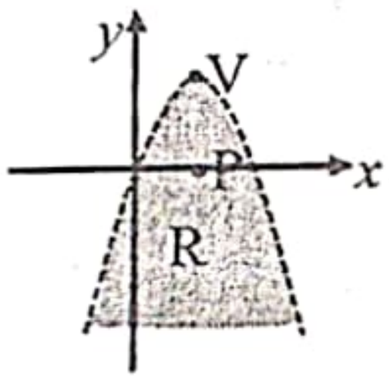
23. Given: functions $f(x)= x^2+4x+1$ and $g(x)= (x+2)^2$

$f(x)= x^2+4x+1= x^2+4x+4-3= (x+2)^2-3=g(x)-3$

$\therefore f(x)=g(x)-3 \Rightarrow$ the graph of f is obtained by shifting the graph of $g(x)$ by 3 units downward.

Answer:- C

24. Given: A relation $R=\{(x, y): x, y \in \mathbb{R} \text{ and } y < 2x - x^2\}$



As shown in the figure in left, the boundary curve is abroken open downwardparabola $y=2x-x^2$ with $a=-1, b=2$ & $c=0$

Testing at a point P(1, 0),

$y < 2x - x^2 \Rightarrow 0 < 2(1) - 1^2 \Rightarrow 0 < 1$ which is true so that graph of R is the shaded region.

Since range of a relation is the interval of y along its region vertically, the range of R is set of real numbers less than y-value at vertex V,

i.e. $\{y \in \mathbb{R}: y < \frac{4ac-b^2}{4a}\} = \{y \in \mathbb{R}: y < \frac{4(-1)(0)-2^2}{4(-1)}\} = \{y \in \mathbb{R}: y < 1\}$

Answer:- C

25. Parallel lines have the same slope.

Slope of the given line $y+2x=1$, i.e. $y=-2x+1$ is $m=-2$

A. Line $\frac{1}{2}y - x = 0$, i.e. $y=2x \Rightarrow$ Slope $=2 \neq m=-2$

B. Line $2y=1-x$, i.e. $y=-\frac{1}{2}x+\frac{1}{2} \Rightarrow$ Slope $=-\frac{1}{2} \neq m=-2$

C. Line through points (2, -1) & (0, -3) $\Rightarrow$ Slope $= \frac{-3-(-1)}{0-2} = 1 \neq m=-2$

D. Line through points (2, -1) & (0, 3) $\Rightarrow$ Slope $= \frac{3-(-1)}{0-2} = -2 = m = 2$

Answer:- D

26. Distance formula gives the distance between points

i) (-1, 1) & (2, 1) $\Rightarrow d = \sqrt{(2-(-1))^2 + (1-1)^2} = \sqrt{9} = 3$

ii) (-1, 1) & (-1, 3) $\Rightarrow d = \sqrt{(-1-(-1))^2 + (1-3)^2} = \sqrt{4} = 2$

iii) (2, 1) & (-1, 3) $\Rightarrow d = \sqrt{(-1-2)^2 + (3-1)^2} = \sqrt{13}$

$\therefore$ Length of the longest side of this triangle is $\sqrt{13}$. **Answer:- B**

27. Given: Two sets A and B such that $A \setminus B = \emptyset$

A. For instance: Let $A=\{0, 1\}$ & $B=\{0, 1, 2\}$
Then, $A \setminus B = \emptyset$ but $B \setminus A = \{2\}$ so that $A \setminus B \neq B \setminus A \Rightarrow$ A is False

B. $A \setminus B = \emptyset \Rightarrow A \subseteq B$ so that $A \cup B = B \Rightarrow$ B is True

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C. For instance: Let A={0, 1} & B={0, 1, 2}
Then, A\B= ∅ but B\A={2} so that B\A≠ ∅ ⇒ C is False

D. For instance: Let A={0, 1} & B={0, 1, 2}
Then, A∩B={0, 1} but B−A={2} so that A∩B≠ B−A⇒ D is False

More: A, B, C & D are all true only if A\B= ∅ where A= ∅ =B.

Answer:- B

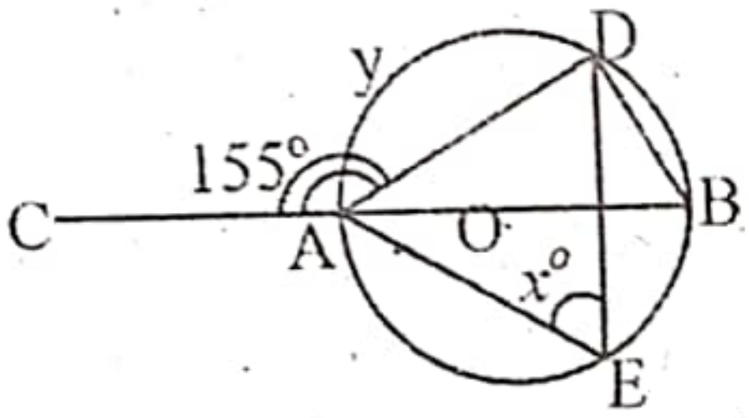
28. $\sqrt{a}\sqrt[n]{a} = a^{\frac{1}{2}}a^{\frac{1}{n}} = a^{\frac{1}{2}+\frac{1}{n}} = a^{\frac{n+2}{2n}} = a^{(n+2)\times\frac{1}{2n}} = (a^{n+2})^{\frac{1}{2n}} = \sqrt[2n]{a^{n+2}}$

Answer:- B

29. Now, we consider only in set of natural numbers.
A. 77=7×11⇒ 1, 7, 11 & 77 are the factors of 77 ⇒ 77 is not prime
B. 71=1×71⇒1 & 71 are the only factors of 71 ⇒ 71 is prime
C. 57=3×19⇒ 1, 3, 19 & 57 are the factors of 57 ⇒ 57 is not prime
D. 51=3×17⇒ 1, 3, 17 & 51 are the factors of 51 ⇒ 51 is not prime

Answer:- B

30. Given $m(\angle CAD)=155^o$ and $m(\angle AED)= x^o$ as in the figure below.

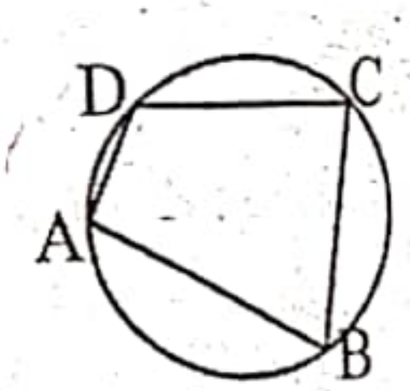


Let point y be at as in the figure. Then,
First, $m(\angle DAB)=180^o-155^o=25^o$so that

$$x^o = \frac{1}{2}m(\widehat{AyD}) = \frac{1}{2}(180^o - m(\widehat{DzB}))$$
$$= \frac{1}{2}(180^o - 2m(\angle DAB))$$
$$= \frac{1}{2}(180^o - 2(25^o)) = 65^o$$

Answer:- A

31. A. In the figure below ABCD is cyclic quadrilateral which is not an example of a regular polygon because measures of its interior angles may not be equal and the lengths of its sides may not be equal.

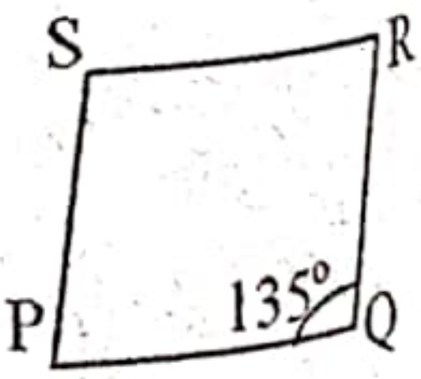


∴Choice A is NOT necessarily true.

A regular polygon is a plane figure with $n \geq 3$ sides where the measures of all its interior angles are equal and the lengths of all its sides are equal. And if its radius is r, it can be inscribed in circle of radius r.

∴Choices B, C and D are necessarily true. **Answer:- A**

32. Given a parallelogram PQRS with perimeter 32 units, PQ=6 units long and $m(\angle PQR)=135^o$,as in the figure below.



From definition of parallelogram,
RS=QP& PS=RQ
⇒RS+QP+PS+RQ=32← given perimeter 32 units
⇒2QP+2QR=32⇒2(6)+2QR=32⇒QR=10 units
Thus, Area=QP·QR·sin 135°

$$= (6 \text{ units})(10 \text{ units})\frac{\sqrt{2}}{2} \leftarrow \sin 135^o = \sin 45^o = \frac{\sqrt{2}}{2}$$

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$=30\sqrt{2}$ sq. units

Answer:- C

33. P∪S={-1, 0, 1, 2, 3} and P∩S={0, 2} so that

$$n(P\Delta S)= n(P\cup S)-n(P\cap S)= 5-2=3$$

Or, $P\Delta S=(P\cup S)\setminus(P\cap S)=\{-1, 1, 3\}\Rightarrow n(P\Delta S)=3$ **Answer:- C**

34. $S=\{x \in \mathbb{N}: -1\leq x \leq 4\}$ is the set of all natural numbers between -1 and 4 (including) so that S={1, 2, 3, 4} **Answer:- C**

35. Given:Set B={3, {3, $\sqrt{3}$}} and P(B) as a power set of B.
Power set of B, P(B) is the set containing all subsets of B.

Thus, P(B)={∅, {3}, {{3, {3, $\sqrt{3}$}}, B} so that

A. $\{\{3,\sqrt{3}\}\}\in P(B)\Rightarrow$ A is True

B. $\{\sqrt{3}\}\notin P(B)$ and $\{\sqrt{3}\}\notin P(B) \Rightarrow$ B is False

C. $\{3,\sqrt{3}\}\notin P(B)$ and $\{3,\sqrt{3}\}\notin P(B) \Rightarrow$ C is False

D. $\{3\}\notin P(B)$ and $\{3\}\notin P(B) \Rightarrow$ D is False **Answer:- A**

36. Given a solid formed by a cylinder with radius r =3 cm and height h =5 cm and by two hemispheres as shown below.The two surmounted hemispheres each with radius 3 cm are combined to form a sphere of radius 3 cm.The volume of the formed solid is

V= volume of the cylinder−Volume of the sphere

$$= \pi r^2 h - \frac{4}{3}\pi r^3 = \pi(3\text{ cm})^2(5\text{ cm}) - \frac{4}{3}\pi(3\text{ cm})^3$$

$=45\pi\text{cm}^3 - 36\pi\text{cm}^3 = 9\pi\text{ cm}^3$ **Answer:- D**

37. $V = \frac{\pi h}{3}(r^2 + rR + R^2)$ where $\frac{h}{H}+\frac{r}{R} = 1$ for H =10, h =5 &R =4

First, $\frac{h}{H}+\frac{r}{R}=1\Rightarrow \frac{5}{10}+\frac{r}{4}=1\Rightarrow r=4(1-\frac{1}{2})=2$ cm

Thus, $V = \frac{\pi h}{3}(r^2 + rR + R^2) = \frac{\pi(5)}{3}(2^2+2(4)+4^2)= \frac{140}{3}\pi$ cubic cm

Answer:- A

38. A. $(\frac{b}{a^r})^r = (a^{-r}b)^r = (a^{-r})^r b^r = a^{-r^2}b^r \neq a^{-r}b^r \Rightarrow$ A is False

B. $\frac{a^r}{b^r} = (\frac{a}{b})^r = (ab^{-1})^r \Rightarrow$ B is True

C. $a^{-r}\times a^s = a^{-r+s} \neq a^{-r\times s} \Rightarrow$ C is False

D. $a^r\times a^s = a^{r+s} \neq a^r + a^s \Rightarrow$ D is False **Answer:- B**

39. A. For b >1, since g(x)= $\log_{\frac{1}{b}} x$ is decreasing

$$x>1\Rightarrow \log_{\frac{1}{b}} x < \log_{\frac{1}{b}} 1 \Rightarrow g(x)<0\Rightarrow \text{A is False}$$

B. For b >1, since $f(x)= \log_b x$ is increasing

$$x>1\Rightarrow \log_b x > \log_b 1 \Rightarrow f(x)>0\Rightarrow \text{B is False}$$

C. For b >1, since g(x)= $\log_{\frac{1}{b}} x$ is decreasing

$$0< x <1\Rightarrow \log_{\frac{1}{b}} 0 > \log_{\frac{1}{b}} x > \log_{\frac{1}{b}} 1$$

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$\Rightarrow \infty > g(x) > 0 \Rightarrow g(x)$ [illegible]

D. For $b > 1$, since $f(x) = \log_b x$ is increasing

$0 < x < 1 \Rightarrow \log_b 0 < \log_b x < \log_b 1$

$\Rightarrow -\infty < f(x) < 0 \Rightarrow f(x) < 0 \Rightarrow$ D is False **Answer:- C**

40. Given: Initial number, $A_o = 100$, Rate, $r = 4\% = 0.04$ and

Final number, $A = 10{,}000$

Then, $A = A_o(1 + r)^t \Rightarrow 10{,}000 = 100(1 + 0.04)^t$

$\Rightarrow 100 = (1.04)^t \Rightarrow t = \log_{1.04} 100$ minutes

Answer:- A

41. $\log_3 x - 2\log_x 9 = 0 \Rightarrow 1 \neq x > 0$ so that

$\log_3 x - 2\log_x 9 = 0 \Rightarrow \log_3 x - \log_x 9^2 = 0 \leftarrow$ by $k \log_a b = \log_a b^k$

$\Rightarrow \log_3 x - \log_x 3^4 = 0$

$\Rightarrow \log_3 x - \frac{1}{\log_{3^4} x} = 0 \leftarrow$ by $\log_a b = \frac{1}{\log_b a}$

$\Rightarrow \log_3 x - \frac{1}{\frac{1}{4}\log_3 x} = 0 \leftarrow$ by $\log_{a^k} b = \frac{1}{k}\log_a b$

Let, $y = \log_3 x$.

Then, $\log_3 x - \frac{1}{\frac{1}{4}\log_3 x} = 0 \Rightarrow y - \frac{4}{y} = 0 \Rightarrow y^2 - 4 = 0 \Rightarrow y = \pm 2$

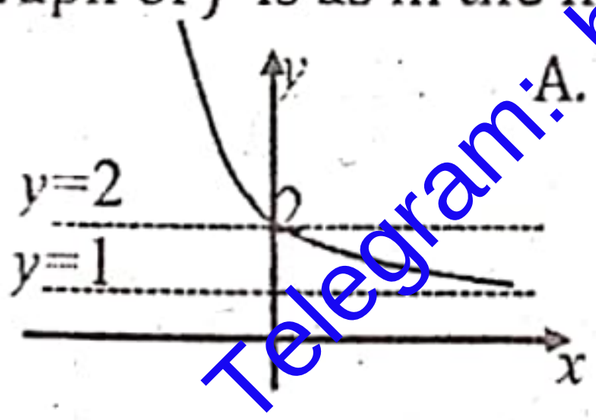
$\log_3 x = 2 \Rightarrow x = 3^2 = 9 > 0$ and $\log_3 x = -2 \Rightarrow x = 3^{-2} = \frac{1}{9} > 0$

$\therefore$ Solution Set $= \{9, \frac{1}{9}\}$ **Answer:- C**

42. $(\frac{25}{9})^{x-1} = \sqrt{0.6} \Rightarrow ((\frac{5}{3})^2)^{x-1} = (\frac{3}{5})^{\frac{1}{2}}$

$\Rightarrow (\frac{5}{3})^{2x-2} = (\frac{5}{3})^{-\frac{1}{2}} \Rightarrow 2x - 2 = -\frac{1}{2} \Rightarrow x = \frac{3}{4}$ **Answer:- A**

43. Graph of f is as in the figure below so that



A. $x > 0 \Rightarrow 0 < (\frac{1}{5})^x < (\frac{1}{5})^0$

$\Rightarrow 0 < (\frac{1}{5})^x < 1$

$\Rightarrow 1+0 < 1+(\frac{1}{5})^x < 1+1 \Rightarrow 1 < f(x) < 2$

$\therefore$ For $x > 0$, f lies between lines $y = 1$ & $y = 2 \Rightarrow$ A is True

B. For $x \in \mathbb{R}$, $(\frac{1}{5})^x > 0 \Rightarrow 1 + (\frac{1}{5})^x > 1$

$\Rightarrow f(x) > 1$

$\Rightarrow$ The line $y = 1$ is its asymptote $\Rightarrow$ B is False

C. $f(0) = 2 \Rightarrow$ Graph of f intersects the y-axis at $(0, 2) \Rightarrow$ C is False

D. Let $x_1, x_2 \in [0, \infty)$ and let $x_1 < x_2$. Then

$(\frac{1}{5})^{x_1} > (\frac{1}{5})^{x_2} \Rightarrow 1 + (\frac{1}{5})^{x_1} > 1 + (\frac{1}{5})^{x_2} \Rightarrow f(x_1) > f(x_2)$

$\therefore f$ is decreasing on $[0, \infty) \Rightarrow$ D is False

Or, as in the graph above f is decreasing on $\mathbb{R}$, so does on $[0, \infty)$

Answer:- A

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44. Mean$\Rightarrow \bar{x} = \frac{-2+(-1)+1+3+4}{5} = 1$ so that standard deviation will be

$$\sigma = \sqrt{\frac{\sum_1^n (x_i-\bar{x})^2}{n}} = \sqrt{\frac{(-2-1)^2+(-1-1)^2+(1-1)^2+(3-1)^2+(4-1)^2}{5}}$$

$$= \sqrt{\frac{26}{5}} = \sqrt{5.2}\text{Ans}$$ **Answer:- D**

45. Let: Missed score$= x$, Wrong mean$= \bar{x}_w$ and Corrected mean$= \bar{x}_c$
Then, for $n = \sum_1^6 f_i = 15$, $x = n(\bar{x}_c - \bar{x}_w) = 15(8-7.6) = 6$ **Answer:- D**

46. A. Let $x_1, x_2 \ \ldots \ x_n$ has a mean $\bar{x}$.
Then, $x_1 - c, x_2 - c \ \ldots \ x_n - c$ has a new mean, $\bar{x}_N$ where

$$\bar{x}_N = \frac{(x_1 - c) + (x_2 - c) + \ \ldots + (x_n - c)}{n}$$

$$= \frac{x_1 + x_2 + \ \ldots + x_n}{n} - \frac{nc}{n} = \bar{x} - c$$

∴Subtracting a constant from each values of a data also subtracts the constant from the old mean ⇒ A is False

B. Let $x_1, x_2 \ \ldots \ x_n$ has a mean $\bar{x}$. Then, $x_1 + x_2 + \cdots + x_n = n\bar{x}$ so that the sum of the deviation of all values is

$(x_1 - \bar{x}) + (x_2 - \bar{x}) + \cdots + (x_n - \bar{x}) = (x_1 + x_2 + \cdots + x_n) - n\bar{x}$
$= n\bar{x} - n\bar{x} = 0 \Rightarrow$ B is True

C. It doesn't exist for *qualitative* data like gender, colors...⇒ C is false

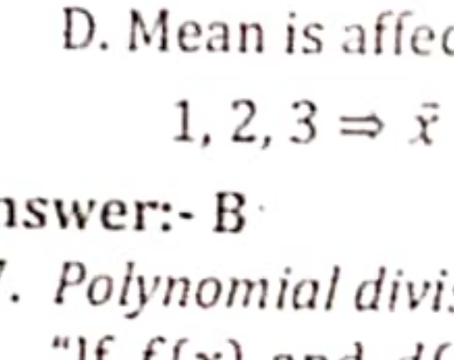
D. Mean is affected by extreme values. For instance:

$1, 2, 3 \Rightarrow \bar{x} = \frac{1+2+3}{3} = 2$ and $1, 2, 96 \Rightarrow \bar{x} = \frac{1+2+96}{3} = 33 \Rightarrow$ D is False

Answer:- B

47. *Polynomial division theorem*:
"If $f(x)$ and $d(x) \neq 0$ are polynomials, and degree of $d(x)$ less than or equal to the degree of $f(x)$ then there exist unique polynomials $q(x)$ and $r(x)$ such that $f(x) = q(x)d(x) + r(x)$ where $r(x) = 0$ or degree of $r(x)$ is less than or equal to the degree of $d(x)$."
For instance, consider the constant polynomials $f(x) = 7$ and $d(x) = 3$.
Then, by long division, $f(x) = q(x)d(x) + r(x) \Rightarrow f(x) = 2(3) + 1$, where $q(x) = 2$ and $r(x) = 1$ so that all f, d, q and r are of zero degrees. Thus, A, B, C and D are False
Answer: No answer is given

48. A. $y = 3x^2 - 3x^4 = 3x^4(\frac{1}{x^2} - 1)$



$$\Rightarrow |y| = \left|3x^4(\frac{1}{x^2} - 1)\right| = |3x^4| \left|\frac{1}{x^2} - 1\right|$$

But as $x \to +\infty$, $\frac{1}{x^2} \to 0 \Rightarrow \left|\frac{1}{x^2} - 1\right| \to |0 - 1| = 1$

Thus, $|y| = |3x^4| \left|\frac{1}{x^2} - 1\right| \to |3x^4| \to +\infty$

⇒ A is true

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B. Multiplicity of zero 0 of $f(x)=3x^2(1-x)(1+x)$ is even; i.e. 2
$\Rightarrow f$ doesn't change signat (0, 0), i.e. $f>0$ around $x=0$.
$\Rightarrow$ Graph of f touches x-axs but doesn't cross it at (0, 0) as in the figure above.$\Rightarrow$B is true

C. Multiplicity of zeros -1 &1 of $f(x)=3x^2(1-x)(1+x)$ is odd, i.e. 1
$\Rightarrow f$ changes signat $(\pm1, 0)$, i.e. -ve$\rightarrow$+ve at $x=-1$ & +ve$\rightarrow$-ve at $x=1$
$\Rightarrow$Graph of f crosses$x$-axis at$(\pm1, 0)$ as in the figure above.
$\Rightarrow$C is true

D. Graph of f has three turning points
, i.e. between -1 and 0, at 0 and between 0 and 1$\Rightarrow$ D is False

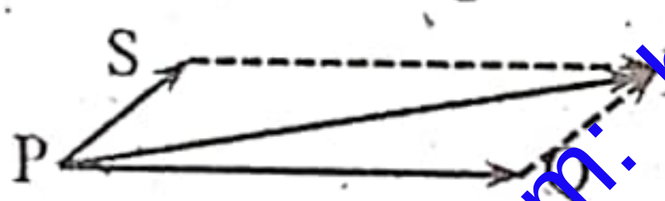
Answer:- D

49. $f(x)=x(1-x^2)^2+2x^3=x(1-2x^2+x^4)+2x^3$
$=x-2x^3+x^5+2x^3=x^5+x$ so that

A. The constant term is 0$\Rightarrow$ A is True
B. Coefficient of x^3 is 0$\Rightarrow$ B is NOT true
C. The leading coefficient is 1$\Rightarrow$ C is True
D. The degree of f is 5$\Rightarrow$ D is True **Answer:- B**

50. By remainder theorem, if a polynomial $f(x)$ is divided by $x-c$, then the remainder is $f(c)$. Thus, if $f(x)=x^5-6x+2$ is divided by $x-c$, the remainder is 7 $\Rightarrow f(c)=7$ so that $f(-1)=7\Rightarrow c=-1$ **Answer:- D**

51. Given a parallelogram PQRS which is determined by vector $\overrightarrow{PS}$ and $\overrightarrow{PQ}$ as shown below.



Then, by triangle law of addition of vectors

$\overrightarrow{PR}-\overrightarrow{PQ}=(\overrightarrow{PQ}+\overrightarrow{QR})-\overrightarrow{PQ}$
$=(\overrightarrow{QR}+\overrightarrow{PQ})+(-\overrightarrow{PQ})\leftarrow$Commutativity
$=\overrightarrow{QR}+(\overrightarrow{PQ}+(-\overrightarrow{PQ}))\leftarrow$Associativity
$=\overrightarrow{QR}+0\leftarrow\overrightarrow{PQ}+(-\overrightarrow{PQ})=0$(Zero vector)
$=\overrightarrow{QR}=\overrightarrow{PS}\leftarrow$Opposite sides of parallelogram

Answer:- D

52. Given: A sphere whose surface area is $144\pi cm^2$.
Let r, A and V be radius, surface area & volume respectively. Then,
$A=4\square r^2=144\pi cm^2\Rightarrow r=6$ cm so that
$V=\frac{4}{3}\square r^3=\frac{4}{3}\square(6\text{ cm})^3=288\pi cm^3$
A **Answer:-**

53. Given:A right triangular prism with isosceles triangle base of sides 7 cm, 6 cm, and 7 cm and height, $h=9$ cm.
Let A_b be area of the base.

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Then, by Heron's formula of area of a triangle with side lengths a, b and c is $A_b = \sqrt{s(s-a)(s-b)(s-c)}$ where $s = \frac{a+b+c}{2}$ so that

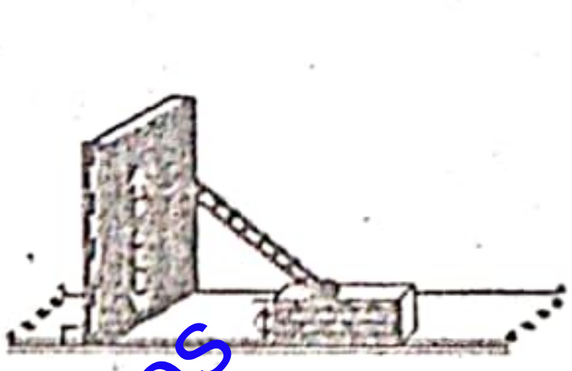
$$V = \frac{1}{3}hA_b = \frac{1}{3}h\sqrt{s(s-a)(s-b)(s-c)}$$

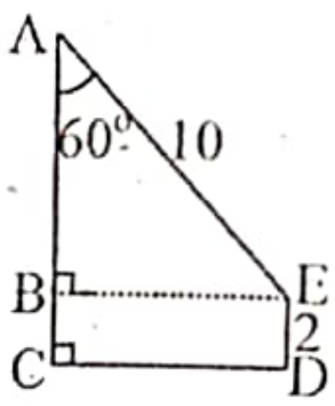
$$= \tfrac{1}{3}(9)\sqrt{10(10-7)(10-6)(10-7)} \leftarrow \text{for } s = \tfrac{7+6+7}{2} = 10$$

$$= 18\sqrt{10}\text{cm}^3$$

Answer: No Answer is given

54. Given: From a stand which is 2 meters high, a 10 meters long ladder leans against a vertical wall and makes an angle of 60^o with the wall as shown in the figure below.From the given figure, we have the following free diagram





Given: AE=10, ED=2 and $m(\angle BAE) = 60^o$

Then, the top of the ladder (point A) from the foot of the wall (point C) on the ground is h =AC so that

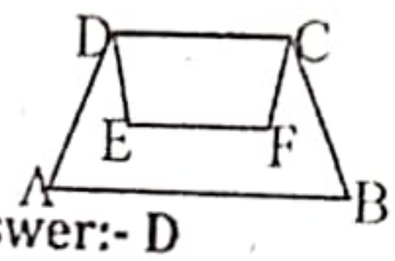
h =AC=AB+BC

=AEcos 60^o +ED

$=(10(\frac{1}{2})+2)$ meters=7 metere

Answer:- B

55. Given: similar quadrilaterals ABCD and CDEF with AB=10 cm, CD=8 cm, AC=12 cm and DE=6 cm as in the figure below. Then, by similarity



Answer:- D

$$ABCD \sim CDEF \Rightarrow \frac{AB}{CD} = \frac{AC}{CE}$$

$$\Rightarrow \frac{10\text{ cm}}{8\text{ cm}} = \frac{12\text{ cm}}{CE} \Rightarrow CE = 9.6\text{ cm}$$

56. Let $\bar{x}$ be the mean and σ^2 be variance of the data $x_1, x_2, x_3, \ldots, x_{10}$.

Let $\sigma_N{}^2$ be the new variance. Note that $\sigma^2 = \frac{\sum_1^{10}(x_i-\bar{x})^2}{10}$.

Then, A) Dividing each value by $\frac{1}{2}$ gives a new data

$2x_1, 2x_2, 2x_3, \ldots, 2x_{10}$ with new mean $2\bar{x}$ so that

$$\sigma_N{}^2 = \frac{\sum_1^{10}(2x_i - 2\bar{x})^2}{10} = \frac{\sum_1^{10} 2^2(x_i-\bar{x})^2}{10} = 2^2\frac{\sum_1^{10}(x_i-\bar{x})^2}{10}$$

$$= 2^2\sigma^2$$

$$\Rightarrow \sigma_N = \sqrt{2^2\sigma^2} = 2\sigma \Rightarrow \text{A is True}$$

B) Subtracting σ from each value gives a new data

$x_1 - \sigma, x_2 - \sigma, x_3 - \sigma, \ldots, x_{10} - \sigma$ with new mean $\bar{x} - \sigma$ so that

$$\sigma_N{}^2 = \frac{\sum_1^{10}((x_i-\sigma)-(\bar{x}-\sigma))^2}{10} = \frac{\sum_1^{10}(x_i-\bar{x})^2}{10} = \sigma^2$$

$$\Rightarrow \sigma_N = \sqrt{\sigma^2} = \sigma \Rightarrow \text{B is False}$$

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C) Multiplying each value by $\frac{1}{4}$ gives a new data $\frac{1}{4}x_1, \frac{1}{4}x_2, \frac{1}{4}x_3, \ldots, \frac{1}{4}x_{10}$ with new mean $\frac{1}{4}\bar{x}$ so that

$$\sigma_N{}^2 = \frac{\sum_1^{10}\left(\frac{1}{4}x_i - \frac{1}{4}\bar{x}\right)^2}{10} = \frac{\sum_1^{10}\left(\frac{1}{4}\right)^2(x_i-\bar{x})^2}{10} = \left(\frac{1}{4}\right)^2\frac{\sum_1^{10}(x_i-\bar{x})^2}{10} = \left(\frac{1}{4}\right)^2\sigma^2$$

$$\Rightarrow \sigma_N = \sqrt{\left(\frac{1}{4}\right)^2\sigma^2} = \frac{1}{4}\sigma \Rightarrow \text{C is False}$$

D) Adding 3to each value gives a new data $x_1+3, x_2+3, x_3+3, \ldots, x_{10}+3$ with new mean $\bar{x}+3$ so that

$$\sigma_N{}^2 = \frac{\sum_1^{10}((x_i+3)-(\bar{x}+3))^2}{10} = \frac{\sum_1^{10}(x_i-\bar{x})^2}{10} = \sigma^2$$

$$\Rightarrow \sigma_N = \sqrt{\sigma^2} = \sigma \Rightarrow \text{D is False}$$

MORE: *Adding or subtracting a constant to each value doesn't change the standard deviation.*

Answer:- A

57. Given: rational number r and an irrational number t

A. False. For instance: $r=3 \Rightarrow \sqrt{1+r} = \sqrt{1+3} = 2$ is rational.

B. False. For instance: $r=0$ &$t=\sqrt{2} \Rightarrow r\times t = 0\times\sqrt{2} = 0$ is rational.

C. False. For instance: $r=3$ &$t=\sqrt{2}$

$\Rightarrow r+t^2 = 3+\sqrt{2}^2 = 3+2 = 5$ is rational.

D. True. Because, Rational−Irrational=Irrational$\Rightarrow r-t$ is irrational

Answer:- D

58. Given: A function $f(x) = \sqrt{x-1}$ where

$f = \{(a, 1), (b, \sqrt{2}), (c, 2), (d, \sqrt{5})\}$.

Clearly, Domain of $f = \{a, b, c, d\}$ so that for $y = f(x)$

$(a, 1) \in f \Rightarrow 1 = \sqrt{a-1} \Rightarrow 1^2 = a-1 \Rightarrow a = 2$

$(b, \sqrt{2}) \in f \Rightarrow \sqrt{2} = \sqrt{b-1} \Rightarrow \sqrt{2}^2 = b-1 \Rightarrow b = 3$

$(c, 2) \in f \Rightarrow 2 = \sqrt{c-1} \Rightarrow 2^2 = c-1 \Rightarrow c = 5$

$(d, \sqrt{5}) \in f \Rightarrow \sqrt{5} = \sqrt{d-1} \Rightarrow \sqrt{5}^2 = d-1 \Rightarrow d = 6$

$\therefore$ Domain of $f = \{2, 3, 5, 6\}$ **Answer:- D**

59. A set with n elements has 2^n subsets.

Then, $A = \{-1, \emptyset, \{0, -1\}\} \Rightarrow n = 3$ so that A has $2^3 = 8$ subsets. **Answer:- D**

60. $|1-3x| = 2|x-2|$ holds iff $1-3x = \pm 2(x-2)$ so that

$1-3x = -2(x-2) \Rightarrow x = -3$ or $1-3x = 2(x-2) \Rightarrow x = 1$

$\therefore$ Solution Set $= \{-3, 1\}$ **Answer:- A**

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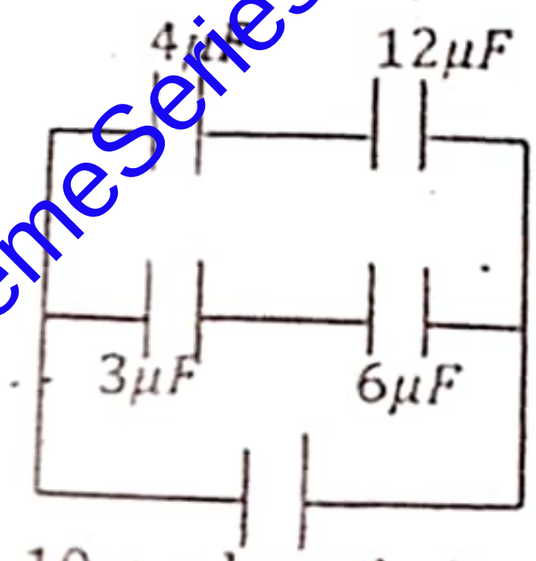
1. Which of the following is true about an absolute electric potential? It is
   A. an electric potential at a postion in an electric field relative to that at infinity from the field source.
   B. an electric potential at a postion in an electric field relative to that at another postion in the field.
   C. the work done to move a unit charge from an infinity to a postion in an electric field.
   D. the work done per unit charge to move it from one point to another in an electric field.

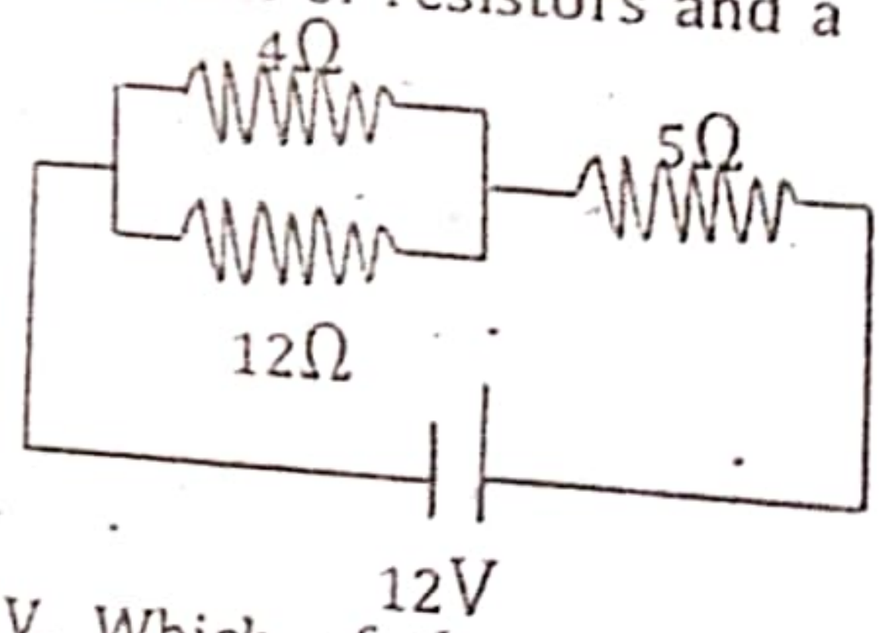
2. The following circuit diagram shows a combination of different capacitors and a voltage sources.
   What is the equivalent capacitance of the circuit?



   A. 16.00μF
   B. 27.00μF
   C. 5.00μF
   D. 5.76μF

3. A uniform cylindrical wire of 1 cm radius and 10 m length has a resistance of $5\times10^{-4}\Omega$ What is the resistivity of the wire?
   A. $1.00\times10^{-4}\Omega m$
   B. $1.57\times10^{-4}\Omega m$
   C. $1.00\times10^{-8}\Omega m$
   D. $1.57\times10^{-8}\Omega m$

4. The circuit diagram shown below is a combination of resistors and a voltage source.
   The voltage drop across the 5 Ω resistor is



   A. 4.5V
   B. 2.0V
   C. 7.5 V
   D. 6.0V

5. A dry cell's voltage is labeled to be 1.5V. Which of the following statements is correct?
   A. The terminal voltage of the cell is equal to 1.5 V for closed circuit.
   B. The electromotive force (emf) of the cell is equal to 1.5V
   C. The terminal voltae of the cell is greater than 1.5 V.
   D. The emf of the dry cell is less than 1.5 V.

6. A resistor of resistance R is connected to a battery as shown in the figure below.

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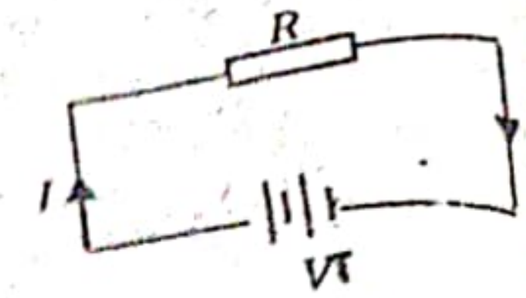

If the potential difference and the current are V and I, respectively, what is the electrical power P delivered to the circuit?

A. $P = R \times V^2$ C. $P = I^2/R$

B. $P = I \times V$ D. $P = (I \times V)/R$

7. Magnetic substances such as steel can be permanently magnetized by properly lining up the magnetic domains. If a nail made of steel is to be magnetized permanently, which of the following processes would allow the magnetic domain in the nail to align in same direction?
   A. Heating up the nail until it becomes red hot by placing it on a bar magnet and then cooling it to room temperature.
   B. Heating up the nail until it becomes red hot and then cooling it to room temperature.
   C. It is not possible to make the nail permanently magnetized.
   D. Heating up the nail until it becomes red hot and cooling it to room temperature by placing it on a bar magnet.
8. A wire carries a steady current of 2.4 A. A straight section of the wire is 75 cm long and lies along the x axis within a uniform magnetic field B Perpendicular to the wire. If the force on the section of the wire is 2.88N, in the negative y direction, the magnetic field strength will be
   A. $0.625T$ B. $1.600T$ C. $0.00625T$ D. $0.016T$
9. The magnetic force that acts on a charge q moving with a velocity v in a magnetiv field B
   A. decreases as the angle between v & B increases from 90^0 to 180^0.
   B. increases as the angle between v & B increases from 90^0 to 180^0.
   C. can vary with the magnitude of q and not with that of v and B
   D. can vary with the magnitudes of v and B not with that of q.
10. The distance between two nearby crests of a water wave in a pond is measured to be 0.40m. If six creats are pass a point every second without considering the reference crest, what is the speed of the wave?
    A. $2.4m/s$ B. $1.2m/s$ C. $24.0m/s$ D. $4.8m/s$
11. In an open field there is a big concret wall. Sound waves that are produced in front of the wall can be heard behind the wall. This is possible mainly due to.
    A. Refraction B. Reflection C. Diffraction D. Interference
12. A certain normal conversation detected an intensity of $10^{-6}W/m^2$ at a distance of 2 m away from the source. What was the power emitted by the source?
    A. $5.02\times10^{-5}Watt$
    B. $2.5\times10^{-5}Watt$
    C. $4\times10^{-6}Watt$
    D. $2\times10^{-6}Watt$

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13. Which of the following is correct about a ball which is projected at an angle of $\theta = 60^0$ above the horizontal?
    A. At the maximum height, the directions of the velocity and acceleration are parallel.
    B. At the maximum height, the velocity of the ball is equal to zero.
    C. The horizontal component of the velocity is constant throughout its motion.
    D. The vertical component of the velocity is maximum at the maximum height.

14. Which statement is correct about the quantities in rotational motion?
    A. Tangential velocity is the rate of change of angular postion per unit time.
    B. Angular velocity is the rate of change of linear position per unit time.
    C. Angular velocity is the rate of change of angular postion per unit time.
    D. Tangential velocity is the rate of change of angular postion per unit time per unit radius.

15. A small ball of mass m is attached to one end of a thin rod of length L, and the other end of the rod is fixed on pivot. What is its rotational moment of inertia, I? (assume that mass of the rod is negligible)
    A. $I = mL$ B. $I = \frac{1}{2}mL^2$ C. $I = \frac{M}{L}$ D. $I = mL^2$

16. If the value of the gravitioanl acceleration on the furface of the earth is g, then what will be its value at a distance of $h = 6.38\times10^6 m$ from the surface of the earth? (the radius of the earth is $R_E = 6.38\times10^6 m$)
    A. $g/2$ B. g C. $g/8$ D. $g/4$

17. A thick wire connects the lighting rod on the roof a house to the ground and in normal good weather ensures that the rod is electrically neutral. If the weather becomes stormy and a negatively changd clound floats overhead, what will be the charge on the rod to protect the house from being struck by the lightning?
    A. The rod will remain electrically neutral unless struck by lightning.
    B. The rod becomes positively charged.
    C. The charge depends on whether the cloud is approaching or moving away.
    D. The rod becomes negatively charged.

18. Charges q_1 & q_2 are located at postions (3m, 0) on the x axis & (0,3m) on the y axis respectively as shown in the following figure.

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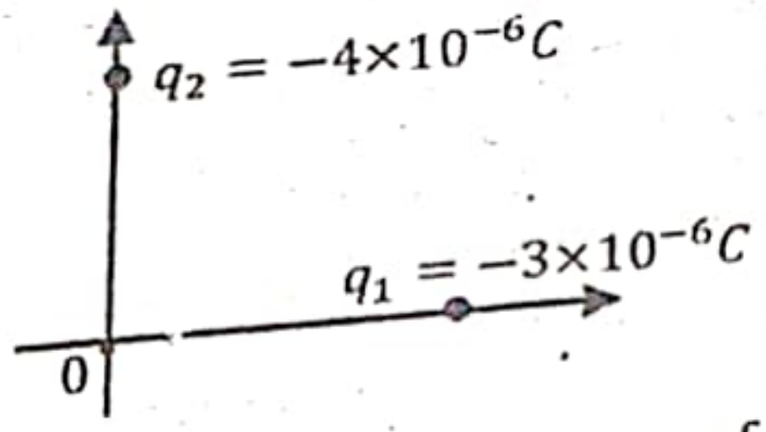
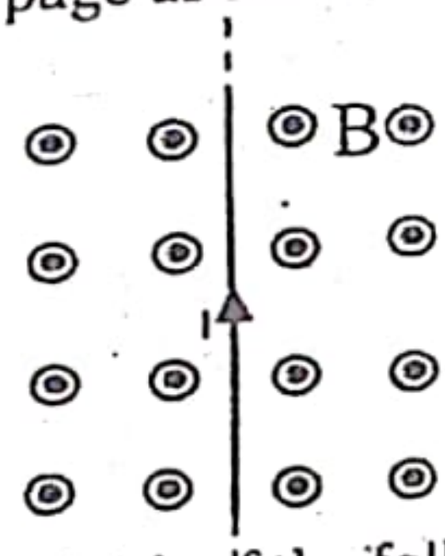


What should be the x and y components of the net electric force on a $1\mu c$ charge placed at the origin? ($k = 1/(4\pi\varepsilon_0) = 9\times10^9 Nm^2C^{-2}$)

A. $F_x = 4\times10^{-3}N$ *and* $F_y = 3\times10^{-3}N$
B. $F_x = 3\times10^{-3}N$ *and* $F_y = 4\times10^{-3}N$
C. $F_x = -4\times10^{-3}N$ *and* $F_y = -3\times10^{-3}N$
D. $F_x = -3\times10^{-3}N$ *and* $F_y = -4\times10^{-3}N$

19. A straight long wire carries an electric current I to the top of the page. The current is placed in a uniform magnetic field B directed out the page as shown in the figure below.



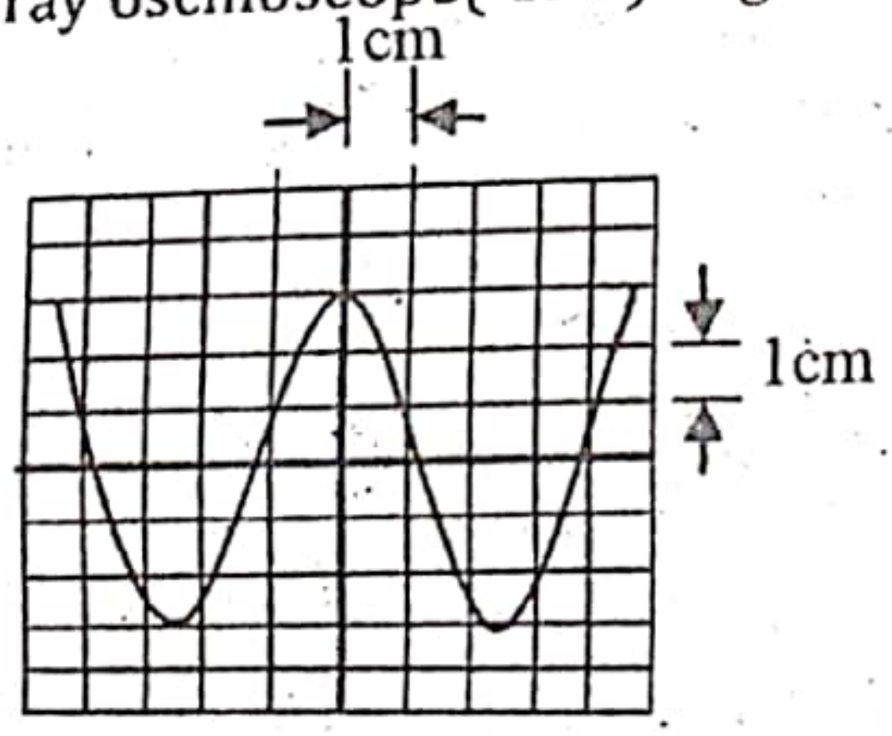
What is the direction of the magnetic force on the current?

A. To the right of the page
B. To the left of the page
C. To the top of the page
D. To the bottom of the page

20. Which of the following is correct about magnetic flux?

A. The magnetic flux in a coil increases when the strength of the magnetic field decreases.
B. The SI unit of magnetic flux is $\frac{T}{m^2}$
C. The magnetic flux in a coil does not depend on the strength of the magnetic field.
D. The SI unit of magnetic flux is $T.m^2$

21. A 12 V peak value of an alternate voltage source is connected to the y-plates of a cathode ray oscilloscope(CRO). It gives the following graph.



The same CRO shows the following graph for unknown alternate voltage source connected to y-plates of CRO.

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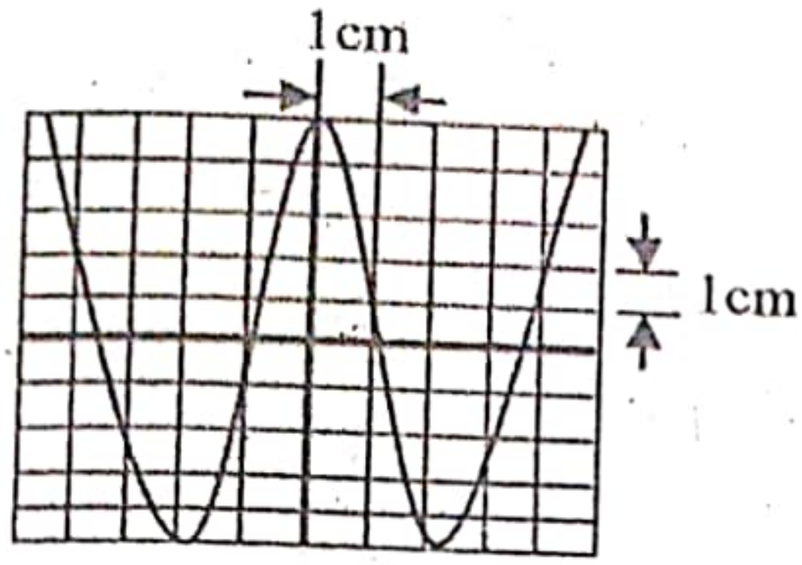
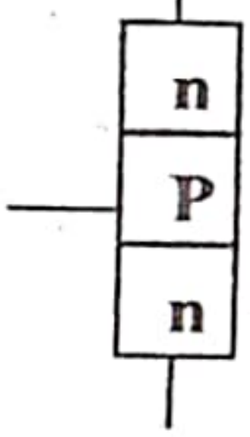
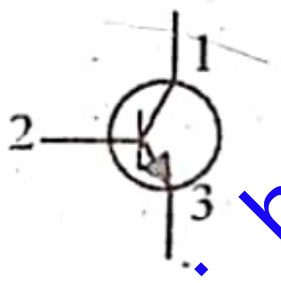


What must be the value of peak voltage for the unknown voltage source?

A. 12 V B. 5 V C. 36 V D. 20 V

22. Which of the following is true about types of semiconductors? A semiconductor
    A. doped with acceptor impurity atoms is called a p-type
    B. doped with donor impurity atoms is called a n-type
    C. in which the majority carriers are holes due to doping in an n-type
    D. doped with acceptor impurity atoms is called an n- type
23. The following figures show the symbol and nomenclature for n-p-n transistor.

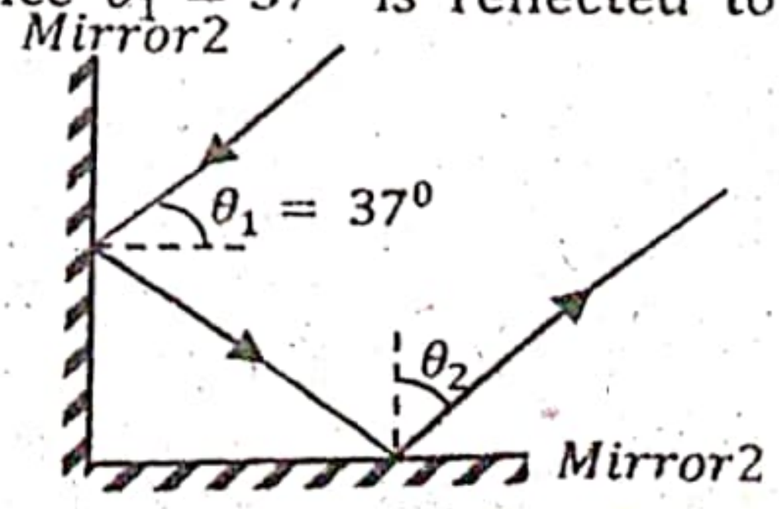




Which numbers represent emitter and base, respectively?

A. 2 and 3
B. 3 and 2
C. 3 and 1
D. 1 and 3

24. Which of the following stements is NOT correct about electromagnetic waves? Electromgnetic waves
    A. produced by accelerations of charge in and electric and magnetic field.
    B. consists of compressions and rarefactions.
    C. can travel in both empty space and material media.
    D. carries energy and momentum.
25. Two plane ,mirrors, mirrors 1 and mirror 2, are placed perpendicular to each other as shown in the figure given below. A ray of light that is incident on mirror 1 at angle of incidence $\theta_1 = 37^0$ is reflected to mirror 2.



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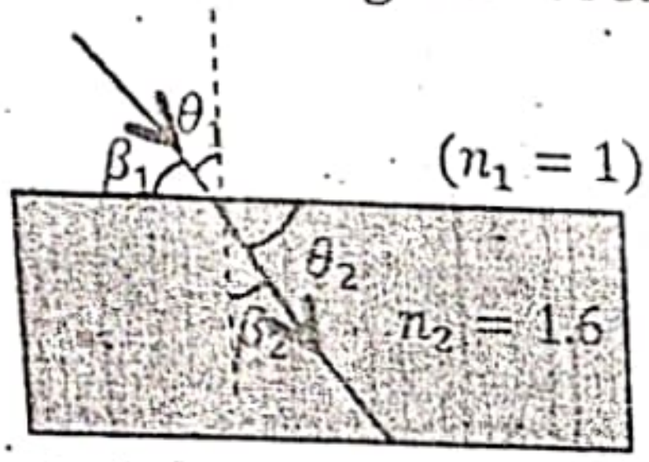
What is the angle of reflection θ_2 at which the ray will be reflected from mirror 2?

A. 37^0 B. 45^0

C. 106^0 D. 53^0

26. Light rays refract when travelling from air into glass. What is the cause (reason) for the refraction? It is because that
    A. the frequency of light is larger in air than in glass.
    B. light travels at the same speed in air and in glass.
    C. light travels slower in glass than in air.
    D. the frequency of light is larger in glass than in air.
27. The following figure shows a light ray passing from air of refractive index n_1 into glass of refractive index n_2.

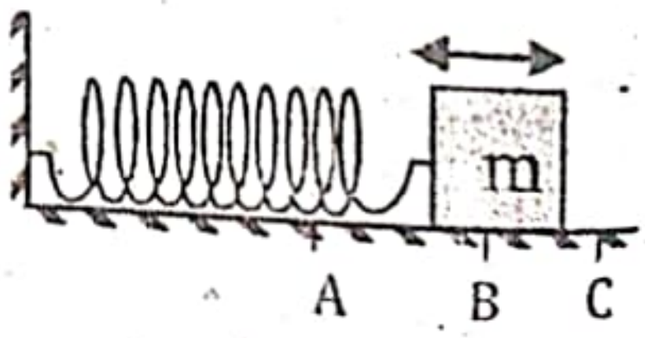


If θ_1 is 53^0 what will be the value of β_2? ($sin53^0 = cos37^0 = 0.8$ $sin37^0 = cos53^0 = 0.6, sin30^0 = cos60^0 = 0.5, sin60^0 = cos30^0 = 0.87$)

A. 53^0 B. 60^0 C. 30^0 D. 37^0

28. A man lifts a 0.2 kg brick from the first floor to the second floor of a building. The first floor is 2.8m above the ground and 2.8m below the second floor. How much work is done on the brick by the gravitational force of the Earth? ($g=10m/s^2$)
    A. $-5.6J$ B. $-11.2J$ C. $11.2J$ D. $5.6J$
29. The following figure shows the oscillation of mass- spring system attached to a vertical wall.

Assuming that the horizontal surface is frictionless and the block oscillates forever between two limited postions A and C. The total mechanical energy of the system is



    A. larger when the block is at postion ***B*** than ***C***.
    B. larger when block is at postion ***A*** than ***B***.
    C. the same at all postions of its oscillations.
    D. larger when the block is in between ***C*** than ***B***.
30. A metallic wedge of thickness t = 1.5 cm and lenght L = 9 cm is used to split a wooden log (wood). If a force F_E= 80 N applied on the wedge results to a load of F_L= 320 N on the log, then which of the following statements is correct about the wedge.
    A. The velocity ratio is 1/6

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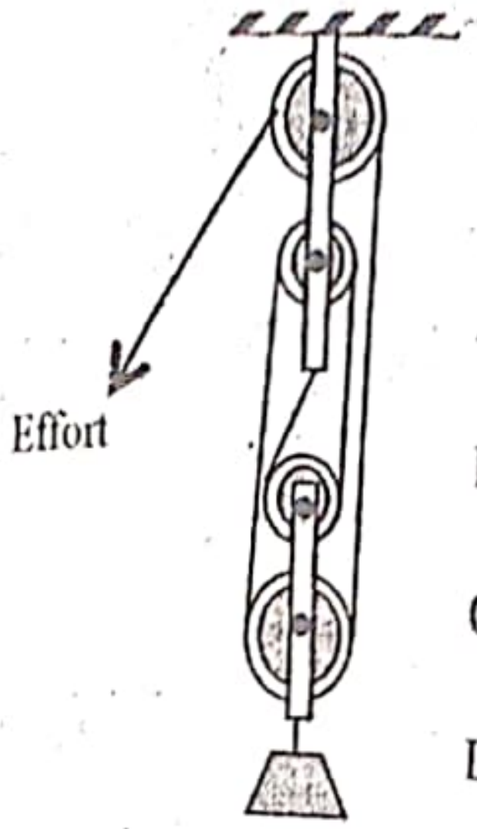
B. The mechanical advantage (MA) of the wedge is ¼
C. The input work is 480J
D. The efficiency of the wedge is 66.7%

31. Consider the pulley system shown in the figure below.



Which of the following is correct about the pulley system?
(mass of pulleys and frictional forces are negligible)

A. Distances moved by the load and moved by the effort are equal.
B. If the effort moves 8cm. then the load will move 2 cm
C. The mechanical advantage of the system is ¼
D. The mechanical advantage of the system is 5

32. If you feel a pressure three times that of atmospheric pressure when you are deep under water. How high is the water level from your postion? (Atmospheric pressure $=10^5$Pa, Density of water $=\frac{10^3kg}{m^3}$, $g = 10m/s^2$)

A. 20m B. 10m C. 40m D. 30m

33. An object of mass m is immersed in a liquid whose density is P_L. If the volume of the liquid displaced by the object is V_L, then what is the apparent weight, W_{App}, of the object? (The acceleration due to gravity is g)

A. $W_{App}= mg + \rho_L g V_L$
B. $W_{App}= mg$
C. $W_{App}= mg - \rho_L g V_L$
D. $W_{App}= \rho_L g V_L$

34. A square metal plate has area of $0.09m^2$ at a temperature of 30^0C. If the temperature is raised to 60^0C the area increases by
(The coefficient of linear expansion for metal is $1.1\times10^{-5}\,^0C$)

A. $1.0000594\ m^2$
B. $1.0001188m^2$
C. $0.0000297m^2$
D. $0.0000594m^2$

35. How much heat energy is required to change an ice cube of mass $m = 720g$ from ice at a temperature of $T_i= 10^0C$ to water at a temperature of $T_f= 15^0C$? (Specific heat capacity of ice is $c_i=2.220\times10^3$ J/(kg. K), specific heat capacity of water is $c_w=4.187\times\frac{10^3J}{kg.K}$, and the latent heat of fusion of ice is $L_F = 3.330\times10^5$J/k

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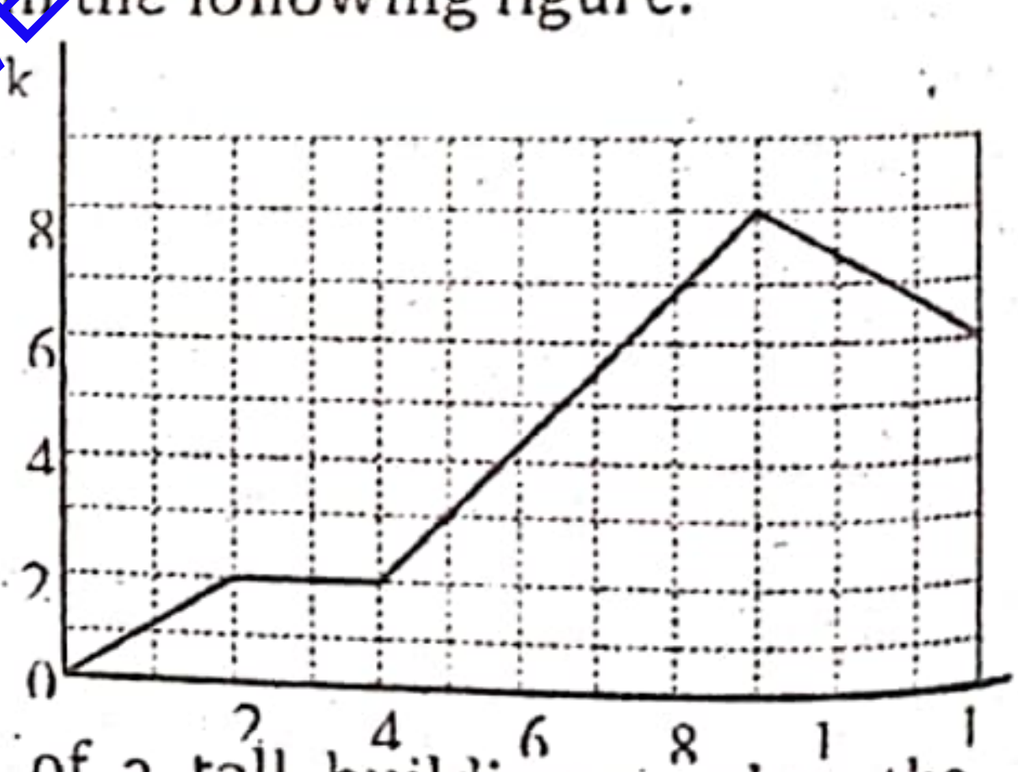
A. 2.40×10^4J B. 6.12×10^4J C. 3.00×10^5J D. 2.68×10^5J

36. Which of the following statements is correct about waves?
    A. A wave that causes the elements of the medium to move perpendicular to the direction of movement of the wave is a longitudinal wave.
    B. A wave that causes the elements of the medium to move parallel to the direction of movement of the wave is transverse wave
    C. Both sound waves and light waves are longitudinal waves.
    D. Sound waves are longitudinal waves whereas light waves are transverse waves.
37. Which one of the following is a vector quantity?
    A. Electric current B. Distance C. Pressure D. Weight
38. What are the x- and y- components of a displacement vector **S** in the xy plane if its direction is 37^0 measured from the negative x- axis to the positive y- axis and its magnitude is 8.0m?
    (Cos 37^0= sin 53^0=0.8 and sin 37^0=cos 53^0=0.6)
    A. $S_x= -6.4m$ *and* $S_y = 4.8m$
    B. $S_x= -4.8m$ *and* $S_y = 6.4m$
    C. $S_x= 6.4m$ *and* $S_y = -4.8m$
    D. $S_x= -4.8m$ *and* $S_y = -6.4m$
39. A car started from x=0 at t= 0 and moves along a straight line according to the displacement- time graph shown in the following figure. What is the magnitude of its average velocity during the first 12 minutes?



    A. 1.20km/min
    B. 4.33km/min
    C. 0.26km/min
    D. 0.50 km/min
40. If a ball dropped freely from the top of a tall building reaches the ground after 3 s, what should be the height of the building?
    (Air resistance is negligible, and g= 10m/s²)
    A. 45 m B. 30m C. 90m D. 60m
41. Two bodies A and B are moving with speed $V_A = V$ and $V_B = 2V$ relative to the ground, respectively. If they are moving in the same direction, what will be the velocity of B with relative to A?
    A. 2V B. V C. $\sqrt{5}V$ D. $3V$
42. Two forces $\mathbf{F_1}$ and $\mathbf{F_2}$ act perpendicular to an object of mass m = 5.0kg as shown in the figure below. The magnitudes of the forces are $F_1 = 20.0N$ and $F_2 = 15.0N$

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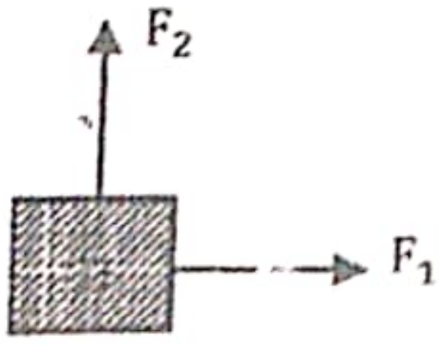

What is the magnitude of its acceleration?

A. $1m/s^2$
B. $7m/s^2$
C. $1.6m/s^2$
D. $5.0m/s^2$

43. Suppose that a man is pushing a car forward horizontally and the car is moving. According to Newton's third law the car is pushing the man backward horizontally by a force.
    A. of zero magnitude since both are moveing forward.
    B. less in magnitude than that the man exerted on it.
    C. larger in magnitude than that the man exerted on it.
    D. equal in magnitude with that the man exerted on it.
44. An object of mass $m_1 = 3.0kg$, moving with a velocity of 20.0m/s in the positive x-direction, collides with and sticks to an object of mass m_2=2.0 kg moving with a veloicity of 10.0m/s in the negative x-direction. What is the final velocity of the composite system?
    A. $16m/s$, negative x- direction
    B. $16m/s$, postive x- direction
    C. $8m/s$, negative x- direction
    D. $8m/s$, positive x- direction
45. An object is moving and accelerating to the right because of a net force of 50N. Then, another 50N force directed to the left is applied to this same object. Which of the following is the correct statement about its state of motion? The object
    A. slows down and eventually stops.
    B. reverses its direction of motion from right to left.
    C. accelerates at a greater speed to the right.
    D. continues traveling to the right at a constant speed.

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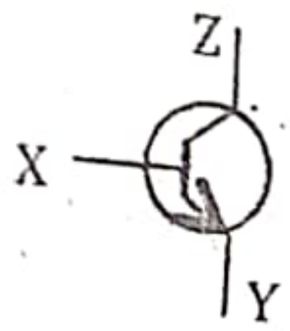
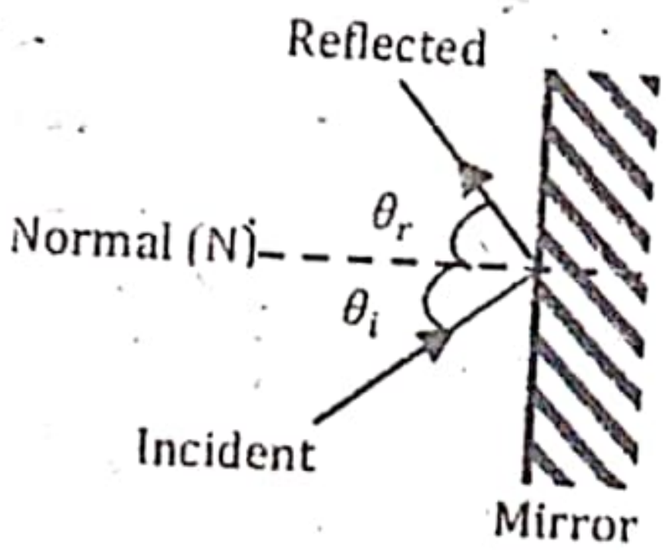
1. The SI unit of magnetic flux is
   A. T m^2 B. T/m^2 C. T/s D. T m
2. The operation of a transformer can be explained based on
   A. The right-hand rule. C. Faraday's law of induction
   B. Fleming's left-hand rule. D. Lenz's law
3. Cathode ray tubes are used to measure all of the following except
   A. period of alternating currents. C. electric field strength
   B. frequency of alternating voltages D. unknown voltage
4. A diode is used in a half-wave rectification ac circuit, because it
   A. reduces the current in a circuit by half.
   B. absorbs the current in the second half of the ac cycle.
   C. generates only half of the current in a circuit.
   D. allows current to flow in one direction only.
5. The following symbol is an *n-p-n* transistor.



What do the letters X, Y and Z represent respectively?
   A. base, collector, emitter
   B. base, emitter, collector
   C. collector, base, emitter
   D. emitter, collector, base
6. an electromagnetic wave can be produced by
   A. a steady current C. any moving charge
   B. electrostatic fields D. any accelerating charge
7. To test the law of reflection a physics student sends a beam of light onto a plane mirror. He then measures the angle of incidence θ_i as shown in the figure below.



Which of the following possibly indicates the result of his measurement? *(where θ_r Represent reflected angle).*
   A. $A\theta_i + \theta_r = 90^0$
   B. $\theta_i = \theta_r$
   C. $\theta_i > \theta_r, always$
   D. $\theta_i < \theta_r, always$

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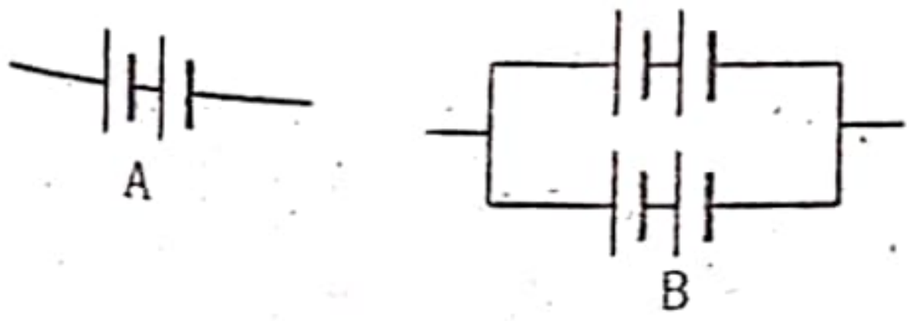
8. Refraction is the
   A. bending of light as it passes through a small gap.
   B. change in the intensity of light as it crosses the boundary between two transparent media.
   C. bending of light as it crosses the boundary between two transparent media.
   D. change in the frequency of light as it crosses the boundary between two transparent media.
9. Which one of the following is not the consequence of refraction?
   A. A stick immersed in water appears to be broken.
   B. Formation of image by mirror.
   C. The apparent depth of water is smaller than the actual depth.
   D. The star looks higher in the sky than in really is.
10. The electric potential at a point in space is the
   A. potential energy per unit charge to brought a charge at that point.
   B. potential energy of a charge brought to that point.
   C. Product of potential energy and a unit charge at that point.
   D. force per unit charge to brought a charge at that point.
11. A 1-μF and a 2-μF capacitors are connected in series and a potential difference of 6 V is applied across the combination. Which of the following is true about the 2-μF capacitor? It has
   A. twice the potential difference of the 1-μF capacitor.
   B. half the potential difference of the 1-μF capacitor.
   C. twice the charge of the 1-μF capacitor.
   D. half the charge of the 1-μF capacitor.
12. A constant electric current of 16A flows through a resistor in 4 minutes. What is the number of electrons that pass through a point?
   (An electron charge is $1.60\times10^{-19}C$)
   A. 6.8×10^{-22} B. 4.2×10^{-22} C. 2.4×10^{22} D. 4×10^{22}
13. The effective resistance of two resistors R_1 and R_2 connected in parallel is
   A. $\frac{(R_1R_2)}{R_1+R_2}$ B. R_1+R_2 C. $\frac{(R_1+R_2)}{R_1R_2}$ D. R_1+R_2
14. A number of cells each with emf, ε are connected in combinations as shown in figures A and B below:



The total *emf* delivered to external resistors in each case is
   A. ε for A and 2ε for B.
   B. 2ε for A and 0.5ε for B.
   C. 2ε for A and ε for B.
   D. 2ε for both A and B.

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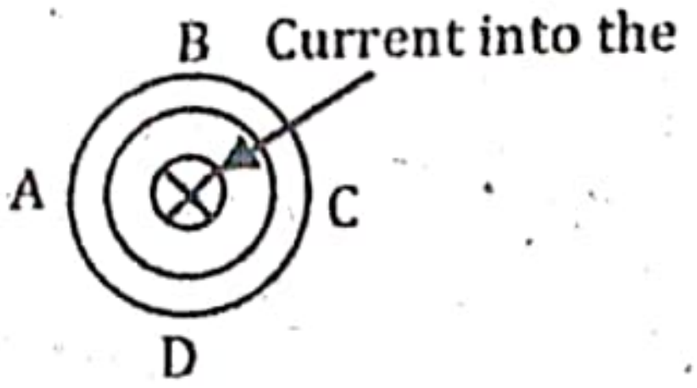
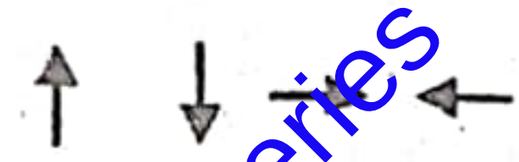
15. Which one of the following is not yet used as a source of electricity in Ethiopia?
A. Wind B. Nuclear C. Hydroelectric D. Geothermal

16. A student determined the existence of a magnetic field experimentally by putting a compass needle near a current carrying wire as shown in the figure below.

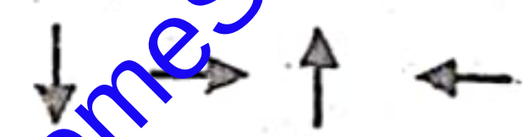


If she put the compass needle at points A, B, C and D which of the following set of arrows correctly indicate the direction of the compass needle at the points A, B, C & D respectively?

A.

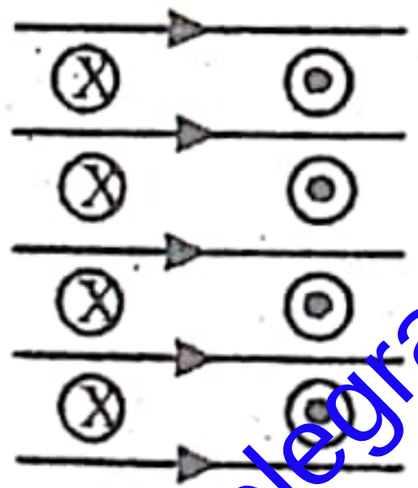
C. 

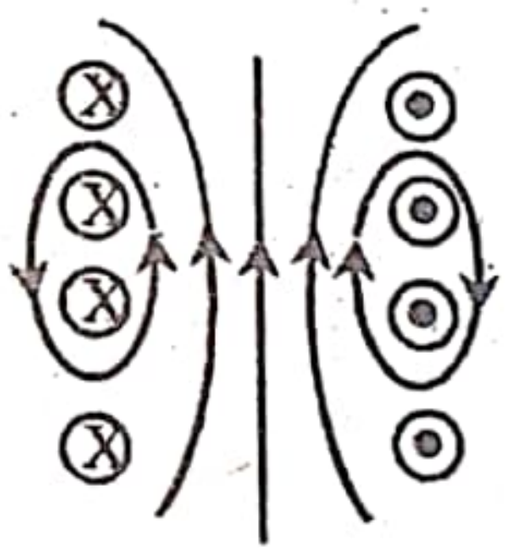
B.

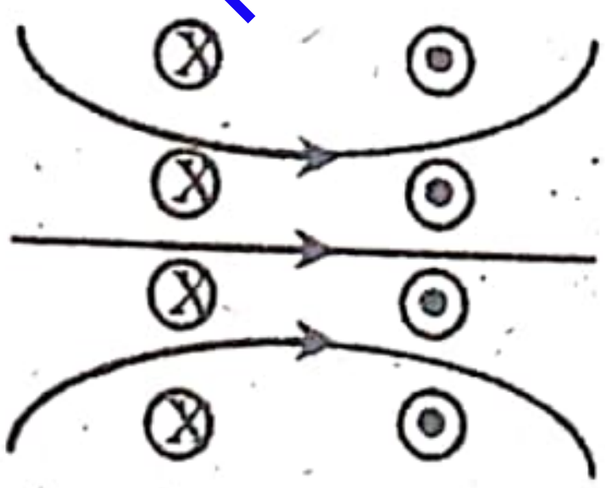
D. 

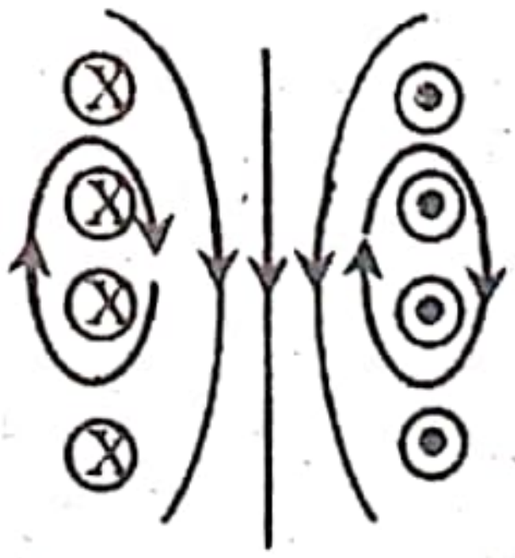
17. Which one of the following diagrams correctly shows the pattern of the magnetic field lines of a solenoid?

Where ⊗ Current into the page. ⊙ Current out of the page. } Are half longitudinal sections of Solenoid

A. 

C. 

B. 

D. 

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18. The equation $F = BIl\sin\theta$ (where F is the force acting on current I carrying wire of length l in a magnetic field B, and θ is the angle between field an current carrying wire). The equation can be derived from $F = qvB\sin\theta$ (where F is the force acting on a charge (q) moving at a velocity (v) in a magnetic field (B), and is the angle between B and v)) by using a definition relating
A. force and velocity C. current and velocity

B. force and current D. current and magnetic field

19. The magnitude of the resultant of two vectors $\vec{S}$ and $\vec{T}$ is 12 units. If the magnitude of vector $\vec{S}$ is 6 units, then magnitude of vector $\vec{T}$
A. cannot be greater than 12.
B. can be less than 6 units
C. is between 6 and 18 units inclusive
D. can be greater than 18 units.

20. If the x-component, A_x, of a vector $\vec{A}$ in the xy plane, is $A_x = \frac{1}{2}A$, where A is the magnitude of the vector $\vec{A}$, the y-component, A_y, of the vector $\vec{A}$ is
A. $\frac{\sqrt{3}}{2}A$ B. $\frac{1}{2}A$ C. $\sqrt{3}\,A$ D. $\frac{3}{2}A$

21. Which one of the following statements does not describe a property of uniform rectilinear motion?
A. The velocity doesn't vary with time.
B. The average speed is equal to the speed at any instant of time.
C. Distance is proportional to the square of time.
D. The slope of distance-time graph has the same value at all times.

22. A body moves along the x-axis with acceleration $-20\frac{m}{s^2}$. This means the
A. speed of the body decreases by $20\frac{m}{s}$ during each second.
B. Acceleration of the body decreases by $20\frac{m}{s^2}$ during each second.
C. body moves 20m during each second.
D. speed of the body increases by $20\frac{m}{s}$ during each second.

23. A boat is able to move through still water at $20\frac{m}{s}$. If the boat moves up stream in a river that flows at $5\frac{m}{s}$, what is the speed of the boat relative to the ground?
A. $20\frac{m}{s}$ B. $15\frac{m}{s}$ C. $100\frac{m}{s}$ D. $25\frac{m}{s}$

24. Newton's second law states that
A. no force is needed to stop a moving body.
B. force is needed to keep a moving object in uniform motion.
C. action and reaction are equal and opposite.
D. acceleration is directly proportional to net force.

25. Rocket propulsion and flying of birds can be explained by
A. Newton's first law. C. Archimedes' principle
B. Newton's third law. D. Pascal's principle

26. The total linear momentum of a system is conserved if
A. the system is under the influence of internal forces .
B. no net external force is acting on the system.

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C. the particles of the system do not exert forces on each other.
D. a net external force is acting on the system.

27. Which one of the following statements is true about elastic and inelastic collisions?
A. Linear momentum is conserved in both elastic and inelastic collisions, but kinetic energy is conserved only in elastic collision.
B. Kinetic energy is not conserved in both elastic and inelastic collisions, but linear momentum is conserved in both types of collisions.
C. Linear momentum and kinetic energy are conserved in both elastic and inelastic collisions.
D. Linear momentum is not conserved but kinetic energy is conserved in elastic collisions.

28. A constructive interference is a wave phenomenon that occurs when two waves move in
A. the same directions in phase.
B. opposite directions out of phase.
C. the same directions out of phase.
D. opposite directions in phase

29. When sound generated by a source travels from air to water, it
A. travels slower.
B. Stops suddenly
C. travels faster.
D. continues with the same speed.

30. When the elevation angle of a projectile is increased above 45^0 then
A. range decreases and maximum height increases.
B. range increases and maximum height decreases.
C. both range and maximum height increase.
D. both range and maximum height decrease.

31. Angular velocity is defined as the time rate
A. of change of angular position.
B. at which the moment of inertia changes.
C. of change of distance
D. of change of tangential distance.

32. A dancer spins around with her arms stretched out at an angular speed of $1\frac{rad}{s}$. Her moment of inertial is $50kgm^2$. When she pulls in her arms the moment of inertial decreased to $40kgm^2$. What is her final angular velocity?
A. $1.50\frac{rad}{s}$ B. $1.25\frac{rad}{s}$ C. $2.0\frac{rad}{s}$ D. $2.5\frac{rad}{s}$

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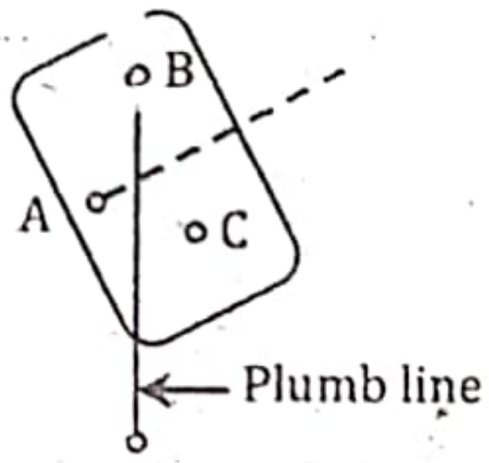
33. Students in a physics laboratory want to determine the center of gravity of a two dimensional object by hanging it from the holes made at the edges of the object as shown in the figure below.



If they drew lines on the object following the plumb lines from the holes *A*, *B*, and *C*, the center of gravity will be at

A. midway between holes *B* and *C*.
B. a point outside the object shown.
C. the intersection of the plumb lines.
D. the geometric center of the object.

34. A planet is four times as massive as the Earth and has twice the radius of the Earth. What is the escape velocity at the surface of the planet, if the escape velocity on Earth is 11 km/s?

A. $10\sqrt{2}\ km/s$ B. $11\sqrt{2}\ km/s$ C. $5\sqrt{2}\ km/s$ D. $11\ km/s$

35. Consider the following procedural steps

I. Ground an electroscope
II. Remove the ground from the electroscope
III. Bring a charged rod near, but not touching, the electroscope.
IV. Remove the charged rod.

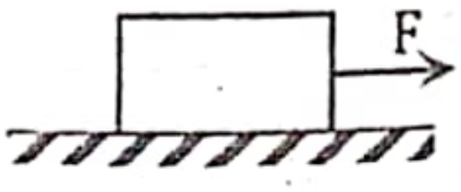
Which steps are correct sequences to charge an electroscope by induction?

A. III, II,I and IV
B. III, I, IV, and II
C. I, III, IV and II
D. III, I, II and IV

36. Consider a spherical solid conductor carrying a net positive charge *Q*: Which of the following is not true about the electrostatic concept?

A. The electric field inside the conductor is zero.
B. The electric field is parallel to the surface.
C. The net charge resides only on the outer surface.
D. The net charge inside the conductor is zero.

37. A crate with a weight of 50N rests on a horizontal surface. A person pulls horizontally on it with a force F as shown below.



If the coefficient of static friction is 0.4, what is the smallest force F need to start motion?

A. 25N B. 20N C. 50N D. 35N

38. A ball is kicked up into the air. Over its entire upward motion, the work done by gravity and the work done by air resistance are respectively

A. negative, positive
B. negative, negative
C. positive, positive
D. positive, negative

39. The mechanical advantage of any machine is the ratio of the

A. force exerted by the machine to the force applied on it.

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B. force applied to the machine to the force exerted by it.
C. work done on the machine to the work done by it.
D. work done by the machine to the work done on it.

40. The relative density of a substance is defined as the
A product of the density of water and the density of a substance.
B. Difference between the density of a substance and the density of water.
C. ratio of the density of a substance to the density of water.
D. ratio of the density of water to the density of a substance.

41. A physics student derived the formula for the surface expansion, ΔA, of a rectangular plate in terms of its linear expansion coefficient, α .Which of the following formulas is possibly derived by the student? ***(Assume that ΔT is the change in temperature of the plate and A_o is its initial area.)***

A. $\Delta A = 2A_0\alpha\Delta T$
B. $\Delta A = 3A_0\alpha\Delta T$
C. $\Delta A = \frac{1}{2}A_0\alpha\Delta T$
D. $\Delta A = A_0\alpha\Delta T$

42. A hot alloy with a specific heat capacity of 200J/kg^{0}C cooled by 50^{0}C when added in to water. What amount of energy is given off by the alloy if its mass is 0.3kg?
A. 3000J B. 5000J C. 1000J D. 1500J

43. A certain mass of ice at 0^{0}C is placed in a thermos bottle containing 0.1kg of water 6^{0}C. How many grams of ice will melt if the final temperature of the system at is 0^{0}C? ***(The heat of fusion of water is 334,000J/kg and the specific heat is 4,200J/kg.^{o}C)***
A. 1.7×10^{-3}kg
B. 5.0×10^{-3}kg
C. 7.5×10^{-3}kg
D. 2.0×10^{-3}kg

44. A sinusoidal transverse wave is traveling on a string. Any point on the string
A. moves in circular motion with a different angular speed than the wave.
B. moves in circular motion with the same angular speed as the wave.
C. oscillates with a different frequency than that of the wave.
D. oscillates with the same frequency as the wave.

45. The frequency of a wave is the
A. Number of complete waves passing a given point per second.
B. Time taken for one complete wave to pass a given point.
C. Distance the wave travels in one second.
D. Minimum distance between identical points on adjacent waves.

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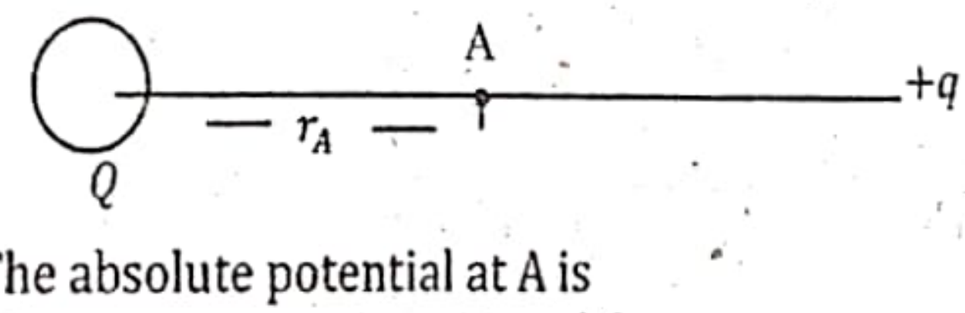
SOLUTION OF PHYSICS EXAMINATION 2009/MAY 2017

BOOKLET CODE 023

1. **Solution:** An absolute electric potential is defined as the amount of work done to move a unit positive test charge from an infinity position to a position in an electric field.



The absolute potential at A is

$$V_A = kQ\left(\frac{1}{r_A} - \frac{1}{r_\infty}\right) = \frac{kQ}{r_A} \text{ because}$$

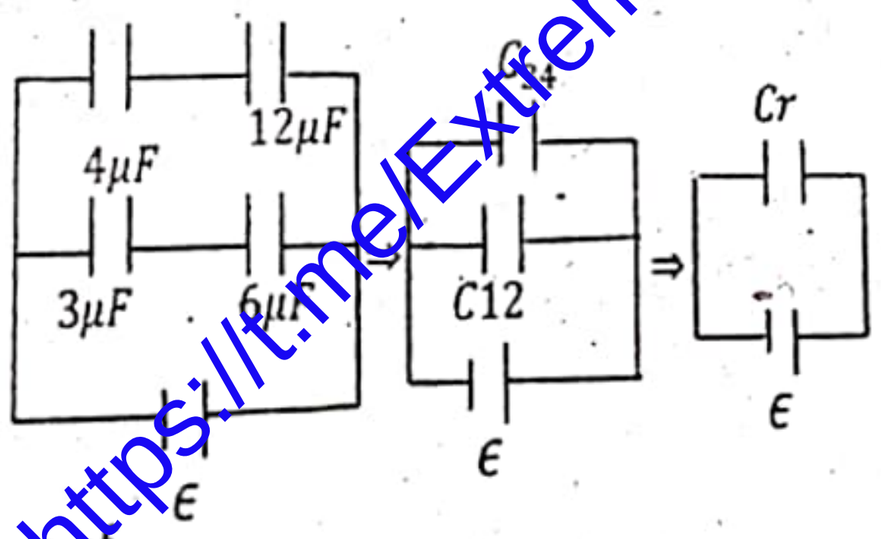
$$V_\infty = \frac{kQ}{r_\infty} \approx 0$$

Answer C

2. **Solution:** Let $C_1 = 3\mu F, C_2 = 6\mu F$.

$C_3 = 4\mu F$ *and*

$C_4 = 12\mu F$



The equivalent capacitance of $C_1 and C_2$ joined in series is

$$C_{12} = \frac{C_1 \times C_2}{C_1 + C_2} = \frac{3\mu F \times 6\mu F}{3\mu F + 6\mu F} = 2\mu F$$

The equivalent capacitance of C_3 and C_4 that are joined in series is

$$C_{34} = \frac{C_3 \times C_4}{C_3 + C_4} = \frac{4\mu F \times 12\mu F}{4\mu F + 12\mu F} = 3\mu F$$

$C_{12} and\ C_{34}$ are joined in parallel and their combined capacitance gvies the equivalent capacitance of the circuit-

Thus,

$C_{eq} = C_{12} + C_{34} = 2\mu F + 3\mu F = 5\mu F$ **Answer C**

3. **Solution:** Given $r = 1cm = 10^{-2}m$, $\ell = 10m$ $R = 5\times10^{-4}\ell$

Required : $\rho = ?$

The area of the wire is $A = \pi r^2 = 3.14(10^{-2}m)^2$

$= 3.14\times10^{-4}m^2$

We also know that the resistance of a thin metal wire given by

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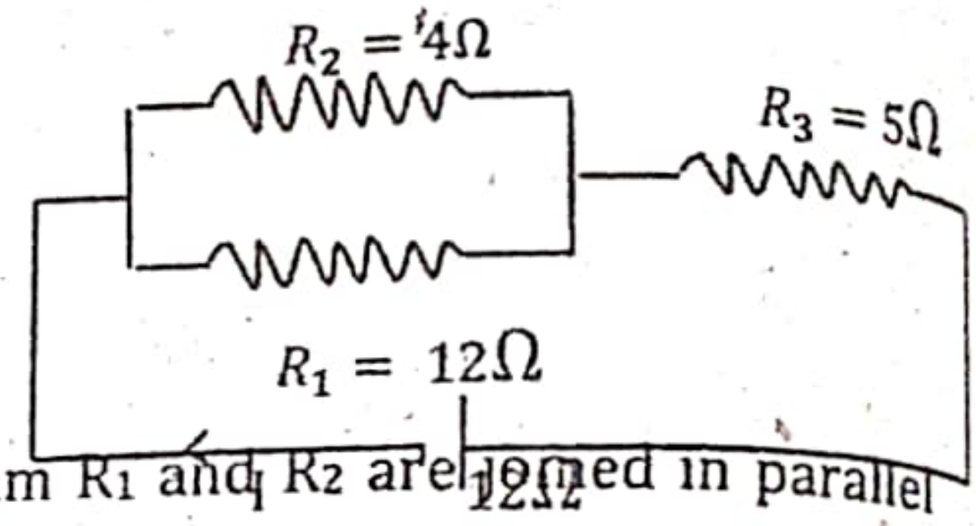
$$\rho = \frac{RA}{R} = \frac{5 \times 10^{-4}\Omega(3.14 \times 10^{-4}m^2}{10m}$$

$$\rho = 15.70 \times 10^{-9}\Omega m = 1.57 \times 10^{-8}\Omega m$$

Answer D

4. Let $R_1 = 12\Omega$

$\mathcal{R}_2 = 4\Omega$ *and* $\mathcal{R}_3 = 5\Omega$



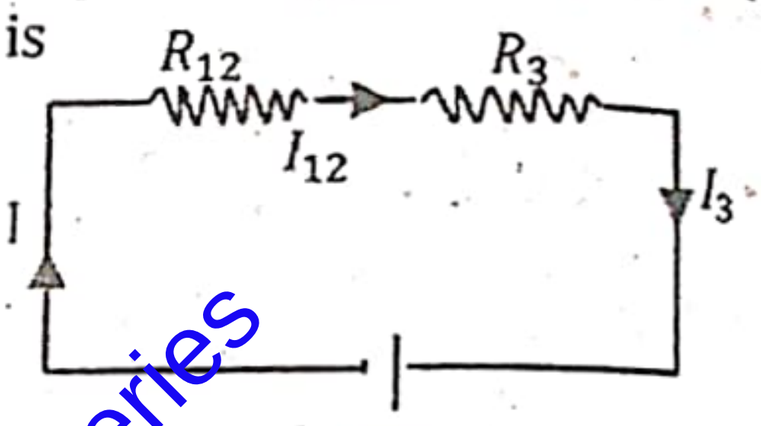
Required : The voltage

Drop across $\mathcal{R}_3 = 5\Omega$

Solution: From the circuit diagram R_1 and R_2 are joined in parallel and their combined resistance is

$$R_{12} = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{12\Omega \times 4\Omega}{12\Omega + 4\Omega}$$

$$R_{12} = 3\Omega$$



R_{12} *and* R_3 are joined in series so the equivalent resistance of the circuit becomes $R_{eq} = R_{12} + R_3 = 3\Omega + 5\Omega = 8\Omega$

The total currents flowing in the circuit is

$$I = \frac{V}{Req} = \frac{12V}{8\Omega} = 1.5A$$

$$I = I_{12} = I_3 = 1.5A$$

The voltage drop across $R_3 = 5\Omega$ is

$$V_3 = I_3 R_3 = 1.5A(5.\Omega) = 7.5V$$

Answer: C

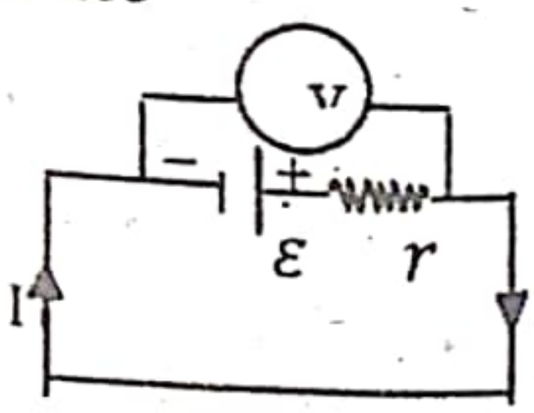
5. For a closed circuit the terminal potential difference is less than emf of the cell because of the presence of an internal resistance

When current flows in the circuit

$V = V_{ter} = \varepsilon - Ir$

The electromotive force (emf) of the cell is the maximum voltage of the cell provides to the circuit which is the same as the short circuit (open circuit) voltage.

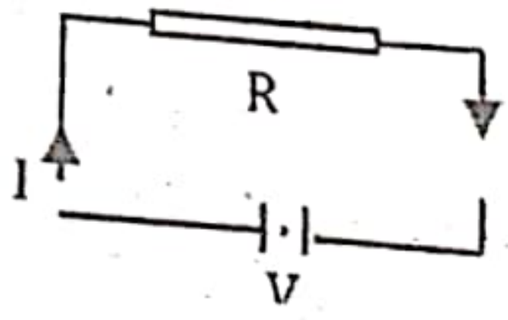
$\varepsilon = 1.5V$



Answer B

6. For the circuit shown to the right the power delivered to the circuit is given by

$$P = I \times V = I^2R = \frac{V^2}{R}$$



Answer B

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$P = I \times V = I^2R = \frac{V^2}{R}$ - **Answer B**

7. In order to magnetize the nail made of still permanently and the magnetic domains to align in the same direction heating up the nail until it becomes red hot and cooling it to room temperature by placing it on a barmagnet.

Answer D

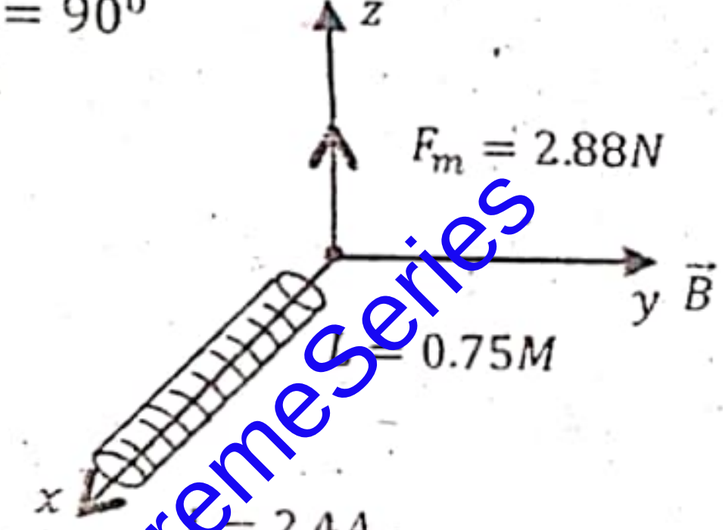
8. The magnetic force exerted on this section of the wire is given by

$F_m = I\ell B \sin\theta$ *Since* $\ell \perp B$ *and* $\theta = 90^0$

$F_m = I\ell B \sin\theta$ Solving for B we get

$$B = \frac{Fm}{I\ell} = \frac{2.88N}{2.4A(0.75m)}$$

$$B = \frac{1.6N}{AM} = 1.6T$$



Answer : B

9. The magnitude of the magnetic field neither increases nor decrease when the angle between the velocity and the magnetic field increases from 90^0 to 180^0 because

$F_m = qvB \sin\theta$

For a given charge q the magnetic force vary with the magnitudes of $\vec{v}$ *and* B.

AnswerD

10. Given: Distance between two nearby creasts of a water wave equals to the wave lenght (λ) of the wave $\lambda = 0.40$ m

When six crests are passing a point every second with out considering the reference crest the frequency of the water wave is equal to

$f = 6/sec$

The speed of the wave becomes

$V = \lambda f = 0.4m \times \frac{6}{sec} = 2.4m/s$ **Answer C**

11. Sound waves produced infront of a big concrete wall can be heard behind the wal due to diffraction becauses there is no chance for the wave to be heard behind the wall by means of reflection, or refraction.

Answer C

12. Given The intensity of sound at a distance of 2m from the source is equal to $\frac{10^{-6}w}{m^2}$. The power of sound is related to its intensity according to the equation.

$P = I \times A$ where $A = 4\pi r^2$

$P = 10^{-6}\frac{W}{m^2} \times 4\ (3.14)(2m)^2 = 50.4 \times 10^{-6} Watt$

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$= 5.04\times10^{-5} Watt$

Answer A

13. In an inclined projectile (Projectile at an angle)
The vertical motion is non uniform because it is affected by gravity.
- The velocity in the y-direction decreases uniformly when the ball ascends.
- The vertical component of its velocity becomes zero when the ball has reached the maximum height.
- The horizontal motion is uniform because it is independent of gravity.
- The horizontal component of the velocity remains constant through out its motion.

Answer C

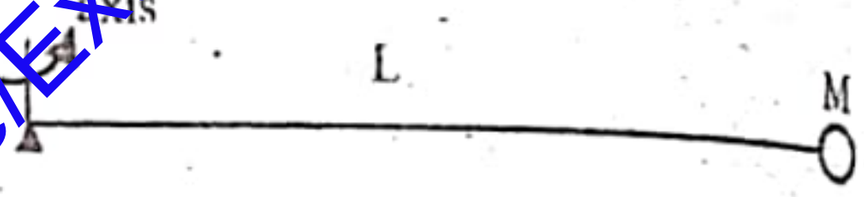
14. The angular velocity of the rotating body is defined as the rate of change of angular postion per unit time.

$$\text{Angular velocity} = \frac{\text{Change in angular positon}}{\text{time}}$$

$$\vec{w} = \frac{\Delta\vec{\theta}}{t} = \frac{(\vec{\theta}_f - \vec{\theta}_i)}{t}$$

Answer C

15. The rotational moment of inertia of the ball about an axis that passes through the pivot is given by



$$I = ML^2$$

Answer:D

16. Acceleration due to gravity at a distance
$R = R_E = 6.38\times10^6 m$ *is given by*

$$g_E = g = \frac{GM_E}{R_E^2}$$

At a point P where $h = R_E$ accalertion due to gravity is expressed as

$$g_P = \frac{GM_E}{(R_E + h)^2} = \frac{GM_E}{(R_E + R_E)^2}$$

$$g_P = \frac{GM_E}{4R_E^2} = \frac{1}{4}\left(\frac{GM_E}{R_E^2}\right) = \frac{1}{4}(g)$$

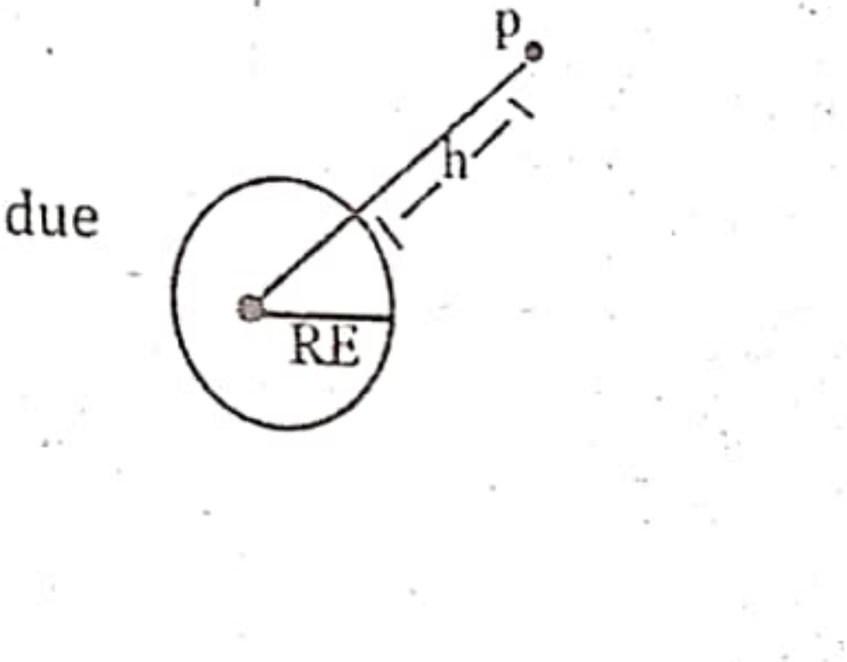
Answer D

17. The charge carried by the cloud floating over the lightning rod is negative and to protect the house from being struck by the lightning the rod must be positively charged.

Answer B

18. The charge $q_1 = -3\times10^{-6}C$ attracts the charge q placed at the origin. The force it exerts on $q = 1\mu C$ is

$$F_1 = F_x = \frac{kq_1q}{r_1^2}$$



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$$= \frac{9\times10^{9}\frac{Nm^2}{c^2}(3\times10^{-6}c)(10^{-6}_{C})}{(3m)^2}$$

$$=3\times10^{9}\times10^{-12}N = 3\times10^{-3}N$$

The charge $q_2 = -4\times10^{-6}C$ attracts the charge $q=1\mu C$ placed at the origin towards it by exerting a force

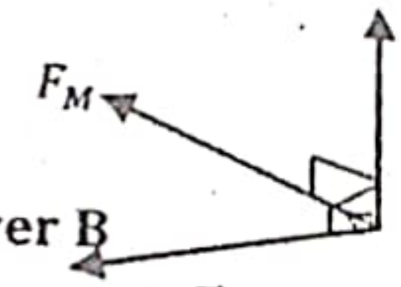
$$F_2 = F_y = \frac{kq_2q}{r_2^2}$$

$$= \frac{9\times10^{9}\frac{NM^2}{C^2}\times4\times10^{-6}C\times10^{-6}C}{(3M)^2} = 4\times10^{9}\times10^{-12}N$$

$$F_y = 4\times10^{-3}N$$

19. The direction of the magnetic force on the current carrying conductor is to the left of the page

Answer B

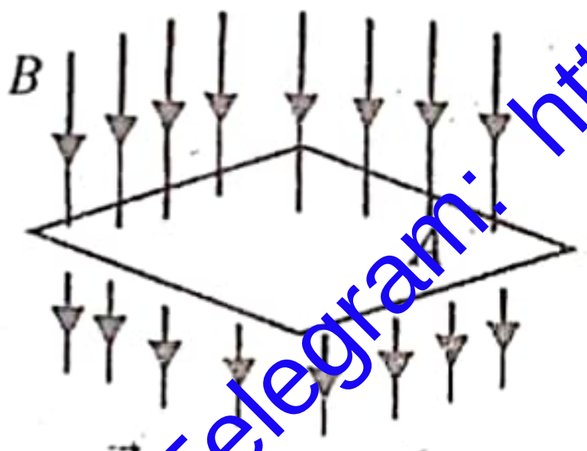


I directed to the top of the page

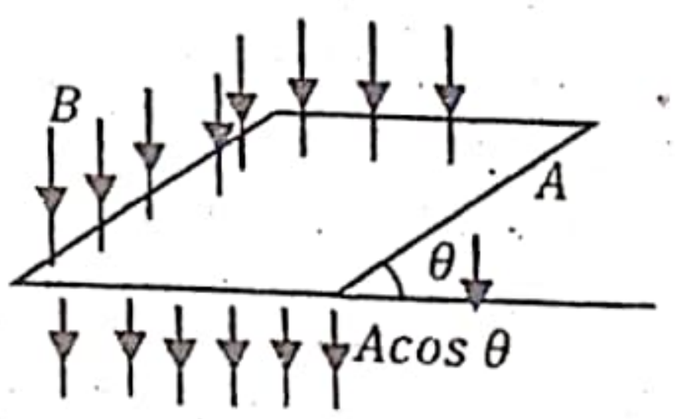
Answer B

$\vec{B}$ *directed out of the page*

20. Magnetic Flux is created when magnetic field lines penetrate (cross or thread) a given surface area



$\vec{B}$, A, and $A\cos\theta$



$\phi = BA\cos\theta$

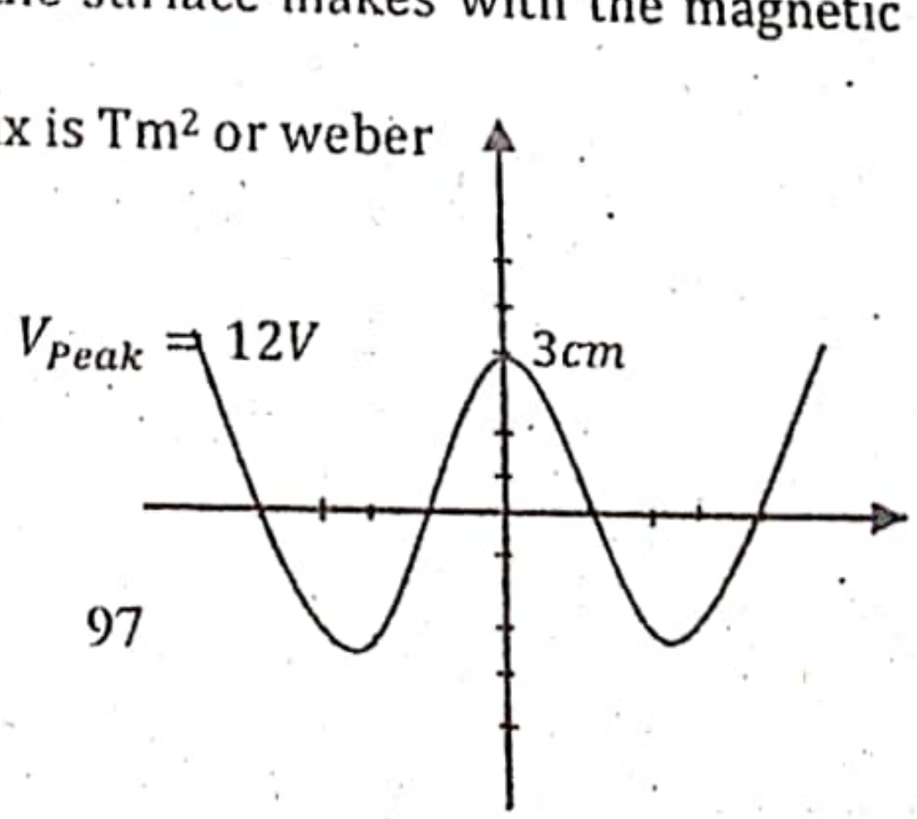
The magnetic flux depends on $\phi_{max} = BA$

a) the strength of the magnetic field
b) the cross-sectional area (A) of the surface
c) the angle the normal to the surface makes with the magnetic field and
d) The SI unit of magnetic flux is Tm^2 or weber
   1 Weber= 1 Tm^2

Answer D

21. $\Delta x = 3cm$

Voltage gradent $= \frac{V_{Peak}}{\Delta x} = \frac{12V}{3cm}$



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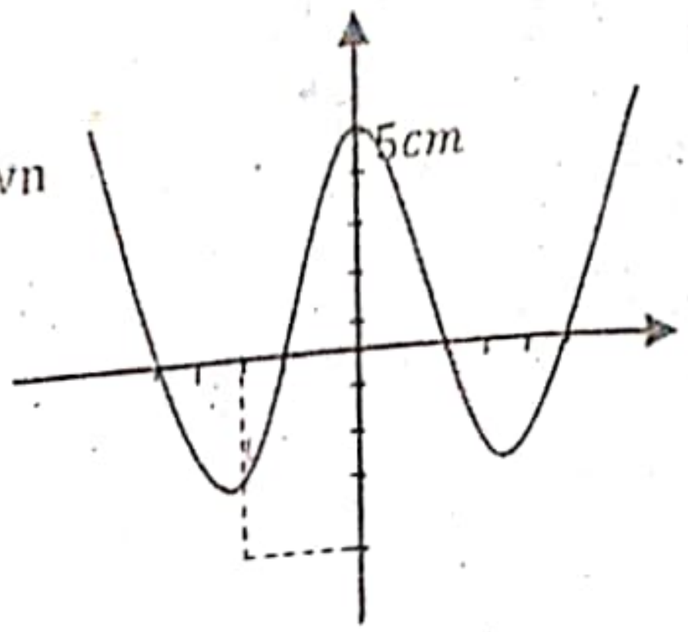
$= \frac{4V}{m}$



The peak voltage for the unknown voltage source is

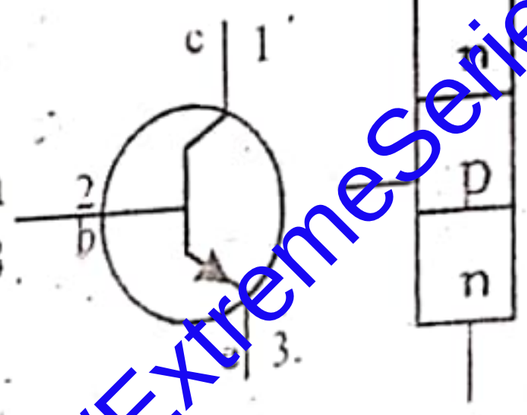
$V_{Peak} = \text{Voltage gradent} \times \text{Vertical displacement}$

$= \frac{4v}{cm}(5cm) = 20v$

Answer D

22. A pure semiconductor conducts electric current when an impure atom is added to its crystal struction. Such a process is called Doping. A semiconductor doped with acceptor impurity atoms is called a p-type.

Answer A



23. The emitter and the base junction are represented by the numbers 3 and 2 respectively

Answer B

24. Electromagnetiv waves are produced by charged particles that accelerate in a region of electric and magnetic field.
    - They can travel through matter and invaccuum.
    - They carry energy and momentum.
    - They are transverse in nature and do not consists compressions and rarefactions.

Answer B

25. From the law of reflection

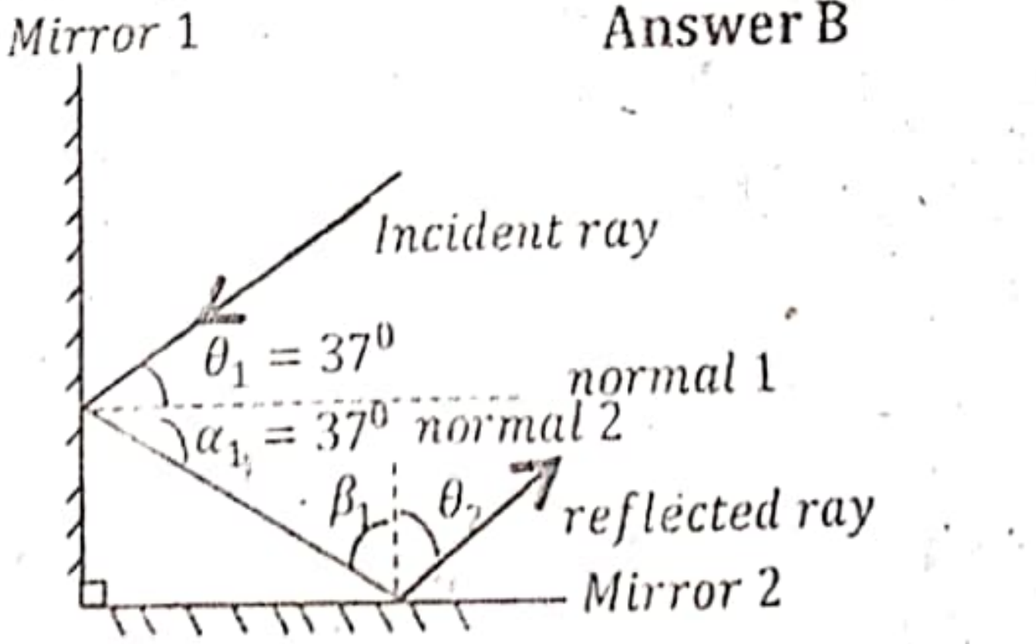
$\theta_1 = \alpha_1 = 37^0$

$\alpha_1 + \beta_1 = 90^0$

$\beta_1 = 90^0 - 37^0 = 53^0$

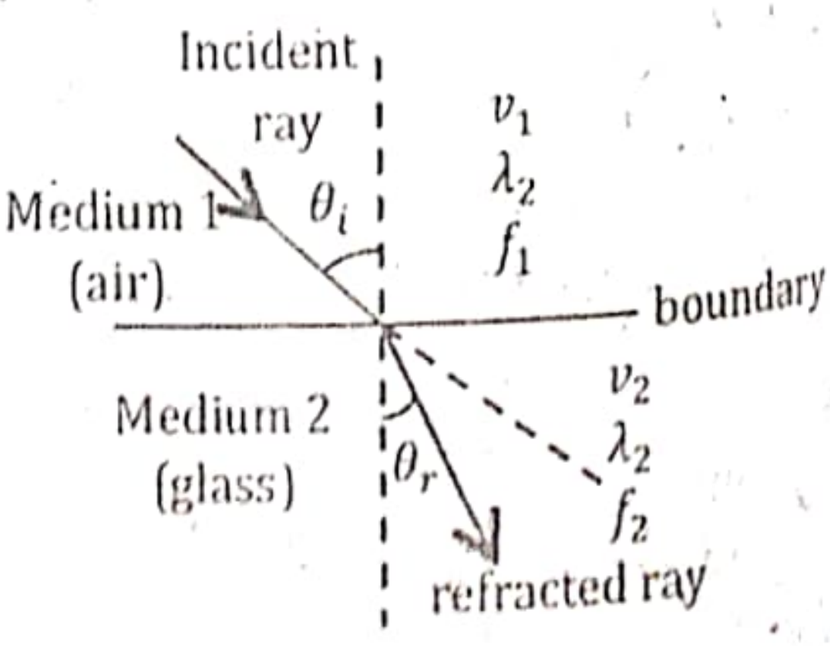
$\beta_1 = \theta_2 = 53^0$

From the law of reflection



Answer: C

26. When a ray of light incident from air enters glass it refracts and bends toward the normal $\theta_i > \theta_r$. During refraction the frequency of light in both mediums remains the same $(f_1 = f_2)$ but both speed and wave length change $v_{air} > v_{glass}$ and $\lambda_{air} > \lambda_{glass}$ light travels slower in glass than in air.



Answer:- C

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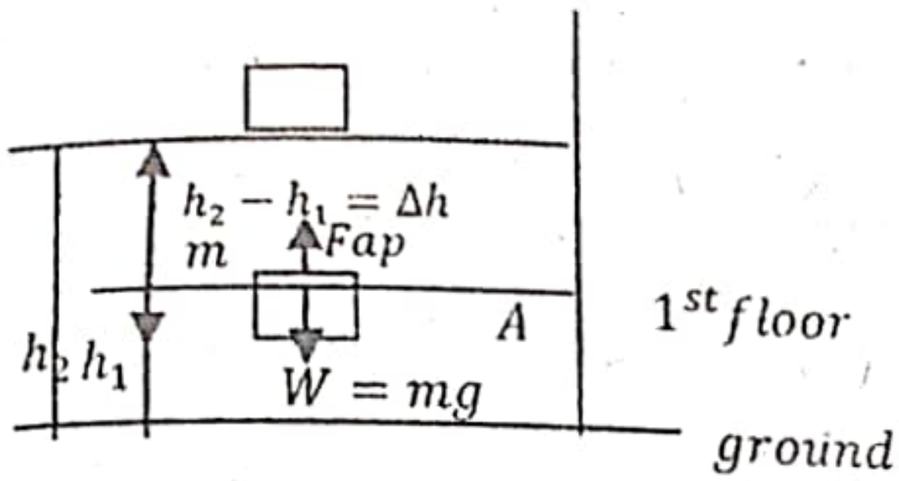
27. Given

Angle of incedince $=\theta_1^i = 53^0$

Requered:- $\beta_2 = ?$ B

Applying snell's law we have $2^{nd} floor$



$$\frac{\sin\theta_1}{\sin\beta_2} = \frac{n_2}{n_1} \Rightarrow \frac{\sin 53^0}{\sin\beta_2} = \frac{1.6}{1}$$

$$\frac{0.8}{\sin\beta_2} = 1.6 \Rightarrow \beta_2 = \frac{0.8}{1.6} = 0.5$$

$\beta_2 = \sin^{-1}\ (0.5) = 30^0$

Answer C

28. When the man lifts the brick from position A to postion B the work done on the brick by the gravitational force is given by

$W_{gravity} = -mg(h_2 - h_1) = -mg\Delta h$

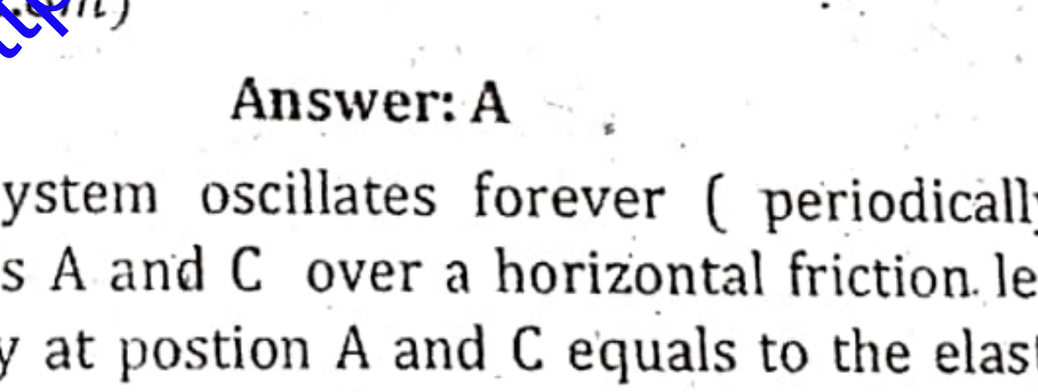
$= -0.2kg \times 10m/s^2(2.8m)$

$= -\frac{5.6kgm^2}{s^2} = -5.6J$ **Answer: A**

29. When the mass- spring system oscillates forever (periodically) between two limited postions A and C over a horizontal friction less surface its mechanical energy at postion A and C equals to the elastic P.E stored in the mass- spring system and at postion B its mechanical energy is only k.E.

According to the law of conservation of mechanical energy the total mechanical energy is the same every where.

Thus, $E_{MA} = E_{MB} = E_{MC}$



Answer C

30. Given: thickness $=t= S_L = 1.5cm$

Length$= L=S_E = 9cm$

$F_E = 80N$ *and* $F_L = 320N$

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The velocity ration of the metallic wedge is

$$V_R = \frac{S_E}{S_L} = \frac{9cm}{1.5cm} = 6$$

The AMA of the metallic wedge is

$$AMA = \frac{F_L}{F_E} = \frac{320N}{80N} = 4$$

The efficiency of the wedge is

$$\eta = \frac{AMA}{IMA} \times 100\% = \frac{4}{6} \times 100\%$$

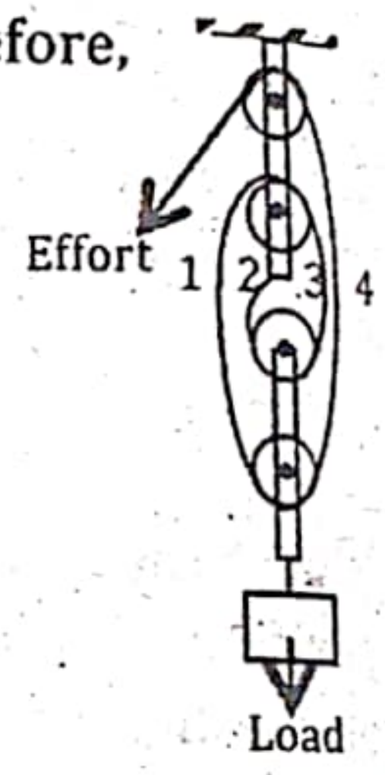
$$\eta = 66.7\%$$

$$W_{inpul} = F_E . S_E = 80N(9\times10^{-2}M) = 7.2J$$

Answer: D

31. The number of strands supporting the load is 4, therefore,

$$VR = 4 = \frac{S_E}{S_L}$$



- The distance moved by the effort is four times the distance moved by the load.

  $S_E = 4S_L$

- The load is four times the effort, (L=4E) and the mechanical advantage of the system is 4.
- When the effort moves through a distance of 8cm the distance moved by load is

$$S_L = \frac{S_E}{4} = \frac{8cm}{4} = 2cm$$

Answer:- B

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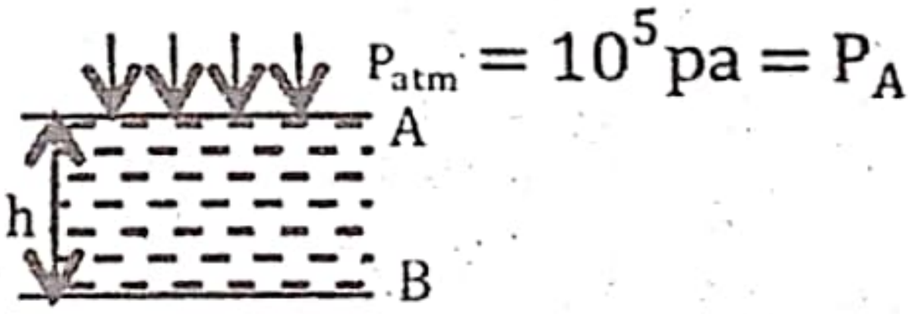
32. The pressure at B is

$P_B = 3P_A = 3\times10^5 Pa$

$\rho_\omega = 10^3 \frac{kg}{m^3}$

$g = 10\frac{m}{s^2}$



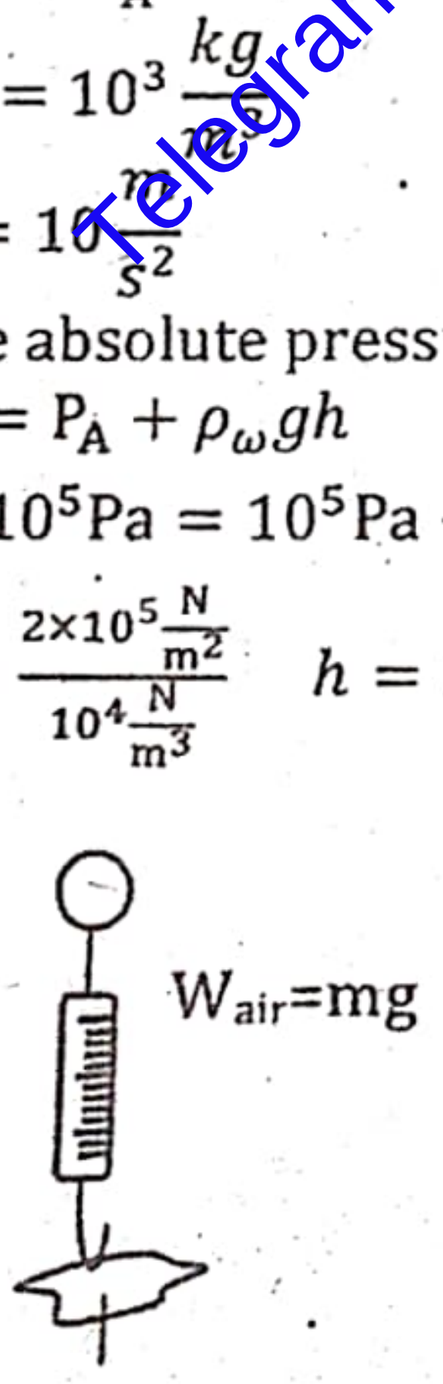
The absolute pressure at B is given by

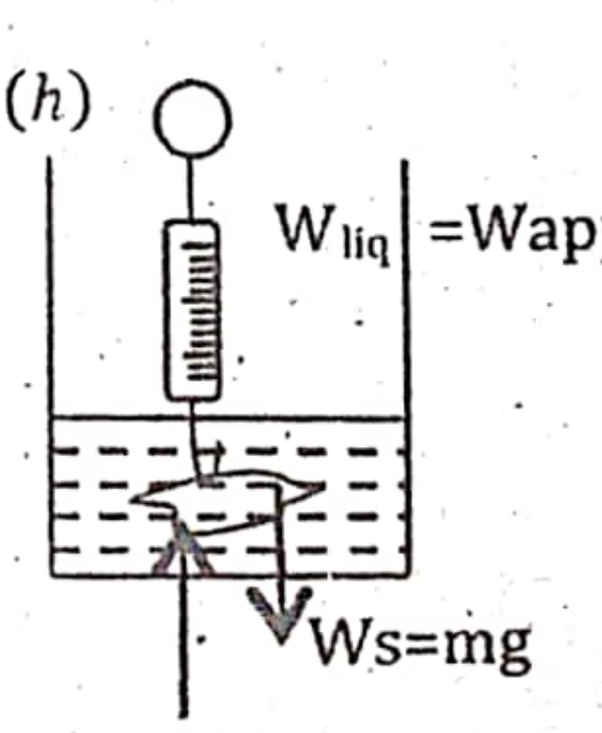
$P_B = P_A + \rho_\omega gh$

$3\times10^5 Pa = 10^5 Pa + 10^3 \frac{kg}{m^3}\left(10\frac{m}{s^2}\right)(h)$

$h = \frac{2\times10^5 \frac{N}{m^2}}{10^4 \frac{N}{m^3}}$ $h = 20m$

33.





Answer:- A

The up thrust of the liquid is the same as the weight of liquid displaced and is equal to the apparent loss of weight.

$W_{app\ loss} = W_{air} - W_L = W_{air} - W_{app} = W_{Ld}$

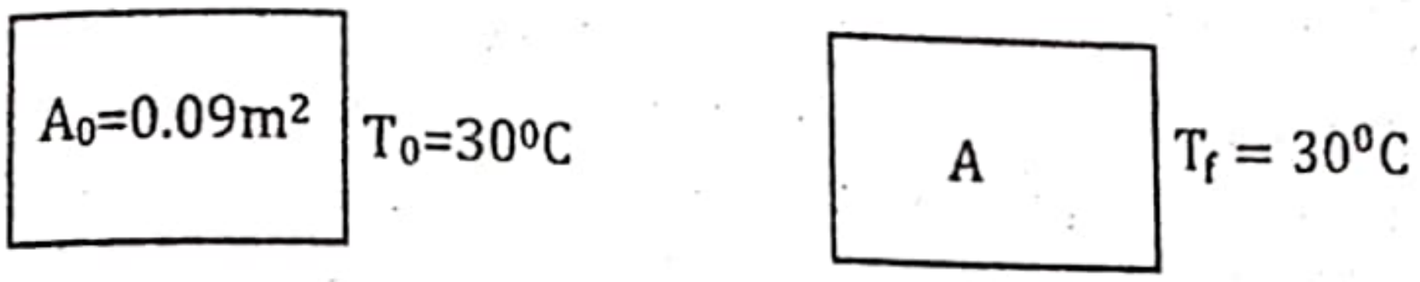
$W_{at.} - W_{app} = \rho_L V_L g$

$mg - W_{app} = \rho_L V_L g$

$W_{app} = mg - \rho_L V_L g$

Answer:- C

34.



Coefficient of linear expansion for metal= $\alpha = 1.1\times10^{-5}/\,^0C$

The change in area in area of the square metal sheet is

$\Delta A = 2\alpha A_0 \Delta T \quad \Delta T = T_f - T_o = 60^0C - 30^0C = 30^0C$

$= 2(1.1\times10^{-5}/^0C)(0.09m^2)(30^0C)$

$= 5.94\times10^{-5}m^2$

$\Delta A = 0.0000594m^2$

Answer:- D

35. Given:- $M_i = 720g,\ T_i = 10^0,\ C_i = 2200\frac{J}{kgk}$

$C_w = 4.187\times10^3\frac{J}{kgk}, L_f = 3.33\times10^5\frac{J}{kg}$

$T_f = 15^0C$

Step 1:- When the temperature of ice is raised from -10⁰C to 0⁰C of ice the heat added to ice is

$Q_1 = M_i C_i \Delta T_i,\ \Delta T_i = 0^0C - (-10^0C) = 10^0C = 10k$

$= 0.72kg\times2200\frac{J}{kg\,k}\times10^k = 15840J$

Step 2:- When ice at 0^0C melts and changes to water at 0^0C the heat added is

$Q_2 = M_i L_f = 0.72kg\times333000\frac{J}{kg} = 239.760J$

Step 3:- The heat added to raise the temperature of water from 0^0C to 15^0C is

$Q_3 = M_\omega C_\omega \Delta T_\omega = 0.72kg\times4187\frac{J}{kgk}\times15k = 45219.6J$

The total heat needed to convert 720g of ice at -10^0C to water at 15^0C is

$Q_{Tot} = Q_1 + Q_2 + Q_3$

$= 15840J + 239{,}769J + 45219.6J = 300819.6J$

$= 3.00\times10^5J$

Answer:- C

36. A wave that causes the elements of the medium to move perpendicular to the direction of movement of the wave is a transverse wave.

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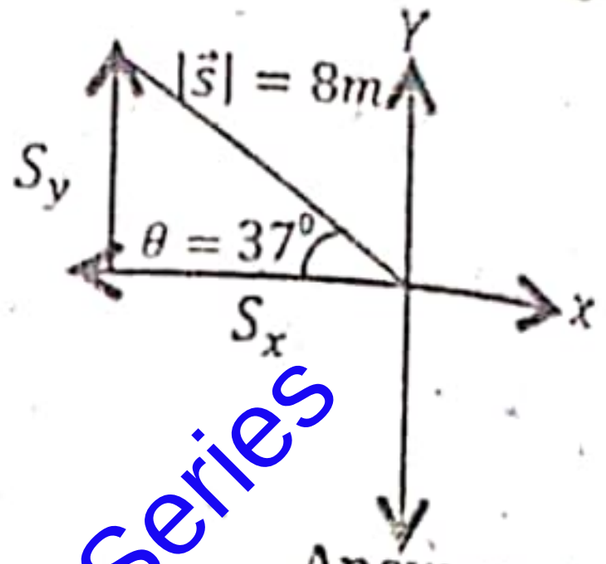
A wave that causes the elements of the medium to move parallel to the direction of movement of the wave is a longitudinal wave. Sound wave are longitudinal waves where as light waves are transverse waves.

Answer:- D

37. Electric current is a scalar quantity because it has no direction. In addition both distance and pressure are also scalars. Weight is a force and it is a vector quantity.

Answer:- D

38. The displacement vector ($\vec{s}$) lies in the second quadrant of the xy plane as shown in the figure.



The x-component of $\vec{s}$ is

$$S_x = -|\vec{s}|cos\theta$$
$$= -(8m)cos37^0 = -6.4m$$

The y component of $\vec{s}$ is

$$\vec{S}_y = |\vec{S}|\sin\theta = (8m)sin37^0 = 4.8m$$

Answer:- A

39. From the graph of displacement against time. The initial displacement of the body when t=0 is $\vec{S}_0 = 0$.

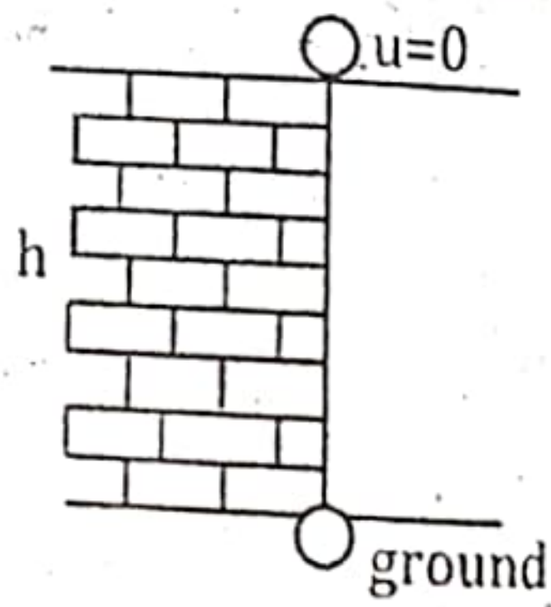
At the end of t=12min its position is $\vec{S}_f = 6km$. North.

The average velocity of the body in magnitude after 12min is

$$V_{ave} = \frac{\vec{S}_f - \vec{S}_0}{t_f - t_0} = \frac{6km - 0}{12min - 0} = 0.5\frac{km}{min}$$

Answer:- D

40. Given:- Initial velocity=$u = 0$ time of descent=t_d=3s acceleration $g = 10\frac{m}{s^2}$.

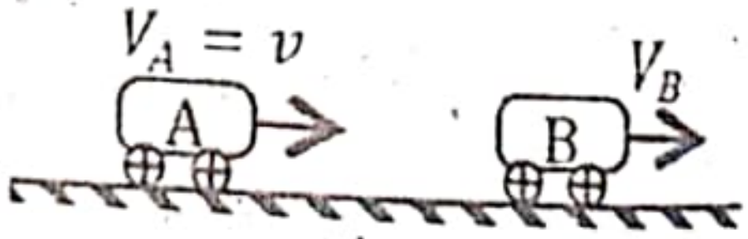


The height (h) of the building is

$$h = \frac{1}{2}gt^2 = \frac{1}{2}\left(10\frac{m}{s^2}\right)(3s)^2$$
$$= 5\frac{m}{s^2}(9s^2) = 45m$$

Answer:- A

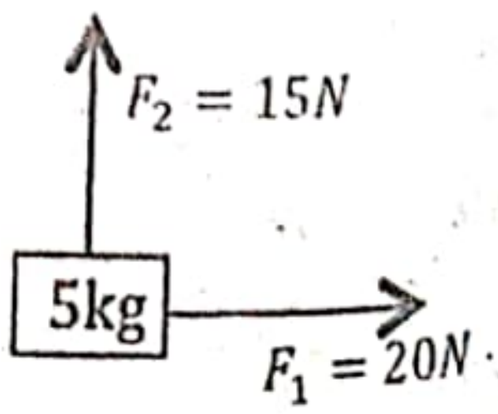
41. Speed of body A is $V_A = V$
Speed of body B is $V_B = 2V$
When the two bodies are moving in the same direction the velocity of B with relative to A is



$$V_{BA} = V_B - V_A = 2v - v = v$$

Answer:- B

42. Given:- $F_1 = 20N$, $F_2 = 15N$
$m = 5kg$ and
$F_2 \perp F_1$
The magnitude of their resultant force is.



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$$F_R = \sqrt{F_1^2 + F_2^2} = \sqrt{(20N)^2 + (15N)^2} = \sqrt{625N^2}$$

$F_R = 25N$

From Newton's 2nd law of motion the magnitude of the acceleration is

$F_R = ma$

$25N = 5kg(a) \Rightarrow a = \frac{25N}{5kg} = 5\frac{m}{s^2}$

Answer:- D

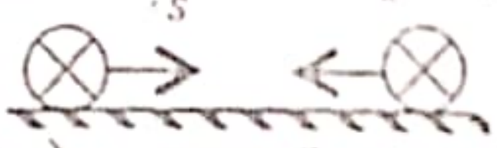
43. According to Newton's third law of motion the car is pushing the man back ward horizontally by a force equal in magnitude with that the man exerted on it.

Answer:- D

44. $m_1 = 3kg$ $m_2 = 2kg$

$v_1 = 20\frac{m}{s}$ $v_2 = 10\frac{m}{s}$

Before collision the total momentum of the system is



$(\vec{P}_{tot})_{bef} = m_1\vec{v}_1 + m_2\vec{v}_2$

$= 3kg\left(20\frac{m}{s} in\ the + ve\ x\ axis\right) + 2kg\left(10\frac{m}{s} in\ the - ve\ x\ axis\right)$

$= 60\ kg\frac{m}{s} to\ the + ve\ x\ axis - 20kg\frac{m}{s} to\ the + ve\ x\ axis$

$= 40kg\frac{m}{s} positive\ x - direction$

Since the collision is inelastic collision the total momentum of the system after collision is

$(\vec{P}_{tot})_{after} = (m_1 + m_2)\vec{v} = (3kg + 2kg)\vec{v} = (5kg)\vec{v}$

From conservation of momentum

$\vec{P}_{tot_{after}} = \vec{P}_{tot_{bef}}$

$(5kg)\vec{v} = 40kg\frac{m}{s}\ positive\ x - direction.$

$\vec{v} = \frac{40kg\frac{m}{s}}{5kg}\ positive\ x - direction$

$= 8\frac{m}{s}\ positive\ x - direction.$

Answer:- D

45. When $\vec{F}_1 = 50N$ to the right is applied to the object it accelerates in the direction of the force. When a force $\vec{F}_2 = 50N$ to the left is applied to the object the net force acting on it becomes zero and the object continues travelling to the right at a constant speed according to Newton's first low of motion.

Answer:- D

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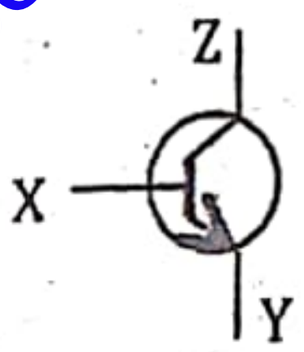
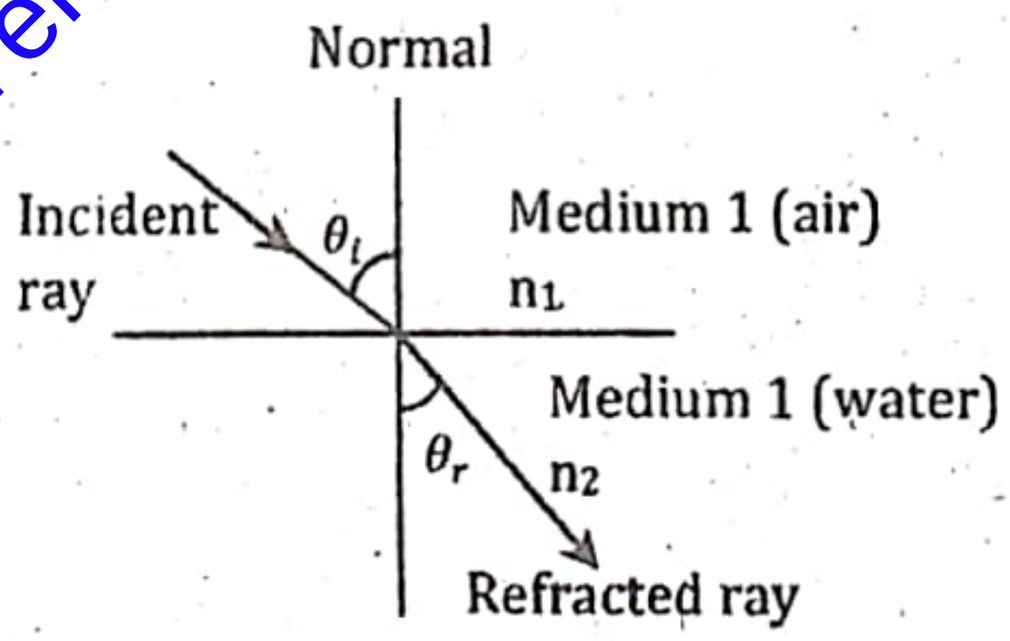
BOOKLET CODE:- 24

1. The SI unit of magnetic flux is Weber (Wb) or Tm^2.
   1 Weber = $1m^2 \times 1Tm^2$
   Answer:- A
2. A transformer is a device use to change voltage by stepping up or stepping down. The working principle of a transformer is based on Faraday's law of electromagnetic induction.
   Answer:- C
3. Cathode ray tubes are not used to measure electric field strength.
   Answer:- C
4. A diode is to rectify voltage by changing ac current to dc current, because a diode only allows current to flow only in one direction.
   Answer:- D
5. The letter X, y and Z respectively, represent the base terminal, the emitter terminal and the collector terminal.



   Answer:- B
6. Any electromagnetic wave can be produced by an accelerating change in a region of electromagnetic field.
   Answer:- D
7. For a regular reflection that takes place on a smooth boundary the law of reflection states that angle of incidence (θ_i) is equal to the angle of reflection (θ_r). Thus, $\theta_i = \theta_r$
   Answer:- B
8. Refraction is bending of light as it crosses the boundary between two transparent media having different indices of refraction



   Answer:- C
9. Formation of image by mirror is the consequence of reflection not the consequence of refraction.
   Answer:- B
10. The electric potential at a point in space is defined as potential energy (electrical work done) per unit charge to brought a charge at that point.

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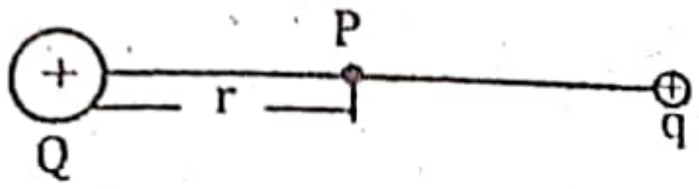


$$V_p = \frac{\Delta U}{q} = \frac{kQ}{r}$$

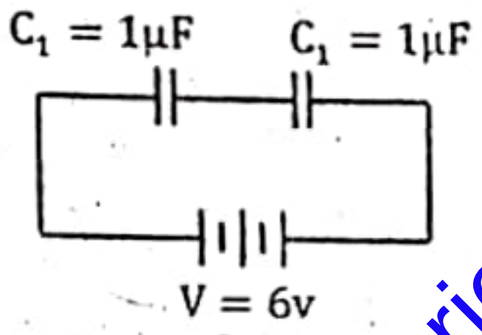
Answer:- A

11. $C_T = \frac{C_1 \times C_2}{C_1 + C_2}$

$$C_T = \frac{1\mu F \times 2\mu F}{3\mu F} = \frac{2}{3}\mu F$$

$$Q_T = C_T V = \frac{2}{3}\mu F \times 6v = 4\mu C$$

$$Q_T = 4\mu C$$



For series connection of capacitors $Q_T = Q_1 = Q_2 = 4\mu C$ the potential differece across the capacitors is

$$V_1 = \frac{Q_1}{C_1} = \frac{4\mu F}{1\mu F} = 4v \text{ and } V_2 = \frac{Q_2}{C_2} = \frac{4\mu F}{2\mu F} = 2v$$

The potential difference of the 2μF capacitor is half the potential difference of the 1μF capacitor.

Answer:- B

12. Given I=16A, t=4min=240sec

$$Q = It = 16A \times 240sec = 3840c$$

$$1\ coulomb = 6.25 \times 10^{18} electrons$$

3840C=?

No of electrons that pass a point=$3840C \times 6.25 \times 10^{18}\ ele/c$

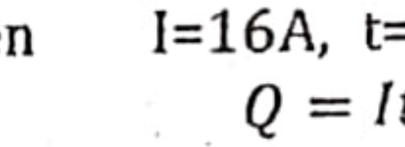
No of e= $2.38 \times 10^{22} \approx 2.4 \times 10^{22}$

Answer:- B

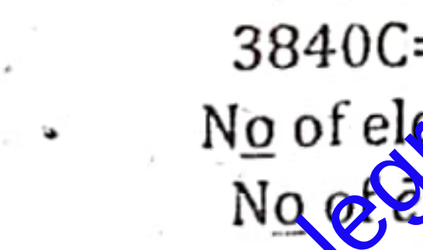
13. The effective resistance of two resistors R_1 and R_2 when connected in parallel is given by

$$\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{R_1 + R_2}{R_1 R_2} \Rightarrow R_{eff} = \frac{R_1 \times R_2}{R_1 + R_2}$$

Answer:- A

14. A) 

The total emf of the combination is=$\varepsilon_A = 2\varepsilon$

B) 

The total emf of the parallel $=\varepsilon_B = 2\varepsilon$ combination is

Answer:- D

15. Nuclear energy is not yet used as a source of electricity in present day Ethiopia.

Answer:- B

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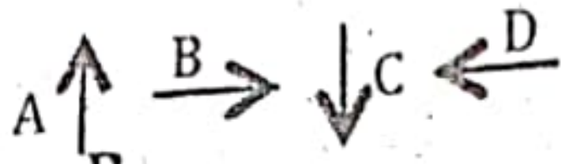
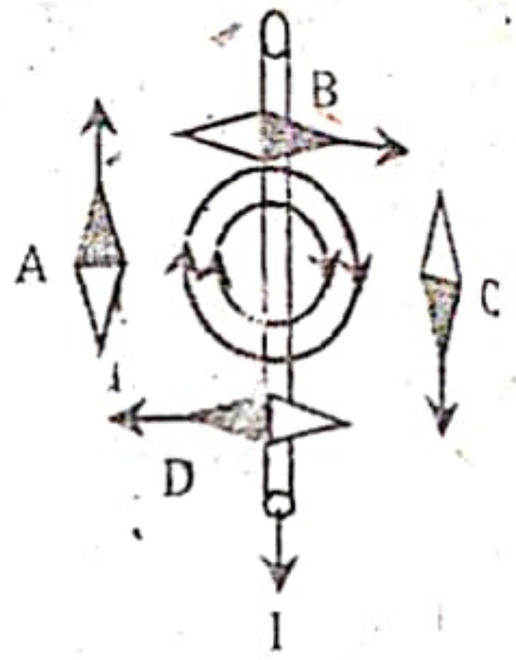
16. Applying the right hand rule the mggneti flux lines are concetric circles as shown in the figure.
When a compass (small magnet) is placed around the wire at positions A, B, C and D the direction of the compass needle is



Answer:- B

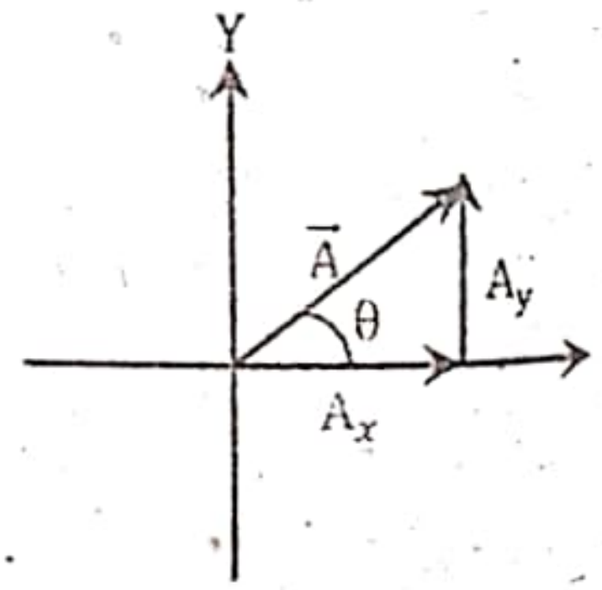


17. The pattern of the magnetic field lines of a solenoid is correctly represented by the diagram shown in choice D. **Answer:- D**
18. The equation of the magnetic force exerted on a current carrying wire is obtained from the equation $F = qvBsin\theta$ as follows.
We know $q = I\Delta t$ *and* $F = I(\Delta tv)Bsin\theta$ *but* $v\Delta t = L$, $F = ILBsin\theta$
This equation is derived by relating current in the wire with the velocity of the charged particle.
Answer:- C
19. Given $|\vec{S} + \vec{T}| = 12\ units$, $|\vec{S}| = 6\ units$ when $\vec{S}$ and $\vec{T}$ are parallel and have the same direction $|\vec{S} + \vec{T}| = |\vec{S}| + |\vec{T}| = 18\ units$ *and* $\vec{T} = -18\ units$. If $|\vec{T}| > 18\ units$, then $|\vec{S}| + |\vec{T}| > 12 units$ which is contrary to the given information. Therefore, the magnitude of vector $\vec{T}$ is between 6 and 18 units inclusive is the possible answer.
Answer:- C
20. Given:- $A_x = \frac{1}{2}A$ *and* $\vec{A}$ is a first quadrant vector on xy plane as shown $|\vec{A}| = A$ applying pythagoras theorem we get

$$A^2 = A_x^2 + A_y^2 = \left(\frac{1}{2}A\right)^2 + Ay^2$$

$$Ay^2 = A^2 - \frac{1}{4}A^2 = \frac{3}{4}A^2$$

$$A_y = \frac{\sqrt{3}}{2}(A)$$



Answer:- A
21. When the motion of a body is uniform rectilinear motion
a) The velocity remains the same in magnitude and direction and it does not vary with time.
b) The speed of th eobject and the average speed are equal at all times.
$v_i = v_f = v_{ave} = constant$
c) The slope under distance-time graph describes the speed which is constant.

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d) Distance increases uniformly in equal time interval and it is proportional to the time't' $s \sim t$ *but* $s \sim t^2$ is for accelerated motion.

Answer:- C

22. $a = -20\frac{m}{s^2}$ means motion with uniform deceleration.In uniform decelerated motion the velocity of the body decreases uniformly.

$v\left(\frac{m}{s}\right)$	80	60	40	20	0
$t(s)$	0	1	2	3	4

The speed of the body drops by $20\frac{m}{s}$ for every 1 second.

Answer:- A

23. Given $\vec{v}_B = 20\frac{m}{s}$ up the stream
$\vec{v}_R = 5\frac{m}{s}$ down the stream
The velocity of the boat relative to the ground is $\vec{v}_{BG} = \vec{v}_B + \vec{v}_R$

$$= 20\frac{m}{s} up + 5\frac{m}{s} down$$
$$= 15\frac{m}{s} up\ the\ stream$$

Answer:- B

24. According to the statement of Newton's second law the acceleration of a body is directly proportional to net force acting on it, inversely proportional to its mass and the direction of the acceleration is in the direction of the net force. $\vec{a} \sim \vec{F}_{net}$ *and* $\vec{a} \sim \frac{1}{m}$.

Answer:- D

25. Rocket propulsion and flying of birds is explained by Newton's 3rd law of motion.

Answer:- B

26. The total linear momentum of a system is conserved only in the absence of net external force. Thus, when $\vec{F}_{net} = 0, \Delta\vec{P} = 0 \Longrightarrow \vec{P}_{total\ bef} = \vec{P}_{total\ aft}$

Answer:- B

27. During elastic and inelastic collision linear momentum is always conserved, but kinetic energy is conserved only for elastic collision. During in elastic collision there is loss in KE.

Answer:- A

28. When two waves move in the same direction in phase they interfere each other constructively.

Answer:- A

29. When sound generated by a source travels from air to water it gets refracted and its speed as well as wave length changes. The sound ttavels faster as it enters water.

Answer:- C

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30. When the angle of elevation of a projectile is increased above 45^0 The maximum height decreased because range is maximum $=h_{max} = \frac{U^2 sin^2\theta}{q}$ increases but its-range $R = \frac{U^2 sin2\theta}{g}$ because range is maximum when $\theta = 45^0$.

Answer:- A

31. Angular velocity of a body in rotational motion is defined as the rate of change of angular position.

$$Angular\ velocity = \frac{change\ of\ angular\ position}{time}$$

$$\vec{\omega} = \frac{\Delta\vec{\theta}}{t}$$

Answer:- A

32. Given:- $\omega_0 = 1\frac{rad}{s},\ I_0 = 50kgm^2, I_f = 40kgm^2$

Required:- $\omega_f = ?$

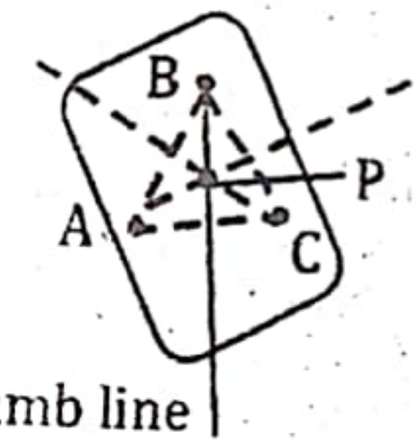
From conservation of angular momentum

$$L_f = L_0 \Longrightarrow I_f\omega_f = I_0\omega_0$$

$$\omega_f = \frac{I_0\omega_0}{I_f} = \frac{50kgm^2 \times 1\frac{rad}{s}}{40kgm^2} = 1.25\frac{rad}{s}$$

Answer:- B

33. The center of gravity will be at the intersection of the plumb lines (at P) as shown in the diagram.



Answer:- C

34. Given:- $M_P = 4M_E$

$R_P = 2R_E$

$V_E = 11\frac{km}{se}$ Required:- $V_P = ?$

$$mv_E^2 = \frac{GM_E m}{R_E^2} \Longrightarrow V_E = \sqrt{\frac{GM_E}{R_E}} = 11\frac{km}{s}$$

For a point mass m on the planet

$$\frac{mv_p^2}{R_p} = \frac{GM_p m}{R_p^2} \Longrightarrow V_p = \sqrt{\frac{GM_p}{R_p^2}} = \sqrt{\frac{G(4M_E)}{2R_E}}$$

$$V_p = \sqrt{2}\sqrt{\frac{GM_E}{R_p}} = \sqrt{2}\left(11\frac{km}{s}\right) = 11\sqrt{2}\frac{km}{s}$$

Answer:- B

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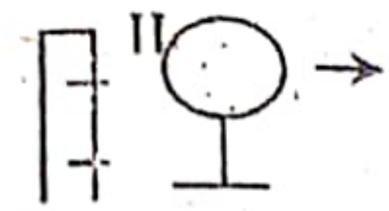
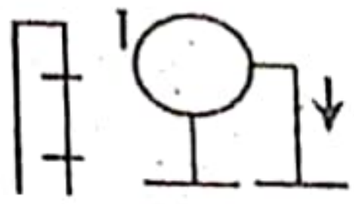
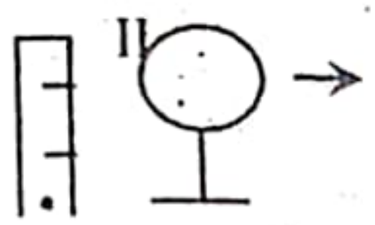
35.



Bring the charged rod



Ground the



Remove the ground from the

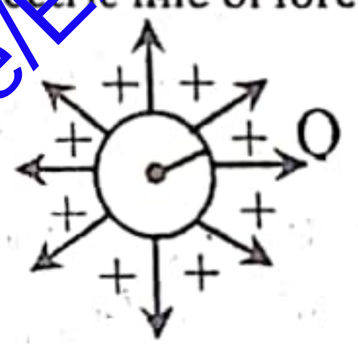
Remove the charged rod

The correct sequences to charge an eletroscope by induction is III, I, II and IV.

Answer:-

D

36. The electric field in side the conductor is zero because there is no charge inside the conductor. The electric field is perpendicular to the surface, not parallel to the surface.



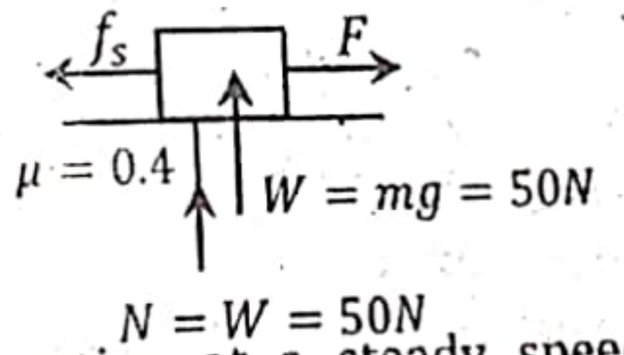
A spherical solid conductor with a net charge Q that reside on its outer surface

Answer:- B

37. The static friction force is

$f_s = \mu_s N$

=0.4(50N)=20N



The smallest force F needed to start motion at a steady speed is

$F_{net} = F - f_s = 0 \Longrightarrow F - 20N = 0 \Longrightarrow F = 20N$

Answer:- B

38. For a ball kicked up in to the air the work done by gravity is negative because the force of gravity is acting in opposite direction to the displacement. Thus, $W_{by\ gravity} = -mgh$

The work done by air resistance is also negative because air resistance is a dissipative force.

Answer:- B

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$$Mechanical\ advantage = \frac{Force\ applied\ to\ the\ machine}{Force\ exerted\ by\ it} = \frac{Load}{Effort}$$

Answer:- B

40. The relative density of a substance is defined as

$$Relative\ density = \frac{density\ of\ a\ substance}{density\ of\ water}$$

Answer:- C

41. Let A_0 be the original area at a temperature T_0 and A be the final area at a temperature T_f.

$\Delta A = A - A_0 = change\ in\ area\ \Delta T = T_f - T_0$. From law surface expansion

$\Delta A \sim A_0$ *and* $\Delta A \sim \Delta T$ which implies $\Delta A \sim A_0 \Delta T$.

$$\frac{\Delta A}{A_0 \Delta T} = constant = \beta = 2\alpha$$

$$\Delta A = 2A_0 \alpha \Delta T$$

Answer:-A

42. Given:- $C_A = 200\frac{J}{kg^0C}$, $\Delta T = 50^0C$, $m_A = 0.3kg$

The heat lost by the alloy=$Q_{lost} = m_A C_A \Delta T$

$Q_{lost} = 0.3kg \times 200\frac{J}{kg} \times 50^0C = 3000J$ **Answer:-C**

43. Given:- $T_{ice} = 0^0C$, $m_w = 0.1kg$, $T_w = 6^0C$, $T_f = 0^0C$

$L_f = 334{,}000\frac{J}{kg}$, $C_w = 4200\frac{J}{kg\,^0C}$

$\Delta T_w = 6^0C$ Required:- m_i =?

Heat lost by water=Heat gained by ice

$m_w C_w \Delta T_w = m_i L_f$

$0.1kg\left(4200\frac{J}{kg\,^0C}\right)(6^0C) = m_i\left(334{,}000\frac{J}{kg}L_f\right)$

$\frac{2520}{334000}kg = m_i$

$m_i = 0.754 \times 10^{-2}kg \approx 7.5 \times 10^{-3}kg$ **Answer:-C**

44. Where sinusoidal transverse wave is traveling on a string, any point on the string oscillates with the same frequency as the wave.

Answer:-D

45. The frequency of a wave is defined as the number of complete waves passing a given point in a unit of time. Thus,

$$frequency\ of\ a\ wave\ motion = \frac{number\ of\ complete\ waves\ passing\ a\ point}{time\ taken}$$

Answer:-A

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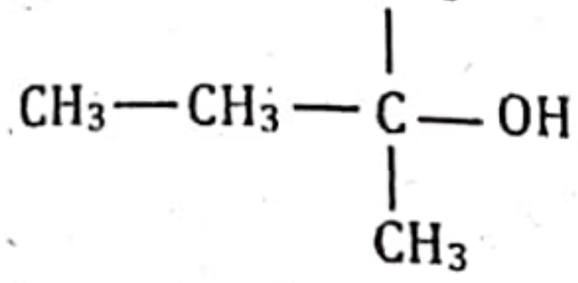
ETHIOPIAN GENERAL SECONDARY EDUCATION CERTIFICATE CHEMISTRY EXAMINATION GINBOT 2009/ MAY 2017

1. Coal is mainly composed of

1. Coal is mainly composed of
   A. phosphorus and sulfur
   B. fluorine and chlorine
   C. carbon and hydrogen
   D. hydrogen and oxygen
2. The molecule with the structural formula is an example of a



   A. tertiary alcohol
   B. quaternary alcohol
   C. primary alcohol
   D. secondary alcohol
3. Which of the following methods can be used to produce ethanol in industry?
   A. Hydrolysis of esters with potassium hydroxide solution.
   B. Dehydration of ammonium carbamate.
   C. Hydration of ethene with steam at 573k and 60 atm in the presence of a catalyst
   D. Hydrolysis of alkyl halides by heating with sodium hydroxide.
4. Which of the following reactions is an esterfication?
   A. $R-COOH+R'O \longrightarrow R-COO-R'+H_2O$
   B. $R-CH-OH \xrightarrow{[O]} R-CHO$
   C. $R-CHOH-R' \xrightarrow{[O]} R-CO-R'$
   D. $2R-CH_2-OH+2N \longrightarrow 2R-CH_2O^-Na^+$
5. What is the molecular formula of pentanoic acid?
   A. $C_5H_{10}O$ B. $C_5H_{10}O_2$ C. $C_5H_{11}O_2$ D. $C_5H_9O_2$
6. A real gas closely approaches the behavior of an ideal gas under the conditions of
   A. low pressure and low temperature
   B. high pressure and high temperature
   C. high pressure and low temperature.
   D. low pressure and high temperature.
7. Which of the following phase changes is an endothermic process?
   A. Condensation
   B. sublimation
   C. Freezing
   D. Deposition

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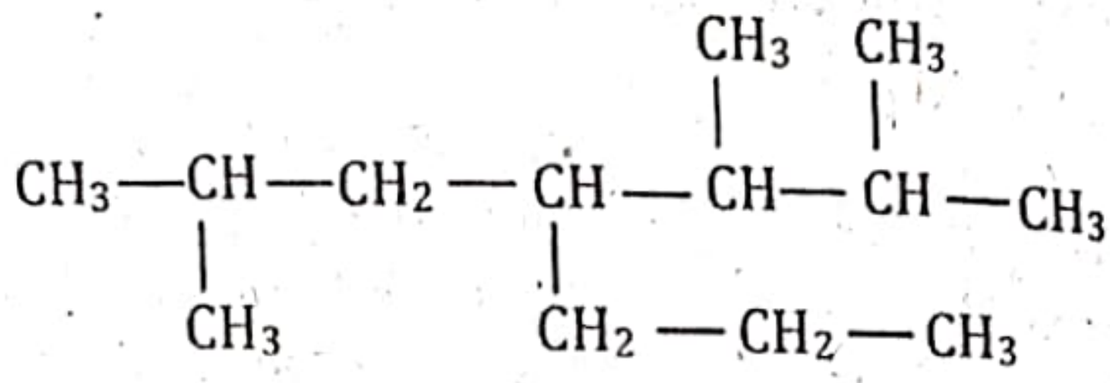
8. a part of an organic molecule that determines much of the chemistry a compound is known as _____ group.
   A. a functional
   B. an alkyl
   C. a homologous
   D. an aryl
9. The IUPAC name for the molecule having the following structure is



   A. 2- isopropyl -3- propyl -5- methyl hexane
   B. 2, 3, 6- trimethyl- 4- propylheptane.
   C. 2, 5, 6- trimethyl-4- propyleheptane
   D. 2- methyl – 4- n- propyl-5-isopropyl hexane.
10. Which of the following reactions could produce alkanes in laboratory?

   A. $R-CH=CH_2+H_2 \xrightarrow[Heat]{Pt}$

   B. $R-CH_2-CH_2-OH \xrightarrow[Heat]{H_2S_4}$

   C. $R-CH_2-CH_2-X+KOH \longrightarrow$

   D. $R-CH=CH_2+H_2O \longrightarrow$

11. Urea is commonly used as a
   A. fuel B. disinfectant C. fertilize D. pesticide
12. The term is that used to describe chemicals that are mainly obtai from the Earth and are not associated with living things is known _____.
   A. homologous
   B. isomer
   C. organic
   D. inorganic
13. Nitric acid is an example of
   A. triprotic acid
   B. polyprotic acid
   C. monoprotic acid
   D. diprotic acid
14. What is a strong acid? It is an acid that
   A. contains a relatively large amount (percentage) of acid.
   B. contains a relatively small amount of acid dissolved in amount of water.
   C. releases few hydrogen ions in aqueous solution.

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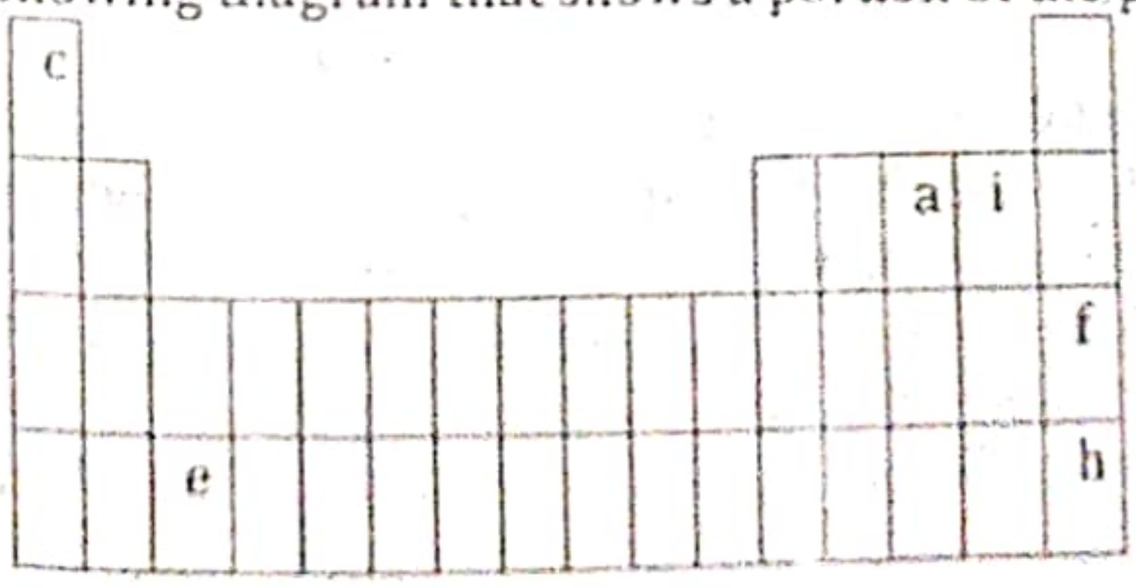
D. ionizes almost completely in aqueous solution.

15. Which of the following statements is true about the pH scale?
    A. The larger the pH value the more acidic, the smaller the pH the more alkaline.
    B. The hydrogen ion concentration of an acid of pH=4 is ten times greater than that of an acid of pH=3
    C. it ranges from 0 to 7.
    D. it is used to measure acidity and alkalinity.
16. The formula for an alkene containing three (3) carbon atoms is given by _____.
    A. C_3H_4 B. C_3H_7 C. C_3H_6 D. C_3H_8
17. Which of the following molecules does not exhibit geometrical isomerism?
    A. 3-hexene B. 2-pentene C. 2-hexene D. 1-butene
18. Acetylene is commonly used
    A. in the production of ethanol.
    B. as a fuel in oxy-acetylene torches for cutting and welding metals.
    C. in the production of polymers.
    D. for making detergents and plastics.
19. Which of the following statements is true about the properties of ethane, ethene and ethyne?
    A. Ethane undergoes substitution reaction whereas ethene and ethyne do not undergo substitution reaction.
    B. Ethane is an unsaturated hydrocarbon and ethene and ethyne are saturated hydrocarbons.
    C. Ethane decolorizes purple basic $KMnO_4$ whereas ethene and ethyne do not decolorize purple basic $KMnO_4$.
    D. Ethane, ethene and ethene show cis-trans (gemometrical) isomerism.
20. What is the fraction obtained from fractional distillation of petroleum at a boiling point range of (175-275)^{0}C
    A. Kerosene C. Petroleum ether
    B. Gasoline (Petrol) D. Fuel oil (Diesel oil)
21. Given the following diagram that shows a portion of the periodic table.



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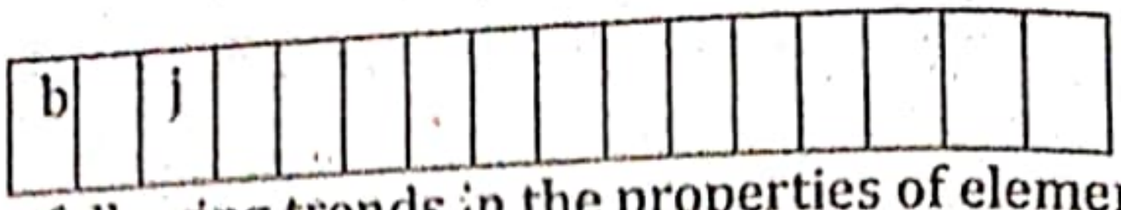

Which of the following trends in the properties of elements is true?

A. Electronegativity: b>j>e>c
B. Metallic character: h<f<i<a
C. Atomic radius: b>j>e>c
D. Ionization energy: h>f>i>a

22. The periodic table is not useful for predicting the
    A. behavior of many compounds.
    B. speed of a chemical reaction.
    C. formulas of compounds.
    D. physical and chemical properties of elements.
23. the force that binds atoms together in a molecule is called
    A. chemical bonding
    B. intermolecular force
    C. electrostatic force
    D. hydrogen bonding
24. given the reaction: $2Na + F_2 \longrightarrow 2NaF$ which of the following statement is true?
    A. Na^+ is larger in size than Na.
    B. F^- is larger in size than F.
    C. Na gains electron to form Na^+.
    D. F loses electron to form F-.
25. Which of the following substances is an ionic compound?
    A. SO_2 B. Al_2O_3 C. H_2O D. CO_2
26. Rutherford's α –particles experiment showed for the first time that an atom has
    A. a neutron B. an electron C. a proton D. a nucleus
27. What are the relative charges of an electron, a proton and a neutron respectively?
    A. -1, +1, 0
    B. 0, +1, -1
    C. +2, -1, 0
    D. 0, -1, +1
28. How many protons, neutrons and electrons respectively are there in $^{200}_{80}Hg$?
    A. 280, 120, 80
    B. 120, 280, 80
    C. 120, 80, 80
    D. 80, 120, 80
29. How many neutrons are there in the element having a chemical symbol $^{13}_{6}C$?
    A. 19 B. 7 C. 6 D. 13
30. Which of the following statements about atomic models is true?
    A. Rutherford model of an atom is a sphere in which electrons are embedded uniformly.
    B. in the Bohr atomic model, electrons are located in orbits.
    C. Dalton model of the atom is a positively charged solid sphere.
    D. Thomson model of the atom is a small hard ball which is indivisible.

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31. Covalent bonding is
    A. a bond in which one atom donates both electrons of the bond.
    B. a bond that results from the attraction between the metal ions and the surrounding sea of electrons.
    C. the force of attraction between oppositely charged ions.
    D. the binding force frormed by sharing of electrons in a molecule.
32. A chemist performs the same tests on two homogenous white crystalline solids, A and B. The results are shown in the table below

	Solid A	Solid B
Melting point	High, 801^{0}C	Low, decompose at 186^{0}C
Solubility in H_2O (grams per 100.0 g H_2O at 0^{0}C)	35.7	3.2
Electrical conductivity (in aqueous soloution)	Good conductor	Non conductor

The result of these tests suggest that
    A. both solids contain an ionic bond
    B. both solids contain a covalent bond.
    C. solid A is an ionic compound, whereas solid B is a covalent molecule.
    D. solid A is a covalent molecule, whereas solid B is an ionic compound.
33. What are dispresion forces? They are
    A. forces that act between all atoms and molecules.
    B. ion-dipole attraction forces.
    C. strong inter-molecular forces between polar molecules.
    D. ion-ion attraction forces.
34. In which of the following molecules the dispersion force most important?
    A. Ne and CH_4
    B. HCl and HBr
    C. H_2O and NH_3
    D. $CHCl_3$ and CO
35. According to the law of definite proportions,
    A. a compound always contains the same elements in the same propotion by mass.
    B. the masses of oxygen in N_2O, NO, N_2O_3, and N_2O_5 that combine with one gram of nitrogen are 0.57, 1.14, 1.71, 2.28 and 2.86.
    C. when two elements combine to form more than one compound, the masses of one element combined with a fixed mass of the second element are in the ratio of small whole numbers.
    D. matter is neither created nor destroyed in a chemical reaction.
36. What is an electron configuration? It is the
    A. arrangement of electrons in an atom.

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B. maximum number of electrons in a main energy level.
C. highest energy arrangement of electrons in an atom.
D. coefficient which shows the main energy level.

37. If a certain element has an atomic number of 35, what will be its electron configuration?
A. $[Ar]4s^23d^{10}4p^5$
B. $[Ar]4s^23d^94p^6$
C. $[Ar]\ 4s^24p^53d^{10}$
D. $[Ar]\ 4s^24p^63d^9$

38. If an element has an electron configuration of $1s^22s^22p^63p^1$, to which block does this element belong in the periodic table?
A. d B. f C. s D. p

39. What are the common names of group IA and IIA elements respectively?
A. Halogens, transition metals.
B. Alkali metals, alkali earth metals.
C. Alkali metals, halogens.
D. Halogens, noble gases.

40. How many groups and periods are there in the modern periodic table?
A. 7, 16 B. 16, 7 C. 7, 18 D. 18, 7

41. Electrolysis can not be used for
A. extraction of metals
B. refining of petroleum
C. electroplating
D. electrorefining

42. Which of the following is a non renewable natural resource?
A. Soil B. Plants C. Coal D. Animals

43. What is a chemical industry? It is a firm that
A. consumes relatively small quantities of energy during the manufacturing process.
B. converts a raw material taken from the environment into a usable product by physical means.
C. use unsafe operation methods in their manufacturing processes.
D. involves chemical reactions to transform raw materials into finished and semi-finished products.

44. Which of the following reactions is not involved in the extraction of aluminum?
A. $2Al_2O_3 + 3C \longrightarrow 4Al + 3CO_2$
B. $Cr_2O_3 + 2Al \longrightarrow Al_2O_3 + 2Cr$
C. $Al_2O_3 + 2NaOH \longrightarrow 2NaAlO_2 + 3H_2O$
D. $2Al(OH)_3 + \text{heat} \longrightarrow Al_2O_3 + 3H_2O$

45. Which of the following is a use of aluminum? It is used
A. in the manufacturing of household cooking utensils.
B. in the rock drills, cutting edges and parts of machinery.
C. as structural metal in the construction of bridges.

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D. to make coins, medals and bells.

46. What is the pH of a 0.00001M HCl solution?
A. 5 B. 9 C. 1 D. 4

47. When there is a small amount of base in a solution, it is known as
A. concentrated
B. supersaturated
C. dilute
D. saturated

48. If the pH of a solution is 4.0, what is the concentration of hydroxide ion in this solution?
A. 1×10^{-6}M
B. 1×10^{-14}M
C. 1×10^{-4}M
D. 1×10^{-10}M

49. Which of the following bases is used as a household cleaning agent?
A. NaOH B. KOH C. NH_3 D. $Ca(OH)_2$

50. Which of the following reactions does not produce a salt?
A. $2Na(s) + Cl_2(g) \longrightarrow$
B. $CaO(s) + 2HCl(aq) \longrightarrow$
C. $N_2O_5(s) + H_2O(l) \longrightarrow$
D. $HCl(aq) + NaOH(aq) \longrightarrow$

51. Which of the following are the correct steps in the production of sugar?
A. collecting the harvest → cleansing & *grinding* → *jucing* → *clarifying* → *evaporation* → *crystallization* → *refinery* → *separation and packaging.*
B. collecting the harvest → clarifying → evaporation → jucing → crystallization → refinery → cleansing & *grinding* → *separation and packaging.*
C. collecting the harvest → jucing → clarifying → cleansing & grinding → crystallization → evaporation → refinery → separation and packaging.
D. collecting the harvest → refinery → crystallization → evaporation → jucing → clarifying → cleansing & grinding → separation and packaging.

52. Which of the following is the modern method for processing and preserving foods?
A. Canning B. Smoking C. Freezing D. Drying

53. Of the following, which one is the effect of air pollutants?
A. Particulates- enhance greenhouse effect leading to global warming.
B. Excess carbon dioxide-cause drop in pH of rivers and lakes.
C. Hydrocarbons and ozone- formation of smog.
D. CFCs- cause an acid rain.

54. Which of the following is not a major factor for land pollution?
A. Leaching of heavy metals.

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B. The spillage of oil.
C. Dumping non-biodegradable wastes.
D. The use of organic fertilizers.

55. What are some of the negative effects of industries on the environment? They
A. process waste products.
B. reduce carbon dioxide emission.
C. discharge pollutants into the environment.
D. treat water before discharge.

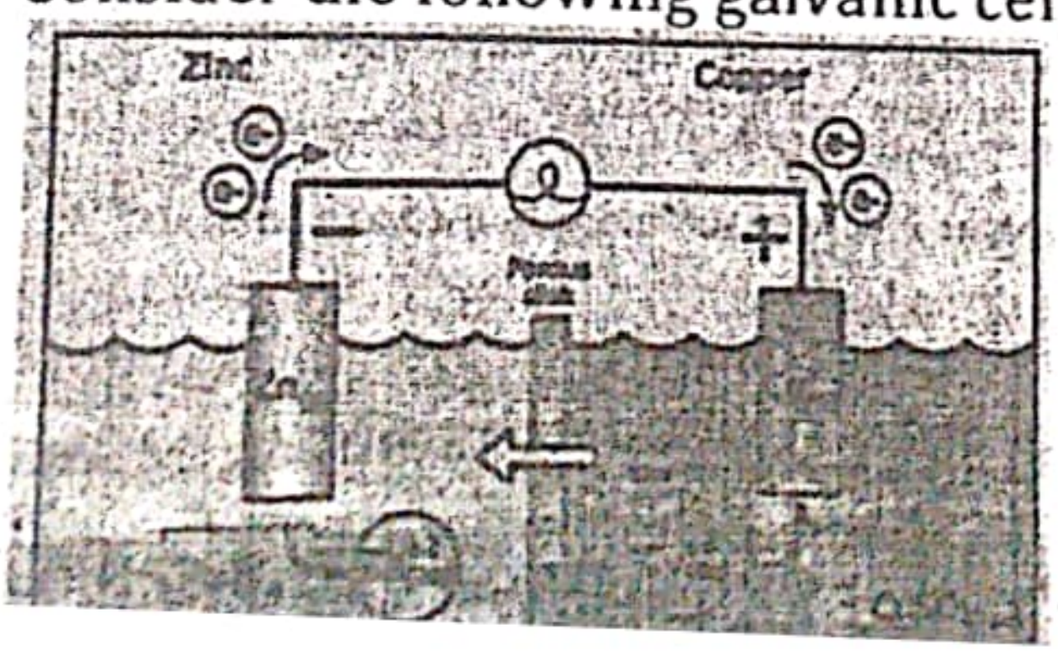
56. Which of the following substances is used as a pesticide?
A. $Cu(NO_3)_2$ B. $(NH_4)_2HPO_4$ C. KNO_3 D. NH_4NO_3

57. What is the difference between metallic conduction and electrolytic conduction?
A. In metallic conduction passage of charge brings about physical and chemical changes, whereas in electrolytic conduction brings about only physical changes.
B. Metallic conduction involves transfer of matter in the form of ions, where as electrolytic conduction does not involve transfer of matter.
C. In metallic conduction the ions are able to move in the solution, whereas in electrolytic conduction delocalized valence electrons are able to move in the solution.
D. In metallic conduction electrical charge is carried by electrons, whereas in electrolytic conduction electrical charge is carried by ions.

58. The reaction that takes place at each electrode in an electrochemical cell is known as_____.
A. oxidation B. half-cell C. reduction D. cell reaction

59. As opposed to a galvanic cell, in an electrolytic cell
A. electrons flow from cathode to anode.
B. the anode is positive.
C. spontaneous redox reaction generates electricity.
D. oxidation takes place at the cathode.

60. Consider the following galvanic cell:



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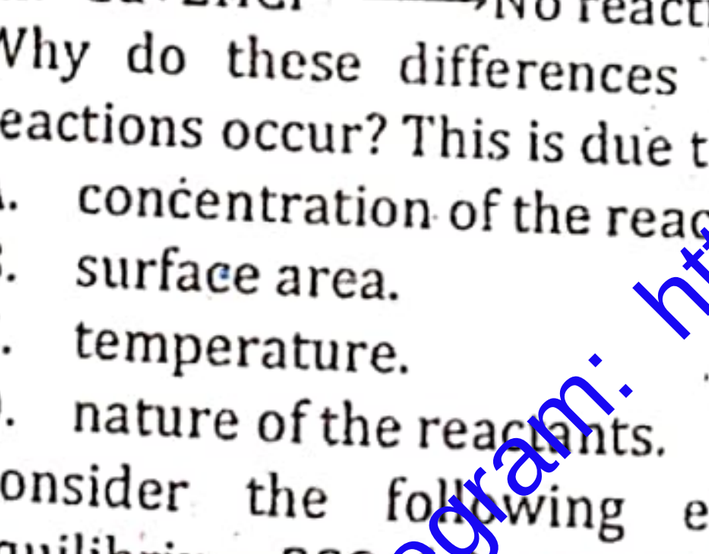
which of th efollowing statement about this cell is true?

A. The anode reaction is: $Zn\ (s) \longrightarrow Zn^{2+} + 2e^-$
B. The cell reaction is: $Cu\ (s) + Zn^{2+} \longrightarrow Cu^{2+} + Zn(s)$
C. Zinc is the cathode.
D. Copper is the anode.

61. In which of the following processes is nitrogen reduced?

A. $NO_2 \longrightarrow NO_3^-$
B. $NO_2^- \longrightarrow NO_2$
C. $NH_4^+ \longrightarrow N_2$
D. $NO_3^- \longrightarrow NO$

62. The following figure shows the rate changes during a chemical reaction.



At which point(s) of the graph does the reaction go to completion? At point(s)

A. I and III
B. III
C. I and II
D. II

63. Given the following reactions:

I. $Mg + 2HCl \longrightarrow MgCl_2 + H_2$ (very fast reaction)
II. $Fe + 2HCl \longrightarrow FeCl_2 + H_2$ (slow reaction)
III. $Cu + 2HCl \longrightarrow$ No reaction.

Why do these differences in the rate of the reaction among the reactions occur? This is due to the difference in

A. concentration of the reactants.
B. surface area.
C. temperature.
D. nature of the reactants.

64. Consider the following equation representing the reaction at equilibrium: $2SO_2(g) + O_2 \rightleftharpoons 2SO_3 + \text{heat}$. Which change causes the equilibrium to shift to the right?

A. Addition of a suitable catalyst.
B. Increasing the temperature.
C. Increasing the concentration of SO_2 or O_2.
D. Decreasing the pressure.

65. Which state of matter has no definite shape and no definite volume?

A. Plasma B. Gas C. Solid D. Liquid

66. Given the unbalanced equation for the complete combustion of ammonia: $NH_3(g) + O_2(g) \longrightarrow NO(g) + H_2O(g)$ What would be the coefficient of O_2 when the equation is correctly balanced?

A. 4 B. 5 C. 2 D. 3

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67. Which of the following is true about energy changes in chemical reactions?
    A. A chemical reaction involves the forming of bonds in the reactants and the breaking of bonds in the products.
    B. If two substances react and the temperature of the mixture decreases, the reaction is exothermic.
    C. If two substances react and the temperature of the mixute increases, the reaction is endothermic.
    D. If a reaction is exothermic; more energy is released when the bonds of the products are formed than it takes to break the bonds of the reactants.
68. An enothermic reaction is a reaction in which,
    A. the temperature of the mixute increases.
    B. there is an increase in enthalpy therefore the change in enthalpy is positive.
    C. heat is given out to the surroundings.
    D. enthalpy of the reactants is more than the enthalpy of the products.
69. A reaction that involves the breaking of a single compound intotwo or more simple substances is known as _____.
    A. decomposition
    B. double displacement
    C. comination
    D. single displacement
70. Which of the following reactions is an example of a double displacement?
    A. $ZnCO_3 + heat \longrightarrow ZnO + CO_2$
    B. $Ca + 2H_2O \longrightarrow Ca(OH)_2 + H_2$
    C. $2NaHCO_3 + heat \longrightarrow Na_2CO_3 + CO_2 + H_2O$
    D. $BaCl_2 + Na_2SO_4 \longrightarrow BaSO_4 + 2NaCl$
71. According to the kinetic moleculuar theory of matter, the particles in
    A. solid are not in fixed positions but able to move.
    B. liquid have more energy than in a gas.
    C. gas are in fixed positions.
    D. liquid are closely packed than in a gas.
72. Which one of the following statements is not consistent with the kinetic molecular theory of gases?
    A. The average kinetic energies of different gases are different at the same temperature.
    B. The theory explains the observed behavior of gases at ordinary temperatures and pressures.
    C. Individual gas molecules are relatively far apart.

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D. The actual volume of the gas molecules is very small compared to the volume occupied by the gas at ordinary temperatures and pressures.

73. The statement : The volume of a given mass of gas is inversely proportional to its pressure provided that the temperature remains constant." is th elaw of
A. Combined gas
B. Graham's
C. Charles
D. Boyle's

74. If a sample of nitrogen gas occupies 5.0L at 298K, at what temperature will it occupy 10.0L at a constant pressure?
A. 596 K B. 124K C. 149K D. 298K

75. The ideal gas equation, PV=nRT, could be derived from which of the following equations?
A. $V \alpha n, \frac{PV}{T} = k, \text{ and } r\alpha \frac{1}{\sqrt{D}}$
C. $V \alpha T, \frac{PV}{T} = k, \text{ and } r\alpha \frac{1}{\sqrt{D}}$
B. $V \alpha \frac{1}{P}, \frac{PV}{T} = k, \text{ and } V\alpha T$
D. $V \alpha \frac{1}{P}, V\alpha T \text{ and } V\alpha n$

76. Given the following balanced equation representing the reaction: $Zn + 2HCl \longrightarrow ZnCl_2 + H_2$ how many liters of hydrogen gas at STP is liberated when 6.54g of zinc is dissolved in an excess hydrochloric acid?
A. 1.12L B. 2.24L C. 22.4L D. 11.2L

77. Given the following balanced equation representing the reaction: $S + 3F_2 \longrightarrow SF_6$ how many grams of SF_6 will be produced when 3.2g of sulfur reacts with 3.8g of fluorine? (Atomic masses: S=32, F=19)
A. 4.9g B. 14.6g C. 17.3g D. 4.09g

78. Given the following equation representing the reaction: $2SO_2 + O_2 \longrightarrow 2SO_3$ if 20g of sulphur dioxide reacts with excess oxygen to produce 23g of sulphur trioxide, what is the percent yield? (Atomic masses: S=32, O=16)
A. 25 B. 10 C. 92 D. 50

79. Reduction is a process in which,
A. there is no net hange in the oxidation number of element(s).
B. a substance loses electron(s) in a chemical reaction.
C. there is an increase in the oxidation number of element (s).
D. a substance gains electron(s) in a chemical reaction.

80. Which statement is true about an oxidizing agent? It is a substance that
A. contains an element that increases its oxidation number.
B. contains an element that undergoes oxidation.
C. brings about oxidation but itself reduced.
D. brings about reduction but itself oxidized.

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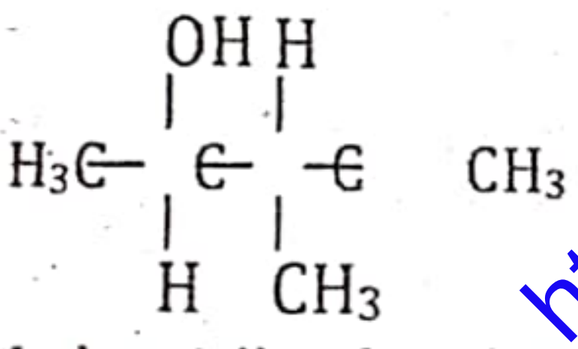
ETHIOPIAN GENERAL SECONDARY EDUCATION CERTIFICATE
CHEMISTRY EXAMINATION GINBOT 2010/MAY 2018

BOOKLET CODE:-29

1. What is the name of a naturally occurring unrefined petroleum product composed of hydrocarbon deposits and other organic materials?
   A. Kerosene
   B. Gasoline
   C. Crude oil
   D. Petroleum ether
2. What are the composition of coal? It is mainly composed of
   A. methane, ethane, propane, butane and small quantities of higher alkanes.
   B. alkanes, cycloalkanes and aromatic hydrocarbons.
   C. element carbon, low percentage of solid, liquid and gaseous hydrocarbons.
   D. low percentage of carbon, compounds of nitrogen and sulphur.
3. Which of the following is the functional group of alcohols?

   A. $-\overset{\overset{O}{\|}}{C}-$ B. $-\overset{\overset{O}{\|}}{C}-H$ C. $-\overset{|}{C}-OH$ D. $-\overset{\overset{O}{\|}}{C}-OH$

4. What is the IUPAC name of the following organic molecule?



   A. 3-methyl-2-butanol
   B. 2, 2-dimethyl-2-butanol
   C. 2- methyl-3-butanol
   D. 3, 3-dimethyl-2-butanol
5. Industrially ethanol can be prepared by
   A. dehydrohalogenation of ethyl halide with KOH.
   B. fermentation of carbohydrates such as sugar.
   C. the reaction between calcium carbide and water.
   D. the reaction between CH_3COONa and soda lime.
6. Consider the following table that shows different liquids with their boiling points.

Liquid	Boiling point (^0C)
Water	100
Ethanol	78
Diethyl ether	35
Toluene	120

   Which liquid has the highest vapor pressure?
   A. Diethyl ether B. Ethanol C. Toluene D. Water
7. The following figure shows the change in temperature as a solid substance is heated.

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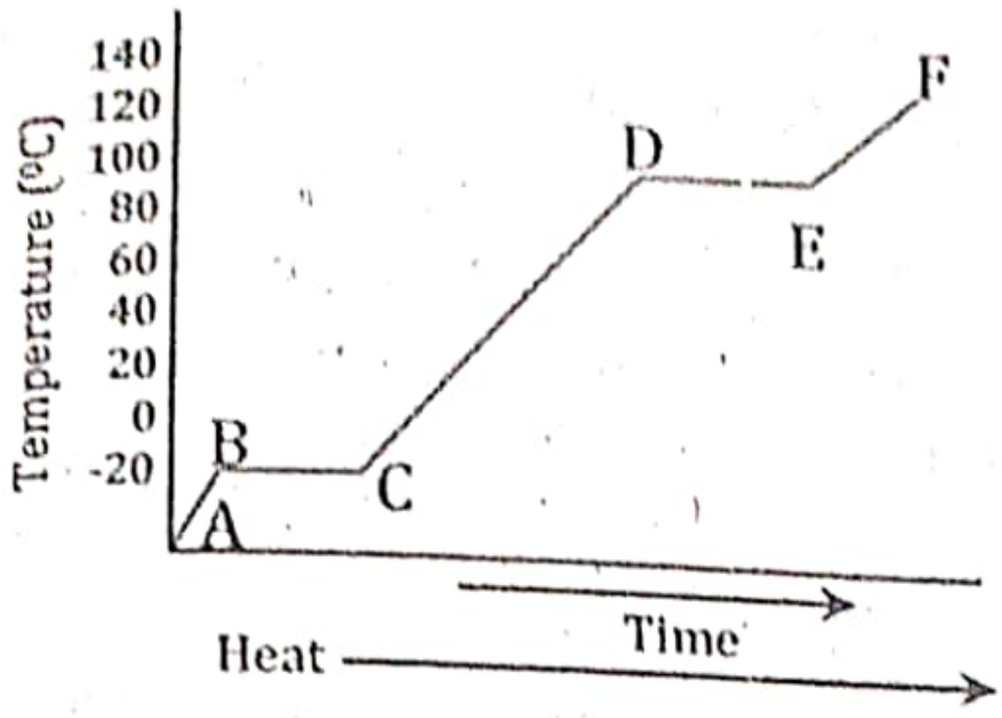


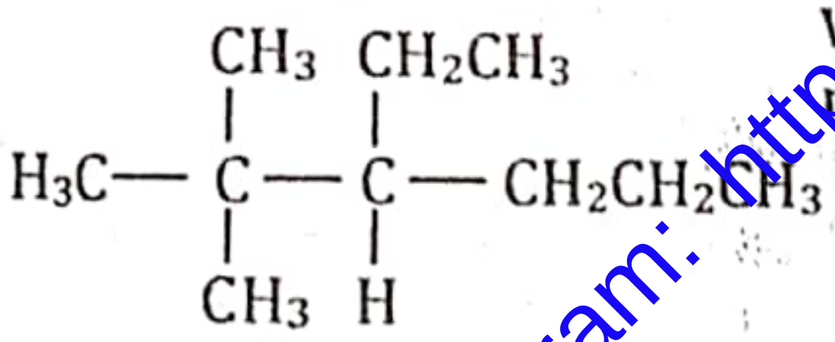
Which path represents the co-exlstence of liquid and gas phases?

A. Path C to D
B. Path D to E
C. Path A to B
D. Path B to C

8. Which of the following is true about the homologous series? The homologous series is

A. the existence of chemical compounds that have the same molecular formula but different structural formulas.
B. a structure in which the isomers have different position of groups with respect to the double bond.
C. the property of an element to form chains with repetition of the same atom.
D. a group of molecules each differing from the next member by a certain group.

9. The following is the structural formula for a certain hydrocarbon



2, 2-dimethyl-3-ethylhexane

What is the correct IUPAC name of the molecule?

A. 2, 2-dimethyl-3-ethylhexane
B. 4-ethyl-5, 5-dimethylhexane
C. 2,2-dimethyl-3-propylpentane
D. 2, 2-dimethyl-3-ethylpentane

10. In which solvent will both alkenes and alkynes dissolve?

A. Alcohols B. Acetic acid C. Water D. Benzene

11. What is the molecular formula for butanal?

A. $C_4H_8O_2$ B. $C_4H_9O_2$ C. C_4H_8O D. $C_4H_{10}O$

12. What are oxides? They are

A. base anhydrides that react with water to form bases.
B. binary compounds that contain oxygen in the -2 oxidation state.
C. acid anhydrides that react with water to form acids.
D. substances that have the link –O-O-

13. Which of the following is not true about the chemical property of acid anhydrides? They react with

A. both acids and bases to form salt and water.
B. basic oxides to give salts.
C. water to form acids.

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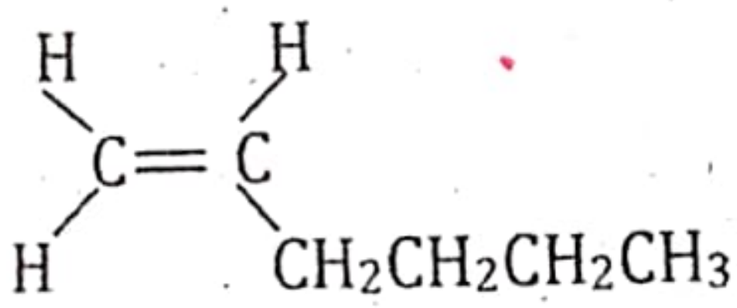
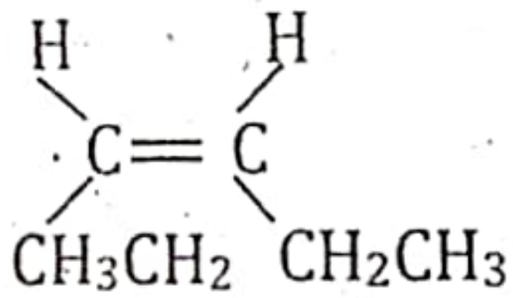
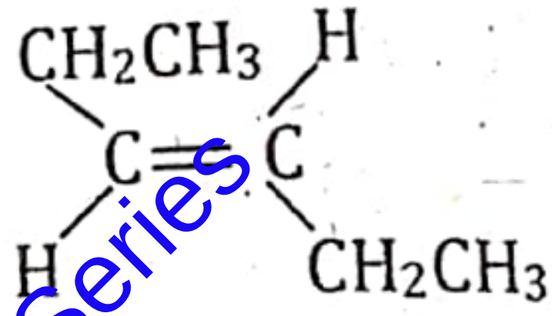
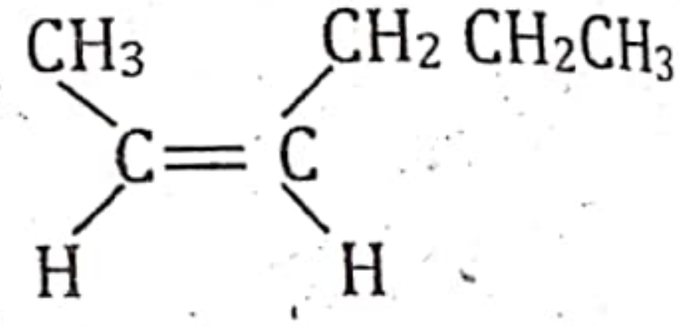
14. Amphoteric oxides
    A. can be represented by CuO, FeO, Fe_2O_3 and Cr_2O_3.
    B. are all non- metallic oxides.
    C. have neither acidic nor basic properties.
    D. have both acidic and basic properties.
15. Oxides that have neither acidic nor basic properties are _____
    A. amphotheric B. neutral C. acidic D. basic
16. The spatial arrangement of atoms of groups about a double bond is known as _____ isomerism.
    A. chain B. functional C. geometrical D. position
17. Which one of the following structures represents of trans -3-hexene?

A. 

B. 

C. 

D. 

18. What is the addition product formed when a mixture of benzene vapor and hydrogen is passed over finely divided nickel catalyst at 200°C?
    A. C_6H_{14} B. C_6H_{12} C. C_6H_{10} D. C_6H_{13}
19. A student took three different test tubes and added 5mL of toluene (methylbenzene) to each test tube. Then, added 2 drops of Br_2 in CCl_4 in the 1st test tube, 2 drops of $KMnO_4$ in the 2nd test tube and 2 drops of concentrated H_2SO_4 in the 3rd test tube. In which test tube does a reaction occur? In the _____ test tube (s).
    A. 1st and 3rd B. 1st, 2nd & 3rd C. 2nd D. 3rd
20. Natural gas is mainly composed of
    A. gases such as oxygen and hydrogen disulphide.
    B. methane gas
    C. all hydrocarbons.
    D. gases such as carbon dioxide and nitrogen

21. The following chart shows the position of different elements in the periodic table

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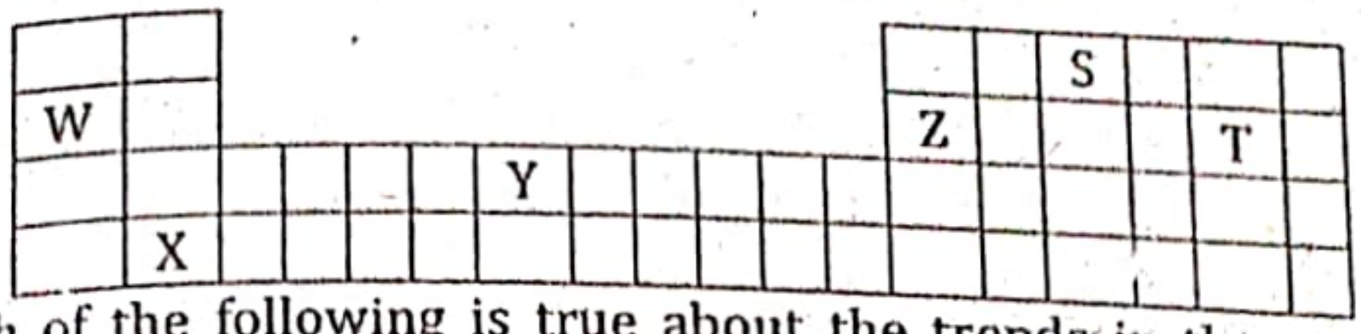

Which of the following is true about the trends in the properties of these elements?

A. Ionization potential T< X< Y
B. Electro negativity X< Z <T
C. Nuclear charge W> Z > T
D. Atomic radius T> Y > S

22. Which of the following information can easily be obtained by just looking at the periodic table?
    A. Number of shells of an element.
    B. Number of valence electron(s) of an element
    C. Nature of the oxides of the elements.
    D. Names, and symbols of the element.
23. Why atoms form a chemical bond? To
    A. transfer one or more electrons
    B. share one or more pairs of electrons
    C. attain a state of higher potential energy.
    D. achieve noble gas electronic configuration.
24. Which of the following statement is true?
    A. Metals tend to gain electrons to form positively charged ions or cations.
    B. Non-metals tend to lose electrons to form negatively charged ions or anions.
    C. Group VIA elements form ions of type X^{2+} by losing 2 electrons.
    D. Group VA elements form ions of the type X^{3-} by gaining 3 electrons.
25. What is an ionic bond? It is
    A. a bond whose formation is favored by high ionization energy and low electron affinity of the combining elements.
    B. a force of attraction between oppositely charged ions.
    C. a bond whose formation is favored by low ionization energy and low electron affinity of the combining elements.
    D. a force of attraction between positively charged nuclei of atoms and the shared electrons.
26. Which of the following is not the postulate of modern atomic theory? Atoms
    A. of the same element have identical chemical properties.
    B. of different element have different chemical properties.
    C. are the smallest particles of elements.
    D. are indivisible and indestructible.

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27. One of the following statements is accepted by both Dalton's and Modern atomic theories. Which one is it?
A. All elements are made up of very small particles called atoms.
B. Atoms of a given element are identical in mass.
C. Atoms of the same element may not be exactly alike.
D. An atom can be further subdivided into electrons, protons and neutrons.

28. The existence of neutrons was discovered by
A. Ernest Rutherford
B. Robert Millikan
C. James Chadwick
D. Joseph John Thomson

29. What is the relative electric charge of an electron?
A. -1 B. $+1.6\times10^{-19}C$ C. +1 D. $-1.6\times10^{-19}C$

30. If a chemical symbol of iodine is given by ${}^{127}_{53}I$, what are the number of electrons, protons and neutrons respectively?
A. 74, 53, 53 B. 53, 53, 74 C. 53, 127, 53 D. 53, 53, 127

31. Which of the following is true about the properties of ionic compounds? They
A. have crystalline nature.
B. are insoluble in polar solvents like water.
C. have low melting and boiling points.
D. conduct electricity in solid state.

32. A bond formed when atoms in a molecule share electron pairs is _____.
A. coordinate covalent
B. metallic
C. ionic
D. covalent

33. The inter-molecular attraction between a partially positive hydrogen atom in one molecule & partially negative atom in another molecule is
Van der waals force
hydrogen bonding
C. London force
D. dispersion force

34. The following table contains noble gases with their boiling point values

Noble gase	He	Ne	Ar	Xe.
Boiling point (^0C)	-269	-246	-186	-107

In which of the following noble gas does the dispersion force is the strongest?
A. Ne B. Xe C. He D. Ar

35. The statement "Matter is neither created nor destroyed during a chemical reaction." is the law of
A. conservation of mass
B. definite proportion
C. multiple proportions
D. constant composition

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36. The quantum mechanical description for the arrangement of electrons of an atom in atomic orbital's in order of increasing energy is known as _____

A. valence shell
B. valence electron
C. electronic configuration
D. diagonal rule

37. Which of the following is the ground state electron configuration for ion (Z=26)?

A. $[Ar]3d^74S^2$ B. $[Ar]4S^13D^7$ C. $[Ar]3d^64s^2$ D. $[Ar]4s^23d^6$

38. The common name for group VIIA elements is

A. noble gases
B. halogens
C. alkali metals
D. alkaline earth metals

39. What is the number of groups and periods in the modern periodic table respectively ?

A. 7, 18 B. 7, 16 C. 18, 7 D. 16, 7

40. If a certain element has an electron configuration of $1s^22s^22p^63s^23p^2$, to which block and group of the periodic table does it belong?

A. p, IVA B. p, IIA C. s, IVA D. s, IIA

41. Which of the following is not the industrial application of electrolysis? It is used for the

A. production of non- metals
B. purification of metals
C. purification of non-metals
D. production of metals

42. Natural resources are

A. chemicals that are obtained from the industries
B. organic chemicals that are used to prepare a huge range of agriculture products.
C. mineral nutrients those are important in plant growth
D. materials those are available on earth for people to use.

43. Which one of the following natural resources is renewable?

A. Petroleum B. Water C. Coal D. Natural gas

44. Which of the following is not the general characteristic of industries? They

A. use large amounts of energy in their manufacturing process.
B. test their products during and after manufacturing in their quality control laboratories.
C. involve mostly chemical processes to transform the raw materials to the desired products.
D. use synthetic materials as raw materials for the production process.

45. Which step in the manufacturing of pulp and paper is use to remove coloring matter from wood pulp?

A. Kraft process
B. Bleaching
C. Pulping
D. Sulphite process

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46. What is the negative logarithm of hydrogen (hydronium) ion concentration?
A. pH B. PKa C. pOH D. pKw

47. What is the pH of a solution containing 0.0002 mole H^+ per liter at 25°C? (log2=0.3)
A. 3.7 B. 10.3 C. 2 D. 4

48. What is a strong base? It is a base that
A. contains relatively large amount of solute in a given volume of solution.
B. contains only a small amount of solute in a given volume of solution.
C. ionizes partially or slightly in aqueous solution.
D. ionizes completely or almost completely in aqueous solution.

49. Which of the following is a strong a base?
A. $Mg(OH)_2$ B. $Ca(OH)_2$ C. $Ba(OH)_2$ D. $Zn(OH)_2$

50. What is the mathematical relationship between pH and pOH for a dilute water solution at 25°C?
A. pOH=pKw+pH
B. pKw=pH-pOH=14
C. $Kw=[H_3O^+][OH]^-$
D. pKw=pH+pOH=14

51. Of the following methods, which one is a traditional food preservation method?
A. Vacuum packing
B. Freeze drying
C. Salting
D. Freezing

52. Which of the following pollution is not common?
A. Sound B. Air C. Water D. Land

53. Which of the following method can be used to reduce water pollution?
A. Lowering temperature furnaces and engines that reduce the amount of nitrogen oxides produced.
B. Banning of CFCs aerosol propellant gases to be replaced by less harmful alternatives.
C. Using moderate amounts of agricultural chemicals and increasing the use of organic fertilizers.
D. Burring or recycling plastic and other non degradable materials.

54. Converting potentially dangerous chemical wastes into harmless substances, by combustion or by other chemical means is one of the main methods used to reduce
A. air pollution
B. sound pollution
C. water pollution
D. land pollution

55. Suppose that your teacher gave you a chance to visit a nearby cement industry. After the visit, you are expected to write a report about its environmental impact. What will be your main concern in your report?
A. How the production efficiency could be enhanced

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B. How the environment is affected by the industry.
C. What raw materials are used
D. What the industry does

56. What is the function of phosphorus in the growth of plants? It is used
A. to activate enzymes in photosynthesis
B. for the synthesis of chlorophyll
C. for the synthesis of amino acids
D. to form nucleic acids.

57. What is the difference between metallic and electrolytic conductivity?
A. In metallic conduction new substance is produced.
B. Conductivity of electricity by electrolytic conductivity is a simple physical change.
C. In metallic conduction there is net movement of the metal atoms.
D. In electrolytic conduction the charge carriers are positive and negative ions.

58. Electrolysis is a
A. process in which electrical energy is used to produce chemical changes.
B. process which consists of two electrodes inside the cell.
C. process in which there is a selection of the electrodes for one kind of ion.
D. system consisting of electrodes that dip into an electrolyte and in which chemical reaction generates electrical energy.

59. The strips of metal or graphite that allow electrons to leave or enter the electrolyte are known as _____.
A. anions B. electrodes C. charge carriers D. cations

60. Which one of the following statements is true about electrolytic and galvanic cells?
A. In electrolytic cells, electrons flow from anode to cathode where as in voltaic cells from cathode to anode.
B. In electrolytic cells, chemical reactions generate electricity where as in voltaic cells electricity brings about chemical reactions.
C. In both electrochemical cells, anode is positive and cathode is negative.
D. I both electrochemical cells, oxidation takes place at anode and reduction takes place at cathode.

61. Why does an increase in temperature affect the rate of a reaction? Because, reactant molecules collide
A. more frequently with greater energy.
B. less frequently with greater energy.
C. more frequently with less energy.

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D. less frequently with less energy.

62. Given the equation representing the reaction at equilibrium

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g) + 92.5kJ$$

Which statement is true about the reaction?

A. Increasing pressure will increase the yield of PCl_3 or Cl_2.
B. Decreasing the volume of the system will shift the equilibrium to the right.
C. Increasing the temperature will shift the equilibrium to the right.
D. Removing PCl_3 as it is formed will shift the equilibrium to the right.

63. Consider the following reaction that takes place inside a sealed reaction chamber. $2KClO_3(s) \longrightarrow 2KCl(s) + O_2(g)$. Which of the following will not increase the rate of reaction?

A. Decreasing the pressure
B. Increasing the amount of $KClO_3$.
C. Increasing the temperature.
D. Adding a catalyst.

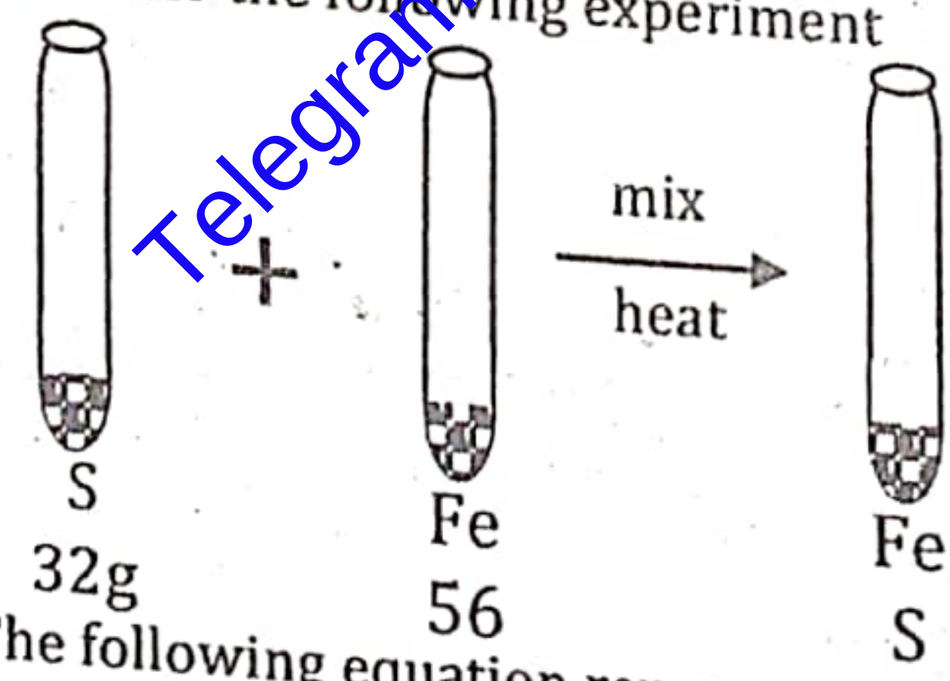
64. Reactions which can take place in both forward and backward directions are called _____

A. one way B. reversible C. irreversible D. equilibrium

65. According to the kinetic molecular theory,

A. gases do not diffuse spontaneously into each other.
B. it is assumed that there is a strong attractive force between the gas molecules, because gas molecules are very close to each other.
C. molecules in a gas have different velocities with different kinetic energies.
D. gases are denser than solids and liquids.

66. Consider the following experiment



The experiment demonstrates the law of

A. conservation of mass
B. constant composition
C. mass action
D. multiple proportions

67. The following equation represents the oxidation of ammonia $NH_3 + O_2 \longrightarrow NO + H_2O$ If the equation is correctly balanced, what are the balancing coefficients respectively?

A. 4, 5, 4 and 6 B. 4, 6, 4 and 5 C. 2, 2, 2 and 6 D. 6, 2, 2 and 2

68. In an endothermic reaction

A. the value of ΔH is negative.
B. heat is released to surroundings.

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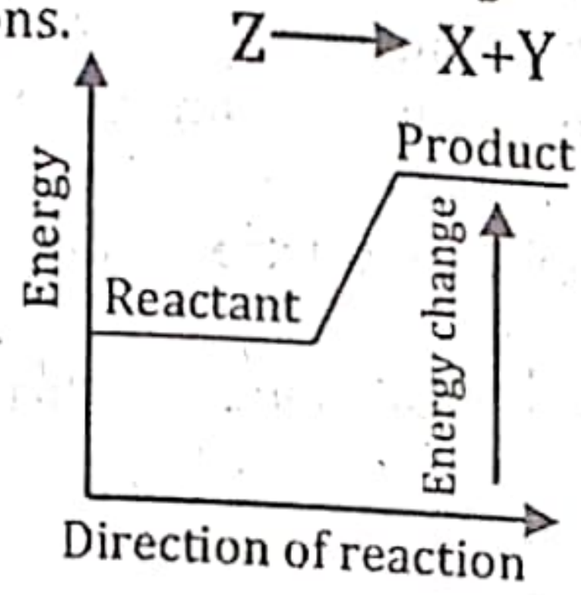
C. heat is absorbed from the surroundings.
D. the internal energy of the reactants is more than the internal energy of the products.

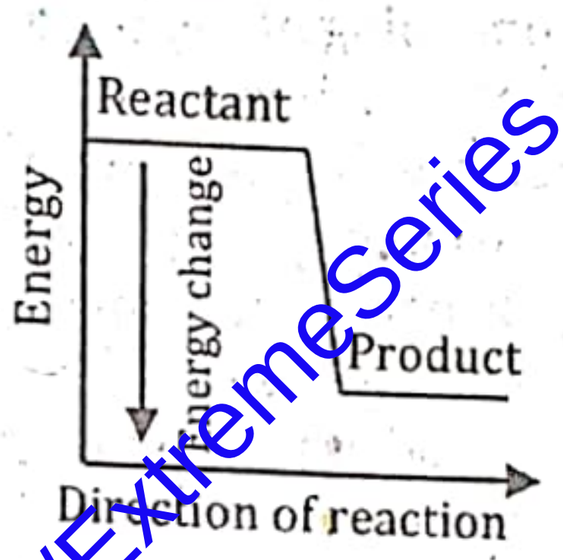
). Which of the following reaction is an exothermi reaction?
A. $C_2H_2+O_2+heat \longrightarrow CO_2+H_2O$
B. $N_2+O_2+181\ kJ/mol \longrightarrow 2NO$
C. $CaCO_2(s) \longrightarrow CaO+CO_2$
D. $HNO_3+H_2S \longrightarrow NO+S+H_2O+heat$.

). The following energy diagrams represent the energy change for two reactions.

$Z \longrightarrow X+Y$



$A+B \longrightarrow C$



Which statement is true about these reactions?
A. The change in enthalpy for the reaction $A+B \longrightarrow C$ is positive.
B. The change in enthalpy for the reaction $Z \longrightarrow X+Y$ is negative.
C. The reaction $Z \longrightarrow X+Y$ is endothermic.
D. The enthalpy of C is greater than the sum of the enthalpies of A and B.

. Charles' law states that
A. the volume of a fixed mass of a gas at a constant pressure is directly proportional to the temperature measured on the Kelvin scale.
B. equal volumes of all gases, measured at the same temperature and pressure contain equal number of molecules.
C. the volume of a fixed mass of a gas is inversely proportional to its pressure when temperature is kept constant.
D. at constant volume, the pressure of a given mass of gas is directly proportional to the Kelvin temperature.

2. The combined gas law equation $\frac{PV}{T} = K$ can be derived from
A. Avogadro's and Graham's laws.
B. Charles' and Graham's laws.
C. Boyle's and Charles' laws.
D. Boyle's and Avogadro's laws.

3. What is the pressure exerted by 48g of CH_4 in a 30L container at 300K? (R=0.082 L.atm/mol.K, molar mass of CH_4=16 g/mol)
A. 7.38 atm B. 8.2 atm C. 0.82 atm D. 2.46 atm

4. The migration or intermingling of molecules of different gases as a result of random molecular motion is known as _____

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A. convection B. diffusion C. evaporation D. osmosis

75. The evaporation of a liquid depends on
A. surface area B. catalysts C. concentration D. volume

76. Which of the following is a decomposition reaction?
A. $2NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4$
B. $2KNO_2 \longrightarrow 2KNO_2 + O_2$
C. $Zn + 2HCl \longrightarrow ZnCl_2 + H_2$
D. $CaCO_3 + 2HCl \longrightarrow CaCl_2 + CO_2 + H_2O$

77. How do you define oxidation in terms of electron transfer? It is the
A. process whereby electrons are removed from atoms or ions.
B. process whereby electrons are added to atoms or ions.
C. combining of oxygen with other substances or the addition of oxygen to substances.
D. process in which there is a decrease in the oxidation number of an element.

78. Which one of the following statements is true about oxidizing and reducing agents?
A. An oxidizing agent causes reduction to occur.
B. A reducing agent gains electron(s).
C. A reducing agent is reduced.
D. An oxidizing agent gains electron(s)

79. Which one of the following reactions is a non-redox?
A. $3NO_2 + H_2O \longrightarrow 2HNO_3 + NO$
B. $CuO + H_2 \longrightarrow Cu + H_2O$
C. $HCl + KOH \longrightarrow KCl + H_2O$
D. $Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2$

80. Consider the following experimental results:

	Experiment 1	Experiment 2
Reactants	Ca(s) and HCl (aq)	Zn(s) and HCl (aq)
Temperature	20^0C	20^0C
Concentration	1.0M	1.0 M
Rate of reaction	Fast	slow

Which factor would account for the faster rate of the reaction in experiment -1?
A. Nature of reactants
B. Concentration
C. Temperature
D. Surface area

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1. C Coal is a non-renewable natural resource which is black in color. It mainly contains carbon, hydrogen and oxygen plus small quantities of nitrogen, sulfur and phosphorus.
2. A In tertiary alcohols the functional group (-OH) is bonded to a carbon that is itself directly bonded to three other carbon atoms.
3. C Ethanol is industrially prepared in two ways
   1) By fermentation of sugar

$$\text{Starch} \xrightarrow{\text{Diastase}} \text{Maltose} \xrightarrow{\text{Maltase}} \text{Glucose}$$

$$\text{Glucose} \xrightarrow{\text{Zymase}} \text{Ethanol} + \text{Carbondioxide}$$

$$C_6H_{12}C_6 \xrightarrow{\text{Zymase}} 2C_2H_5OH + 2CO_2$$

   2) By hydration of ethene

$$CH_2 = CH_2 + H_2O \longrightarrow CH_3CH_2OH$$

4. A Esterfication is the reaction of carboxylic acid with alcohol to form ester and water.
   - Carboxylic acid +Alcohol $\longrightarrow$ Ester +Water

$$R - COOH + R' - OH \longrightarrow R - COO - R' + H_2O$$

   Example:-$CH_3COOH + CH_3CH_2OH \longrightarrow CH_3COOCH_2CH_3 + H_2O$

   Ethanoic acid Ethaol Ethylethanoate
5. B The general formula for alkanoic acids is $CnH_{2n+1}COOH$ or $C_nH_{2n}O_2$.
   - Pentanoic acid contains 5 carbons (n=5)

$C_4H_{2.4+1}COOH = C_4H_9COOH = C_5H_{10}O_2$

6. D Ideal gas condition is favored at low pressure and high temperature because gases are liquified at high pressure and low temperature.
7. B When a solid directly changes to gas (sublimation) energy is added and it is an endothermic process.

$$\text{Solid} \xrightarrow{\text{sublimation}} \text{Gas}$$

   - Freezing, condensation and depostion are exothemic processes because energy is released.
8. A A functional group is the part of a molecule that determines the physical and chemical prperties of an organic compound For example, OH is the functional group of alcohols.
9. B Choose the longest chain. The straight chain is the longest chain. The 2nd , 3rd and 6th carbons contain CH_3(methyl) groups and the

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4th carbon contains $-CH_2-CH_2-CH_3$ (propyl) group (start numbering from the left).

- 2, 3, 6 trimethyl-r-propylheptane.

10. **A** Alkanes are formed by hydrogenation of alkenes which is addition reaction.

- $R-CH=CH_2 + H_2 \xrightarrow[heat]{pt} R-CH_2-CH_3$

  Alkene Alkane

$$R-CH_2-CH_2-OH \xrightarrow[heat]{H_2SO_4} R-CH=CH_2 + H_2O$$

$$R-CH_2-CH_2-X + KOH \longrightarrow R-CH_2-CH_2-OH + KX.$$

$$R-CH=CH_2 + H_2O \longrightarrow R-\underset{\displaystyle OH}{\underset{|}{CH}}$$

11. **C** Urea is a common organic fertilizer and usually found in the urine of animals.

12. **D** Inorganic chemicals are also known as minerals and are mainly obtained from the Earth rather than from plants and animals.

13. **C** Monoprotic acids contain only one hydrogen in their formula.

Example:- HNO_3, HCl, HCN, $HClO_4$

Diprotic acids contain two hydrogens in their formula.

Example:- H_2SO_4, H_2CO_3

Triprotic acids contain three hydrogens in their formula.

Example:- H_3PO_4

14. **D** Strong acids ionize almost completely in aqueous (water) solutions. They release much hydrogen ions in water solutions.

- A concentrated acid contains large amount of acid dissolved in small amount of water. The opposite is true for weak acids.

15. **D** pH is a measure of acidity and alkalinity. $pH = -\log[H^+] = \log\frac{1}{[H+]}$.

- The larger the pH value the less acidic and the more alkaline and ranges from 0 to 14. The hydrogen ion concentration of pH=3 is ten times greater than that of an acid of pH=4 because, pH=3 means $[H^+]=10^{-3}$.

  pH=4 means $[H^+]=10^{-4}$.

  Hence, $10^{-3}=10\times10^{-4}$

16. **C** The general formula for alkenes is C_nH_{2n} n=3 means $C_3H_{2\times3} = C_3H_6$.

17. **D** Geometrical isomerism occurs in alkenes. It is also known as cis-trans isomerism cis-means on the same side of the double bond and trans means on opposite side of the double bond.

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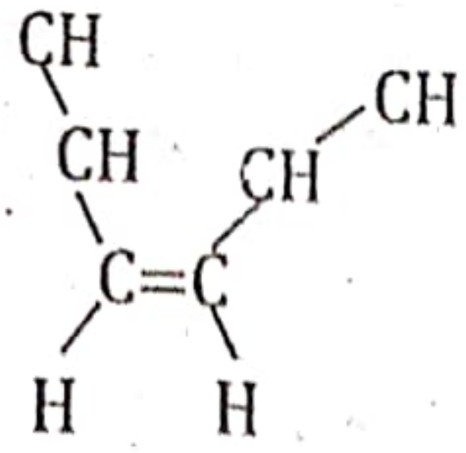
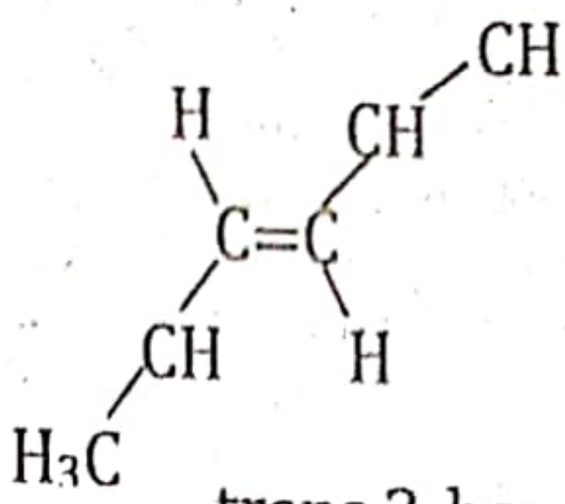
Example:-



cis- 3-hexene



trans 3-hexene

18. B Acetylene is mainly used as a fuel in oxy-acetylene torches for cutting and welding metals.
    - It can also be used in chemical synthesis such as vinylchloride, acetaldehyde, neoprene and 1, 1, 2, 2 -tetrachloroethane.
19. A Ethane undergoes substitution reaction while ethene and ethyne undergo additon reaction.
    - Ethane is saturated hydrocarbon and ethene and ethyne are unsaturated hydrocarbons.
    - Ethane, ethene and ethyne do not show cis –trans (geometrical) isomerism.
    - Ethene and ethyne decolorize purple basic $KMnO_4$ and bromine water whilt ethane doesnot.
20. A Kerosene is obtained between a boiling point range of 175-275^{0}C.
    - Gasoline=40-200^{0}C
    - Diesel oil=250-350^{0}C
21. C Atomic radius increases down a group and decreases across a period. Hence, b>j>e>c
    - Electronegativity decreases down a group and increases across a period. Hence, b<j<e<c
22. B Periodic table is useful to predict the physical and chemical prperties of elements and the behavior of compounds the elements can form as well as their formulas. It cannot predict the speed of a chemical reaction.
23. A A chemical bond is the attraction force that holds atoms together in a molecule.
    - Intermolecular force is the attraction force between molecules.

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- Hydrogen bonding is a bond formed when hydrogen is bonded to fluorine, oxygen and nitrogen.

24. B Na loses electrons to form Na^+.
- F gains electrons to form F^-.
- Na^+ is maller in size than Na.
- F^- is larger in size than F.

25. B Ionic compound is formed between metals and non-metals. Al_2O_3 is formed between Al(metal) and O (non metal) and it is ionic compound SO_2, H_2O and CO_2 are covalent compounds.

26. D Rutherford conducted a series of experiments using thin gold foils and other metals as targets for alpha particles. Alpha particles are positively charges and showed a large deflection when they passed nearest the positive mass because like charged repel each other. The central positive mass is the nucleus.

27. A Electron is negatively charged (-).
- Proton is positively charged (+1)
- Neutron has no charge (0).

28. D Atomic number=Number of protons=Number of electrons since the atomic number is 80, there are 80 protons and 80 electrons.
Number of neutrons=Mass number-Atomic number
=200-80=120

29. B Number of neutrons=Mass number-Atomic number =13-6=7

30. B Thomson's atomic model is a sphere of positive charges in which electrons are embedded.
- Dalton's atomic model is a solid sphere which is indivisible.
- In Bohr's atomic model, electrons are located in orbits.
- Rutherford's model assumed on atom has a tiny, dense positively charged nucleus around which electrons are whirled.

31. D A covalent bond is the electrostatic attraction between positive nuclei and shared electrons in a molecule. It is formed by sharing of electrons between atoms of non-metals.
- One atom donates both electrons of the bond in coordinate covalent bond.
- Metallic bond is the attraction between metal ions and sea of electrons. Ionic bond is the attraction force between oppositely charged ions.

32. C Ionic compounds have high melting points, are soluble in water and are good conductors of electricity. Covalent compounds have low melting points, are usually soluble in non polar solvents, and are poor conductors of electricity.

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33. A London dispersion forces are forces that exist between non-polar molecules and atoms due to temporary fluctuations of dipoles.

34. A Dispersion forces exist in non-polar molecules Ne and CH_4 are non polar molecules and the london force is important.

35. A The law of definite proportion states that a compound always contains the same elements in the same proportion by mass.
- The formation of N_2O, NO, N_2O_3, NO_2 and N_2O_5 between N and O is in line with the law of multiple proportion. That is, two elements can form more than one compound.
- The law of conservation of mass states that matter is neither created nor destroyed in a chemcal reaction.

36. A The way electrons are distributed among the orbitals of anatom or the arrangement of electrons in an atom is called electronic configuration.

37. A Atomic number=35 means
$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^5 = [Ar] 4s^2 3d^{10} 4p^5$

38. C The last electron enters the 3s orbital and it is an s-block element.

39. B Group IA elements are Alkali metals.
- Group IIA elements are Alkaline earth metals.
- Group VIIA elements are Halogens.
- Group VIIIA elements are noble gases.

40. B There are 18 columns in the perioic table but 16 groups since Group VIIIB includes columnscontaining Fe, Co and Ni. There are 7 periods in the modern periodic table.

41. B Petroleum is refined by fractional distillation which is based on boiling point differences of the petroleum fractions. Electrolysis is employed in the extraction of metals, electroplating, and electrorefining.

42. C Soil, plants and animals are renewable natural resources and they can be replenshed Coal, petroleum, natural gas and minerals are non renewable resources becuause they cannot be replenished once finished.

43. D A chemical industry is an institution involved in manufacturing activity of various chemical products. It transforms raw materials into finished and semi - finished products using chemical reactions.

44. B Aluminum is extracted from bauxite ($Al_2O_3.2H_2O$) by the Hall's process.
- Bauxite is treated with NaOH to remove impurities $Al_2O_3 + 2NaOH \longrightarrow 2NaAlO_2 + H_2O$
- $NaAlO_2 + 2H_3O^+ \longrightarrow Al(OH)_3 + NaOH$
- $Al(OH)_3$ is heated to remove water.

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$$2Al(OH)_3 \xrightarrow{heat} Al_2O_3+3H_2O$$

Al_2O_3 is electrolysed to give Al metal.

$$2Al_2O_3+3C \xrightarrow{electrolysis} 4Al + 3CO_2$$

45. A Al is used to make household cooking utensils due to its high heat conductivity. It can also be used to make window frames, roofs, wrappers, air plane wings , light alloys, etc.

46. A $pH=-\log[H^+]$

$HCl \longrightarrow H^+ + Cl^-$

$0.00001M \quad 0.00001M$

$0.00001 = 1\times10^{-5}$

$pH = -\log 1\times10^{-5} = 5 - \log 1 = 5 - 0 = 5$

47. C A dilute solution contains small amount of solute (in this case base) in a solution.

- A concentrated solution contains large amount of solute (base) in a solution.

48. D $[H^+] = 10^{-pH} = 10^{-4} = 1\times10^{-4}M.$

$$[OH^-] = \frac{kw}{[H^+]} = \frac{1\times10^{-14}}{1\times10^{-4}} = 1\times10^{-10}M$$

49. C NH_3 is used as a house hold cleaninig agent. NaOH, kOH and $Ca(OH)_2$ cannot be used as a cleaning agent since they are corrosive.

50. C $N_2O_5(s) + H_2O(l) \longrightarrow 2HNO_3(aq)$ (nitric acid)

- $CaO(s)+2HCl\ (aq) \longrightarrow CaCl_2(s) + H_2O(l)$
- $2Na(s) + Cl_2(g) \longrightarrow 2Nacl(s)$
- $HCl\ (aq) + NaOH\ (aq) \longrightarrow NaCl(s) + H_2O(l)$

51. A The procedures in the production of sugar are in the following steps.

1. Collecting the harvest
2. Cleansing and grinding
3. Chopping to pieces
4. Juicing
5. Clarifying by adding CO_2 and lime water.
6. Evaporation
7. Crystallization
8. Refinery which involves using charcoal and SO_2 to bleach the raw sugar.
9. Packaging

52. A Canning is used to preserve a wide veriety of foods such as soups, sauces, fruits, vegetables, juices, meats, fish, and dairy products.

53. C Particulates affect crop production, buildings and health.

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- Excess CO_2 Causes global warming.
- Hydrocarbons and ozone cause smog due to photochemical reactions.

54. D Heavy metals, spillage of oil, chemical fertilizers, non biodegradable wastes cuase land pollution. Organic fertilizers such as compost have little or no effect on land pollution.

55. C Industries are the major source of envoronmetnal pollution, that is air pollution. Water pollution, land pollution and noise pollution.

56. A $(NH_4)_2HPO_4$ and NH_4NO_3 are used as fertilizers. KNO_3 is used to make gunpowder and used as a fertilizer Copper (II) salts such as $Cu(NO_3)_2$ and $CuSO_4$ are used as pesticides.

57. D Metallic conduction is a physical process and the electrical charges are carried by electrons. Electrolytic conduction is a chemical process and the electrical charges are carried by ions.

58. B Half-cell reaction is the reaction that takes place at the anode and at the cathode. Oxidation takes place at the anode and reduction takes place at the cathode. The overall reaction is cell reaction.

59. B In an electrolytic cell:-
- Electrons flow from anode to cathode.
- The anode is positive and the cathode is negative.
- Non spontaneous redox reaction generates electricity.
- Oxidation takes place at the anode and reduction takes place at the cathode.

60. A Anode reaction: $Zn(s) \longrightarrow Zn^{2+}+2e^-$
- Cathode reaction: $Cu^{2+}+2e^- \longrightarrow Cu(s)$
- Cell reaction: $Zn(s)+Cu^{2+} \longrightarrow Zn^{2+}+Cu(s)$
- Zinc is anode and copper is cathode.

61. D $NO_2 \longrightarrow NO_3^-$ ($N^{+4} \longrightarrow N^{+5}$, that is, N is oxidized)
- $NO_2^- \longrightarrow NO_2$ ($N^{+3} \longrightarrow N^{+4}$, that is, N is oxidized)
- $NO_3^+ \longrightarrow NO$ ($N^{+5} \longrightarrow N^{+2}$, that is, N is reduced)
- $NH_4^+ \longrightarrow N_2$ ($N^{+3} \longrightarrow N^+$, that is, N is oxidized)
- Oxidation is increase in oxidation number.
- Reduction is decrease in oxidation number.

62. D Beyond point II ther is no change in the amount of product and hence the reaction goes to completion at point II.

63. D The nature of reactants affects the rate of a chemical reaction. Mg>Fe>Cu in reactivity.

64. C $2SO_2(g)+O_2(g) \rightleftharpoons 2SO_{3(g)}$. The reaction is exothermic. The equilibrium shifts to the right:
i) by increasing the concentration of SO_2 or O_2.

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ii) by decreasing the temperature.
iii) by increasing the pressure.
A catalyst has no effect on the state of equilibrium.

65. **B** A solid has definite shape and volume.
- A liquid has definite volume but not definite shape.
- A gas has no definite shape and definite volume.

66. **B** By using the inspection method. $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$

67. **D** In exothermic reaction, energy is released and the heat of products is smaller than the heat of reactants and ΔH is negative.

68. **B** In endothermic reaction, energy is absorbed and the heat of products is greater than the heat of reactants and ΔH is positive.

69. **A** A decomposition reaction has only one reactant.
Examples:- $CaCO_3 \longrightarrow CaO + CO_2$
$2KClO_3 \longrightarrow 2KCl + 3O_2$

70. **D** A double displacement reaction involves exchange of ions between two ionic compounds.
Examples:- $BaCl_2 + Na_2SO_4 \longrightarrow BaSO_4 + 2NaCl$

71. **D** The particles in solid are in fixed positions.
- The particles in liquid have less energy than in a gas and are more closely packed.
- The particles in a gas are randomly distributed.

72. **A** If different gases are at the same temperature, their average kinetic energies must be the same.

73. **D** Boyle's law:- volume is inversely proportional to pressure at constant temperature.
- Charles's law: volume is directly proportional to temperature at constant pressure.
- Graham's law: The rate of diffusion of gases is inversely proportional to the square root of their densities or molar masses at the same temperature and pressure.
- Combined gas law is derived from Boyle's, Charles' and Gay-Lussac's laws.

74. **A** Using Charles' law,
$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \Rightarrow T_2 = \frac{T_1 V_2}{V_1} = \frac{298k \times 10L}{5L} = 596k$$

75. **D** Ideal gas law can be derived from $v\alpha\frac{1}{p}$ (Boyle's law), $v\alpha T$ (Charles' law) and $v\alpha n$ (Avogadro's law) .

76. **B** $\overset{65g}{\underset{6.54g}{Zn}} + 2HCl \longrightarrow ZnCl_2 + \overset{22.4L}{\underset{x}{H_2}}$
$$\frac{65g}{6.54g} = \frac{22.4L}{x}$$

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$$\Rightarrow x = \frac{6.54\times22.4L}{65} = 2.25L$$

77. **A** First determine the limiting reactant.

$$n_s = \frac{3.2g}{32\frac{g}{mol}} = 0.1mol$$

$$n_{F2} = \frac{3.8g}{38\frac{g}{mol}} = 0.1\ mol$$

From the balanced chemical equation,

$1mol\ S \longrightarrow 3mol\ F_2$

$0.1\ mol\ s \longrightarrow x$

$$x = \frac{0.1\times3}{1} = 0.3mol\ F_2$$

$0.3mol\ F_2$ is required but there is only $0.1\ molF_2$

Hence, F_2 is the limiting reactant.

From the balanced chemical equation

$114g\ F_2 \longrightarrow 146g\ SF_6$

$3.8g\ F_2 \longrightarrow x$

$$\Rightarrow x = \frac{3.8\times146}{114} = 4.9g\ SF_6$$

78. **C** Calculate the theoretical yield from the balanced chemical equation,

$$^{128g}_{20g}2SO_2 + O_2 \longrightarrow {}^{160g}_{x}2SO_3$$

$$\frac{128g}{20g} = \frac{160g}{x}$$

$$x = \frac{20\times160g}{128} = 25gSO_2$$

$$\text{Percent yield} = \frac{\text{Actual yield}}{\text{Theoretical yield}}\times100 = \frac{23}{25}\times100 = 92\%$$

79. **D** Reduction is gain of electrons and decrease in oxidation number.
Oxidation is loss of electrons and increase in oxidation number.

80. **C** Oxidizing agent brings about oxidation but itself reduced.
Reducing agent brings about reduction but itself oxidized.

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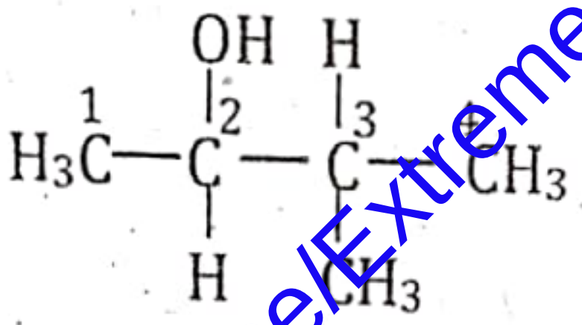
1. C Crude oil (petroleum) is a naturally occurring oil, bituminous liquid composed of various hydrocarbons and other organic chemicals. It is separated into its components by fractional distillation. Its major components are naphtha, gasoline, kerosene, diesel oil, lubricating oil, fuel oil and tar.

2. C Coal is mainly composed of carbon, hydrogen and oxygen and is rich in hydrocarbons.

3. C

Ketones Aldehydes Alcohols Carboxylic acids

4. A Start numbering nearest the hydroxyl (-OH) group .



3 methyl – 2 butanol

5. B Industrially, ethanol can be prepared by

Fermentation of sugar

$$\text{Starch} \xrightarrow{Diastase} \text{Maltose} \xrightarrow{Maltase} \text{Glucose}$$

$$\underset{\text{Glucose}}{C_6H_{12}O_6} \xrightarrow{yeast\ (zymase)} \underset{\text{Ethanol}}{2C_2H_5OH} + 2CO_2$$

Catalytic hydration of ethene

$$\underset{\text{Ethene}}{H_2C = CH_2} + H_2O \xrightarrow{acid} \underset{\text{Ethanol}}{CH_3CH_2OH}$$

Note:-

$$CH_3COONa + NaOH \xrightarrow[heat]{CaO} \underset{\text{Methane}}{CH_4(g)} + Na_2CO_3$$

$$CaC_2(s) + H_2O \longrightarrow Ca(OH)_2 + \underset{\text{Ethyne}}{C_2H_2}$$

$$\underset{\begin{matrix}|\\X\end{matrix}\quad\begin{matrix}|\\X\end{matrix}}{CH_2 - CH_2} + 2KOH \longrightarrow \underset{\text{Ethyne}}{C_2H_2} + 2H_2O + 2KX$$

(X = Cl, Br)

6. A The higher the boiling point of a liquid the lower its vapor pressure.

- Dithylether > Ethanol > Water > Toluene in vapor pressure.

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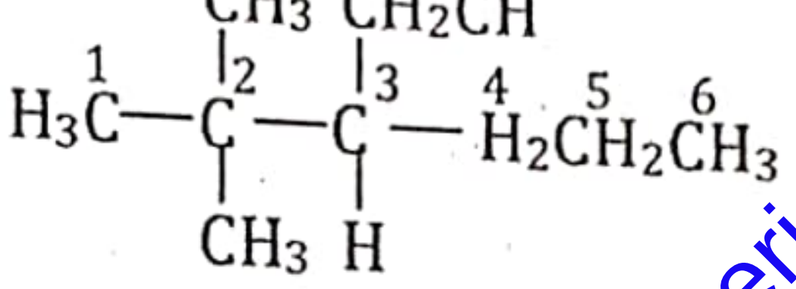
7. B A is solid Path B to C is solid and liquid (melting point because temperature is constant). Path C to D is liquid. Path D to E is liquid and gas (boiling point because temperature is constant) . Path E to F is gas.

8. D Homologous series is a group of molecules each differing from the next member by a certain group such as CH_2 group. In hydrocarbons, it is successive addition of CH_2 group.
Example:- CH_4, C_2H_6, C_3H_8, C_4H_{10}, etc

9. A Choose the longest chain as the parent name and the other groups become branches



2, 2-dimethyl-3-ethylhexane

10. D In line with 'like dissolves like, polar compounds dissolve in polar solvents, and non polar compounds dissolve in non-polar solvents.
- Alkenes and alkynes are non polar and dissolve in the non polar benzene solvent. Water, acetic acid and alcohols are polar compounds.

11. C Aldehydes have the IUPAC name Alkanal. Their general formula is $C_nH_{2n}O$.
-Butanal has 4 Carbon atoms (n = 4) $C_nH_{2n}O = C_4H_{2\times4}O = C_4H_8O$

12. B Oxides are binary compound containing oxygen in the- 2 oxidation state.
Example:- MgO, Na_2O, Al_2O_3, CO_2

13. A Acid anhydrides react with bases to form salt and water, react with water to form acids, react with basic oxides to give salts.
Example:- $CO_2 + 2KOH \longrightarrow K_2CO_3 + H_2O$

$CO_2 + H_2O \longrightarrow H_2CO_3$

$CO_2 + MgO \longrightarrow MgCO_3$

14. D Amphoteric oxides have both acidic and basic properties. They act as acids when they react with bases and as bases when they react with acids to form salt and water. They are metallic oxides.
Example:- ZnO, PbO, SnO, SnO_2, Be l_2O_3

$ZnO + 2HCl \longrightarrow ZnCl_2 + H_2O$

Base

$ZnO + 2NaOH \longrightarrow Na_2ZnO_2 + H_2O$

Acid

15. B Neutral oxides have neither acidic nor basic properties.

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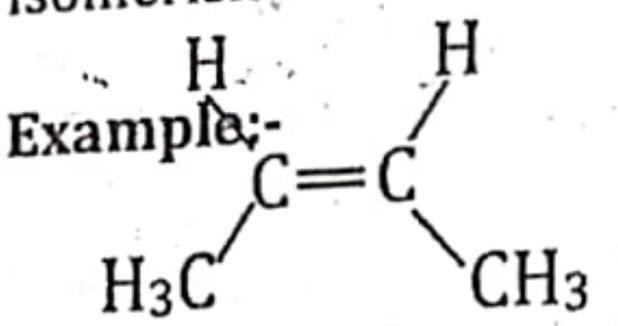
Example:- CO, N_2O, H_2O, NO

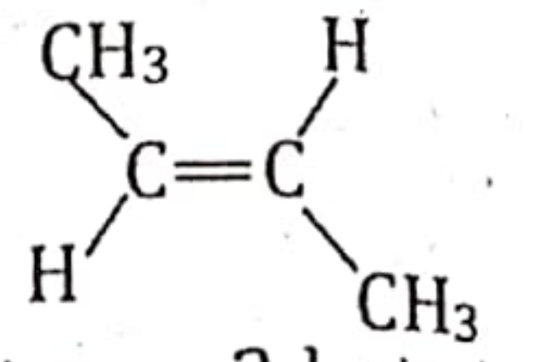
16. C The Spatial arrangement of atoms about the double bond in alkenes is called Cis-tans (geometric) isomerism.

-The difference in the position of the double bond in molecules is position isomerism.

-The difference in the position of branch in molecules is chain isomerism.



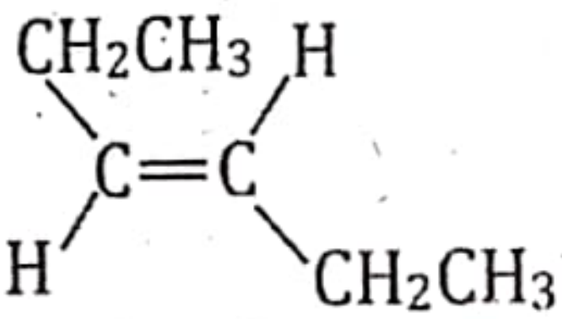
C is -2-butene



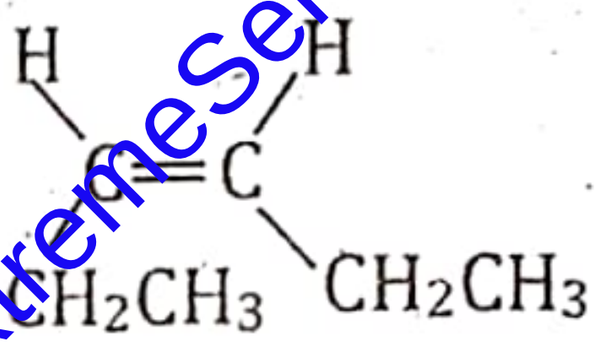
trans -2-butene

-Butene (C_4H_8) exhibits geometric isomerism.

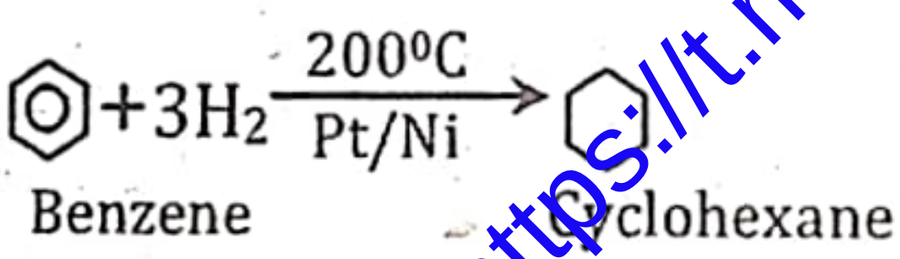
17. C



trans-3-hexene



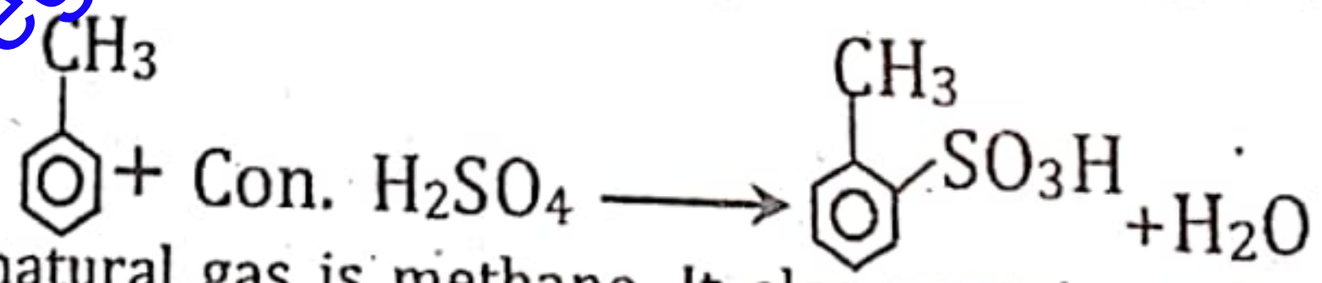
C is -3-hexene



Benzene Cyclohexane

18. B or C_6H_{12}

19. D The benzene ring is not attacked by ordinary reagents like Br_2 in CCl_4 and $KMnO_4$ solution. But it reacts with concentrated H_2SO_4.



20. B 90% of natural gas is methane. It also contains ethane, propane, butane, N_2, H_2, H_2S, CO_2, O_2, He, etc.

21. B Atomic radius increases down a group and decreases across a period from left to right. $\Longrightarrow$ $Y > T > S$

-Nuclear charge increases both down a group and across a period from left to right $\Longrightarrow$ $W < Z < T$

-Electro negativity decreases down a group and increases across a period from left to right. $\Longrightarrow$ $X < Z < T$

-Ionization potential (Ionization energy) decreases down a group and increases across a period from left to right. $\Longrightarrow$ $T > Y > X$

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22. D The names and symbols of elements can just easily be obtained from the periodic table. But nature of oxides of the elements, number of valence electrons, and number of shells are trends in the periodic table and cannot be determined just by looking at the periodic table.

23. D Atoms form a chemical bond by losing, gaining or sharing of electrons thereby achieving noble gas (stable) electronic configuration and attaining a state of lower potential energy (stability).

24. D Group VA elements form X^{3-} type ions by gaining 3 electrons.
Example:- $N + 3e^- \longrightarrow N^{3-}$
-Group VIA elements for X^{2-} type ions by gaining 2 electrons.
Example:- $S + 2e^- \longrightarrow S^{2-}$
-Metals tend to lose electrons to form positive ions.
Example:- $Na - e^- \longrightarrow Na^+$
-Non metals tend to gain electrons to form negative ions.
Example:- $Cl + e^- \longrightarrow Cl^-$

25. B Ionic bond is a force of attraction between two oppositely charged particles (ions).
-Covalent bond is a force of attraction between positively charged nuclei of the atoms and the shared electrons.
-Ionic bond formation is favored between atoms of elements having low ionization energy and high electron affinity.

26. D According to the modern atomic theory, atoms are divisible and indestructible. But Dalton's atomic theory states that atoms are indivisible and indestructible.

27. A According to both modern atomic theory and Dalton's Atomic theory, all elements are made up of very small particles called atoms.
-An atom can be subdivided into electrons, protons and neutrons and atoms of the same element may not be exactly alike is the modern atomic theory.
-Atoms of a given element are identical in mass is Dalton's Atomic theory.

28. C Joseph John Thomson discovered the electron.
-James Chadwick discovered the neutron.
-Robert Millikan determined the charge of the electron by the oil drop experiment.
-Ernest Rutherford discovered the nucleus.

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29. A The electric charge of the electrons is $-1.6\times10^{-19}C$, but that of the proton is $+1.6\times10^{-19}$C.The relative electric charge of the electron is -1 and that of the proton is +1.

30 B Number of electrons=Number of protons=Atomic number=53
$-{}^{127}_{53}I$ means atomic number (Z) equals 53 and mass number (A) equals 127. Hence, Number of neutrons= $A - Z = 127 - 53 = 74$

31. A Ionic compounds:-
- conduct electricity in the molten state.
- have high melting and boiling points.
-are soluble in polar solvents like water.
-are insoluble in non-polar solvents like carbon tetra-chloride or benzene.
-are crystalline solids.

32. D Covalent bond is formed when atoms share electron pair in a molecule.
-Ionic bond is formed by transfer of electrons from one atom to the other.
-Metallic bond is formed by the electrostatic attraction between the positively, charged metallic cations that have lost electrons and a sea of free (mobile) electrons.
-Coordinate covalent bond is a bond in which one atom furnishes both of the shared electrons in a compound.

33. B When a partially positive hydrogen atom in one molecule is bonded to a partially negative atom of another molecule, the resulting inter-molecular attraction is known as hydrogen bonding.
-London force is the force that holds non polar molecules together. It is the attraction force between the temporarily, induced dipoles.
-Van der Waals force is the collective name for London force and dipole-dipole force.

34. B The higher the dispersion force the greater the boiling point. $Xe > Ar > Ne > He$ in boiling point means, Xe has the strongest dispersion force.

35. A The law of conservation of mass states that matter is neither created nor destroyed during a chemical reaction.
-The mass of the reactants is exactly equal to the mass of the products.
-The law of definite proportion or constant composition states that in a compound the elements are always present in definite proportions by mass.
-The law of multiple proportions states that two elements can form more than one compound.

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36. C Electronic configuration is the systematic distribution of electrons in the various energy levels or shells of an atom. It is the arrangement of electrons in an atom with the lowest possible energies to attain stability.
-The diagonal rule is a systematic approach to remember the electronic configuration in line with the Aufbau principle.
-Valence electron is the outermost shell electron in an atom.
-Valence shell is the outermost shell of an atom.

37. D The electronic configuration of iron (Fe) (Z=26) is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$. The noble gas electronic configuration is $[Ar]4s^2 3d^6$.

38. B Group VIIA are halogens.
Group VIIIA are noble gases.
Group IA are alkali metals
Group IIA are alkaline earth metals.

39. B There are 7 periods and 16 groups in the periodic table. The periods are from 1 through 7. The groups are from IA through VIIIA and from IB through VIIIB.

40. A $1s^2 2s^2 2p^6 3s^2 3p^2$. The main shell electronic configuration is 2, 8, 4. Since the valence electron is 4, the element belongs to IVA. Since the last electron enters the last orbital, it belongs to p-block.

41. C Electrolysis is used:
in the production of metals and non-metals.
in refining (purification) of metals.
in electroplating of metals.

42. D Natural resources are materials that are available on earth to be used by people. They can be grouped into renewable and non renewable natural resources.
Renewable natural resources can be replenished/replaced by natural processes or human supported processes.
Example:- Plants, water, soil, solar energy.
Non-renewable natural resources – cannot be replaced /replenished once finished.
Example:- Fossil fuels, minerals

43. B Fossil fuels such as natural gas, coal and petroleum are non-renewable natural resources.

44. D Water is a renewable natural resource.

Industries :-
use raw materials for production process.
use chemical processes to transform the raw materials into the desired products.

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undergo quality assurance for their products.
use large amount of energy in the manufacturing process.

45. B Bleaching is a process of making substances white. It can be effected by using Cl_2 (chlorine) and $HClO_2$ (chlorous acid) solution.
The sulfite process is used to make the chemical pulp liquor.

46. A PH is the negative logarithm of hydrogen ion (hydronium ion) concentration. Mathematically, $P^H = -\log^{[H+]} = -\log^{[H_3O^+]}$

Note:- $P^{KW} = -\log^{KW}$

$P^{OH} = -\log^{[OH^-]}$

$P^{Ka} = -\log^{Ka}$

47. A $P^H = -\log^{[H^+]}$

$$[H^+] = \frac{\text{no of moles of } H^+}{\text{Volume in liters}} = \frac{0.0002\text{mole}}{1L} = 0.0002M = 2\times10^{-4}M$$

$P^H = -\log^{2\times10^{-4}} = 4 - \log^2 = 4 - 0.3 = 3.7$

48. D A strong base or acid ionizes completely or nearly completely in aqueous (water) solutions.
A concentrated base or acid contains large amount of base or acid in a given volume of solution.
A weak base or acid ionizes partially in aqueous solutions.
A dilute base or acid contains small amount of solute in a given volume of solution.

49. C Bases like NaOH, KOH and $Ba(OH)_2$ are strong bases.
Bases like $Zn(OH)_2$, $Ca(OH)_2$, $Mg(OH)_2$, and LiOH are weak bases.
Note:- Generally, basicity increases down a group and decreases across a period from left to right in the periodic table.

50. D $P^{KW} = P^H + P^{OH} = 14$

$P^H = 14 - P^{OH} \Rightarrow P^{OH} = 14 - P^H$

51. C Salting is a traditional food preservation method.
Freezing is keeping food items at low temperatures to reduce the rate of growth of microorganisms.
Freeze drying is a process in which frozen food is placed in a vacuum chamber to draw out the water.

52. A Land, air and water pollutions are common in the environment. But sound pollution is not recognized by many people and it is a new type of pollution.

53. C Agricultural chemicals are the main sources of water pollution. These include fertilizers, pesticides and herbicides. They can easily pollute surface and ground waters. Using bio pesticides and bio fertilizers (organic fertilizers) can reduce water pollution.

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Burning of materials contributes to air pollution. But recycling of plastics and other non biodegradable materials can reduce air and land pollutions.

Banning of CFCs and reducing the amount of nitrogen oxides(NO_x) can reduce greenhouse emissions thereby reducing ozone depletion.

54. D Land pollution can be reduced by using thermal treatment (combustion or incineration), Chemical treatment (neutralization, redox reaction, hydrolysis, substitution) and biological treatment (aerobic and anaerobic decomposition)

55. B Environmental impact assessment (EIA) deals with the impact of any human activity, including industry on the environment. It is all about how the environment is affected by the activity.

56. D Phosphorus is an essential constituent of plants and animals. It is the constituent of teeth, bones and nucleic acids (DNA and RNA). It also stimulates root development and assists early growth of plants.

57. D In metallic conductivity, the charge carriers are electrons. No new substance is formed and it is a physical change.

In electrolytic conductivity, the charge carriers are ions (positive and negative ions). New substance is formed and it is a chemical change.

58. A Electrolysis is a process in which electricity (electrical energy) is used to produce chemical reactions.

In electrolysis, electrodes (positive and negative electrodes) are dipped into an electrolytic solution and the electricity generates chemical reactions.

Electrolysis involves selection of the electrodes for one kind of ions called preferential discharge.

Two electrodes are found inside a galvanic cell not in an electrolysis cell.

59. B Electrons leave the electrolyte through the anode (positive electrode).

Electrons enter the electrolyte through the cathode (negative electrode). Generally, electrodes are strips of metal or graphite that allow electrons to leave or enter the electrolyte.

60. D Electrons always flow from Anode to cathode in both voltaic and electrolytic cells.

Oxidation always takes place at the anode and reduction always takes place at the cathode in both voltaic (galvanic) and electrolytic cells.

Anode is negative and cathode is positive in voltaic (galvanic cells)

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In electrolytic cells, electric current (electricity) generates chemical reactions.

In voltaic (galvanic) cells, chemical reactions generate electricity.

61. A As temperature increases, the motion of the reacting molecules increases thereby colliding more frequently with greater energy and consequently the rate of the reaction increases.

62. D The forward reaction is exothermic. For this exothermic reaction, Removing PCl_3 or Cl_2 at it is formed shifts the equilibrium to the right.
Increasing the temperature shifts the equilibrium to the left (towards the reverse reaction or the endothermic reaction).
Decreasing the volume of the system means increasing the pressure of the system. Increasing the pressure of the system shifts the equilibrium towards smaller number of molecules, that is, to the left and more pCl_5 is formed.

63. A For the reaction $2KClO_{3(s)} \longrightarrow 2KCl_{(s)} + 3O_{2(g)}$, the rate of the reaction can be increased by adding a catalyst, increasing the temperature and increasing the amount of the reactant ($KClO_3$).

64. B Pressure affects the rate of only gaseous reactants.
A reversible reaction is a reaction that can undergo in both forward and reverse directions. It is a two-way reaction.
An irreversible reaction undergoes only in one direction. It is a one-way reaction.
Examples:- $2KClO_{3(s)} \longrightarrow 2KCl_{(s)} + 3O_{2(g)}$ (irreversible reaction)
$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ (reversible reaction)
Note:- Irreversible reation is denoted by a single arrow while reversible reaction is denoted by a double arrow.

65. C According to the kinetic molecular theory,
Gases are less dense than solids and liquids.
Molecules in a gas have different velocities with different kinetic energies.
There is a weak attractive force between the gas molecules since the gas molecules are very far apart from each other.
Gases diffuse spontaneously into each other.

66. A The law of conservation of mass states that the mass of reactants is exactly equal to the mass of the products or mass is neither created nor destroyed in a chemical reaction. It is also known as the law of indestructibility of matter which states that matter is neither created nor destroyed in a chemical reaction.
Example:- 32g of S reacts with 50g of Fe to 88g of FeS.

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67. A $\overset{-3+1}{NH_3} + \overset{0}{O_2} \longrightarrow \overset{+2-2}{NO} + \overset{+1-2}{H_2O}$ (Oxidation number change method)

$N^{-3} - 5e^- \longrightarrow N^{+2}$

$\overset{0}{O_2} + 4e^- \longrightarrow 2O^{2-}$

Multiply by a suitable integer to cancel the electrons and then add the equations

$$\left.\begin{array}{l} N^{-3} - 5e^- \longrightarrow N^{+2} \\ \overset{0}{O_2} + 4e^- \longrightarrow 2O^{2-} \end{array}\right\} \begin{array}{l} \times 4 \\ \times 5 \end{array}$$

$$4N^{-3} - 20e^- \longrightarrow 4N^{+2}$$

$$5O_2 + 20e^- \longrightarrow 10O^{2-}$$

$$4N^{-3} + 5O_2 \longrightarrow 4N^{+2} + 10O^{2-}$$

$\Longrightarrow 4NH_3 + 5O_2 \longrightarrow 4NO + 6H_2O$

(Finish by inspection method)

68. C In endothermic reaction,

The internal energy of the products is greater than that of the reactants.

Heat is absorbed from the surrounding.

The value of ΔH is positive.

In exothermic reaction

The energy of the reactants is greater than the energy of the products.

Heat is released to the surrounding.

ΔH is negative

Note:- $\Delta H = H_p - H_r$ where

P = product, r = reactant

69. D An exothermic reaction is represented by Reactant $\longrightarrow$ product + heat or Reactant $\longrightarrow$ Product, ΔH = Negative

Example:- $HNO_3 + H_2S \longrightarrow NO + S + H_2O$ + heat. An endothermic reaction is represented by

Reactant + heat $\longrightarrow$ Product

Reactant $\longrightarrow$ Product, ΔH = Positive

Example:- $C_2H_2 + O_2$ + heat $\longrightarrow CO_2 + H_2O$

$N_2 + O_2 + 181 \frac{KJ}{mol} \longrightarrow 2NO$

Decomposition reaction is endothermic because it absorbs heat.

Example:- $CaCO_3 \xrightarrow{heat} CaO + CO_2$

70. C The reaction $Z \longrightarrow X + Y$ is endothermic.

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The reaction $A + B \longrightarrow C$ is exothermic. Hence, the enthalpy of C is less than the enthalpy of $A + B$.
The enthalpy change for the reaction $Z \longrightarrow X + Y$ is positive since it is endothermic.
The enthalpy change for the reaction $A + B \longrightarrow C$ is negative since it is exothermic.

71. A Charles law:- Volume is directly proportional to Kelvin temperature at constant pressure $\left(\frac{V_1}{T_1} = \frac{V_2}{T_2},\ P = \text{constant}\right)$.
Boyle's law:- Volume is inversely proportional to pressure at constant temperature. $(P_1V_1 = P_2V_2,\ T = \text{constant})$
Gay-Lussac's law:- Pressure is directly proportional to Kelvin temperature at constant volume. $\left(\frac{P_1}{T_1} = \frac{P_2}{T_2}\ V = \text{constant}\right)$.
Avogadro's law of combining volumes:- Equal volumes of all gases have equal number of molecules at constant temperature and pressure. $(V\alpha\, n)$.

72. C The combined gas law is derived from Boyle's law $(P_1V_1 = P_2V_2)$ and Charles' law $\left(\frac{V_1}{T_1} = \frac{V_2}{T_2}\right)$.
Boyle's law:- $V\alpha\frac{1}{P}$ (at constant $T\& n$).
Charles' law:- $V\alpha T$ (at constant $P\& n$)

$$\Rightarrow V\,\alpha\, T\left(\frac{1}{P}\right) \Rightarrow PV\,\alpha\, T$$

$$\Rightarrow PV = KT \Rightarrow \frac{PV}{T} = K \Rightarrow \frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

73. D $\frac{P=\frac{nRT}{V}=\frac{48}{16}\text{mol}\times 0.082\frac{\text{Latm}}{\text{molk}}\times 300\text{k}}{30\text{L}} = 2.46atm$

74. B Osmosis is the movement of solute particles from an area of low concentration to an area of high concentration across a semi permeable membrane.
Evaporation is escaping of molecules from the liquid surface to the vapor phase.
Diffusion is the mixing of molecules of gases as a result of random molecular motion.

75. A The rate of evaporation of a liquid depends on:-
Surface area-the larger the surface area the faster the rate of evaporation.
Temperature-the higher the temperature the faster the rate of evaporation.
Intermolecular force -the weaker the intermolecular force the faster the rate of reaction.

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76. B A decomposition reaction is a reaction having only one reactant and usually splits into two or more products.

Example:- $2KNO_3 \longrightarrow 2KNO_2 + O_2$

$CaCO_3 \longrightarrow CaO + CO_2$

$CaCO_3 + 2HCl \longrightarrow CaCl_2 + CO_2 + H_2O$

(Double displacement reaction)

$Zn + 2HCl \longrightarrow ZnCl_2 + H_2$

(Single displacement reaction)

$2NH_3 + H_2SO_4 \longrightarrow (NH_4)2SO_4$

(Composition/synthesis/ addition reaction)

Note:- Composition reaction is a reaction in which two or more products form a single product and it is the opposite of decomposition reaction.

77. A Oxidation is removal of electrons from an atom or an ion. It is an increase in oxidation number and is an addition of oxygen to a substance.

Reduction is addition of electrons to an atom or an ion. It is a decrease in oxidation number and is a removal of oxygen from a substance or addition of hydrogen to a substance.

78. D Oxidizing agent :- gains electrons
reduced itself
causes reduction to occur.
decreases in oxidation number.

Reducing agent:
loses electrons
oxidized it self
causes oxidation to occur
increase in oxidation number.

79. C Double displacement reactions and acid base reactions are not redox reactions as there is no change in oxidation number.

Example:- $HCl + KOH \longrightarrow KCl + H_2O$

$NaCl + KNO_3 \longrightarrow NaNO_3 + KCl$

80. A The factor affecting reaction rate are:-
- Natural of reactants
- Concentration
- Temperature
- Surface area

In this experiment, temperature and concentration are the same. The fast rate of the reaction in experiment is due to the nature of the reactants, that is Ca is more reactive than Zn.

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ETHIOPIAN GENERAL SECONDARY EDUCATION CERTIFICATE BIOLOGY EXAMINATION GINBOT 2009/MAY 2017

1. A renowned Ethiopian biomedical scientist and a long serving professor at Addis Ababa University is:
   A. Prof Gebisa Ejeta
   B. Prof Yalemtsehay Mekonnen
   C. Prof Sebsebe Demissew
   D. Prof Beyene Petros
2. Which one of the following research institutes works on crop and livestock improvement?
   A. Ethiopian Institute of Agricultural Research
   B. Ethiopian Health and Nutrition Research Institute
   C. The Institute of Biodiversity Conservation
   D. Armauer Hansen Research Institute
3. What is the function of ribosomes?
   A. Food production
   B. Link the nucleus
   C. Protein synthesis
   D. Carry respiration
4. Turgor pressure and plasmolysis happen in plants when:
   A. excess water enters into the cells and excess water is lost from the cells respectively
   B. excess water is lost from the cells and excess water enters into the cells, respectively
   C. the cell is hypotonic, and the environment is hypertonic, respectively
   D. the cell and the environment are isotonic, and the environment is hypotonic respectively
5. A grade 9 student demonstrated the effect of osmosis on plant cells using onion epidermal cells. He put the onion epidermal cells in distill water for 30 minutes and then observed it under the microscope. The cells he observed became:
   A. plasmolysed B. burst C. turgid D. unchanged
6. What is the measure of the variety of species in a particular area, including everything from the smallest microbes to the largest animals?
   A. Biodiversity
   B. Conservation
   C. Monoculture
   D. Polyculture
7. What is the importannce of protecting the natural environment? Because it:
   A. increases the effect of global warming
   B. insures the presence of healthy whether and climate
   C. increases the accumulation of CO_2 in the atmosphere
   D. reduces the availability of decomposers for waste removal
8. Which of the following statements is true about vegetation? It:
   A. cause global warming
   B. causes air pollution
   C. is a source of clothes and foods
   D. reduces the amount of O_2 in the air

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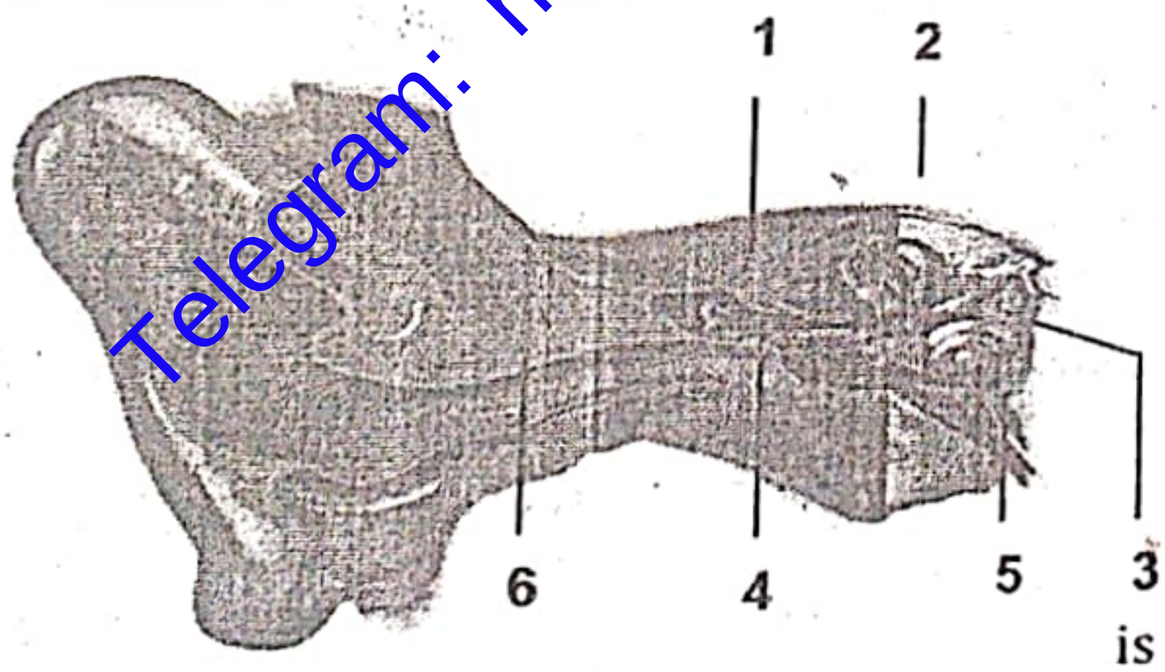
9. Which of the following animals is found in Bale Mountains National Park?
   A. Zebra
   B. Mountain Nyala
   C. Walya
   D. Flamingo
10. The burning of fossil fuels causes air pollution by:
   A. releasing unburnt hydrocarbon particles into the air
   B. releasing burnt hydrocarbon particles into the air
   C. producing and releasing carbon dioxide into the air
   D. producing and releasing nitrogen dioxide into the air
11. What is the function of dendrites? They transport impulse towards the:
   A. muscle
   B. axon
   C. sense organ
   D. cell body
12. Which one of the following statements is correct about the reflex arc? is:
   A. unidirectional flow of information from receptor to spinal cord
   B. flow of impulse from the sensory neuron to the brain and spinal cord
   C. flow of impulse from the spinal cord to the brain and back to sense organs
   D. flow of information from receptors to the spinal cord and to the effectors
13. Which one of the following eye defects is correctly matched with its corrective measure?
   A. Short sight – irregular lens
   B. Astigmatism – any eye glass
   C. Long sight – diverging lens
   D. Long sight – convex lens

This item is only for non-blind candidates and it is based on the following diagram of the human ear.



14. What is the name of the structure of the ear indicated by number 4 in the above diagram?
   A. Ear drum
   B. Semicircular canal
   C. Chochlea
   D. Oval window
15. Which area of the tongue is used to taste salty foods? The:
   A. middle
   B. back
   C. tip
   D. sides

This item is only for non-blind candidates and it is based on the following diagram which shows the carbon cycle

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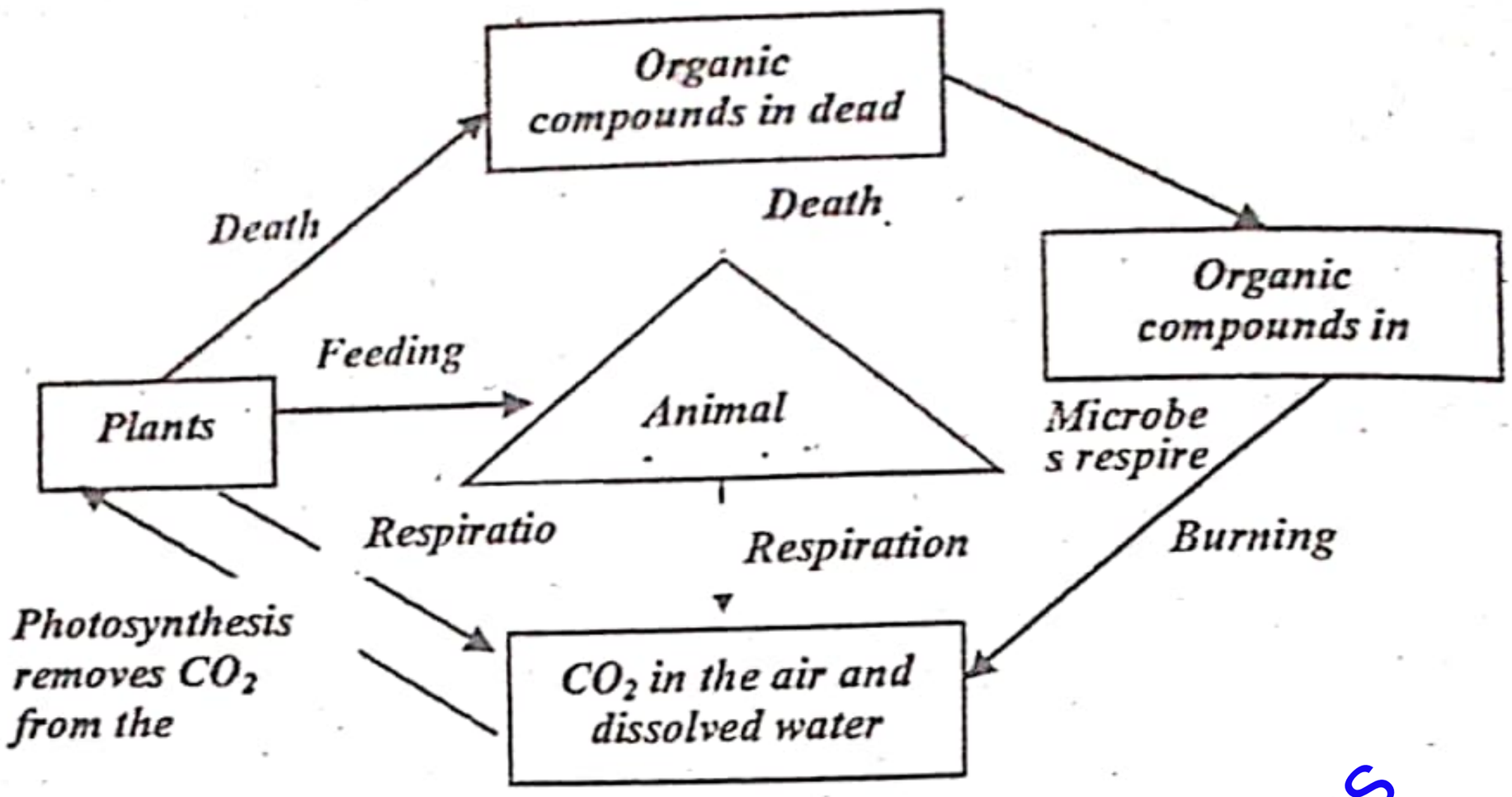

16. Which of the following statements is true about the above carbon cycle?
    A. Carbon dioxide is removed from the atmosphere by animals.
    B. Carbon dioxide is removed from the atmosphere by plants.
    C. Photosynthesis contributes CO_2 to the atmosphere.
    D. Burning removes CO_2 from the atmosphere.
17. Which of the following is an area of biotechnology application? Production of:
    A. plastic bottles B. soft drinks C. computers D. medicines
18. What is mitosis? It is:
    A. forming of zygote C. copying of chromosomes
    B. formation of sex cells D. division of somatic cells
19. Which one of the following statements is correct about the work of Mendel on pea plants? Mendel:
    A. removed both the stigma and pollen from the plant
    B. opened his pea plants before the pollen matured
    C. observed that round shape was a recessive trait
    D. observed that axial flower was a recessive trait
20. What is the stage of meiosis where a pair of chromatids moves to the opposite poles?
    A. Anaphase II C. Prophase I
    B. Metaphase II D. Anaphase I

This item is based on the following table, which shows the process of food test for starch, protein, fat, and vitamin C.

Food test used	Solution A	Solution B	Solution C	Solution D
Iodine solution	No change	No change	No change	Blue black
Biuret reagent	No change	Mauve	No change	No change
Emulsion test	No change	No	White	No

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		change	cloudy	change
Dicchlorophenol indophenol	Fading of blue color	No change	No change	No change

Based on the above table, what is the food in solution B?

A. Fat B. Starch C. Vitamin D. Protein

22. Food travels from mouth to stomach through:

A. oesophagus B. wind pipe C. trachea D. epiglottis

23. Which of the following blood groups has no antigen on the red blood cells?

A. A B. B C. O `D. AB

24. The valves of the heart play a significant role to:

A. transport deoxygenated blood into the lung
B. transport oxygenated blood in the rest of the body
C. allow the transport of blood in the right direction
D. supply blood to the cardiac muscle and atrium

25. During delivery, a pregnant woman whose blood type "**O**" has lost a lot of blood. The doctor ordered her family to get a blood donor. What should be the blood type of the donor?

A. A B. B C. O D. AB

26. A grade 9 student wanted to prove the importance of CO_2 for photosynthesis. She prepared two bell jars, in the first jar (Jar 'A') she kept plant 'A' and potassium hydroxide solution, and in the second jar (Jar 'B') she kept plant 'B' without potassium hydroxide. Then she exposed both jars to the sun light for 12 hours. The student then tested for the presence of starch in leaves of plant 'A' and 'B'. What was the result of the student's experiment?

A. Plant 'A' produced more starch than plant 'B' because potassium hydroxide increased the CO_2 concentration.
B. Plant 'B' produced more starch than plant 'A' because potassium hydroxide decreased the CO_2 concentration.
C. Both plants produced equal amounts of starch as potassium hydroxide has no effect.
D. Plant 'A' produced more starch as there was more potassium hydroxide in the bell jar.

27. How does photosynthesis balance concentration of O_2 and CO_2? It:

A. releases carbon dioxide and absorbs oxygen from the atmosphere
B. releases oxygen and absorbs carbon dioxide from the atmosphere
C. oxidizes and removes the carbon dioxide from the atmosphere
D. converts oxygen to ozone and carbon dioxide to carbonic acid

28. Water uptake by roots is facilitated by:

A. high surface area of the root hairs
B. active transport of water molecules

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C. high concentration of water in the root
D. the large vacuoles in the root and leaves

29. Which of the following transport processes takes place in the phloem?
The movement of:
A. water form soil to the root
B. minerals from the roots to the leaves
C. organic minerals from the roots to other parts
D. CO_2 from the air to the palisade mesophyll

30. What is the effect of removing apical dominance? It:
A. prevents fruit formation
B. increases water loss
C. increases stem length
D. cause side shoot growth

31. What is the importance of ATP to the body? It is used to transport:
A. ions from high concentration to low concentration
B. carbon dioxide through the cell membrane
C. water molecule through cell membrane
D. some materials against concentration gradient

32. Which of the following methods helps to keep the hygiene of breathing structures?
A. Smoking cigarettes
B. Drinking alcohol
C. Cleaning teeth
D. Breathing deeply

33. **This item is based on the following list of events about transport of oxygen into the cell.**
1. Oxygen enters into the cells where it is needed.
2. Haemoglobin picks up oxygen and form oxyhaemoglobin.
3. Oxyhaemoglobin splits to give purple-red haemoglobin and oxygen.
4. Oxygen enters into mitochondria for aerobic respiration.

34. Which of the following sequence of events is correct about transport of oxygen into the cell?
A. 1 → 2 → 4 → 3
B. 2 → 3 → 1 → 4
C. 2 → 3 → 4 → 1
D. 4 → 3 → 2 → 1

35. On a clean glass slid, a nurse mixed donor's 'AB' blood group with recipient's 'B' blood group. What result could she observe after mixing the two blood groups?
A. The mixture becomes agglutinated.
B. There was no change in appearance.
C. The red blood cells increased in size.
D. The color of the mixture changed to blue.

36. Which of the following blood components transports waste products to the kidneys?
A. Red blood cells
B. Plasma
C. White blood cells
D. Platelets

37. How do the poikilotherms adapt to a cold weather? By:

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A. wallowing in mud
B. bathing in water
C. basking in the sun
D. licking their body parts

38. Dilation of the blood vessels supplying blood to the capillaries in the skin is a response to:
A. cold temperature
B. high acidity
C. low oxygen concentration
D. hot temperature

39. What happens during ovulation in a woman?
A. The levels of pituitary hormones begin to increase.
B. The follicle forms the yellow body called corpus luteum.
C. A new egg matures in the ovary.
D. The mature egg is released from the ovary.

40 Which one of the following ear structures collects sound waves and directs them to the eardrum?
A. Middle ear B. Pinna C. Cochlea D. Malleus

41. What is the function of the palisade mesophyll? It is used:
A. to make food by photosynthesis
B. for gas exchange between leaves and the environment
C. for protection of water loss from the leaf surfaces
D. for absorption of water and minerals

42. What is the importance of a vaccine? It:
A. kills disease causing agents
B. facilitates cell division
C. triggers the immune system
D. inhibits movement of viruses

42. **This item is based on the following methods required for growing microorganisms.**
1. Sterilization of Petri dishes and nutrient agar
2. Dipping the sterilized loop in a suspension of the bacteria you want to grow
3. Sterilize the inoculating loop used to transfer microorganisms
4. Make zigzag streaks across the surface of the agar.
5. Incubate the Petri dishes to allow the microorganisms to grow

Which of the following are the correct sequences of methods to grow microorganisms?
A. 1 → 3 → 2 → 4 → 5
B. 1 → 2 → 3 → 4 → 5
C: 5 → 4 → 2 → 3 → 1
D. 1 → 3 → 4 → 5 → 2

43. What is the risk of self-prescribed antibiotics? They:
A. facilitate the functions of liver and kidneys
B. effectively destroy disease causing agents
C. may leads to the use of wrong medicine
D. have no apparent risk as long as properly used

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44. Which of the following chemicals is applied on living tissues to kill microorganisms?
A. Disinfectant B. Antiseptic C. Antibody D. Hormone

45: Which of the following diseases is correctly matched with its symptom?
A. Malaria ⟶ sweating
B. Tuberculosis ⟶ shivering
C. Tapeworm ⟶ fever
D. Diarrhea ⟶ running nose

46. What is accommodation? It is the ability of the eye to:
A. change the shape of retina to focus objects
B. focus far objects than close objects
C. focus objects during the night time
D. change the shape of the lens to focus the objects

47. Astigmatism is one of the common eye defects which is caused by:
A. the egg shaped eye instead of round shape
B. regular shape of the eye which becomes more rounded
C. lens that is too strong and curved while ciliary muscles contracted

48. Lens that is too weak and flat while ciliary muscles contracted When there is no pregnancy after ovulation,
A. oestrogen level increases
B. progesterone level decrease
C. the egg is released from the ovary
D. follicle is formed in the ovary

49. What is the difference between endocrine and exocrine glands? Endocrine glands:
A. have ducts whereas exocrine glands do not have ducts
B. do not have ducts whereas exocrine glands do have ducts
C. release their secretions through ducts to site of destination
D. are more in number in our body than exocrine glands

50. Which of the following is the effect of female genital mutilation? It:
A. avoids genital infection
B. increases sexual excitement
C. increases the risk of HIV infection
D. reduces pain during delivery

51. What is the impact of HIV and AIDS on the society? It:
A. increases the immune system through time
B. reduces stigma and discrimination
C. makes people loyal to their parents
D. leads to significant social cost

52. A doctor prescribed abdominal medicine to Ato Hailu. Ato Hailu bought the medicine and wanted to check if it is the right medicine. Which of the following information should he read on the leaflet?
A. Name of the medicine
B. Dosage of the medicine
C. Side effects of the medicine
D. Illness which the medicine treats

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53. What is the name of the lymph nodes located in the armpit?
A. Auxiliary lymph nodes C. Cervical lymph nodes
B. Mesenteric lymph nodes D. Inguinal lymph nodes

54. According to the system of the Linnaeus nomenclature, species name is written in:
A. abbreviation C. small letter
B. capital letter D. local language

55. This item is based on the following table which shows classification of different groups of organisms.

№	Organisms	Kingdom Fa	Family	Genus	Species
	Human	Animalia	Huminidae	Homo	Sapiens
	Teff	Plantae	Poaceae	Canis	Familiaris
	Mushroom	Fungi	Poaceae	Agaris	Campestris
	Honeybee	Animalia	Apidae	Musca	Domestica

Which one of the above rows shows the correct classification of organisms?
A. 1 B. 2 C. 3 D. 4

56. What causes goiter?
A. Excessive salt intake C. Vitamin deficiency
B. Iodine deficiency D. Excessive protein intake

57. Which one of the following methods is used to treat diabetes?
A. Insulin injection C. Taking more glucose
B. Eating red meat D. Taking antibiotics

58. How does the Intrauterine Device (IUD) work? By:
A. preventing ovulation C. preventing implantation
B. killing sperm cells D. preventing fertilization

59. What is homeostasis? It is:
A. adaptation of human beings to their environment
B. production of red blood cells from the bone marrow
C. maintenance of constant internal environment
D. absorption of food in the digestive system

60. Which of the following is correct about the adaptation of snakes to high temperature?
A. Moving into the shade C. Vasoconstriction
B. Basking in the sun D. Sweating

61. Which one of the following molecules is made up of amino acids?
A. Sucrose B. Vitamins C. Proteins D. Minerals

62. Why do we need a balanced diet? To:
A. prevent infection by HIV C. gain excessive weight
B. become very tall D. support healthy growth

63. Which one of the following structures produces hydrochloric acid that kills most of the bacteria taken with food?
A. Liver B. Stomach C. Gall bladder D. Small intestine

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64. What makes the lining of the small intestine adapted to rapid absorption of digested foods?
A. Sphincter B. Villi C. Bladder D. Pancreas

65. Which one of the following statements is correct about anaerobic respiration? It:
A. produces more ATP than aerobic respiration
B. is the main source of energy for warm blooded animals
C. breaks down food to release energy without oxygen
D. produces carbon dioxide and water as a byproduct

66. What is the importance of turgor pressure for plants? It:
A. prevents loss of water from plants
B. keeps plant parts rigid and firm
C. helps roots to lose excess animals
D. makes plant cells to plasmolyse

67. **This question is based on the following table which shows the comparison of plant and animal cells.**

№	Plant cells	Animal cells
	have chloroplasts	lack chloroplast
	have endoplasmic reticulum	lack endoplasmic reticulum
	lack cell wall	have cell wall
	lack cell membrane	have cell membrane

Which of the above comparisons in the table is correctly matched?
A. 1 B. 2 C. 3 D. 4

68. What is the cellular structure that forms a very thin 'skin' like barrier on the outside of the cell?
A. Endoplasmic reticulum C. Cell membrane
B. Cytoplasm D. Nucleus

69. Why do we stain cells? To make them:
A. decay faster C. easier to see
B. bigger in size D. move around

70. Which one of the following statements is correct about the function of the electron microscope? It is used to see:
A. living materials suspended in a vacuum
B. details of objects invisible under light microscope
C. the movement of very small organisms
D. how mosquitoes transmit disease

71. Which one of the following is a renewable resource?
A. Fossil fuel B. Gold C. Coal D. Livestock

72. The tendency of plants to grow towards water is:
A. hydrotropism C. chemotropism
B. phototropism D. geotropism

73. What is the role of chlorophyll photosynthesis? To absorb:
A. oxygen gas C. carbon dioxide
B. water molecules D. light energy

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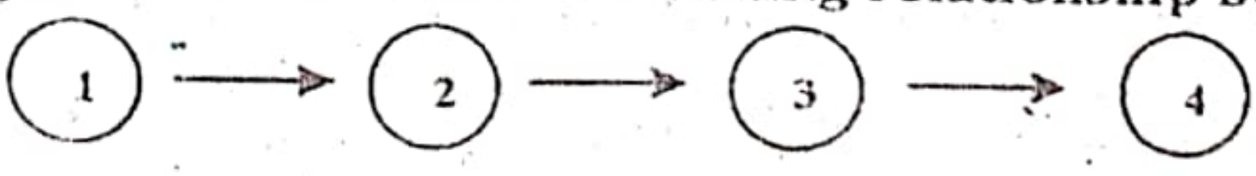
74. How is deforestation directly related with carbon building that leads to global warming? By:
   A. the loss of so many plants that can trap the excess CO_2 in the atmosphere
   B. the loss of so many animals that can trap the excess CO_2 in the atmosphere
   C. increasing the forest cover that can trap the excess CO_2 in the atmosphere
   D. increasing the oceans area that can trap the excess CO_2 in the atmosphere
75. How do plants absorb minerals ions?
   A. Without using energy from the plant
   B. By the action of enzymes as carriers
   C. Without using the action of enzymes
   D. By using energy from the plant
76. What is the result of a cross between pea plants with heterogeneous alleles for round seeds (Rr) and those with wrinkled seeds (rr)?
   A. The first generation produces only round seeds.
   B. 50% of the first generation produces round seeds.
   C. 75% of the first generation produces wrinkled seeds.
   D. The first generation produces only wrinkled seeds.
77. The type of breeding done in order to get a particular trait is:
   A. cross breeding
   B. multi breeding
   C. selective breeding
   D. true breeding
78. Production of biogas is an application of biotechnology in the area of:
   A. medicine B. food C. energy D. water
79. The main objective of Gregor Mendel's work was to proof how:
   A. inheritance works
   B. cell division works
   C. blood circulation works
   D. breathing system works
80. Abrham has three sisters and both of his parents have homozygous alleles for straight thumbs. How many of his sisters will inherit the allele for straight thumbs?
   A. One of his sisters inherits the allele for straight thumbs.
   B. Two of his sisters inherit the allele for straight thumbs.
   C. None of his sisters inherits the allele for straight thumbs.
   D. All of his straight sisters inherit the allele for straight thumbs.
81. **This item is only for non-blind candidates and it is based on the following diagram, which shows the feeding relationship between organisms.**



Which one of the above members in the food chain represents the primary consumers?

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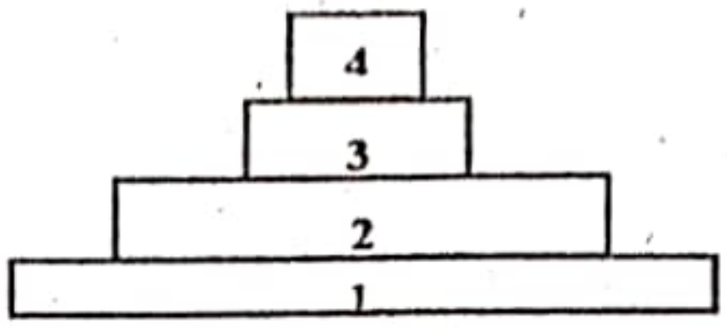
A. 1 B. 2 C. 3 D. 4

The item is only for non-blind candidates and it is based on the following diagram, which shows a pyramid of biomass.



82. Which organism has the highest biomass?

A. 4 B. 1 C. 2 D. 3

83. **This item is based on the following table, which shows adaptation of plants to specific conditions and their examples.**

№	Adaptation	Example
	Thick leathery leaves	Lobelia
	Leaves with large surface area	Cactus
	Broad and flat leaves	Tumble weeds
	Winged seeds	Maize

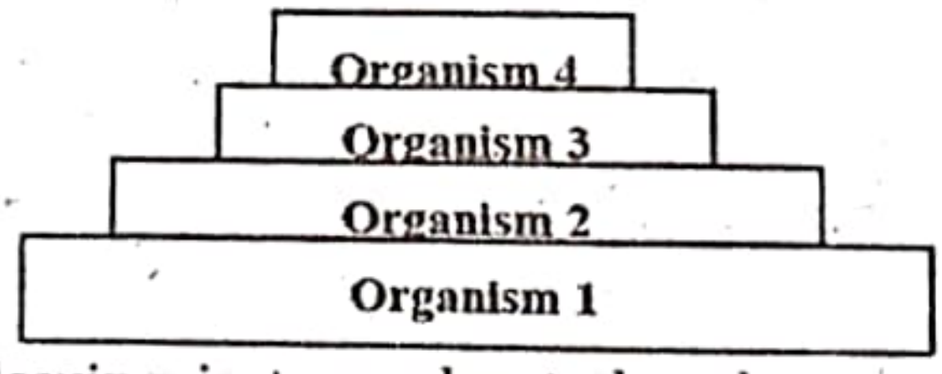
Which one of the above plant adaptations and the examples given is correctly matched?

A. 1 B. 3 C. 4 D. 2

84. What is the significant of planting trees? To:

A. reduce the number of soil microorganisms
B. reduce the rate of soil erosion
C. increase the amount of carbon dioxide
D. reduce the number of wild life

85. **This item is only for non-blind candidates and it is based on the following diagram, which shows the pyramid of energy.**



Which of the following is true about the above energy of pyramid? Organism:

A. 1 has a very low energy from the rest
B. 1 has a very high energy from the rest
C. 2 is a producer and has high energy from the rest
D. 4 is consumer and has high energy from the rest

86. Which one of the following features best describes members of kingdom monera? They:

A. are eukaryotic C. have nucleus

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B. are multicellular D. are all single celled

87. Which one of the following statements is correct about the characteristics of monocots? They:
A. are generally insect pollinated
B. have leaves with network of veins
C. have stem with a ring of vascular tissue
D. have embryo with single seed leaf

88. A grade 9 student collected some animals from a forest ecosystem and classified all of them as vertebrates. Which of the following features were used by the student to group the organisms as vertebrates? The:
A. presence of endoskeleton
B. presence of open circulatory system
C. absence of reproductive structures
D. absence of kidneys as excretory organ

89. How do we distinguish phylum Chordate from the other animal phyla? Animals in phylum chordate have:
A. ring of tentacles with singing cells
B. flatted body with mouth but no anus
C. notochord and hollow nerve cord
D. segment bodies and joined limbs

90. Which one of the following is the biotic component of an ecosystem?
A. Soil type C. Amount of sunlight
B. Level of oxygen D. Amount of food

ITEM 91 – 95 ARE ONLY FOR BLIND CANDIDATES. START WITH NUMBER 91 ON YOUR ANSWER SHEET.

91. In a given food chain cow is found at the:
A. 1st trophic level C. 3rd trophic level
B. 2nd trophic level D. 4th trophic level

92. According to the pyramid of biomass, which of the following groups of animals has the highest biomass?
A. Sheep B. Dog C. Lion D. Hyena

93. In a given pyramid of energy, which of the following groups of organisms has the lowest energy?
A. Grass B. Horse C. Tiger D. Ox

94. Which of the following processes removes CO_2 from the atmosphere?
A. Burning C. Decomposition
B. Respiration D. Photosynthesis

95. What is the function of semicircular canals of the ear? They:
A. are concerned with the detection of motion
B. filter the air entering the ear canal
C. collect sound waves from the air
D. lubricate the ear ossicles

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CamScanner

ETHIOPIAN GENERAL SECONDARY EDUCATION CERTIFICATE BIOLOGY EXAMINATION GINBOT 2010/MAY 2018

1. Professor Yalemtsehay Mekonnen is a renowned biologist known for her contribution on
   A. the discovery of endod as a control for schitosomiasis.
   B. the development of hybrid sorghum which is resistant to drought.
   C. assessment of the impact of chemical pesticide hazards on humans.
   D. preservation of different traditional crop varieties.
2. The Ethiopian biological research institute which is conducting research on leprosy and other diseases is
   A. Armauer Hansen Research Institute.
   B. Aklilu Lemma Institute of Pathobiology.
   C. The Institute of Biodiversity con
   D. Ethiopian Institute of Agricultural Research
3. The ability of the microscope to separate two objects close to each other is called
   A. magnification
   B. condensation
   C. reflection
   D. resolution
4. Which of the following stains may be used to study only plant cells?
   A. Iodine
   B. Acetocarmine
   C. Haematocylin
   D. Methylene blue
5. The cell theory states that cells
   A. Originate spontaneously
   B. are the basic units of life
   C. have central nucleus
   D. reproduce sexually
6. Which of the following indicates the advantage of conserving natural resources?
   A. Reduces crop pests
   B. Protects biodiversity
   C. Control population growth
   D. Reduces animal production
7. Which of the following methods helps in biodiversity conservation?
   A. Monoculture agriculture
   B. Farming with tractors
   C. Planting trees for fire wood
   D. Setting national parks
8. Which of the following plants is endemic to Ethiopia?
   A. Teff B. Bahir zaf C. Woira D. Mango
9. Which of the following methods helps to conserve the vegetation of our country?
   A. Substitution of indigenous trees
   B. Planting only wind pollinated trees
   C. Planting of bahir zaf
   D. Replanting endemic trees
10. Which of the following National parks contains endemic wildlife of Ethiopia such as Ethiopian wolf, walia ibex, and gelada baboon?
    A. Gambella National Park
    B. Simien Mountains National Park
    C. Nechsar national Park

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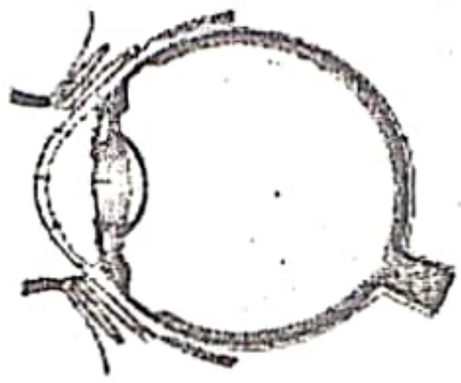
D. Bale Mountains National park

11. Which of the following structures is part of a neuron?
    A. Nephron B. Ossicle C. Medulla D. Axon
12. What is the function of the axon of a nerve cell?
    A. Transport of the nerve impulse
    B. Connection of neighboring cells
    C. Increasing speed of the impulse
    D. Receiving signal from other cells.
13. Which of the following is an example of reflex action?
    A. Washing our face with the hands
    B. Peeling a banana with the hands
    C. Removing a finger from a hot plate
    D. Writing a note on the exercise book
14. This item is only for non-blind candidates and it is based on the following diagram which shows the human eye.



In the diagram letter "A" represents A

    A. cornea B. lens
    C. iris D. retina

15. Which part of the eye regulates the size of the pupil?
    A. Retina B. Sclera C. Iris D. Lens
16. Which of the following steps is correct about planting and growing of trees?
    A. Get a 3 year sapling ⟶ add water ⟶ dig big hole ⟶ plant the sapling ⟶ return the soil
    B. Get a 2 year sapling ⟶ dig big hole ⟶ add water ⟶ plant the sapling ⟶ return the soil
    C. Get a 1 year sapling ⟶ dig big hole ⟶ plant the sapling ⟶ add water ⟶ return the soil
    D. Get a 3 year sapling ⟶ dig big hole ⟶ add water ⟶ plant the sapling ⟶ return the soil
17. Which of the following is an example that shows the significance of biotechnology?
    A. Using genetic engineering to improve plant growth
    B. Digging large holes to conserve rain water
    C. Building terraces to minimize soil erosion
    D. Using dung to fertilize the soil
18. In which of the following areas of the present day application of biotechnology could human and animal waste be used?
    A. Medicine production C. Steel production

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B. Biogas production D. Plastic production

19. During which stage of mitosis do sister chromatics separate?
A. Prophase B. Anaphase C. Metaphase D. Telophase

20. This item is based on the information provided in the table below.

Mitosis	Meiosis
Takes place in somatic cells	Takes place in germ cells
Daughter cells have the same number of chromosomes with parent	Daughter cells have half number of chromosomes of the parent
Two daughter cells	"A"
Daughter cells are identical with the parent	Daughter cells are different from the parent

In the above table "A" represents
A. Two daughter cells C. eight daughter cells
B. Six daughter cells D. four daughter cells

21. Which of the following list is different from the rest in relation to the six classes of food?
A. Carbohydrate, mineral, water C. Fat, vitamin, water
B. Carbohydrate, protein, fat D. Protein, fat, alcohol.

22. Which of the following group of food items is a good source of fats and oil?
A. Butter, ground nut, coconut
B. Injera, rice, potato
C. Meat, cheese, chick peas
D. Carrots, green vegetables, fruits

23. A balanced diet is composed of
A. more protein and carbohydrate and small fats and vitamins.
B. more carbohydrate and fats and small protein and minerals.
C. enough quantity of all of the major types of food items.
D. more fat and protein and small carbohydrates and minerals.

24. The blue colour of dichlorophenol indophenol (DCPIP) reagent turned colourless upon addition to unknown food sample is
A. starch B. protein C. vitamin C D. fat

25. Which of the following is part of the digestive system?
A. Bladder B. Duodenum C. Lung D. Scrotum

26. What is the main function of spongy mesophyll? It
A. controls the size of stomata.
B. is involved in water conservation.
C. protects the leaf from damage.
D. facilitates gaseous exchange.

27. What is the role of chlorophyll in photosynthesis?
A. Absorbing CO_2 C. Splitting water
B. Capturing light D. Releasing O_2

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28. How does photosynthesis balance CO_2 and O_2 concentration in the atmosphere? By
A. absorbing O_2 and releasing CO_2
B. absorbing CO_2 and O_2
C. releasing CO_2 and O_2
D. absorbing CO_2 and releasing O_2

29. Which of the following conditions demonstrates the effect of deforestation? Increase in
A. O_2 concentration followed by global cooling
B. CO_2 concentration followed by global cooling
C. CO_2 concentration followed by global warming
D. O_2 concentration followed by global warming

30. Which of the following structures increase surface area for absorption of water from the soil?
A. Vacuoles B. Root hairs C. Cell wall D. Xylem

31. If a person with blood group "B" loses blood due to an accident, he may receive blood donation from group
A. "AB" only B. "A" & "B" C. "B" & "O" D. "B" only

32. Even though anaerobic respiration results in much less amount of ATP, cells are forced to use it when there is shortage of
A. N_2 B. CO_2 C. HCO_3^- D. O_2

33. This items is based on the comparison between aerobic and anaerobic respiration in animals provided in the table below.

Aerobic respiration	Anaerobic respiration
Oxygen dependent	Oxygen independent
Produces more ATP	Produces less ATP
Metabolizes glucose	Metabolizes glucose
Produces H_2O and energy	Produces "A"

In the above table "A" Stands for
A. Amino acid and energy.
B. CO_2 and energy
C. Lactic acid and energy.
D. Uric acid and energy

34. The black substance found in tobacco smoke which causes chronic obstructive pulmonary disease is
A. nicotine
B. tar
C. carbon monoxide
D. caffeine

35. Which of the following occurs during inhalation of air?
A. Relaxation of the diaphragm
B. Relaxation of the external inter coastal muscles
C. Movement of the ribs up and out
D. The diaphragm becomes dome shaped

36. How does antiretroviral medicine help in the treatment of HIV/AIDS? It
A. controls the reproduction of the virus.
B. makes T-cells more susceptible to the virus.
C. fights against respiratory infection.

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D. kills the virus by attacking is DNA.

37. What do you call the body's ability to maintain normal function and stability?
    A. Thermoregulation
    B. Excretion
    C. Osmosis
    D. Homeostasis

38. What do you call animals which have body temperature that varies with the external temperature?
    A. Poiklotherms
    B. Homoiotherms
    C. Hibernators
    D. Aestivators

39. Which of the following parts of the body is different from the rest in relation to homeostasis?
    A. Kidney B. Heart C. Skin D. Liver

40. This item is only for non-blind candidates and it is based on the following diagram of cross section of the leaf.

Letter "A" represents
    A. Stomata
    B. Spongy mesophy ll
    C. palisade
    D. guard cell

41. Vaccines that help to protect the body from deadly diseases are produced from
    A. antibiotics
    B. extracts of tablets
    C. medicinal plants
    D. weakened pathogens

42. A chemical agent that is applied on a living tissue to kill microorganisms is
    A. disinfectant
    B. antiseptic
    C. autoclave
    D. sterilizer

43. Which of the following processes requires the activity of yeast?
    A. Producing wine
    B. Preparing coffee
    C. Cooking shero wot
    D. making orange juice

44. In a petri dish used to grow microorganisms, agar is added
    A. to provide moisture
    B. to lower the temperature
    C. to maintain PH.
    D. as a source of food.

45. Which of the following is the moist appropriate support that can be provided to people living with HIV/AIDS (PLWHA)?
    A. Counseling on how to live longer with HIV/AIDS
    B. Washing the clothes of PLWHA
    C. Giving money to PLWHA
    D. Building the moral of PLWHA by giving them authority

46. Which of the following is the function of thyroxin?
    A. Conversation of glucose to glycogen
    B. Conversion of gluycogen to glucose
    C. Control of the metabolic rate

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D. Development of sexual characteristics

47. What do you call glands that produce secretions which are directly released to the blood stream?
A. Exocrine B. Mammary C. Salivary D. endocrine

48. **This item only for non-blind candidates and it is based on the following diagram.**

Letter "A" represents
A. thyroid gland.
B. adrenal gland.
C. Pituitary gland.
D. salivary gland.

49. Which of the following is a physical method of contraception?
A. Condom B. Pill C. Implant D. Injection

50. Which of the following statements is correct about female genital mutilation (FGM)? It.
A. Helps to make girls pure.
B. makes girls acceptable by men.
C. Increases exposure to HIV/AIDS
D. Helps women to give birth easily.

51. Which of the following is advisable with regard to the use of medicine?
A. Children can be given medicine prescribed for adults for similar symptoms.
B. Self-prescription of medicine based on previous history.
C. Finishing the prescribed does even if the symptoms are gone.
D. It is bad to take overdose of a medicine but taking under doe is not a problem.

52. Which of the following sexually transmitted diseases is caused by Haemophilus ducreyi?
A. Chancroid C. Gonorrhea
B. Syphilis D. AIDS

53. Which of the following diseases is caused by Haempophilus ducreyi?
A. Chncroid B. typhoid C. tuberculosis D. Cholera

54. Classification of organisms into groups is useful because it helps to
A. protect endangered animals.
B. attract tourists that come to visit animals.
C. understand the great diversity of living things.
D. Understand larger organisms better than smaller ones.

55. Which of the following represents the correct use of the binomial nomenclature developed by Linnaeus?
A. Order name followed by species name
B. Genus name followed by order name
C. Species name followed by genus name
D. Genus name followed by species name

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56. Which of the following takes place during image formation?
    A. Refraction of light by sclera
    B. Refraction of light by lens
    C. Divergence of light by the cornea
    D. Divergence of light by the aqueous humour
57. Which of the following defects is different from the rest?
A. Deafness B. Astigmati C. Short sightedness D. Long sightedness
58. If a person has problem of focusing the image behind the retina, which type of leans can be prescribed to correct the problem?
    A. Convave B. Flat C. Scattering D. Convex
59. How are the sensory cells of the semicircular canals in the inner ear stimulated to initiate a never impulse? They are stimulated by
    A. tilting of otolighs.
    B. Vibration of the ear drum.
    C. tilting of the cupula.
    D. vibration of the ossicles.
60. What do you call structures which produce secreations that have an effect on other parts of the body?
    A. Vesicles B. vessels C. Glands D. Gonads
61. The digestion of proteins takes place in the
    A. mouth and stomach
    B. stomach only
    C. small intestine only
    D. stomach & small intestine
62. Which of the following methods helps to maintain oral hygiene?
    A. Regular tooth brushing
    B. Taking sweet food
    C. Regular body exercise
    D. Taking lots of acidic food
63. Which of the following enzymes digests starch in the mouth?
    A. Lactase B. Maltase C. Sucrase D. Amylase
64. Part of the respiratory system where gaseous exchange takes place is
    A. tracheas. B. alveolus. C. bronchus. D. bronchiole.
65. Blood vessels that carry deoxygenated blood and return it to the heart are
    A. capillaries B. lymph vessels. C. veins. D. arteries.
66. **This item is only for non-blind candidates and it is based on the following diagram which shows a generalized animals cell.**

In the diagram, letter "A" represents
    A. cytoplasm.
    B. mitochondrion
    C. smooth endoplasmic reticulum.
    D. rough endoplasmic reticulum
67. A cellular structure where protein synthesis takes place is
    A. nucleus.
    B. mitochondrion.
    C. ribosome.
    D. endoplasmic reticulum.
68. This items is based on the information provided in the table below.

No	Cellular structure
1.	Nucleus
2.	Cell wall

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3.	Cell membrane
4.	Mitochondria
5.	Chloroplast
6.	Large vacuole
7.	Endoplasmic reticulum

Which of the following combinations consists of cellular structures which are found only in plants?

A. 2,3,5 B. 1,2,6 C. 1,4,7 D. 2,5,6

69. A plant cell has 10% sucrose concentration in its cytoplasm. If the cell is immersed in 1% sucrose solution, it will

A. become turgid.
B. burst.
C. become plasmolysed
D. become big.

70. If a plant cell is plasmolysed, the

A. Cell membrane will be firmly attached on the cell wall.
B. Cytoplasm shrinks away from the cell wall.
C. Vacuole pushes the cytoplasm towards the cell wall.
D. Pressure inside its vacuole increases.

71. How is the water in the xylem transported upwards to the top of the plant?

A. By using cellular energy
B. By pulling of transpiration stream
C. Due to high mineral concentration in the soil
D. By using only root pressure

72. Which of the following do you recommend to minimize the negative effect of transpiration in agriculture?

A. Cutting the trip of stems
B. Fertilizing the soil
C. planting crops in line
D. Irrigate the land

73. Which of the following plant hormones stimulates cell division?

A. Ethylene B. Cytokinin C. Auxin D. Gibberelic acid

74. Why is apical dominance removed when the growing tip is cut? Because the

A. negative effect of auxin on side shoots stops.
B. positive effect of auxin on side shoots stops.
C. positive effect of auxin on main shoots is enhanced.
D. amount of auxin produced at the tip increases.

75. Which of the following is a non-renewable resource?

A. Crop B. Tree C. Oil D. Cattle

76. According to Mendel's law of inheritance, round seeds (R) are dominant over wrinkled sees (r). which of the following proportions will be obtained in the F1 generation if homozygote patents with round (RR) and wrinkled (rr) seeds are breed?

A. All round C. 1 round: 1 wrinkled

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B. 3 round: 1 wrinkled D. All wrinkled

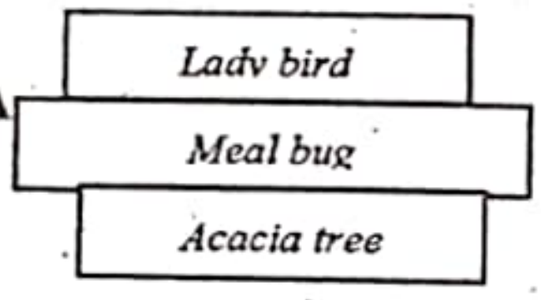
77. In humans, dangly ear lobe is dominant over attached ear lobe. If two parents heterozygote for dangly ear lobes produce children, what will be the probability of producing offspring with attached ear lobes?
A. 50% B. 75% C. 25% D. 100%

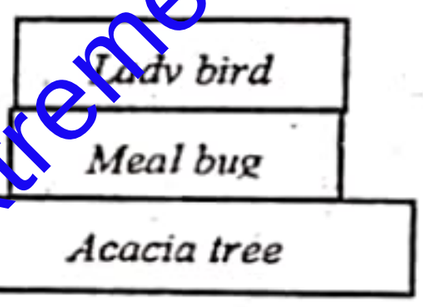
78. In the chromosomes, what links the two DNA stands?
A. Sugars B. Bases C. Phosphate D. Nucleotides

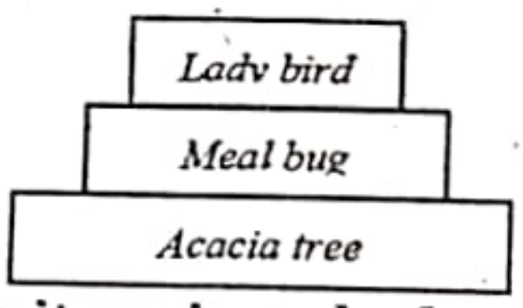
79. Which of the following breeds of cattle has Ethiopian origin?
A. B orene B. Holstein C. Friesian D. Hareri

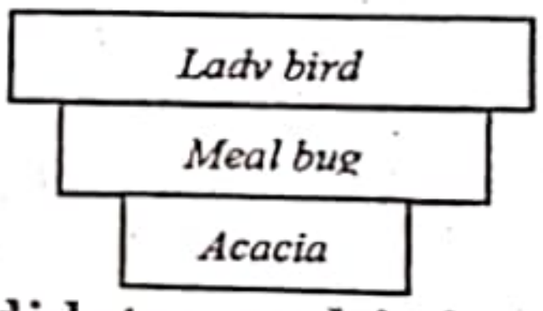
80. What do you call a breeding technique which helps to get a combination of good traits from two different parents?
A. Pure breeding C. Group breading
B. Selective breading D. Cross breading

81. **This item is only for non-blind candidates and it is based on the following diagram of pyramid of biomass.**
Which of the following diagrams represents a pyramid of biomass?

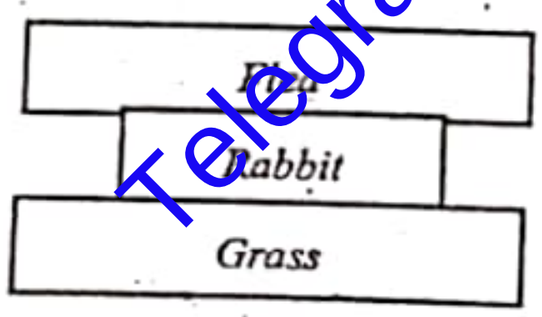
A


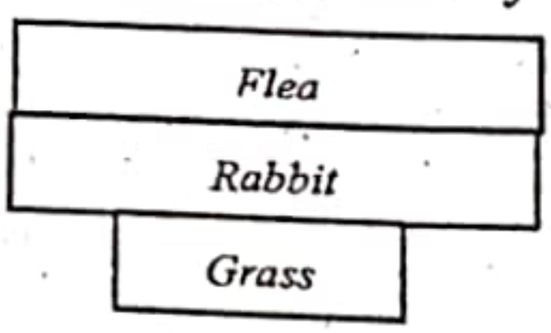
C


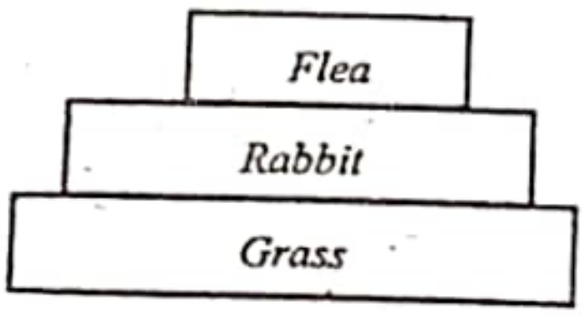
B


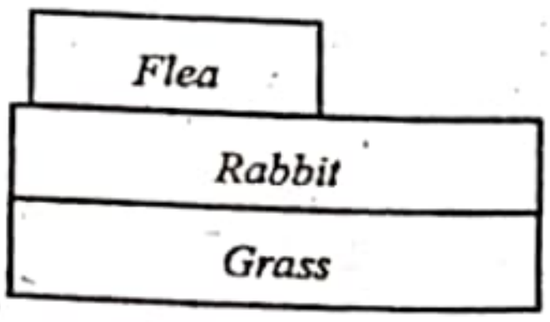
D


82. **This item is only for non-blind candidates and it is based on the following diagram of energy transfer.**
The energy transfer in pyramid of energy is represented by

A


C


B


D


83. Which of the following options indicates the correct pathway of the conversion of nitrogen (N_2) into nitrates?
A. N_2 ⟶ putrefying bacteria ⟶ ammonium compounds ⟶ nitrifying bacteria ⟶ nitrates
B. N_2 ⟶ denitrifying bacteria ⟶ ammonium compounds ⟶ nitrifying bacteria ⟶ nitrates

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C. N_2 ⟶ nitrogen fixing bacteria ⟶ ammonium compounds ⟶ nitrifying bacteria ⟶ nitrates

D. N_2 ⟶ nitrogen fixing bacteria ⟶ ammonium compounds ⟶ denitrifying bacteria ⟶ nitrates

84. Which of the following is an adaptation by desert plants such as cacti to conserve water?
A. Production of seeds that can be transported by water
B. Having broad leaves
C. Having leaves which are not covered by cuticle
D. Having narrow and spiny leaves

85. Which of the following is adaptation by desert animals to lose excess heat?
A. Large and thin ear
B. Thick layer of fat
C. Large body size
D. Thick fur coat

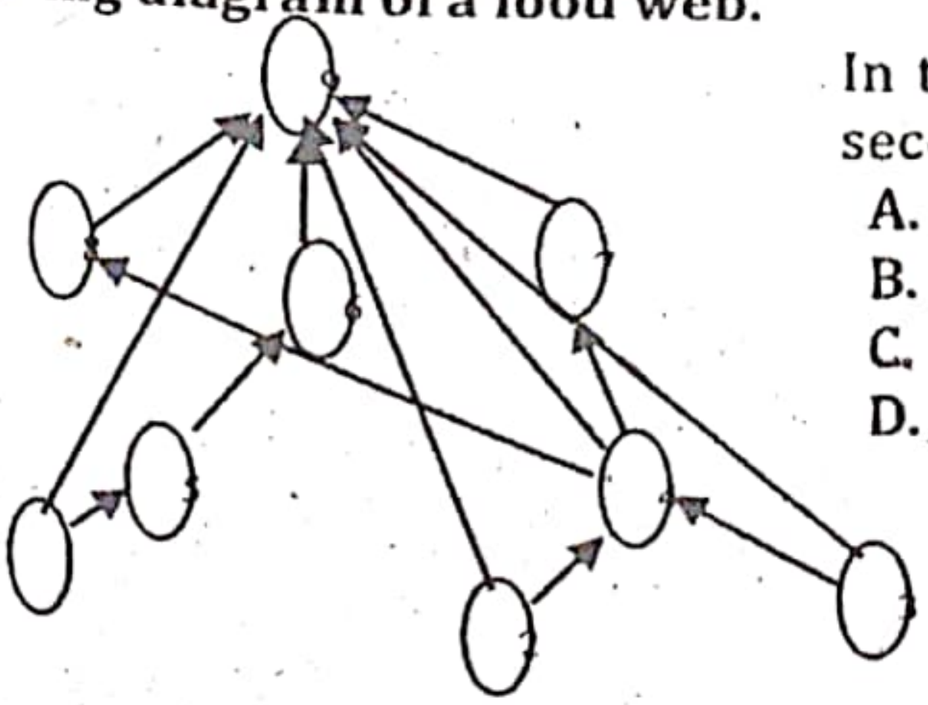
86. Which of the following is a scientific name?
A. Abyssinian wolf
B. Canis simensis
C. Gelada baboon
D. Mountain nyala

87. All microscopic single-celled organisms that do have nucleus are grouped under kingdom
A. Protista B. Animalia C. Monera D. Plantae

88. In which of the following divisions are evergreen seed plants that have cones grouped?
A. Pteridophyta
B. Angiospermae
C. Gymnospermae
D. Bryophyta

89. Which of the followings belongs to kingdom Monera?
A. Mycobaterium tuberculosis
B. Entamoeba histolytica
C. Plasmodium falciparum
D. Candida albicans

90. **This item is only for non-blind candidates and it is based on the following diagram of a food web.**



In the food web given above, the secondary consumers are
A. 6,7,8
B. 1,2,3
C. 4,5,6
D. 5,7,8

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ANSWER TO EGSECE BIOLOGY EXAMINATION, GINBOT 2009/ MAY 2017

1. D: **Prof Beyene Petros** is a biomedical scientist and long serving professor at Addis Ababa University.
2. B: **Institute of Agricultural Research** is the institute where biologists work to improve crops and livestock, and soils fertilities.
3. C: Ribosome takes part in the manufacture of proteins.
4. A: The pressure develops because of entrance of water is **turgor pressure**, whereas **plasmolysis** is the loss of this turgor pressure.
5. C: **Turgidity** is a pressure builds up until no more water can enter the plant cell.
6. A: **Biodiversity** is the range of living organisms in an area.
7. B: Protecting the natural environment is important to preserve biodiversity, and to insure the presence of healthy weather and climate.
8. C: Vegetation is used for food, clothing, building materials and medicines.
9. B: Gelada Baboon, **Mountain Nyala** and Ethiopian Wolves are some very rare animals in Bale Mountain National Park.
10. C: Air pollution is caused by smoke, by **carbon dioxide** and carbon monoxide from the burning of fossil fuel.
11. D: Dendrite is a branched extension of neuron that receives electrical signal from other neurons and conducts towards the **cell body**.
12. D: The reflex arc is the neural pathway leads from stimulus to effector via the spinal cord.
13. D: Long sightedness can be corrected using **convex** (converging) lenses.
14. A
15. B: The salty receptors of tongue are located at the back upper surface of the tongue.
16. B: Plants and other organisms turn **carbon dioxide** and water into carbohydrates and oxygen, using energy trapped by chlorophyll.
17. D: Modern biotechnology has an application in production of food, livestock, **medicine** and energy.
18. D: Mitosis is cell division that takes place in the normal body cells and produces identical cells.
19. B: Mendel removed immature **stamens** from the flowers of the pea plant so that these flowers could not pollinate themselves.
20. D: During **anaphase I**, a pair of chromatids from each homologous pair moves to opposite poles of the cell.
21. D: In the presence of protein in the foodstuff, the Biuret reagent turns into **mauve** or **purple**.
22. A: Oesophagus is the passage down which food moves between the throat and the stomach.
23. C: Blood group A and B has antigen A and B, respectively, whereas; blood group O lacks antigen on the surface red blood cells.
24. C: Valves control the flow of blood in the heart.

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25. C: To avoid blood agglutination, the blood type of the donor must be "O" blood type.

26. B: The plant that kept in Jar 'A' produced lesser amount of starch compared to Jar 'B' because the KOH in the Jar absorbed CO_2, which is the raw material for synthesis of carbohydrate.

27. B: For photosynthesis turns CO_2 and H_2O into carbohydrate and O_2, it balances concentration of CO_2 and O_2 in the atmosphere

28. A: The root hairs are hair-like extensions that increase surface area for **osmosis** to take place in plant root.

29. C: Phloem is a plant tissue that conducts synthesized nutrients to all parts of the plant.

30. D: Removing apical dominance enhances rapid growth of side shoot.

31. D: ATP is important to the body for production of macromolecules, **active transport**, muscle contraction, and nerve impulse transmission.

32. C: **Hygiene of breathing:** clean teeth and tongue, avoid smoking, cover mouth during coughing and sneezing, and good ventilation.

33. B

34. A:

Blood group	Antigen of RBC	Antibodies in the plasma	Donate to	Receive from
A	A	B	A and AB	A and O
B	B	A	B and AB	B and O
AB	A and B	None	AB only	AB group
O	None	A and B	All groups	O only

35. B: Plasma transports dissolved food, carbon dioxide, and urea.

36. C: **Poikilotherms** adapt to a cold weather by behavioral mechanisms, such as **basking in the sun**, pressing their body against a warm surface, and erect special sails.

37. D: **Temperature control in homoiotherms:** sweating, **vasodilation,** panting and licking, vasoconstriction, piloerection, shivering etc.

38. D: **Ovulation** is the expulsion of an ovum from ovary.

39. B: The **pinna** helps to trap and funnel sound into the ear.

40. A: **Palisade mesophyll** is the major photosynthetic tissue of a plant.

41. C: Vaccine is a substance given to stimulate the body's production of antibodies and provide immunity against disease.

42. A

43. C: Self-prescribed medicines increase the risk of people will become seriously ill and even death. Avoid self-medication except for the simplest conditions.

44. B: **Antiseptic** is a chemical applied to living tissue to kill microorganisms.

45. A

46. D: **Accommodation** is ability of the human eye to focus on objects at different distances.

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47. A: **Astigmatism** is an eye defect resulted from irregular eye shape (more egg-shaped eye).

48. B: When no pregnancy has occurred after ovulation, the ovary reduces the level of both oestrogen and progesterone production.

49. B: Endocrine glands are ductless glands, whereas exocrine glands have ducts or tubes.

50. C: Female genital mutilation put girls at great risk of bleeding when a girl cut and give birth, infection of HIV/AIDS, and sexual displeasure.

51. D: The social and economic impacts of AIDS include the loss of young adults in their most productive years, and high expenditure for medical care, drugs etc.

52. D: It is important to **read the instruction** before use medicines, take the right dose, keep them at the right temperature, and keep them away from children.

53. A:

Types of lymph nodes	Location
Auxiliary lymph nodes	In armpit
Mesenteric lymph nodes	Back of abdominal cavity
Cervical lymph nodes	In neck
Inguinal lymph nodes	In groin

54. C: In Car Linnaeus nomenclature, the genus name is written with a capital letter, whereas the species name is written with a small letter.

55. A

56. B: Goiter is enlargement of the thyroid gland and resulted in deficiency of iodine.

57. A: **Insulin** injections treat diabetes successfully but they do not cure it.

58. C: Intrauterine Device interferes with and prevents the implantation of the early embryo.

59. C: **Homeostasis** is the body's ability to maintain normal function and stability.

60. A: **Poikilotherms**, such as snakes adapt to high temperature by moving into the shade or moving into water or mud.

61. C: Amino acids are building blocks of proteins.

62. D: Balanced diet is a diet that contains adequate amounts of all the necessary nutrients required for healthy growth and activity.

63. B

64. B: **Microvilli** are minute hair-like structures that increase the surface area of a cell.

65. C: **Anaerobic respiration** breaks down food to release energy without oxygen.

66. B: **Turgor pressure** is a pressure developed because of entrance of water and useful for rigidity of plants.

67. A

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68. C: The **cell membrane** forms a barrier like a very thin 'skin' around the outside of the cell.

69. C: **Stains** are chemicals added to slide tissues to make the cells easier to see.

70. B: Electron microscope achieves much greater resolving power to study a lot more about cells.

71. D: **Renewable resource** is any natural resources (mainly leaving things and their products) that can be replenish naturally with the passage of time.

72. A: The tendency of plants grows towards water – **hydrotropism**; light – **phototropism**, chemicals - chemotropism and gravity or earth – geotropism.

73. D: Chlorophyll is green pigment in plants that captures the light energy required for photosynthesis.

74. A

75. B: The mineral ions in the soil are usually found in very dilute solutions. By using active transport, plants can absorb these mineral ions.

76. B:

	R	r
r	Rr	Rr
r	Rr	rr

Progeny genotype ratio: 50% Rr : 50% rr

77. C: **Selective breeding** is the process of breeding plants and animals for particular traits.

78. C: Biotechnology is used to generate **biogas** from animal and human wastes.

79. A: Gregor Mendel work puts fundamental concept of law of inheritance.

80. D

81. B: In the food chain, 1 represents producer, whereas 2 represents primary consumer because it eats producer.

82. B: **Pyramids of biomass** represents the total biomass of organisms at each trophic level at a moment in time, irrespective of their numbers.

83. A

84. B: Trees are important for many reasons including taking carbon dioxide out of the atmosphere, holding the soil together and providing us with food and many other products.

85. B A **pyramid of energy** represents the energy in the producers and how much of that energy is passed on at each stage along the food chain.

86. D: Members of Kingdom Monera include all of bacteria and cyanobacteria. They are single-celled organism without nucleus. They are microscopic and reproduce by binary fission.

87. D: A monocot has a single cotyledon.

88. A: **Vertebrates** are creatures with a vertebral column/ backbone. They have an internal skeleton (endoskeleton) made of bone or cartilage.

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89. C: **Chordates** are creatures characterized by a notochord, nerve chord an gill slits

90. D: The biotic components of an ecosystem are the living organisms within a ecosystem, which affect the ability of an organism to survive there.

91. B: **Trophic levels** are levels in a food chain to which organism belongs. Co belongs to 2nd trophic level for it eats 1st trophioc level.

92. A: Sheep are primary consumers. There is a progressive less energy availabl for organisms further down a food chain.

93. C (Consult question number 92)

94. D: **Photosynthesis** is the process a plant uses to combine sunlight, wate and **carbon dioxide** to produce oxygen and sugar (energy).

95. A: The semicircular canals in your inner ear are concerned with the detectio of motion.

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ANSWER KEY TO EGSECE BIOLOGY EXAMINATION, GINBOT 2010/ MAY 2018

1. C **Prof Yalemtsehay Mekonnen**'s research is mainly focused on:
   -the assessment of the *impact* of chemical pesticide hazard on humans
   -the use of plants as *medicine* against human and animal disease.
2. A AHRI is conducting research on **leprosy, tuberculosis, leishmaniasis, malaria**, and **HIV/AIDS.**
3. D **Resolution** is the ability of an optical instrument to show two objects clearly as separate.
4. A **Application of commonly used stains**

Types of stain	Types of cells	Main organelles stained
Iodine	**Plant cells**	Any material containing starch
Haematoxylin	Animal and plant cells	Nuclei stained blue/ purple or brown
Methylene blue	Animal cells	Nuclei stained blue
Acetocarmine	Animal and plant cells	Staining the chromosomes in dividing nuclei

5. B The **cell theory** states that:
   All living things are made of one or more cells.
   Cells are the basic living units of structure and function in organisms.
   All cells come from preexisting cells.
6. B Conserving natural resources protect biodiversity.
7. D Careful management in National Parks conserve biodiversity of many animals and plants.
8. A Teff (*Eragrostis teff*), many *Euphorbia* spps, noug or niger seed (*Guizotia abyssinica*), enset (*Ensete ventricosum*, etc are examples of endemic species.
9. D
10. B
11. D A neuron is made up of **dendrites**, a **cell body,** and an **axon.**
12. A The **axon** carries impulses *away* from the cell body.
13. C Breathing, blinking, sneezing, knee jerk, and **withdrawal** are examples of reflex action.
14. D **Retina** is a light-sensitive tissue lining the inner surface of the eye.
15. C **Iris** is a membrane in the eye, which responsible for controlling the amount of light reaching the retina.
16. B

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17. A The **modern applications** of biotechnology include using genetic engineering to modify animals and plants, producing new medicines and helping to provide new energy sources.

18. B

19. B During anaphase stage, the spindle fibers shorten and *separate* the sister chromatids to opposite end of the cell.

20. D

21. D The six classes of food are carbohydrates, proteins, fats, vitamins, minerals, and water.

22. B Butter, beef fat, sesame oil, niger seed oil (nug), ground nut, coconut and olive oil are all lipids.

23. C **Balanced diet** is taking food from all major food groups in order to maintain a healthy body.

24. C

25. B: The human digestive system includes mouth, oesophagus, stomach, small intestine (duodenum, ileum, and jejunum), large intestine and rectum.

26. D Spongy mesophyll is the main gas exchange tissue of a leaf.

27. B Chlorophyll is the green pigment in plants that captures the energy from the sun for photosynthesis.

28. D

29. C Deforestation leads to a CO_2 build-up in the atmosphere and finally to global warming.

30. B **Root hairs** are hair-like extensions that increase the surface area.

31. C "√" – **Compatible blood; "×" – non- compatible blood**

D / R	O	A	B	AB
O	√	√	√	√
A	×	√	×	√
B	×	×	√	√
AB	×	×	×	√

32. D

33. C

34. B Tar is a sticky black chemical in tobacco smoke that makes smokers to develop bronchitis.

35. C During inhalation of air:
Diaphragm muscles contract – diaphragm flattens
External intercostals muscles contract, pulling ribs up and out

36. A Antiretroviral drugs can slow down the progress of HIV/AIDS.

37. D **Homeostasis** is the body's ability to maintain normal function and stability.

38. A **Poikilotherms** animals whose internal temperature varies along with that of the ambient environmental temperature.

39. B Kidney, liver and skin are organs of homeostasis, but not heart.

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40. C
41. D Vaccine is produced from dead or weakened strains of pathogens.
42. B Antiseptics are chemicals that can be used to kill micro-organisms or stop them growing on living tissue.
43. A Yeasts are used in making bread and alcoholic beverages.
44. A
45. A
46. C **Thyroxine** is hormone produced by the thyroid glands to regulate metabolism by controlling the rate of oxidation in cells.
47. D: Endocrine gland is any gland of the body that secretes hormones directly into the blood.
48. C
49. A **Physical or barrier methods of contraception** prevent the meeting of the ovum and the spermatozoa. E.g. **condoms**
50. C FGM increases risk of: HIV infection, infertility, painful sexual intercourse, and giving birth.
51. C
52. A Chancroid is caused by the bacterium *Haemophilus ducreyi*.
53. B Symptom of typhoid: high fever, a painful abdomen, **enlarged spleen and liver**, constipation or diarrhea etc.
54. C
55. C
56. B In human eye, light is refracted as it passes from the air through the cornea and the lens.
57. A Deafness is hearing disorder, others are common eye defects.
58. D Focusing the image behind the retina can be corrected using convex lenses that bring the light rays together more before they reach the eye.
59. C The moving fluid causes the cupula to tilt, thus pressing the hairs of the sensory cells that creates nerve impulses in the sensory nerve endings. The nerve impulses are transmitted to the brain. The brain then interprets the direction and speed of motion of the body or head.
60. B Glands are structures which produce hormones that regulates the function of a specific tissue or organ.
61. D
62. A Cleaning teeth and tongue avoid bad breath.
63. D The amylase in saliva begins the digestion of the carbohydrates in the mouth.
64. B **Alveolus** is tiny air sacs in a lung that used for the exchange of gases.
65. C **Veins** carry deoxygenated blood towards the heart.
66. B

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67. C **Ribosomes** are organelles involved in protein synthesis.
68. D
69. A Cell placed in dilute solution absorbs water by osmosis and becomes turgid.
70. B Excessive loss of water by osmosis causes the cell to become plasmolysed.
71. B The transpiration stream is pulling a column of water in the xylem up to the top of the plant.
72. D
73. B **Cytokinin** is plant hormones that promote cell division.
74. A Removing the growing tip enhances the growth of side shoots.
75. C **Non-renewable resources** cannot be replaced when they are used up.
76. A:

Let R = round, r = wrinkled

Parent genotypes: RR x rr

Gametes: R R r r

F_1 genotypes:

F_1 phenotypes: **All round**

	R	R
r	Rr	Rr
r	Rr	Rr

77. C:

Let D = dangly earlobe, d = attached earlobe

Parent genotypes: Dd x Dd

Gametes: D d D d

Offspring genotypes:

x	D	d
D	DD	Dd
d	Dd	dd

DD : 2Dd : dd

Offspring phenotypes: **75% dangly : 25% attached**

78. B
79. A
80. D **Cross breeding** is the type of breeding used to improve a breed of organisms through combining good traits from two different parents.
81. B The total amount of biomass in the living organisms at each stage of the food chain can be drawn to scale and shown as a **pyramid of biomass.**
82. B A **pyramid of energy** represents the energy in the producers and how much of that energy is passed on at each stage along the food chain.
83. C
84. D Cacti have leaves that have been reduced to spines with a very small surface area. Their leaves lose a tiny amount of water.

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85. A **Desert animals** often have large, thin ears as well to increase their surface area for losing heat.

86. B *Canis simensis* is scientific name while others are common names.

87. A The **protista** are all microscopic single-celled organisms that do have a nucleus.

88. C **Gymnospermae** are plant division of **conifers** whose reproductive structures are found in cones.

89. A Kingdom Monera include all of the bacteria, as well as the blue-green algae.

90. A Secondary consumer eats the plant eater.

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ETHIOPIAN UNIVERSITY ENTRANCE EXAMINATION(EUEE)
ENGLISH, YEKATIT 2012/13/ MARCH2020/21

6.

Booklet code:54

I. Writing

DIRECTIONS: Questions 1-18 are related to different types of writing. Read each of them very carefully and choose the best answer from the four alternative answer, A-D given.

Section One: Forms of writing

DIRECTIONS: Items 1 and 2 contain extracts taken from different writings. Choose the best alternative answer.

1. A story written by a person about his/ her experience of traveling from one place to another would most likely be_____
   A. argumentative B. process C. narrative D. descriptive
2. ...His ears are turned back while his noble looking head is held high .His all black coat glistens in the late afternoon sun. this is taken from a piece of writing that is most likely :
   A. descriptive B. expository C. narrative D. argumentative

Section Two :paragraph Coherence

DIRECTIONS : When put in the correct order , sentences in questions 3-6 make up a complete paragraph .from the alternatives A-D given in each case, choose the letter that contains the correct order of the sentences .

3. a. The fastest computers can handle millions of problem in a few seconds .
   b. it is fairly a recent invention .
   c: in a few minutes a computer can perform calculations that humans would need years to complete.
   d. computers is one of the greatest advances in modern technology.
   e. computers are capable of doing extremely complicated works.
   A. a ,e, d, c, b B. b ,c, a ,e, d C. d ,b ,e, c, a D. c ,e, d, a b
4. a. John lived with his mother and father in a village on the edge of the forest.
   b. John always woke up early in the morning and went to school.
   c. He was a good boy and the helped his parents in the fields when he got home from school.
   d. Life was quiet in the village and most of the people who lived there were farmers.
   e. At night ,he usually slept very well after he had studied hard.
   A e, c, b, d. a B. a, d, b, c ,e C. b, e, a, d ,c D. a, d, c, e, b
5. a. Milk teeth loosen and fall out when a child is about 6 years old.
   b. Humans grow two sets of teeth in their lives.
   c. When a person is around 20-25, four additional back teeth grow called wisdom teeth.
   d. The first set has 20 teeth called milk teeth.

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e. The second set has 28 teeth that replace the milk teeth and are called permanent teeth.

A. b, d, a ,e c, B. a, d, e, c, b C. c ,b, d ,e, a D. b, d, e, a, c

6. a. people are used to taking aspirin whenever they feel pain.
   b. It's scientifically proven that excessive use of aspirin turns it in to a toxin.
   c. It's true that aspirin is an effective pain killer; for example, in headache cases.
   d. However, aspirin like in other medicine can be dangerously harmful.
   e. Aspirin can be a fatal poison.

A. d, a, c, b, e B. c, e, d, b, a C. b, a, d, e, c D. e, a, c, d ,b

Section Three: Letter writing

DIRECTIONS: Items 7-9 are based on letter writing .read each question carefully and choose the best answer according to the given context .

7. I am W/O Almaz and I am writing this letter to inform you that my 5 year old- son Belete will remain absent for class the next two days because of sickness. This short extract is most likely taken from a letter written to

A. a football coach C. given an invitation
B. ask information D. a school principal

8. Could you please ship it to me by mail? I will immediately transfer the cost involved to a bank in your area. This is most likely taken from

A. the body of formal letter C. the body of letter to a friend
B.the beginning of formal letter D. the ending of formal letter

9. A letter with the opening salutation 'Dear sir or Madam' most likely ends in:

A. Lots of love C. Best wishes
B. Yours sincerely D. Yours faithfully

Section Four: Word Spelling

DIRECTIONS: For items 10 and 11 , choose the alternative with a correct spelling

10. A. communication B. cooperative C. optimistic D. distinguished
11. A . earthquake B. critical C. depression D. illegible

Section Five: Word Order

DIRECTIONS: In questions 12-14, the words are Not in the proper order. When put in the right order, they make correct English sentences. From the given alternatives, choose the one with a correct order.

12. environment/a/of/ affect /can/ quality/polluted/our/a
    A. A polluted affect environment of quality can our life.
    B A polluted quality environment of our life can affect.
    C. A polluted environment can affect our quality of life.
    D. A polluted can environment affect our quality of life.
13. now/books/on/ studying/out/of/is/print/of/the /one/ best
    A. Now books on studying out of print are one of the best.

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B. Books on study out of print is now one of the best.
C. One of the best books on studying is now out of print.
D. Studying on books now is one of the best of print.

14. babies/the/people/throughout/love/world
A. people love babies throughout the world.
B. People the babies throughout love world.
C. babies world throughout love the people.
D. Throughout babies the world love people.

Section six: punctuation and capitalization

DIRECTIONS: For items15-18, choose the best alternative which is written in a correct punctuation and capitalization.

15. A. The high level symposium was organized by Ethiopian Ministry of education.
B. The high level symposium was organized by Ethiopian ministry of education.
C. The high level symposium was organized by Ethiopian Ministry of Education.
D. The high level symposium was organized by Ethiopian ministry of education.

16. A. What an incredible situation, it must make you nervous.
B. What an incredible situation; it must make you nervous.
C. What an incredible situation? It must make you nervous.
D. What an incredible situation: it must make you nervous.

17. A. The OAU. Usually meets in Addis Ababa.
B. The O.A.U usually meets in Addis Ababa.
C. The OA.U usually meets in Addis Ababa.
D. The O.A.U. usually meets in Addis Ababa.

18. A. When I get older, I will be able to drive.
B. When I get older I, will be able to drive.
C. When I get older I will be able to drive.
D. When I get older.I will be able to drive.

II. Communicative Activities

DIRICTIONS: Questions 19-44 are presented in the form of a dialogue. The part said one of the speakers is given, and a blank space is left for the other. For each of the blank spaces, four alternative answers, A-D, are given,

19. **Interviewer:**_____?
Interviewee: I would like to be a specialized worker.
A. Could you begin by telling me where you can see yourself in a year
B. Can you tell me your greatest weakness.
C. .Could you tell me how you handle a stress
D. Can you tell me about a difficult situation and how you overcame it

20. **Mesele:** it seems we are in a difficult time at the moment.
Kuma:_____
A. Where are we?

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B. Please arrive here as early as you can
C. We have informed them already.
D. I believe the situation will improve soon.

21. **Dibabe:**_____
Muleta: As I see it, it has both pros and cons.
A. I think globalization is a threat to developing countries like ours.
B. Globalization is relatively a recent phenomenon.
C. It is not easy to define globalization.
D. Where did you first hear the term globalization?

22. **Melaku:** Many students don't follow their lessons attentively.
Senait:_____
A. We found it interesting to watch.
B. Did you see them this morning?
C. I don't think I can help you.
D. Another problem is that they don't come to class regularly.

23. **Reservationist:** Can I have the dates for your departure and return?
Customer: Yes, I want to leave on July 27 and return on 30.
Reservationist:_____ Did you say you wanted to return on July 30 Customer: sorry I didn't make myself clear I want to return on August 30
A. That is your first tome to book a ticket
B. I' am sorry; I didn't follow what you side.
C. Do you have a flight phobia?
D. Why did you confuse me?

24. **Teacher:**_____
Student: He has gone to Nazreth to attend his sister's marriage.
A. Can you tell me where petros is sitting?
B. Do you know much about your friend?
C. Can you tell me why your friend is absent?
D. Did your friend do the assignment?

25. **Marketing manager :** our clients are not pleased with our service .what suggestion have you got?
Marketing officer_____
Marketing manager: That's an excellent idea .it'll show them our concern.
A. I'd like to give some suggestion.
B. shall we invite them to meet with us?
C. What are we going to change?
D. I don't understand why they are not happy.

26. **Meseret:**_____
Muhaba: That's a good idea. I'm really getting bored at home.
A. I'm afraid we can't get all the job done today.
B. we my go for a walk this evening. What do you think ?
C. we couldn't do anything about it

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D. Do you really think you can get along with him?

27. **Alemayehu:** can you ride a motorbike?
Sintayhu:_____.
A. No, I can ride a bicycle
B. No, I don't
C. Yes I can't ride a horse
D. Yes I, do

28. **Rawda :** Would you like something to drink?
Fatuma :_____
A. I need time to determine.
B. That is exactly what I see.
C. Interesting to be here with you.
D. I would like tea if you don't mind.

29. **Surafel :** They should know that we can't help them because we don't have the time .
Tolra :_____
A. We can find out the information from our friend.
B. We don't clearly know why he is so interested in the matter.
C. They insist on going out for dinner this evening.
D. Besides, we don't have the required resources.

30. **Teklu:** Some people turn deaf ear to the plight of others.
Digafe: _____
A. I here that you are form Bahir Dar.
B. Shell we have a break?
C. Excuse me, but I didn't understand what you were saying
D. I think we should go for a walk.

31. **Stranger :** I am having trouble with my car.
Renitent :_____.
A. That sounds grate
B. Perhaps you could go to a garage
C. Let's not forget the brand
D. At lest you should know about cars

32. **Dawit :** I like watching television in the evenings.
Fanuael : _____
A. We are all happy that shura won the marathon race .
B. For me , films are the most enjoyable pastime.
C. We do exercise every morning.
D. Television has a long history.

33. **Selam :** _____
Kidist: An Italian person named Guglielmo Marconi.
A. Where was a radio invented
B. Why was a radio invented
C. How was a radio invented
D. Who invented a radio

34. **Kasa:** What do you think of selam ?
Henok:_____
A. I feel the same way sometimes
B. She is patient in class, but on the other hand, she isn't patient in at home
C. I work with her.

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D. Alright! I will let you know if you need any information. Just ask me

35. Driver: What's the matter?
Policeman: you have violated a traffic rule ._____
A. When you see a red light, you should stop.
B. How can allow people like you?
C. I am sorry. The road is free. I failed to notice you
D. It is not my business. It is the minimum penalty.

36. Mother : _____
Son : Well I will do as you said.
A. Do you really finish your homework by your own?
B. I have put some soft drinks in the refrigerator.
C. You won't go out unless you do your homework.
D. What sort of things did you buy?

37. Abush:_____
Helen :Yes, I do.
A. Have you ever compared beauty against virtue.
B. Do you think beauty is more important than virtue
C. Can you say virtue is more important than beauty.
D. What do you think is more important? Virtue or beauty

38. Masersha: _____
Samuel : I hope so.
A. By the end of the lesson we will have all understood the topic thoroughly.
B. The teacher advised us to work very hard in preparation for the national exam.
C. What do you think about the national exam?
D. Talk to the teacher about it.

39. Tigist : Are there any flowers in your house ?
Abera :_____
A. My pleasure B. In my bedroom C. Both D. A few

40. Musema : _____?
Tiruye : Dad we always enjoy listing to him.
A. Are you parents strict or funny
B. Who is the funniest person in your family
C. Do you usually have family gathering
D. What do you like the most about your family

41. Patient :_____
Doctor : That is what I told you several times to stay health.
A. How about eating a lot of fat every day?
B. Sometimes, I don't know the best foods.
C. I should have avoided caffeine and sugar.
D. Is it okay to drink alcohol at weekends?

42. Doul: What kinds of websites are very popular?
Ochala : _____

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A. Take for example face book and twitter
B. They have famous in many countries.
C. Popular people use twitter most often
D. Many people like to use them .

43. **Merid :** Improvement in our team have been observed as a result of good trainings.
Thomase :_____.
A. it is fun to watch sports with a lot of people
B. it is because that is where the players are.
C. Football is loved by many people every where
D. I am quite sure that our team will win

44. **Gudina :** What's required to be a merchant /
Moges:_____
A. It's the best way of giving service for a small business
B. It is a good idea.
C. Having a variety of asset would be an essential matter.
D. How could you open business?

III. Comprehension

Section one: sentence Comprehension

DIRECTIONS: For items 45 and 46, choose the sentence that is closest in mining to the head sentence. Each question has for choices, A-D.

45. unlike the buses in the operation .the ones which are to be imported have significant improvements
A. both the buses in operation and those to be improved have similar improvements.
B. The buys which are to be imported are more updated than the once that are more update than the ones that are in operation.
C. Neither the buys in operation nor the ones to be improved have significant improvements.
D. The buys in operation are to be replaced by the buys imported with little improvement.

46. There is no reason why our customers can't use or service as long as we provide regularly service with quality.
A. There is a reason for customers not use our service as far as service with quality
B. There is a tendency that excellent services drive away customers.
C. Giving gunine service bring customers' satisfaction.
D. Our customers can't use our service even if we provided regular service with quality

Section two: Vocabulary

A.Substitution

DIRECTION: Questions 47-57, each has an underlined word or phrase. There are for alternative A-D given after each sentence. Choose one word or phrase that best keeps the mining of the underlined word or phrase

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47. Workers in big firms receive a substantial part of their pay in the form of attractive bonuses and overtime.
A. distinct B. considerable C. fixed D. sensitive
48. the economy requires a stable environment in order to growth rapidly
A. Precious B. successful C. solitary D. firm
49. The price of the good are known for their variability because they increase and decrease every other day.
A. evenness B. fluctuation C. transition D. certainty
50. I have to unzip my jacket because it is getting very hot.
A. undo B. zip up C. unlock D. fasten
51. as a parent, I have the discretion determine which TV program my children may watch.
A. wealth B. alternative C. right D. spirit
52. despite a cancer diagnoses, Melt would remain resilient and fight without losing hope until the very end.
A. Clam B. rude C. desperate D. strong
53. On her return from a trip, she got an accident resulting in the amputation on her right leg.
A. sickness B. fitness C. impairment D. scratch
54. He has promised his on a new bicycle for Christmas and nothing will break his resolution.
A. decision B. ability C. talent D. relation

DIRECTION: Question 55-62 are incomplete sentence. There are four alternative words or phrases, A-D, given below each question .Choose one word or phrase that best complete the sentence.

55. You can get advice from some _____students, as they have more experience.
A. junior B. underage C. ambitious D. senior
56. the disease is _____ and you can catch it easily if you are not more careful.
A. Sever B. tropical C. infection D. malnourishment
57. Never allow yourself to get discourage and think that your life is____ and can't make a change.
A. insignificant B. independent C. faithful D. worth
58. my grandfather's face was_____ with age.
A. surprised B. wrinkled C. frowned D. confuse
59. sleep disorders can cause serious _____and physiological problem.
A. mood B. difficulty C. metrological D. psychological
60. There was a significant_____ in crop production because of the shortage rain.
A. rise B. reduction C. proportion D. amount
61. He couldn't have been successful without my help, but I don't want to _____anything from his achievements.
A. take away B. take of C. take out D. take back

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62.. Applicant should have _____ experience in the relevant area to apply for the job.

A. access B. prior C. popular D. sector

DIRECTION: Read the following passage carefully and answer question 63-82 according to the information therein. Each question has for choice, A-D choose the best alternative.

Section Three: Reading passage I

1. where would modern society be without advertising? Individual advertiser might think they are just trying to sell a particular product but advertising as a whole sells us an entire lifestyle. If it weren't for advertising the whole of society would be quite different. The economy, for instance, would be plunged into a crisis without the adverts and all the publicity that fuel our desire for limitless consumption.
2. As John Berger observed in his books "Ways of seeing", all advertising conveys the same simple message: my life will be richer, more fulfilling once I make the next crucial purchase. Adverts persuade us with their image of others who have apparently been transformed and are, as a result, enabled. The purpose is to make me marginally dissatisfied with my life – not with the life of society, just with my individual life. I am supposed to imagine my self-transformed after the purchase into an object of envy for others- an envy which will then give me back my love of myself.
3. The prevalence of this social envy is a necessary condition if advertising is to have any hold on us whatsoever. Only if we have got in to the habit of comparing ourselves with others and finding ourselves lacking, will we fall prey to the power of advertising.
4. While fanning the films of our envy advertising keeps us preoccupied with ourselves, our house, our cars, our holiday and the endless line of new electronic gadgets that suddenly seem indispensable. Tension in society and problem in the rest of the world, if attended to at all quickly fade into the background. They are certainly nothing to get particularly worked up about. After all, there can't be any winners without losers. That's life.
5. People like John Berger are also not entirely over the moon about the impact that advertising and shopping have on the value of political freedom. Freedom is supposed to be highest value in our societies, but in the age of the consumer that freedom is all too readily identified with the freedom to choose between Pepsi and Coke, McDonald's and Kentucky Fried Chicken, Toyota and Ford, and people lose interest in the various political freedoms and our ability to participate in the process of exercising democratic control. There are lots of criticisms that could be made of modern democracies, But no one is going to pay much attention to them if they are more interested in becoming happy shoppers.

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6. In all these ways advertising helps to keep the whole socio economic show on the road. We are rarely aware of this because we are too busy working to earn the money to pay for the objects of our dreams_ dreams that play on the screen of our mined like the little clips of film we see in the commercial breakers.

63. The writer is more concerned about
A. the criticism made of modern democracy
B. the happiness of shoppers as a result of good advertising.
C. The lack of attention given to the criticism made of modern democracies.
D. The attention given to modern democracies

64. one benefit of advertising according to the information in the passage is that
A. consumers will buy products of their choice.
B. It prevents the economy from plummeting into crisis.
C. Adverts limit the endless desire for consumption.
D. People need publicity to fuel their needs.

65. The writer of the passage
A. describes how advertising works.
B. says that advertising should be used more carefully.
C. is critical of advertising.
D. argues against the use of advertising.

66. The main idea of paragraph 3 is that
A. The influence of advertising lies in its power to make people envious of others.
B. People compare themselves with others in order to have what they think they need and what they lack.
C. The prevalence of social envy is a necessary condition for advertising.
D. Social envy creates the need for advertising.

67. Which one of the following statements is true according to the passage?
A. Advertising adds to our existing problems.
B. Advertising advise us to give more attention to things that are of concern to us collectively.
C. New electronic gadgets are truly indispensable.
D. Advertising forces us to be less concerned about tensions in society and other social problems.

68. Political freedom is mentioned in the passage to illustrate that
A. Freedom is equated with the choice that people make between different products.
B. Societies value political freedom more than advertising.
C. Advertising and political freedom are equally valued by societies in the age of consumerism.

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D. People need political freedom in order to be able to make choices between different products.

69. In the passage, the writer says that
    A. society would have no place in the modern world without advertising.
    B. advertising is a key factor in human existence.
    C. advertising has little influence on our lifestyle.
    D. advertising is more than merely trying to sell a product.
70. The word them (in paragraph 5) refers to
    A. modern democracies B.shoppers C. criticisms D. advertisers
71. Advertising influences us by
    A. Making us desire to have what we lack but others have.
    B. Helping us transform our individual lives.
    C. Showing us how we will become richer in life.
    D. Making us compare ourselves with other members of the society.
72. The passage can be summarized as
    A. the impact of advertising on the economy.
    B. the power of advertising.
    C. advertising and consumerism.
    D. advertising and the freedom of choice.

Section Four: Reading Passage II

(1) Elizabeth Black well was the first woman to receive an M.D. degree since the Renaissance, graduating from Geneva Medical Collage, in New York State, in 1849. She supported women's medical education and helped many other women's careers. By establishing the New York Infirmary in 1857, she offered a practical solution to one of the problems facing women who were ejected from internships elsewhere but determined to expand their skills as Physicians. She also published server important books on the issue of women in medicine, including Address on the Medical Education of Women in 1864 and Medicine as a Profession for women in 1860.

(2) Elizabeth Black well was born in Bristol, England in 1821, to Hannah Lane and Samuel Black well. Both for financial reasons and because her father wanted to help abolish slavery, the family moved to America when Elizabeth was eleven years old, Her father died in 1838. As adults, his children campaigned for women's rights and supported the anti-slavery movement.

(3) In her book pioneer Work in Opening the Medical Profession to Women, Published in 1895, Dr. Black well wrote that she was initially repelled by the idea of studying medicine. She said she had" hated everything connected with the body, and could not bear the sight of a medical book..... My favorite studies were history and metaphysics, and the very thought of dwelling on the physical structure of the body and its various ailments filled me with disgust." Instead she went into

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teaching, then considered more suitable for a woman. She claimed that she turned to medicine after a close friend who was dying suggested the would have been spared her worst suffering if her physician had been a woman.

(4) Blackwell had no idea how to become a physician, so she consulted with several physicians known by her family. They told her it was a fine idea, but impossible; it was too expensive and such education was not available to women. Yet Blackwell reasoned that if the idea were a good one, there must be some way to do it, and she was attracted by the challenge. She convinced two physician friends to let her read medicine with them for a year, and applied to all the medical schools in New York and Philadelphia, She also applied to twelve more schools in the northeast states and was accepted by Geneva Medical Collage in 1847. The faculty, assuming that the all- male student body would never agree to a woman joining their ranks, allowed them to vote on her admission. As a joke, they voted "yes," and she gained admittance, despite the reluctance of most students and faculty. Two years later, in 1849.Elizabeth Blackwell become the first woman to receive an M.D. degree from an American medical school.

(5) Dr. Elizabeth Blackwell established a practice in New York City, but had few patients and few opportunities for intellectual exchange with other physicians and " the means of increasing medical knowledge which dispensary practice affords." She applied for a job as physician at the women's department of a large city dispensary, but was refused. In 1853, with the help of friends, she opened her own dispensary in a single rented room, seeing patients three afternoons a week.

73. Which one of the following paragraphs shows that Blackwell was reluctant to study medicine?
A. Paragraph4 B. Paragraph 3 C. Paragraph2 D. Paragraph 1

74. According to paragraph 3, why did Blackwell change her mind to study medicine? Because (of)
A. It was her childhood desire
B. The suggestion of her dying close friend.
C. The job was suitable for a woman .
D. She wanted to study the structures of the body.

75. What does the word physicians mean in paragraph1?
A. Nurses B. Doctors C. Medical assistances D. physics graduates

76. In paragraph 3,... she would have been spared her worst suffering if her physician had been a woman. What does this idea imply?
A. Women who study medicine were not successful.
B. Women physicians by nature are efficient.
C. The medical profession was dominated by men.
D. Men physicians were not experienced.

77. They in paragraph 4 refers to

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A.Several people B.. Black well's family C. Women D. physicians

78. The passage mainly focuses on the_____
    A. Campaign for women's right
    B. Availability of medicine education for women
    C. Social barriers women could face.
    D. Work ethics of physicians in the 19th century
79. Which one of the following statements is WRONG about Elizabeth Black well?
80. The action of the all-male student body_____
    A. Voted yes honestly to admit Blackwell in the college.
    B. Found evidence of Blackwell's reluctance to join the college.
    C. Appreciated Blackwell's ambition to be a physician.
    D. Let Black well gain admittance in Geneva Medical College.
81. What do paragraphs 4 and 5 have in common?
    A. Biological information of a woman
    B. The challenges and successes of Blackwell
    C. Opportunities one could miss in life.
    D. The admission of Blackwell into the college.
82. What contribution did Blackwell make to women interested in medicine as a profession?
    A. She supported the medical education.
    B. She helped the poor financially.
    C. She made medicines accessible.
    D. She gave free medical service.

IV. Grammer

DIRECTIONS: Questions 83-120 are incomplete sentences. There are four alternative words or phrases. A-D given below each question. Choose the word or phrase that best completes sentence.

83. The more people come to the party,_____food will be needed.
    A. He less B. the more C. the most D. the least
84. My classmate goes to school by bus_____I walk every day.
    A. Whereas B. because C. since D. so
85. Ten years have passed_____my mother died.
    A. When B. during C. since D. for
86. We have to hurry up, or we_____be able to catch the bus.
    A. May not B. oughtn't to C. shall D. do not
87. The hospital has a backup generator,_____there is a power.
    A. In the fact of B. in the case of C. in case D. instead
88. Sleep is not her friend_____the day.
    A. since B. as C. during D. while
89. If I_____her, I would learn more from her.
    A. met B. would have met C. have met D. meet
90. I_____I studied harder in school.
    A. will wish B. would wish C. am wishing D. wish

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91. I will _____a decision next week about whether or not to buy a ring.
A. Be done B. be made C. make D. do

92. It is a matter of time._____that good thing will come and when it comes, it will be a surprise.
A. We can't be certain
B. I have a doubt
C. There is no doubt
D. There is no use

93. Peter has reorganized his files_____he can easily find what he is looking for.
A. so that B. such that C. that is D. at that

94. Biology_____is very interesting to study.
A. ,which is my favorite subject,
B. which is my favorite subject
C. that is my favorite subject
D. ,that is my favorite subject

95. **Thomas:** Constant use of social media leads to low reading habits.
Gebre:_____.
A. I don't have reading habits
B. I have the right to say that
C. I agree with you entirely
D. I won't say that

96. The committee_____the issue thoroughly at the meeting yesterday.
A. discusses B. is discussed C. was discussed D. discussed

97. It took us_____hour and a half to arrive at the tourist site.
A. for B. the C. a D. an

98. I went shopping;_____, my sister cleaned the house.
A. during B. the time being C. meanwhile D. in a while

99. The exam _____at 8:00 a.m. tomorrow morning.
A. is started B. starting C. is starting D. starts

100. Speaking on the phone, if you press the red button, the call_____.
A. ends B. ended C. would end D. is ending

101. Many accidents occur_____drivers are often reckless.
A. Because B. however C. So that D. despite

102. The professor said that he_____in a university
A. Would taught B. had taught C. will teach D. is teaching

103. **Ababe:**_____car was blocking the street?
Tesfaye: It was my uncle's car.
A. Who B. Whose C. Whom D. What

104. _____she_____that soup without reading a recipe?
A. Has/taste B. Has/tasted C. Has/made D. Has/make

105. I kept on looking for a sign, but I_____see anything for it was so dark.
A. might B. couldn't C. may D. can't

106. She wondered if_____ a new laptop.
A. You have been bought
B. You had bought
C. I have been bought
D. I had bought

107. You_____try paying them more money to get the job done.
A. are to B. were to C. have to D. should

108. _____telephones first appeared they were called talking telegraphs.

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A. When B. Next C. Then D. Later on

109. He_____the accident if he had been more careful.
A. Would have avoided C. avoided
B. will avoid D. would avoid

110. The hens_____plenty of eggs in there weeks' time.
A. have laid C. have been lied
B. have been laid D. have lied

111. I will raise your case_____I see the manager.
A. Immediate B. for C. when D. since

112. I whish I_____to your advice about spending my money wisely.
A. Would listen B. had listened C. have listened D. listen

113. Meron decided_____home as soon as possible.
A. go B. to going C. to go D. going

114. You_____get professional advice before buying very expensive materials.
A. had B. ought C. would better D. had better

115. I finished work,_____to the beach and found a nice place to swim.
A. was walking B. is walking C. walked D. walks

116. The whole story_____by the kids in a few days.
A. was forgotten B. had forgotten C. forgets D. forgot

117. I can't sleep! That noise is really_____.
A. annoyance B. annoying C. annoyed D. annoy

118. You wouldn't have lost your job if you _____your boss.
A. haven't insulted C. don't insult
B. hadn't insulted D. didn't insult

119. I think she did an excellent job on her presentation.
A. neither do I B. neither did I. C. So did I D. So do I.

120. The marketing plan_____I was talking about was the one that may be implemented recently.
A. would B. Can C. which D. how

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1. C That is retelling a story.
2. A This piece of writing is used to describe the person or create mental picture.
3. C The sentence given under 'd' can be the topic sentence and the correct order is d, b, e, c, a.
4. B Those five sentences should be rearranged as a, d, b, c, and e to make up a complete paragraph.
5. A The paragraph talks about the two sets of teeth, the milk teeth and the permanent teeth.
6. D As asprin is effective painkiller, it is also harmful if taken excessive.
7. D This extract is taken from a letter written to school principle to ask for permission.
8. D This is most likely taken from the ending part of letter of order (business letter)
9. D If the greeting begins with Dear sir/madam, the right closing would be yours faithfully.
10. B Comunication ×
    Optmistic ×
    distnguished ×
    communication✓
    optimistic✓
    distinguished✓
11. A crtical ×
    deprsion ×
    illegble ×
    critical✓
    depression✓
    illegible✓
12. C Noun phrase+HV+MV+Obj(noun phrase)
13. C one of the best books on studying is now out of print.
14. A S+V+Complement
    People love babies throughout the world.
15. C Name of international organizations, governments and ministries are written in capital letter.
16. A comma is used to separate introductory phrases.
17. D Abbreviations of names of international organization us full stop.
18. A When a subordinate clause comes at the beginning of a complex sentence, it is separated by comma.
19. A Asking about the person's plan for the year.
20. D The person's feeling or opinion about the situation of that time.
21. A That is to mean it has both advantages and disadvantages.
22. D Giving additional opinion on what has been said before.
23. B Asking for clarification.
24. C Asking the reason of the absence.
25. B To express the person's suggestion concerning what is to be done.
26. B The person's comment to relax.
27. No it should be either yes, I can or No, I can't
28. D Inviting some one to have something.

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29. D That is to mean in addition.
30. C Asking for clarification
31. B Suggestion to find solution to the problem
32. B That is telling what he enjoys most while watching Tv
33. D Asking about a person who invented a radio.
34. B That is to show the person's personality (describing the person)
35. A Explaining the traffic violated.
36. C Giving order to the child.
37. B Asking for someone's choice.
38. A Expressing the speaker's expectation at the end of the lesson.
39. D To show as there are a few flowers in the house.
40. B Asking who the funniest person is in the family.
41. C To show what was necessary to be done.
42. A To mention some examples of popular web sites.
43. D The person is sure about the success of the team due to good training.
44. C Explaining what is needed to begin business.
45. B
46. C
47. B considerable –fair
48. D firm-not liked to change.
49. B fluctuation-changeable
50. C unlock-unzip means unfasten or open by pulling a zip.
51. C right-discretion means judge and decide for oneself.
52. D strong-resilient means getting strong and a bit powerful.
53. D scratch-cut off on arm or leg by surgery
54. A decision
55. D senior- who have been of higher level students.
56. A Severe- strict and causes serious sickness.
57. D worth- having a certain value.
58. B wrinkled- small fold or line of the skin on the face of a person because of age.
59. D psychological- a problem of the mind or related to the mind.
60. B reduction –decreasing
61. C takeout – obtaining
62. B prior- former
63. B
64. B para 1 line 5
65. C
66. B para 3 line B
67. B
68. B
69. D
70. C para 5 line 8

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71. B
72. B
73. B
74. B para 3 line 8-19
75. C para 1 line 7
76. C para 3 line 8
77. D
78. C
79. D
80. D
81. B
82. A
83. B parallel comparison- we use the + double Comparative (the more..... the more.....)
84. A To present two different /opposite/ ideas to mean on the contrary.
85. C Since + starting time(beginning from that time upto now)
86. A That is to mean other wise there is less probability to catch the bus.
87. C To mean may be if the condition of power interruption happens.
88. C During + now (flood, war, peace, election etc)
89. A In conditional II/ Improbable or unlikely condition/ the if clause would be simple past, met.
90. D Expressing wish- present wish which is " I don't study harder " is expressed.
91. C make- decision, agreement , arrangement etc.
92. C That is to mean it is exactly true.
93. A It expresses for what purpose or what for he reorganized his documents.
94. A Non- defining relative clause is used to give extra information and separated by commas.
95. C Expressing agreement to mean I completely agree.
96. D The simple past tense expresses short quickly completed action in the past i.e. yesterday.
97. D Indefinite article 'an' uses before singular noun beginning with 'h' when it is not sounded e.g. an hour, an heir, an honest etc.
98. C mean while means during the time or between.
99. D Simple present is used to express action scheduled to take place at point of time in the future.
100. A It is conditional truth and the main clause is in simple present.
101. A Because, since and as are conjunctions which show the idea expressing cause or reason of an event.
102. B It is how- the professor said," I taught in a university."
103. B Wh / information question about possession uses relative pronoun 'whose '
104. C. present perfect question is formed using has / have +s+v3.

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105. B. It was not possible for me to see any sign because of the darkness.
106. D Reporting question- First the direct question is rearranged the statement form and the simple past is changed to the past perfect tense.
107. D should expresses personal suggestion.
108. A When- means by the time, shows paint of time.
109. A Conditional III- impossible condition in the past is using perfect conditional(would+ have+v3) in the main cluse.
110. A That is to mean the hens have produced eggs. Lay (present) and laid (past participle)
111. C When- by the time or the moment I see the manager.
112. B past with- where the reality was "I didn't listen." Is changed to I has listened.
113. C Decide- is a verb followed by infinitive with "to"
114. D Expresses advice- should, ought to and had better.
115. C List of past actions in sequence are all expressed used the simple past.
116. A Past passive voice to express what was done.
117. B present participle (the ing adjective) describes the nature or characteristic of someone or something S+V+ing adjective.
118. B Conditional III expresses a condition which was totally impossible in the past, past perfect tense is used.
119. C short addition to avoid unnecessary repetition. i.e positive addition to positive remark with SO+ HV+ subject.
120. C Relative pronoun 'which' is used for a thing

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ETHIOPIAN UNIVERSITY ENTRANCE EXAMINATION (EUEE)
MATHEMATICS FOR NATURAL SCIENCE
YEKATIT 2012/13; MARCH 2020/21
BOOKLET CODE: 48

1. A Geometric Series with first term a and common ratio r is convergent if
   A. $r \geq 1$ B. $r > 1$ C. $r \leq 1$ D. $r < 1$

2. Which one of the following is an upper bound of the sequence $\{\frac{n+1}{n}\}_{n=1}^{\infty}$?
   A. 1 B. 0 C. 2 D. 4/3

3. The general term of a sequence is given by $a_n = (\frac{1}{2})^{1-n}$. Then, which one of the following is true about this sequence?
   A. 0 is one of its terms.
   B. Its terms are non-positive.
   C. Its terms are odd numbers.
   D. Its terms are natural numbers.

4. The distance between point (4, -3) and the line $\ell: x + y - 7 = 0$ is
   A. $4\sqrt{2}$ B. $2\sqrt{2}$ C. $\frac{3\sqrt{2}}{2}$ D. $3\sqrt{2}$

5. Let $f(x) = \frac{x^2 - x}{2x^2 - x - 1}$. Which one of the following is **NOT** true about f and its graph?
   A. The line $x = -\frac{1}{2}$ is a vertical asymptote.
   B. Graph of f has hole at a point $(1, \frac{1}{3})$.
   C. f is an even function.
   D. The line $y = \frac{1}{2}$ is a horizontal asymptote.

6. Which one of the following statements is NOT true?
   A. When a variance of a data is close to zero, the data shows less variability.
   B. Variance indicates the variability of a set of numerical data items.
   C. If two data have equal mean, the data with less variance shows less variability.
   D. The large value of variance shows that the data values are closer to the mean.

7. The following is the frequency distribution of a grouped data.

Class Intervals	Frequencies (f_i)
3 – 7	2
8 – 12	2
13 – 17	10
18 – 22	6

   What is the mean of the data?
   A. 15 B. 14 C. 12.5 D. 13

8. If p: "the rainy season is very good this year" and q: "rivers are rising", then the statement "It is not true that neither the rainy season is very good this year nor rivers are rising" is denoted by:

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A. $(\neg p \Rightarrow \neg q) \vee \neg q$
B. $(p \Rightarrow q) \vee (\neg p \Rightarrow \neg q)$
C. $\neg p \wedge \neg q$
D. $\neg(\neg p \vee \neg q)$

9. $\lim_{x\to\infty} (\frac{x-2}{x})^{-2x+1} =$ _____.
A. e^4
B. e^{-4}
C. e^2
D. e^{-2}

10. Given $0 < a_n < \frac{1}{n}$ for each $n = 1, 2, 3, \ldots$, which one of the following is true?
A. $\lim_{n\to\infty} a_n < 0$
B. $\lim_{n\to\infty} a_n = 0$
C. $\lim_{n\to\infty} a_n$ does not exist
D. $\lim_{n\to\infty} a_n > 0$

11. Let $f(x) = x^2 + 2x + 3$. For what a number $c \in (1, 3)$, $f'(c) = \frac{f(3)-f(1)}{2}$?
A. -6
B. 6
C. -2
D. 2

12. For what value of c the conclusion of Rolle's Theorem is satisfied for the function $f(x) = 2x - x^2 - x^3$ on $[-2, 1]$?
A. $\frac{-1\pm\sqrt{5}}{3}$
B. $\frac{1\pm\sqrt{7}}{3}$
C. $\frac{1\pm\sqrt{5}}{3}$
D. $\frac{-1\pm\sqrt{7}}{3}$

13. Let $z_1 = 2+i$, $z_2 = 3+i$ and $z_3 = 1-i$. What are the real and imaginary parts of $z_1 z_2 z_3$ respectively?
A. 6 and 3
B. -6 and -3
C. -1 and 1
D. 10 and 0

14. What are the values of a and b if $\frac{a+2i}{3i} = \frac{b+i}{4}$?
A. $a = 3/4, b = -8/3$
B. $a = 3/4, b = 8/3$
C. $a = -3/4, b = 8/3$
D. $a = -3/4, b = -8/3$

15. The table shown below is a simple frequency distribution of data with variable x.

x	1	3	4	5	7
Frequency	2	5	6	5	2

What is the variance of the data?
A. 3
B. $\sqrt{3}$
C. 3.2
D. 2.3

16. The second derivative of the function $f(x) = xe^{-x}$ is _____.
A. $-xe^{-x}$
B. $(2-x)e^{-x}$
C. $(x-2)e^{-x}$
D. $(x+2)^2 e^{-x}$

17. If p and q are propositions, then which one of the following pairs of compound propositions are equivalent?
A. $p \Rightarrow \neg q$ and $\neg p \wedge q$
B. $\neg(p \wedge q)$ and $\neg p \vee q$
C. $p \Leftrightarrow q$ and $\neg p \Leftrightarrow \neg q$
D. $\neg p \Rightarrow q$ and $q \Rightarrow \neg p$

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18. The volume of a cube is increasing at a rate of 9 cm^3/sec. How fast is the surface area increasing when the length of the edge is 10 cm?

A. 3.6 cm^2/sec C. 90 cm^2/sec

B. 36 cm^2/sec D. 6 cm^2/sec

19. The product of two positive numbers is 100 such that the sum of two times the first number and eight times the second number is minimum. Which one of the following pairs of numbers are the first and the second numbers respectively?

A. 25 and 4 B. 20 and 5 C. 1 and 99 D. 50 and 2

20. The maximum profit that a company can make if the profit function is given by $P(x)=36+72x-18x^2$ is __.

A. 124 B. 31 C. 2232 D. 108

21. If $f(x)=\begin{cases} k(2^x), \text{if } x>2 \\ x^3-x+1, \text{ if } x\leq 2 \end{cases}$, then for what value of k is f continuous at $x=2$?

A. 7/4 B. 4 C. 2 D. 7/2

22. Which one of the following is equal to $\sin^{-1}(\sin\frac{3\pi}{4})$?

A. $\frac{\pi}{4}$ B. $\frac{5\pi}{4}$ C. $-\frac{\sqrt{2}}{2}$ D. $\frac{\sqrt{2}}{2}$

23. Which one of the following is true about $f(x)=-2+3\sin(\frac{1}{2}x+3)$?

A. The period is 2π C. The range is [-5, 1]

B. The Amplitude is 1 D. The phase shift is 6

24. If the 1st and 4th terms of a geometric progression are 3 and 81 respectively, what is the common ratio of this progression?

A. 1/3 B. 9 C. 1/9 D. 3

25. If the inverse of the matrix $\begin{pmatrix} x & 2x \\ -x & x \end{pmatrix}$ is $\begin{pmatrix} -1 & -2 \\ 1 & 1 \end{pmatrix}$, then which one of the following is the value of x?

A. -1 B. 1 C. -1/3 D. 1/3

26. What is the set of values of k so that the system $\begin{cases} x-y+z=3 \\ 2x+y-z=4 \\ kx-2y-z=5 \end{cases}$ has a solution?

A. $\mathbb{R}$ B. $\mathbb{R}\setminus\{1\}$ C. $\mathbb{R}\setminus\{-1\}$ D. $\mathbb{R}\setminus\{-1, 1\}$

27. Let $A=\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B=\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$. Then which one of the following is equal to $3A+2B$?

A. $\begin{pmatrix} 7 & 4 \\ 7 & 18 \end{pmatrix}$ B. $\begin{pmatrix} 7 & 1 \\ 7 & 7 \end{pmatrix}$ C. $\begin{pmatrix} 3 & 1 \\ 2 & 7 \end{pmatrix}$ D. $\begin{pmatrix} 3 & 4 \\ 2 & 18 \end{pmatrix}$

28. Which one of the following is an interval at which the function $f(x)=\frac{1}{x-1}-\sqrt{4-x^2}$ is differentiable?

A. $(-2, 1)\cup(1, 2)$ C. $(-\infty, -2)\cup(2, \infty)$

B. $(-2, 2)$ D. $[-2, 1)\cup(1, 2]$

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29. The function $f(x)= (x-1)^{\frac{2}{3}}$ is differentiable on:

A. $[1, \infty)$
B. $(-\infty, 1]$
C. $(-\infty, -1) \cup (-1, \infty)$
D. $(-\infty, 1) \cup (1, \infty)$

30. What is the slope of the line tangent to the graph of $f(x)= x^2 + \tan x$ at a point $(\pi, f(\pi))$?

A. 2π
B. 2
C. $2\pi + 1$
D. $2\pi - 1$

31. If $f(x)=4-x^2 \sin x$, then which one of the following expressions describes $f'(x)$?

A. $-x(x \cos x + 2\sin x)$
B. $1+\cos x + 2\sin x$
C. $x(x \cos x - 2\sin x)$
D. $1 - \cos x - 2 \sin x$

32. Consider a sequence whose terms are $a_1 = 6$, and $a_{n+1} = -\frac{1}{3}a_n$ for $n \in \mathbb{N}$. The, what is the sum of the first ten terms?

A. $9(1-(-\frac{1}{3})^{10})$
B. $\frac{9}{2}(1-(-\frac{1}{3})^{10})$
C. $9(1-(-\frac{1}{3})^{11})$
D. $\frac{9}{2}(1-(-\frac{1}{3})^{11})$

33. Which one of the following is a row reduced echelon form of the matrix

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 4 & -1 & 4 \end{pmatrix}?$$

A. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$
B. $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}$
C. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
D. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

34. For what value of k does the system $\begin{cases} 2x - y + 4k = 6 \\ kx - y - z = 2 \\ x + y + 2z = 8 \end{cases}$ has a unique solution?

A. $k \neq -\frac{1}{2}$
B. $k \neq -\frac{1}{3}, \frac{1}{2}$
C. $k \neq -\frac{1}{6}$
D. $k \neq \frac{1}{2}$

35. Let $A = \begin{pmatrix} 2 & 3 \\ 2 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 5 & -4 \\ 3 & 2 & 6 \end{pmatrix}$. Then, $AB =$ ____.

A. $\begin{pmatrix} 8 & 13 & -18 \\ -1 & 8 & -2 \end{pmatrix}$
B. $\begin{pmatrix} 13 & 1 \\ 16 & 8 \\ 10 & -14 \end{pmatrix}$
C. $\begin{pmatrix} 8 & -1 \\ 13 & 8 \\ 18 & -2 \end{pmatrix}$
D. $\begin{pmatrix} 13 & 16 & 10 \\ 1 & 8 & -14 \end{pmatrix}$

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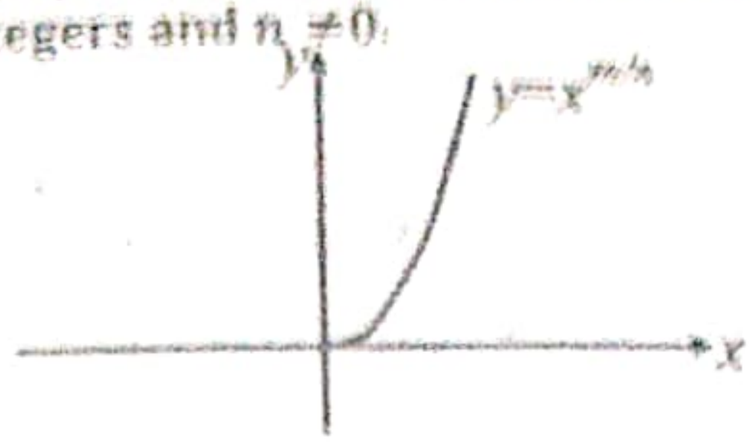
36. The following graph is the graph of the function $y = x^{\frac{m}{n}}$, where m and n are positive integers and $n \neq 0$.



Which one of the following is true about m and n?

A. m is odd, n is even and $m > n$

B. m is even, n is odd and $m < n$

C. m is odd, n is even and $m < n$

D. m is even, n is odd and $m > n$

37. Consider the experiment of rolling a die whose sample space is {1, 2, 3, 4, 5, 6}. If two dies are rolled simultaneously, what is the probability that a prime number turns up on one of the dice and a composite number on the other?

A. 1/3 B. 2/3 C. 5/6 D. 1/6

38. Consider the following frequency distribution of grouped data.

Values	Frequency
95 – 99	4
90 – 94	6
85 – 89	10
80 – 84	16
75 – 79	9
70 – 74	5

Which one of the following statements is true about this frequency distribution?

A. The class interval is 4.

B. 95.5 is the upper limit of the second class.

C. 85.5 – 90.5 is one of the class boundaries.

D. 82 is the class midpoint of the fourth class.

39. Let $f(x) = \frac{1}{(1-\sqrt{x})^2}$. Then, $f'(4)$ is equal to

A. 1/2 B. -1/2 C. 1 D. 2

40. Given the function $f(x) = 2x^2 - 3x + 1$, what is the slope of the line passing through the point $(-1, f(-1))$ and $(2, f(2))$?

A. -1 B. -1/3 C. 1/3 D. -1

41. Let $u = (2, 0)$ and $v = (-1, 3)$ be two vectors in a plane and

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$w = -3u + 2v$. Which one of the following is the unit vector in the direction of w, in terms of the standard unit vectors i and j?

A. $\frac{4}{5}i - \frac{3}{5}j$

B. $-\frac{4}{5}i + \frac{3}{5}j$

C. $\frac{3}{5}i - \frac{4}{5}j$

D. $-\frac{3}{5} + \frac{4}{5}j$

42. Which one of the following is **NOT** true about the scalar product of vectors?
   A. The scalar product is commutative.
   B. The scalar product of non-zero parallel vectors is zero.
   C. The scalar product is distributive over addition of vectors.
   D. The scalar product of perpendicular vectors is zero.
43. Which one of the following is **NOT** true about transformation?
   A. Rotation moves triangles in to congruent triangles.
   B. Translations move angles to congruent angles.
   C. Rotation is a rigid motion.
   D. Reflection is not a rigid motion.
44. If $z = -1+i$ and $w = 1-3i$, what is the simplified form of $\frac{5z}{w} + |z|^2\bar{w}$?
   A. $4i$ B. $5i$ C. $4+5i$ D. $5 + 4i$
45. A real number a is the limit of a sequence $\{a_n\}$ if
   A. a is closer to every term of the sequence.
   B. $a \leq |a_n|$ for all n.
   C. a is very close to all terms a_n from some arbitrary big n onwards.
   D. $|a_n| \leq a_n \leq |a|$ for all .
46. Which one of the following is the standard form of equation of an ellipse with vertices: (3, 1), (3, 9); and minor axis length 6?
   A. $\frac{(x-3)^2}{4} + \frac{(y-5)^2}{16} = 1$
   B. $\frac{(x+2)^2}{4} + \frac{(y-3)^2}{9} = 1$
   C. $\frac{(x-3)^2}{4} + \frac{(y-3)^2}{36} = 1$
   D. $\frac{(x-1)^2}{16} + \frac{(y-9)^2}{64} = 1$
47. What is the image of the ellipse $\frac{(x-3)^2}{6} + \frac{(y-2)^2}{4} = 1$ when it is rotated through 90^o about the origin?
   A. $\frac{(x-2)^2}{4} + \frac{(y+3)^2}{6} = 1$
   B. $4(x+3)^2 + 6(y-2)^2 = 24$
   C. $\frac{(x+2)^2}{6} + \frac{(y-3)^2}{4} = 1$
   D. $6(x+2)^2 + 4(y-3)^2 = 24$
48. Let p and q be propositions with truth values True and False respectively. Then, which one of the following is correct?
   A. $(p \vee \neg q) \Rightarrow q$ is True
   B. $\neg p \Rightarrow q$ is False
   C. $p \vee (q \wedge \neg p)$ is True
   D. $(p \Rightarrow q) \wedge p$ is True
49. Let $f(x) = \sqrt{x+2}$ and $g(x) = x^2 - 1$. What are the domain and range of the composition of f with g, $f \circ g$ respectively?
   A. $\mathbb{R}$ and $[1, \infty)$
   C. $[0, \infty)$ and $[1, \infty)$

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B. $\mathbb{R}$ and $[0, \infty)$ D. $[0, \infty)$ and $[0, \infty)$

50. Given a rational function $f(x)=\frac{2x+4}{x-1}$, which one of the following is NOT true about its graph? The graph has
A. a horizontal asymptote, $y=2$.
B. a vertical asymptote, $x=1$.
C. y-intercept (0, -4).
D. x-intercept (-4, 0).

51. $\lim_{x\to\infty}(1+\frac{5}{x})^x$ is equal to
A. $1/e$ B. e^5 C. $5e$ D. e^{-5}

52. Which one of the following is true about limit of a function?
A. Limit of a constant function is always zero.
B. Right hand and Left hand limits are the same for every function.
C. Initiatively, the limit of a function f is a y-value L to which $f(x)$ approaches as x approaches to some specified number.
D. Whenever the limit exists for a function it is the same as functional value.

53. Which one of the following is equal to $\lim_{n\to\infty}\frac{\sin n}{n^2}$?
A. -1 B. 1 C. π D. 0

54. If the functions f and g are continuous at $x=c$, then which one of bthe following combinations is continuous at $x=c$?
A. $\sqrt{g}$ B. $\frac{f}{g}$ C. f^{-1} D. f^2

55. What is the value of $\lim_{x\to 0}\frac{\sin 4x}{\sin 6x}$?
A. 3/2 B. 2/3 C. 24 D. 6

56. Which one of the following is zero for the function $f(x)=x^3-3x^2+3x-1$?
A. 0 B. 1 C. -2 D. -1

57. If $f(x)=\frac{\ln 2}{x^2}$, then $f'(x)$ is equal to
A. $\frac{1}{2x^2}$ B. $\frac{-2\ln 2}{x^3}$ C. $\frac{-2}{x^4}$ D. $\frac{1+2\ln x}{x^3}$

58. Let $x^2+y^2+kx=1$ be equation of a circle for some $k\in\mathbb{R}$. What is the radius of the circle if its center is at (2, 0)?
A. 5 B. $\sqrt{5}$ C. $\sqrt{2}$ D. 2

59. The equation of a parabola is given by $y^2+6y+8y+25=0$. Which one of the following is NOT true about this parabola?
A. Its vertex is (-2, -3)
B. Its directrix is the y-axis
C. It is open to the right
D. Its focus lies at (-4, -3)

60. What is the slope of the line tangent to the graph of $f(x)=x^2+\tan x$ at $(\pi, f(\pi))$?

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A. 2π B. 2 C. $2\pi+1$ D. $2\pi-1$

61. Suppose the measurement of height (in meter) of twelve students is:
1.72, 1.65, 1.70, 1.56, 1.72, 1.70, 1.65, 1.70, 1.65, 1.65, 1.70, 1.72
Which one of the following is true about the mode of the data?
A. The data is Unimodal with mode 1.65.
B. The data is Unimodal with mode 1.65.
C. The data is bimodal with modes 1.65 and 1.70.
D. The data is multimodal.

62. Which one of the following is correct about $f(x)=\cot x$?
A. Its range is $(-\infty, 1]\cup[1, \infty)$.
B. Its domain is $\{x \in \mathbb{R}: x \neq n\pi, n \in \mathbb{Z}\}$.
C. Its domain is $\{x \in \mathbb{R}: x \neq \frac{n}{2}\pi, n \in \mathbb{Z}\}$.
D. Its period is 2π.

63. Let $f(x)=3-2x$. What is the inverse of f?
A. $f^{-1}(x)=2x-3$ C. $f^{-1}(x)=\frac{3}{2}-\frac{x}{2}$
B. $f^{-1}(x)=3+2x$ D. $f^{-1}(x)=\frac{x}{2}-\frac{3}{2}$

64. Let $\{a_n\}$ be a sequence of real numbers. Then, which one of the following is true about this sequence?
A. M is an upper bound of $\{a_n\}$ if $M \leq a_n$ for all a_n.
B. m is a lower bound of $\{a_n\}$ if $a_n \leq m$ for all a_n.
C. The sequence $\{a_n\}$ is bounded if it has both an upper and a lower bound.
D. The sequence $\{a_n\}$ is bounded if there exists $k>0$ such that $k \leq |a_n|$ for all a_n.

65. Which of the following is lower bound for the sequence $\{\frac{n}{n-1}\}_{n=2}^{\infty}$?
A. 9/5 B. 1 C. 7/4 D. 5/3

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, For Geometric Sequence $(G_n)_{n\in\mathbb{N}}$ with first term G_1 and common ratio r, the Geometric Series

$$S_n = \sum_{k=1}^{n} G_k \ \left(\text{i.e. } S_n = \frac{G_1(1-r^n)}{1-r}\right)$$

, converges to $\frac{G_1}{1-r}$ as $n \to \infty$ if $|r| < 1$, i.e. $-1 < r < 1$

Answer: No Answer

By Listing: Initial terms for $a_n = \frac{n+1}{n}$ are 2, 3/2, 4/3, 5/4, ...

And since $\frac{n+1}{n} = 1 + \frac{1}{n}$, the sequence goes to 1 because $\frac{1}{n}$ goes to 0 as $n \to \infty$

Since 2, 3/2, 4/3, 5/4, ..., 1 is strictly decreasing, it shows that $a_1 = 2$ is the largest term so that every real number greater than or equal to 2 is an upper bound of $(\frac{n+1}{n})_{n=1}^{\infty}$.

By Algebra: Note that $a_n = \frac{n+1}{n} = 1 + \frac{1}{n}$ so that

$$n \geq 1 \Rightarrow 0 < \frac{1}{n} \leq 1 \Rightarrow 1+0 < 1+\frac{1}{n} \leq 1+1 \Rightarrow 1 < a_n \leq 2$$

Thus, 2 is the largest value of a_n so that every real number greater than or equal to 2 is an upper bound of $(\frac{n+1}{n})_{n=1}^{\infty}$.

Answer: C

3. By the exponents rule $(\frac{a}{b})^m = (\frac{b}{a})^{-m}$, we have that

$$a_n = (\tfrac{1}{2})^{1-n} = 2^{-(1-n)} = 2^{n-1}, \text{ i.e. } a_n = 2^{n-1}$$

Thus, listing some terms of the sequence as:

$2^{1-1}, 2^{2-1}, 2^{3-1}, 2^{4-1}, \ldots$, i.e. 1, 2, 4, 8, 16,...

, clearly the terms are positive (so A and B are False), containing both odd and even natural numbers (so C is False) and elements of set of natural numbers (so D is True).

Answer: D

4. The distance between a point (h, k) and a line ℓ: $Ax + By + C = 0$ in a coordinate plane is given by: $d = \frac{|Ah+Bk+C|}{\sqrt{A^2+B^2}}$.

Hence, for point (4, -3) and the line ℓ: $x + y - 7 = 0$, we have $h = 4$, $k = -3$, $A = 1$, $B = 1$, and $C = -7$ so that

$$d = \frac{|Ah+Bk+C|}{\sqrt{A^2+B^2}} = \frac{|1(4)+1(-3)+(-7)|}{\sqrt{1^2+1^2}} = \frac{|-6|}{\sqrt{2}} = \frac{6}{\sqrt{2}} = 3\sqrt{2}.$$

Answer: B

5. $f(x) = \frac{x^2+x}{2x^2+x-1} = \frac{x(x+1)}{(2x-1)(x+1)} = \frac{x}{2x-1}$, $x \neq -\frac{1}{2}, 1$

A in simplest form f is given as: $f(x) = \frac{x}{2x-1}$, where the denominator $2x - 1$ is 0 if $x = \frac{1}{2}$

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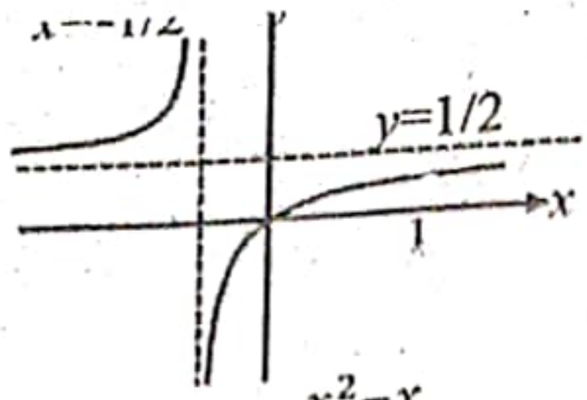

$$f(x)=\frac{x^2-x}{2x^2-x-1}$$

which shows the line $x=-\frac{1}{2}$ is a vertical asymptote of graph of f, i.e. as shown in the figure:

$f \to +\infty$, as $x \to -\frac{1}{2}^-$, and

$f \to -\infty$, as $x \to -\frac{1}{2}^+$

Or using calculus: $\lim\limits_{x\to-\frac{1}{2}^-}\frac{x}{2x+1}=+\infty$

Therefore, the line $x=-\frac{1}{2}$ is a vertical asymptote of graph of f so that A is true.

B. The factor $x-1$ is completely cancelled out in denominator in simplest form so that graph of f has hole at $x=1$, i.e. at the point the point $(1,\frac{1}{3})$.

Or using calculus: $\lim\limits_{x\to1}\frac{x}{2x+1}=\frac{1}{3}$ but $1\notin$ Domain of f

Therefore, graph of f has hole at a point $(1,\frac{1}{3})$ so that B is true.

C. $f(-x)=\frac{-x}{2(-x)+1}=\frac{-x}{-2x+1}\neq\frac{x}{2x+1}=f(x)$, i.e. $f(-x)\neq f(x)$ which implies f is not even function so that C is NOT true

D. The line $y=\frac{1}{2}$ is the Horizontal Asymptote of graph of f, because Degree of (x)=1=Degree of $(2x+1)$ so that

$$y=\frac{\text{Leading coafficient of }(x)}{\text{Leading coafficient of }(2x+1)}=\frac{1}{2}$$

Or using calculus: $f(x)=\frac{x}{2x+1}=\frac{1}{2}-\frac{1}{4x+2}$ so that $f\to\frac{1}{2}$ as $x\to\pm\infty$, i.e. since $\frac{1}{4x+2}\to0$ as $x\to\pm\infty$ we have that

$$\lim_{x\to\pm\infty}\frac{x}{2x+1}=\lim_{x\to\pm\infty}\left(\frac{1}{2}-\frac{1}{4x+2}\right)=\frac{1}{2}$$

Therefore, the line $y=\frac{1}{2}$ is a Horizontal Asymptote of graph of f so that D is true.

Answer: C

6. Variance is a non-negative value which measures and indicates the variability or dispersion or scattering of values of items in a data. Hence, the smaller (i.e. approaching 0) the value of variance the closer data items' values to the mean (and also to each other) so that the statement only in (D) is NOT true.

Answer: D

7. The distribution with class marks is as in table below.

Class Intervals	Class marks, x_i	Frequencies, f_i	f_ix_i
3 – 7	5	2	10
8 – 12	10	2	20
13 – 17	15	10	150
		250	

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18 – 22	20	6	120
SUM		$\sum_{i=1}^{4} f_i = 20$	$\sum_{i=1}^{4} f_i x_i = 300$

Then, from the results in bottom row: $\bar{x} = \frac{\sum_{i=1}^{4} f_i y_i}{\sum_{i=1}^{4} f_i} = \frac{300}{20} = 15.$

nswer: A

Given p: "the rainy season is very good this year" and

q: "rivers are rising"

Then, "It is not true that neither the rainy season is very good this year nor rivers are rising"

$\equiv$"It is not true that the rainy season is not very good this year and rivers are not rising"

$\equiv$"It is not true that $(\neg)$ the rainy season is not very good this year $(\neg p)$ *and* $(\wedge)$ rivers are not rising $(\neg q)$"

$\equiv \neg (\neg p \wedge \neg q)$

$\equiv p \vee q$, by De Morgan's Law

Therefore, "It is not true that neither the rainy season is very good this year nor rivers are rising" $\equiv p \vee q$

A. $(\neg p \Rightarrow \neg q) \vee \neg q \equiv (\neg(\neg p) \vee \neg q) \vee \neg q$, Implication Equiv.

$\equiv (p \vee \neg q) \vee \neg q$, Negation Equiv.

$\equiv p \vee (\neg q \vee \neg q)$, Associativity

$\equiv p \vee \neg q$

But, $p \vee \neg q \not\equiv p \vee q$

B. $(p \Rightarrow q) \vee (\neg p \Rightarrow \neg q)$

$\equiv (\neg p \vee q) \vee (p \vee \neg q)$, Implication Equiv.

$\equiv (q \vee \neg p) \vee (p \vee \neg q)$, Commutativity.

$\equiv q \vee ((\neg p \vee (p \vee \neg q))$, Associativity.

$\equiv q \vee ((\neg p \vee p) \vee \neg q)$, Associativity.

$\equiv q \vee (T \vee \neg q)$, By $\neg p \vee p \equiv T$.

$\equiv q \vee T$, By $T \vee Q \equiv T$.

$\equiv T$, By $Q \vee T \equiv T$.

But, $p \vee q$ need not be T, true.

C. $\neg p \wedge \neg q \equiv \neg(p \vee q) \not\equiv p \vee q$

D. $\neg(\neg p \vee \neg q) \equiv p \wedge q \not\equiv p \vee q$

iswer: No Answer

Recall that $\lim_{x \to \pm\infty} (1 + \frac{1}{x})^x = e$. Then, $\frac{x-2}{x} = 1 + \frac{-2}{x}$ so that let $\frac{-2}{x} = \frac{1}{y}$. Then, $x = -2y$ and $y \to -\infty$ as $x \to \infty$ so that

$$\lim_{x \to \infty} (\frac{x-2}{x})^{-2x+1} = \lim_{x \to \infty} (1 + \frac{-2}{x})^{-2x+1}$$

$$= \lim_{y \to -\infty} (1 + \frac{1}{y})^{-2(-2y)+1}$$

$$= \lim_{y \to -\infty} (1 + \frac{1}{y})^{4y+1}$$

$$= \lim_{y \to -\infty} (1 + \frac{1}{y})^{4y} (1 + \frac{1}{y})^{1}$$

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$$= \lim_{y\to+\infty}(1+\tfrac{1}{y})^{4y} \lim_{y\to-\infty}(1+\tfrac{1}{y})^{1}$$
$$= \lim_{y\to-\infty}((1+\tfrac{1}{y})^{y})^{4}, \text{ by } \lim_{y\to-\infty}(1+\tfrac{1}{y})^{1}=1$$
$$= (\lim_{y\to-\infty}(1+\tfrac{1}{y})^{y})^{4} = e^{4}$$

Answer: A

10. Recall the following theorem

Sandwich or Squeezing Theorem

"If $b_n \le a_n \le c_n$ for each n =1, 2, 3,....., and $\lim_{n\to\infty} b_n = L = \lim_{n\to\infty} c_n$, then $\lim_{n\to\infty} a_n = L$."

Thus, for $b_n = 0$ and $c_n = \frac{1}{n}$ since $b_n < a_n < c_n$ implies $b_n \le a_n \le c_n$, given $0 < a_n < \frac{1}{n}$ for each n =1, 2, 3,... implies $0 \le a_n \le \frac{1}{n}$ for each n =1, 2, 3,..., and $\lim_{n\to\infty} 0 = 0 = \lim_{n\to\infty} \frac{1}{n}$

Therefore, by *Squeezing Theorem*: $\lim_{n\to\infty} a_n = 0$.

Answer: B

11. $f(x)= x^2 +2x +3 \Rightarrow f'(x)=2x +2$ so that $f'(c)=2c +2$.

Then, for $f(3)= 3^2 +2(3)+3= 18$ and $f(1)= 1^2 +2(1)+3=6$

$f'(c)= \frac{f(3)-f(1)}{2} \Rightarrow 2c +2= \frac{18-6}{2} \Rightarrow 2c +2= 6 \Rightarrow c = 2 \in (1, 3)$

Answer: D

12. Given $f(x)=2x - x^2 - x^3$

i) f is continuous on [-2, 1]

ii) f is differentiable on (-2, 1) and $f'(x)=2-2x -3x^2$

iii) $f(-2)=0= f(1)$

Thus, by Rolle's Theorem there is $c \in (-2, 1)$ such that $f'(c)=0$

$\Rightarrow 2-2c -3c^2 = 0 \Rightarrow c = \frac{2\pm\sqrt{28}}{-6} \Rightarrow c = \frac{-1\mp\sqrt{7}}{3}$

Since $\frac{-1\mp\sqrt{7}}{3} \in (-2, 1)$, the value is $c = \frac{-1\mp\sqrt{7}}{3}$.

Answer: D

13. Recall: $(a + bi)(c + di)=(ac - bd)+(ad + bc)i$. Then,

$z_1z_2 = (2+i)(3+i) =(2(3)-1(1))+(2(1)+1(3))i = 5+5i$ so that

$z_1z_2z_3 =(z_1z_2)z_3 = (5+5i)(1-i)$

$= 5(1+i)(1-i)$

$= 5(1^2 + 1^2)$, by $(a + bi)(a - bi)= a^2 + b^2$

$= 10+0i$

Hence, $z_1z_2z_3 =10+0i$ implies 10 is the real and 0 is the imaginary parts of $z_1z_2z_3$.

Answer: D

14. By cross multiplication:

$\frac{a+2i}{3i} = \frac{b+i}{4} \Rightarrow 4(a +2i)=3i(b + i)$

$\Rightarrow 4a +8i =3bi +3i^2 \Rightarrow 4a +8i =3bi -3$, by $i^2 =-1$

Hence, by definition of equal complex numbers

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$4a + 8i = 3bi - 3 \Rightarrow 4a = -3$, and $8 = 3b$

$\Rightarrow a = -3/4$, and $8/3 = b$

Answer: C

15. Given a frequency distribution of data with variable x as follows.

x_i	1	3	4	5	7
Frequency, f_i	2	5	6	5	2

Then, $\sum_{i=1}^{5} f_i = 2+5+6+5+2 = 20$ and

Mean, $\bar{x} = \frac{\sum_{i=1}^{5} f_i x_i}{\sum_{i=1}^{5} f_i} = \frac{2(1)+5(3)+6(4)+5(5)+2(7)}{20} = \frac{80}{20} = 4.$

Then, build the following table of deviations.

x_i	1	3	4	5	7	**SUM**
f_i	2	5	6	5	2	$\sum_{i=1}^{5} f_i = 20$
$x_i - \bar{x}$	-3	-1	0	1	3	-
$(x_i - \bar{x})^2$	9	1	0	1	9	-
$f_i(x_i - \bar{x})^2$	18	5	0	5	18	$\sum_{i=1}^{5} f_i(x_i - \bar{x})^2 = 46$

Therefore, the variance of the data is:

$$s^2 = \frac{\sum_{i=1}^{5} f_i(x_i - \bar{x})^2}{\sum_{i=1}^{5} f_i} = \frac{46}{20} = 2.3$$

Answer: D

16. Given $f(x) = xe^{-x}$, by product rule of differentiation:

$f'(x) = (xe^{-x})'$

$= (x)'e^{-x} + x(e^{-x})'$

$= 1e^{-x} + xe^{-x}(-x)'$, by $(e^x)' = e^x$ & Chain Rule

$= e^{-x} + xe^{-x}(-1)$, by $(-x)' = -1$

$= (1 - x)e^{-x}$, factoring out e^{-x}

Hence, $f'(x) = (1 - x)e^{-x}$

Again, by product rule of differentiation

$f''(x) = (f'(x))'$

$= ((1 - x)e^{-x})'$

$= (1 - x)'e^{-x} + (1-x)(e^{-x})'$

$= (-1)e^{-x} + (1-x)e^{-x}(-x)'$, by Chain Rule

$= -e^{-x} + (1-x)e^{-x}(-1)$

$= -e^{-x} + (x - 1)e^{-x}$

$= (-1 + x - 1)e^{-x}$, factoring out e^{-x}

$= (x - 2)e^{-x}$

Hence, $f''(x) = (x - 2)e^{-x}$

Answer: C

17. We may use either truth table or rules of equivalencies with laws of operators.

A.	p	q	$\neg p$	$\neg q$	$p \Rightarrow \neg q$	$\neg p \wedge q$
	T	T	F	F	F	F
	T	F	F	T	T	F
	F	T	T	F	T	T

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	F	F	T	T	F	F

From the shaded columns we can see that the truth values are not the same so that $p \Rightarrow \neg q$ and $\neg p \wedge q$ aren't equivalent.

B.

p	q	$\neg p$	$\neg q$	$p \wedge q$	$\neg (p \wedge q)$	$\neg p \vee q$
T	T	F	F	T	F	T
T	F	F	T	F	T	F
F	T	T	F	F	T	T
F	F	T	T	F	T	T

From the shaded columns we can see that the truth values are not the same so that $\neg (p \wedge q)$ and $\neg p \vee q$ aren't equivalent.

C.

p	q	$\neg p$	$\neg q$	$p \Leftrightarrow q$	$\neg p \Leftrightarrow \neg q$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

From the shaded columns we can see that the truth values are all the same so that $p \Leftrightarrow q$ and $\neg p \Leftrightarrow \neg q$ are equivalent.

D.

p	q	$\neg p$	$\neg q$	$\neg p \Rightarrow q$	$q \Rightarrow \neg p$
T	T	F	F	T	F
T	F	F	T	T	T
F	T	T	F	T	T
F	F	T	T	F	T

From the shaded columns we can see that the truth values are not the same so that $\neg p \Rightarrow q$ and $q \Rightarrow \neg p$ aren't equivalent.

Answer: C

18. Let the edge lengths, the total surface area, and the volume of the cube be x, A, and V respectively. We need $\frac{dA}{dt}$. Then,

$$A(x)=6x^2, \quad \text{and} \quad V(x)= x^3 \Rightarrow \frac{dA}{dx} = 12x, \quad \text{and} \quad \frac{dV}{dx} =3x^2$$

Then by Cain Rule on $A(x)=6x^2$ and $V(x)= x^3$ we have

$$\frac{dA}{dt} = \frac{dA}{dx}\frac{dx}{dt} \Rightarrow \frac{dA}{dt} =12x\frac{dx}{dt} \Rightarrow \frac{dx}{dt} = \frac{1}{12x}\frac{dA}{dt} \quad \ldots\ldots(1)$$

$$\frac{dV}{dt} = \frac{dV}{dx}\frac{dx}{dt} \Rightarrow \frac{dV}{dt} =3x^2\frac{dx}{dt} \Rightarrow \frac{dx}{dt} = \frac{1}{3x^2}\frac{dV}{dt} \quad \ldots\ldots(2)$$

Combining (1) and (2),

$$\frac{1}{12x}\frac{dA}{dt} = \frac{1}{3x^2}\frac{dV}{dt} \Rightarrow \frac{dA}{dt} = \frac{4}{x}\frac{dV}{dt}$$

Thus, given $\frac{dV}{dt} = 9\ \text{cm}^3/\text{sec}$ and $x = 10\ \text{cm}$

$$\frac{dA}{dt} = \frac{4}{x}\frac{dV}{dt} \Rightarrow \frac{dA}{dt} = \frac{4}{10\ \text{cm}} \times 9\ \text{cm}^3/\text{sec} = 3.6\ \text{cm}^2/\text{sec}$$

Therefore, the surface area is increasing at a rate:

$$\frac{dA}{dt} = 3.6\ \text{cm}^2/\text{sec} \qquad \textbf{Answer: A}$$

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19. Let the first number be x and the second number be y. Then, the sum of two times the first number $(2x)$ and eight times the second number $(8y)$ is $S(x,y) = 2x + 8y$ and we need values of x and y minimizing S.

Then, given $xy = 100$ for $x, y > 0 \Rightarrow y = \frac{100}{x}$ .....(1)

Using (1), S in one variable is:

$$S(x) = 2x + 8(\frac{100}{x}) \Rightarrow S(x) = 2x + \frac{800}{x}$$

Then, $S'(x) = 2 - \frac{800}{x^2}$ for $x > 0$

And, $S'(x) = 0 \Rightarrow 2 - \frac{800}{x^2} = 0 \Rightarrow x = 20$

And, by (1), $y = \frac{100}{x} = \frac{100}{20} = 5 \Rightarrow y = 5$.

Therefore, the numbers are 20 and 5

Answer: B

20. The maximum profit occurs at x where $P'(x) = 0$

$\Rightarrow (36 + 72x - 18x^2)' = 0 \Rightarrow 72 - 36x = 0 \Rightarrow x = 2$

Thus, the maximum profit is: $P(2) = 36 + 72(2) - 18(2)^2 = 108$

Answer: D

21. (i) $2 \in$ Domain of f, i.e. $f(2) = 2^3 - 2 + 1 = 7$

(ii) $\lim_{x \to 2} f(x)$ exists so that $\lim_{x \to 2^+} f(x)$ and $\lim_{x \to 2^-} f(x)$ exist and

$$\lim_{x \to 2^+} f(x) = \lim_{x \to 2^-} f(x) \Rightarrow \lim_{x \to 2^+} k(2^x) = \lim_{x \to 2^-} (x^3 - x + 1)$$

$$\Rightarrow k(2^2) = 2^3 - 2 + 1 \Rightarrow k = 7/2$$

Thus, f is continuous at $x = 2$ if $k = 7/4$

Answer: A

22. Recall that $\sin^{-1}(\sin x) = x$ for $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$ but $\frac{5\pi}{4} \notin [\frac{\pi}{2}, \frac{\pi}{2}]$

[*This question must assume* $x \in [\frac{3\pi}{2}, \frac{5\pi}{2}]$ *or* $x \in [\frac{5\pi}{2}, \frac{7\pi}{2}]$ *for* $y = \sin x$ *and if so the answer is* $\frac{5\pi}{2}$]

, otherwise $\sin\frac{5\pi}{4} = -\sin(\frac{5\pi}{4} - \pi) = -\sin\frac{\pi}{4} = -1$ so that

$\sin^{-1}(\sin\frac{5\pi}{4}) = \sin^{-1}(-1) = -\sin^{-1}(1)$, by $\sin^{-1}(-x) = -\sin^{-1}(x)$

$$= -\frac{\pi}{4}$$

Answer: A

23. Recall that $f(x) = c + a\sin(kx + b)$ has:

Period: $p = \frac{2\pi}{|k|}$, Amplitude $= |a|$,

Range $= [c - |a|, c + |a|]$, and Phase Shift: $-\frac{b}{k}$.

Hence, $f(x) = -2 + 3\sin(\frac{1}{2}x + 3) \Rightarrow c = -2, a = 3, k = \frac{1}{2}$ & $b = 3$ so that

Period: $p = \frac{2\pi}{|k|} = \frac{2\pi}{|\frac{1}{2}|} = 4\pi \Rightarrow$ (A) is False

Amplitude $= |a| = |3| = 3 \Rightarrow$ (B) is False

Range $= [c - |a|, c + |a|]$

$= [-2 - |3|, -2 + |3|] = [-5, 1] \Rightarrow$ (C) is **True**

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Phase Shift: $-\frac{b}{k} = -\frac{\frac{3}{1}}{2} = -6 \Rightarrow$ (D) is False

Answer: C

24. Given a Geometric sequence $\{G_n\}$ with $G_1 = 3$ and $G_4 = 81$,

Then, the common ratio is: $r = \sqrt[4-1]{\frac{G_4}{G_1}} = \sqrt[3]{\frac{81}{3}} = \sqrt[3]{27} = 3$

Answer: D

25. Given inverse of the matrix $\begin{pmatrix} x & 2x \\ -x & x \end{pmatrix}$ as $\begin{pmatrix} -1 & -2 \\ 1 & 1 \end{pmatrix}$ so that

$\begin{pmatrix} x & 2x \\ -x & x \end{pmatrix}\begin{pmatrix} -1 & -2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, by $AA^{-1} = I$; Identity matrix

$\Rightarrow \begin{pmatrix} (x & 2x)\begin{pmatrix} -1 \\ 1 \end{pmatrix} & (x & 2x)\begin{pmatrix} -2 \\ 1 \end{pmatrix} \\ (-x & x)\begin{pmatrix} -1 \\ 1 \end{pmatrix} & (-x & x)\begin{pmatrix} -2 \\ 1 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (Matrix Product)

$\Rightarrow \begin{pmatrix} x(-1) + 2x(1) & x(-2) + 2x(1) \\ -x(-1) + x(1) & -x(-2) + x(1) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (Matrix Product)

$\Rightarrow \begin{pmatrix} x & 0 \\ 2x & 3x \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

But by equality of Matrix: $x = 1$, $2x = 0$, and $3x = 1$ which is wrong because we found three different values of x.

<u>ANOTHER TRY!</u> inverse of $\begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}$ is $\begin{pmatrix} 1 & 2 \\ -1 & -1 \end{pmatrix}$ so that by uniqueness of inverse of matrix and definition of equal matrices

$\begin{pmatrix} x & 2x \\ -x & x \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ -1 & -1 \end{pmatrix}$ gives $x = 1$ and $x = -1$ which is wrong because we found two different values of x. **Answer: No Answer**

26. The system $\begin{cases} x - y + z = 3 \\ 2x + y - z = 4 \\ kx - 2y - z = 5 \end{cases}$ has solution (unique or infinitely many)

if the determinant of the coefficient matrix is not zero,

, i.e. $\begin{vmatrix} 1 & -1 & 1 \\ 2 & 1 & -1 \\ k & -2 & -1 \end{vmatrix} \neq 0$. Then expanding along the first row

$\Rightarrow 1\begin{vmatrix} 1 & -1 \\ -2 & -1 \end{vmatrix} - (-1)\begin{vmatrix} 2 & -1 \\ k & -1 \end{vmatrix} + 1\begin{vmatrix} 2 & 1 \\ k & -2 \end{vmatrix} \neq 0$

$\Rightarrow 1(-3) - (-1)(-2+k) + 1(-4-k) \neq 0$

$\Rightarrow -3 - 2 + k - 4 - k \neq 0$

$\Rightarrow -9 \neq 0$ which is true statement and k is canceled out so that the system has solution for every real number k.

Answer: A

27. Given $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$,

$3A + 2B = 3\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + 2\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$

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$$= \begin{pmatrix} 3(1) & 3(2) \\ 3(3) & 3(4) \end{pmatrix} + \begin{pmatrix} 2(2) & 2(-1) \\ 2(-1) & 2(3) \end{pmatrix}$$

Multiplying matrix by scalar

$$= \begin{pmatrix} 3 & 6 \\ 9 & 12 \end{pmatrix} + \begin{pmatrix} 4 & -2 \\ -2 & 6 \end{pmatrix}$$

$$= \begin{pmatrix} 3+4 & 6+(-2) \\ 9+(-2) & 12+6 \end{pmatrix}$$, Matrix addition

$$= \begin{pmatrix} 7 & 4 \\ 7 & 18 \end{pmatrix}$$ **Answer: A**

28. $f(x)=\frac{1}{x-1}-\sqrt{4-x^2} \Rightarrow f'(x)=\frac{-1}{(x-1)^2}+\frac{x}{\sqrt{4-x^2}}$ so that domain of f' is (-2, 1)∪(1, 2). Thus, f is differentiable on (-2, 1)∪(1, 2)

Answer: A

29. $f(x)=(x-1)^{\frac{2}{3}} \Rightarrow f'(x)=\frac{2}{3(x-1)^{\frac{1}{3}}}$ so that domain of f' is $\mathbb{R}\setminus\{1\}$. Thus, f is differentiable on $(-\infty, 1)\cup(1, \infty)$ **Answer: D**

30. The slope of the line tangent to the graph of $f(x)=x^2+\tan x$ at a point $(\pi, f(\pi))$ is $f'(\pi)$. First we find $f'(x)$ as $f'(x)=2x+\sec^2 x$.

Hence, the required slope is $f'(\pi)=2\pi+\sec^2\pi=2\pi+1$

Answer: C

31. Given $f(x)=4-x^2\sin x$

$$\Rightarrow f'(x)=(4-x^2\sin x)'$$
$$=(4)'-(x^2\sin x)', \text{ Difference Rule}$$
$$=0-[(x^2)'\sin x+x^2(\sin x)'], \text{ Product Rule}$$
$$=-[2x\sin x+x^2\cos x]$$
$$=-x(2\sin x+x\cos x)$$ **Answer: A**

32. Given $a_1=6$, and $a_{n+1}=-\frac{1}{3}a_n$ implies $\frac{a_{n+1}}{a_n}=-\frac{1}{3}$ so that the given sequence is a geometric sequence with first term $a_1=6$ and common ratio $r=-\frac{1}{3}$. Thus, by n^{th} partial sum formula

$$S_n=\frac{a_1(1-r^n)}{1-r} \Rightarrow S_{10}=\frac{6(1-(-\frac{1}{3})^{10})}{1-(-\frac{1}{3})}=\frac{6(1-(-\frac{1}{3})^{10})}{\frac{4}{3}}=\frac{9}{2}(1-(-\frac{1}{3})^{10})$$

Answer: B

33. Given $\begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 4 & -1 & 4 \end{pmatrix}$

To make $a_{21}=0=a_{31}$ pivoting R_2 and R_3 as:

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 4 & -1 & 4 \end{pmatrix} \xrightarrow[R_3\to R_3-4R_1]{R_2\to R_2-2R_1} \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & -5 & 0 \end{pmatrix}$$

To make $a_{22}=1$ and $a_{23}=1$, rescaling R_2 and R_3 as:

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & -5 & 0 \end{pmatrix} \xrightarrow[R_3\to -\frac{1}{5}R_3]{R_2\to -R_2} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

To make $a_{32}=0$, pivoting R_3 as:

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$$\begin{pmatrix}1 & 1 & 1\\ 0 & 1 & -1\\ 0 & 1 & 0\end{pmatrix} \xrightarrow{R_3 \to R_3 - R_2} \begin{pmatrix}1 & 1 & 1\\ 0 & 1 & -1\\ 0 & 0 & 1\end{pmatrix}$$

(Row echelon form)

To make $a_{12} = 0$, pivoting R_1 as:

$$\begin{pmatrix}1 & 1 & 1\\ 0 & 1 & -1\\ 0 & 0 & 1\end{pmatrix} \xrightarrow{R_1 \to R_1 - R_2} \begin{pmatrix}1 & 0 & 2\\ 0 & 1 & -1\\ 0 & 0 & 1\end{pmatrix}$$

To make $a_{23} = 0$, pivoting R_2 as:

$$\begin{pmatrix}1 & 0 & 2\\ 0 & 1 & -1\\ 0 & 0 & 1\end{pmatrix} \xrightarrow{R_2 \to R_2 + R_3} \begin{pmatrix}1 & 0 & 2\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}$$

To make $a_{13} = 0$, pivoting R_1 as:

$$\begin{pmatrix}1 & 0 & 2\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix} \xrightarrow{R_1 \to R_1 - 2R_3} \begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}$$

(Row reduced echelon form)

Answer: D

34. $\begin{cases}2x - y + 4k = 6\\ kx - y - z = 2\\ x + y + 2z = 8\end{cases} \Rightarrow \begin{cases}2x - y + 0z = 6 - 4k\\ kx - y - z = 2\\ x + y + 2z = 8\end{cases}$ so that the augmented matrix is $\left(\begin{array}{ccc|c}2 & -1 & 0 & 6-4k\\ k & -1 & -1 & 2\\ 1 & 1 & 2 & 8\end{array}\right)$

Swapping R_1 and R_3:

$$\left(\begin{array}{ccc|c}2 & -1 & 0 & 6-4k\\ k & -1 & -1 & 2\\ 1 & 1 & 2 & 8\end{array}\right) \xrightarrow{R_1 \leftrightarrow R_3} \left(\begin{array}{ccc|c}1 & 1 & 2 & 8\\ k & -1 & -1 & 2\\ 2 & -1 & 0 & 6-4k\end{array}\right)$$

Pivoting R_2 and R_3 using R_1 to make $a_{21} = 0 = a_{31}$ as:

$$\left(\begin{array}{ccc|c}1 & 1 & 2 & 8\\ k & -1 & -1 & 2\\ 2 & -1 & 0 & 6-4k\end{array}\right) \xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - kR_1} \left(\begin{array}{ccc|c}1 & 1 & 2 & 8\\ 0 & -1-k & -1-2k & 2-8k\\ 0 & -3 & -4 & -10-4k\end{array}\right)$$

Rescaling R_2 to make $a_{22} = 1$ as:

$$\left(\begin{array}{ccc|c}1 & 1 & 2 & 8\\ 0 & -1-k & -1-2k & 2-8k\\ 0 & -3 & -4 & -10-4k\end{array}\right) \xrightarrow{R_2 \to \frac{1}{1+k} R_2} \left(\begin{array}{ccc|c}1 & 1 & 2 & 8\\ 0 & 1 & \frac{1+2k}{1+k} & \frac{8k-2}{1+k}\\ 0 & -3 & -4 & -10-4k\end{array}\right)$$

Pivoting R_3 using R_2 to make $a_{32} = 0$ as:

$$\left(\begin{array}{ccc|c}1 & 1 & 2 & 8\\ 0 & 1 & \frac{1+2k}{1+k} & \frac{8k-2}{1+k}\\ 0 & -3 & -4 & -10-4k\end{array}\right)$$

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$$\xrightarrow{R_3 \to R_3+3R_2} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & \frac{1+2k}{1+k} & \frac{8k-2}{1+k} \\ 0 & 0 & \frac{-1+2k}{1+k} & \frac{-4k^2+10k-16}{1+k} \end{array}\right)$$

Rescaling R_3 to make $a_{33}=1$ as:

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & \frac{1+2k}{1+k} & \frac{8k-2}{1+k} \\ 0 & 0 & \frac{-1+2k}{1+k} & \frac{-4k^2+10k-16}{1+k} \end{array}\right)$$

$$\xrightarrow{R_3 \to \frac{1+k}{-1+2k}R_3} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & \frac{1+2k}{1+k} & \frac{8k-2}{1+k} \\ 0 & 0 & 1 & \frac{-4k^2+10k-16}{-1+2k} \end{array}\right)$$

One non-zero coefficients in row 3 in echelon form

$\Rightarrow$ It has a unique solution if $\frac{-4k^2+10k-16}{-1+2k}$ is defined

$\Rightarrow$ It has a unique solution if $-1+2k \neq 0 \Rightarrow k \neq 1/2$ **Answer:**

D

Given $A=\begin{pmatrix} 2 & 3 \\ 2 & -1 \end{pmatrix}$ and $B=\begin{pmatrix} 2 & 5 & -4 \\ 3 & 2 & 6 \end{pmatrix}$

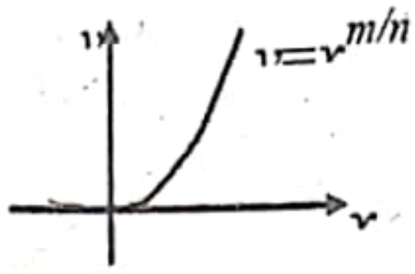
$$AB=\begin{pmatrix} 2 & 3 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} 2 & 5 & -4 \\ 3 & 2 & 6 \end{pmatrix}$$

$$=\begin{pmatrix} (2 \quad 3)\begin{pmatrix} 2 \\ 3 \end{pmatrix} & (2 \quad 3)\begin{pmatrix} 5 \\ 2 \end{pmatrix} & (2 \quad 3)\begin{pmatrix} -4 \\ 6 \end{pmatrix} \\ (2 \quad -1)\begin{pmatrix} 2 \\ 3 \end{pmatrix} & (2 \quad -1)\begin{pmatrix} 5 \\ 2 \end{pmatrix} & (2 \quad -1)\begin{pmatrix} -4 \\ 6 \end{pmatrix} \end{pmatrix}$$

$$=\begin{pmatrix} 2(2)+3(3) & 2(5)+3(2) & 2(-4)+3(6) \\ 2(2)+(-1)(3) & 2(5)+(-1)(2) & 2(-4)+(-1)(6) \end{pmatrix}$$

$$=\begin{pmatrix} 13 & 16 & 10 \\ 1 & 8 & -14 \end{pmatrix}$$ **Answer: D**

Given graph of the function $y=x^{\frac{m}{n}}$, where m and n are positive integers and $n \neq 0$ as follows.



Domain of the given function is $[0, \infty)$ so that m is odd and n is even

Since the graph is open up and strictly increasing, $\frac{m}{n}>1 \Rightarrow m>n$

Thus, m is odd, n is even and $m>n$

Answer: A

Sample Space, S={1, 2, 3, 4, 5, 6}

Let the event of prime number be P={2, 3, 5} and the event of composite number be C={4, 6}.

Then, $n(S)=6$, $n(P)=3$ and $n(C)=2$.

If two dies are rolled simultaneously, then the probability that a prime number turns up on one of the dice and a composite number on the other by *product rule* is:

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. $p(\text{P and C}) = p(\text{P})p(\text{C}) = \frac{n(\text{P})}{n(S)} \times \frac{n(\text{C})}{n(S)} = \frac{3}{6} \times \frac{2}{6} = \frac{1}{6}$.

Answer: D

38. First build class boundaries and class midpoints as follows.

Values	class boundaries	class midpoints	Frequency
95 – 99	94.5 – 99.5	97	4
90 – 94	89.5 – 94.5	92	6
85 – 89	84.5 – 89.5	87	10
80 – 84	79.5 – 84.5	82	16
75 – 79	74.5 – 79.5	77	9
70 – 74	69.5 – 74.5	72	5

A. The class interval is the positive difference between consecutive lower limits (for instance 95-90=5) or upper limits (for instance 99-94=5) or size of each class boundaries, i.e. 99.5-94.5= 5 so that A is False.

B. The upper limit of the second class is 94 so that B is False.

C. 85.5 – 90.5 is not one of the class boundaries so that C is False.

D. 82 is the class midpoint of fourth class as shown in the table.

Answer: D

39. First $f(x) = \frac{1}{(1-\sqrt{x})^2}$

$$\Rightarrow f'(x) = \left(\frac{1}{(1-\sqrt{x})^2}\right)' = \frac{((1-\sqrt{x})^2)'}{((1-\sqrt{x})^2)^2}, \quad \text{by } \left(\frac{1}{f}\right)' = \frac{-f'}{f^2}$$

$$= \frac{2(1-\sqrt{x})(1-\sqrt{x})'}{(1-\sqrt{x})^4}, \text{ by Chain Rule}$$

$$= \frac{2(1-\sqrt{x})(1'-(\sqrt{x})')}{(1-\sqrt{x})^4}, \text{ by Difference Rule}$$

$$= \frac{2(1-\sqrt{x})(0-\frac{1}{2\sqrt{x}})}{(1-\sqrt{x})^4}, \text{ by Difference Rule}$$

Thus, $f'(4) = \frac{2(1-\sqrt{4})(0-\frac{1}{2\sqrt{4}})}{(1-\sqrt{4})^4} = \frac{2(1-2)(0-\frac{1}{2(2)})}{(1-2)^4} = \frac{-2(-\frac{1}{4})}{(-1)^4} = \frac{1}{2}$

Answer: A

40. First $f(-1) = 2(-1)^2 - 3(-1) + 1 = 6$ and

$f(2) = 2(2)^2 - 3(2) + 1 = 3$.

Thus, the slope of the line passing through the point $(-1, f(-1))$ and $(2, f(2))$ is: $m = \frac{\Delta y}{\Delta x} = \frac{f(2)-f(-1)}{2-(-1)} = \frac{3-6}{3} = -1$

Answer: D

41. Given vectors $u = (2, 0)$, $u = (-1, 3)$ and $w = -3u + 2v$ in a plane.

Hence, $w = -3u + 2v = -3(2, 0) + 2(-1, 3) = (-6, 0) + (-2, 6) = (-8, 6)$

The unit vector in the direction of w is:

$$\frac{w}{|w|} = \frac{(-8,6)}{|(-8,6)|} = \frac{(-8,6)}{\sqrt{(-8)^2+6^2}} = \frac{(-8,6)}{10} = \frac{1}{10}(-8, 6) = \left(-\frac{4}{5}, \frac{3}{5}\right)$$

Therefore, $\frac{w}{|w|} = \left(-\frac{4}{5}, \frac{3}{5}\right) = -\frac{4}{5}i + \frac{3}{5}j$

Answer: B

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42. Let u, v and w be three vectors in a plane. Then,
A. $u \cdot v = v \cdot u$ so that scalar product is commutative.
B. Let u and v be non-zero parallel vectors. Then the angle between these vectors is either 180^o or 0^o so that either
$$u \cdot v = |v||u| \cos 180^o = -|v||u| \text{ or}$$
$$u \cdot v = |v||u| \cos 0^o = |v||u|$$
Thus, scalar product of non-zero parallel vectors is non-zero.
C. $u \cdot (v + w) = u \cdot v + u \cdot w$ so that scalar product is distributive over addition of vectors.
D. If the angle between vectors u and v is $\theta = 90^o$, then
$u \cdot v = |v||u| \cos 90^o = |v||u| \times 0 = 0$ so that scalar product of perpendicular vectors is zero.

Answer: B

43. Translation, Reflection, and Rotation are transformations which are rigid motion because they moves geometric figures in to congruent figures (for instance triangles in to congruent triangles, angles to congruent angles,...). Hence only D is NOT true.

Answer: D

44. Given $z = -1+i$ and $w = 1-3i$. Then,
$$\text{» } \frac{5z}{w} = \frac{5(-1+i)}{1-3i} = \frac{5(-1+i)}{1-3i} \times \frac{1+3i}{1+3i}$$
$$= \frac{5(-1+i)(1+3i)}{(1-3i)(1+3i)}$$
$$= \frac{5(-4-2i)}{1^2+3^2} = \frac{-4-2i}{2} = -2-i \ldots\ldots(1)$$
$$\text{» } |z|^2 = z^2 = z \cdot z = (-1+i)(-1+i) = -2i \ldots\ldots(2)$$
$$\text{» } \overline{w} = \overline{1-3i} = 1+3i \ldots\ldots(3)$$
Combining (1), (2), and (3):
$$\frac{5z}{w} + |z|^2 \overline{w} = (-2-i) + (-2i)(1+3i)$$
$$= (-2-i) + (-2i+6) = 4-3i$$

Answer: No Answer

45. A real number a is the limit of a sequence $\{a_n\}$ if a_n gets closer and closer to a as n goes to infinity, i.e. a is very close to all terms a_n from some arbitrary big n onwards.

Answer: C

46. Given ellipse with vertices: (3, 1), (3, 9); and minor axis length 6
Then, Center $C(h, k) = \left(\frac{3+3}{2}, \frac{1+9}{2}\right) = (3, 5)$, and
Vertices: (3, 1), (3, 9) $\Rightarrow a = \frac{|9-1|}{2} = 4 \Rightarrow a^2 = 16$, and
Length of minor axis $= 6 \Rightarrow 2b = 6 \Rightarrow b^2 = 9$.
Thus, $x_{V_1} = x_{V_2} = x_C = 3 \Rightarrow$ Vertical ellipse $\frac{(x-3)^2}{9} + \frac{(y-5)^2}{16} = 1$

Answer: A

47. Given an ellipse $\frac{(x-3)^2}{6} + \frac{(y-2)^2}{4} = 1$ with center $C(3, 2)$

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For a rotation R about $(0, 0)$ through angle θ: $R_\theta(x, y) = (-y, x)$ if $\theta = \frac{\pi}{2}$ so that the center of the image ellipse is:

$$C' = R_\theta(3, 2) = (-2, 3)$$

The given ellipse is horizontal since $a^2 = 6$ is denominator of the x-part so that rotating it through $\theta = \frac{\pi}{2}$ gives a vertical ellipse:

$$\frac{(x+2)^2}{4} + \frac{(y-3)^2}{6} = 1 \Rightarrow 6(x+2)^2 + 4(y-3)^2 = 24$$

Answer: D

48. Given propositions $p \equiv$T and $q \equiv$F so that

A. $(p \vee \neg q) \Rightarrow q \equiv (T \vee \neg F) \Rightarrow F \equiv (T \vee T) \Rightarrow F \equiv T \Rightarrow F \equiv F$

B. $\neg p \Rightarrow q \equiv \neg T \Rightarrow F \equiv F \Rightarrow F \equiv T$

C. $p \vee (q \wedge \neg p) \equiv T \vee (F \wedge \neg T) \equiv T \vee (F \wedge F) \equiv T \vee F \equiv T$

D. $(p \Rightarrow q) \wedge p \equiv (T \Rightarrow F) \wedge T \equiv F \wedge T \equiv F$

Answer: C

49. Given $f(x) = \sqrt{x+2}$ and $g(x) = x^2 - 1$ so that

$$(f \circ g)(x) = f(g(x)) = \sqrt{g(x)+2} = \sqrt{x^2 - 1 + 2} = \sqrt{x^2 + 1}$$

Then, Domain of $f \circ g$

= Domain of $g \cap$ Domain of f in g

= Domain of $x^2 - 1 \cap$ Domain of $\sqrt{x^2 + 1}$

$= \mathbb{R} \cap \mathbb{R} = \mathbb{R}$

Since $x^2 + 1 \geq 1$, also $\sqrt{x^2 + 1} \geq 1$ for all $x \in \mathbb{R}$ so that range of $(f \circ g)(x) = \sqrt{x^2 + 1}$ is $[1, \infty)$

Answer: A

50. Given a rational function $f(x) = \frac{2x+4}{x-1}$, then the graph has

A. a horizontal asymptote, $y = 2$ because $\frac{2x+4}{x-1} = \frac{2+\frac{4}{x}}{1-\frac{1}{x}}$ so that $\frac{4}{x}$ and $\frac{1}{x}$ go to 0 as x goes to $\pm\infty$ and hence $\frac{2+\frac{4}{x}}{1-\frac{1}{x}}$ goes to $\frac{2+0}{1-0} = 2$

B. a vertical asymptote, $x = 1$ because f goes to $-\infty$ as x approaches 1 from left and goes to $+\infty$ as x approaches 1 from right.

C. y-intercept $(0, -4)$ because $x = 0 \Rightarrow f(0) = \frac{2(0)+4}{0-1} = -4$.

D. y-intercept $(0, -2)$ because $y = 0 \Rightarrow 2x + 4 = 0 \Rightarrow x = -2$.

Answer: D

51. Let $\frac{5}{x} = \frac{1}{y}$. Then, $x = 5y$ and $y \to \infty$ as $x \to \infty$ so that

$$\lim_{x\to\infty}(1 + \frac{5}{x})^x = \lim_{y\to\infty}(1 + \frac{1}{y})^{5y} = \lim_{y\to\infty}((1 + \frac{1}{y})^y)^5$$

$$= (\lim_{y\to\infty}(1 + \frac{1}{y})^y)^5 = e^5$$

Answer: B

52. A. False; only limit of a zero constant function is always zero. For instance: $f(x) = 3$ is a constant function but

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$$\lim_{x\to 0} f(x) = \lim_{x\to 0} 3 = 3 \text{ (Not zero)}$$

B. False; For instance: for $f(x)=\begin{cases} x & \text{if } x > 0 \\ x+1 & \text{if } x \le 0 \end{cases}$

$$\lim_{x\to 0^-} f(x) = 1 \text{ but } \lim_{x\to 0^+} f(x) = 0$$

C. True; because $\lim_{x\to a} f(x) = L$ means $f \to L$ as $x \to a$ which means the limit of a function f is a y-value L to which $f(x)$ approaches as x approaches to some specified number, a.

D. False; For instance: for $f(x)=\begin{cases} 1 & \text{if } x = 0 \\ x & \text{if } x \neq 0 \end{cases}$

$$\lim_{x\to 0} f(x) = 0 \text{ (limit value) but } f(0) = 1 \text{ (functional value)}$$

Answer: C

53. $-1 \le \sin n \le 1$ for all n. Multiplying all sides by $\frac{1}{n^2}$ gives

$$-\frac{1}{n^2} \le \frac{\sin n}{n^2} \le \frac{1}{n^2}$$

Thus, $\lim_{n\to\infty} -\frac{1}{n^2} = 0 = \lim_{n\to\infty} \frac{1}{n^2}$ so that by Squeezing Theorem

$$\lim_{n\to\infty} \frac{\sin n}{n^2} = 0 \qquad \textbf{Answer: D}$$

54. Given functions f and g continuous at $x = c$. Then,

A. $\sqrt{g}$ may fail to be continuous, for g may be non-positive around $x = c$. For instance $g(x) = x$ is continuous at $x = -1$ but $\sqrt{x}$ isn't continuous at $x = -1$.

B. $\frac{f}{g}$ may fail to be continuous at $x = c$, for may be $g(c) = 0$.

C. f^{-1} may fail to be continuous at $x = c$, for may be $f(c) = 0$ (f^{-1} may not even be a function).

D. Since the product of continuous functions is also continuous the product of f and f i.e. f^2 is continuous at $x = c$.

Answer: D

55. $\lim_{x\to 0} \frac{\sin 4x}{\sin 6x} = \lim_{x\to 0} \frac{\frac{\sin 4x}{4x}\times 4x}{\frac{\sin 6x}{6x}\times 6x} = \frac{4}{6} \lim_{x\to 0} \frac{\frac{\sin 4x}{4x}}{\frac{\sin 6x}{6x}}$

$$= \frac{2}{3} \frac{\lim_{x\to 0} \frac{\sin 4x}{4x}}{\lim_{x\to 0} \frac{\sin 6x}{6x}}$$

$$= \frac{2}{3} \frac{\lim_{t\to 0} \frac{\sin t}{t}}{\lim_{z\to 0} \frac{\sin z}{z}} \text{ for } t = 4x \text{ and } z = 6x$$

$$= \frac{2}{3} \times \frac{1}{1} = \frac{2}{3}$$

Answer: B

56. $x = c$ is zero of f if $f(c)=0$ so that among the given choices only 1 satisfies this, i.e. $f(1) = 1^3 - 3(1^2) + 3(1) - 1 = 0$.

Thus, 1 is zero for f.

Answer: B

57. $f(x) = \frac{\ln 2}{x^2}$

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$$\Rightarrow f'(x) = \left(\frac{\ln 2}{x^2}\right)' = \ln 2\,(x^{-2})' \text{ by } (kf)' = kf' \text{ and } \frac{1}{x^2} = x^{-2}.$$
$$= \ln 2\,(-2x^{-2-1}), \text{ by Power Rule}$$
$$= \ln 2\,(-2x^{-3})$$
$$= \ln 2\left(\frac{-2}{x^3}\right) = \frac{-2\ln 2}{x^3}$$

Answer: B

58. Given equation of a circle as $x^2 + y^2 + kx = 1$ for some $k \in \mathbb{R}$
, i.e. $x^2 + y^2 = 1 - kx$.....(1)

Let r be the radius of the circle.

Given the center at (2, 0) implies the standard equation of the circle is: $(x-2)^2 + y^2 = r^2 \Rightarrow x^2 - 4x + 4 + y^2 = r^2$

$\Rightarrow x^2 + y^2 = r^2 - 4 + 4x$.....(2)

Since the left sides of (1) and (2) are the same, the right sides must be equal so that

$r^2 - 4 + 4x = 1 - kx$

Then, by fundamental theorem of polynomials (equal polynomials)

$r^2 - 4 = 1$ and $4x = -kx$

$\Rightarrow r^2 = 5$ and $4 = -k \Rightarrow r = \sqrt{5}$ and $k = -4$

Answer: B

59. $y^2 + 6y + 8y + 25 = 0 \Rightarrow y^2 + 14y + 25 = 0$ so that by quadratic formula this is equation of two horizontal lines, i.e. $y = -7 \pm 2\sqrt{6}$

If we assume the given equation as $y^2 + 6y + 8x + 25 = 0$, then by completing the square with respect to y, in standard form we have

$y^2 + 6y + 8x + 25 = 0 \Rightarrow (y+3)^2 = -8(x+2)$

, where vertex: $(h, k) = (-2, -3)$ so that A is True

Focal length: $p = 2$ and so the directrix is the line

$x = h + p = -2 + 2 \Rightarrow x = 0$ (y-axis) so that B is True

By $4p = -8 < 0$, it is open to the left so that B is **False**

Focus: $(h - p, k) = (-2-2, -3) = (-4, -3)$ so that D is True

Answer: Wrong Question

60. The slope of the line tangent to the graph of f at $(a, f(a))$ if it exists is $f'(a)$ so that for $f(x) = x^2 + \tan x$ at $(\pi, f(\pi))$, first

$f'(x) = (x^2 + \tan x)' = (x^2)' + (\tan x)' = 2x + \sec^2 x$

Hence the slope is: $f'(\pi) = 2\pi + \sec^2 \pi = 2\pi + 1$, since $\sec \pi = 1$

Answer: C

61. First we build frequency table as follows

Height (in m)	1.56	1.65	1.70	1.72
Frequency	1	4	4	3

From the table values 1.65 and 1.70 are with highest frequency of 4 so that the data is bimodal with modes 1.65 & 1.70.

Answer: C

62. Given $f(x) = \cot x$

Range of f is $\mathbb{R}$ so that **A is True** because $(-\infty, 1] \cup [1, \infty) = \mathbb{R}$

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Domain: $\cot x = \frac{\cos x}{\sin x} \Rightarrow \sin x \neq 0 \Rightarrow x \neq n\pi$ for $n \in \mathbb{Z}$

Thus, domain of $f = \{x: x \neq n\pi$ for $n \in \mathbb{Z}\}$ so that B is True and C is False

Period: $\cot x = \frac{1}{\tan x} = \frac{1}{\tan(x+n\pi)}$ ← Period of $y = \tan x$ is π

$= \cot(x + n\pi)$

$\Rightarrow f(x) = f(x + n\pi) \Rightarrow$ Period of f is $p = \pi \Rightarrow$ D is False

Answer: A and B

63. Given $f(x) = 3-2x \Rightarrow f: y = 3-2x$, Given definition of f

$\Rightarrow f^{-1}: x = 3-2y$, Switching x & y in f^{-1}

$\Rightarrow f^{-1}: 2y = 3-x$

$\Rightarrow f^{-1}: y = \frac{3}{2} - \frac{x}{2}$, Making y a subject

$\Rightarrow f^{-1}(x) = \frac{3}{2} - \frac{x}{2}$, Replacing y by f^{-1}

Answer: C

64. A. M is an upper bound of $\{a_n\}$ if $M \geq a_n$ for all n so that A is False.

B. m is a lower bound of $\{a_n\}$ if $a_n \geq m$ for all n so that B is False.

C. A sequence of real numbers $\{a_n\}$ is bounded if and only if it has both an upper and a lower bound so that C is True.

D. The sequence $\{a_n\}$ is bounded if there exists $k > 0$ such that $k \geq |a_n|$ for all n so that D is False.

Answer: C

65. Given a sequence $\{\frac{n}{n-1}\}_{n=2}^{\infty}$, i.e. $a_n = 1 + \frac{1}{n-1}$ for $n \geq 2$

Then, listing some terms: 2/1, 3/2, 4/3, 5/4,...goes to 1 since $\frac{1}{n-1}$ goes to 0 as $n \to \infty$. Hence, 1 is the smallest value to which the sequence converges so that every $x \leq 1$ is lower bound for the sequence $\{\frac{n}{n-1}\}_{n=2}^{\infty}$. Therefore, only 1 is the lower bound.

Answer: B

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You may refer to the information given below when you need to work on some of the questions.

Name of constants	Symbol	Value
Acceleration due to gravity	g	$10 m/s^2$
Mass of proton	m_p	$1.67\times10^{-27} kg$
The charge of an electron	$e-$	$1.6\times10^{-19} C$
Universal gravitational constant	G	$6.67\times10^{-11} Nm^2/kg^2$
Density of water	ρ	$1000 kg/m^3$
Permittivity of vacuum	ε_o	$8.85\times10^{-12} F/m$
Magnetic permeability of free space	μ_o	$4\pi\times10^{-7} Tm/A$
Coulomb's constant	k	$9\times10^9 Nm^2/C^2$

$\sin 15^o = 0.16, \cos 15^o = 0.97$
$\sin 25^o = 0.42, \cos 25^o = 0.91$
$\sin 30^o = 0.5 = \cos 60^o$
$\sin 37^o = 0.6 = \cos 53^o$
$\sin 60^o = \cos 30^o = 0.87$
$\cos 79^o = 0.19, \sin 79^o = 0.98$
$\cos 104^o = 0.24, \sin 104^o = 0.97$

1. Which one of the following is correct about diffraction of light?
   A. Diffraction of light passing through a circular aperture results in parallel bright and dark fringes.
   B. Diffraction of light passing through a single slit produces bright and dark fringes of equal widths.
   C. As the width of the slit decreases, the width of the central bright fringe decreases too.
   D. As the number of slits in diffraction grating increase, the bright fringes appear brighter, taller and narrower.
2. In Millikan's oil drop experiment,
   A. the mass of the droplet was determined by measuring its terminal velocity in the absence of electric field.
   B. tiny droplets of negatively charged mineral oil where balanced under the action of downward electric force and weight.
   C. the electric field between the anode and cathode can be increased by decreasing the distance and the potential difference between them.

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D. the charge of the droplet was found to be half - integer multiple of electron charge.

3. Many chemical reactions release energy. Suppose that at the beginning of a reaction, an electron and a proton are separated by 0.110 nm, and their final separation is 0.100 nm. The change in the electric potential energy in this reaction (in eV) is

A. $14.4eV$ B. $27.5eV$ C. $1.30eV$ D. $13.1eV$

4. When an insulator is placed in an external electric field,

A. the external electric field induced within the insulator is greater than the external electric field, but has the same direction as the external field.

B. electric dipoles are induced within the insulator and produce induced electric field opposite to the external field.

C. the induced electric field within the insulator is perpendicular to the external field.

D. the net electric field within the insulator is greater than the external electric field.

5. Consider a parallel plate capacitor with plate area **A**, plate separation **d**, charge **Q**, capacitance **C**, potential difference **V** and electric field **E** between the plates. The electrical energy density stored by the capacitor

A. increases as the capacitance C and potential difference V increase.

B. depends on the plate area A and plate separation d.

C. is directly proportional to the square of the electric field E between the plates.

D. is directly proportional to the electric field E between the plates.

6. A potential difference **V** is applied across the ends of a uniform conducting wire of length **l**, cross-sectional area **A**, and containing n free charge carriers per unit volume.

Which one of the following is true?

A. The electric current in the conductor flows arbitrarily in all directions.

B. The free electrons in the conductor flow in the direction of the electric field in the conductor.

C. The current density in the conductor is directed in the direction of the drift velocity of the free charge carriers.

D. The electric current in the conductor flows perpendicularly to the drift velocity of the free charges.

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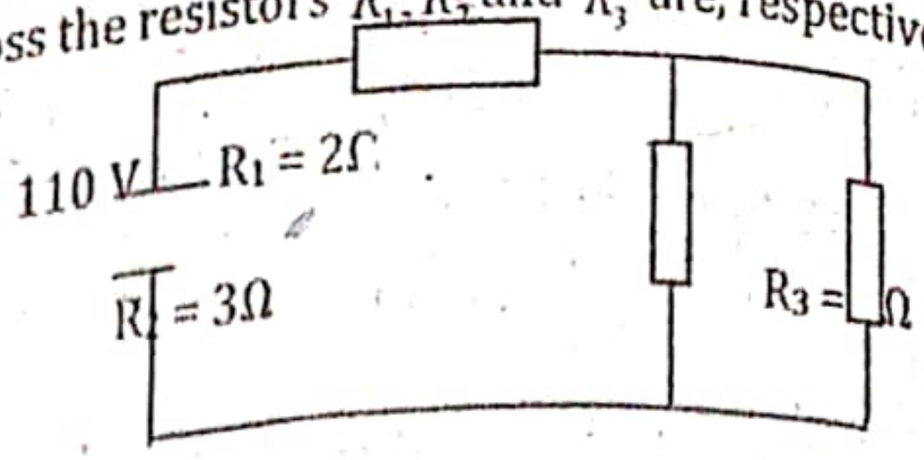
7. Three resistors are connected across a battery as shown below. The potential difference across the resistors R_1, R_2 and R_3 are, respectively,



A. 55 V, 82.5 V, 165 V.
B. 50 V, 30 V, 30 V.
C. 110 V, 25 V, 30 V.
D. 55 V, 55 V, 55 V.

8. According to Kirch0ff's junction rule,
   A. the amount of electric charge flowing into a junction is less than the amount flowing out of the junction.
   B. the amount of electric charge flowing into a junction is greater than the amount flowing out of the junction.
   C. the amount of electric charge entering a junction is equal to the amout leaving the junction.
   D. charge can disappear or appear at the junction.
9. Which of the following is true about galvanometer?
   A. A galvanometer can be converted to a voltmeter by connecting a small resistance in parallel with the galvanometer.
   B. To use a galvanometer as an ammeter that can measure large currents, we need to increase its resistance by connecting a large resistance in series with the galvanometer.
   C. A galvanometer can be converted to an ammeter that can read large currents by connecting a small resistance in parallel with it, and can be converted to a voltmeter by connecting a large resistance in series with the galvanometer.
   D. To use a galvanometer as an ammeter that can read large voltage, we connect a small resistance in series with the galvanometer.
10. A galvanometer has an internal resistance of 100 Ω and deflects full scale for 1 mA. The shunt resistance required to use it as an ammeter to read currents up to 1.0 A, and the resistance required to use it as a voltmeter to give full scale deflection of 100 V are, respectively,
    A. $1m\Omega, and 9.9\times10^{4}\Omega$
    B. $10m\Omega, and 10^{4}\Omega$
    C. $0.1\Omega, and 100\Omega$
    D. $1.00\Omega, and 10^{5}\Omega$
11. Equilibrium is the state of an object when
    A. the net force acting on it is constant.
    B. the object has a constant linear acceleration.
    C. the net force and net torque are both zero.
    D. the object has constant angular acceleration.
12. The fulcrum of a uniform 20 kg seesaw that is 4.0 m long is located 2.5 m from one end. A 30 kg child sits on the long end. The weight of a person sitting on the other end in order to balance the seesaw is

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A. 750 N B. 850 N C. 433 N D. 567 N

13. A material rod that is 4.00 m long and 0.50 cm^2 in cross-sectional area is found to stretch 0.20 cm under a tension of 500 N. The Young's Modulus of this material is.....

A. $1.0\times10^{4}\,N/m^2$ B. $1.0\times10^{8}\,N/m^2$ C. $2.0\times10^{11}\,N/m^2$ D. $5.0\times10^{8}\,N/m^2$

14. Which one of the following is **NOT** true about Pascal's principle and its application?

A. A. It has applications in hydraulic brakes for vehicles, where confined oil connects a small piston (brake pedal) to a large piston connected to the brake drums on the wheel of the car.

B. It has applications in hydraulic lift in which a small force applied to a smaller piston is usedto lift heavy object on large piston.

C. In pneumatic application, air is used for pressure transmission instead of liquid.

D. The hydraulic lift is a speed multiplying device in which a large force lifts a light object over a long distance.

15. Which one of the following is **NOT** correct about factors that affecting the streamlining of objects moving in a fluid?

A. Compressibility of the fluid.

B. Density of the fluid.

C. The mass of the object.

D. Shape and surface area of the object moving in a fluid

16. Which of the following is correct?

A. Temperature is the measure of the amount of energy that flows from hot to colder region.

B. Heat is energy that can be transferred from one object to another at equal temperature.

C. Internal energy of a system is independent of its mass or the number of molecules in the system.

D. Temperature is a measure of the average molecular kinetic energy of a gas.

17. Two moles of an ideal gas at a temperature of 308 K is confined in a volume of $6.00\times10^{-3}m^3$.

The pressure of the gas is (Universal gas constant = 8.314 J/mol.K)

A. $1.02\times10^{6}\,Pa$ B. $1.00\times10^{6}\,Pa$ C. $1.23\times10^{6}\,Pa$ D. $5.12\times10^{6}\,Pa$

18. A glass contains 12 mole of water molecule (H_2O). The number of water molecules, the mass of water,and the mass of each water molecule in the glass are, respectively

(Molar mass of H is 1.0 g and O is 16 g)

A. $90.32\times10^{23}\,molecules, 0.192kg, 18\times10^{-26}\,kg$

B. $72.24\times10^{23}\,molecules, 0.216kg, 35.9\times10^{-26}\,kg$

C. $96.32\times10^{23}\,molecules, 0.036kg, 0.09\times10^{-26}\,kg$

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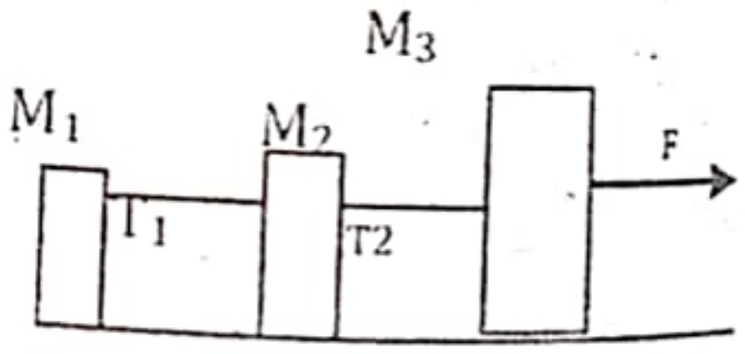
D. $102\times10^{23}\,molecules, 0.024kg, 6.02\times10^{-26}kg.$

19. Which of the following statements is correct about thermodynamic processes?
    A. An isobaric process occurs at constant volume and its p-v diagram is a vertical line.
    B. An isothermal process ocurrs at constant pressure and its p-v diagram is a linearly increasing plot.
    C. An isochoric process occrs at constant pressure and its p-v diagram is horizontal.
    D. In an adiabatic process, there is no heat removed from or absorbed by the system and its p-v diagram is a decreasing plot.
20. In a thermodynamic process, $1.5\times10^{5}J$ of heat is added to an ideal gas to keep the pressure at $2.0\times10^{5}Pa$ while the volume increase from $6.3m^3$ to $7.1m^3$. The change in the internal energy of the gas is
    A. $3.1\times10^{4}J$ B. $1.0\times10^{4}J$ C. $-1.0\times10^{4}J$ D. $14.1\times10^{4}J$
21. Which one of the following is true about fundamental forces?
    A. Gravitational force is a long-range attractive force between particles due to their charges.
    B. Strong nuclear force is a short - range that holds protons and neutrons to form atomic nucleus.
    C. A weak nuclear force is an extremely short-range force that binds protons and neutrons together in the nucleus.
    D. Electromagnetic force is a long-range attractive force between particles due to their masses.
22. Three blocks of mass M_1 = 1 kg, M_2 = 2 kg, and M_3 = 3 kg, placed on a horizontal frictionless table, are connected with strings and pulled to the right by a force F = 9 N, as shown below.

    The acceleration, tension T_1 and T_2 are, respectively
    A. $a=3/2m/s^2, T_1=3N and T_2=9/2N.$
    B. $a=3m/s^2, T_1=3N and T_2=9/2N.$
    C. $a=2/3m/s^2, T_1=2/3N and T_2=2N.$
    D. $a=3/2m/s^2, T_1=3/2N and T_2=9/2N.$



23. Which of the following is correct about an astronaut inside an international space station orbiting the Earth?
    A. Since the Earth exerts equal forces on both the astronaut and the space, the astronaut experiences weightlessness.
    B. The astronaut experiences weightlessness since the Earth's gravitational force on the astronaut is less than the Earth's force on the space station.
    C. The astronauts experiences weightlessness because the space station and the astronauts are freely falling towards the center of the earth.

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D. Because the Earth exerts a large force on the astronaut than on the space station,the astronaut experienced weightlessness.

24. Two masses of m_1 and m_2 are balanced on a horizontal stick of negligible mass pivoted at the center of mass of the system. If the distance of m_1 to the center of mass is 3.0 m and the distance from the mass m_2 to the center of mass is 1.0 m, the ratio of m_1 to m_2(m_1 / m_2) is

A. $\frac{4}{3}$ B. $\frac{3}{4}$ C. 3 D. $\frac{1}{3}$

25. Identify the correct statement about energy, work, and force.

A. Work is said to be bone by a force whenever the force is exerted on the object.

B. Force is a vector quantity, whereas work and energy are scalars.

C. Machine can multiply energy, but not force.

D. Energy cannot be possessed by an object at rest.

26. Distinguish between conservative and non-conservative forces.

A. Work done by a conservative force along any closed path is zero, whereas the work done by non-conservative force along a closed path is not zero.

B. Work done by a conservative force is path dependent, whereas work done by non-conservative force does not depend on the path followed.

C. Friction force is an example of conservative force, whereas spring force and gravitational force are typical examples of non-conservative force.

D. A conservative force dissipates energy, whereas a non-conservative force does not.

27. A crane raises 75.0 kg of bricks from the ground to a height of 45.0 m in 52 s at a constant speed. The average power of the crane is.....

A. 3,375 W B. 144 W C. 14.4 W D. 649 W

28. For a rigid body rotating about a fixed axis,

A. a point closer to the axis has smaller linear speed than a point farther from the axis.

B. a point closer to the axis has greater linear speed than a point farther from the axis.

C. for the same time interval, a point closer to the axis a point farther from the axis travel the same linear distance.

D. for the same time interval, a point closer to the axis turn through a smaller angle than a point farther from the axis.

29. A disk of radius 0.10 m and mass 0.1 kg is set into motion by a tangential force of 2.0 N applied at the edge of the disk. If the angular acceleration of the disk is 400 rad/s², the torque and angular velocity at 2.0 s are, respectively

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A. 0.2 N.m and 400 rad /s
B. 0.1 N.m and 200 rad/s
C. 0.5 N.m and 100 rad /s
D. 0.2 N.m and 800 rad/s

30. Identify the correct statement about torque and angular acceleration.
A. For the same time interval, a small torque and a large torque change angular momentum by the same amount.
B. A small torque applied for a longer time and a large torque exerted for a shorter time have the same effect on the change of angular momentum.
C. No torque acts on an object when its angular momentum increases.
D. The change in angular momentum is inversely proportional to the torque exerted for a given time interval.

31. A scientific method is a scientific procedure which ensures a rigorous evidence-based structure in which only ideas that have been carefully tasted are accepted as scientific theory.
B. is a scientific procedure in which a researcher begins with performing experiments.
C. is a scientific procedure where a researcher first predicts the result of a problem before proposing the possible hypothesis.
D. begins from an accepted scientific theory and concludes with designing an experiment.

32. Which one of the following is correct about accuracy and precision of a measurement?
A. Accuracy describes the closeness of repeatedly measured data to each other, whereas precision describes about the closeness of a data to a true value..
B. Accuracy refers to the closeness of repeatedly measured data to a true value, whereas precision is related to the closeness of the measured data to each other.
C. Accuracy is related to the number of significant figures, whereas precision describes how well we minimize random errors.
D. A more accurate measurement has larger uncertainty, whereas a more precise data has smaller uncertainty.

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39. Choose a correct statement about centripetal force and centripetal acceleration.

A. Centripetal force on an object moving in a circular path is perpendicular to the radius of the path.

B. Centripetal force on an object moving in a circular path tends to pull the object outward from the path.

C. For an object moving in a circular path, its centripetal force and centripetal acceleration are parallel and directed toward the center of the path.

D. Centripetal acceleration and centripetal force are constant in direction, but vary in magnitude.

40. According to Newton's law of Universal Gravitation,

A. the period of a satellite orbiting the Earth in a circular orbit increases as the radius of its orbit increases.

B. the gravitational attraction between two objects is directly proportional to the square of the distance between them.

C. the speed of a satellite moving around the Earth in a circular orbit depends on the mass of the satellite.

D. for two interacting objects, the gravitational force exerted on a light object is less than the force on the heavy one.

41. According the kinetic theory of gases, the

A. average translational kinetic energy of the particles of an ideal gas increases as the gas gets cooled.

B. internal energy of an ideal gas is equal to the potential energy due to interaction between the particles of the gas.

C. diffusion rate of a gas is inversely proportional to the square root of the molar mass because it is directly proportional to the molecular speed.

D. particles of an ideal gas undergo inelastic collision between themselves and with the the walls of the container.

42. At t = 0, an object undergoing SHM has maximum displacement of 0.650 m and angular frequency of 740 rad/s. Which of the following is correct about the motion of the oscillator?

A. The position of the object is $x = 0.650\cos(7.40t)$ and its acceleration is $a = -3.59\sin(7.40t)$.

B. The position of the object is $x = 0.650\sin(7.40t)$ and its velocity is $v = 4.81\cos(7.40t)$.

C. The frequency of the oscillator is 7.40 Hz and its period is 0.135 s.

D. The position of the object is $x = 0.650\cos(7.40t)$ and its acceleration is $a = -35.59\cos(7.40t)$.

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43. A 0.35 kg mass attached to the end of a spring oscillates 2.5 times per second with amplitude of 0.15 m. Which of the following is true about the motion of the mass?
A. The total energy of the system is 97.0 J.
B. The velocity when it passes the equilibrium is 2.355 m/s.
C. The acceleration when it is 0.10 m from equilibrium is 25.00 m/s².
D. The velocity when it is 0.10 m from equilibrium is 3.0 m/s.

44. Identify a true statement about a mechanical wave.
A. The velocity of a traveling wave is equal to the velocity of the particles through which the wave propagates.
B. As a wave travels from one point to another along a medium, it transmits energy and particles of the medium from point to point.
C. The speed of a wave traveling along a string increases as the tension in the string increases and as the linear density of the string decreases.
D. The speed of a wave traveling across a thin string is equal to its speed when traveling across a thick string of the same length and tension.

45. Which of the following is a correct statement about the intensity of sound?
A. It is energy transmitted per unit time.
B. It is the product of power of a sound and area through which the sound is transmitted.
C. It is inversely proportional to the square of the distance from the source of sound.
D. It is directly proportional to the square of the distance from the sound source.

46. Which of the following is correct about Doppler effect?
A. When a source of sound wave moves toward stationary observer, the wavelength of the wave increases while its frequency decreases.
B. It is the change in the velocity of a wave due to relative motion between the source of the wave and an observer.
C. Doppler effect is the change in the frequency of a wave due to relative motion between the wave source and an observer.
D. When a sound source moves away from stationary observer, the frequency detected by the observer increases, whereas the wavelength decreases.

47. Huygens's principle states that
A. the secondary wavelets combine to produce a new wave front which has a shape different from the original wave front.
B. all points along a wave front acts as a source of plane secondary wavelets that travel at speeds and frequencies greater than that of the original wave front.
C. the secondary wavelets are parallel to the original wave front.
D. all points along a wave front behave as a source of secondary wavelets that have the same frequency and speed as the original wave front.

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48. Which one of the following is correct about the law of reflection and the law of refraction?
A. The speed and wave length of a reflected wave are different from that of an incident wave.
B. The incident and refracted rays make equal angles with the boundary of the two interface.
C. The ratio of the sine of incident angle to the sine of the refracted angle is equal to the ratio of the speed of the incident wave to the speed of the refracted wave.
D. If the speed of the incident wave is less than the speed of the refracted wave, the refracted wave bends towards the normal line.

49. Choose the correct statement about interference pattern due to diffraction of light through single and double slits.
A. Constructive interference due to double slit and single slit interference occur when the path difference between interfering waves is half-integer multiple of the wavelength.
B. The bright and dark fringes formed when light passes through double slit have equal widths, whereas the central bright fringe formed in single slit diffraction is twice wider as the first order maxima.
C. The light intensity of interference pattern due to double and single slits remain the same as the distance from the central bright fringe increases.
D. The central bright fringes in both double and single slits interference are twice as wide as the first order bright fringes.

50. In order for interference of light waves to occur, the interfering light waves should
A. have constant phase difference and the same frequency and speed.
B. be of different types
C. be coherent.
D. be in phase

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BOOKLET CODE; 40

1. During the diffraction of light
   a. light passing through a circular aperture results circular bright and dark fringes.
   b. light passing through a single slit produces bright and dark fringes of different widths.
   c. decreasing the width of a single slit produces a wider bright fringe.
   d As the number of slits of a diffraction grating increase, slit width decrease and the bright fringes formed appear brighter, taller and narrower.

Answer; D

2. According to Millikan's oil drop experiment in the absence of an electric field the droplet moves with its terminal velocity and the weight of the droplet is balanced by the drag force which isdirectly proportional to the terminal velocity. This fact helps him in determining the mass of the droplet.

Answer; A

3. Given; e = p = q = 1.6×10^{-19} C, $r_1 = 0.110$ nm = 11×10^{-11} m, $r_2 = 0.100$ nm = 10×10^{-11} m

Solution; The change in electrical potential energy is given by;

$$\Delta U = kq^2\left(\frac{1}{r_2} - \frac{1}{r_1}\right) = 9\times10^9\, Nm^2/C^2 (1.6\times10^{-19}\, C)^2\left(\frac{10^{11}}{10m} - \frac{10^{11}}{11m}\right)$$

$$= 9\times(1.6)^2\times10^{-18}\left(\frac{11-10}{110}\right)J = 9\times(1.6)^2\times10^{-19}\left(\frac{1}{11}\right)J$$

Since; 1 eV = 1 v x 1.6×10^{-19} C = 1.6×10^{-19} J, therefore;

$$\Delta U = \frac{9\times1.6}{11} eV = 1.309 eV$$

Answer; C

4. When an insulator such as a dielectric is placed in an external uniform electric field between the plates of oppositely charged parallel plate capacitor the external electric field decrease in size and electric dipoles are induced within the insulator. This induced electric field opposes the external electric field.

Answer; B

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5. For a parallel plate capacitor, the capacitance, the voltage and the energy stored are given as follows.

$C = \frac{\varepsilon_o A}{d}$, $V = Ed$ and energy stored

$= U = \frac{1}{2}CV^2 = \frac{1}{2}\left(\frac{\varepsilon_o A}{d}\right)(Ed)^2 = \frac{1}{2}\varepsilon_o AdE^2$

Energy density is given by. $u = \frac{U}{Volume} = \left(\frac{\varepsilon_o AdE^2}{2Ad}\right) = \frac{1}{2}\varepsilon_o E^2$

The energy density stored by the capacitor is directly proportional to the square of the electric field.

Answer; C

6. When voltage is applied between the ends of a conductor of length 'L' cross-sectional area 'A' and containing 'n' free electrons per unit volume of the conductor, the total charge that flow through the conductor is given by $Q = n(AL)q$. The current, I is given by

$I = \frac{Q}{t} = \frac{nALq}{t} = nAvq$. The current density, 'J' is a vector quantity and is given by,

$J = \frac{I}{A} = nqv$. From this the direction of the current density is in the direction of the drift velocity of the free charge carriers.

Answer; C.

7. Given; three resistors with resistance, $R_1 = 2\Omega$, $R_2 = 3\Omega$ and $R_3 = 6\Omega$ are connected across a battery that supplies a voltage, V = 110 v as shown in the circuit. From the circuit R_2 and R_3 are joined in parallel and their combined resistance is

$R_{23} = \frac{R_2 \times R_3}{R_2 + R_3} = \frac{3\Omega \times 6\Omega}{3\Omega + 6\Omega} = 2\Omega$. The total resistance of the circuit is $R = R_1 + R_{23} = 4\Omega$

The total current is, $I = \frac{V}{R} = \frac{110V}{4\Omega} = 27.5A = I_1 = I_{23}$ and the voltage drop across R_1 and R_{23} is

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$V_1 = IR_1 = 27.5A \times 2\Omega = 55V, V_{23} = IR_{23} = 27.5A \times 2\Omega = 55V = V_2 = V_3.$

Thus, $V_1 = V_2 = V_3 = 55V.$

Answer; D

8. According to Kirchhoff's junction or point rule the sum of currents (charge entering) entering into a junction is equal to the sum of currents (charge leaving) leaving the junction.

Answer; C

9. A galvanometer can be modified as an ammeter to read large current by connecting a smallresistance known as shunt resistor in parallel to the coil resistance of the galvanometer.

A galvanometer can be modified as a voltmeter by connecting a very large resistor known as multiplier resistor in series to the coil resistance of the galvanometer.

Answer; C

10. Given; $R_g = 100\Omega, \quad I_g = 1mA = 0.001A,$ Full scale deflection current = $I = 1.0A$

Full scale deflection voltage = V = 100V.

Solution; the shunt resistor required is;

$$R_s = \frac{I_g R_g}{I - I_g} = \frac{10^{-3} A \times 100\Omega}{1A - 0.001A} = \frac{0.1V}{0.999A} = 0.1\Omega.$$

The multiplier resistor required is;

$$R_m = \frac{V - I_g R_g}{I_g} = \frac{100V - 0.1V}{0.001A} = 9.99 \times 10^4 \Omega.$$

-No answer.

11. A body under the action of a number of forces is said to be in equilibrium if it satisfies the two conditions. Thus,

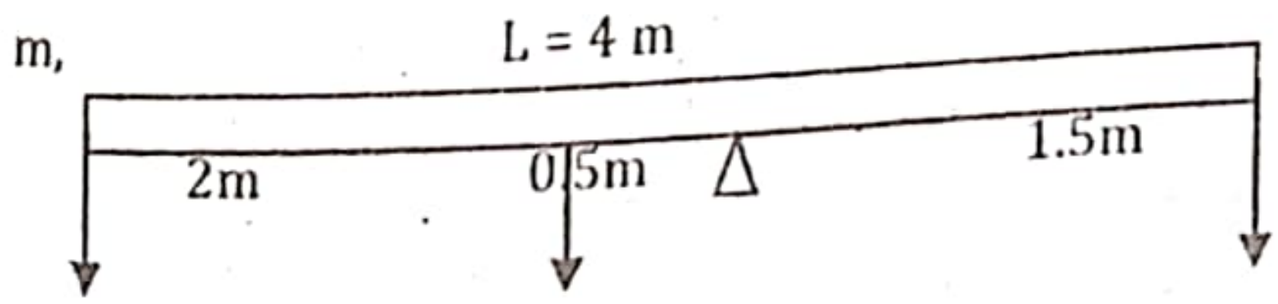
According to the first condition the net force acting on it must be zero.

$d\Sigma\vec{F} = 0.$

According to the second condition ; 'the net torque acting on it must be zero'. $\Sigma\vec{\tau} = 0$

Answer; C

12. Given; $W_{boy} = m_b g = 30kg \times 10m/s^2, W_{sea} = m_{sea} g = 20kg \times 10m/s^2$, L= 4



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$W_{boy} = 300\ N$ $W_S = 200\ N$ W_{man}

Taking moment at the pivot point, the weight of the boy the sea saw produce

a counter clock wise torque given by;

$\Sigma\tau_{anti} = W_{boy}x2.5m + W_{sea}x0.5m = 300Nx2.5m + 200Nx0.5m = 850Nm$

The weight of the man produces a clock wise torque given by;

$$\Sigma\tau_{clo} = W_{man}x1.5m$$

From the second condition of equilibrium we have;

$$\tau_{clock} = \Sigma\tau_{anti} \Rightarrow W_{man}x1.5m = 850Nm \Rightarrow W_{man} = \frac{850Nm}{1.5m} = 567N.$$

Answer; D

13. Given;L=4m,

$\Delta L = 0.20cm = 2x10^{-3}m, F = 5000N, A = 0.50cm^2 = 5x10^{-5}m^2.$

Solution; The tensile stress of the material is,

$$T = F/A = \frac{5000N}{5x10^{-5}m^2} = 10^8\ Pa.$$

The tensile strain is given by; $E = \frac{\Delta L}{L} = \frac{2x10^{-3}m}{4m} = 5x10^{-4}$

The Young's modulus of the material is given by;

$$Y = \frac{Tensilestress}{Tensilestrain} = \frac{T}{E} = \frac{10^8\ N/m^2}{5x10^{-4}} = 2x10^{11}\ N/m^2.$$

Answer; D

14. Pascal's principle is applied in hydraulic brakes for vehicles, in a hydraulic lift as a force multiplying machine and in pneumatic application. Since a hydraulic machine cannot multiply force and speed at the same time so it multiplies only force.

Answer; D

15. The streamlining of objects moving in a fluid depends on

i. compressibility of the fluid ii. density of the fluid,
iii. shape and surface area of the object moving in a fluid;
for example birds and air planes flying in air, ships sailing in water have pointing nose.

but, the streamlining of objects moving in a fluid does not depend on the mass of the object.

Answer; C

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CamScanner

16. Heat is energy that can be transferred one object to the other due to temperature difference.Temperature indicates the direction of heat flow. The internal energy of a system depend on its mass or on the number of molecules in the system.
Temperature is the measure of the average molecular kinetic energy of a gas.

Answer; D

17. . Given; n = 2 moles, T = 308 K, V = 5.00 x 10^{-3} m^3, R = 8.314 J/molK,
Required; P = ?

Solution; From the general gas equation we have,

$$PV = nRT \Rightarrow P = \frac{nRT}{V} = \frac{2mol \times 8.314J/molK \times 308K}{5 \times 10^{-3} m^3}$$

$$P = 1{,}024 \times 10^3 Pa = 1.02 \times 10^6 Pa$$

Answer; A

18. Given; n = 12 mol, Molar mass of H_2O molecule = 18 g, N_A = 6.02 x 10^{23} par/mol.

Solution; The number of water molecules is,

$$N = n \times N_A = 12mol \times 6.02 \times 10^{23} molecules/mol = 72.24 \times 10^{23} molecules$$

$$Massofwater = m = n \times M = 12mol \times 18g/mol = 216g = 0.216kg$$

$$Massofeachwatermolecule = m_p = \frac{M}{N_A} = \frac{18g/mol}{6{,}02 \times 10^{23} molecules/mol} = 2.99 \times 10^{-26} kg$$

No answer;

19. . During an isobaric process the pressure of the gas is constant and the p-v diagram is a horizontal line.
During an isothermal process the temperature of the gas is constant and the p-v diagram is a parabola.
An isochoric process occurs at constant volume of a gas and the p-v diagram is a vertical line.
During an adiabatic process there is no heat added or removed from the system and the p-v diagram is a decreasing(hyperbolic) plot.

Answer; D

20. Given; heat entering the system = $\Delta Q = 1.5 \times 10^5 J$, $P = 2.0 \times 10^5 Pa$

$$V_1 = 6.3m^3, V_2 = 7.1m^3 \Rightarrow \Delta V = V_2 - V_1 = 7.1m^3 - 6.3m^3 = 0.8m^3$$

$$\Delta W = -P\Delta V = 2 \times 10^5 Pa \times 0.8m^3 = -1.6 \times 10^5 J$$

$$\Delta U = \Delta Q + \Delta W = 1.5 \times 10^5 J - 1.6 \times 10^5 J = -0.1 \times 10^5 J = -1.0 \times 10^4 J.$$

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Answer; C

21. -Gravitational force is a long range attractive force between particles due their masses.

Weak nuclear force is a force which is responsible for radioactive decay.

-Electromagnetic force is attractive or repulsive force and acts on charged bodies.

-Strong nuclear force is a short range force that binds protons and neutrons in the nucleus.

Answer; B

22. Given; $M_1 = 1$ kg, $M_2 = 2$ kg, $M_3 = 3$ kg, F = 9 N, Required; a = ?, T_1 = ?, T_2 = ?

Solution; Applying Newton's 2nd law to each mass gives;

$$F_{net}(m_1) = M_1 a \Rightarrow T_1 = M_1 a = 1kg(a).....1$$

$$F_{net}(m_2) = M_2 a \Rightarrow T_2 - T_1 = 2kg(a) \Rightarrow T_2 - 1kg(a) = 2kg(a) \Rightarrow T_2 = 3kg(a)....2$$

$$F_{net}(m_3) = M_3 a \Rightarrow F - T_2 = 3kg(a) \Rightarrow 9N - 3kg(a) = 3kg(a)$$

$$6kg(a) = 9N \Rightarrow a = 9N/6kg = 1.5m/s^2$$

$$T_1 = 1kg(1.5m/s^2) = 1.5N, T_2 = 3kg(1.5m/s^2) = 4.5N$$

Answer; D

23. For an astronaut inside an international space orbiting the Earth the astronaut experiences weightlessness because the Earth's gravitational force on the astronaut is less than the Earth's force on the space station.

Answer; B

24. Given; distance of m_1 from the center of mass = r_1 = 3.0 m

distance of m_2 from the center of mass = r_2 = 1.0 m

Solution; When the system balances we have

$$\tau_{anti} = \tau_{clock} \Rightarrow m_1 g(r_1) = m_2 g(r_2) \Rightarrow m_1(3m) = m_2(1m)$$

$$\frac{m_1}{m_2} = \frac{1m}{3m} = \frac{1}{3}$$

Answer; D

25. Work is done when a net force is applied in the direction of the displacement.

- An object at rest may have energy depending on its position relative to Earth.

- A machine can multiply force or speed, but not energy.

-Both work and energy are scalars, whereas force is a vector quantity.

Answer; B

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26. Work done by a conservative force is

i. path independent, ii. the work done along any closed path is zero and

iii. recoverable. In general field forces are examples of conservative forces.

The work done by a non-conservative force is path dependent, irrecoverable and there is energy dissipation. Forces like, friction, air resistance and drag forces are good examples.

Answer; A

27. Given; m = 75.0 kg, height = h = 45 m, t = 52.0 s, g = 10m/s², Required; P = ?

Solution; the work done in lifting the mass is equal to the GPE stored in it and given by;

$$W = \Delta PE = mgh = 75kg \times 10m/s^2 \times 45m = 33,750J$$

$$P = \frac{W}{t} = \frac{33,750J}{52s} = 649J/s = 649W$$

Answer; D

28. For a rigid body that undergo rotation about a fixed axis;

-A point closer to the axis of rotation and farther away from the axis will have the same angular speed and angle turned in the same time, but a point farther away from the axis will has greater linear speed than a point closer to the axis.

Answer; A

29. Given; R = 0.10 m, m = 0.1 kg, F = 2 N, t = 2 sec, $\alpha = 400rad/s^2$.

Required; $\tau = ? and \omega = ?$

Solution; The torque exerted on the disk is;

$$\tau = FR = 2N \times 0.1m = 0.2Nm$$

The angular velocity of the disk is;

$$\omega = \omega_o + \alpha t = 0 + 400rad/s^2 \times 2s = 800rad/s.$$

Answer; D

30. When a net torque is exerted on a rigid body for some time interval the angular momentum of the rigid body changes. When a small torque is applied for a longer time and a large torque is exerted for a short period of time the change in angular momentum is the same.

Answer; B

31. A scientific method is a scientific procedure which ensures a rigorous evidence-based structure in which only ideas that have been carefully tasted are accepted as scientific theory.

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Answer; A

32. Precision is the degree of closeness or exactness of measured values to each other, whereas accuracy is the closeness of a repeatedly measured value to the true or accepted value.

Answer; B

33. In table of choice D, the displacement of the rolling ball increase by 2 m for every 1 sec increase in the time. Therefore;

$$v = v_{ave} = \frac{S_f - S_i}{t_f - t_i} = \frac{10m - 0}{5s - 0} = 2m/s = cons\tan t$$

Answer; D

34. Position vector describes the distance covered by a body in a specified direction.

Answer; A

35. Given; $F_1 = 10N, \theta = 30^o, F_2 = 20N, \beta = 60^o$. Required; R_{1x} = ?, and R_{2x} = ?

Solution; The x and y - components of each are given by;

$$F_{1X} = F_1 \cos\theta = 10N \times \cos 30^o = 10N(0.866) = 8.66N$$
$$F_{1Y} = F_1 \sin\theta = 10N \times \sin 30^o = 10N(0.5) = 5N$$
$$F_{2X} = F_2 \cos\beta = 20N \times \cos 60^o = 20N(0.5) = 10N$$
$$F_{2Y} = F_2 \sin\beta = 20N \times \sin 60^o = 20N(0.866) = 17.32N$$
$$R_X = F_{1X} + F_{2X} = 8.66N + 10N = 18.66N$$
$$R_Y = F_{1Y} + F_{2Y} = 5N + 17.32N = 22.32N$$

Answer; A

36. Given; $\vec{A} = 2\vec{i} - \vec{j}, \vec{B} = 4\vec{i} - 2\vec{j}, \vec{C} = \vec{i} - \vec{j}, and \vec{D} = 2\vec{i} + 3\vec{j}$

Solution; $|\vec{A}| = \sqrt{5}, |\vec{B} = 2\sqrt{5}, |\vec{C}\| = \sqrt{2}, and |\vec{D}| = \sqrt{13}.$

If two vectors are collinear, then the angle between the vectors should be 0° or 180°. Thus,

$$\vec{A} \bullet \vec{B} = A_X B_X + A_Y B_Y = 2(4) + (-1)(-2) = 10 = |\vec{A}\|\vec{B}|\cos 0^o = \sqrt{5}x2\sqrt{5} = 10.$$

Answer; D

37. Given; $\vec{S}_1 = 3km, East, \vec{S}_2 = 4km, North, and t = 2hrs.$

Solution; The resultant displacement is given by;

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$$\vec{S} = \vec{S}_1 + \vec{S}_2 = 3km, E + 4km, N \Rightarrow |\vec{S}| = \sqrt{(3km)^2 + (4km)^2} = 5km$$

$$|\vec{v}_{ave}| = \frac{S}{t} = \frac{5km}{2hr} = 2.5km/hr$$

$$S_{tot} = |\vec{S}_1| + |\vec{S}_2| = 3km + 4km = 7km$$

$$v_{ave} = \frac{S_{tot}}{t} = \frac{7km}{2hr} = 3.5km/hr$$

Answer; C

38. Given; $u = 20\ m/s, a = g = 10\ m/s^2$.

Solution; When the ball ascends its velocity uniformly decrease, by 10 m/s in every one second and as it reaches the maximum height its velocity becomes zero.The time taken to reach the maximum height is,

$$t = \frac{u}{g} = \frac{20m/s}{10m/s^2} = 2s.$$

The velocity after t = 1sec is,

$$v = u - gt = 20m/s - 10m/s^2(1s) = 10m/s$$

The maximum height raised in air is, $h_{max} = \frac{u^2}{2g} = \frac{(20m/s)^2}{2(10m/s^2)} = 20m.$

Answer; A

39. Centripetal force is a center seeking force that pulls the mass toward the center of the circle. According to Newton's second law of motion, both centripetal force directed to the center of the circle and both are also perpendicular to the tangential velocity.

Answer; C

40. According to Kepler's third law of planetary motion the square of period of revolution of the satellite is directly proportional to the cube of the mean distance between the satellite and the earth. Newton formulated his law of universal gravitation based on Kepler's third law. Therefore, the period of a satellite orbiting the earth increases as the radius of its orbit increases.

Answer; A

41. According to the kinetic theory of gases,

- the average KE of the particles of an ideal gas increase, as the temperature of the gas increases.
- the collision between the molecules is perfectly elastic.
- the internal energy of an ideal gas is only due to molecular KE of the molecules, because in an ideal gas molecular interaction is negligible.

Answer; C

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42. Given; A = 0.650 m, $\omega = 7.40 rad/s$.

$$f = \frac{\omega}{2\pi} = \frac{7.40 rad/s}{6.28} = 1.18 Hz$$

When t = 0, the oscillation begins from the amplitude position. The position of the object is,

$$x(t) = A\cos\omega t = A\cos(2\pi ft). = 0.650\cos(7.40t)m.$$

the acceleration of the object is given by;

$$a(t) = -\omega^2 A\cos(7.40t) = -35.59\cos(7.4t)m/s^2$$

$$v(t) = -\omega A\sin(\omega t) = -4.81\sin(7.4t)m/s$$

Answer; D

43. Given; m = 0.35 kg, f = 2.5 Hz, A = 0.15 m

Solution; The maximum speed of the SHM occurs when it passes the equilibrium position. Thus,

$$v_{max} = \omega A = 2\pi fA = 2(3,14)(2,5Hz)(0.15m) = 2.355m/s$$

$$ME = \frac{1}{2}mv^2{}_{max} = \frac{1}{2}(0.35kg)(2,355m/s)^2 = 0.97J$$

Answer; B

44. The speed of a transverse wave produced by a vibrating string fixed at both ends is affected by

i. the tension, T in the string and ii. on the linear density of the string according to the equation

$v = \sqrt{\frac{T}{\mu}}$ The speed of the wave increases as the tension in the string increases and as the linear density of the string decreases.

Answer; C

45. The intensity of sound depends on the power of the source and area covered by the sound.

It is directly proportional to the rate of sound energy (power of sound) transmitted and inversely proportional to the square of the distance between the source and the reciever.

Thus;

$$I = \frac{P}{A} = \frac{P}{4\pi R^2}.$$

Answer; C

46. Doppler effect is the alternate variation or the change in the frequency of a wave due to the relative motion of the source and the observer away or toward each other.

Answer; C

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47. Huygen's principle states that;
- every point on a wave front acts as a source of spherical secondary wavelets.
- the secondary wavelets spread in all directions with the same frequency, speed and wavelength as the original wave.
- a new wave front is formed when these wavelets combine.

Answer; D

48. During reflection of wave the speed, wavelength and frequency of the wave is the same before and after reflection.
- the angle of incidence is equal to the angle of reflection.
- During refraction of wave, if the speed of the incident wave is less than the speed of the refracted wave, the refracted wave bends away from the normal.
- during refraction of a wave both speed and wavelength change whereas frequency will remain the same.
- Snell's law states that the ratio of the sine incident angle to the sine of the refracted angle is equal to the ratio of the speed of the incident wave to the speed of the refracted wave.

Answer; C

49. The interference pattern due to diffraction when light pass through a double slit produces bright and dark fringes of equal widths.
- The interference pattern due to diffraction when light pass through a single slit, the central bright fringe appears twice as wide as the first order maxima.

Answer; B

50. In order for interference of light waves to occur coherent source of light must be used.
In order for two waves to be coherent, they must
- be the same type of wave,
- have same frequency and maintain constant phase relationship.

Answer; A

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1. Which of the following does a closed system allow to happen?
   A. Neither matter nor energy to cross a system boundery
   B. Only matter to cross a system boundery
   C. Only energy to cross a system boundery
   D. Matter and energy to cross a system boundery
2. The standard enthalpies of formation of HBr(g) and $Br_2(g)$ are -36 and +31 kJ mol^{-1} (at 298 K), respectively. What is the enthalpy change, ΔH for the following reaction? $H_2(g) + Br_2(g) \rightarrow 2HBr(g)$
   A. -103 kJ mol^{-1} B. + 103 kJ mol^{-1} C. -67 kJ mol^{-1} D. + 67kJ mol^{-1}
3. Which of the following statements about the standard state of a substance is correct?
   According to their observation, which one of the following elements has
   B. the highest melting and boiling point?
   A. The standard state of a substance is its most kinetically stable state under a pressure of 1.00×10^5 pa and at specified temperature, T
   B. The standard state of a substance is its most thermodynamically stable state under a pressure of 1.00×10^5 pa and at specified temperature, T
   C. The standared state of a substance is its most thermodynamically stable state under a pressure of 1.00×10^5 kpa an at specified temperature, T.
   D. The standard state of a substance is its most kinetically stable state under a pressure of 1.00×10^5 kpa and at specified temperature, T
4. Which of the following statements best describes th Second Law of Thermodynamics?
   A. The internal energy of the universe is constant.
   B. Enegy can neither be created nor destroyed.
   C. When an isolated system under goes a spontaneous change, the entropy of the system will increase.
   D. At absolute zero, the entropy of a perfect crystal is considered to be zero.
5. Colligative property is defined as a property that depend on
   A. The amount of solute present in the solution.
   B. the nature of pure liquid.
   C. the amount of pure liquid.
   D. the nature of solute present in the solution.
6. Which of the following is NOT the correct geometric configuration for for the given molecule?
   A. BF_3, trigonal planar C. NF_3, trigonal planar
   A SF_6, octahedral D. CH_4 , tetrahedral

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7. Which one of the folloing substances has London dispersion forces as its only inter molecular force? (No hydrogen bonding, no dipol-dipole forces).
   A. CH_4 B. H_2S C. NH_3 D. CH_3OH
8. Which of the following is a basic chemical bonding theory?
   A. Valance Shell Electron Repulsion (VSER) theory
   B. Ionic bonding theory
   C. Covelent Boning Theory.
   D. Molecular orbital theory.
9. Which type of intermolecular forces are present molecular crystalline solids?
   A. Metallic, ispesion, dipole diole, hydrogen bonding
   B. Covelent, dispersion, dipole-diole, hydrogen bonding
   C. Dispersion, dipole-dipole, hydrogen bonding
   D. Ionic, dispersion, dipole-dipole, hydrogen bonding
10. Which statement is NOT correct about the valance bond theory?
   A. According to the valance bond theory, each bond only involves two atoms.
   B. Valance bond theory treats electrons as localized between pairs of atoms.
   C. According to valance bond theory, molecular orbitals are formed by combining atomic orbitals.
   D. According to valance bond theory, hybride orbitals are formed by overlapping of atomic orbitals.
11. Which of the following statements is correct about fats and oils?
   A. They are soluble in polar solvents
   B. They burn without leaving ash
   C. They are heavier than water
   D. They are readly volatile
12. Which of the following is the correct procedure during the preparation of saop?
   A. Placing animal fats or vegetable oil in a reflex set up and boiling with concentrated solution of brine.
   B. Placing vegetable oil and animal fat in a reflux set up and boiling after addition of boiling chips.
   C. When saponification is complete, pouring the mixture quickly into a saturated NaOH solution.
   D. When saponification is complete, pouring the mixture quickly in to saturated NaCl solution
13. Which of the following discretion about a suspension is correct?
   A. A suspension contains particles which will not settle out if the suspension is left standing.

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B. A suspension is a "cloudy" mixture containg particles which will easly settle out when the suspension is left standing.
C. A suspension is a homogenous mixture.
D. A suspension is a heterogeneous mixture.

14. What is meant by the rate of dissolution of solute?
A. The dynamic equilibrium attained between the solute and the solvent
B. The maximum amount of solute that a pure solvent can hold in solution
C. The ability of solute to go into solution
D. The amount of solute that goes into solution per unit tome.

15. How can you prepare a super saturated solution of sodium thiosulfate, if the solubility of the salt at 25°C is 50 g per 100 g water?
A. By dissolving 70 g of sodium thiosulfate in hot 100 g water, carefully cooling the solution to 25°C and adding a seed crystal of the solute.
B. By dissolving 45 g of sodium thiosulfate in 95 g hot water and then cooling the solution to 25°C.
C. By dissolving 70 g of sodium thiosulfate in hot 100 g water and then carfully cooling the solution to 25°C.
D. By dissolving 45 g of sodium thiosulfate in 100 g water.

16. What is dynamic chemical equilibrium?
A. Dynamic chemical equilibrium is the state where the forward and backward reaction rates are equal to zero.
B. Dynamic chemical equilibrium is the state of equal opposite rates with equal concentrations of reactants and products.
C. At dynamic chemical equilibrium, reactants are converted to products and products are converted to reactants at an equal and constant rate.
D. Dynamic chemical equilibrium is the state where all reactants and products in the reaction are at rest.

17. The Reaction Quotient (Q) is
A. the ratio of concentration of the products to the concentrations of reactants to the power of their respective coefficients under equilibrium conditions.
B. the ratio of concentrations of the products to the concentrations of reactants to the power of their respective coefficients at any stage after the start of the reaction.
C. a quantity that does not change as a reaction system approaches equilibrium.
D. a measurement of the relative quantities of products and reactants present in a reaction mixture.

18. What factors affect chemical equilibrium in a chemical reaction?

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A. Change in reactant or product concentrations, pressure change, temperature, presence of catalyst.
B. Change in reactant or product concentrations, volume and pressure changes, temperature, presence of catalyst
C. Change in reactant(s) concentration(s), volume changes, temperature, presence of catalyst
D. Change in reactant or product concentrations, volume and pressure changes, temperature

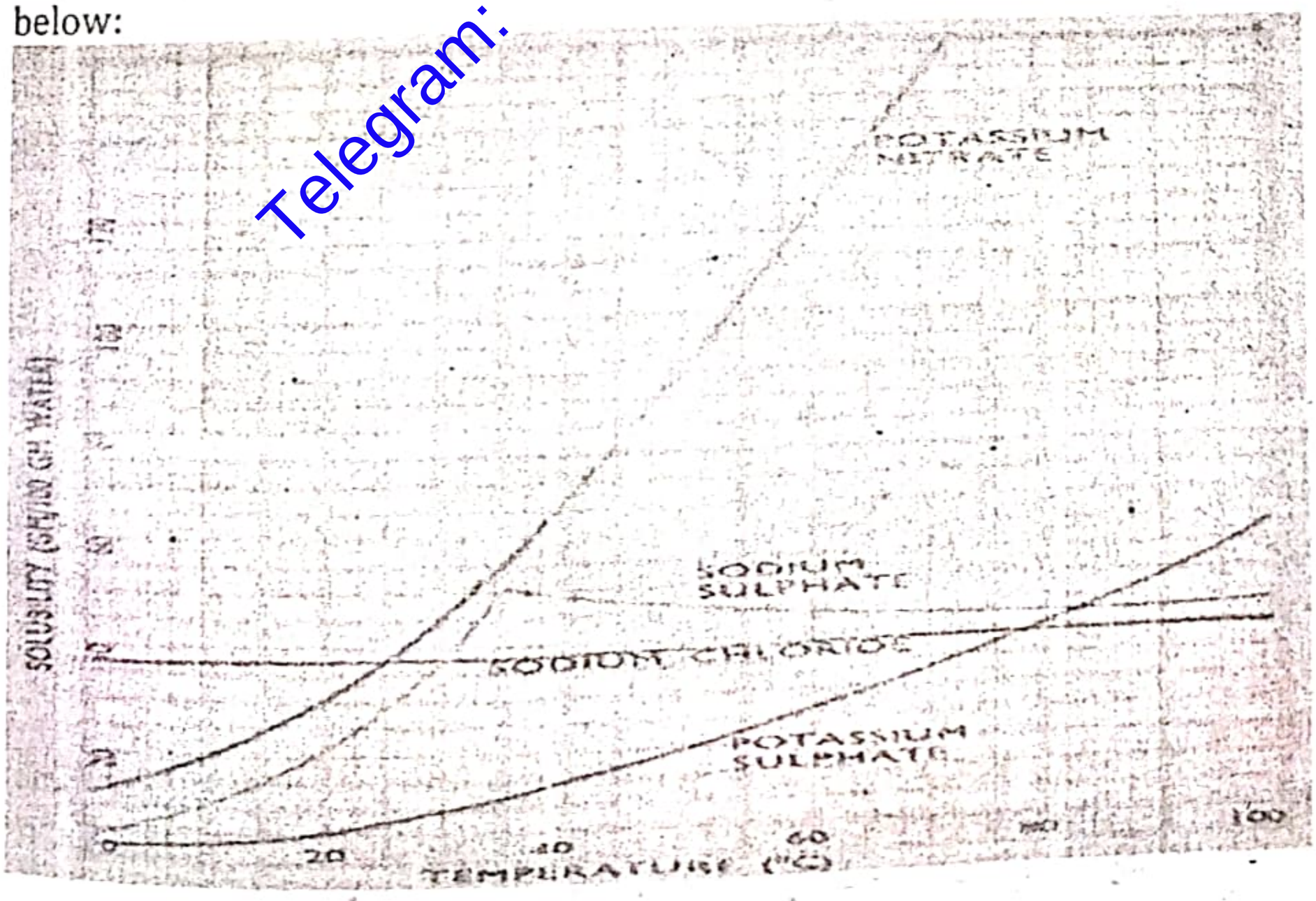
19. Which of the following are equal for a chemical system at equilibrium?
A. The rate of the forward and reverse reaction
B. The time that a particular atom or molecule stays as a reactant or product
C. The rate constant for the forward and reverse reactions
D. The concentrations of reactant and products

20. The equilibrium constant for the formation of calcium carbonate from the ions in solution is 2.2×10^{8} according to the reaction:
$Ca^{2+}(aq) + CO_3^{2-}(aq) \rightleftharpoons CaCO_3(s) \quad K = 2.2\times10^{8}$
What is the value of the equilibrium constant for the reverse of this reaction?

$$CaCO_3(s) \rightleftharpoons Ca^{2+}(aq) + CO_3^{2-}(aq)$$

A. 2.2×10^{8} B. 4.55×10^{-9} C. 4.4×10^{-8} D. 2.2×10^{-8}

21. The solubility curves of some common salts in water at 30°C are shown below:



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When 30 grams of sodium sulfate, Na_2SO_4, is dissolved in 100 grams of water at 20°C, the solution can be correctly escribed as

A. diluted
C. saturated
B. supersaturated
D. unsaturated

22. What is mole fraction? Mole fraction is
    A. mass of solute divided by mass of solution.
    B. moles of solute divided by kilogram of solvent.
    C. mass of a component divied by the total mass of all components.
    D. number of moles of a component divided by the total number of moles of all components.
23. A 5 g sample of ground water was found to contain 10.5 μg of Pb^{2+}. What is the concentration of Pb^{2+} in ppm?

    A. 3.4 B. 2.1 C. 1.7 D. 0.48

24. How can you prepare a 0.53 molal solution that contains 8.6% glucose $C_6H_{12}O_6$(w/w) in water?

    (Molecular mass of glucose = 180g/mol)

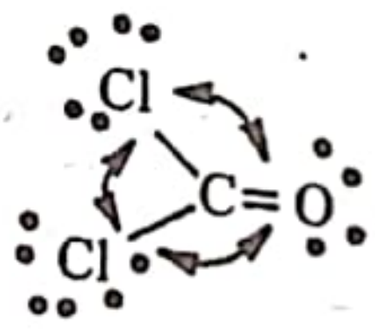
    A. By dissolving 8.6 g sugar in 914 g of water
    B. By dissolving 8.6 g sugar in 91.4 g of water
    C. By dissolving 86 g sugar in 100 g of water
    D. By dissolving 8.6 g sugar in 100 ml water
25. The thermal decomposition of potassium hydrogen carbonate yields potassium carbonate, carbon dioxide and water.

    $2KCO_3 \rightarrow K_2CO_3 + CO_2 + H_2O$

    How many moles of potassium carbonate can be roduced from 550 grams potassium hydrogen carbonate?

    A. 27.8 B. 2.78 C. 0.36 D. 3.60

26. Which of the following properties is true about covelent molecules?
    A. High electrical and thermal conductivity
    B. Low melting point and boiling points
    C. Low melting point but high boiling points
    D. High melting but low boiling points
27. For the molecule (phosgene) shown below, why do the lone pairs and double bonds affect the bond angles?



A. Lone pairs occupy less space, but multiple bonds occupy more space than single bonds.
B. Both multiple bons and lone pairs occupy less space.

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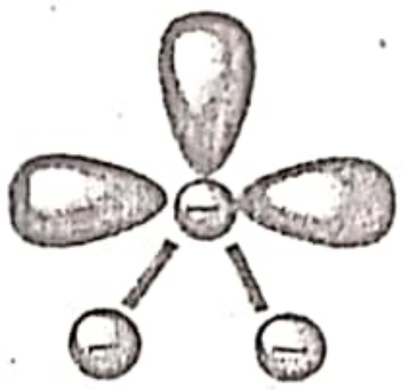
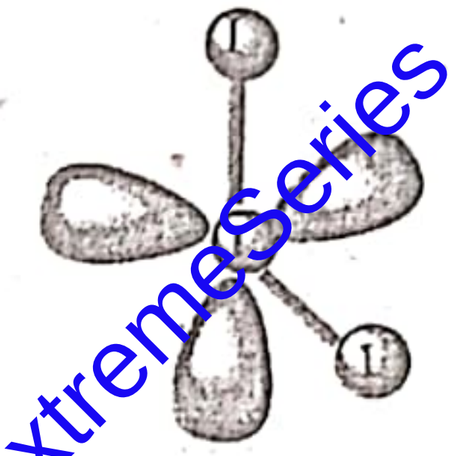
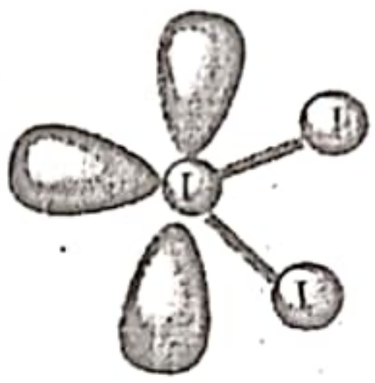
C. Lone pairs repel smaller than bonded pairs and multiple bonds have a higher charge density around the central atom.
D. Lone pairs repel more due to a greater charge density than bonding pairs.

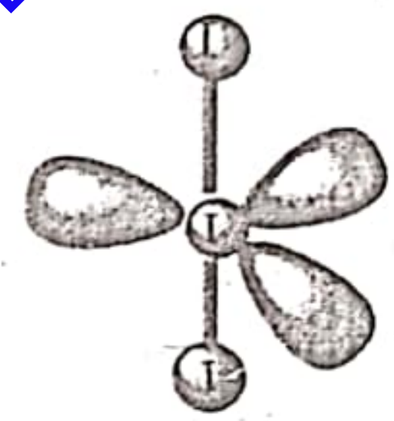
28. If you are provided with four covelent compounds namely naphthalene, graphite, iodine and ethanol to investigate their electrical conductivity, which one would you expect to conduct electricity either in the solid or in the liquid state?
A. Ethanol B. Iodine C. Graphite D. Naphthalene

29. Based on the VSEPR model, what shape woud you expect for the molecule $[I_3]^-$?

A. 

C 

B 

D 

30 Given the reaction:

$$NH_3 + H^+ \rightarrow NH_4^+$$

The bond formed between NH_3 and H^+ is
A. Nonpolar covelent C. Simple covelent
B. Coordinate covelent D. Ionic bond

31. If 3.423 grams of sucrose is dissolved in 200.0 g of water, what is molality of the sucrose solution? (molar mass of sucrose = 342.3 g/mol)
A. 0.01 m B. 0.5 m C. 0.1 m D. 0.05 m

32. What is the Arrhenius concept of base?
A. A substance that releases hydrogen ion in water is base.
B. A base furnishes a pair of electrons.
C. A substance that increases hydrogen ion in water is a base.
D. A base is protone acceptor.

33. The Brønsted-Lowry concept defines an acid as a substance that
A. furnishes proton to a molecule

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B. gives hydrogen ion in water.
C. accepts proton from a molecule.
D. produces hydroxide ion in water.

34. Which of the following explains the cleaning action of soap?
A. Cleaning action is not related to hydrophilic and hydrophobic nature of soap.
B. Oily dirt is absorbed into both hydrophilic and hydrophobic parts but not washed away.
C. Oily part is absorbed into the hydrophilic part of soap and washed away.
D. Oily dirt is absorbed into the hydrophobic part of soap and washed away.

35. Which of the following statement is WRONG about entropy and change of entropy (ΔS)?
A. ΔS decreases with randomness.
A. Entropy is a measure of randomness.
B. $\Delta S_{rxn} = \sum mS_{reactant} - \sum nS_{product}$
D. $\Delta S > 0$ indicates an increase in randomness.

36. Which of the following states the law of definite proportions?
A. Chemical compounds are formed when atoms combines in whole number ratios.
B. The mass of products in a chemical reaction must be equal to the mass of the reactants.
C. The total mass in the universe is always constant.
D. Every compound always contains the same elements in the same proportions by weight.

37. What did J.J Thomson contiribute with his experiments in cathode-ray tubes in understanding the atomc structure of the atom?
A. Catthode rays and electric charge were separate entities, and and not combined together.
B. The electric charge deposited was negative.
C. Cathode rays were actually electrons.
D. Cathoe rays and charge were combined together, and not separate entities.

38. An element X has two naturally occurring isotopes. X (10.013 amu) and X (11.01 amu). The first accounts for 19.9% of the total and the second accounts for remaining 80.1%. What is the average atomic mass of element X?
A. 10.125 B. 10.512 C. 10.211 D. 10.812

39. What are isotopes? Isotopes are atoms with the same number of protons but different numbers of
A. neutrons B. ions C. protons D. electrons

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40. Which statement is correct for the emission spectrum of hydrogen atom?
A. The line corresponding to the greates emission of energy is in the ultraviolet region.
B. The lines are produced when electrons move from lower to higher energy levels.
C. The lines in the visible region involve electron transitions into the energy level closest to the nucleus.
D. The lines are produced when electrons move from second to fourth.

41. Which of the following physical properties of a solvent are affected by the formation of solution?
A. Vapour pressure lowering and boiling point lowering.
B. Vapour pressure elevation, boiling point lowering, freezing point lowering, osmotic pressure.
C. Vapour pressure lowering, boiling pointlowering, freezing point lowering, osmotic pressure
D. Vapour pressure lowering lowering, boiling point elevation, freezing point lowering , osmotic pressure

42. A solution is made by dissolving 3.75g of pure non-volatile solute in 95g of octane. The BP of pure octane was observed to be 55. 95°C and that of solution was 56.50°C . Which of the following is correct based on the given evidence? If the dissolved solute is electrolyte. The boiling point of the solution
A. will be slightly higher than 55. 95°C
B. will be slightly lower than 56.50°C
C. will be slightly lower than 56.50°C
D. remains to be 56.50°C

43. In the following reaction, which is BrØnsted-Lowry base?
$HC_2O_4^-(aq) + H_2O(l) \rightarrow H_3O^+(aq) + C_2O_4^{2-}(aq)$
A. $C_2O_4^{2-}$ B. H_2O C. H_3O^+ D. $HC_2O_4^-$

44. In the aotoionization of water
A. There is no transfer of hydrogen ion from the acid or the base.
B. one water molecule acts as a Lewis base and the other water molecule acts as Lewis acid.
C. a proton is tranisfered from one water mololecule to another.
D. an electron is transferred from one water molecule to another.

45. Given the following acid-base reaction
$Ag^+ + NH_3 \leftrightarrows [Ag(NH_3)_2]^+$, the Lewis base is
A. $[Ag(NH_3)_2]^+$ B. $Ag^+$and NH_3 C. Ag^+ D. NH_3

46. Given a general reaction at equilibrium: $aA + bB \leftrightarrows mM + nN$
The equilibrium constant (K_c) expression is

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A. $K_c = \frac{[M]^m[N]^n}{[A]^a[B]^b}$

B. $K_c = [A]^a[B]^b - [M]^m[N]^n$

C. $K_c = [M]^m[N]^n - [A]^a[B]^b$

D. $K_c = \frac{[A]^a[B]^b}{[M]^m[N]^n}$

47. The first law of thermodynamics states that
    A. energy can be created or destroyed.
    B. the total amount of energy in the universe is constant.
    C. in absolute zero the entropy of a perfect crystal is considered to be zero.
    D. the entropy of universe increases in spontaneous process.
48. Given the production of hydrogen iodide: $H_2(g) + I_2(g) \rightleftarrows 2HI(g)$
    The equilibrium constant (K_p) expression is

A. $K_p = \frac{(P_{H_2})(P_{I_2})}{(P_{HI})^2}$

B. $K_p = (P_{H_2})(P_{I_2})$

C. $\frac{1}{(P_{H_2})(P_{I_2})}$

D. $K_p = \frac{(P_{HI})^2}{(P_{H_2})(P_{I_2})}$

49. Which of the following is a homogeneous reaction?
    A. $Fe^{3+}(aq) + 3OH^-(aq) \rightleftarrows Fe(OH)_2(s)$
    B. $N_2(g) + 3H_2(g) \rightleftarrows NH_3(g)$
    C. $CaCO_3(s) \rightleftarrows CaO(s) + CO_2(g)$
    D. $NH_4Cl(s) \rightleftarrows NH_3(g) + HCl(g)$
50. Which of the following procuder indicates a system which CANNOT exchange both energy and matter with surroundings?
    A. Baking *injera*
    B. Cooking shiro-wet in a closed pan
    C. Keepining hot water in thermostate
    D. Water boiling in an open beaker
    E. In III, the color of red litmus will be changed to blue.
26. The major fields of chemistry that study all the elements and their comounds with the exception of carbon and its compounds are known as
    A. analaytical
    B. inorganic
    C. physical
    D. organic
27. Which of the following correctly defines accuracy of a measurements?
    A. The degree to which reeated measurments under the same conditions ar unchanged.
    B. Measurments that are close to the known value.
    C. Measurments that have been repeated several times.
    D. Measurments that are close each other.
3. How many significant figures does a measurement of 0.00134 m^3 have?
    A. Six
    B. Five
    C. Four
    D. Three

What is a scientific method? It is a method of
A. quantitative observations and experiments.

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B. investigation involving experimentation and observation to aquire new knowledge, solve roblems, and answer questions.
C. searching for information based only on past scientific research.
D. gaining knowledge through theory modified as needed.

55. A hypothesis step in the scientific mothod is
A. the step at which data collected by observations and experiments.
B. an explanation of the general principles of certain phenomena.
C. a ossible explanation for an observation.
D. a verbal statement or a mathematical equation that summarizes observations.

56. Which of the following factors will affect the rate of a chemical reaction?
A. Humidity, presence of catalyst, concentration or pressure of reactants, surface area of solid reactant
B. Temperature, volume of container, presence of catalyst, concentration or pressure of reactants
C. Temperature, presence of catalyst, concentration or pressure of reactants, surface area of solid reactant
D. Temperature, volume of container, presence of catalyst, concentration or pressure of reactants, surface area of solid reactant

57. Which statement is correct for the reaction: $2A + B \rightarrow C$?
A. The rate of disappearance of B is twice that of A.
B. The rate of appearance of C equals the rate of disappearance of A.
C. The rate of disappearance of A is twice that of B.
D. C appears at a rate equal to the sum of the rates of the disappearance of A and B.

58. The theory that states "the collision between two reacting species results in the formation of an activated complex" is known as
A. intermediate state theory.
B. rate theory.
C. transition state.
D. collision theory.

59. For the reaction: $A + B \rightarrow C$, the ate equation is r=k[A] Which of the following statement is NOT correct?
A. The unit of the rate constant is s^{-1}
B. The rate depends on the concentration of A.
C. A lot of [A] againist time is linear.
D. The rate is independent of the concentration of B.

60. What is mwant by rate-determing step in chemical kinetics?
A. The rate determing step is the mechanism step with the average activation energy.
B. The rate determing step is the mechanism step with the least activation energy.

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C. The rate determing step is the fastest step of a chemical reaction that determines the speed (rate) at which the overall reaction proceeds.

D. The rate determing step is the slowest step of a chemical reaction that determines the speed (rate) at wich the overall reaction proceeds.

61. How much energy would be released as electron is moved from the n=4 to the n=3 energy level? ($(RH = 2.18\times10^{-8}\ J)$

A. 3.981×10^{-18} J

B. 2.043×10^{-18}J

C. 1.938×10^{-19} J

D. 1.050×10^{-19}J

62. Which set of quantum numbers uniquely defines one of the electrons in an atomic orbital with n=2 and l=0?

A. n=2, l=0, m_l=1 m_s=$+\frac{1}{2}$

B. n=2, l=0, m_l=1 m_s=+1

C. n=2, l=0, m_l=0 m_s=$+\frac{1}{2}$

D. n=2, l=0, m_l=0 m_s=+1

63. What is Pauli Exclusion Principle? The Pauli Exclusion Principle states that

A. an orbital can hold a maximum of two electrons with the same quantum number ms.

B. two elctros in an atom can have the same set of the four quantum nubers.

C. no two electrons in an atom can have the same set of the four quantum numbers.

D. all electrons in an atom are in the same energy state.

64. Which of the following is correct relationship between the given parameter and the property of an EMR?

A. The lower the frequency of an EMR, the higher is its wave length.

B. The higher the value of the wave length of an EMR,the greater is its energy.

C. The lower the value of the frequency of an EMR, the higher is is the energy of an EMR.

D. The higher the value of the wave length of an EMR, the greater is its frequency.

65. When equal amount of sodium chloride lead iodide are dissolved in water, which of the following statements are true regarding the electrical conductivity of the dissolved ionic solution?

A. The electrical conductivity depends on the type of electrodes.

B. Both solutions will have the same electrical conductivity.

C. The lead iodide solution will have higher electrical conductivity.

D. The sodium chloride solution will have higher electrical conductivity.

66. Which of the following explanation is correct about vapour pressure and its relation with phase changes?

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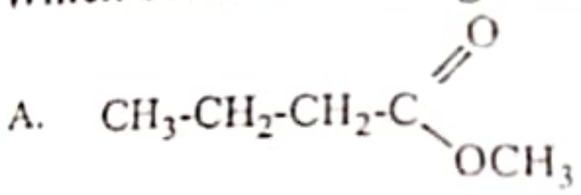
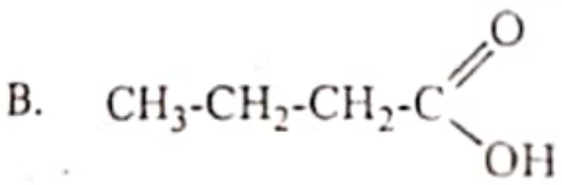
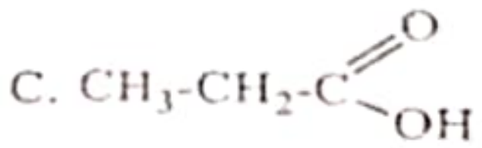
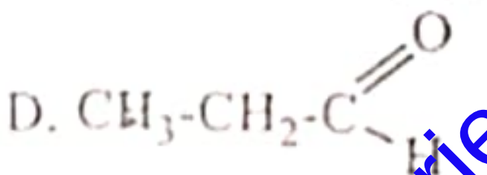
A. Rate of evaporation increases as higher number of molecules go into the vapor phase.
B. Rate of condensation increases as lower number of molecules go into the vapour phase.
C. As the temperature decreased more molecules go from liquid phase to the vapor phase.
D. As the temperature is increased; more molecules go from liquid phase to the vapor phase.

67. Which of the following structural formula represents propanoic acid?

A. 

B. 

C. 

D. 

68. Acetic acid is manufactured industrially by the fermentation of
A. Methanal B. ethanal C. ethanol D. methanol

69. Which of the following DOES NOT represent an ester with molecular formula of $C_5O_2H_{10}$?
A. pentyl formate
B. Buthyl methanoate
C. Ethyl propanoate
D. Propyl ethanoate

70 What is the characteristic smell of an ester?
A. Fish like smell
B. Fruity like smell
C. Alcoholic smell
D. Rotten egg small

71. What is the pH and $[H^+]$ of a 0.2M solution of HCN? ($K_a = 4.9\times10^{-10}$)
A. pH=5; $[H^+] = 4.9\times10^{-10}$
B. pH=5; $[H^+] = 9.9\times10^{-6}$
C. pH=3; $[H^+] = 4.9\times10^{-10}$
D. pH=3; $[H^+] = 4.9\times10^{-6}$

72. A common ion effect in a certain week acid will make the solution less acidic. This is because the presence of a common ion
A. enhances the ionization of salt of weak acid.
B. suppresses the ionization of weak acid.
C. enhances the ionization of the weak acid.
D. surppresses the ionization of the weak acid.

73. A salt is formed from weak acid (HCN) whose $K_a = 4.9\times10^{-10}$ and a weak base whose $K_b = 1{,}8\times10^{-5}$. What would you expect abaout the solution of the salt formed from HCN and NH_3?
A. The solution can be either acidic or basic at the same time.
B. The solution will be neutral, because the concentration od OH^- equals that of H^+
C. The solution will be basic, because the concentration of OH^- is less than that of H^+.
D. The solution will be basic, because the concentration of OH^- is greater than that of H^+.

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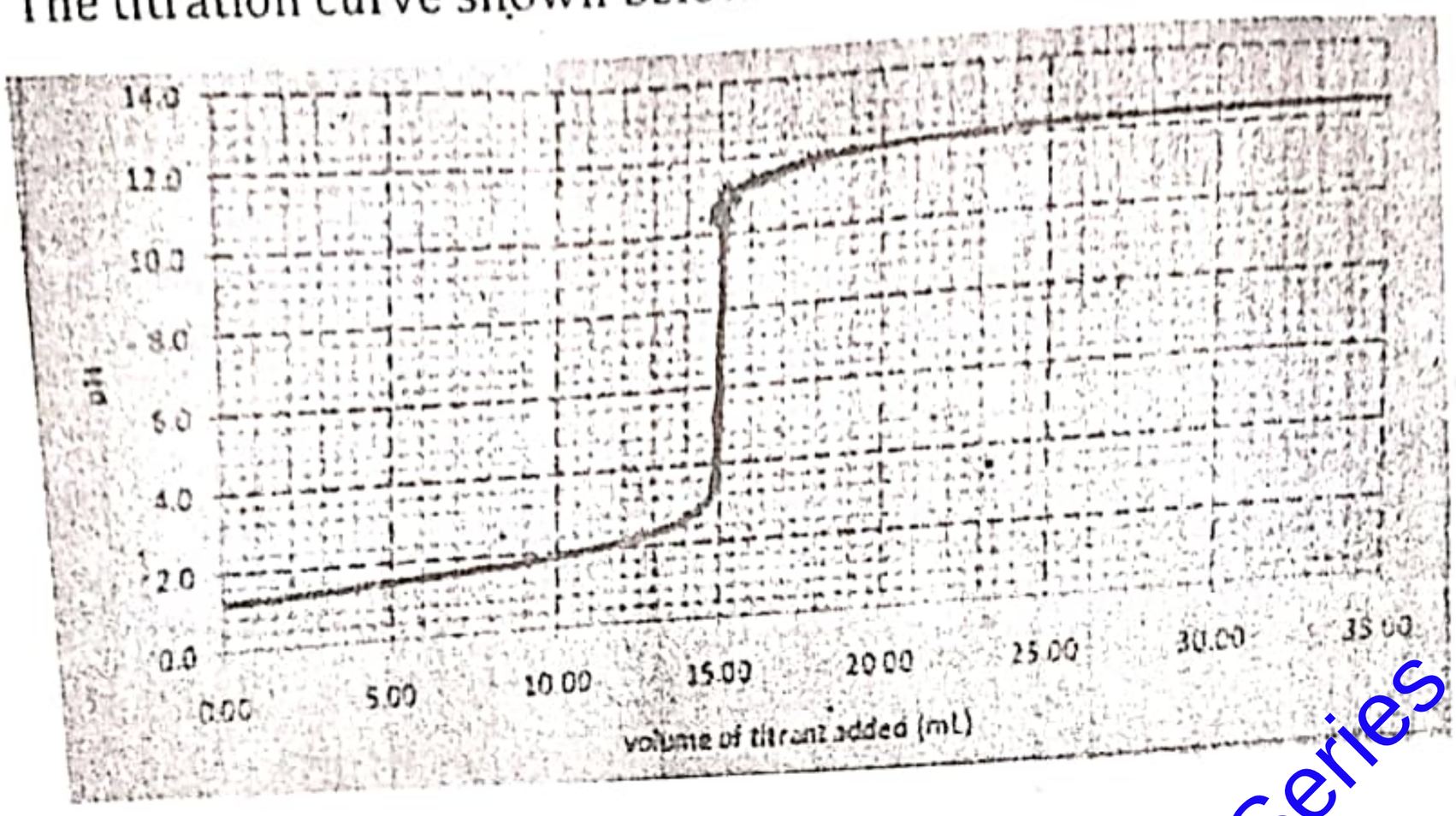
74. The titration curve shown below shows the titration of



A. a weak acid with strong base.
B. a weak base with strong acid.
C. a weak acid with a weak base.
D. a strong base with a strog acid.

75. What is the difference between equivalence point and end point in an acid base titration?
A. At the equivalence point, the pH is always 7 and at the end point there is a color change.
B. At the end point, the pH is always 7 and at the equivalence point there is a color change.
C. Equivalence oint is where stoichiometrically equivalent amounts of the analyte and titrant react and end point where an observable physical change occurs in a titration.
D. Equivalence point is where an oservable physical change occurs in a titration.

76. Which of the following represents a net ionic equation?
A. $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$
B. $Cr_2O_7^{2-} + 6Fe^{2+} + 14H^+ \rightarrow 2Cr^{3+}6Fe^{3+} + 7H_2O$
C. $AgNO_3(aq) + NaCl(aq)AgCl(s) + NaNO_3(aq)$
D. $Na^+(aq) + OH^-(aq) + H^+(aq) + Cl^-(aq) \rightarrow NaCl(s) + H_2O$

77. Which of the following is correct about the solution process?
A. A dilution rocess is carried out by adining a solute.
B. Addining solute to a stock solution gives less concentrated solution.
C. Addining solvent to a stock solution to get less concentrated solution.
D. A dilution process gives a concentrated solution.

78. The number of equivalent per liter of solution is known as
A. mole fraction
B. molality
C. normality
D. molarity

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79. When silver nitrate solution is mixed sodium chloride solution it gives an insoluble silver chloride. Which of the following is correct?
A. Silver and chloride ions are reacting species.
B. Sodium ion is spectator ion while nitrate is reacting ion.
C. Silver and chloride ions are spectator ions.
D. Nitrate and sodium ions are reacting species.

80. Which of the following explains the change of colligative properties of a solvent upon addition of a non-volatile solute?
A. The nature of the solute determines the volatility of the solvent.
B. The solvent shows weak intermolecular interactions.
C. The solute covers the surface of the solvent and results in lowering of vapour pressure.
D. Lower amount of a non-volatile solute results in a higher vapour pressure change.

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1. B In a closed system energy can exchange between the system and surrounding but not matter.

1. A $\Delta H_{rxn} = \sum nH_{formation}(product) - \sum nH_{formation}(reactant)$
$= 2N(-36\,kJ/mol^{-1}) - (31kJ/mol^{-1} + 0)$
$= -72\,kJ\,mol^{-1} - 31kJ/mol^{-1}$
$= -103kJ/mol^{-1}$
Note that standard enthalpy of formation of the most stable form of an element is zero.

2. B Standard state of a substance is the form most stable at the particular temperature of interest (usually 25°C) and at standard atmospheric pressure (1 atm= 1.01325×10^5pa=101.325 kPa)

3. C The second law of thermodynamics can be stated in different ways. The most common statement in chemical contexts is that *in any spontaneous process the entropy of universe increases.*

4. A Colligative properties are properties that depend on the quantity (amount) or concentration of the solute particle.

5. C NH_3 has one lone pair and three bonding pairs, and take the form AL_3E form and the geometric shape of the molecule is *trigonal pyramidal.*

6. A All the given molecules are polar except CH_4. London dispersion forces exist in all kinds of molecules, but more important in nonpolar molecules.

7. D Valance bond theory and Molecular orbital theories are the two basic chemical bonding theories.

8. C Metallic covelent and ionic bonds are intramolecular forces not intermolecular forces.

9. C Molecular orbitals are formed by combining atomic orbitals according to *molecular orbital (MO)* theory not according valance bond theory.

10. B Fats and oils are:
- soluble in non polar solvents.
- not readly volatile.
- are lighter than water.
- odorless and colorless.

11. D The main procedure is: Setup a reflux apparatus→place vegetable oil or animal fat in ditilation flask→ add 20% NaOH solution→ add few boiling chip→ when saponification is complete pour the mixture into brine (concentrate NaCl solution)→ collect the precipitate and wash it with ice-cold distilled water and dry it.

12. *B and D* Suspensions are heterogeneous mixtures whose particles

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appear cloudy and settle down upon standing.

13. D Rate of dissolution is the speed with which solute grese into solution.

14. C As the solubility of the given salt is 50g/100g of water, the solution which contain more than 50 g of the salt at the given temperature is supersaturated solution.

15. C At dynamic chemical equilibrium, rates of forward and back ward are equal.

17. B For the general reaction: $aA + bB \rightleftharpoons cC + dD$, the reaction quotient (Q) is given as $Q = \frac{[C]^c[D]^d}{[A]^a[B]^b}$ where the concetrations are concentrations at any stage of the reaction.

18. A Factors that can affect chemical equilibrium are: Temperature, concentration, pressure and volume.

19. A See explanation in number 16 above.

20. B If the equilibrium constant for a reaction is K, the equilibrium constant for the reverse reaction is $\frac{1}{K}$. Therefor the equilibrium constant for the reverse reaction is $\frac{1}{4.3\times10^{8}} = 0.23\times10^{-8}$
$= 2.33\times10^{-9}$

21. B From the solubility curve given, at 20 °C the solubility of the salt is 20 g per 100 g of water. That the solution is saturated and the solution containg more than 20 g (i.e. 30 g) is supersaturated solution.

22. D Number of moles of component divided by the toltal numbers of moles of all components is said to be mole fraction of the component.

23. B $ppm = \frac{mass\ of\ solute}{mass\ of\ solution} \times 10^{6}$
$= \frac{10.5\times10^{-6}\ g}{5\ g} \times 10^{6}$
$= 2.1$ ppm

24. B From the given percentage (%), if mass of the solution is 100 g, mass of glucose is 8.6 g and mass of water is 100-8.6=91.4 g=0.0914 kg.

Then the number of moles of glucose is:

$$n = \frac{8.6\ g}{180\ g/mol} = 0.048\ mol$$

$$molality = \frac{n\ of\ solute}{mass\ in\ kg\ of\ solvent} = \frac{0.048\ mol}{0.0914\ kg} = 0.525\ molal$$

25. B First calculate number of mole of $KHCO_3$:

$$n = \frac{given\ mass}{molar\ mass} = \frac{550\ g}{100\ g/mol} = 5.50\ mol$$

Now calculate mole of K_2CO_3 from the balanced chemical calculation: $2KHCO_3 \rightarrow K_2CO_3 + CO_2 + H_2O$

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$$\frac{x}{1\ mol} = \frac{5.5\ mol}{2\ mol} \Rightarrow x = 2.75\ mol$$

26. B Covelent compounds have generally low melting and boiling points and are non conductors of heat and electricity.

27. The shape of the given molecule is trigonal planar and the bond angles are 120°. There is no lone pair on the centeral atom (has no effect on the bond angles). The three sets of electrons repeal each other nearly equally.

28. C Graphite is the only non metal that can cundut electricity.

29. D The molecule $[I_3]^-$ takes the form AX_2E_3 type and has linear shape.

30. B Amonia molecule (NH_3) donate lone pair to H^+ and coordinate covalent bond is formed.

31. D Given: mass of sucrose=3.423 g, molar mass of sucrose=342.3 g/mol and mass of water=200.0 g=0.2kg. First calculate number of mole of sucrose

$$n = \frac{\text{given mass}}{\text{molar mass}} = \frac{3.423g}{342.3\ g/mol} = 0.01\ mol$$

Now calculate molality

$$\text{molarity} = \frac{\text{number of mole of solut}}{\text{mass in kg of solvent}} = \frac{0.01\ mol}{0.2kg} = 0.05\ molal$$

32.C Accordining to Arrhenius acid-base concept, acid is the substance that produce H^+ in water and base is a sabstance the produce OH^- in water.

33. A According to BrØnsted-Lowry, acid is proton donar and base is proton acceptor.

34. D. Soap has two ends; polar and non-polar ends. The oily dirt is dissolved by the non-polar part (hydrophobic end) of the the soap which is in the middle of micells that washed away by water.

35. C. $\Delta S_{rxn} = \sum nS_{product} - \sum nS_{reactant}$

36.D Accoding to the law of definite proportion, a pure compound always contains the same elements in the same proportion by mass.

37.C J.J. Thomson in his discharge tube experments showed that cathode rays consist of negatively charged particles called electros.

38.D Given: $A_1 = 10.013$, $f_1 = 19.9\%$, : $A_2 = 11.01$ and $f_2 = 80.1\%$.Then

$$\text{average atomic mass} = \frac{A_1 f_1}{100} + \frac{A_2 f_2}{100} = \frac{10.013 \times 19.9}{100} + \frac{11.01 \times 80.1}{100}$$

$$= 1.993 + 8.819$$

$$= 10.812$$

39. A Isotopes are atoms of the same element with the same atomic number (number of proton) but different number of neutron or atomic mass.

40. A When electrons drop from any energy level to n=1, Lyman series which is ultraviolate (Uv) emission spectrum is formed.

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41. D The colligative properties of solution are: Vopor pressure lowering, boiling point elevation, freezing point depression and osmotic pressure.

42. C Colligative properties of electrolyte solution are higher than that of non electrolyte solutions.

43. B In the reaction the H_2O accept proton (H^+) and changed to H_3O^+.

44. C One water molecule act as Brnsted-Lowry acid and the one water molecule act as Brnsted-Lowry base (Proton tranisfer from one water molecule to to another water molecule).

45. D Ag^+ is electron deficient an act as Lewis acid, while NH_3 has lone pair of electron and it is electron rich and act as Lewis base.

46. A Accoring to the law of mass action, the equilibrium constant K_c is the ratio of concentrations of the products to the concentrations of reactants to the power of their respective coefficients at equilibrium.

47. B According to the second law of thermodynamics energy of the universe is conserved.

48. A K_p and K_c are difined similarly, but K_p is defined for gaseous reactants and products interms of partial pressures.

49. B Homogeneous reaction is a reaction whose both the reactant and product are in gaseous or liquid phase.

50. C Thermostat is example of isolated system.

51. B Inorganic chemistry is branch of chemistry which study about all compounds except organic compounds.

52. B Accuracy is degree to which measurments close to true or actual value.

53. D Zeros that are not proceeded by non zero digit are not significant.

54. B A scientific method is a general approach to solve probles which involves making observation, formulating hypothesis and testing hypothesis.

55. C Hypotheis is tentative explanation for observation.

56. C Factors that can affect rate of chemical reaction are: Concentration, pressure, temperature, surface area of reactant and catalyst.

42. C Rate of reaction $= -\frac{\Delta[A]}{2\Delta t} = -\frac{\Delta[B]}{\Delta t} = \frac{\Delta[C]}{\Delta t}$, from this

Rate of disappearance of B is half of ratr of disappearance of A.

Rate of appearance of C is half of of rate of disappearance of A.

Rate of appearance of C is equal to rate of disappearance of B.

Rate of disappearance of A is twice the rate of disappearance of B.

43. C According to Transition state Theory or Activated-Complex Theory, collision between two reacting species results in the formation an activated complex or transition state.

44. C For first order reaction the plote [A] against time is not linear, but

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ln[A] against time is linear.

45. **D** In a reaction mechanism, the slowest step (elementary reaction) determines the rate of the overall reaction.

46. **D** Given that $n_i = 4$ and $n_f = 3$

$$\Delta E = -2.18\times10^{-18}J\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$
$$= -2.18\times10^{-18}J\left(\frac{1}{3^2} - \frac{1}{4^2}\right)$$
$$= -2.18\times10^{-18}J\,(0.1111 - 0.0625)$$
$$= -2.18\times10^{-18}J\times0.0486$$
$$= -1.059\times10^{-19}J$$

47. **C** For l=0, ml=0 and the value of ms is either $+\frac{1}{2}$ or $-\frac{1}{2}$.

48. **C** According to Pauli exclution rinciple, no two electrons in an atom can have the same four quantum nubers.

49. **A** For a given EMR, the higher the frequency, the lower the wave lengeth and the higher the energy.

50. **D** Sodium chloride is more soluble than lead iodide. The sodium chloride solution contain more ions than the lead iodide solution.

51. **D** The number of molecules go to gas phase from liquid form increases as temperature increases.

52. **C** Propanoic acid is carboxylic acid with three carbon atoms.

53. **C** Large quantities of acetic acid is prepared industrial by fermentation of ethanol.

54. **A** Pentayl formate contain 9 carbon atoms and cannot be isomer of the remaining compounds which have 5 carbon atoms.

55. **B** Pleasant small of many esters is responsible for the flavor and fragrance of many fruits.

56. **B** HNC is a weak acid and the pH can be calculated as follows:

$$HCN(aq) \rightleftharpoons H^+(aq) + CN^-(aq)$$

	$[HCN]$	$[H^+]$	$[CN^-]$
Initial	0.2	0	0
Change	$-x$	$+x$	$+x$
Equilibrium	$0.2 - x$	x	x

$$K_a = \frac{[H^+][CN^-]}{[HNC]}$$

$$4.9\times10^{-10} = \frac{x^2}{0.2-x} = \frac{x^2}{0.2}\ (0.2 - x \approx 0.2)$$

$$x^2 = 4.9\times10^{-10}\times0.2 = 0.98\times10^{-10}$$

$$x = \sqrt{0.98\times10^{-10}} = 0.9899\times10^{-5} = 9.9\times10^{-6}$$

$[H^+] = x = 9.9\times10^{-6}$ and

$pH = -\log[H^+] = -\log 9.9\times10^{-6}$

$pH = 6 - \log 9.9 = 6 - 0.99 = 5.01$

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37. D Presence common ion surpresses the ionization of weak acid and the solution becomes less acidic.

38. D Scince K_b is greater than K_a, the solution becomes basic.

39. D From the given titration curve: The pH is low at the beging and at the equivalent point the pH is 7 and these are characterstics of titration curve of strong acid with strong base.

40. C End point is the point at which color of indicator changes and equivalence point is the point at which stoichiometrically equivalent amounts of the acid and the base react (neutralization reaction compelet).

41. B Net ionic equation is a chemical equation with no spectator ions.

42. C Dilution process is preparing less concentrated solution from more concentrated solution by adding solvent.

43. C $\text{Normality} = \frac{\text{Number of equivalents of solute}}{\text{Volume of solution in L}}$

44. A The molecular equation is:

$AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$

The ionic equation is:

$Ag^+(aq) + NO_3^-(aq) + Na^+(aq) \rightarrow AgCl(s) + Na^+(aq) + NO_3^-(aq)$

The net ion equation is: $Ag^+(aq) + NO_3^-(aq) \rightarrow AgCl(s)$

45. C The solute takes the area from where solvent can evaporate (decreases surface area from where solvantes can evaporate).

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A. 2lactate, $2CO_2$, 2ATP
B. 2ethanol, $2CO_2$, 2ATP
C. 2ethanol, $2NAD^+$, 2ATP
D. 2lactate, $2NAD^+$, 2ATP

16. Which stage of cellular respiration is correctly paired with its site?
A. Glycolysis – cristae
B. Pyruvate oxidation cytoplasm
C. Chemiosmosis – outer membrane
D. Krebs cycle – matrix

17. The table below compares the two types of competition.

№	Intra-specific completion	Inter-specific competition
1.	Completion between different species	Completion between members of the same species
2.	Occurs between individuals with different adaptation	Occurs between individuals with similar adaptation
3.	Occurs for types of requirements	Occurs for specific requirements
4.	Directly affect the population size	Either both or one species can be suppressed

In the above table, which rows contain **WRONGLY** stated statements with respect the type of competition?
A. Row 2 and 4
B. Row 1 and 3
C. Row 1 and 2
D. Row 2 and 3

18. What happens during the process of electron transport chain? Protons
A. diffuse against concentration gradient through ATP synthase.
B. build up in the matrix than the inter- membrane space.
C. diffuse down concentration gradient to reduce NAD.
D. build up in the inter-membrane space than the matrix.

19. Identify the correct statement about the role of NAD and FAD during aerobic respiration. They
A. are the terminal electron acceptors of electron transport chain.
B. carry hydrogen atoms that will be split into protons and electrons.
C. oxidize intermediate products of cellular respiration.
D. are electron carriers that pump protons into inter-membrane space.

20. Which one of the following options represents the correct catabolic pathway of the given respiratory substrates?
A. Lipids → fatty acids → acetyle-CoA → TCA
B. Polysaccharide → glucose → acetyle-CoA → pyruvate → TCA
C. Nucleic acids → amino acids → Nucleotide → acetyle-CoA → TCA
D. Proteins → ammonia → amino acid → pyruvate → TCA

21. The chromosomal mutation that results Down's syndrome is
A. inversion.
B. translocation.
C. chromosome non-disjunction.
D. insertion.

22. When a piece of one chromosome is transferred to another non-homologous chromosome, such kind of mutation is called
A. inversion. B. translocation. C. duplication. D. insertion.

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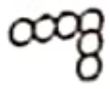
23. Consider the following diagram of the shapes of bacteria.

I

II 

III

IV

Which of the following shapes of bacteria belongs to spirochetes?

A. I B. III C. II D. IV

24. Which of the following diseases have water as their reservoir?

A. Pneumonia and tuberculosis C. Trachoma and diphtheria

B. Amoeba and cholera D. Salmonella and malaria

25. A geographic area with specific climate, soil type, plants and animals with similar adaptation is _____.

A. ecosystem B. habitat C. biome D. niche

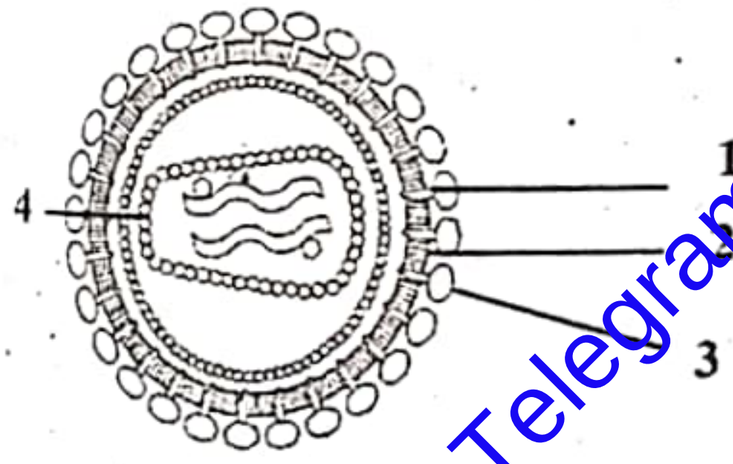
26. The following table compares viruses with free-living cells.

№	Feature	Viruses	Free-living cells
1.	Size	10 - 100µm	10 - 100µm
2.	Nucleus	Absent	Present
3.	DNA or RNA	Present	Present
4.	Mitochondria	Present	Present

Which rows of the above table contain the correct comparison?

A. 1 and 4 B. 2 and 3 C. 3 and 4 D. 1 and 2

27. Consider the following diagram of HIV virus.



What does number 1 represent?

A. Genetic material C. Central core

B. Enzyme molecule D. Antigen

28. What is the economic impact of AIDS?

A. Increase the resources for public expenditure

B. Enhance the life expectancy and GDP

C. Encourage savings for medical treatment

D. Seriously affects the taxable population

29. As the energy from the sun enters the ecosystem in the form of light, one of the following happens to it. It

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A. goes on decreasing along the trophic levels from primary producers to decomposers.
B. depends on the type of the trophic levels and changes at each trophic level irregularly.
C. goes on increasing along the trophic levels from primary producers to decomposers.
D. remains much of the time the same but increases as it reaches decomposers.

30. Suppose a self-fertilizing organism has AaBb genotype, how many different types of gametes would it produce if there is no linkage?
A. 12 B. 4 C. 16 D. 8

31. Which one of the following viruses is an RNA virus?
A. Bacteriophage B. HIV C. Herpes simplex D. H1N1

32. Which one of the following statements is true regarding the need for recycling nutrients through an ecosystem?
A. It is energy consuming process hence nutrients are constantly created and destroyed.
B. It is the key to saving the earth hence nutrients become available.
C. It depletes the resources hence infinite resources become limited by the process.
D. It pollutes the environment hence waste products are accumulated during the process.

33. What is the impact of humans on the sulphur cycle? Production of
A. hydrogen sulphide through the oxidation of sulphur
B. sulphur through decomposition of dead matter
C. sulphur dioxide through combustion of fossil fuels
D. sulphate through weathering and oxidation of rocks

34. Answer this question based on the following steps of the nitrogen cycle.

$$N_2 \xrightarrow{W} NH_4^+ \xrightarrow{X} NO_2^- \xrightarrow{Y} NO_3^- \xrightarrow{Z} N_2$$

The bacteria that are represented by the letters W, X, Y and Z are, respectively
A. Azotobacter, Nitrosomonas, Nitrobacter and Pseudomonas.
B. Thiobacillus, Nitrosomonas, Nitrobacter and Pseudomonas.
C. Klebsiella, Nitrobacter, Pseudomonas and Rhizobium.
D. Rhizobium, Pseudomonas, Nitrobacter and Nitrosomonas.

35. As succession progresses, the bare land turns out to be productive due to the fact that the
A. species diversity and richness falls.
B. amount and depth of the soil increases.
C. environment become less stable.
D. abiotic conditions remain unchanged.

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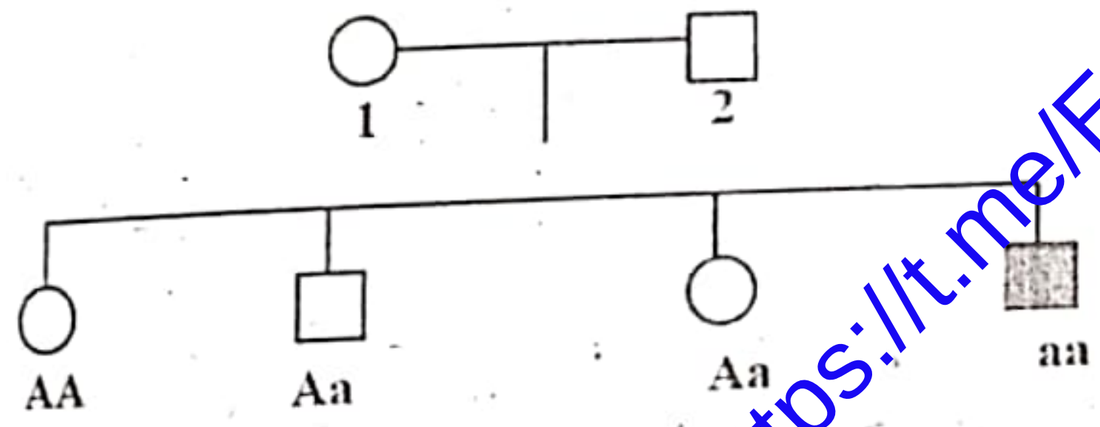
36. What will happen to the population size if the rate of natality is less than the rate of mortality?
A. Unchanged B. Increase C. Fluctuate D. Decrease

37. Which one of the following statements is correct about the stages of population growth rate curve?
A. Natality exceeds mortality in stationary phase.
B. Nutrient exhausted in decline phase.
C. Toxic excretory products accumulated in log phase.
D. Organisms adapted to the environment in lag phase.

38. During meiosis, crossing over is important to increase variation of gametes with regard to alleles that are
A. dominant and heterozygote.
B. independently segregated.
C. recessive and homozygote.
D. linked and not assorted independently.

39. Consider the following pedigree of albinism.



What will be the possible genotypes of the parents?
A. Aa, aa B. AA, Aa C. aa, AA D. Aa, Aa

40. In which phase of meiosis does pairing of chromosomes and exchange of segments take place?
A. Prophase II B. Prophase I C. Anaphase II D. Anaphase I

41. Among the following statements, which one is correct about mutations?
Mutations
A. are accidents that occur spontaneously and randomly during DNA replication.
B. that occur in somatic cells can pass on the next generation.
C. are changes in genetic composition of a population.
D. that occur in sex cells are completely harmless.

42. If the particular triplet of base in the sense strand of DNA is AGT, then the corresponding codon for the transcribed mRNA willbe _____.
A. ACU B. UGA C. UCA D. AGU

43. What is the theory that states disease is caused by microorganisms?
A. Spontaneous generation C. Endosymbiotic theory
B. Biogenesis theory D. Germ theory

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44. In the bare bottom of a newly formed pond a biologist observed emergence of new algae. The form of succession he observed can be best described as
A. early succession.
B. late succession.
C. secondary succession.
D. primary succession.

45 The extent to which variability exists among living organisms, with in .species and even between ecosystem is known as
A. biosphere. B. biomass. C. bioaccumulation. D. biodiversity.

46. Which one of the following is the significance of biodiversity?
A. Worsening of social relationships
B. Increasing vulnerability to environmental stress
C. Reducing the level of material wealth
D. Increasing security of food supply

47. The carrying capacity of a population can be defined as the
A. minimum population size supported by its habitat.
B. maximum population size supported by its habitat.
C. maximum population growth rate in a habitat.
D. average population size supported by its habitat.

48. If pea plants with round yellow heterozygous seeds are self-fertilized and produced 4800 seeds, how many round green seeds will be produced?
(Hint: round yellow seed is dominant),
A. 2700 B. 4800 C. 300 D. 900

49. The nucleotide that build up the DNA molecule contain the following bases **EXCEPT**
A. guanine. B. adenine. C. cytosine. D. uracil.

50. A grade 12 student extracted an insulin gene from insulin producing cells of the pancreases and plasmid from bacteria. Then he transferred the insulin gene into the plasmid and the hybrid plasmid was put back into bacteria for insulin production. What enzyme did the student used to join the insulin gene and bacteria plasmid?
A. DNA helicase
B. ATP synthase
C. RNA polymerase
D. DNA ligase

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51. Which one of the following options contains the main steps of scientific method in their correct order?
A. Doing background research → Asking question → Testing the hypothesis → Formulating the hypothesis → Analyzing the result Reporting the result
B. Asking question → Formulating hypothesis → Doing background research → Testing the hypothesis → Analyzing result → Reporting the result
C. Doing background research → Formulating hypothesis → Asking question → Testing the hypothesis → Analyzing the result → Reporting the result
D. Asking question → Doing background research → Formulating hypothesis → Testing the hypothesis → Analyzing the result → Reporting the result

52. Francesco Redi disproves the belief that rotting meat produces flies. He used wide mouth jars containing meat. He left some jars open to the air and others were covered with a piece of gauze. What do you think might have been Redi's most plausible hypothesis?
A. Flies would be formed in the covered jars.
B. Only flies could produce more flies.
C. Flies could be produced through spontaneous generation.
D. Flies could be produced from rotting meat.

53. Suppose a grade 12 student wants to investigate the need of CO_2 for photosynthesis. He destarched plants A and B then he placed plant A in a bell jar which has no CO_2 in the surrounding air, while plant B in a bell jar which has normal level of CO_2. He placed the jars in a place with light for 12 hours and kept other variables constant. Finally, he tested for the presence of starch. Less starch was produced by plant A than plant B. What are the dependent and independent variables of this experiment, respectively?
A. Amount of starch, presence of light
B. Presence of light, amount of starch
C. Amount of starch, presence or absence of CO_2
D. Presence or absence of CO_2, amount of starch

54. Helen was conducting iodine test for starch. She took two different food samples and placed one of the samples in test tube A and the other in test tube B. She added a few drop of iodine solution to both test tubes. Finally, she observed blue-black color in test tube B, but the sample in test tube A remained unchanged. If Helen wants to write the report of this experiment, she should write
A. the importance of starch in the procedure part.
B. her name in the acknowledgement part.

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C. the sample in test tube B was starch containing as her conclusion.
D. the general test for starch is iodine test as her result.

55. A field biologist can measure the slope of an area by using
A. theodolite. B. flow meter. C. scalpels. D. data logger.

56. Which organic molecules in our body have functions as antigens and antibodies of the immune system?
A. Carbohydrates B. Proteins C. Nucleic acid D. Lipids

57. If α-glucose reacts with fructose molecule in condensation reaction, what will be the chemical formula of disaccharide formed in this reaction?
A. $C_6H_{12}O_6$ B. $C_{12}H_{24}O_{12}$ C. $C_{10}H_{24}O_{10}$ D. $C_{12}H_{22}O_{11}$

58. In grade 11 laboratory class, the teacher gave the students a sample of food and asked them to investigate the presence of glucose. What kind of test and result will enable the students to confirm the presence of glucose?
A. Emulsion test and milky white color
B. Iodine test and blue black color
C. Benedict's test and orange color
D. Biuret test and purple color

59. A group of grade 12 students carried out four biochemical tests to investigate the composition of four different food samples. The results with respective type of tests are shown in the table below.

Type of test	Results			
	Sample A	Sample B	Sample C	Sample D
Iodine test	Positive	Positive	Positive	Positive
Benedict's test	Positive	Negative	Positive	Negative
Emulsion test	Negative	Positive	Negative	Positive
Biuret test	Negative	Positive	Positive	Positive

Which food samples contained both lipid and protein?
A. Sample A and D C. Sample B and C
B. Sample B and D D. Sample A and C

60. Identify the correct statement about enzymes. They
A. from an intermediate complex which dissociates into the product and free enzyme.
B. function as organic catalysts by irreversibly combining with the substrate.
C. are globular proteins with a multiple shaped active site.
D. function by affecting the nature of the product.

61. A medical laboratory technologist at Black Lion Hospital placed a solution of stool in a centrifuge to separate the particles quickly for the

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stool test. Which one of the following will happen when he spins the centrifuge at high speed?

A. The solid particles of the stool will float on the top of the mixture.
B. The gravitational forces on the solid particles force the solid particles of the stool to the bottom of the tube.
C. The solid particles of the stool will mix up highly and suspended freely.
D. The liquid particles will evaporate but the solid particles will remain in the centrifuge tube.

62. The relevance and promise of biology as a field of science can be best explained by its application in

A. agriculture and genetic engineering.
B. sport and culture.
C. oil and mineral production.
D. economic growth and urbanization.

63. Biologists played a role in the fight against AIDS because they developed

A. anti-retroviral drugs that target and aim to break different stages of the HIV life cycle.
B. anti-retroviral drugs that provide lasting immunity against reinfection by HIV.
C. a vaccine from people who have been repeatedly exposed to HIV but never developed AIDS.
D. a vaccine from a monkey that develops a very similar disease to AIDS which is caused by a very similar virus.

64. Which one of the following statements shows willingness of participating in a community undertaking against HIV and AIDS? A person who

A. takes volunteer counseling and testing
B. decides to abstain from any relation with the community
C. discriminates HIV positive from HIV negative people
D. isolates himself from HIV infected people

65 One benefit of water related to its high specific heat capacity is its importance

A. to move particles inside the cell more easily.
B. for the normal functioning of enzymes in living cells.
C. for the transport of blood cells around circulatory system.
D. to support the weight of some insects while they walk on it.

66. Among the parts of the cell, which organelles are bounded with double membrane?

A. Golgi body, chloroplast and lysosome
B. Vacuole, endoplasmic reticulum and ribosome
C. Nucleus, mitochondria and chloroplast

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D. lysosome, vacuole and nucleus

67. Why are cells that secret hormones and enzymes rich in rough endoplasmic reticulum? Because they produce more
A. lipids. B. carbohydrates. C. ATP. D. proteins.

68. What is the role of cholesterol in the cell membrane? To
A. anchor integral proteins with the membrane
B. maintain the membrane fluidity
C. make the membrane selectively permeable
D. serve as signals to other membrane of cells

69. If potato cells immersed in a hypertonic solution, they will
A. lose water and plasmolysed. C. lose water and become turgid.
B. gain water and haemolysed. D. gain water and become larger.

70. Which molecule of the plasma membrane contains carbohydrate chains and allows identification of cell by the immune system?
A. Integral protein C. Channel protein
B. Peripheral protein D. Glycoprotein

71. Which one of the following options contains only a group of organic biochemical molecules?
A. Ribose, methane, fatty acid, lactose and sodium chloride
B. ATP, deoxyribose, sulfuric acid, lactic acid and amino acid
C. Fatty acid, glycine, DNA, maltose and triglyceride
D. Lactic acid, hydrochloric acid, sucrose, water and ATP

72. Unbalanced hydrogen bonding at the surface of the water is responsible for high
A. surface tension of water.
B. latent heat of vaporization of water.
C. density of water in its liquid form.
D. specific heat capacity of water.

73. In a structure of amylose molecule that contains one hundred α-glucose monomers, how many α-1, 4-glycosidic bonds are formed?
A. 99 B. 100 C. 50 D. 200

74. Which part of phospholipids is considered as hydrophobic region?
A. Phosphate group C. Carboxyl group
B. Hydrocarbon tails D. Glycerol head

75. Identify the organic molecule which is correctly paired with its function.
A. Glycogen – formation of microfibrils
B. Phospholipids – control metabolism
C. Triglycerides – respiratory substrate
D. Cellulose – component of cell membrane

76. Where does the production of ATP and reduced NADP take place?
A. Cytoplasm B. Thylakoid membrane C. Fluid matrix D. Stroma

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77. The biochemical mechanism by which chlorophyll absorbs light energy is efficient because
A. the molecules of the photosystems and ETCs are floating around in a solution.
B. the same pigment molecules in the antenna complex absorb different wave length of light.
C. the molecules of photosystems and ETCs are fixed in the membranes of thylakoid.
D. the different pigment molecules in the antenna complex absorb the same wave length of light.

78. The leaves of C3 plants in the tropics close their stomata to minimize water loss, as a result concentration of CO_2 in the leaves falls. When C3 plants face such problem, ribulose bisphosphate reacts oxygen and reduce
A. hydrogen peroxide and CO_2 in the mitochondrion.
B. glycertae phosphate and phosphoglycolate in the chloroplast.
C. triose phosphate and sugar in the mitochondrion.
D. oxaloacetate and water in the peroxisome.

79. An enzyme which is designated as EC 2.7.1.2 can be grouped to which class of enzyme?
A. Isomerases B. Transferases C. Oxidoreductases D. Ligases

80. Sorghum and wheat are C4 plants which are well adapted to photosynthesize efficiently in hot bright days. Peas and carrots would not photosynthesize efficiently in such conditions because they are C3 plants. In which of the following features does those plants are similar? The
A. enzyme used to fix carbon dioxide
B. cells where the Calvin cycle takes place
C. presence of chloroplast in the bundle sheath cells
D. cells where carbon dioxide fixation take place

81. Plants that follow CAM photosynthetic pathway carry out the two stages of photosynthesis at different times. Which of the following products are formed during the night?
A. Malate and sugar C. ATP and NADPH
B. Oxaloacetate and malate D. CO_2 and sugar

82. Using enzyme technology in industrial production helps to reduce global warming. This is because enzymes
A. absorbs much carbon dioxide from the environment.
B. allow reaction to be carried out at much lower temperature.
C. increase the heat required for reactants.
D. give high energy for industrial reactants.

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83. How do enzymes lower the activation energy of substrate molecules according to the induced-fit model? By
A. increasing the kinetic energy and probability of collusion.
B. providing extra energy for the conversion of reactants into a product.
C. creating a tension in the bond of substrate molecules.
D. forming strongly bonded enzyme-substrate complex.

84. If an enzyme depends on a cofactor for its activity, then the enzyme becomes active and functional when it is found in the form of
A. mineral ion. B. apoenzyme. C. holoenzyme. D. coenzyme.

85. Which one of the following enzymes requires copper ions for its catalytic activity?
A. Flavin adenine dinucleotide
B. Nicotinamide adenine dinucleotide
C. Alcohol dehydrogenase
D. Cytochrome oxidase

86. A change in temperature can affect the rate of enzymes controlled reaction by
A. increasing or decreasing the probability of collusion.
B. altering the charge amino acids that form the active site.
C. changing the shape of substrate and enzyme molecules.
D. decrease the turnover rate of enzymes.

87. If the amount of enzyme is kept constant and the substrate concentration is gradually increased, what will happen to the rate of the reaction? The reaction velocity
A. decreases first but increases after the active sites are occupied with the substrate.
B. increases until all the available enzyme active sites become saturated.
C. increases further as the substrate concentration increases further.
D. remains unchanged; hence it isn't influenced by substrate concentration.

88. The following reaction shows enzymes controlled metabolic pathway in a living cell.

Substance W $\xrightarrow{E_1}$ Substance X $\xrightarrow{E_2}$ Substance Y $\xrightarrow{E_3}$ Substance Z

What will happen if the concentration of substrate Z increases beyond the cell's requirement?
A. The reaction will take place in the reverse direction.
B. Substance Z will inhibit E_1 and switch off the pathway.
C. E_3 will convert substance Y into another substance rather than substrate Z.
D. E_1 and E_2 will be inactive until substance Z equals to the normal concentration.

89. How is the process of end-product inhibition triggered? By

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A. low concentration of the end product
B. inhibition of the last enzyme
C. excess production of the last product
D. change of the active site of the enzyme

90. Identify the WRONG statement about the modern cell theory.
A. Cells arise only from other pre-existing cells by division.
B. The process of metabolism and heredity occur within cells.
C. Cell is the basic unit of organization of all organisms.
D. All cells have basically different chemical composition.

91. During practice of genetic engineering, bacteria are important to
A. provide plasmids.
B. produce restriction endonucleases.
C. breakdown the recombinant DNA molecule.
D. produce ligase enzymes.

92. Consider the following diagram of a bacteriophage.

In the above diagram, the viral capsid is represented by number ____.
A. 5 B. 2 C. 1 D. 4

93. Which one of the following steps of HIV life cycle comes first?
A. The viral DNA is transcribed into RNA and produces viral proteins including reverse transcriptase.
B. The RNA, proteins and reverse transcriptase molecules assembled and bud out of the host cell to infect a new cell.
C. The HIV viron attached to the plasma membrane of the T-helper.
D. The reverse transcriptase converts RNA into DNA and incorporates into the DNA of the host cells.

94. Which of the following antiretroviral drugs prevents the assembly of new virus particles?
A. Protease inhibitors
B. Entry inhibitors
C. Non-nucleotide reverse transcriptase inhibitors
D. Nucleotide reverse transcriptase inhibitors

95. Which of the following statements is correct about the reproductive cycles of lytic and lysogenic viruses?
A. The DNA of lysogenic viruses duplicates as the host cell divides.
B. Only lytic viruses require the host DNA for their replication.
C. Lytic viruses are released by exocytosis.
D. Lysogenic viruses do not form a protein coat.

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96. Red-green color blindness is due to X-linked gene and determined by the recessive alleles. Which one of the following is true about this condition?
A. It affects male due to alleles inherited from the father.
B. It is much more common in females.
C. Expressed in both sexes equally if the parents are carriers.
D. Females inherit one allele from each parent and becomes affected
97. Which one of the following statements is true about a chromosome? It is
A. clearly visible in the nucleus when the cell is not dividing.
B. made up of DNA which is tightly coiled many times around histones.
C. loosely organized throughout the nucleus when the cell is dividing.
D. divided into two equal sections by the centromere.
98. What is the complementary strand for a DNA strand having the following sequence of bases?
5'-AGGTTACTGAGCT-3'
A. 3'-TCCAATGACTCGA-5'
B. 3'-TCCAATGACTCGA-3'
C. 3'-UCCAAUGACUCGA-5'
D. 5'-UCCAAUGACUCGA-3'
99. Consider the following disordered events of transcription.
I. RNA polymerase crosses a stop sequence in the gene.
II. RNA polymerase reads the antisense DNA strand and builds mRNA.
III. mRNA molecule detaches from DNA and DNA rejoin.
IV. RNA polymerase binds to promoter which signals the DNA to unwind.
A. I → IV → II → III
B. IV → II → I → III
C. IV → I → II → III
D. I → IV → II → III
100. Which of the following bases are the triplets of codes in the mRNA that cease translation and the ribosome fold the polypeptide into its final structure?
A. ATT, ATC and ACT
B. CCC, UUA and GUU
C. UAA, UAG and UGA
D. TAA, TAG and TGA

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ANSWER TO EUEE BIOLOGY EXAMINATION, MEGABIT 2012/13 (MARCH 2020/21)

1. D:

60 division on eyepiece graticule = 20 division on stage

= 15 x20 µm = 300 µm

= 300 µm = 6 cells

= ?X = 1 cell

= X = 50 µm = 0.05 mm

1. A: Major functions of cells: provide structure and support, repair damaged cells through **mitosis**, cellular respiration and aid in reproduction.
2. D:

Features	Prokaryotic	Eukaryotic
Size	1-10 µm	10-100 µm
Ribosomes	Smaller than in eukaryotic cells (70S)	Larger than in prokaryotic cells (80S)
Cell wall	Always present Not made from cellulose	Present in some Cellulose in plant cell

3. C: Presence of membrane bounded organelles in eukaryotic cells enhances cellular activities to be carried out more efficiently in eukaryotes than prokaryotes.
4. C: The lipid bilayer is structural backbone of a cell membrane that selectively permeable water soluble substances.
5. B: Peptidoglycan is the chemical composition of a bacterial cell wall. It is a complex molecule made from sugars and amino acids.
6. A: Gram's staining is a test for classifying bacteria. Bacteria can be classified as Gram-positive or Gram-negative by their response to the Gram stain.
7. B: Functional disease is a disease due to malfunction of organ or organ system. For examples, heart disease, and Irritable Bowel Syndrome.
8. D: Yeast cells produce ethanol (ethyl alcohol) when ferment glucose.
9. C: Vinegar is a dilute solution of ethanoic acid in water. It is produced by fermenting alcohol for a second time using *Acetobacter* bacteria.
10. D: The taiga is a forest of the cold, subarctic region. It lies between the tundra to the north and temperate forests to the south. It is characterized by adequate precipitation, cool year round temperature and conifer trees.
11. D: Most epiphytes are found in moist tropical areas. An epiphyte is an organism that grows on the surface of a plant.

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12. A: Ethiopia is rich in biodiversity due to wide range in altitude and climate, and existence of so many different biomes in the country.
13. B: Habitat destruction is the greatest threat to global biodiversity. So maintaining habitat is significant in conserving biodiversity
14. D;

    In animal cell, anaerobic respiration converts one glucose molecule into two lactate, two ATPs, and 2NAD+.
15. D:

Stages of cellular respiration	Site
Glycolysis	Cytoplasm
Pyruvate oxidation	Mitochondrial matrix
Krebs cycle	Mitochondrial matrix
Chemiosmosis	Cristae

16. B: Intra-specific competition is a competition between members of the same species, and inter-specific competition of different species.
17. B: Because of the action of the proton pumps, protons accumulate in the inter-membrane space creating a higher concentration than in the matrix.
18. B: NAD and FAD are molecules that transfer hydrogen ions from glucose molecule to ATP synthase.
19. A: Catabolism is the set of metabolic pathways that breaks down molecules, such as lipids into smaller units, such as fatty acids.
20. C: Down's syndrome is the result of an additional copy of all, or a specific part of chromosome 21. This occurs when homologous chromosome do not separate during meiosis.
21. B: Translocation is the transfer of part of a chromosome to another non- homologous chromosome.
22. A: Spirochetes are any of a group of spiral or corkscrew-shaped bacteria.
23. B: Reservoir of infection is any person, animal, plant, soil, air and water that housed disease causing organisms.
24. C: A biome is a large area characterized by its vegetation, soil, climate, and wildlife.
25. B: Viruses are non-living infectious particles, much smaller than a cell, and lack organelles that found in free living cells.
26. B:Along with the RNA, HIV contains three key enzymes: reverse transcriptase, integrase, and protease.
27. D: AIDS-related illnesses and deaths to employees affect a firm by both increasing expenditures and reducing revenues.
28. A: As energy moves through trophic levels in an ecosystem the amount that is available decreases.
29. B:

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F_1 self-pollination **AaBb X AaBb**
F_1 gametes (n) **AB, Ab, aB, ab**

30. D: RNA viruses are viruses that have RNA as their genetic material..
31. B: Nutrient cycle recycles valuable nutrients, such as nitrogen, sulphur, carbon and phosphorus that are important for organisms to exist.
32. C: The burning of large quantities of fossil fuels releases larger amounts of hydrogen sulfide gas into the atmosphere that can result in acid rain.
33. A: Various bacteria involve in the conversion of nitrogen gas.
34. B: Primary succession turns out to productive due to increment of amount and depth of the soil.
35. D: If birth rate is less than death rate, the population size decreases.
36. B: In decline phase, nutrients exhausted, a new disease strikes or toxic excretory products accumulate.
37. D:Crossing-over is the exchange of genetic material between homologous chromosomes. When cells divide during meiosis, homologous chromosomes are randomly distributed during anaphase I, separating and segregating independently of each other.
38. D:

Parental genotypes: Aa x Aa

Gametes: A a x A a

Offspring genotypes:

x	A	a
A	AA	Aa
a	Aa	aa

39. B:During prophase I, homologous chromosomes pair with each other and exchange genetic material in a process called chromosomal crossover.
40. A: A mutation is a change that occurs in DNA sequence, either due to mistakes when the DNA is copied or as the result of mutagens.
41. C: In DNA, A corresponds to T and C corresponds to G, the only difference in RNA is that T is replaced with U, so the mRNA sequence that corresponds to the codon AGT would be UCA.
42. D: The theory that disease can be caused by micro-organisms is called the germ theory.
43. D: Where a succession starts from bare or from a newly formed pond with no life, the succession is a primary succession.
44. D: The term biodiversity refers to the variety of life on Earth at all its levels, from genes to ecosystems.

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45. D: Biodiversity provides functioning ecosystems that supply oxygen, clean air and water, pollination of plants, pest control, wastewater treatment and many ecosystem services.

46. B: Carrying capacity can be defined as a species' average population size in a particular habitat.

47. D: *F_1 self-pollination* RrYy X RrYy
F_1 gametes (n) RY, Ry, rY, ry (*1st parent*) RY, Ry, rY, ry (*2nd parent*)

	RY	Ry	rY	ry
RY	*RRYY*	*RRYy*	*RrYY*	*RrYy*
Ry	*RRYy*	*RRyy*	*RrYy*	*Rryy*
rY	*RrYY*	*RrYy*	*rrYY*	*rrYy*
ry	*RrYy*	*Rryy*	*rrYy*	*rryy*

F_2 phenotype ratio: 9 round yellow:3 round green: 3 wrinkled yellow: 1 wrinkled green seeds
Therefore, round green seeds = 4800x3/16 = <u>900</u>

48. D: Uracil is one of the four nucleobases in the nucleic acid RNA that are represented by the letters A, G, C and U.

49. D: DNA ligase is a specific type of enzyme that facilitates the joining of DNA strands together by catalyzing the formation of a phosphodiester bond.

50. D: Scientific method is the process by which scientists approach their work.

51. B: Francesco Redi hypothesised that only flies could produce more flies.

52. C: Independent variable (IV) is the factor that the scientist changes, or manipulates, while dependent variable (DV) is the factor that the scientist measures to see if it changes when the IV is changed.

53. C: A blue-black color results if starch is present. If starch amylose is not present, then the color will stay orange or yellow.

54. A: A theodolite is an item of field equipment that measures slope.

55. B: Proteins function as antigens and antibodies in immune system.

56. D:

$$\underset{\text{Monosaccharides}}{C_6H_{12}O_6 + C_6H_{12}O_6} \underset{\text{Hydrolysis}}{\overset{\text{Condensation}}{\rightleftharpoons}} \underset{\text{Disaccharide}}{C_{12}H_{22}O_{11}} + H_2O$$

57. C: Benedict's test is used to test for the presence of reducing sugar, such as glucose.

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58. B: The emulsion test is a method to determine the presence of lipids while a Biuret test is a chemical test used to determine the presence of a peptide bond in a food substance.

59. A: The enzyme substrate complex is a temporary molecule formed when an enzyme comes into perfect contact with its substrate.

60. B: Large centrifuges are used to simulate high gravity or acceleration environments.

61. A: Advances in medicine, agriculture, biotechnology, and many other areas of biology have brought improvements in the quality of life.

62. A: Antiretroviral treatment is the drugs that treat HIV. It is not a cure for HIV, but it keeps HIV under control.

63. A:Voluntary counseling and testing serves as an entry point ***for*** prevention and care strategies.

64. B: High specific heat capacity of is important as the functioning of enzymes in living cells is affected by temperature.

65. C: Generally, eukaryotic cells have double-membrane-bounded organelles, such as the nucleus, mitochondria, and chloroplasts.

66. D: Ribosomes on the rough endoplasmic reticulum are called 'membrane bound' and are responsible for the assembly of many proteins.

67. B: The main function of cholesterol is to maintain integrity and fluidity of cell membrane.

68. A: In the hypertonic solution, the cells lose water by osmosis and shrink.

69. D: Glycoproteins are molecules that comprise protein and carbohydrate chains that are involved in many physiological functions including immunity.

70. C: Carbohydrates, lipids, proteins, and nucleic acids are the four major classes of biological macromolecules.

71. A: Water has a high surface tension because the water molecules on the surface are pulled together by strong hydrogen bonds.

72. A: A ***glycosidic bond*** or glycosidic linkage is a type of covalent bond that joins a carbohydrate (sugar) molecule to another group.

73. B: The phosphate group is the negatively-charged polar head, which is hydrophilic. The fatty acid chains are the uncharged, nonpolar tails, which are hydrophobic.

74. C:

Substrate	Function
Glycogen	Storage carbohydrate in animal cell
Phospholipids	Basic component of a cell membrane
Triglyceride	Respiratory substrate
Cellulose	Primary component of plant cell wall

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75. **B:** In plants, the production of ATP and NADPH take place in the thylakoid membranes of chloroplasts.
76. **C:** The biochemical mechanism by which chlorophyll absorbs light energy is efficient due to the molecules of photosystems and ETCs are fixed in the membranes of thylakoid.
77. **B:** When C3 plants face deficiency of CO_2, ribulose bisphosphate reacts oxygen and reduce glycerate phosphate and phosphoglycolate in the chloroplast.
78. **B:** Enzymes are generally classified on the basis of the type of reactions that they catalyse.
79. **D:** In C3 and C4 plants, CO_2 fixation take place in mesophyll cells.
80. **B:** In CAM plant, the stomata are opened at night and allow CO_2 reacts with PEP in mesophyll cells to form oxaloacetate and then malate.
81. **B:** Enzymes help to reduce global warming in industrial production for they allow reaction to be carried out at much lower temperature.
82. **C:** According to the induced-fit model of enzyme action, reacting molecules enter a transition state in which bonds in reacting molecules are put under tension. Accordingly, the activation energy is lowered.
83. **C:** The cofactor combine with an apoenzyme to produce an active enzyme called holozenyzme.
84. **D:** The mineral ion in Cytochrome oxidase is copper ions.
85. **A:** The rate of an enzyme catalyzed reaction increases as the temperature increases and vice versa.
86. **B:** Increasing substrate concentration increase enzymatic reaction until it reaches maximum velocity.
87. **B:** End-product inhibition is a form of negative feedback by which metabolic pathways can be controlled. In end-product inhibition, the final product in a series of reactions (Z) inhibits an enzyme from an earlier step (E1) in the sequence.
88. **C:** The end-product inhibition is triggered by excess production of last product.
89. **D:**

In general, the modern **cell theory** includes:
1) all living organisms are composed of one or more cells.
2) the cell is a unit of structure and function of all living organisms.
3) all cells come from pre-existing cells by cell division.
4) cells carry hereditary materials that can pass to daughter cells during cellular division.
5) all cells have essentially the same chemical composition.
6) all energy flow (metabolism and biochemistry) occurs within the cells.

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91. A: Plasmids are important tools used in genetic engineering.
92. C: Viral capsid is an outer protein coat a virus.
93. C: The major stages of HIV life cycle include fusion, entry, transcription of RNA to DNA, integration of viral DNA to host DNA, transcription of DNA to RNA, assemblage of virions, infection of other T – cells.
94. A: Protease enzyme of HIV is useful for assembling the viral particles.
95. A: In lysogenic life cycle, infection causes the virus to enter a latent state where its DNA is reproduced with the host DNA, but no new viruses are formed.
96. D: Red-green color blindness is X-linked and recessive alleles. It affects female that inherit one allele from each parent.
97. B:Chromosome is a long strand of DNA that bounds with histone proteins.
98. A: DNA base sequence pairing rule: A-T, G-C. The DNA molecule is a double helix in which the base sequence on the sense strand is complementary to the base sequence on the antisense strand.
99. B: Transcription is the process that converts genetic information from a DNA code into an mRNA code.
100. D: Three triplets (TAA, TAG and TGA) do not code for any amino acid because they are 'stop' codes.

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ETHIOPIAN UNIVERSITT ENTRANCE EXAMINATION (EUEE) SCHOLASTIC APTITUDE TEST YEKATIT 2012/13 (FEBRUARY 2020/21)

SECTION ONE: Verbal Reasoning (Question 1-13)

This section consists of a total of 35 verbal reasoning question with sub-sections of analogy, synonyms, antonyms, sentence completion, reading comprehension, and logical reasoning. Each sub- section has its own directions.

Antonyms(1-3)

Directions: Questions 1-3 consist of words printed in bold letters , followed by four other words. Choose the word that is mostly most nearly Similar in mining to the word in capital letters .

1. He has reached across the word , fomenting violence against the west.
   A. Subduing B. negating C. instigating D. planning
2. My sister came back to find her house in shambles,
   A. neat B. disorder C. mess D. ruins
3. Which of the following words cannot go together with the others?
   A. Rain B. Snow C. Wind D. Sky

Analytical and Logical Reasoning Question(4-13)

Direction: Question 4-13 deal with logical and analytical reasoning ,Read the given set of condition and assumption for each question. Choose the answer that you think is most appropriate among the given alternatives.

4. Which two statements prove that " Lemlem and Samuel are married".
   A. Lemelm is married to Alemu's only brother.
   B. Lemelm married her high school sweet heart
   C. Samuel and Lemelm loved each other.
   D. Alemu has the same parents as Samuael .
   A. C and D B. A and C C. Band C D. A and B
5. Which two statements prove that "Senaite is a child" ?
   A. Senaite lives with her parents . C. Bekele has just turned 13
   B. Senaite is the little sister of Bekele. D. Bekele is still in school
6. Which two statement prove that Eleni is a doctor ?
   A. Eleni loves people. C. Eleni's brother's is doctor.
   B. Eleni's father is teacher D. Eleni has the same job as her brother
   A. B and C B. A and C C. C and D D. A and D
7. Which two statement prove that Hana is young ?
   A. Hana lives with her brother. C. Alemu is still a student
   B. Hana is the little sister of Alemu. D. Alemu has just turned 14
   A. B and D B. B and C C. C and D D. A and C
8. Which statements prove that "Mulualem is a lawyer "?
   A. Mulualem lives with her parents
   B. Mulualem's mother is an accountant.

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C. Mulualem's sister is a lawyer
D. Mulualem has the same job as her sister
A. B and C B. A and C C. C and D D. A and D

9. Read the following three statements which are followed by four conclusion .choose the conclusion which logically follow from the given
1. All birds are black
2. All crows are black
3. All birds are loud
4. All crows are loud
A. Only(2)and (4)
B. Only (3) and (4)
C. Only (2) and (3)
D. Only (1) and (4)

10. Statements : some question are answer. Some answers are writers. all the writers are poets .

Conclusion:

1. Some writers are answers
2. some poets are question
3. All question are poets
4. Some poets are answer
A. Only(2)and (3)
B. Only (3) and (4)
C. Only (1) and (4)
D. Only (1) and (2)

11. Some envelops are gums .Some games are sales are adhesives.

Conclusion

1. Some envelops are seals
2. Some gums are adhesives
3. Some adhesives are seals.
4. Some adhesives are gums
A. Only 4
B. Only 2
C. Only 1
D. Only 3

Answer the following question the information given

12. Mike finished ahead of paulos and Berhanu both Finished before Lema. Orion did not finish last Who was the last to finish ?
A. Paulos
B. Berhanu
C. Lema
D. Orion

13. At university three girls , Alemaz, Sophia, and Samra are dating three boys ,Michael ,Thomas, and Maru ,Alemaz is dating Michael . Sophia is not dating the physics major . Samra is not dating the theater major.Michale is a biology major. Maru is not majoring in theater. Which of the following statements is correct , based on the above information.
A. Only Alemaz and Sophia are dating two of the students.
B. Alemaz is dating theater major
C. Samra is dating Maru ,who is majoring in theater.
D. Samra is dating Maru ,who is physician major.

Reading comprehension (14-21)

Direction: Read the passages carefully and answer question 14-21. Each question will be based on the information stated or implied in the passage .
Reading passage II

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Marie Curie was one of the most accomplished scientists in history. Together with her husbands , Pierre . The discovered radium, an element widely used for treating cancer, and studied uranium and other radioactive substance .Pierre and Marie's amicable collaboration latter helped to unlock the secrets of the atom

Marie was born in 1867 in Warsaw, Poland ,were her fathers was a professor of physics. At an early age , she displayed a brilliant mined and a blithe personality. Her great exuberance for learning prompted her to continue with her studies after high school. She became disgruntled. However, when she learned that the University in Warsaw was collected to women. Determined to receive a higher education ,she defiantly lift Poland and in 1891 entered the Sorbonne , a French University , where she earned her master ' s degree and doctorate in Physics.

Marie was fortunate to have studied at the Sorbonne with some of the greatest scientists , one of whom was Pierre Curie . Marie and Pierre were married in 1895 and spent many productive years working together in the physics laboratory .A short time after they discovered radium, Pierre was killed by a horse drawn wagon in 1906.Marie was stunned this horrible misfortune and endured heartbreaking anguish. Despondently she recalled their close relationship and the joy that they had shared in scientific research . the fact that she had two young daughters to raise by herself greatly increased her distress.

Curie's feeling of desolation finally began to fade when she was asked to succeed her husband as a physics professor at the Sorbonne. She was the first women to be given professorship at the world –famous University . in 1911 she received the Noble Prize in the chemistry for isolating radium. Although Marie Curie eventually suffered a fatal illness from her long exposure to radium. She never became disillusioned about her work. Regardless of the consequence ,she had dedicated herself to science and to reviling the my sisters of the physical world.

14. When she learned that she could not attend the University in Warsaw, she felt _____
A. Worried B. delegated C. annoyed D. hopeless

15. Marie _____by living Poland and traveling to France to enter he Sorbonne.
A. Was distressed C. showed intelligence
B. behaved D. Changed authority

16. _____she remembered their joy together.
A. Happily B. Tearfully C. Worriedly D. Dejectedly

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17. Her_____ began to fade when she returned to the Sorbonne to succeed her husband .
A. disappointment B. wretchedness C. anger D. misfortune

Reading passage I

Regular physical activity provided numerous health benefits.: from learner bodies and lower blood pressure to improved mental health and cognitive functioning. As the school Physical education program promotes physical activity and can teach skills as well as from or change behavior, it holds an important key to influencing health and wellbeing across life span. To improve the fitness of students, we need to rethink the design and delivery of school- based physical education program. Adults in some countries think that information about health was more important for students to learn than continent in language arts, mathematics , science, history or any other subject. Despite this high raking, most skills devote minimum curriculum time teaching students how to lead healthy lives. Our first step might be to consider ways to increase curriculum time devoted to physical education .In addition, school need to thoughtfully analyze the design and delivery of school physical education program to ensure that they are engaged ,developmentally appropriate, inclusive and instructionally powerful.

18. According to this passage regular physical activity is needed to
A. Improve ones' physical as well as mental health.
B. Improve ones' cognitive skills.
C. Lose weight .
D. Control; one's blood pressure.

19. In order to tone up the physical education program,
A. The program should be reoriented and implemented.
B. A committee should be set up in every school.
C. An assessment of existing program should be made .
D. It should be compulsory at school.

20. According to some countries, health education is more important than teaching.
A. Liberal arts B. any subjects C. natural science D. social science

21. The author wants the reoriented physical education program to be
A. Thoughtful B. relevant C. comprehensive D. short

Sentence Completion (22-25)

Direction: each sentence below has now blank space indicating that set of words or statements have been omitted. Choose the set of words or statements that best fits the mining of each sentence as a whole frome the given alternatives.

22. Which continuation is a right?
I don't truest social medias._____.

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A. Never can I and never I well
B. Never I have and never I well
C. I have never and I well never
D. I never have and I never well

23. If I were in your position, I _____
A. Would have done such silly things.
B. Would never do such silly things.
C. Will never have done such silly things .
D. Would never had don such silly things.

24. She contracted COVED 19 easily because she was obese and had other preexisting diseases. The doctor side that if she _____ obese ,her chances of getting the disease would have been slim.
A. were not B. were C. had been D. had not been

25. I: Haven't you see this film?
_____
A. Yes, have I
B. Yes, I haven't
C. No, I have
D. Yes; I have

Analogy.(26-32)

Direction: To answer question 26-32 read each related pair of words written in capital letters followed by four pairs of words. Select the pair (A,B,C, or D) that best express a relationship similar to that expressed by the original capitalization pair.

26. EGELE is to BIRD as ROSE is to _____
A. FOWER B. ANIMAL C. BEAUTIFUL D. GENTELMAN

27. CAR is to TRANESPORTATION as GAME is to _____
A. FOOTBOLL
B. MANIPULATION
C. INFORMATION
D. ENTERTAINMENT

28. SPEAD is to FAST as TEMPERATURE is to _____
A. CLIAMET B. WARME C. ICE D. THERMOMETER

29. FAST is to RUNNING as SLOW is to _____
A. WALKING B. SPEAD C. LAYZ D. PACE

30. CANDLE: WAX
A. PAPER:PULP
B. BAMBOO: FURENATURE
C. THREE: LAVELES
D. BRANCH :WOOD

31. CHEF: FOOD
A. BRICK:CARPENTER
B. SCULPTOR:STONE
C. BRUSH: WALL
D. PAINT : PAINTER

32. BOARD:TRAIN
A. MOUNT:HORSE
B. BICYCLE:DRIVE
C. WALK;SHOE
D. CHAIR:SIT

Synonyms.(33-35)

Direction : question 33-35 consists of words printed in capital letters, Followed by four other words . choose the word that is most nearly similar in meaning to the word in capital letter.

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5) The man vehemently denied ale the charges of killing that were brought against him

A. fervently B. carelessly C. hysterically D. forcefully

6) Which word means the some as"counterfeit?"

A. copy B. match C. robbery D. fake

7) He inherited a large amount of fortune from his parents. So, he lives a very lavish lifestyle.

A. Spartan B. humble C. stingy D. extravagant

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ETHIOPIAN UNIVERSITT ENTRANCE · EXAMINATION (EUEE) SCHOLASTIC APTITUDE TEST ANSWER KEY AND BIREFLY EXPLANATION, YEKATIT 2012/13 (2020/21 FEBRUARY)

1. A. Fomenting- cause or increase disorder subduing –overcoming ,bringing under control .
2. A. Shambles- state of confusion ,dirt or disorder net – clan
3. D. sky –But the other three are weather condition
4. B. If Alemu has the some parents as Samuel , they are brothers . Lemelm is married to Alemu's only brother means ,she married Samuel.
5. B. senait is the little sister of Bekele ,who is just tarned 13,that means senait is the younger brother of Bekele, there for, she is a child less than 13 years.
6. C. Eleni's brother is doctor and Eleni has the same job as her brother. Therefor, Elenis is a doctor
7. A. hana is the little sister of Alemu , and Alemu has just turned 14 proves that Hana is younger.\
8. C. Mulualem's sisters is a longer, and Mulualem has the same job as her sister proves that Mulualem is a longer .
9. A. crows are black –all black bards are loud. All crows are birds.Therefore, all crows are black and loud.
10. C. Some question are answer-some answers are writers .all the writers are poets. Therefore some writers and some poets are answers.
11. A. Some envelops are gums , some gums are seals. some seals are adhesives. therefore some adhesives are gums.
12. C. The correct order to finish is making paulos and Berhanu and lastlyLema.
13. D. Alemaze dating Michael who is biology major .Sophia dating Thomas who is physics major.
14. C. Annoyed –discontent, dissatisfaction.
15. A. Distressed- great discomfort or sorrow.
16. D. Dejected to be in stat of low spirit ,filling sad.
17. B. wretchedness – to be unfortunate and miserable.
18. A. Regular physical activity is needed to improve one's physical as women will as mental health
19. A. in order to tone up the physical education program , the program should be reoriented and implemented.
20. B. According to some countries, health education is more important than teaching.
21. C. The author wants the reoriented physical education program to be comprehensive.
22. B. Never I have and never I will.

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23. B. For unreal /imaginary condition would +v1 is used in the main clause

24. D. It is the third conditional, and we use had + v3(past perfect tense) in the if clause.

25. D. Yes, I have. positive short answer (Yes, s+ HV)

26. A. Eagle and Bird have part and class relation and we say Rose and Flower.

27. D. Car and Transportation shows the relation of purpose, and we say Game and Entertainment.

28. B. Fast is used to describe speed, like wise warm describe temperature.

29. A. Fast show speed of doing something, similarly slow is speed of working.

30. A. Candle is made of wax, and paper is made of pulp.

31. D. Chef and food have worker and product relation, and a painter paints.

32. A. Board and Tran- is getting on train, Mount- Horse-is getting on horse.

33. C. Vehemently means strong eager felling similar to historically.

34. D. Counterfeit is something made or done in limitation of another thing in order to deceive Fake- what seems genuine. But not

35. D. Lavish –excessive and extravagant-excessively wasting money.

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For Questions 1 – 5 Use The Following Information.

The following table gives the expenditure (in millions of Birr) of a certain company per year over five years.

Year	Items of Expenditure				
	Salary	Fuel and Transport	Bonuses	Interest on Loan	Taxes
2008	288	98	3.1	23.4	83
2009	342	112	2.52	32.5	108
2010	324	101	3.84	41.2	74
2011	336	133	3.68	36.4	88
2012	422	142	3.96	49.4	98

1. What is the average amount of interest on loan per year that the company has to pay during the five year period?
   A. 36.58 B. 34.18 C. 33.72 D. 33.43
2. What percent of the total amount of salary was paid as bonus by the company during the given period?
   A. 1.25 % B. 1.00 % C. 0.5 % D. 0.10 %
3. In which year did the company have the highest expenditure?
   A. 2010 B. 2012 C. 2009 D. 2008
4. What is the total expenditure of the company in the year 2012?
   A. 456.36 B. 586 C. 715.36 D. 1710
5. For what item or purpose was the company's expenditure the least during the given period?
   A. Taxes B. Transport and Fuel C. Bonus D. Salary
6. If $x = 15\%$ of 30 and $y = 30\%$ of 15, then which of the following is true?
   A. The relationship between x and y cannot be determined.
   B. y is less than x.
   C. x and y are equal.
   D. y is less than x.
7. Student X and student Y have Birr 2000. If 4/15 of X's share is equal to 2/5 of Y's share, how much amount does Y have?
   A. Birr 1200 B. Birr 1100 C. Birr 800 D. Birr 1500
8. What is the simplified form of $(2x+3)(x+6)-(2x-5)(x+10)$?
   A. 72 B. 68 C. 16 D. 32

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9. If the square of a number increased by two times the number is equal to six times the number increased by twelve, which one of the following is the possible value of the number?

A. - 4 B. - 6 C. 4 D. 6

10. The monthly telephone bill of Getachew was Birr 400 and it increases by 10% after 12 months, and another 20% increase is applied six months later. What is the price of Getachew's telephone bill after 18 months?

A. Birr 489
B. Birr 430
C. Birr 520
D. Birr 528

11. The sum of the ages of the father and his son is equal to sixty years. Six years ago the age of the father was five times the age of the son. What will be the age of the son six years from now?

A. 22 years
B. 20 years
C. 19 years
D. 23 years

12. What is the sum of the following infinite series

$$S = 2 - \frac{1}{3} + 1 - \frac{1}{9} + \frac{1}{2} - \frac{1}{27} + \frac{1}{4} - \frac{1}{81} + \cdots$$

A. 41/9 B. 9/2 C. 7/2 D. 32/9

13. The Triangular Number Sequence T_n can be defined recursively as $T_1 = 1$, $T_n = T_{n-1} + n$ for $n > 1$. What is the eleventh term of the sequence?

A. 78 B. 55 C. 66 D. 45

14. The next term of the sequence 162, 54, 18, 6,...equals to

A. 1/3 B. 3 C. 2 D. 1

15. The first four terms of an arithmetic sequence are 39, 33, 27 and 21. What is the tenth term of the sequence?

A. -33 B. -27 C. -21 D. -15

16. The sum of the first n terms of an arithmetic progression is $2n^2 + n$. What is the tenth term of the sequence?

A. 38 B. 41 C. 39 D. 40

17. What is the probability of getting exactly two tails when three coins are tossed at the same time?

A. 3/4 B. 3/8 C. 1/4 D. 1/8

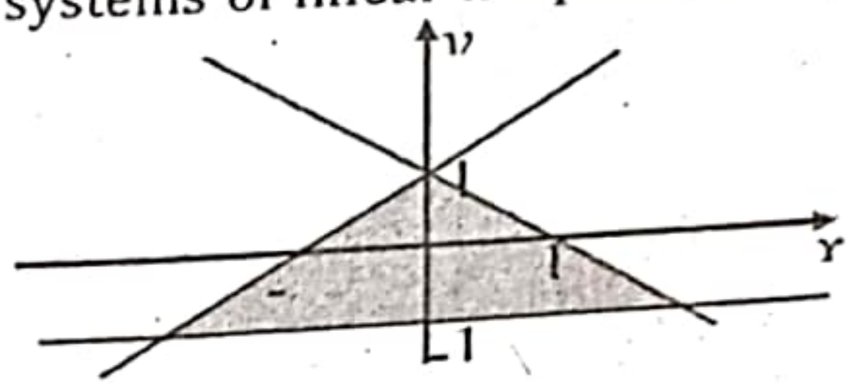
18. What is the probability that a number selected at random from the first 50 natural numbers is prime and odd number?

A. 8/25 B. 3/5 C. 3/10 D. 7/25

19. If $3x + 7y = 80$ and $x - y = 10$, then what is the value of $x + y$?

A. 25 B. 20 C. 10 D. 15

20. Which of the following systems of linear inequalities is represented by the graph given below



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A. $\begin{cases} y \leq x+1 \\ y \leq x-1 \\ y \geq -1 \end{cases}$

C. $\begin{cases} y \leq x+1 \\ y \leq x-1 \\ y \leq -1 \end{cases}$

B. $\begin{cases} y \leq x+1 \\ y \leq -x+1 \\ y \geq -1 \end{cases}$

D. $\begin{cases} y \leq x+1 \\ y \leq -x+1 \\ y \leq -1 \end{cases}$

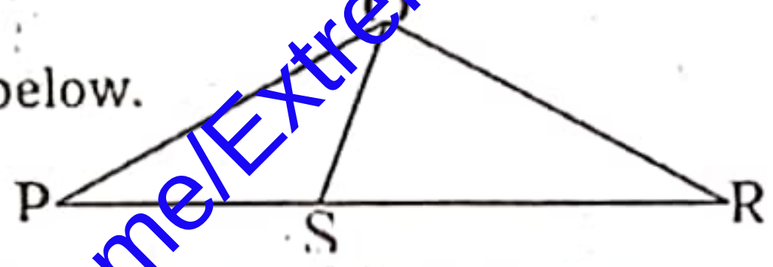
21. The square of the 7^{th} term of an arithmetic progression with positive common difference is equal to the product of the 3^{rd} and 17^{th} terms. What is the ratio of the first term to the common difference?

A. 4:3 B. 3:2 C. 3:4 D. 2:3

22. If $1^2 + 2^2 + 3^2 + \cdots + 10^2 = a$, then what is the value of $2^2 + 4^2 + 6^2 + \cdots + 20^2$?

A. $a+4$ B. $a+2$ C. $4a$ D. $2a$

23. Consider the figure given below.



It is given that PQ=RQ. Compare the following two quantities X and Y whose information are given below.

Quantity X	Quantity Y
PS	SR

A. The relation cannot be determined with the information provided.
B. The two quantities are equal.
C. Quantity Y is greater than Quantity X.
D. Quantity X is greater than Quantity Y.

24. If $y > 2$, compare the following two quantities X and Y whose information are given below.

Quantity X	Quantity Y
$\frac{3y-2}{5}$	y

A. The relation cannot be determined with the information provided.
B. The two quantities are equal.
C. Quantity Y is greater than Quantity X.
D. Quantity X is greater than Quantity Y.

25. The rectangular garden measuring 12 meters by 16 meters is to have a pedestrian pathway that is w meters wide installed all way around so that the total area including the pedestrian pathway is 252 m^2. What is the width of the pathway?

A. 1.2 m B. 15 m C. 12 m D. 1.5 m

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For Questions 1 – 5 Use The Following Information.
The following table gives the expenditure (in millions of Birr) of a certain company per year over five years.

Year	Items of Expenditure				
	Salary	Fuel and Transport	Bonuses	Interest on Loan	Taxes
2008	288	98	3.1	23.4	83
2009	342	112	2.52	32.5	108
2010	324	101	3.84	41.2	74
2011	336	133	3.68	36.4	88
2012	422	142	3.96	49.4	98

First build a table containing each year's total expenditure and total budget for each item of expenditure.

Year	Items of Expenditure					
	Salary	Fuel and Transport	Bonuses	Interest on Loan	Taxes	**Total Expenditure in a year**
2008	288	98	3.10	23.4	83	**495.50**
2009	342	112	2.52	32.5	108	**597.02**
2010	324	101	3.84	41.2	74	**544.04**
2011	336	133	3.68	36.4	88	**597.08**
2012	422	142	3.96	49.4	98	**715.36**
Total	**1712**	**586**	**17.10**	**182.9**	**451**	

1. The average amount of interest on loan per year that the company has to pay during the five year period is total amount of interest on loan divided by five, i.e. $\frac{182.9}{5} = 36.58$
Answer: A
2. Percent of total amount of salary paid as bonus is:
$\frac{\text{Total amount paid for bonus}}{\text{Total amount paid for salary}} \times 100 = \frac{17.10}{1712} \times 100\% \approx 1\ \%$
Answer: B
3. 715.36 million spent in 2012 is the highest expenditure.
Answer: B
4. The total expenditure of the company in the year 2012 is 715.36 million.

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Answer: C

5. The Company's least expenditure during the given period was 17.10 million for bonus.

Answer: C

6. $x = 15\%$ of $30 = \frac{15}{100} \times 30 = 45$, and

$y = 30\%$ of $15 = \frac{30}{100} \times 15 = 45$ so that x and y are equal.

Answer: B

7. Let x and y be the respective amount of share of X and Y.

Given $x + y = 2000$, and

4/15 of X's share is equal to 2/5 of Y's share

$$\Rightarrow \frac{4}{15}x = \frac{2}{5}y \Rightarrow x = \frac{3}{2}y \text{ so that}$$

$$x + y = 2000 \Rightarrow \frac{3}{2}y + y = 2000 \Rightarrow \frac{5}{2}y = 2000$$

$$\Rightarrow y = 800$$

Answer: C

8. $(2x+3)(x+6) - (2x-5)(x+10)$

$$= (2x^2 + 15x + 18) - (2x^2 + 15x - 50)$$

$$= \cancel{2x^2 + 15x} + 18 \cancel{- 2x^2 - 15x} + 50 = 68$$ **Answer: B**

9. Let the number be x. Then,

The square of a number increased by two times the number $(x^2 + 2x)$ is equal to (=) six times the number increased by twelve $(6x + 12)$

$$\Rightarrow x^2 + 2x = 6x + 12 \Rightarrow x^2 - 4x - 12 = 0$$

$$\Rightarrow (x+2)(x-6) = 0$$

$$\Rightarrow x = -2 \text{ or } x = 6$$ **Answer: D**

10. Birr 400 increases by 10% after 12 months and become

Birr (400+10%400)= Birr 440

Again Birr 440 increases by 20% after 6 months and become

Birr (440+20%440)= Birr 528

Therefore, Getachew's telephone bill after 18 months is Birr 528

Answer: D

11. Let x and y be the present ages of the son and father respectively.

The sum of the ages of the father and his son is equal to sixty years, i.e. $x + y = 60$ so that $y = 60 - x$....(1)

Six years ago the age of the father was five times the age of the son, i.e. $y - 6 = 5(x - 6)$

Then, using (1), $y - 6 = 5(x-6) \Rightarrow (60 - x) - 6 = 5(x-6)$

$$\Rightarrow 54 - x = 5x - 30$$

$$\Rightarrow 6x = 84 \Rightarrow x = 14$$

Thus, six years from now son's age will be:

$x + 6 = 14 + 6 = 20$ years

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Answer: B

12. Regrouping the series $S = 2-\frac{1}{3}+1-\frac{1}{9}+\frac{1}{2}-\frac{1}{27}+\frac{1}{4}-\frac{1}{81}+\cdots$ as

$$S = (2+1+\frac{1}{2}+\frac{1}{4}+\cdots)-(\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+\frac{1}{81}+\cdots)$$

We got the difference between two infinite geometric series with common ratios $r = \frac{1}{2}$ and $\frac{1}{3}$ where $|r| < 1$ so that by $\frac{G_1}{1-r}$

$$S = (2+1+\frac{1}{2}+\frac{1}{4}+\cdots)-(\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+\frac{1}{81}+\cdots)$$

$$= \frac{2}{1-\frac{1}{2}} - \frac{\frac{1}{3}}{1-\frac{1}{3}} = 4-\frac{1}{2} = \frac{7}{2}$$ **Answer: C**

13. $T_1 = 1, T_n = T_{n-1} + n$ for $n > 1$ so that

$T_2 = T_{2-1} + 2 = T_1 + 2 = 1+2 = 3$
$T_3 = T_{3-1} + 3 = T_2 + 3 = 3+3 = 6$
$T_4 = T_{4-1} + 4 = T_3 + 4 = 6+4 = 10$
$T_5 = T_{5-1} + 5 = T_4 + 5 = 10+5 = 15$
$T_6 = T_{6-1} + 6 = T_5 + 6 = 15+6 = 21$
$T_7 = T_{7-1} + 7 = T_6 + 7 = 21+7 = 28$
$T_8 = T_{8-1} + 8 = T_7 + 8 = 28+8 = 36$
$T_9 = T_{9-1} + 9 = T_8 + 9 = 36+9 = 45$
$T_{10} = T_{10-1} + 10 = T_9 + 10 = 45+10 = 55$
$T_{11} = T_{11-1} + 11 = T_{10} + 11 = 55+11 = 66$ **Answer: C**

14. The sequence 162, 54, 18, 6,... is Geometric with $r = \frac{1}{3}$ so that the next term is $r \times 6 = \frac{1}{3} \times 6 = 2$ **Answer: C**

15. 39, 33, 27 and 21 is arithmetic with $d = -6$ and $a_{10} = 39$ so that by $a_n = a_1 + (n-1)d$ we have

$$a_{10} = 39+(10-1)(-6) = -15$$ **Answer: D**

16. The sum of the first n terms of an arithmetic progression $\{a_n\}$ is $S_n = 2n^2 + n$. Thus, $a_1 = S_1 = 2(1)^2 + 1 = 3$

Then, by $S_n = \frac{n}{2}(a_1 + a_n)$ for $n = 10$

$$S_{10} = \frac{10}{2}(a_1 + a_{10}) \Rightarrow 2(10)^2 + 10 = 5(3+a_{10})$$
$$\Rightarrow 210 = 5(3+a_{10})$$
$$\Rightarrow 42 = 3+a_{10} \Rightarrow a_{10} = 39$$

Answer: C

17. In tossing three coins at the same time the sample space is S={HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}and $n(S)=8$
The event "exactly two tails" is E= {HTT, THT, TTH} and $n(E) = 3$

Thus, the probability of getting exactly two tails is: $p = \frac{n(E)}{n(S)} = \frac{3}{8}$

Answer: B

18. Considering the first 50 natural numbers the sample space S is with $n(S) = 50$, and the event "prime and odd number" is

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E={3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47} and $n(E)=$ 14

Thus, the probability of getting prime and odd number is:

$$p = \frac{n(E)}{n(S)} = \frac{14}{50} = \frac{7}{25}$$ **Answer: D**

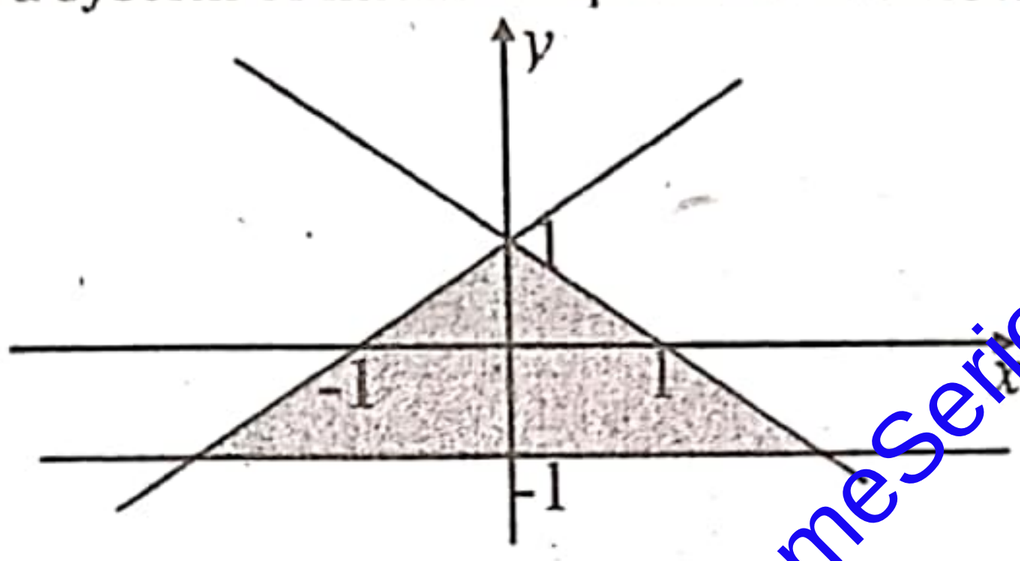
19. $x - y = 10 \Rightarrow y = x - 10$ in to $3x + 7y = 80$ gives

$3x + 7(x - 10) = 80 \Rightarrow 10x = 150 \Rightarrow x = 15$

Then, $y = x - 10 = 15 - 10 = 5$ so that $x + y = 15 + 5 = 20$

Answer: B

20. Given graph of a system of linear inequalities as follows.



Since the origin (0, 0) is in the region, testing the inequalities only the system in choice C satisfies, i.e. at (0, 0)

$$\begin{cases} y \le x + 1 \\ y \le -x + 1 \\ y \ge -1 \end{cases} \Rightarrow \begin{cases} 0 \le 0 + 1 \\ 0 \le -0 + 1 \text{ which is true.} \\ 0 \ge -1 \end{cases}$$

Answer: C

21. Given an arithmetic sequence $\{a_n\}$ be with common difference $d > 0$ and $a_7^2 = a_3 a_{17}$.

$\Rightarrow (a_1 + 6d)^2 = (a_1 + 2d)(a_1 + 16d)$

$\Rightarrow a_1^2 + 12da_1 + 36d^2 = a_1^2 + 18da_1 + 32d^2$

$\Rightarrow 4d^2 = 6da_1$

$\Rightarrow 4d = 6a_1 \Rightarrow \frac{a_1}{d} = \frac{4}{6} = \frac{2}{3}$

Thus, the ratio of the first term to the common difference is 2:3

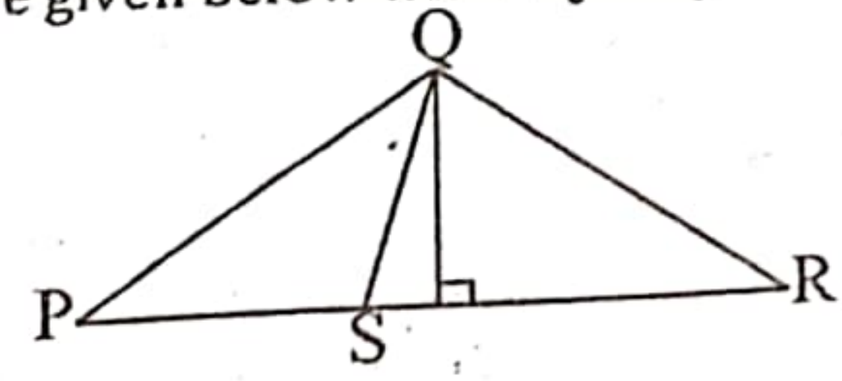
Answer: D

22. Given $1^2 + 2^2 + 3^2 + \cdots + 10^2 = a$. Then, factoring out 2^2

$2^2 + 4^2 + 6^2 + \cdots + 20^2 = 2^2(1^2 + 2^2 + 3^2 + \cdots + 10^2)$

$= 2^2 a = 4a$ **Answer: C**

23. Given the figure given below with PQ=RQ.



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We need to compare X and Y as follws.

Quantity X	Quantity Y
PS	SR

Since S is left of the foot of the perpendicular from Q, SR>PS so that Quantity Y is greater than Quantity X.

Answer: C

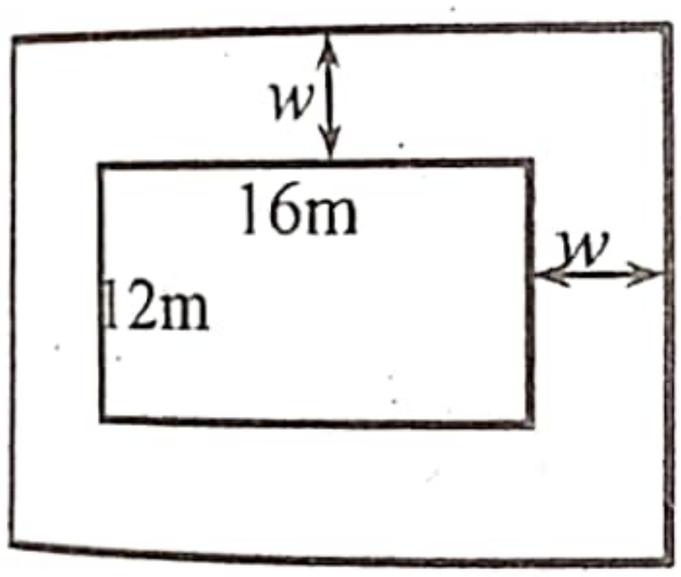
24. Given $y>2$, we need to compare the following two quantities X and Y whose information are given below.

Quantity X	Quantity Y
$\frac{3y-2}{5}$	y

$y>2 \Rightarrow 3y>6 \Rightarrow 3y-2>4 \Rightarrow \frac{3y-2}{5}>\frac{4}{5}$ but $y>2$ implies the relation cannot be determined with the information provided.

Answer: A

25. Given a rectangular garden measuring 12 meters by 16 meters with a pedestrian pathway that is w meters wide installed all way around as shown in the figure below so that the total area including the pedestrian pathway is 252 m^2.



The larger rectangle is with dimension: $16+2w$ by $12+2w$ so that

$$(16+2w)(12+2w)=252$$

$$\Rightarrow 192+56w+4w^2=252$$

$$\Rightarrow 4w^2+56w-60=0$$

$$\Rightarrow w^2+14w-15=0 \Rightarrow w=\frac{-14\pm\sqrt{14^2-4(1)(-15)}}{2(1)}$$

$$=\frac{-14\pm16}{2}=-15 \text{ or } 1$$

Since length is non-negative, $w=1$ m

Answer: No Answer

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DIRECTION : Items 1-18 are related to different aspects of writing. Read each of the questions carefully and choose the best answer.

Section One : Letter Writing

DIRECTION : Items 1-3 are about letter writing. Read each question carefully and choose the best answer according to the context.

1. The sender's address in an informal letter appears
   A. near the right margin
   B. below the date
   C. below the closing salutation
   D. near the left margin
2. the sentence, "I'm extremely delighted to her that you are going to join the University next year"is possibly taken from
   A. the body of a formal letter
   B. the closing remarks of an informal letter
   C. the beginning of an informal letter
   D. the end of a formal letter
3. which of the following is an appropriate opening salutation in an informal letter?
   A. Dear Mrs.Tsehay
   B. Dear Dr.Jemal
   C. Dear Sarah
   D. Dear Mr.Samuel

Section Two: Identifying the topic sentence of paragraph

DIRECTION: for items 4 and 5 choose the correct topic sentence of the paragraph

There are several reasons why my father is in excellent health condition. First, he is very healthy because he has stopped smoking cigarettes. He quit smoking because whenever he climbed stairs, he would stop several times on his way and cough loudly. He also has good health as a result of stopping eating the wrong kinds of food. Third, he is in good physical shape for he dose a lot of exercise. He swims three times a week at the local gym. In short, my father is in a better shape than some of his children.

4. A. There are several reasons why my father is in excellent healt condition
   B. He quit smoking because whenever he climbed stairs, he would stop several times on his way and cough loudly.
   C. In short , my father is is in a better shape than some of his children
   D. First, he is very healthy because he has stopped smoking cigarettes.

Electric cars are the future in the automobile industry. As climate change gets wors, governments are going to start limiting the number of gasoline cars being produced. The cost of gas will continue to rise, which will make it harder for people to afford to run gas powered cars. As battery technology

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improves, electric cars will be able to travel longer distances and cost less to buy. Eventually most gas powered cars will be replaced by electric cars.

5. A. As climate change gets worse, governments are going to start limiting the number of gasoline cars being produced.
   B. The cost of gas will continue to rise, which will make it harder for people to afford to run gas powered cars.
   C. Electric cars are the future in the automobile industry.
   D. Eventually most gas powered cars will be replaced by electric cars.

Section Three: paragraph Coherence

DIRECTION : when put in the correct order, the sentences items 6-9 make complete and coherent paragraphs.

6. a. One problem in relation to city buses is that a bus that should arrive at 7:30 in the morning may not arrive until 8:00 or later.
   b. Unfortunately, many people are often late for work or miss important appointments.
   c. Poor public transportation system is a big problem in our city.
   d. Moreover, bus drivers often pass up groups of people waiting for the bus at the bus stops.

   A. b c d a C. a b c d
   B. a c d b D. c a d b

7. a. Similarly, smoking is linked to other lung diseases such as bronchitis.
   b. The link between smoking and cancer is well known.
   c. Furthermore, there is substantial research evidence that even passive smoking can have long term effects on health
   d. In addition to lung diseases, smokers have a greater risk of heart diseases later in life.

   A. b d c a C. a c d b
   B. b a d c D. b c d a

8. a. At the beginning, my elder sister used to go to gymnastics lessons and showed me how to do some gymnasticed at home.
   b. Now, I take part in gymnastics when she practiced every Saturday.
   c. Thus, gymnastics is very special to me, especially because it is some thing I share with my sister.
   d. I started gymnastics is when I was six years old and it has been my favourite hobby since then.

   A. d a b c C. d b a c
   B. a d c b D. b c d a

9. a. Some peoples move to find better jobs.
   b. Other want to move to a place with a lower cost of living.

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c. There are many reasons why people go to other places to live.
d. Still others want to move to a place with less crime.

A. a b d c C. c a b d
B. d a b c D. a d b c

Section four : Punctuation and Capitalization

DIRECTION : For items 10-13, Choose the best alternative which is written with correct punctuation and capitalization.

10. A. The teacher said answer the question in complete sentences.
    B.The teacher said, "answer the question in complete sentences."
    C. The teacher said, Answer the question in complete sentences.
    D. The teacher said, Answer the question in complete sentences."
11. A. We are going to Guba to see the Great Renaissance Dam in September.
    B. We are going to Guba to see the great renaissance Dam in september.
    C. We are going to guba to see the great renaissance Dam In September.
    B. We are going to guba to see the Great Renaissance Dam in september.
12. A. She went to the shop and bought, a kilo of sugar a litre of milk a kilo of bananas and some vegetables .
    B. She went to the shop and bought, a kilo of sugar ,a litre of milk a kilo of bananas and some vegetables .
    C. She went to the shop and bought, a kilo of sugar, a littre of milk, a kilo of bananas and some vegetables .
    D. She went to the shop and bought a kilo of sugar a litre of milk a kilo of bananas and some vegetables .
13. A. He wanted to find out how the accident occurred!
    B. He wanted to find out how the accident occurred
    C. He wanted to find out how the accident occurred.
    D. He wanted to find out how the accident occurred?

Section Five : word Order

DIRECTION : in item **14-16**, the words are given in the **wrong** order. When put in the right order, the words form a correct sentences . from the given alternatives, choose the one with the correct order.

14.If/have/us/phoned/we/joined/them/had/would/
    A. If phoned them would they joined us have we had.
    B. They would have joined us if we had phoned them.
    C. We have if would them us had they joined phoned.
    D. They would joined them have phoned if had we as.
15. They/home/are/today/early/coming

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A. Coming are they early today home?
B. They are coming early home today?
C. Are they coming home early today?

16. read/ notes/ you/to/the/must/your/ test/do
A. You must read your notes to do the test.
B. Read notes must to your do the test you.
C. The test notes to read you must your do.
D. You must to read the test notes to your.

Section Six: word Spelling

DIRECTION : for items **17and 18**, choose the word with the correct spelling

17. A. simultaneousls B. for cast C. lisence D. accomodate
18. A. approach B. immature C. playright D. courageous

II.Communicative Activities

DIRECTION: Items 19-44 are presented in the form of dialogues. This part side by one of the speakers is given, and blank space is left for the other. For each of the blanck space, for alternative answers,**A-D** are given

19. **Mubarak :**_____

Semir: in my cases, there were more primary and secondary schoolteachers

A. There are many difference between primary and secondary school teachers.
B. There were many more female teachers than male teachers in my primary school
C. The in my primary school were more help full than those in my secondary school.
D. The number of teachers in my secondary school was greater than those in my primary school

20. **Bikila:** Could you tell me where the Federal Ministry of education buildingis?

Olani:_____

A. Why do you want to know the location?
B. You mean the education beauro of Addis Ababa.
C. The Ministry of education has many offices.
D. It is at Arat Kilo,close to the Miazia 27 Monument.

21. **Foreigner:**_____

Wolde: Oh,sorry, I don't know. I don't live in this part of the city.

A. Can you tell me where the Capital Hotel is ?
B. How long have you been living here?
C. Can I talk to you for a minute, please?
D. Excuse me. Do you live near here, if I may ask?

22. **Ahemed:** I was very hungry and actually ate a lot at lunch time.

Muhaba:_____

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A. I and my best friend eat lunch together frequently.
B. They say hunger is the best sauce.
C. I usually can't find a good place to eat.
D. Food has become expensive recently.

23. ***Tariku:*** _____
Barnabas: I think that is what we need in order to get rid of poverty.
A. I do not understand why some people like to cause trouble to others.
B. Many young people are in need to jobs.
C. We must prevent conflicts and live together peacefully.
D. There are a lot of challenges that our country is facing.

24. ***Dinku:*** Traffic accidents continue to be real problems in this city.
Fikru : _____
A. Many people use buses because they are cheaper.
B. Both drivers and pedestrians must be much more careful on the road.
C. The new roads that are being built in different parts of the city are very wide.
D. the population of the city is growing fast and the city is expanding in all direction

25. ***Teacher:*** _____
Student: Earthquake is one type of natural disaster.
A. Given an example of a natural disaster.
B. when natural disaster occur in Ethiopia?
C. where do natural disaster occur?
D. what do you mean natural disaster

26. ***Duressa:*** are you the eldest in your family?
Demitu: _____
A. I am the youngest of the children of my family.
B. we are a large family with four children
C. My sister is older than me
D. My brothers are all older than me,

27. ***Father :*** please let me Know that you have arrived there safely.
Daughter: _____
A. I am sure we will be there by this time tomorrow.
B. It is going to be a long journey.
C. I will call you as soon as I reach there.
D. I' am sure I will enjoy my fight.

28. ***Daughter*** are you ready, Mum?
Mother: _____
A. Yes I am coming.

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B. What are you doing there?
C. Yes, we will ready to buy a lot of things.
D. No, there is no one else coming.

29. *Jourunalist:* Do you think there is enough fish for you to catch every day in this lake?

*Fisherman :*_____

A. Most of us who live around here do fishing both for food and support.
B. There are more fisherman nowadays there are n't enough for every one.
C. The lake is home to different kinds of fish and other aquatic animals
D. My livelihood depends on fishing and I love fishing more than anything else.

30. *Fekadu :* which radio station do you listen to?

*Alemu:*_____

A. I didn't listen to any radio programmes these last two days.
B. I am not sure how many radio programmes there are at present.
C. Radio Fana programs are my favourite.
D. These days there are many radio programs to listen

31. *Aman:* Talking on the mobile phone is not free from problems.

Abdul: How ?

*Aman:*_____

A. Every body has a mobile phone these day.
B. I don't pick up on the phone if I don't want to
C. There may be sudden connection problem.
D. Smart phones are expensive for many people.

32. *Seifu:* Surely, the internet is not perfect invention.

Biruk: What do you mean?

*Seifu:*_____

A. If you want, I can show you how to use it.
B. The internet is the art of the technology.
C. To put it another way, it has many shortcomings
D. Using the internet speed up our work.

33. *Student:* I am sorry but I haven't understood your last point.

*Teacher:*_____

A. Do you have any other question?
B. You will have some point about it.
C. I don't think you understand.
D. Ok, let me rephrase what I just said.

34. *Chairperson:* Let us discuss how inflation in our country is affecting our work

*Participant:*_____

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A. Excuse me for interrupting but we should first discuss the problem in our organization.
B. I think inflation is difficult to control.
C. There is no doubt that inflation affects the lives of our citizens in many ways.
D. There is inflation all over the world and there is need for global solution to control it

35. **Farmer 1:** Why are we not able to get enough crops from the land recently?
Farmer 2: It is because we have been farming the land for many years without making any changes on it
Farmer 3:_____
A. We may not even food our families.
B. We should continue to work as hard as we used to
C. Plus the fact that we have never used fertilizers is another cause.
D. We always work very hard but we don't produce enough.

36. ***Tesfu:*** I think we are going to face to face serious social problems in our city.
Bereket: How?
Tesfu:_____
A. The solution to our problems is for as to sit together and work towards a common goal.
B. The administration is doing its best to create jobs to reduce the number of unemployed people.
C. All of as an responsibility to tackle our social problems.
D. To gave you an idea, there are too many unemployed young people in the city

37. ***Fikirte:***_____
Hiwa: I am your friend and I am always ready to do any thing that makes you happy
A. Can you guess what I want to do for your?
B. I have some good news to share with you, but don't be in a hurry.
C. If you like me, you will go to my parents' house with me tomorrow.
D. Is there any thing I can do for you?

38. ***Father:***_____
Son: But Dad,I have agreed to study with my friends on Saturday.
A. You know that your grandmother loves you very much and talks about you always.
B. I have decided that you will go and visit your grandmother on Saturday.
C. The weekend is approaching, and you must be looking forward to it
D. What are you going to do next Saturday?

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47. a. A typical example is what happened in the late 1990s.
b. Dependence on coffee has thrown a lot of nations into crisis.
c.This usually happens when the world commodity price falls.
d.Coffee farmers unions should be set up to resist the problem.

A. b c a d
B. d b c a
C. a b d c
D. c d a b

Section Four: Paragraph Development Techniques

Direction: Questions **48** and **49** are related to paragraph development techniques. From the alternatives **A-D** given for each question, choose the correct answer and **blacken** the letter of your choice on the separate answers net provided.

48. Which one of the following sentences is most likely taken from a paragraph
   A. There are lots of things that make life in the area very difficult.
   B. Whenever she gets nervous and depressed, I try to calm her down.
   C. Robots are one of the most amazing inventions of the human mind.
   D. Every morning, we clean the compound and make everything very neat.

49. Which one of the following sentences is most likely taken from a descriptive paragraph?
   A. Paul says injustice kills your work spirit and demoralizes those who work hard
   B. Paul was such a quiet chap who was always silent regardless of the bitter injustice.
   C. Paul finally convinced himself that he should rebel and then went straight to his boss.
   D. Paul understands that injustice is a practice that everybody should fight and eliminate.

Section Five: Punctuation and Capitalization

Direction: Items **50-53** are about punctuation and capitalization. From the given alternatives A-D, choose the statement which is correctly punctuated and capitalized, and blacken the letter of your choice on the separate answer sheet.

50. A. We bought them blankets, bed sheets, cloths and, some rice.
   B. We bought them blankets, bed sheets, cloths and some rice.
   C. We bought them blankets, bed sheets, cloths, and some rice.
   D. we bought them,blanket,bed sheets,cloths and some rice.

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51. A. At St Paul Hospital, they train doctors in a variety of fields, e.g., cardiology, neurology, etc.

B.At St. Paul Hospital, they train doctors in a variety of fields, eg., cardiology, neurology, etc.

C. At St. Paul Hospital, they train doctors in a variety of fields, e.g. cardiology, neurology, etc.

D. At St. Paul Hospital, they train doctors in a variety of fields, e.g.,cardiology, neurology, etc.

52. A. We used to stay at the Hilton hotel and dine at Kokeb Restaurant.

B. We used to stay at the Hilton Hotel and dine at Kokeb restaurant.

C. We used to stay at the Hilton hotel and dine at Kokeb restaurant.

D. We used to stay at the Hilton Hotel and dine at Kokeb Restaurant.

53. A. They donated money;gave out food; clothing and medication; and pledged a lot more.

B. They donated money; gave out food; clothing and medication, and pledged a lot more.

C. They donated money; gave out food, clothing; and medication; and pledged a lot more.

D. They donated money; gave out food, clothing; and medication; and pedged a lot more.

Section Six: Letter Writing

Direction: Questions **54-56** are about letter writing. Read each item carefully and answer it accordingly. Then, **blacken** the letter of your choice on the separate answer sheet provided.

54. What is your opening sentence in a formal letter supposed to do?

A. State clearly why you are writing the letter.

B. Greet the person you are writing the letter to.

C. Introduce yourself to the person you are writing to.

D. Make your address known to the person you are writing to.

55. When do we use "Dear Sir/Madam" in a formal letter?

A. When we want to write about something that is very serious.

B. When we want to show respect to the person we are writing to

C. When we do not know the name of the person to whom we write.

D. When the person we are writing to is the head of an organization.

56. Which of the following could be a typical ending to your letter of complaint to a company?

A. Yours truly,
B. Many thanks,
C. Kind regards,
D. Yours faithfully,

III. COMPREHENSION

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Directions: There are two passages under this section. Read each of them carefully and answer questions **57-76** according to the information therein. From the four choices, **A-D**, given for each question, choose the correct answer and **blacken** the letter of your choice on the separate answer sheet provided.

Section One: Reading Passage I

Grown-ups on Campus

1. Many professors enjoy teaching adults because they frequently work harder and have a better idea of what they want out of school than their younger classmates. "A lot of young people who come in here at 18 don't have the foggiest notion of why they're here," says Art Gallaher, vice president of academic affairs at the University of Kentucky.

2. The adults tend to be better prepared, too. Because they went to high school at a time when learning how to write was considered more important than advanced student leadership, their compositions have fewer errors in syntax, grammar and spelling than the essays of some of today's brightest undergraduates. And older students usually get their work in on time. "I'm going to get the assignment done because I know I might have to stay home one day with my sick child," says Phyllis Dunham, 27, a psychology major at the University of Florida. "Younger students seem to wait until the last minute and have to have an extension for this reason or that." A few teachers even find their older students intimidating. "If you're in your early 30's with a fresh PhD. and a lot of book knowledge, it's rather threatening to be teaching someone who's been working in your field for years and is questioning your theories," says Arthur Chickering, director of the Center for the Study of Higher Education at Memphis State University.

3. But more often than not it is the adult students who are insecure. Friends and family may not like or understand their new status. Jean Byrd, 46, a Hood sophomore says, returning to school at first created some friction with her 12-year-old daughter. "She resented it because my grades were better than hers." Younger classmates aren't always friendly either. "My first couple of years it was, like, what are these women doing here?" says a young Hood junior. "I'm in college. I thought I left my mother at home." The sense of alienation is heightened at places like Florida, which is still very much geared toward youngsters though about one-fifth of its students are older. "They're putting a strain on the place because we're not used to dealing with them." confesses Mike Rollo, assistant dean for student services.

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4. The feeling can be mutual. "In the beginning I felt very old," says Juliet Margolin, 28, a Florida pre-vet student who lived in a dorm for three months and felt "like a counselor in a sleep-away camp." Appalled by ketchup in the showers and jelly on the mirrors in the communal bathroom, Margolin soon found a roommate of her own age and moved off campus. Kate Erickson, a 35-year-old English major at Rice University, says she is so out of place that she is trying to organize older students into a group she might call "Anachronisms Anonymous." At Hood, Dixie Miller has set up a hide-away cottage for her continuing-ed students- a place she says, "to meet somebody else who is gray and grumpy."

Source: Reading and Study Skills (Second Edition)

57. What, according to Mike Rollo, was the problem at the University of Florida?
    A. Older students feel so unfit to the context.
    B. Younger girls embarrass older women calling them 'mothers".
    C. The University administration is new to handling older students.
    D. Older students account for only about one-fifth of the student community.

58. "... because we're not used to dealing with them" (Paragraph 3). What does the word ***them*** refer to?
    A. Older students
    B. Younger students
    C. Places like Florida
    D. The student community

59. What does "The feeling can be mutual" mean (Paragraph 4)?
    A. Both groups think it is hard to live with the other.
    B. Both the grownups and their roommates feel the same way.
    C. Both the university and the youngsters think the grownups are old.
    D. Both the university and the youngsters think those grownups are unfit.

60. What is the central idea of the last paragraph?
    A. It is not very wise for older students to go back to university.
    B. Youngsters do not like to see older students on university campus.
    C. It is very hard to have young and old students together in a university.
    D. University is a place where distinctions should not be made because of age.

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61. *"I thought I left my mother at home"* (paragraph 3). My mother in this sentence refers to:
A. Women classmates
B. The speaker's mom
C. Elderly women on campus
D. Older women in the group

62. Why do university professors prefer teaching adults to teaching young people?
A. Eighteen year-olds join universities with foggy notions.
B. Adult learners have a clear purpose for being on campus.
C. Younger students are much more careless about their life.
D. Adults work much harder than their younger counterparts.

63. Why do adult learners writing better than some of the best younger students?
A. Because they liked writing more than they did the other skills.
B. Because their teachers preferred teaching writing to the other skills.
C. Because today's students attach more importance to leadership only.
D. Because during their days, the system had given more emphasis to writing.

64. Comparatively, why do older students tend to get their work done in time?
A. They do not want to be rushed up when it is too late.
B. Some of them have sick children whom they have to look after.
C. The old school system shaped them in a way that does not procrastinate duties.
D. It is to have time for unforeseen situations that may unexpectedly drag them away.

65. What makes adult students feel insecure?
A. Younger students sometimes swear at them.
B. Friends and family members may dislike them.
C. Some of the students may envy their performance
D. Socially, it is hard to compete with younger students.

66. Why did Juliet Margolin eventually decide to evacuate her dorm?
A. The little dorm mates made her feels very old.
B. She wanted to live with someone more matured.
C. She couldn't bear the way her dorm mates behaved.
D. She didn't like the ketch up her dorm mates were using.

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Section Two: Reading Passage II

Laziness or Dishonesty is Ruining Reputation

1. A study by the Center for Academic Integrity found that almost 80% of college students admit to cheating at least once, while this is between 75% and 98% of high school students. You might not believe this, but the statistics of 2018 show that. Although, it is not astonishing news when there are fears of failing, desires to get better grades, poor time management, or a lack of recognition of plagiarism among students.
2. Also, there is another kind of plagiarism that is stealing other's quotes, arts, recipes, and ideas without mentioning the name of the owner, and it is non- academic plagiarism. There are some similarities between academic and non-academic plagiarism, particularly, in motivations or reasons, methods, and effects. In academic plagiarism, students have reasons and methods to cheat and obviously, they will result in negative effects. Plagiarism in academic situations is nothing new, and there has been a high percentage of college students and high schools who have plagiarized for several years. The most important reason should be getting better grades without enough effort, due to laziness or lack of talent, confidence, and knowledge to stand on their own feet. Most cheaters would like to have a fun social life so. they don't spend time studying as much as they should.
3. Students steal others' research, ideas, and statistics, and use other students' lab reports and fake signatures on official documents for their financial aid applications. Ultimately, the consequence shows that most of the cheaters get disciplinary actions, alerts, and notifications. They can be removed from all classes for one or more terms, and this affects the student's ability to attend graduate schools and will be noted on their academic transcript. On the other hand, hold and pending may be placed on students' records, and the record will be kept for a minimum of five years. And the worst result of that trauma and shock is the psychological and mental impact on them which remains for years. However, all the sanctions and policies are to protect the safety and integrity of the university community, and to learn the students, new behaviors.
4. Non-academic plagiarism is similar to academics in terms of motivation, methods, and effects. For instance, the pressure and fear could be the motivations for the politicians to plagiarize, because it would be a safer way to steal a confirmed presentation. On the other hand, other reasons would be lack of preparation, or confusion. The same as academics, the politicians repost some

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sentences in their speeches or books without mentioning the name of origin, and without respecting the real author and mentioning her/his name. As a matter of fact, the result will be losing their reputation, trust, dignity, expulsion, losing their positions, quitting their job, getting impeached, just like the college students when be caught of plagiarism.

5. In summary, the main similarity between academic and non-academic plagiarism should be the personal, professional, ethical, and legal consequences. If in academics, plagiarism causes a student to be suspended, the same can cause to a politician to be impeached. In the case where a journalist works for a magazine, newspaper or other publishers, or even a student is found plagiarizing in school, the offending plagiarist could have to pay monetary penalties. Both types mean a destroyed reputation and the end of the academic position. Following as a result, laziness or dishonesty can lead to ruined reputation, the loss of a career, and legal problems.

67. What does the word **they** in "... obviously, they will result in negative effects" refer to (Paragraph 2)?
   A. Reasons and methods of cheating
   B. Students behind the act of cheating
   C. Similarities between types of cheating
   D. Both academic and non-academic cheating

68. Which one of the following, in your opinion, belongs to non academic cheating?
   Using others' lab reports for submission in a course
   A. Using fake signatures in applications for financial aid
   B. Taking other students' statistics for report in assignments
   C. Stealing others' research plans and designs to write a thesis

69. What, according to Paragraph 3, is the most undesirable consequence of academic cheating?
   A. Suspension from the university for one or more terms
   B. A disciplinary note recorded on and academic transcript
   C. The psychological and mental trauma inflicted on the cheater
   D. A five-year hold and pending remark on the cheater's records

70. If you send a plagiarist to jail for one-year service, the most likely, what is the consequence of the decision?
   A. legal C. personal

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B. ethical D. Professional

71. According to paragraph one, which one of the following is true about cheating?
    A. Almost 80% of college students in the US cheat in their exams.
    Recently, cheating has become part of every student's life in the US.
    B. The statistics of 2018 show that cheating is very common among students
    C. Cheating is more severe among university students than high school students.

72. Which one of the following is **NOT** a possible cause of cheating?
    A. The interest to score good grades.
    B. Recognition of the role of plagiarism
    C. The fear to fail in a test one is taking.
    D. Lack of awareness that cheating is wrong.

73. Which one of the following is **NOT** among the similarities between academic and non-academic plagiarism (Paragraph 2)?
    A. The actors of plagiarism
    B. The way the act is committed
    C. The anticipated rewards of the act
    E. The specific interest behind the act

74. What do you think is the most important reason behind secondary school leavers' attempt to plagiarize in this exam?
    A. Lack of self-confidence
    B. Laziness and making little effort
    C. The interest to score good grades
    D. The feeling of dependence on others

75. Which one of the following sentences is true about politicians in relation to plagiarism?
    A. Politicians avoid plagiarism due to fear of pressure from the public.
    B. Confirmed presentations tempt politicians and push them to plagiarize.
    C. Politicians may resort to plagiarism as a means of covering up their laziness.
    D. Consequences of plagiarism could be comparatively less harsh on politicians.

76. What is the main purpose of taking measures against cheaters in a university environment?
    A. To discourage undeserved merits
    B. To safeguard the academic community

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C. To teach new behaviours to the cheaters
D. To protect students' safety and integrity

Section Three: Vocabulary

A. Completion

Directions: Questions **77-84** are incomplete sentences. There are four alternative words or phrases, **A-D**, given below each question. Choose the word or phrase that best completes the sentence and **blacken** the letter of your choice on the separate answer sheet provided.

77. What all the police could find were _____ of glasses from the building but further investigation might reveal the group behind such a heinous move.
A. parts
B. signs
C. evidences
D. fragments

78. As the company is new, we will need to produce some kind of_____ to be distributed to the community around.
A. notice
B. poster
C. leaflet
D. newspaper

79. We all need to work hard to reverse the ongoing moral_____ in this country and breathe life into it.
A. decay
B. challenge
C. state
D. plunge

80. That twenty of their friends died in a crash came to everybody as a_____ and they didn't know what to do.
A. shock
B. worry
C. chance
D.calamity

81. We will need to_____ what they are doing and how they are doing it so as to give our testimonies.
A. observe
B. visualize
C.repots
D. overhear

82. If we are not able to_____ ideas accurately, misunderstandings would naturally crop up.
A. inform
B. enunciate
C. communicate
D. propagate

83. Teachers should_____ the attitude of honesty and integrity in the minds of their learners.
A. inflict
B. cultivate
C. impart
D.consolidate

84. It is _____ that he committed suicide. They said he was a very successful politician.

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A. vivid
B. apparent
C.unjust
D.incredible

B. Substitution

Directions: For questions **85-92**, each sentence has an underlined word or phrase followed by alternatives **A-D**. Choose the word or phrase that best corresponds with the meaning of the underlined word or phrase and **blacken** the letter of your choice on the separate answer sheet provided.

85. The war between Russia and Ukraine cannot be solved through discussion by both countries only. It needs joint negotiations among groups with vested interest.
    A. popular
    B.common
    C. multilateral
    D. multisectoral

86. If we cannot eliminate malaria, we are going to pay dearly: It is claiming a lot of young lives working in the rural areas of the country
    A. remove
    B. fight
    C. dismiss
    D. eradicate

87. Most of us are not modest enough to learn from others. We always think we are the best.
    A. inferior
    B. humble
    C. polite
    D. submissive

88. When we take an entrance examination like this, the results are almost always dual. So, we have to work hard to be in the pass group and avert the fail one.
    A. both
    B. binary
    C. paired
    D. teamed

89. There is a huge increase in the number of newly set up educational institutions during the last thirty years but we need to work on quality as well.
    A. bulky
    B. massive
    C. progressive
    D.comprehensive

90. We cannot be talking about failure of plans every year. What have you done about them and where and when did you succeed?
    A. Shallowness
    B. Impracticality
    C. miscarriage
    D. weakness

91. The main factor that led to the disbanding of the union is unfair distribution of income among members. Some of the leaders were absurdly egocentric.
    A. simple
    B. direct
    C. best
    D. principal

92. The disagreement between the warring parties should be solved amicably. That seems to be the only way that could lead to a win-win situation.

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A. hatred
B. dispute
C. assault
D. contempt

Section Four: Sentence Comprehension.

Directions: For Items **93** and **94**, choose the sentence with the closest meaning to the head sentence from the four alternatives given, and **blacken** the letter of your choice on the separate answer sheet provided.

93. The rapid spread of HIV and AIDS is a consequence of mass denial and social disgrace of persons living with the virus.
   A. The main cause of HIV/AIDS is denial and social disgrace.
   B. Denying the existence of HIV/AIDS leads to eventual disgrace of victims.
   C. The spread of HIV/AIDS gets worse when people deny infection and also alienate victims.
   D. The existence of HIV/AIDS and disgrace of people living with the virus aggravates the spread of the virus.

94. Abusing drugs can only provide a temporary escape from the world reality.
   A. Drug abuse can effectively solve temporary problems.
   B. Drugs are used by people who live in an elusive world.
   C. Drug abuse creates a world free of worries and sufferings.
   D. Drug abuse is a means of avoiding unpleasant facts of life.

IV. Communicative Activities

Direction: Questions **95 - 120** are presented in the form of conversation. The part said by one of the speakers is given, and a blank space is left for the other. Choose the alternative that is most appropriate to complete the dialogue and **blacken** the letter of your choice on the separate answer sheet provided.

95. **Haji:** Why don't we visit Chebera Churchura?
   Gorge:_____
   A. Is the administration under the SNNPR or the federal government?
   B. I think the biggest problem is where to stay. There is no hotel there.
   C. We have contributed money, though I don't know exactly how much.
   D. I've seen it on TV and they're now almost finishing it. Great achievement!

96. **Tesfaye:** What advantage do I have in joining the union?
   Bereket: :_____
   A. Come on! Many of our close friends have joined it already.
   B. You're always against such new ideas, aren't you, brother?

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C. Lots! For instance, you can get a loan of at least birr 50,000.
D. Someone from the union was talking about its details yesterday

97. **Oumer:** Shall I pay you when I come back in June?
Mensur: :_____
A. No! I will have already left for Spain by the end of May
B. Never lend anyone your money. They don't pay you back.
C. I know you have hundreds and thousands in the bank. Bravo!
D. Are you flying or taking a bus? How long will you be there?

98. **Amesu:** When is your sister going to the university?
Yonas: :_____
A. In my opinion, she is not going to do that. She is not interested.
B. Addis Ababa University is her favorite choice.
C. She wants to go to Dubai and work. They have bought her a ticket.
D. None of the universities will start within a month. They are not ready.

99. **Colleague:** We're going out for dinner tonight, aren't we?
You: :_____
A. That restaurant is always busy..
B. I'm not coming, if you don't mind.
C. How much money do we have for that?
D. Are you a member of the Social Committee?

100. **Mimi:** I guess our driver was thrown out. Anyway, it was
Kuku: :_____
A. I don't understand what he threw out.
B. I'm sorry I didn't follow what you said.
C. I think he and his boss were great friends.
D. It was a serious accident and he lost a hand.

101. **Chuchu:** Your friend has got a good job, hasn't she? **Meklit:**
:_____
A. Getting a job is not very easy these days.
B. Yes, she's got a job. I heard it is out of town.
C. She graduated about five years back, you know.
D. I feel that the offer is rather too low, birr 2500 a month.

102. **Telila:** Why didn't you come to my party?
Aster:
A. Did everyone else come?
B. Was it held at your home or...?
C. I could imagine it was so lavish
D. I apologize for that. I was on-duty.

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103. **Teacher:** Where is the boy sitting over there?
Fola: :_____

A. He is locked up. What I mean is,... he is jailed.
B. He always fights with people and no one likes him.
C. He is very intelligent. You don't know that, do you?
D. His name is Zekarias. I don't know his father's name.

104. **Eyob:** What is your grandma like
Sisu: :_____

A. She likes eating *shiro wot*
B. She's now been ill for a year.
C. She's very old and weak. She is 92.
D. She had a checkup yesterday. Right move!!

105. **Woltaji:** Where is the office of the National Farmers Union?
Mbobo: :_____

A. You don't need any bus or taxi for that. Are you a stranger around?
B. I have been to that office only two days ago. It is on a beautiful building.
C. Up there. When you see a tall building with blue glasses, you should turn left.
D. Office of the National Farmers Union? They're easing food prices in this country.

106. **You:** He wouldn't give me back my computer. What should I do?
Sister: :_____

A. Unless you pay him back his money, he won't do it.
B. Is that the computer they gave you from your office?
C. Doesn't he know that computers make life very easy?
D. It is very difficult to do assignments without a computer

107. **Shashe:** I wanted to buy a TV but I couldn't decide the brand.
Melat: :_____

A. The best these days is Sony TLC
B. Most TV programs are really boring.
C. Good TVs cost over 50,000 birr now.
D. I know someone who sells electronics stuff.

108. **Kebede:** Do you think their plan is going to work out? **Nigist:** :_____

A. Good plans can sometimes fail.
B. Who knows? Let's wait and see.
C. Plans should be carefully discussed.
D. By the way, who has masterminded it?

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109. **Leul:** I hear they have all been trapped in a tunnel.
Brother: :_____.
A. I know tunnels could be dangerous.
B. When do they come out, you think?
C. No, no! What can be done to save them?
D. In fact, tunnels should have been out of use.

110. **Christian:** Sir! **:**_____
Mr. Sharma: That is not in my field of studies, but basically it is...
A. Your presentation was finished just on time. Good job!
B. Is it true that you are one of the best astronomers in India?
C. Would you like to say something about Artificial Intelligence?
D. Some people say that you have published over fifty one books so far.

111. **Journalist:** Why is it that you always avoid coming out on mass media such as TV?
Psychologist: :_____
A. The reason I don't want to do that is very personal. You know,....
B. In fact, mass media plays a big role in raising awareness on the issue.
C. After all, mass media in this country should be doing their proper jobs.
D. You know many people who call themselves elites like TVs and radios.

112. **Tesfaye:** The security guard tells me his wife has been hospitalized.
You:_____
A. You know, I have never met his wife, have you?
B. Really? Health care is very expensive these days.
C. What a pity! Who takes care of the children then?
D. Perhaps we could contribute some money and help.

113. **Almaz:** This plate fell off the cabinet but it didn't break.
Husband: :_____
A. These days, plates are not like that.
B. Please, keep the plates in a safe place.
C. How nice! It seems to be made of porcelain.
D. You are being rather too careless these days.

114. **Director:** We are going to suspend that student who broke the window.
Unit leader:_____
A. Such misconducts are becoming a fashion.

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B. I agree. That would be an essential measure.
C. Which section does he belong to? Section 4?
D. Amazing! Why do some students like to misbehave?

115. **Teacher:** I always see Mimi and Bultum studying. They're working haRD
Colleagu :_____
A. Maybe, they'll also need to do other things in between.
B. Bultum is one of the very hardworking students in class.
C. Great! There is a good chance they'll pass the national exam.
D. You know, I teach them two subjects - Biology and Chemistry.

116. **Driver:** What do you mean? How can a robot replace me?
Yo :_____
A. Robots are products of modern technology.
B. To give you an idea, robots can do anything we do.
C. Haven't you seen robots helping people cross roads?
D. I have seen robots on TV and they are really very smart.

117. **Farah:** Is that boy Ali, your classmate?
Nisra: :_____
A. Ali is very gentle and polite.
B. I didn't know you knew Ali.
C. No, Ali is from around Babile.
D. No, Ali is not as tall as this one.

118. **Interviewer:** Was that your first time starring in a film?
Bamlak: _____
A. No, I had acted in another film before that.
B. No, I don't remember the films I starred in.
C. Never, I was involved in three films after that.
D. No, I acted in a film in English-*No Good Death.*

119. **Student:** Sir, We've won the inter-school competition. Our school stood first.
Director: :_____
A. This is our first time to participate.
B. Wasn't the competition stiff as such?
C. Really? I actually knew two of the judges.
D. What great news! I'm pleased to hear that

120. **Interviewer: :**_____
Jorgie: Actually, my interest is in fashion designing.

A. Do you like fashion designing?
B. Did I hear you like fashion designing?
C. Could you tell me about your interest?
D. Why are you interested in fashion designing?

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1. A. In an informal letter the sender's address is written on the right top corner / near the right margin/ above the date
2. C. Such piece of writing is taken from the beginning of an informal letter.
3. C. opening salutations like Dear sir/Medam, Dear Father, Der Sarah are commonly used in an informal letter,

Identifying the topic sentence of a paragraph

4. A. This paragraph mainly talks about the reasons why the writter's father is in excellent health condition,
5. C. The topic sentence is the one given at the beginning of the paragraph,i.e,electric cars are the future in the automobile industry, paragraph coherence
6. D using poor public transportation system is a big problem in our city' as a topic sentence if the others are arranged a,d,b, they can make up a complete paragraph.
7. B. If those four sentence are arranged , b,a,d and c, they can make up a complete paragraph
8. B. The correct arrangment to make a complete paragraph is a,d,c,and,b.
9. C. The complete paragraph can be made up arranging the sentences c,a,b,and ,d

Punctuation and capitalization

10. D In a direct speech, the introductory clause is separated by a command the actual speech is enclosed with in inverted commas./
11. A. proper nouns / names of places, towns,dams and months/are written in capital letter.
12. B. List of items are separated by commas but the least two are joined by conjunction and,
13. C. Statements ends in full stop.

Word order

14. B. When the given jumbled words are rearranged, you get a third conditional sentence.
    They would have joined us if we had phoned them,
15. C when the jumbled words are rearranged a question is formed,i.eAre they coming home early today?
16. A. These groups of words can be arranged to be meaning ful as, you must read your notes to do the test.

Word Spelling

17. A. Simultaneously –Correct spelt
    forcast × forecast ✓
    lisence × license ✓
    accommod × ate accommodate ✓

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18. approach × approach ✓
immature × immature ✓
playright × playwrite ✓
courgeous × correctly spelt ✓

Communicative Activity

19. D. The response of the second speaker gives clue about the number of teacher in primary and secondary school.
20. D. Asking for location of a building, i.e. at Aratkilo.
21. A. From the response of the second speaker we see it is about asking for a place.
22. B. The best response from the given alternative is 'they say hunger is the best Sauce'.
23. C. Suggesting what is better in order to get rid of poverty.
24. B. Suggesting solution to the problem raised.
25. A. Asking to mention example of a natural disaster.
26. A. Asking who the eldest is in the family.
27. C. That is promising as she makes telephone call when she reaches there (her destination)
28. A. Telling as she is ready for the next step.
29. b. Telling the journalist as there are more fishermen nowadays.
30. C. Telling the radio program he prefers and listen to.
31. C. Asking what problems may happen during talking on the mobile phone.
32. C. That is clarifying what has been said.
33. D. Asking for explanation or clarification
34. C. Telling how inflation affects citizens life.
35. C. Giving additional reason to show why farmers do not get enouph crops.
36. D. That is expressing your personal opinion.
37 C. That is a dialogue which shows their intimacy.
38. B. The son told his father as he has another appointment with his friends.
39. B. The second speaker is saying he uses traditional healer because of its less expense.
40. C. This is the possible idea that might be raised a meeting for the issue.
41. A. That is asking about the frequency/repletion/ and 'I spent most of my evenings' is the possible response.
42. D. Asking for the preference of films she likes to watch.
43. A. The second speaker (his friend) advised him to ask his father.
44. D. Telling as disability should not be a reason for discrimination.

Understanding sentence / Comprehension/

45. B. Expressing past wish

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When we express wish the verb moves one step to past.

46. A. The original sentence is to mean rapid population growth is a threat to our survival.

Vocabulary/ completion/

47. D. terrifying- fearful
48. A. Launched-announced
49. B. memories-having something in memory
50. C. pessimistic-tendency to think things are the worst
51. C. declare- make known clearly or formally
52. A. honeymoon- holiday taken by a newly married couple.
53. D. diversity – various or different
54. B. Loan- money lent from a bank

Vocabulary/ substitution/

55. C. The phrasal ' make up' means' form" a family
56. D. ' murderer' means 'killer'
57. B. 'cowardly' means a person unable to control his fear .i.e timed (easely frightened,shy)
58. D. motion- a kind of movement
59. A. job- person responsibility in a certain career.
60. C. Under pressure – intension
61. B. heighten- make grater means raise.
62. A. immobile- not able to move u.e motionless.

Reading passage

63. C. "political leadership is not away of practicing spirituality
64. D. explaining why people should be happy in life.
65. D. ' material things' is not mentioned as a source of happiness.
66. A. Spirituality is about gating happiness by understanding the beauty of nature.
67. C. As the writer believes in order to be happy one should have deeper understanding of friends feeling.
68. D. Volunteering is getting happiness by helping others (the needy)
69. B. people who do not have close relationship are not as those who have.
70. C. The choice we make are partly responsible for our level of happiness.
71. D. being Concerned about your position an aspect of thinking positively.
72. A. Wealth, fame and power are not necessarily the right source of happiness.

Reading passage is

73. A. 'causes and effective of air population'.is the appropriate title for the passage
74. D. Air pollutant is said to be present when we feel suffocated during breathing.

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75. A. 'particulate matter's- a source of indoor pollutants.
76. C It is the result of the combined effect of various diseases.
77. C. Paragraph 3 – nature is uniform.
78. D. Paragraph 3 – The accumulation of carbon dioxide exhaled from lungs increases in an airtight room and could cause suffocation.
79. A. Paragraph 3 – line 2- A feeling of suffocation is indication of pollution.
80. D. paragraph 2-Industrial and ventricular exhausts are the main sources of indoor air pollution.
81. B. The passage of the writer of the passage thinks individuals can contribute in handling indoor air pollution.
82. D. Air pollution affect as all in different ways.

Grammer

83. A. fascinating – present participle used as adjective (complement).
84. D. might –shows probability
85. D. The presence of all committee members is necessary to conduct the meeting .
86. B. 'So' expresses result
87. C. In the first conditional, the main clause would be simple future (will be)
88. D. Has done all her assignments already- means by now.
89. A. Neither ...nor..correlative conjunction which means not this, not that, none.
90. C. I think we should take a taxi " that is expressing suggestion.
91. C. extremely tired(exhausted)
92. B. Simple future passive from (will be + past present)
93. B. Some and compound words formed with some are used with positive verb.
94. B. 'when'- introgative adverb(Wh—word) used to ask question about time.
95. D. think (to house something in mind)
96. B. 'yet' is used in negative statement to mean up to now.'stop'is a verb followed by'-ing'
97. A. Adjective + enough i.e. old enough. At the right age.
98. D. interested +in (adjective followed by preposition)
99. C. a little- adjective of quantity which means some but not much and it has positive meaning
100. D. during – shows period of time followed by noun or ing form.
101. C. The third conditional uses past perfect(had +v_3)-had driven
102. A. advice' ought to + infinitive)
103. B. interrogative adverb / wh – word / used to ask for reason(why)
104. A. To express present wish – we use past verb or 'were'
105. D. Simple past show action completed in the past at a particular time

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106. C. That is to show the interaction some one has what he is going to do.
107. D. She is like mother , similar to or seems, but not really is
108. C. ' for get' is a verb followed by ' to infinitive'.
109. A. I'm certain means I'm sure that things will be changed.
110. C. That is present wish, but not possible because the persons is busy.
111. B. Simple past passive (was/were+v_3). i.e were told
112. B. Simple future continuous (will+ be +ing)shows some thing expected to be progressive in the future.
113. D. Simple past passive (were allowed). The participants are not the real doer.
114. B. in order to / so as to/- to show for what purpose (positive purpose)
115. A. Expressing contrast with despite.

Despite
Inspite of → + noun, noun phrase, objective pronoun or ing form

116. A. with respect to ' introducing new topic or idea. With respect to you advertisement for hotel manger I'm writing to apply for the job.
117. B. Monethless – to means on the contrary
118. D. might have+ v_3 - show past possibility.
119.C. their hard work made the person be sure as they finish in the time given.
120. C. 'How often ' interrogative adverb used to ask about frequency.

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1.C. 'Do' means perform or carry out an action.The following actions are used with 'do' You do homework/gardening exercise/ the washing the house work the shopping / your best/the Cooking/ business /favour and... so on.

2.B. 'Enjoy' is a verb followed by – "ing".
All my friends enjoy watching English football.

3.C. The Correct wh-question that makes the sentence meaningful is 'Why don't you see your doctor?'.

4. C. That is what the speaker wishes would happen to Stop Cleaning the forest.

5. A. The verb 'let' is followed by object plus infinitive without to. She let her son play out with his friends.

6. C. Defining relative clause is introduced with 'which' to qualify the noun 'creature'. The book which was stolen from the library must be returned immediately.

7.C. Parallel Comparison (the more... the more..) ie the + double comparative The more he works, the more money he earns.

8.C. the order of the adjectives is: Opinion + age + size + Colour ugly +old + small + red

9.C. It shows future possibility If you get out early, you may get there on time. i.e It will be possible for you to get there on time.

10. C. 'Both means two of them. Both the teacher and the Students have left the School Compound.(Both...and) - this and that, together.

11. D. Past Continuous tense shows the first longer unfinished action in the past .He was doing his assignment when the bell of the door rang.

12. A. A phrase used to express positive purpose

to in order to So as to	+ infinitive

She gave up working so as to give more time to take care of her newly born baby.

13. C. Past perfect tense shows the first completed action from two in the past: They had already left before it began to rain.

14. A. 'Meanwhile' as adverb of time means in or during the time between.

15. B. "Although' Shows Contrast. Although I studied all they taught me, I still didn't understand everything.

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16. B . "ought to" + "infinitive" is used to mean it is desirable to do Something.
17. A. 'Able to' + 'inf'
18. A. Neither---nor-means not this, not that, none Neither his parents nor his teachers were happy with his poor result.
19. D. Definite article 'the' is used before a noun made definite or a particular one with a Clause.
    The book you lent me is very interesting.
20. C. In case + present = to describe a possible future reason for present action.
    The teacher is always asking his students incase they need his assistance
21. D. If She had known about your problem, She Could have helped you.This is to mean it was possible for her to help you.
22. A. 'Ought to' here is used to Show Strong advice.
    You ought to finish you work very soon because it is time to leave.
23. A. 'must '+ 'infinitive' - to mean it is necessary to do something.
    We Students must obey the rule and regulation of scool.
24. D. past perfect Continuous tense- Shows action began the first, continued for a long time and stopped when the second happened.
    He scored full because he had been studying the Some Course for a long time.
25. C. That is to show past passive (was were +V3)
    Two ware houses were broken last week.
26. B 'To make' means to force. Make + obj +infinitive without to
    The robbers made the passangers give them all what they had in their pockets when they had driven at night.
27. C. Inspite of the fact that
    Despite the fact that + S+v express Contrasting idea
    Despite the fact that she loves him, she will not marry him.
28. A. Suddenly - unexpectedly, being emotional
29. C. 'Can' is used to meon it is possible.
30. A. A clause expressing positive purpose with 'so that'
31. B. positive addition to positive remark. /so + auxiliary + subject/
    She likes watching movies, so do I. This means She likes watching movies, and I like watching movies.
32. C. 'Whose' expresses possession to whom the book belongs.

    He opened the door whose handling is broken'
33. C. 'While' means at the same time and refers to period of time. The sentence expresses two continuous actions which were taking place at the same time

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The Students were listening to her while she was teaching in the class.

34. D. past action is expressed using simple past tense.
35. B. Present wish is expressed using past tense. I wish he didn't miss today's meeting.
36. C. 'Fast' as adverb shows or tells us how he walks.
    He writes fast. (adverb)
    He's fast writer. (adjective)
37. A. 'when' means by the time or by the moment. It refers to point of time.
38. C. a certain amount of fund.
39. A. The word correctly spelt is 'recession' – means withdrawal from business and industrial activity
40. C. 'sought' is the word Correctly spelt and it is the past participle form of 'seek'.
41. B. when the jumbled words are rearranged the meaningful sentence formed can be 'There are many stories about people who Couldn't learn home to be rich'.
42. C. The Correct arrangment of the words is as follows. 'Abebe Bikila was the first African to ever take home an olympic gold medal.
43. A. "passing students who have actually failed in their exams is like killing a generation," is the Correct rearrangment.
44. D. If those four Sentences are arranged in c,a,b and d, they can make up a complete paragraph.
45. B. The complete paragraph can be best made up arranging the sentences as b, c, d and a.
46. C. The Correct arrangment is b, d.c and a
47. A. According to the chronological order of the ideas in those four sentences, what has been given under 'b' is the first sentence and the others c, a and d are used to develop it.
48. D. The idea given under choke 'd' is most likely a good example for a paragraph developed using Cause effect method.
49. B. The idea given under choice 'b' is most likely taken from a descriptive paragraph which is Creating some kind of picture in the mind of the reader.
50. B. List of nouns are separated by commas and the last two are only joined by 'and'.
51. D. -Complete name of a hospital is written in Capital
    -Abbreviation St. (saint) is followed by full stop.
    -They train doctors in a variety of field is explanation and separated by commas.
    -List of names of fields are separated by commas.

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52. D. Complete names of hotels and restaurants are written in capital.
53. B. The list of clauses are separated by Semi-colon, but the last two are joined by 'and'.
54. A. In the opening paragraph of a formal letter, state Clearly why you are writing the letter.
55. C. Dear Sir/Madam is used in the greeting part of a formal letter if you don't know the name of the Person to whom you are writing.
56. D. Letter of Complaint could best end in' yours faithfully
57. C. The University administration is new to handling older Students.
58. A. Pronoun' them' refers to older students.
59. A. In paragraph 4" the feeling can be mutual" means both groups think it is hard to live with the other.
60. C. The main idea of the last paragraph is to show as it is very hard to have young and old students together in a university.
61. B. "I thought I left my mother of home". My mother in this sentence. refers to the speaker's mom.
62. D. university professors prefer teaching adults to teaching young because adults wore much harder than their younger Counter parts.
63. D. Adults learn writing better than some of the best younger students because during their days, the system had given more emphasis to writing.
64. D. Comparatively older students tend to get their work done in time to have time for unseen situations that may unexpectedly drag them away.
65. C. Adult Students feel insecured since some of the Students may have Envey their performance
66. C. She didn't like the way her down mates behaved.
67. A. The pronoun 'they' in paragraph 2 refers to reasons and methods of cheating.
68. B. using fake signature in applications for financial aid belongs to non-academic cheating.
69. C. As stated in paragraph 3 the most undesirable Consequence of academic Cheating is the psychological and mental trauma inflicted on the cheater.
70. A. If a plagiarist is sent to a jail for one-year Service, the consequence of the decision is legal.
71. C. According to paragraph one the idea given under 'C' is true about cheating,
72. B. Recognition of the role of plagiarism is not Possible cause of cheating
73. B. As stated in paragraph 2, the way the act is Committed is not among the similarities between academic and non-academic plagiarism

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74. C. The interest to score good grades is the most important reason behind Secondary School leavers' attempt to plagiarism in the exam.
75. C. Politicians may resort to plagiarism as a means of Covering up their laziness
76. D. The main purpose of taking measures against Cheaters in a university environment is to protect Students' Safety and integrity or the Community.
77. D. Fragment- perks broken off
78. C. leaflet- printed sheet (unbound but sometimes folded) with announcements.
79. A. moral decay- a decline in knowing the difference between right and wrong.
80. D. Calamity-great and serious misfortune or disaster
81. A. Observe-pay attention to watch Carefully
82. C. Communicate- pass information.
83. B. Cultivate- make a person having good manner and education.
84. D. Incredible- difficult to believe, Surprising.
85. C. Multilateral- having many sides.
86. A. Eliminate- remove, take or put away,
87. B. Humble - having a modest opinion on oneself.
88. B. Dual- binary of thus; double
89. B. Huge- massive – Large
90. B. Failure impracticality- that Cannot be put into practice.
91. D. Main-principal- higher in order of importance.
92. B. Disagreement-dispute- quarrel, argument, controversy
93. C. The idea stated here has almost similar meaning to the first one.
94. B. This idea has the closest meaning to the original one
95. B. The respondent thought there is no hotel to staying
96. C. Asking what advantages the person get because Of joining the union.
97. A. The person wanted to get the money back Soon before he flies to Spain.
98. A. Expressing the person's opinion about something.
99. C. The respondent asked how much money is needed to go out for dinner tonight.
100. D. That is to talk about the car accident happened.
101. B. That is Confirming what the first Speaker has said.
102. D. Asking for an apology for not attending the Party.
103. A. To be locked up means to be in prison. Telling where he is.
104. A. Expressing liking and disliking
105. C. That is asking for a location of the National Farmers union, and telling the direction to location of the place.

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106. A. Telling what he should do to get his computer back.
107. A. Telling the best brand these days.
108. B. The person wanted to be sure if the plan is applicable or not.
109. C. Asking what is to be done to help those persons who have been trapped in a tunnel!
110. C. The person wanted to know Something about Artificial Intelligence and asked for explanation
111. A. The person does not want to come on mass media because of his personal Case.
112. C. The Second Speaker felt sad with the news he heard.
113. C. It is to show the strength and durability of the plate.
114. B. Expressing agreement to the measure taken.
115. C. Their working hard helps to predict as they have good chance to pass the national exam
116. B. That is to tell as robots can do what people do.
117. D. Looking at his physical appearance the respondent said that it is not Ali.
118. A. The interviewee replied that was the first film to star in.
119. D. Expressing pleasure for the Success achieved by the school team.
120. C. The response given indicates the Conversation is about asking for some one's interest.

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1. If $f(x) = x^2 + px + 1$ where $f(0)$ and $f(1)$ have opposite signs so that f has a root in (0, 1), then which one of the following numbers is a possible value of p?

A. -2
C. 3
B. -3
D. 2

2. What is the locus all points in a plane in which the sum of the distances from two fixed points is constant?

A. Parabola
C. Circle
B. Hyperbola
D. Ellipse

3. If the volume of a right circular cone is 64π cm^3 and the diameter of its base is 8 cm, what is the height of the cone?

A. 12 cm
C. 3 cm
B. 4 cm
D. 8 cm

4. Which of the following is **NOT** true about argument and validity?

A. An argument which is not valid is fallacy.

B. The validity of an argument can be checked by using truth table.

C. In a valid argument, the conclusion may be true or false whenever all the premises are true.

D. In a valid argument if all the premises are true then the conclusion must also be true.

5. Given matrices $A = \begin{bmatrix} 2 & 0 & 5 \\ 3 & 1 & 4 \\ 0 & 6 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 3 & 0 \\ 6 & 5 & 2 \\ 9 & 7 & 0 \end{bmatrix}$, which one of the following is $A - B$?

A. $\begin{bmatrix} 1 & 3 & 5 \\ -3 & -4 & -2 \\ -9 & 1 & 2 \end{bmatrix}$
C. $\begin{bmatrix} 1 & 3 & 5 \\ -3 & -4 & 2 \\ 9 & 1 & 2 \end{bmatrix}$

B. $\begin{bmatrix} 1 & -3 & 5 \\ -3 & -4 & 2 \\ -9 & -1 & -2 \end{bmatrix}$
D. $\begin{bmatrix} 1 & -3 & -5 \\ -3 & -4 & 2 \\ 9 & 1 & 2 \end{bmatrix}$

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6. Let $R = \{(x, y) \mid y \geq x^2 + 1 \text{ and } y \leq 5\}$ be a relation. Then, which one of the following defines the inverse of R?

A. $\{(x, y) \mid x \geq y^2 + 1 \text{ and } x \geq 5\}$

B. $\{(x, y) \mid x \geq y^2 + 1 \text{ and } x \leq 5\}$

C. $\{(x, y) \mid x \geq y^2 - 1 \text{ and } x \leq 5\}$

D. $\{(x, y) \mid x \geq y^2 + 1 \text{ and } x \geq 5\}$

7. Which one of the following is derivative of $f(x) = \tan x + 3^x$?

A. $-\csc^2 x + 3^x \ln x$

B. $\sec^2 x + 3^x \ln x$

C. $\sec^2 x + 3^x$

D. $-\tan x + 3^x \ln x$

8. What is the surface area of a sphere with radius $r = 6$ cm?

A. 36π cm^2 B. 24π cm^2 C. 144π cm^2 D. 48π cm^2

9. Which one of the following quantities represents a vector?

A. Volume of a box

B. The width of your room

C. Weight of an object

D. Speed of a motorbike

10. If $\vec{a} = (4, 3, 2)$ and $\vec{b} = (1, 2, -3)$ are two vectors in space, then which one of the following is NOT correct?

A. $\vec{a} + \vec{b} = (5, 5, -1)$

B. $\vec{a} + 2\vec{b} = (6, 7, -4)$

C. $\vec{b} - \vec{a} = (-3, -1, -5)$

D. $\vec{a} - 2\vec{b} = (2, 1, 8)$

11. In rolling a fair die, what is the probability of obtaining 3 or 5?

A. 1/6 B. 1/2 C. 1/3 D. 1/4

12. Which one of the following points is inside the sphere with equation $x^2 + y^2 + z^2 = 5$?

A. (1, -2, 2) B. (1, 1, 1) C. (1, 2, 3) D. (0, -1, 3)

13. What are the quotient and remainder respectively, when 75 is divided by 20?

A. 15 and 3 B. 2 and 19 C. 3 and 15 D. 4 and 5

14. The radian measure of an angle 120^o is equal to:

A. $\frac{\pi}{2}$ B. $\frac{\pi}{8}$ C. $\frac{2\pi}{3}$ D. $\frac{\pi}{2}$

15. Which of the following is the measure of an exterior angle of a 10 sided regular polygon?

A. 180^o B. 36^o C. 72^o D. 144^o

16. In order to find point P in space one can start from the origin O(0, 0, 0); move 5 units in the direction of negative x-axis, then move 5 units in the

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direction of positive y-axis and finally move 5 units in the direction of negative z-axis. Which one of the following ordered triple of numbers represented by point P?

A. (5, 5, 5) B. (-5, 5, -5) C. (5, -5, 5) D. (-5. -5, 5)

17. Which of the following pairs of events are dependent?

A. Drawing two cards one after the other from a well shuffled pack of cards without replacement.

B. Tossing two coins simultaneously and obtaining head from one and tail from the other.

C. Any events obtaining from tossing a coin and rolling a die at the same time.

D. Drawing two balls successively from a box containing balls having the same size and shape with replacement.

18. Which of the following is the variance of the data given as: 2, 3, 4, 5, 7?

A. 3.06 B. 3.70 C. 2.46 D. 2.96

19. Which one of the following is the augmented matrix associated to the system $\begin{cases} 2x + y + 3z = 4 \\ x - z = 1 \\ -4x + y - 3 = 3 \end{cases}$?

A. $\left(\begin{array}{ccc|c} 3 & 1 & 2 & 4 \\ -1 & 0 & 1 & 1 \\ 4 & 1 & -3 & 0 \end{array}\right)$

B. $\left(\begin{array}{ccc|c} 3 & 2 & 1 & 4 \\ -1 & 1 & -1 & 3 \\ -1 & -4 & 1 & 4 \end{array}\right)$

C. $\left(\begin{array}{ccc|c} 2 & 1 & 3 & 4 \\ 1 & 0 & -1 & 1 \\ -4 & 1 & 0 & 3 \end{array}\right)$

D. $\left(\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ -1 & 1 & -1 & 1 \\ 1 & -4 & -1 & 3 \end{array}\right)$

20. If A is a 3×3 matrix with $\det(A) = 5$, then what is the determinant of $2A^T$?

A. 40. B. 30 C. 10 D. 2/5

21. The sum of the first 8 terms of a geometric sequence with first term 0.3 and common ratio $\frac{1}{10}$ is equal to:

A. $\frac{1}{3}\left(\frac{10^8-1}{10^7}\right)$

B. $\frac{3}{10}\left(\frac{10^8-1}{10^9}\right)$

C. $\frac{3}{10}\left(\frac{10^8-1}{10^8}\right)$

D. $\frac{1}{3}\left(\frac{10^8-1}{10^8}\right)$

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22. What is the seventh term of an arithmetic sequence whose third term is – 4 and common difference is 5?

A. 16 B. -16 C. 15 D. -15

23. The angle of depression of the top of a flag pole from the top of a building that is $40\sqrt{3}$ m away from the flag pole is 30^o. If the height of the flag pole is 10 m, what is the height of the building?

A. $20\sqrt{3}$ m B. $10(4\sqrt{3}+1)$ m C. 50 m D. 40 m

24. What is the derivative of the function $f(x) = \frac{xe^x}{\cos x}$ at $x = 0$?

A. -1 B. 2 C. 0 D. 1

25. Let f and g be two functions. Which one of the following statements is true about f and g?

A. If f is differentiable at a, then it is continuous at a.

B. The line $y = 0$ is a tangent line to the graph of $f(x) = |x|$ at $(0, 0)$.

C. If f and g are differentiable at a, then $\frac{f}{g}$ is differentiable at a.

D. If $f'(a) = 0$, then the tangent line to the graph of f at $(a, f(a))$ is $y = a$.

26. Suppose f is a twice continuously differentiable function and $f''(3)=5$. If $g(x) = f(2x+3)$, then the second derivative of g at 0 equal to:

A. 20 B. 40 C. 10 D. 5

27. If the perimeter of a rectangle is 120 m, what are the length ℓ and width w of the rectangle that maximizes its area?

A. $\ell = 30$ m and $w = 30$ m C. $\ell = 40$ m and $w = 20$ m

B. $\ell = 50$ m and $w = 10$ m D. $\ell = 35$ m and $w = 25$ m

28. Suppose f is a continuous function on the interval $[a, b]$ and differentiable on (a, b) with property $f(a) = f(b)$. Which one of the following must be true?

A. The function has minimum value in (a, b).

B. The function has a zero in (a, b).

C. The graph of f has horizontal tangent line at $(c, f(c))$ for some $c \in (a, b)$.

D. The function has maximum value in (a, b).

29. Let f be continuous on $\mathbb{R}$ and c be a number in the domain of f. Which one of the following must be true about the function?

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A. If $f(c)$ is a local minimum value of f, then $f'(c) = 0$.

B. An absolute maximum value of f which is attained at a critical number is also a local maximum value of f.

C. If c is a critical number of f, then $f(c)$ is a local extreme value of f.

D. If $f''(c) = 0$, then $(c, f(c))$ is an inflection point of f.

30. What is the product of the two complex numbers $z = 2-3i$ and $w = 5+2i$?

A. $4-11i$

B. $16+11i$

C. 29

D. $16-11i$

31. Which one of the following is the polar form of the complex number $z = 2-2\sqrt{3}i$?

A. $4(\cos\frac{1}{4}\pi + i\sin\frac{1}{4}\pi)$

B. $4(\cos\frac{5}{3}\pi + i\sin\frac{5}{3}\pi)$

C. $4(\cos\frac{1}{3}\pi + i\sin\frac{1}{3}\pi)$

D. $4(\cos\frac{3}{4}\pi + i\sin\frac{3}{4}\pi)$

32. Which one of the following compound statements has a converse with truth value False?

A. If man is immortal, then the earth does not rotate around the sun.

B. If 6 is a prime integer, then 4 is an even number.

C. If man is mortal, then the earth does not rotate around the sun.

D. If athlete Derartu Tulu was born on October 21, 1971, then 6 is prime.

33. Which one of the following is the simplified form of $\frac{4x^4-64}{2-x}$ for $x \neq 2$?

A. $-4(x+2)(x^2+4)$

B. $-4(x-2)(x^2+4)$

C. $4(x+2)(x^2+4)$

D. $4(x-2)(x^2-4)$

34. Which one of the following is an onto function from $\mathbb{R}$ on to $[0, \infty)$?

A. $f(x) = |x| + 2$

B. $f(x) = \sqrt{x^2}$

C. $f(x) = 2^x$

D. $f(x) = x^2 + 1$

35. Which one of the following trigonometric values is correct?

A. $\tan(-120^o) = -1$

B. $\cos(-60^o) = \frac{\sqrt{3}}{2}$

C. $\sin(-120^o) = 0.5$

D. $\cos(-120^o) = -0.5$

36. Which one of the following intervals is the solution set of the inequality

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$x^2 - 5x + 6 \leq 0$?

A. $[3, \infty)$

C. $[0, 2]$

B. $(-\infty, 2]$

D. $[2, 3]$

37. Which one of the following is true about the trigonometric values of the given pairs of angles?

A. $\sin(75^o) = \cos(105^o)$

C. $\sin(120^o) = \sin(60^o)$

B. $\tan(75^o) = \tan(105^o)$

D. $\cos(120^o) = \cos(60^o)$

38. The solution set of the equation given by $\log(x^2 - 3) = 2\log(x - 1)$:

A. $\{4\}$ B. $\emptyset$ C. $\{1/2\}$ D. $\{2\}$

39. What are the coordinates of a point that divides the line segment with end points P(0, 1) and Q(5, 6) in the ratio 2:3 from P?

A. (0, 2) B. (2, 3) C. (0, 3) D. (3, 2)

40. If f and g are continuous functions at a, then which one of the following functions may **NOT** be continuous at a?

A. $\frac{f}{g}$ B. $f - 2g$ C. $2f + 3g$ D. $f \cdot g$

41. Which of the following is true about bounded sequence?

A. Every decreasing bounded sequence converges to the least upper bound of the sequence.

B. Every monotonic bounded sequence converges.

C. Every bounded sequence is convergent.

D. Every increasing bounded sequence converges to the greatest lower bound of the sequence.

42. What is the value of a for which the function defined by

$$f(x) = \begin{cases} ax + 3 & \text{for } x \geq 1 \\ x^2 - 2x + 3 & \text{or } x < 1 \end{cases}$$ is continuous at 1?

A. 3 B. 4 C. 1 D. -1

43. If $\lim_{x \to 2} f(x) = 6$ and $\lim_{x \to 2} g(x) = -9$, then $\lim_{x \to 2} \left(\frac{f-g}{2f-7}\right)(x) =$ ____.

A. 15 B. -15 C. 3 D. -3

44. What is the value of $\int_0^1 xe^x dx$?

A. e B. $-e$ C. -1 D. 1

45. $\int \left(\frac{1}{\sqrt{x}} - e^{-3x} + \frac{5}{x^2}\right) dx$ is equal to:

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A. $2\sqrt{x} - \frac{e^{-3x}}{3} - \frac{5}{x} + C$

B. $2\sqrt{x} - \frac{e^{-3x}}{3} + \frac{5}{x} + C$

C. $2\sqrt{x} + \frac{e^{-3x}}{3} + \frac{5}{x} + C$

D. $2\sqrt{x} + \frac{e^{-3x}}{3} - \frac{5}{x} + C$

46. Let f be continuous function on $[a, b]$. Then, which of the following is NOT true?

A. $\int_a^b f(x)dx = \int_a^c f(x)dx - \int_c^b f(x)dx$ for any c in (a, b).

B. $\int_c^c f(x)dx$ for any c in (a, b).

C. $\int_a^b f(x)dx = -\int_b^a f(x)dx$.

D. $\int_a^b kf(x)dx = k\int_a^b f(x)dx$ for any constant k.

47. In ΔABC, if AB= 10 units, BC= 14 units and AC= 8 units, then the area of the triangle is_____ .

A. $16\sqrt{6}$ square unit

B. $10\sqrt{6}$ square unit

C. $16\sqrt{2}$ square unit

D. $16\sqrt{3}$ square unit

48. Which one of the following sets is the symmetric difference, $A\Delta B$ of the sets A= {1, 3, 5, 6, 8, 10} and B= {2, 5, 8, 11}?

A. {2, 11}

B. {1, 2, 3, 5, 6, 8, 10, 11}

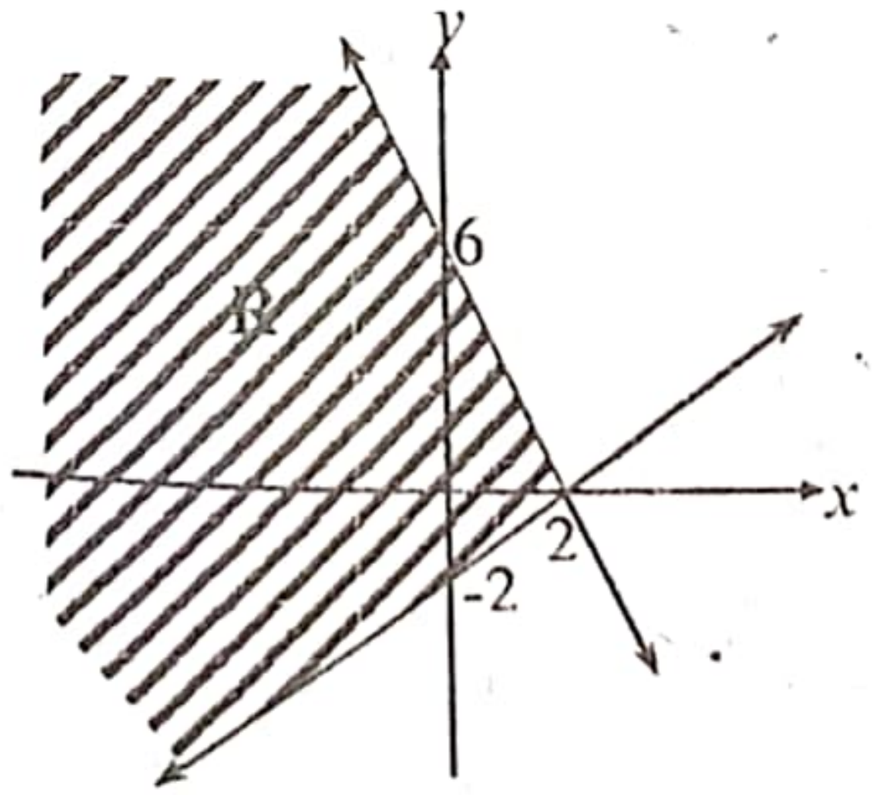
C. {1, 3, 6, 10}

D. {1, 2, 3, 6, 10, 11}

49. The simplified form of the expression $\frac{6\sqrt{20}-3\sqrt{45}}{3\sqrt{75}}$ is:

A. 1/5 B. 5 C. $\sqrt{5}$ D. $\frac{\sqrt{15}}{15}$

50. The graph of a certain relation R is represented by the shaded region shown on the figure below.



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Which one of the following pairs of sets respectively gives the domain and range of this relation?

A. $\{x/\ x \leq 2\}$ and $\mathbb{R}$

C. $\mathbb{R}$ and $\mathbb{R}$

B. $\{x/\ x \leq 2\}$ and $\{y/\ y \leq 6\}$

D. $\mathbb{R}$ and $\{y/\ y \leq 2\}$

51. Which of the following is true about the Arithmetic Mean of a given data?

A. It can also be used for qualitative data.

B. It can be obtained even in the absence of some of the values in the data.

C. It is affected by extreme values.

D. There can be two means for a given data.

52. Which one of the following is **true?**

A. The range of arctangent function is $[-\pi, \pi]$.

B. The range of arcsine function is $[-\infty, \infty]$.

C. The domain of arccosine function is $[-1, 1]$.

D. The domain of arcsine function is $[0, \pi]$.

53. A line segment chord in an ellipse which passes through the center and perpendicular to the major axis is called:

A. Minor axis.

C. Latus rectum.

B. Semi-major axis

D. Eccentricity.

54. The distance from the point P(3, 4) to the line L with equation $3x + 4y - 5 = 0$ is _____.

A. 4 units

B. 7 units

C. 5 units

D. 6 units

55. The following is the distribution of the marks of students obtained in Mathematics test out of 50.

Range of marks	20-25	26-31	32-37	38-43	44-49
Frequency	15	30	25	20	10

What is the first quartile of the distribution?

A. 26

B. 28

C. 27.5

D. 26.5

56. What is the image of the circle with equation $(x + 2)^2 + (y - 2)^2 = 1$, when it is rotated through $\frac{\pi}{6}$ about the origin?

A. $(x + (\sqrt{3} - 1))^2 + (y - (\sqrt{3} + 1))^2 = 1$

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B. $(x-(\sqrt{3}+1))^2+(y+(\sqrt{3}-.1))^2=1$

C. $(x+(\sqrt{3}+1))^2+(y-(\sqrt{3}-1))^2=1$

D. $(x-(\sqrt{3}-1))^2+(y+(\sqrt{3}+1))^2=1$

57. If r_1 and r_2 are the roots of the equation $3x^2+9x+16=0$, which one of the following is true?

A. $r_1+r_2=3$

B. $r_1r_2=-16/3$

C. $r_1{}^2+r_2{}^2=-5/3$

D. $\frac{1}{r_1}+\frac{1}{r_2}=9/16$

58. What is the dot (scalar) product of $u=i+2j$ and $v=-2i+4j$?

A. -8 B. 5 C. 6 D. 7

59. Consider the points A(-2, 1), B(3, 0), C(3, -1) and D(1, 1) in xy-plane. If $u=\overrightarrow{AB}$ and $=\overrightarrow{CD}$, then which one of the following is equal to $u-v$?

A. (3, 1) B. (-7, 3) C. (-3, 1) D. (7, -3)

60. Let p and q be propositions. Which one of the following compound statements is a tautology?

A. $(p\wedge q)\Leftrightarrow p$

B. $(p\Rightarrow q)\Leftrightarrow(\neg p\vee q)$

C. $(p\vee q)\Leftrightarrow p$

D. $(p\Rightarrow q)\Leftrightarrow p$

61. Which one of the following is NOT true about the graph of $f(x)=5+\frac{|x|}{x+1}$?

A. $y=6$ is its horizontal asymptote.

B. $y=5+x$ is its oblique asymptote.

C. $x=-1$ is its vertical asymptote.

D. $y=4$ is its horizontal asymptote.

62. If Birr 20,000 is deposited in a bank at a rate of 12% interest compounded monthly, how long will it take to double the amount?

A. 6 years C. 5 years

B. 7.59 years D. 5.81 years

63. Which one of the following is **true** about the matrix $\begin{pmatrix}1&3&5\\6&4&2\\9&7&0\end{pmatrix}$?

A. The minor of entry 0 is 14.

B. The cofactor of entry 1 is -14.

C. The minor of entry 3 is 18.

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D. The cofactor of entry 5 is -6.

64. What is the point of intersection of the median of ΔABC whose vertices are given by A(0, 0), B(6, 0) and C(0, 4)?

A. (9/5, 6/5)

B. (6/5, 4/5)

C. (2, 4/3)

D. (1, 2/3)

65. $\int \frac{x^3+x^2-x}{x^2+x-2} dx$ is equal to:

A. $\frac{x^2}{2} - \frac{1}{3}\ln|x-1| - \frac{2}{3}\ln|x+2| + C$

B. $\frac{x^2}{2} + \frac{1}{3}\ln|x-1| + \frac{2}{3}\ln|x+2| + C$

C. $\frac{x^2}{2} + \frac{1}{3}\ln|x-1| - \frac{2}{3}\ln|x+2| + C$

D. $\frac{x^2}{2} - \frac{1}{3}\ln|x-1| + \frac{2}{3}\ln|x+2| + C$

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1. Given A = {x∈ ℕ: x < 3} and B is the set of all possible factors of 13. Then, which one of the following is equal to B×A?
   A. {(1, 1), (1, 2), (13, 1), (13, 2)}
   B. {(1, 1), (2, 1), (1, 13), (2, 13)}
   C. {(1, 2), (13, 1), (13, 2)}
   D. {(1, 1), (13, 2)}
2. Which of the following statements is true about $f(x) = x^2 - 6x + 10$?
   A. The line x = 1 is the axis of symmetry of graph of f.
   B. $f(x) \geq 2$ for all x in the domain of f.
   C. The vertex of the graph of f is (3, 1).
   D. The graph of f opens downward.
3. What is the area of a regular hexagon inscribed in a circle of radius 6 cm?
   A. $54\sqrt{3}$ cm²
   B. $48\sqrt{3}$ cm²
   C. $36\sqrt{3}$ cm²
   D. 48 cm²
4. Which of the following defines a prime number?
   A. It is a number whose only factor is 1.
   B. It is a number that has more than two positive factors.
   C. It is a number whose only factor is itself.
   D. It is a number that has only two positive factor.
5. Which one of the following rational numbers is written in its standard form?
   A. $0.\bar{8} \times 10^2$
   B. 10.1×10^{-2}
   C. 5×10^5
   D. 0.09×10^5
6. What is the value of x in the equation $\sqrt{125^x 5^{2x-1}} = 25^{3x+1}$?
   A. $\frac{5}{7}$
   B. -5
   C. $\frac{5}{7}$
   D. 5
7. The solution set of the equation $3|2x - 4| = 6 - 2|1 - 5x|$ is
   A. {-1, 1/2, 5/4}
   B. {1/2, 5/4}
   C. {-1, 1/2}
   D. { }

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8. A regular pyramid has a hexagonal base whose side length is 12 cm. What is the lateral surface area of the pyramid in square centimeters if the length of each lateral edge is 10 cm?

A. 72

C. 720

B. 360

D. 288

9. The object shown in the figure below is made of wood. Its top cover is solid hemisphere and its bottom is a cylinder of base diameter 6 cm and height of 12 cm. From the bottom of the cylindrical part a right circular cone of height 8 cm is cut off.

Thus, what is the volume of the solid in cubic centimeters?

A. 72π

C. 126π

B. 102π

D. 96π

10. The domain of the function $f(x) = 2x^{\frac{2}{3}}$ is:

A. $[0, \infty)$

C. $(0, 2)$

B. $\mathbb{R} \setminus \{0\}$

D. $\mathbb{R}$

11. The slope of a line that makes an angle of 135° with the positive x-axis is equal to:

A. -1

C. 3

B. 2

D. 1

12. If $\tan 31^\circ = 0.81$, then what is the value of $\tan 509^\circ$

A. - 0.81

C. – 0.91

B. 0.19

D. 0.81

13. If $\sin\theta = -\frac{3}{5}$ and θ is 4^{th} quadrant angle, then what is the value of $\sec\theta$?

A. 5/4

C. – 5/4

B. – 5/3

D. 5/3

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14. Suppose ABCD is a parallelogram with P a mid-point of $\overline{AB}$ and Q a mid-point of $\overline{CD}$. Then which of the following is **NOT** necessarily true?

A. $\overline{AQ} \equiv \overline{PC}$

B. $\Delta ADQ \equiv \Delta CBP$

C. APCQ is a rhombus

D. APCD is a trapezium

15. The solution set of the equation $5\log_9 x - 2\log_9(x^2) - \log_4 8 = -1$ is

A. {3}

B. {4}

C. {3/2}

D. {9}

16. What is the value of x that satisfies the equation $5(\frac{125}{8})^{\frac{1}{3}x^2-\frac{2}{3}x} = 2$?

A. -3/2

B. 1

C. 4

D. 3/2

17. The solution set of the inequality $(2x-3)(x+5) \leq 0$ is

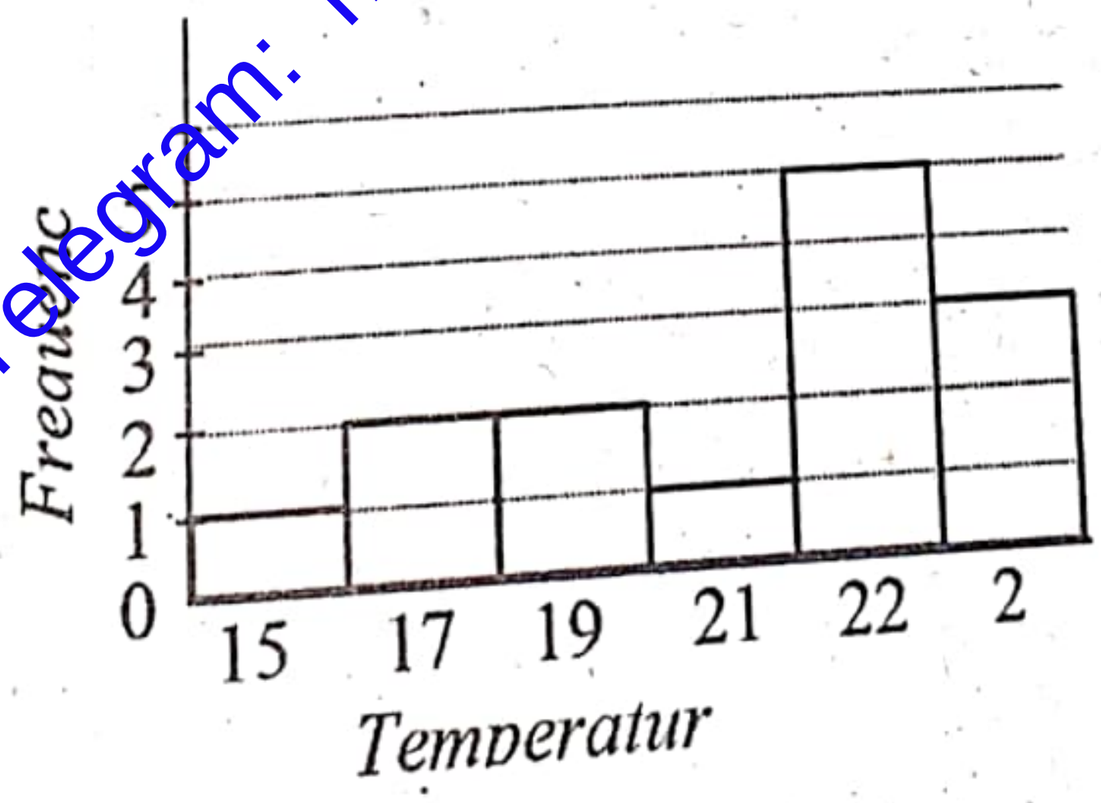
A. $(-\infty, -5] \cup [3/2, \infty)$

B. (-5, 3/2)

C. [-5, 3/2]

D. $(-\infty, -3/2] \cup [5, \infty)$

18. The histogram below shows the recorded temperature (in °C) of certain town in Ethiopia for the first 15 days of March 2023.



Then which one of the following statements is true about the data represented by the histogram?

A. The mode of the data is 22 °C.

B. The temperature was 24 °C for two days.

C. For 45% of the days, the temperature was more than 19 °C.

D. The highest frequency is 3.

19. A 20 m high building casts a shadow of 4 m, at the same time a 35 m tree casts a shadow of x m. What is the value of x?

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A. 11 m

C. 15 m

B. 8 m

D. 7 m

20. A car started moving 9 km to the East then moved 4 km to the North and then 12 km towards West to reach its destination. How far the care travelled from its starting point?

A. 12 km

C. 5 km

B. 25 km

D. 4 km

21. When the polynomial $p(x) = 3x^6+5x^4-7x^3+2kx^2+3$ is divided by $x+1$, the remainder is 4. What is the value of k?

A. -7

C. 6

B. -5

D. 3

22. The equation of a parabola with vertex V(2, -1) and focus F(-1, -1) is

A. $(y + 1)^2 = -12(x-2)$

C. $(x - 2)^2 = -12(y+2)$

B. $(x + 1)^2 = -12(y+2)$

D. $(y + 1)^2 = 12(x-2)$

23. What is the equation of the hyperbola with foci $F_1 = (-2, 1)$, $F_2 = (8, 1)$ and the length of the conjugate axis is 8 units?

A. $\frac{(x-3)^2}{16} - \frac{(y-1)^2}{25} = 1$

C. $\frac{(x-3)^2}{25} - \frac{(y-1)^2}{15} = 1$

B. $\frac{(y+1)^2}{16} - \frac{(x-3)^2}{9} = 1$

D. $\frac{(x-5)^2}{9} - \frac{(y-1)^2}{16} = 1$

24. If the truth value of p is T, then which of the following compound propositions has a truth value T for any proposition q?

A. $(\neg p \wedge q) \Rightarrow q$

C. $(p \Rightarrow q) \wedge \neg p$

B. $\neg p \Leftrightarrow (p \vee \neg q)$

D. $(p \wedge \neg p) \wedge q$

25. Which of the following functions is one to one?

A. $f:(0, \infty) \rightarrow \mathbb{R}, f(x) = |x| - 1$

B. $f = \{(1, 5), (2, 3), (5, 4), (6, 5)\}$

C. $f = \{(x, y): y \text{ is the mother of } x\}$

D. $f:\mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2 - 1$

26. If $\frac{x^2+14}{(x+2)(x-1)^2} = \frac{A}{x+2} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$, then what are the values of A, B and C respectively?

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A. 3, -2, 6

C. 2, -1, 5

B. 1, -1, 3

D. 2, -2, -5

27. Which of the following is true about the intercept and symmetry of the graph of $f(x) = \frac{x^3-3x}{x^2-6}$?

A. Its y-intercept is 3.

B. The graph is symmetrical with respect to the y-axis.

C. The graph is symmetrical with respect to the origin.

D. Its x-intercepts are $\pm\sqrt{6}$.

28. Consider the lines ℓ_1:y =2x+3 and ℓ_2:y =1-3x. Then the angle between ℓ_1 and ℓ_2 is

A. $3\pi/2$

C. $\pi/6$

B. $\pi/4$

D. $\pi/2$

29. Let **u**=(-1, 6) and **v**=(2, 4) be position vectors. Then the value of 4**u**-3**v** is

A. (-10, 11)

C. (-15, 18)

B. (-10, 12)

D. (-4, 24)

30. Which of the following is true about the graph of $y = 3\sin(4x)$?

A. The graph lies below x-axis on $(\pi/4, \pi/2)$.

B. The graph completes one cycle on the interval $[0, \pi]$.

C. The graph rises up on $[0, \pi/4]$.

D. The graph completes one cycle on the interval $[0, \pi/4]$.

31. If a translation T takes a point (-1, 1) to point (3, 2), then the image of the circle $x^2+y^2+2x+4y+1=0$ under T is

A. $(x-2)^2+(y+3)^2=4$

C. $(x+3)^2+(y+3)^2=4$

B. $(x-3)^2+(y-3)^2=4$

D. $(x-5)^2+(y+1)^2=4$

32. Let A be a square matrix of order n and let B be a matrix obtained by multiplying one row of A by 2 and C be a matrix found by adding one row of A to another row of A. If det(A) =10, what are det(B) and det(C)?

A. 20 and 0

C. 10 and 10

B. 20 and 10

D. 10 and 20

33. The solution set of the system $\begin{cases} 2x-y+3z=1 \\ x+y+3z=5 \\ x-y+z=-1 \end{cases}$ is

A. {(k+1, 2k, k-1): k$\in \mathbb{R}$}

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B. $\{(-4-2k, k, 2-k): k \in \mathbb{R}\}$
C. $\{(k, k+1, k-1): k \in \mathbb{R}\}$
D. $\{(2-2k, 3-k, k): k \in \mathbb{R}\}$

34. Let z_1, z_2, and z_3 be complex numbers. Then which one of the following is **NOT** necessarily true?
A. $z_1 - (z_2 - z_3) = (z_1 - z_2) - z_3$
B. $z_1 - (z_2 + z_3) = (z_1 - z_2) - z_3$
C. $(z_1 z_2) z_3 = z_1 (z_2 z_3)$
D. $(z_1 z_2) z_3 = (z_2 z_1) z_3$

35. Which one of the following complex numbers corresponds to the point (-3, -1) in the Argand plane?
A. 2+i
C. -3- i
B. -3+i
D. -1 -3i

36. A group of students, 5 from grade 10, 8 from grade 11 and 2 from grade 12 to be formed. If a student is chosen randomly from this group, what is the probability that the chosen student is either from grade 10 or from grade 12?
A. 5/15
C. 1/7
B. 7/15
D. 12/15

37. If $\begin{pmatrix} 2 & -1 \\ a & 4 \end{pmatrix}\begin{pmatrix} 1 & b \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ -1 & 18 \end{pmatrix}$, then what are the values of a and b respectively?
A. 2, -2
C. 0, -1
B. 3, 2
D. -1, 2

38. Let $A = \begin{pmatrix} 1 & -1 \\ 0 & 2 \\ -3 & 6 \end{pmatrix}$. Then, A^T is
A. $\begin{pmatrix} 1 & -1 \\ -1 & 2 \\ -3 & 6 \end{pmatrix}$
C. $\begin{pmatrix} -1 & 1 \\ 2 & 0 \\ 6 & -3 \end{pmatrix}$
B. $\begin{pmatrix} 1 & 0 & -3 \\ -1 & 2 & 6 \end{pmatrix}$
D. $\begin{pmatrix} -1 & 2 & 6 \\ 1 & 0 & -3 \end{pmatrix}$

39. In a certain month of a year, the temperature of a city was recorded and presented as follows.

Temperature (in °C)	10-14	15-19	20-24	25-29	30-34

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Frequency (Number of days)	2	9	11	5	3

Which one of the following is the cumulative frequency of the fourth class?

A. 27 B. 22 C. 20 D. 18

40. On set of Natural numbers $\mathbb{N}$, which one of the following propositions has a truth value True?

A. $(\forall_x)(\forall_y)(x \neq y \Rightarrow x > y)$

B. $(\exists_x)(\exists_y)(x-y = y-x)$

C. $(\forall_x)(\exists_y)(xy < x)$

D. $(\forall_x)(\exists_y)(x - y = 5)$

41. The following is distribution of the weight of 40 students in a certain class.

Weight (in Kg)	40-48	49-57	58-66	67-75	76-84
Frequency, f	8	18	7	5	2

Then, for the data given above, what is the median weight in Kg?

A. 55 C. 53.8

B. 54.5 D. 53

42. A fair die is going to be rolled once and let events be given as;

E_1 = an even number occurs;
E_2 = a prime number occurs;
E_3 = a multiple of 3 occurs;
E_4 = a multiple of 5 occurs.

Then, which one of the following pairs of events is mutually exclusive?

A. E_3 and E_4 C. E_2 and E_4

B. E_1 and E_3 D. E_1 and E_2

43. Which of the following is true about zero(s) of a function f(x)?

A. If f(a) = 0, then y = a is the y-intercept.
B. If f(a) = 0, then x - a is a factor of f(x).
C. If f(a) = 0, then a is not necessarily in the domain of f.
D. If f(a) = 0, then $(x-a)^2$ is a factor of f(x)

44. What is the value of a number in the interval (0, 1) that satisfies the Mean Value Theorem for the function $f(x)=3x^2 - 5x + 1$?

A. 3/5 B. 1/2 C. 1 D. 5/6

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45. Water is running into a canonical tank which is pointed down at a constant rate of $6\pi\,cm^3/sec$. The cone has a height of 10 cm and base radius 5 cm. How fast is the water level rising when the water reaches 2 cm deep?

A. 6 cm/sec

C. $\frac{6}{\pi}$ cm/sec

B. $\frac{4}{\pi}$ cm/sec

D. 4 cm/sec

46. Let $f(x)=\frac{1}{\sqrt{x}}$ and $g(x)=2-\frac{1}{6}x^3$ be two functions. For what values of x do the graphs of f and g have tangent lines of the same slope?

A. 1

C. 1/4

B. $\sqrt{3}/2$

D. 4

47. Which of the following describes the mechanical meaning of derivative?

A. Area under curve

B. Gradient of graph of functions

C. Slope of tangent line

D. Instantaneous rate of change

48. Let $f(x)=5x^3+2x-\sin x+e^x$. What is he value of $f'(x)$?

A. 1

C. 4

B. 2

D. -1

49. Let f and g be differentiable functions and $h(x) = g(f(3x+1))$. If $f'(1)=f(1)=2$ and $g'(2)=1$, then what is the value of $h'(0)$?

A. 1

C. 4

B. 2

D. 6

50. Which of the following argument is valid?

A. $p, p \Rightarrow q, r \vee \neg q \vdash p \wedge \neg r$

C. $p \Rightarrow q, \neg r \wedge q \vdash p \vee r$

B. $p \vee q, p \Rightarrow q \vdash \neg q$

D. $p \Rightarrow q, q, r \Rightarrow p \vdash r$

51. What is the coordinate of the mid-point of the line segment joining (-1, 3, 3) and (-1, 5, -7) in space?

A. (1, 4, 3)

C. (-1, 3, 5)

B. (-1, 3, 7/2)

D. (-1, 1, -2)

52. Which of the following assertions is true applying the Principle of Mathematical Induction?

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A. $n! \geq 2^n$ for $n \geq 0$

B. $1+2+2^2+\cdots+2^n = 2^{n+1}-1$ for $n \geq 0$

C. $3+5+7+\cdots+(2n+1) = n^2$ for $n \geq 1$

D. $2+3+4+\cdots+n = \frac{n(n+1)}{2}+1$ for $n \geq 2$

53. Let F be the anti-derivative of f. Then which of the following is NOT true about F and f?

A. $F' = f(x)$

B. $\frac{d}{dx}\int F(x)dx = F(x)+c$

C. $\int \frac{d}{dx} f(x)dx = f(x)+c$

D. $\int F(x)dx = f(x)+c$

54. Which of the following is equal to $\int 3\cos 3x\,dx$?

A. $\sin 3x + c$

B. $\frac{1}{3}\sin 3x + c$

C. $-\sin x + c$

D. $\sin x + c$

55. What is the area of the region bounded by the graph of $f(x) = x^2 - 4$ and x-axis in square units?

A. 32

B. 16

C. 32/3

D. 16/3

56. Let f be a continuous function on [0, 1] and let F be an anti-derivative of f with F(1) = -11 and F(0) = 11. Then $\int_0^1 f(x)dx$ is equal to

A. 11

B. -22

C. 22

D. 0

57. Which of the following set of equations define the xz-plane?

A. $\{(x, y, z) \mid x, y, z \in \mathbb{R} \text{ and } y = 0\}$

B. $\{(x, y, z) \mid x, y, z \in \mathbb{R} \text{ and } x = z = 0\}$

C. $\{(x, y, z) \mid x, y, z \in \mathbb{R} \text{ and } z = 0\}$

D. $\{(x, y, z) \mid x, y, z \in \mathbb{R} \text{ and } x = 0\}$

58. Let the sequence $\{a_n\}$ converges to 3 while $\{b_n\}$ converges to -1. Then the sequence $\{a_n - b_n\}$ converges to

A. 4

B. 1

C. 1/3

D. 3

59. What is the value of $\lim_{x\to 1} \frac{x^2-1}{x^2+1}$?

A. 1

C. 0

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B. 3

D. 2

60. Which of the following is the value of α so that the function given by $f(x) = \begin{cases} \alpha x + 1, & x \geq 1 \\ x^2 - 4, & x < 1 \end{cases}$ is continuous at 1?

A. -4

C. 1

B. 4

D. -1

61. Which of the following functions has a graph that completes one full cycle on $[-\frac{\pi}{4}, \frac{11\pi}{4}]$?

A. $f(x) = 3\sin(\frac{4}{3}x + \frac{\pi}{2})$

B. $f(x) = 3\sin(3x + \frac{\pi}{6})$

C. $f(x) = 2\sin(2x + \frac{\pi}{6})$

D. $f(x) = 2\sin(\frac{2}{3}x + \frac{\pi}{6})$

62. Let $f(x) = -\frac{1}{3}\cos(\frac{3}{2}x - \frac{\pi}{3})$. Then what is the period and Phase shift of the graph of f respectively?

A. $\frac{2\pi}{3}$ and $\frac{2\pi}{9}$

C. $\frac{\pi}{9}$ and $\frac{4\pi}{9}$

B. $\frac{4\pi}{3}$ and $\frac{2\pi}{9}$

D. $\frac{3\pi}{2}$ and $\frac{4\pi}{9}$

63. What is the fifth term of the sequence $\{2^n + 4n - 5\}_{n=0}^{\infty}$?

A. 17

C. 15

B. 27

D. 47

64. Which one of the following numbers is the upper bound of the sequence $\{\frac{2n+1}{n+1}\}_{n=1}^{\infty}$?

A. 0

C. 4

B. 3/2

D. 1

65. If the 6^{th} and the 11^{th} terms of an arithmetic sequence are 35 and 75 respectively, then what is the sum of the first 31 terms of the sequence?

A. 724

C. 846

B. 2145

D. 2635

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SOLUTION FOR MATHEMATICS TIKIMT 2014/15;
BOOKLET CODE: 057

1. Given: A quadratic function $f(x) = x^2 + px + 1$ where

(i) $f(0)$ and $f(1)$ have opposite signs, i.e., $f(0)f(1) < 0$

(ii) f has a root in $(0, 1)$, i.e., $f(c) = 0$ for some $c \in (0, 1)$

We need to find possible value of p.

From (i): $f(0)f(1) < 0$ implies that the open upward ($a > 0$) parabolic curve of f crosses x-axis twice so that it has two real roots, and hence $b^2 - 4ac > 0 \Rightarrow p^2 - 4(1)(1) > 0$

$\Rightarrow |p| > 2$

$\Rightarrow p < -2$ or $p > 2$

Hence, the possible candidates are -3 and 3. Then

If $p = 3$, then $f(x) = x^2 + 3x + 1 \Rightarrow r_{1,2} = \frac{-3 \pm \sqrt{5}}{2} \in (0, 1)$

If $p = -3$, then $f(x) = x^2 - 3x + 1 \Rightarrow r_1 = \frac{3 - \sqrt{5}}{2} \in (0, 1)$

Therefore, (i) and (ii) are satisfied by $p = -3$

Therefore, the possible value of p is -3 Answer: B

NOTE: Actually, "*No need to give information (ii)*" since f is continuous on $[0, 1]$ and given (i) so that by ***Intermediate Value Theorem***, there exists $c \in (0, 1)$ such that $f(c) = 0$.

2. An **ellipse** is a locus (or a set of points P in Fig 1) whose sum of distances from two fixed distinct points (called **foci** ; plural of *focus*; points F_1 and F_2 in Fig 1) of the ellipse is constant (See Figure 1 below).

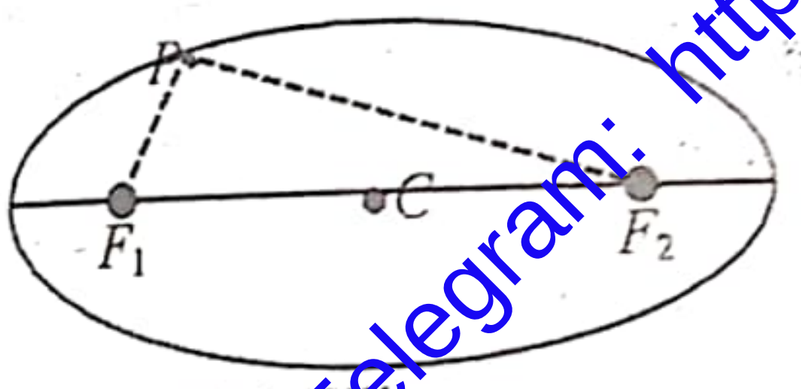

Fig 1 Ellipse

, i.e., $PF_1 + PF_2 =$ constant

Answer: D

3. The volume V of a right circular cone with base radius r and height h is given by $V = \frac{1}{3}\pi r^2 h$. Thus, $h = \frac{3V}{\pi r^2}$.

Given the diameter of its base 8 cm implies $r = \frac{8}{2}$ cm $= 4$ cm so that given $v = 64\pi$ cm^2, its height is:

$$h = \frac{3V}{\pi r^2} = \frac{3(64\pi \text{ cm}^3)}{\pi(4 \text{ cm})^2} = 12 \text{ cm}$$

Answer: A

4. The Argument form $p_1, p_2, \ldots, p_n \vdash q$ is said to be *valid* if q (the conclusion) is true whenever all the premises $p_1, p_2, \ldots, p_n$ are true; otherwise it is ***invalid*** (and called ***fallacy***), so that C is NOT true and A and D are **True.**

Validity of argument form $p_1, p_2, \ldots, p_n \vdash q$ can be checked by:

a) **Formal proof:** By equivalencies and rules of logical connectives.

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b) **Truth table:** It is valid if the conclusion is true in every row where premises are simultaneously true, (B is True).
c) **Tautology:** It is valid if $(p_1 \wedge \ldots \wedge p_n) \Rightarrow q$ is tautology.
d) **Rules of Inference:** This is also formal proof and is defined to be any valid argument. This is advised for large number of component statements.

Answer: C

5. Given matrices $A = (a_{ij})_{m\times n}$ and $B = (b_{ij})_{m\times n}$ be matrices of order $m\times n$, then $A - B = (a_{ij} - b_{ij})_{m\times n}$, i.e. subtracting the corresponding entries and the order of $A - B$ is the same as the order of A and B, i.e. $m\times n$.

Thus, for $A = \begin{bmatrix} 2 & 0 & 5 \\ 3 & 1 & 4 \\ 0 & 6 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 3 & 0 \\ 6 & 5 & 2 \\ 9 & 7 & 0 \end{bmatrix}$

$$A - B = \begin{bmatrix} 2 & 0 & 5 \\ 3 & 1 & 4 \\ 0 & 6 & -2 \end{bmatrix} - \begin{bmatrix} 1 & 3 & 0 \\ 6 & 5 & 2 \\ 9 & 7 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2-1 & 0-3 & 5-0 \\ 3-6 & 1-5 & 4-2 \\ 0-9 & 6-7 & -2-0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -3 & 5 \\ -3 & -4 & 2 \\ -9 & -1 & -2 \end{bmatrix}$$

Answer: B

6. To find inverse of a relation means to interchange x and y either in the order pair or in the given equations or/and inequalities so that given:
$R = \{(x, y) \mid y \geq x^2 + 1 \text{ and } y \leq 5\}$, then
$R^{-1} = \{(y, x) \mid y \geq x^2 + 1 \text{ and } y \leq 5\}$ or
$R^{-1} = \{(x, y) \mid x \geq y^2 + 1 \text{ and } x \leq 5\}$.

Answer: B.

7. $(\tan x)' = \sec^2 x$ and $(3^x)' = 3^x \ln 3$ so that by
$(g + h)' = g' + h'$
$f'(x) = (\tan x + 3^x)' = (\tan x)' + (3^x)' = \sec^2 x + 3^x \ln 3$

Answer: *No Answer*

8. The surface area A of a sphere with radius r is given by:
$A = 4\pi r^2$
Thus, for $r = 6$ cm, $A = 4\pi r^2 = 4\pi(6\text{ cm})^2 = 144\pi\text{ cm}^2$

Answer: C

9. Volume, length (or width), speed are scalar quantities but weight is a vector quantity because it has mass (a scalar quantity) and gravity (direction).

Answer: C

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10. Recall that for vectors $\vec{a} = (a, b, c)$ and $\vec{b} = (d, e, f)$ in space.

(i) $\vec{a} + \vec{b} = (a + d, b + e, c + f)$,

(ii) $k\vec{a} = (ka, kb, kc)$, and

(iii) $\vec{a} - \vec{b} = (a - d, b - e, c - f)$.

Thus, for $\vec{a} = (4, 3, 2)$ and $\vec{b} = (1, 2, -3)$:

A. $\vec{a} + \vec{b} = (4+1, 3+2, 2+(-3)) = (5, 5, -1)$ so A is **True** by (i)

B. $\vec{a} + 2\vec{b} = (4, 3, 2) + 2(1, 2, -3)$

$= (4, 3, 2) + (2(1), 2(2), 2(-3))$, by (ii)

$= (4, 3, 2) + (2, 4, -6)$

$= (4+2, 3+4, 2+(-6)) = (6, 7, -4)$ so B is **True** by (i)

C. $\vec{b} - \vec{a} = (1-4, 2-3, -3-2) = (-3, -1, -5)$ so C is **True** by (iii)

D. $\vec{a} - 2\vec{b} = (4, 3, 2) - 2(1, 2, -3)$

$= (4, 3, 2) - (2(1), 2(2), 2(-3))$, by (ii)

$= (4, 3, 2) - (2, 4, -6)$

$= (4-2, 3-4, 2-(-6)) = (2, -1, 8)$ so D is **False** by (iii)

Answer: D

11. In rolling a fair die, the sample space is $S = \{1, 2, 3, 4, 5, 6\}$ so that $n(S) = 6$. And, obtaining 3 or 5 gives an event $E = \{3, 5\}$ so that $n(E) = 2$. Thus, the probability of obtaining 3 or 5 is

$$p(E) = \frac{n(E)}{n(S)} = \frac{2}{6} = \frac{1}{3}.$$

Answer: C

12. Given a sphere $x^2 + y^2 + z^2 = 5$ implies center O(0, 0, 0) and radius $r = \sqrt{5}$

A. The distance between point P(1, -2, 2) and the center O(0, 0, 0) is $PO = \sqrt{1^2 + (-2)^2 + 2^2} = 3 > \sqrt{5} = r$ so that the point (1, -2, 2) is outside the sphere.

B. The distance between point P(1, 1, 1) and the center O(0, 0, 0) is $PO = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3} < \sqrt{5} = r$ so that the point (1, 1, 1) is inside the sphere.

C. The distance between point P(1, 2, 3) and the center O(0, 0, 0) is $PO = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14} > \sqrt{5} = r$ so that the point (1, 2, 3) is outside the sphere.

D. The distance between point P(0, -2, 3) and the center O(0, 0, 0) is $PO = \sqrt{0^2 + (-2)^2 + 3^2} = \sqrt{13} > \sqrt{5} = r$ so that the point (1, 2, 3) is outside the sphere.

Answer: B

13. Since $75 = 3 \times 20 + 15$ the quotient is 3 and the remainder is 15.

Answer: C

14. 120° in radian measure is: $120^\circ \times \frac{\pi}{180^\circ} = \frac{2\pi}{3}$.

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Answer: C

15. The exterior angle of a regular n-sided regular polygon is $\frac{360^o}{n}$ so that for $n = 10$ we have $\frac{360^o}{n} = \frac{360^o}{10} = 36^o$.

Answer: B

16. For a point $P = (a, b, c)$ in space one can start from origin $O(0, 0, 0)$; move 5 units in the direction of negative x-axis so that $a = -5$, then move 5 units in the direction of positive y-axis so that $b = 5$ and finally move 5 units in the direction of negative z-axis so that $c = -5$. Therefore, P= (-5, 5, -5).

Answer: B

17. Since there is a replacement the occurrence of one event depends on the other so that "*Drawing two balls successively from a box containing balls having the same size and shape with replacement*" is pairs of dependent events.

Answer: D

18. The mean is $\bar{x} = \frac{1}{n}\sum_1^n x_i = \frac{1}{5}(2+3+4+5+7) = 4.2$.

Then, the variance is: $\sigma^2 = \frac{1}{n}\sum_1^n (x_i - \bar{x})^2$

$$= \frac{(2-4.2)^2+(3-4.2)^2+(4-4.2)^2+(5-4.2)^2+(7-4.2)^2}{5}$$

$$= \frac{4.84+1.44+0.04+0.64+7.84}{5} = 2.96$$

Answer: D

19. The system $\begin{cases} 2x + y + 3z = 4 \\ x - z = 1 \\ -4x + y - 3 = 0 \end{cases}$ can be rewritten as:

$$\begin{cases} 2x + y + 3z = 4 \\ x + 0y - z = 1 \\ -4x + y + 0z = 3 \end{cases}$$

Hence, the coefficient matrix is $\begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & -1 \\ -4 & 1 & 0 \end{bmatrix}$ and the constant matrix is $\begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix}$ so that the corresponding augmented matrix is

$$\left(\begin{array}{ccc|c} 2 & 1 & 3 & 4 \\ 1 & 0 & -1 & 1 \\ -4 & 1 & 0 & 3 \end{array}\right)$$

Answer: C

20. If A is a $n \times n$ matrix, then (i) $\det(A^T) = \det(A)$ and
(ii) $\det(kA) = k^n \det(A)$

Thus, for $n = 3$ and $k = 2$:

$\det(2A^T) = 2^3 \det(A^T)$, by (ii)

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$= 2^3\det(A)$, by (i).
$= 2^3(5)$, by given
$= 40$

Answer: A
21. By $S_n = \frac{G_1(1-r^n)}{1-r}$ given $G_1 = 0.3, r = \frac{1}{10}$ and $n = 8$ we have that

$$S_8 = \frac{0.3(1-(\frac{1}{10})^8)}{1-\frac{1}{10}} = \frac{\frac{3}{10}(1-(\frac{1}{10})^8)}{\frac{9}{10}} = \frac{1}{3}(1-\frac{1}{10^8}) = \frac{1}{3}(\frac{10^8-1}{10^8})$$

Answer: D
22. By $a_n = a_m + (n-m)d$ given $n = 7, a_m = a_3 = -4$ and $d = 5$

$$\Rightarrow a_7 = a_3 + (7-3)(5) = -4+20 = 16$$

Answer: A
23. Let angle of depression of (point P) top of 10 m high flag pole which is $40\sqrt{3}$ m away from a point T on top of a building be 30^0 as shown in Figure 2.

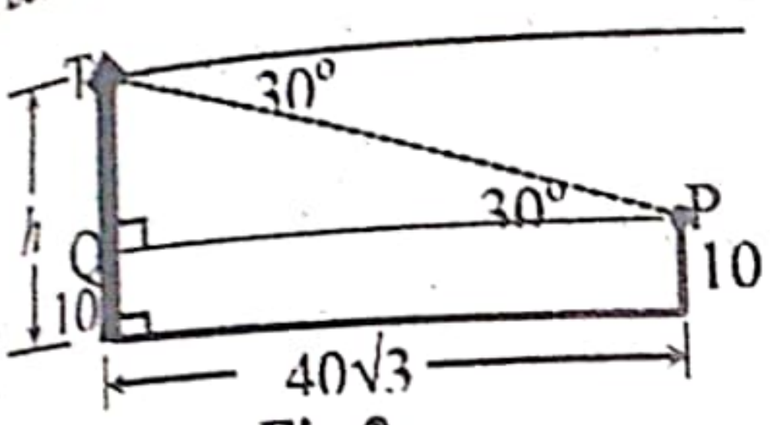

Fig 2

QP = $40\sqrt{3}$ so that in right-angled-triangle ΔTQP by alternate interior angle, we have $m(\angle TPQ) = 30^o$. Then, $\tan 30^o = \frac{TQ}{QP}$ gives

$$\tan 30^o = \frac{TQ}{40\sqrt{3}} \Rightarrow TQ = 40\sqrt{3}\tan 30^o = 40\sqrt{3}\times\frac{\sqrt{3}}{3} = 40$$

If h = height of the building, then $h = TQ + 10$
$= 40 + 10 = 50$ m

Answer: C
24. Given $f(x) = \frac{xe^x}{\cos x}$ implies

$$f'(x) = (\frac{xe^x}{\cos x})' = \frac{(xe^x)'\cos x - (\cos x)'xe^x}{(\cos x)^2}, \text{ quotient rule}$$

$$= \frac{[x'e^x + (e^x)'x]\cos x - (-\sin x)xe^x}{\cos^2 x}, \text{ product rule}$$

$$= \frac{[e^x + e^x x]\cos x + xe^x \sin x}{\cos^2 x}$$

Thus, $f'(0) = \frac{[e^0 + e^0(0)]\cos 0 + (0)e^0 \sin 0}{\cos^2 0} = \frac{[1+0]1+(0)1(0)}{1^2} = 1$

Answer: D
25. A. Since $\lim_{x\to a}\frac{f(x)-f(a)}{x-a}$ exists implies $\lim_{x\to a} f(x) = f(a)$; if f is differentiable at a, then it is continuous at a so that A is **True.**

B. $f(x) = |x|$ is not differentiable at $x = 0$ so that it has no tangent line at (0, 0), and so B is False.

C. If f and g are differentiable at a, then $\frac{f}{g}$ is differentiable at a only if $g(a) \neq 0$ so that C is False.

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D. If $f'(a)= 0$, then the tangent line to the graph of f at $(a, f(a))$ is $y = f(a)$ so that D is False.

Answer: A

26. Given twice continuously differentiable function f with $f''(3)=5$ and $g(x)= f(2x+3)$, then the second derivative of g at any x is:

$$\begin{aligned} g'(x) &= (f(2x+3))' \\ &= (2x+3)'\, f'(2x+3), \text{ by chain rule} \\ &= 2f'(2x+3) \ldots\ldots\ldots(1) \end{aligned}$$

Again $g''(x) = (g'(x))' = (2f'(2x+3))'$, by (1)

$$\begin{aligned} &= 2(f'(2x+3))', \text{ constant multiple rule} \\ &= 2(2x+3)'\, f''(2x+3), \text{ by chain rule} \\ &= 4f''(2x+3) \end{aligned}$$

Thus, by given $f''(3) = 5$ at $x = 0$ we have that

$$g''(0) = 4f''(2(0)+3)= 4f''(3) = 4\times5 = 20$$

Answer: A

27. Given the dimensions of the rectangle as ℓ(length) and w(width) with perimeter 120m implies

$$2\ell + 2w = 120, \text{ for } \ell, w > 0 \text{ (Length is positive)}$$

$$\Rightarrow \ell = 60 - w \ldots\ldots(1)$$

Let the area be A. Then, we need to maximize: $A(w, l) = wl$.

Using (1) we express A in single variable as:

$$A(w) = w(60-w), \text{ i.e. } A(w)= 60w - w^2$$

Since both $w > 0$ and $\ell = 60-w > 0$, we get domain of A as $0< w < 60$.

Then, to find critical numbers: $A'(w)=60-2w$ so that

$$A'(w) = 0 \text{ gives: } 60-2w = 0 \Rightarrow w = 30$$

Then, by Second Derivative Test:

$A''(w) = -2 \Rightarrow A''(30) = -2< 0$, so that A is maximum if $w =$ 30. Then, by (1), $\ell = 60-w = 60-30 = 30$.

Therefore, maximum area is obtained if the dimension is $\ell = 30$ m and $w = 30$ m.

Answer: A

28. If f is a continuous function on the interval $[a, b]$ and differentiable on (a, b) with property $f(a) = f(b)$, then by Rolle's Theorem there exists some $c \in (a, b)$ such that $f'(c) = 0$ so that graph of f has horizontal tangent line at $(c, f(c))$. Hence C is **True.**

Since extreme values may occur at $x = a$ or $x = b$ both A and D are False. f may have no x-intercept at all so that B is False.

Answer: C

29. Let f be continuous on $\mathbb{R}$ and $c \in$ dom f. [*Actually* dom $f = \mathbb{R}$ *since* f *is given to be continuous on* $\mathbb{R}$]. Then,

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A. False; for instance $f(x)= |x|$ has local minimum at $x = 0$, but $f'(0)$ doesn't exist.

B. True; because if $f(c) \leq f(x)\ \forall_x \in$ dom f, then there is very small open interval I containing c such that $f(c) \leq f(x)\ \forall_x \in I$.

C. False; for instance 0 is a critical number of $f(x)= x^3$ but f has no local extrema value at $x = 0$. [*There should be sign change in f'*]

D. False; for instance $f''(0) = 0$ for $f(x) = x^4$ but f has no inflection point at $(0, f(0))$. [*There should be sign change in f''*]

Answer: B

30. By the rule $(a + bi)(c + di) = ac - bd + (ad + bc)i$

$$zw = (2-3i)(5+2i) = 2(5) - (-3)2 + (2(2) + (-3)5)i$$
$$= 16 - 11i$$

Answer: D

31. $2 - 2\sqrt{3}i \Rightarrow x = 2\ \&\ y = -2\sqrt{3} \Rightarrow r = \sqrt{2^2 + (-2\sqrt{3})^2} = 4$

And, $x = 2 > 0 \Rightarrow \text{Arg } z = \tan^{-1}(\frac{y}{x})$

$$= \tan^{-1}(\tfrac{-2\sqrt{3}}{2}) = -\tan^{-1}(\sqrt{3}) = -\frac{\pi}{3}$$

Thus, the principal argument is: $\text{Arg } z = -\frac{\pi}{3}$

And $\arg z = \text{Arg } z + 2n\pi = -\frac{\pi}{3} + 2n\pi$ for $n \in \mathbb{Z}$ is also argument of z. So $n = 1$ gives $\theta = -\frac{\pi}{3} + 2\pi = \frac{5\pi}{3}$ so that the possible polar form of z is:

$$z = 4(\cos\tfrac{5}{3}\pi + i\sin\tfrac{5}{3}\pi)$$

Answer: B

32. A. Convers of A is "If the earth does not rotate around the sun, then man is immortal.", i.e., F⇒F≡ T

B. Convers of B is "If 4 is an even number, then 6 is a prime integer.", i.e., T⇒F≡ F

C. Convers of C is "If the earth does not rotate around the sun, then man is mortal.", i.e., F⇒T≡ T

D. Convers of D is "If 6 is prime, then athlete Derartu Tulu was born on October 21, 1971.", i.e., F⇒T≡ T or F⇒F≡ T. **WHY?**

Answer: B

33. For $x \neq 2$

$$\frac{4x^4-64}{2-x} = \frac{4(x^2-4)(x^2+4)}{2-x} = \frac{4(x-2)(x+2)(x^2+4)}{2-x}$$
$$= \frac{-4(2-x)(x+2)(x^2+4)}{2-x}$$
$$= -4(x+2)(x^2+4)$$

Answer: A

34. A function $f: \mathbb{R} \to [0, \infty)$ is onto if Codomain= $[0, \infty)$ = Range.

A. $|x| \geq 0 \Rightarrow |x| + 2 \geq 2 \Rightarrow f(x) \geq 2 \Rightarrow$ Rang of $f = [2, \infty)$

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Thus, $[2, \infty) \neq [0, \infty) \Rightarrow f$ is not onto

B. $\sqrt{x^2} = |x| \geq 0 \Rightarrow$ Rang of $f = [0, \infty)$

Thus, $[0, \infty) = [0, \infty) \Rightarrow f$ is onto

C. $2^x > 0 \Leftrightarrow f(x) > 0 \Rightarrow$ Rang of $f = (0, \infty)$

Thus, $(0, \infty) \neq [0, \infty) \Rightarrow f$ is not onto

D. $x^2 \geq 0 \Rightarrow x^2 + 1 \geq 1 \Rightarrow f(x) \geq 1 \Rightarrow$ Rang of $f = [1, \infty)$

Thus, $[1, \infty) \neq [0, \infty) \Rightarrow f$ is not onto

Answer: B

35. A. $\tan(-120^o) = -\tan 120^o$, tangent function is odd

$= -(-\tan(180^o - 120^o)$, quadrant II angle

$= \tan 60^o = \sqrt{3} \neq -1$

B. $\cos(-60^o) = \cos 60^o$, cosine function is even

$= \frac{\sqrt{3}}{2}$

C. $\sin(-120^o) = -\sin 120^o$, sine function is odd

$= -\sin(180^o - 120^o)$, quadrant II angle

$= -\sin 60^o = -0.5 \neq -1$

D. $\cos(-120^o) = \cos 120^o$, cosine function is even

$= -\cos(180^o - 120^o)$, quadrant II angle

$= -\cos 60^o = -\frac{\sqrt{3}}{2} \neq -1$

Answer: B

36. $x^2 - 5x + 6 \leq 0 \Rightarrow (x - 2)(x - 3) \leq 0$

Case I: $x - 2 \leq 0 \cap x - 3 \geq 0 \Rightarrow x \leq 2 \cap x \geq 3 \Rightarrow \emptyset$

Case II: $x - 2 \geq 0 \cap x - 3 \leq 0 \Rightarrow x \geq 2 \cap x \leq 3 \Rightarrow 2 \leq x \leq 3$

Thus, Solution Set $= \emptyset \cup 2 \leq x \leq 3 = [2, 3]$

Or simply graph of $f(x) = x^2 - 5x + 6$ is open upward and has x-intercepts 2 and 3 so that it is on and below x- axis for $x \in [2, 3]$.

Answer: B

37. A. $\sin(75^o) = \sin(180^o - 75^o)$, quadrant II angle

$= \sin(105^o) \neq \cos(105^o)$ ∴ A is False

B. $\tan(75^o) = -\tan(180^o - 75^o)$, quadrant II angle

$= -\tan(105^o) \neq \tan(105^o)$ ∴ B is False

C. $\sin(120^o) = \sin(180^o - 120^o)$, quadrant II angle

$= \sin(60^o)$ ∴ C is True

D. $\cos(120^o) = -\cos(180^o - 120^o)$, quadrant II angle

$= -\cos(60^o) \neq \cos(60^o)$ ∴ D is False

Answer: C

38. $\log(x^2 - 3) = 2\log(x - 1)$ is defined if

$x^2 - 3 > 0$ and $x - 1 > 0 \Rightarrow |x| > \sqrt{3}$ and $x > 1 \Rightarrow x > 1$

Then, $\log(x^2 - 3) = 2\log(x - 1)$

$\Rightarrow \log(x^2 - 3) = \log(x - 1)^2$, by $k \log a = \log a^k$

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$\Rightarrow x^2 - 3 = (x-1)^2$, since $y = \log x$ is 1-1 function
$\Rightarrow x^2 - 3 = x^2 - 2x + 1$
$\Rightarrow 3 = -2x + 1$
$\Rightarrow x = -1$

But $x > 1$ so that the solution set is $\emptyset$

Answer: B

39. The coordinates of a point that divides the line segment with end points $P(a, b)$ and $Q(c, d)$ in the ratio $r:t$ from P is

$$R(x, y) = \left(\frac{t(a)+r(c)}{r+t}, \frac{t(b)+r(d)}{r+t}\right).$$

Thus, the coordinates of a point that divides the line segment with end points P(0, 1) and Q(5, 6) in the ratio 2:3 from P is

$$R(x, y) = \left(\frac{3(0)+2(5)}{2+3}, \frac{3(1)+2(6)}{2+3}\right) = (2, 3)$$

Answer: B

40. $\frac{f}{g}$ may not be continuous since maybe $g(a) = 0$

Answer: A

41. Every monotonic bounded sequence converges (∴ B is True); and it converges to its:

(i) least upper bound if it is increasing ∴ A is False
(ii) greatest lower bound if it is decreasing ∴ D is False

$a_n = (-1)^n$ is bounded but it doesn't converge; ∴ C is False

Answer: B

42. Given $f(x) = \begin{cases} ax + 3 & \text{for } x \geq 1 \\ x^2 - 2x + 3 & \text{for } x < 1 \end{cases}$, then

f is continuous at $x = 1$ if

i) $0 \in$ Domain of f. Of course $f(1) = a(1)+3$ so that $0 \in$ Domain of f

ii) $\lim_{x\to 1} f(x)$ must exist $\Rightarrow \lim_{x\to 1^-} f(x)$ and $\lim_{x\to 1^+} f(x)$ exist and be equal

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x) \Rightarrow \lim_{x\to 1^-}(x^2 - 2x + 3) = \lim_{x\to 1^+}(ax + 3)$$
$$\Rightarrow 1^2 - 2(1)+3 = a(1)+3$$
$$\Rightarrow -1 = a$$

Answer: D

43. Given $\lim_{x\to 2} f(x) = 6$ and $\lim_{x\to 2} g(x) = -9$, then

$$\lim_{x\to 2}\left(\frac{f-g}{2f-7}\right)(x) = \lim_{x\to 2} \frac{f(x)-g(x)}{2f(x)-7}, \text{ by definition}$$
$$= \frac{\lim_{x\to 2}(f(x)-g(x))}{\lim_{x\to 2}(2f(x)-7)}, \text{ quotient rule}$$
$$= \frac{\lim_{x\to 2} f(x) - \lim_{x\to 2} g(x)}{2\lim_{x\to 2} f(x) - \lim_{x\to 2} 7}, \text{ difference \& multiple rule}$$
$$= \frac{6-(-9)}{2(6)-7} = 3 \qquad \text{Answer: C}$$

44. First for the indefinite integral $\int xe^x\,dx$ using by *parts* method

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Let $u = x$, and $dv = e^x dx$.
Then, $du = dx$ and $v = e^x$ so that
$\int xe^x\,dx = xe^x - \int e^x dx = xe^x - e^x + C$
Then, by Fundamental Theorem of Calculus
$\int_0^1 xe^x dx = [xe^x - e^x]_0^1$
$= (1)e^1 - e^1 - ((0)e^0 - e^0) = 1$

45. $\int \left(\frac{1}{\sqrt{x}} - e^{-3x} + \frac{5}{x^2}\right) dx = \int \frac{1}{\sqrt{x}} dx - \int e^{-3x} dx + \int \frac{5}{x^2} dx$ **Answer: D**

$= \int x^{-\frac{1}{2}} dx - \int e^{-3x} dx + 5\int x^{-2} dx$

$= \frac{x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} - \left(\frac{e^{-3x}}{-3}\right) + 5\frac{x^{-2+1}}{-2+1} + C$

$= 2\sqrt{x} + \frac{e^{-3x}}{3} + \frac{5}{x} + C$ **Answer: C**

46. B, C and D are True; but A is **NOT** True, because
$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$ for any c in (a, b)
Answer: A

47. The area of ΔABC can be given by
$A = \sqrt{s(s-AB)(s-BC)(s-AC)}$ for $s = \frac{AB+BC+AC}{2}$
Thus, for AB= 10 units, BC= 14 units and AC= 8 units
$s = \frac{AB+BC+AC}{2} = \frac{10+14+8}{2} = 16$ so that
$A = \sqrt{16(16-10)(16-14)(16-8)} = 16\sqrt{6}$ square unit
Answer: A

48. Given A= {1, 3, 5, 6, 8, 10} and B= {2, 5, 8, 11}, then
A-B= {1, 3, 6, 10} and B-A= {2, 11} so that
$A\Delta B = (A-B) \cup (B-A) = \{1, 3, 6, 10\} \cup \{2, 11\}$
$= \{1, 2, 3, 6, 10, 11\}$
Answer: D

49. Since $\sqrt{20} = 2\sqrt{5}$, $\sqrt{45} = 3\sqrt{5}$ and $\sqrt{75} = 5\sqrt{3}$ we have that
$\frac{6\sqrt{20}-3\sqrt{45}}{3\sqrt{75}} = \frac{6(2\sqrt{5})-3(3\sqrt{5})}{3(5\sqrt{3})} = \frac{3\sqrt{5}}{15\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{15}}{15}$
Answer: D

50. As shown in the graph of R below, the shaded region extends horizontally along x-axis from 2 including (since the boundary lines are solid) to -∞ so that:

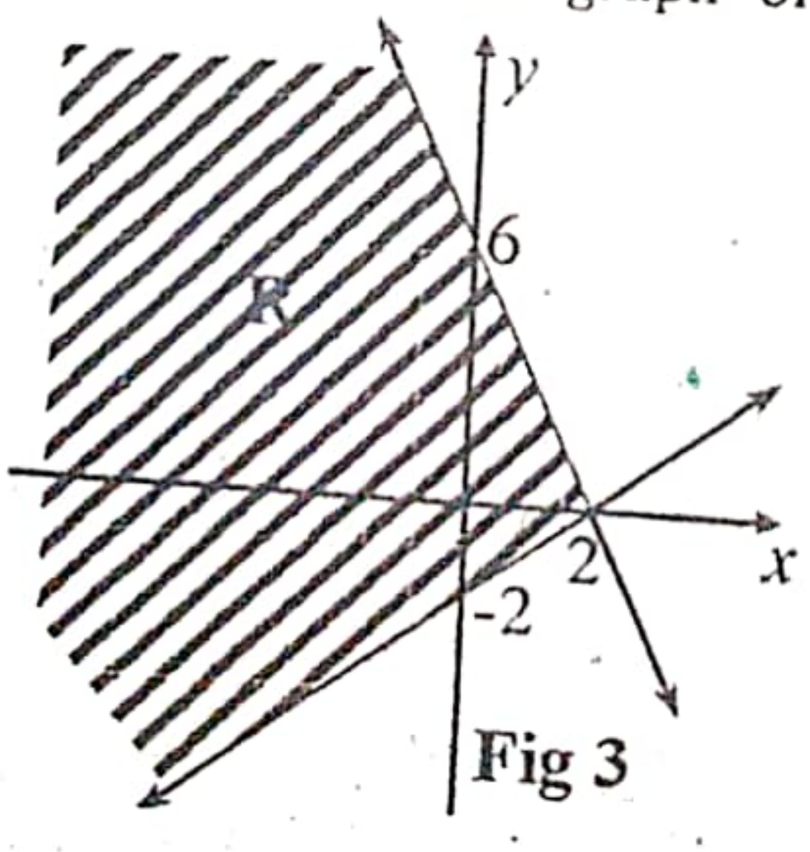

Fig 3

Domain of R = (-∞, 2]
$= \{x / x \leq 2\}$

The shaded region extends vertically along y-axis to -∞ and to -∞ so that

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Range of R = $(-\infty, \infty) = \mathbb{R}$

Answer: A
51. No mean for qualitative data; ∴ A is False. Every member of data must be considered (represented) to calculate the mean; ∴ B is False. Mean is unique; ∴ D is False. One of the limitation of mean is it is affected by extreme values; ∴ C is True.

Answer: C
52. A. arctangent function is defined by $y = \tan^{-1} x$ if and only if $x = \tan y$ for $-\frac{\pi}{2} < y < \frac{\pi}{2}$ so that the range is (π, π).

B. arcsine is defined by $y = \arcsin x$ if and only if $x = \sin y$ for $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$ so that the range is $[\frac{\pi}{2}, \frac{\pi}{2}]$.

C. arccosine is defined by $y = \arccos x$ if and only if $x = \cos y$ for $0 \leq y \leq \pi$ so that the domain is [-1, 1].

D. arcsine is defined by $y = \arcsin x$ if and only if $x = \sin y$ for $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$ so that the domain is [-1, 1].

Answer: C
53. A line segment chord in an ellipse which passes through the center and perpendicular to the major axis is the Minor axis. Latus rectum is the line segment chord in an ellipse which passes through the focus and perpendicular to the major axis.

Answer: A
54. The distance, d from a point (a, b) to a line ℓ: $Ax + By + C = 0$, is given by:

$$d = \frac{|Aa+Bb+C|}{\sqrt{A^2+B^2}}$$

Thus, for P(3, 4) and the line L with equation $3x + 4y - 5 = 0$ we have $a = 3$, $b = 4$, $A = 3$, $B = 4$ and $C = -5$ so that

$$d = \frac{|3(3)+4(4)-5|}{\sqrt{3^2+4^2}} = 4 \text{ units}$$

Answer: A
55. Given a grouped data of a distribution as follows

Range of marks	20-25	26-31	32-37	38-43	44-49
Frequency	15	30	25	20	10

First calculate class boundaries & cumulative frequencies as follows.

Class boundaries	19.5-25.5	25.5-31.5	31.5-37.5	37.5-43.5	43.5-49.5
Cumulative	15	45	70	90	100

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frequencies					

The first quartile Q_1 of the distribution is calculated as follows:

$$Q_1 = B_L + \left[\frac{\frac{1n}{4} - Cf_b}{f_1}\right] i, \text{ where}$$

$Q_1 = (\frac{1(100)}{4})^{th}$ item $= 25^{th}$ item $\Rightarrow$ found in 2^{nd} class, i.e., 25.5 – 31.5 so that

$B_L = 25.5 \leftarrow$ Lower boundary of 2^{nd} class

$Cf_b = 15 \leftarrow$ Cumulative freq. before 2^{nd} class

$f_k = 30 \leftarrow$ Frequency of 2^{nd} class (i.e., Q_1 class)

$i = 6 \leftarrow$ Class size

Thus, $Q_1 = 25.5 + (\frac{\frac{1(100)}{4} - 15}{30})6 = 27.5$

Answer: C

56. To rotate $(x+2)^2 + (y-2)^2 = 1$ through $\frac{\pi}{6}$ about the origin is to rotate the center (-2, 2) through $\frac{\pi}{6}$ about the origin and use the same radius since rotation is a rigid motion.

Given $\theta = \frac{\pi}{6} \Rightarrow \sin\theta = \sin\frac{\pi}{6} = \frac{1}{2}$ & $\cos\theta = \cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$ so that for $(x, y) = (-2, 2)$ if (x', y') is center of the image circle, then

$x' = x\cos\theta - y\sin\theta = -2(\frac{\sqrt{3}}{2}) - 2(\frac{1}{2}) = -(\sqrt{3}+1)$, and

$y' = x\sin\theta + y\cos\theta = -2(\frac{1}{2}) + 2(\frac{\sqrt{3}}{2}) = (\sqrt{3}-1)$.

Thus, $R_{\frac{\pi}{6}}(-2, 2) = (-(\sqrt{3}+1), \sqrt{3}-1)$ so that the image circle is:

$(x + (\sqrt{3}+1))^2 + (y - (\sqrt{3}-1))^2 = 1$

Answer: C

57. If r_1 and r_2 are real roots of $ax^2 + bx + c = 0$, then by Viet's Theorem $r_1 + r_2 = -\frac{b}{a}$ and $r_1 r_1 = \frac{c}{a}$ so that for $3x^2 + 9x + 16 = 0$, $a = 3$, $b = 9$ and $c = 16$ so that

A. $r_1 + r_2 = -\frac{9}{3} = -3$ so that A is False.

B. $r_1 r_2 = \frac{16}{3}$ so that B is False.

C. $r_1^2 + r_2^2 = (r_1 + r_2)^2 - 2r_1 r_2$

$= (-3)^2 - 2(\frac{16}{3})$, from results of A and B

$= -\frac{5}{3}$ so that C is **True**

D. $\frac{1}{r_1} + \frac{1}{r_2} = \frac{r_1 + r_2}{r_1 r_2} = \frac{-3}{16/3}$, from results of A and B

$= -\frac{9}{16}$, so that D is False

Answer: C

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58. The dot product of $u = ai + bj$ and $v = ci + dj$ is given by

$$u \cdot v = ac + bd$$

Thus, for $a = 1$, $b = 2$, $c = -2$ and $d = 4$

$$u \cdot v = ac + bd = (1)(-2) + (2)(4) = 6$$

Answer: C

59. Given A(-2, 1), B(3, 0), C(3, -1) and D(1, 1) in xy-plane imply

$u = \overrightarrow{AB} = (3-(-2), 0-1) = (5, -1)$ and

$v = \overrightarrow{CD} = (1-3, 1-(-1)) = (-2, 2)$ so that

$u - v = (5, -1) - (-2, 2) = (5-(-2), -1-2) = (7, -3)$

Answer: D

60. Let us use Truth table as follows.

Let P: $(p \wedge q) \Leftrightarrow p$, Q: $(p \Rightarrow q) \Leftrightarrow (\neg p \vee q)$,

R: $(p \vee q) \Leftrightarrow p$ & S: $(p \Rightarrow q) \Leftrightarrow p$

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$\neg p \vee q$	$p \Rightarrow q$	P	Q	R	S
T	T	F	T	T	T	T	T	T	T	T
T	F	F	F	T	F	F	F	T	T	F
F	T	T	F	T	T	T	T	T	F	F
F	F	T	F	F	T	T	T	T	T	F

From the table only is with truth value True in all possible rows (cases) so that Q: $(p \Rightarrow q) \Leftrightarrow (\neg p \vee q)$ is Tautology.

Answer: B

61. $f(x) = 5 + \frac{|x|}{x+1} = \begin{cases} 5 + \frac{-x}{x+1} & \text{if } x < 0 \\ 5 + \frac{x}{x+1} & \text{if } x \geq 0 \end{cases} = \begin{cases} \frac{4x+5}{x+1} & \text{if } x < 0 \\ \frac{6x+5}{x+1} & \text{if } x \geq 0 \end{cases}$

Since both pieces of f ar rationals with equal degree of numerator and denominator, graph of f cannot have oblique asymptote so that B is **NOT** true. Clearly A, C and D are True.

Answer: B

62. If amount A_o increases with rate r compounded monthly (i.e., $m = 12$), then in n years the total amount A is given by:

$$A = A_o(1 + \frac{r}{m})^{mn}$$

Thus, for $A_o = 20{,}000$, $r = 12\% = 0.12$, and $m = 12$ the doubling time (i.e., $A = 2A_o$) n (in years) is as follows:

$$A = A_o(1 + \frac{r}{m})^{mn} \Rightarrow 2A_o = A_o(1 + \frac{0.12}{12})^{12n}$$

$$\Rightarrow 2 = (1.01)^{12n}$$

$$\Rightarrow n = \frac{1}{12}\log_{1.01} 2$$

$$= \frac{\log 2}{12 \log 1.01}$$

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Answer: D $\approx \frac{0.3010}{12(0.0043)} \approx 5.81$ years

NOTE: In question 62, givens $\log 2 = 0.3010$ and $\log 1.01 = 0.0043$ or calculator is needed.

63. Given a matrix $\begin{pmatrix} 1 & 3 & 5 \\ 6 & 4 & 2 \\ 9 & 7 & 0 \end{pmatrix}$

A. The minor of entry 0 is: $\begin{vmatrix} 1 & 3 \\ 6 & 4 \end{vmatrix} = -14$ ∴ A is not True.

B. The cofactor of entry 1 is: $(-1)^{1+1}\begin{vmatrix} 4 & 2 \\ 7 & 0 \end{vmatrix} = -14$ ∴ B is True.

C. The minor of entry 3 is: $\begin{vmatrix} 6 & 2 \\ 9 & 0 \end{vmatrix} = -18$ ∴ C is not True.

D. The cofactor of entry 5 is: $(-1)^{1+3}\begin{vmatrix} 6 & 4 \\ 9 & 7 \end{vmatrix} = 6$ ∴ D is not True.

Answer: B

64. Given ΔABC with vertices A(0, 0), B(6, 0) and C(0, 4), the intersection point of medians is called ***Centroid*** and is the arithmetic mean of coordinates of all points making the triangle

, i.e., $(\frac{x_A+x_B+x_C}{3}, \frac{y_A+y_B+y_C}{3}) = (\frac{0+6+0}{3}, \frac{0+0+4}{3}) = (2, 4/3)$

Answer: C

65. By long division and partial fraction decomposition:

$$\frac{x^3+x^2-x}{x^2+x-2} = x + \frac{1}{3(x-1)} + \frac{2}{3(x+2)} \text{ so that}$$

$$\int \frac{x^3+x^2-x}{x^2+x-2}dx = \int (x + \frac{1}{3(x-1)} + \frac{2}{3(x+2)})dx$$

$$= \int x dx + \frac{1}{3}\int \frac{1}{x-1}dx + \frac{2}{3}\int \frac{1}{x+2}dx$$

$$= \frac{x^2}{2} + \frac{1}{3}\ln|x-1| + \frac{2}{3}\ln|x+2| + C$$

Answer: B

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1. Givens sets A and B where $A = \{x \in \mathbb{N}: x < 3\} = \{1, 2\}$, and B is the set of all possible factors of 13 so that $B = \{-13, -1, 1, 13\}$. Then

$B \times A = \{(x, y): x \in B \text{ and } y \in A\}$, By definition

$= \{(-13, 1), (-13, 2), (-1, 1), (-1, 2), (1, 1), (1, 2), (13, 1), (13, 2)\}$

Answer: No Answer is given

2. Given quadratic function $f(x) = x^2 - 6x + 10$ so that for the general form $f(x) = ax^2 + bx + c$; $a = 1$, $b = -6$ and $c = 10$. Then,

Axis of symmetry of graph of f is the line $x = -\frac{b}{2a}$, i.e., $x = -(\frac{-6}{2(1)}) = 3$.

Vertex of graph of f is at a point $(-\frac{b}{2a}, \frac{4ac-b^2}{4a})$ or simply:

$$(-\frac{b}{2a}, f(-\frac{b}{2a})) = (3, 3^2 - 6(3) + 10) = (3, 1).$$

And, $a = 1 > 0$ implies graph of f opens upward and range of f is

$\{y: y \geq f(-\frac{b}{2a})\}$ or $y: y \geq \frac{4ac-b^2}{4a}\}$, i.e., $f(x) \geq 1$

Answer: C

3. The area A of a regular n-sided polygon with radius r (or inscribed in a circle of radius r) is given by

$$A = \frac{1}{2}nr^2 \sin\frac{360^\circ}{n}$$

Given hexagon implies n=6; and given r= 6 cm give

$$A = \frac{1}{2}(6)(6\text{ cm})^2 \sin\frac{360^\circ}{6}$$

$$= 108 \sin 60^\circ \text{ cm}^2 = 54\sqrt{3}\text{ cm}^2, \text{ recall that } \sin 60^\circ = \frac{\sqrt{3}}{2}$$

Answer: A

4. A prime number is a natural number that has only two positive factor (i.e., 1 and itself).

Answer: D

5. A rational number r is in its standard form if it can be put as:

$$r = a \times 10^n \text{ where } 1 \leq a < 10 \text{ and } n \in \mathbb{Z}$$

Thus, only 5×10^5 is in standard form since $1 \leq 5 < 10$ and $5 \in \mathbb{Z}$.

Answer: C

6. For $a > 0$, and integers m, n, recall the exponential rules that

(i) $\sqrt{a} = a^{\frac{1}{2}}$; (iii) $a^n a^m = a^{n+m}$

(ii) $(a^n)^m = a^{nm}$; (iv) $a^n = a^m \Leftrightarrow n = m$

First, by (ii) $125^x = (5^3)^x = 5^{3x}$, and $25^{3x+1} = (5^2)^{3x+1} = 5^{6x+2}$ ...(*)

Then, by (iii) and (*): $125^x 5^{2x-1} = 5^{3x} 5^{2x-1} = 5^{5x-1}$ ...(**)

And, by (i), (ii) and (**): $\sqrt{125^x 5^{2x-1}} = (5^{5x-1})^{\frac{1}{2}} = 5^{\frac{5x-1}{2}}$ ...(***)

Therefore, by (*), and (***):

$$\sqrt{125^x 5^{2x-1}} = 25^{3x+1} \Rightarrow 5^{\frac{5x-1}{2}} = 5^{6x+2}$$

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$\Rightarrow \frac{5x-1}{2} = 6x + 2$, by (iv)

$\Rightarrow 5x - 1 = 12x + 4$

$\Rightarrow x = -\frac{5}{7}$

Since the given equation is defined for every real number x, $x = -\frac{5}{7}$.

Answer: A

7. First, determine the domain of the variable (Universe).

By absolute value definition, in the left $3|2x - 4| \geq 0$ so that

$6 - 2|1 - 5x| \geq 0 \Rightarrow |1 - 5x| \leq 3 \Rightarrow -3 \leq 1-5x \leq 3$

$\Rightarrow -\frac{2}{5} \leq x \leq \frac{4}{5}$

$\Rightarrow x \in [-\frac{2}{5}, \frac{4}{5}]$.....$[I_1]$

Again $3|2x - 4| = 6 - 2|1 - 5x|$ implies $2|1 - 5x| = 6 - 3|2x - 4|$ so that by absolute value definition, in the left $2|1 - 5x| \geq 0$ so that

$6 - 3|2x - 4| \geq 0 \Rightarrow |2x - 4| \leq 2 \Rightarrow -2 \leq 2x-4 \leq 2$

$\Rightarrow 1 \leq x \leq 3$

$\Rightarrow x \in [1, 3]$.....$[I_2]$

Hence, the domain of variable is: $I_1 \cap I_2 = [-\frac{2}{5}, \frac{4}{5}] \cap [1, 3] = \{\}$.

Since the universe is empty set, the solution set is also empty set.

Answer: D

8. The lateral surface area of a regular pyramid with length of lateral edge ℓ, and n-sided regular polygon base of side length s is given by:

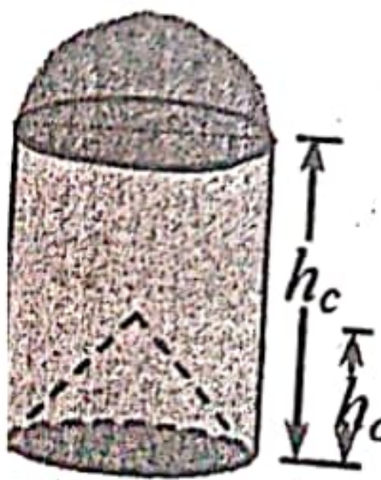
$$A_\ell = \frac{1}{2}ns\ell$$

Given, n = 6 (hexagonal base), s = 12 cm, and ℓ = 10 cm, we have

$$A_\ell = \frac{1}{2}(6)(12\text{ cm})(10\text{ cm}) = 360\text{ cm}^2.$$

Answer: B

9. Suppose an object is given as in the figure below. Before the cut off, the volume of the object V_o was the sum of the volumes of the hemisphere V_H and the cylinder V_C. Diameter of the hemisphere and base of the cylinder are equal, i.e., given d = 6 cm; and height of the cylinder is given by h_C = 12 cm so that:



$$V_o = V_H + V_C = \frac{1}{12}\pi d^3 + \frac{1}{4}\pi d^2 h_C$$

$$= \frac{1}{12}\pi(6\text{ cm})^3 + \frac{1}{4}\pi(6\text{ cm})^2(12\text{ cm})$$

$$= 126\pi\text{cm}^3 \quad \text{(volume before cut off)}$$

The cut off right circular cone has the same base diameter d = 6 cm, and given height h_{Co} = 8 cm so that the volume V_{Co} of the cone is:

$$V_{Co} = \frac{1}{12}\pi d^2 h_{Co} = \frac{1}{12}\pi(6\text{ cm})^2(8\text{ cm}) = 24\pi\text{cm}^3$$

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Therefore, the volume V of the remaining object after cut off is:
$V = V_o - V_{co} = 126\pi cm^3 - 24\pi cm^3 = 102\pi cm^3$.

Answer: B

10. The domain of a power function $f(x) = ax^{\frac{n}{m}}$ where $a \neq 0$, $\frac{n}{m} > 0$ and n is odd integer is $\mathbb{R}$ because f is defined for every real number x so that domain of $f(x) = 2x^{\frac{2}{3}}$ is $\mathbb{R}$.

Answer: D

11. The slope of a line that makes an angle of θ with the positive x-axis (recall this angle is called angle of inclination) is given by

$$m = \tan\theta.$$

Given, $\theta = 135^\circ$ implies:

$m = \tan 135^\circ$

$= -\tan(180^\circ - 135^\circ)$, quadrant II reference angle formula

$= -\tan 45^\circ = -1.$

Answer: A

12. By co-terminal angles relation and periodicity of tan by 180°:

$$\tan\theta = \tan(\theta + n180^\circ) \text{ for every integer n.}$$

For $\theta = 509^\circ$ choose $n = -3$ so that:

$\tan 509^\circ = \tan(509^\circ + (-3)180^\circ)$

$= \tan(-31^\circ)$

$= -\tan 31^\circ$, because tangent function is odd

$= -0.81$, because given $\tan 31^\circ = 0.81$

Answer: A

13. Recall the identities $\sec\theta = \frac{1}{\cos\theta}$ and $\cos\theta = \pm\sqrt{1-\sin^2\theta}$.

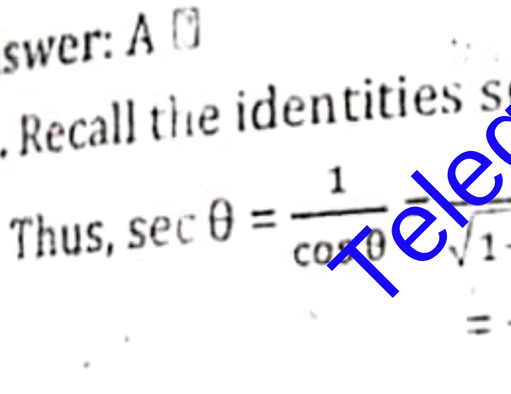
Thus, $\sec\theta = \frac{1}{\cos\theta} = \frac{1}{\sqrt{1-\sin^2\theta}}$, since $\sec\theta > 0$ in 4th quadrant

$= \frac{1}{\sqrt{1-(-\frac{3}{5})^2}}$, given $\sin\theta = -\frac{3}{5}$

$= \frac{5}{4}$.

Answer: A

14. Suppose in the following figure that ABCD is a parallelogram with P a mid-point of $\overline{AB}$ and Q a mid-point of $\overline{CD}$. Then,



First note that:

i) By opposite sides of parallelogram $\overline{AD} \equiv \overline{CB}$

ii) By opposite angles of parallelogram

$\angle ADQ \equiv \angle CBP$, and

iii) By opposite sides of parallelogram and given mid points P and Q, $\overline{DQ} \equiv \overline{BP}$

Therefore, by SAS congruency, $\Delta ADQ \equiv \Delta CBP$ so that B is necessarily true; and by corresponding sides of congruent triangles and symmetric

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property of congruency of a line segment, A is necessarily true. Since we have parallel $\overline{AP}$ and $\overline{DC}$, by definition of trapezium, APCD is a trapezium so that D is necessarily true.

In ΔCBP, $\overline{PB}$ and $\overline{PC}$ may not be congruent since ∠PBC might be obtuse so that $\overline{PC}$ be the longest side. Thus, PC≠ PB = AP (P is mid-point) gives; APCQ isn't a rhombus so that C is **NOT** necessarily true.

Answer: C

15. First $5\log_9 x - 2\log_9(x^2) - \log_4 8 = -1$ is defined if $x > 0$.

Recall that for $1 \neq a > 0$:

(i) $\log_{a^k} x = \frac{1}{k}\log_a x$, (ii) $\log_a x^m = m\log_a x$,

(iii) $\log_a a = 1$, (iv) $\log_a x + \log_a y = \log_a xy$ for $x, y > 0$.

By (i), (ii) and (iii), $\log_4 8 = \log_{2^2} 2^3 = \frac{3}{2}\log_2 2 = \frac{3}{2}$.

And, $x > 0$ implies $\log_9(x^2) = 2\log_9 x$ so that if $\log_9 x = y$, then

$5\log_9 x - 2\log_9(x^2) - \log_4 8 = -1 \Longrightarrow 5y - 2(2y) - \frac{3}{2} = -1 \Longrightarrow y = \frac{1}{2}$

Thus, $\log_9 x = y \Longrightarrow \log_9 x = \frac{1}{2} \Longrightarrow x = 9^{\frac{1}{2}} = 3$.

Therefore, solution set = {3}.

Answer: A

16. See rules under solution of Question 6. Here, $x \in \mathbb{R}$.

$$5\left(\frac{125}{8}\right)^{\frac{1}{3}x^2-\frac{2}{3}x} = 2 \Longrightarrow \left(\left(\frac{5}{2}\right)^3\right)^{\frac{1}{3}x^2-\frac{2}{3}x} = \frac{2}{5}$$

$$\Longrightarrow \left(\frac{5}{2}\right)^{3\left(\frac{1}{3}x^2-\frac{2}{3}x\right)} = \frac{2}{5}$$

$$\Longrightarrow \left(\frac{5}{2}\right)^{x^2-2x} = \frac{2}{5}$$

$$\Longrightarrow \left(\frac{2}{5}\right)^{2x-x^2} = \frac{2}{5}, \text{ by } \left(\frac{a}{b}\right)^n = \left(\frac{b}{a}\right)^{-n}$$

$$\Longrightarrow 2x - x^2 = 1$$

$$\Longrightarrow (x-1)^2 = 0 \Longrightarrow x = 1$$

Answer: B

17. You may use Case or Sign Chart or Graphical method.

Using case method for $(2x-3)(x+5) \leq 0$ implies

Case I: $2x-3 \leq 0 \cap x+5 \geq 0 \Longrightarrow x \leq 3/2 \cap x \geq -5 \Longrightarrow -5 \leq x \leq 3/2$

Case II: $2x-3 \geq 0 \cap x+5 \leq 0 \Longrightarrow x \geq 3/2 \cap x \leq -5 \Longrightarrow \emptyset$

Thus, Solution Set $= \emptyset \cup \{x: -5 \leq x \leq 3/2\} = [-5, 3/2]$

Or graph of $f(x) = (2x-3)(x+5)$ is open upward and has x-intercepts 3/2 and -5 so that it is on and below x- axis for $x \in [-5, 3/2]$.

Answer: C

18. The histogram below shows the recorded temperature (in °C) of certain town in Ethiopia for the first 15 days of March 2023.

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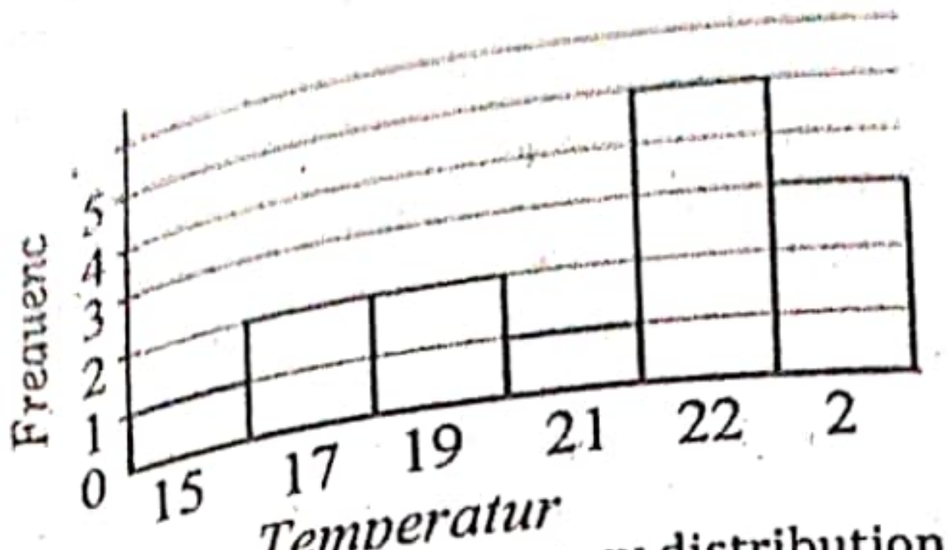


First prepare the following frequency distribution table.

Frequency, f_i (No. of days)	1	2	2	1	5	3
Temperature, T_5	15	17	19	21	22	24

Then, the highest frequency is $f_5 = 5$ of $T_5 = 22^oC$ (the Mode) so that A is true but D is false. The temperature was $T_6 = 24\ ^oC$ for $f_6 = 3$ (three days) so that B is false.

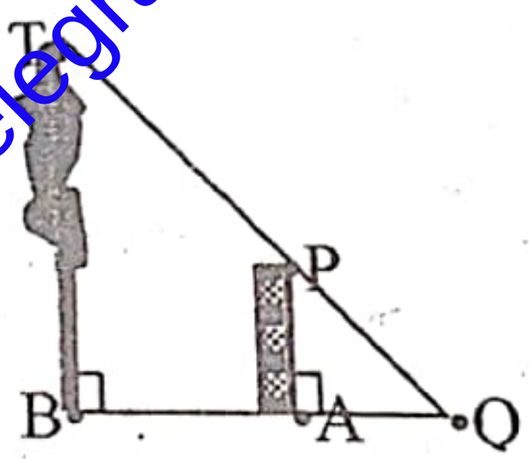
The temperature was more than $T_3 = 19\ ^oC$ for $f_4 = 1$ (i.e., $T_4 = 21^oC$), for $f_5 = 5$ (i.e., $T_5 = 19^oC$), and for $f_6 = 3$ (i.e., $T_6 = 24^oC$). Thus, the temperature was more than $T_3 = 19\ ^oC$ totally for

$$f_4 + f_5 + f_6 = 1 + 5 + 3 = 9 \text{ days.}$$

, i.e., $\frac{f_4+f_5+f_6}{\sum_1^6 f_i} \times 100\% = \frac{9}{1+2+2+1+5+3} \times 100\% = 64.30\%$ of the days was with temperature more than $19\ ^oC$ so that C is false.

Answer: A

19. Given a 20 m high building $\overline{AP}$ casts a shadow of AQ = 4 m, at the same time a BT = 35 m tree casts a shadow of x = BA + AQ m as shown in the figure below.



By AA Similarity, $\Delta TBQ \sim \Delta PAQ$ so that by proportionality of corresponding sides:

$$\frac{BA + AQ}{AQ} = \frac{BT}{AP} \Rightarrow \frac{x}{4m} = \frac{35m}{20m} \Rightarrow x = 7\ m$$

Answer: D

20. How far means the distance so that the total distance traveled by the car is: 9 km + 4 km + 12 km = 25 km.

Answer: B

21. By Remainder Theorem if a polynomial p(x) is divided by x-a, then the remainder is p(a) so that if $p(x) = 3x^6+5x^4-7x^3+2kx^2+3$ is divided by

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$x+1=x-(-1)$, then the remainder is $p(-1)$ so that given a remainder 4 implies:

$$p(-1) = 4 \Longrightarrow 3(-1)^6+5(-1)^4-7(-1)^3+2k(-1)^2+3 = 4$$
$$\Longrightarrow 2k = -14 \Longrightarrow k = -7.$$

Answer: A

22. Let $V(h, k)$= vertex and $F(x_F, y_F)$ = Focus. Then,
Vertex: $V(h, k)=V(2, -1)$, and Focus: $F(x_F, y_F) = F(-1, -1)$ so that
$k = -1 = y_F$, $h = 2$, $x_F = -1$, and $y_F = -1$.
It is horizontal parabola, because $y_F = k = -1$.
It is open left parabola $(y - k)^2 = -4p(x - h)$, because $-1 = x_F < h = 2$.
So $F(h-p, k) = F(2-p, -1) = F(-1, -1) \Rightarrow 2 - p = -1 \Rightarrow p = 3$
Therefore, the parabola is, $(y + 1)^2 = -12(x-2)$

Answer: A

23. We need equation of hyperbola with foci $F_1 = (-2, 1)$, $F_2 = (8, 1)$ and the length of the conjugate axis is $2b = 8$ units. Then, distance between the foci $F_1 = (-2, 1)$ and $F_2 = (8, 1)$ is $2c$

$$\Rightarrow 2c = 8 - (-2) = 10 \Rightarrow c = 5 \Rightarrow c^2 = 25$$

Length of conjugate axis: $8 \Rightarrow 2b = 8 \Rightarrow b = 4 \Rightarrow b^2 = 16$

Center $= \left(\frac{-2+8}{2}, \frac{1+1}{2}\right) = (3, 1) = (h, k)$ ← mid-point between foci, and

$$a^2 = c^2 - b^2 = 25 - 16 = 9.$$

It is a horizontal hyperbola $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$, because $y_{F_1} = 1 = y_{F_2}$

Therefore, the hyperbola is $\frac{(x-3)^2}{9} - \frac{(y-1)^2}{16} = 1$.

Answer: No Answer is given

24. Given proposition p with $p \equiv T$ and any proposition q

A. $(\neg p \wedge q) \Rightarrow q \equiv (\neg T \wedge q) \Rightarrow q$, given
$\equiv (F \wedge q) \Rightarrow q$, rule of negation
$\equiv F \Rightarrow q$, rule of conjunction
$\equiv T$, rule of implication

Similarly, we can show that the (B), (C), and (D) are F.

Answer: A

25. A. Since $|x| = x$ for $x \in (0, \infty)$ we have $f(x) = x - 1$ which is a linear function so that f is one-to-one function.
B. Not one-to-one, because for instance $f(1) = f(6) = 5$ but $1 \neq 6$.
C. Not one-to-one, because y may have two or more children.
D. Not one-to-one, because for instance $f(-1) = f(1) = 0$ but $-1 \neq 1$.

Answer: A

26. Given $\frac{x^2+14}{(x+2)(x-1)^2} = \frac{A}{x+2} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$

$$\Rightarrow \frac{x^2+14}{(x+2)(x-1)^2} = \frac{A(x-1)^2+B(x-1)(x+2)+C(x+2)}{(x+2)(x-1)^2}$$

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$$\Rightarrow \frac{x^2+14}{(x+2)(x-1)^2} = \frac{(A+B)x^2+(-2A+B+C)x+(A-2B+2C)}{(x+2)(x^2+1)}$$

$$\Rightarrow x^2+14 = (A+B)x^2+(-2A+B+C)x+(A-2B+2C)$$

$$\Rightarrow \begin{cases} A+B=1 \quad \dots\dots\dots(1) \\ -2A+B+C=0 \dots\dots\dots(2) \\ A-2B+2C=14 \dots\dots\dots(3) \end{cases}$$

$$\Rightarrow \begin{cases} -3A+C=-1 \\ 3A+2C=16 \end{cases} \leftarrow \text{replacing } B=1-A \text{ from (1) in to (2) \& (3)}$$

$\Rightarrow A = 2$ and $C = 5$ so that by $B = 1-A$, $B = -1$

Answer: C ?

27. Given a rational function $f(x) = \frac{x^3-3x}{x^2-6}$. Then

A. y-intercept is $y = f(0) = \frac{0^3-3(0)}{0^2-6} = 0$.

B & C. $f(x) = \frac{x^3-3x}{x^2-6}$ is odd because it has a form $\frac{\text{odd}}{\text{even}}$ so that graph of f is symmetrical with respect to the origin.

D. x-intercept is the value of x for which $x^3-3x = 0$, i.e., $x = 0, \pm\sqrt{3}$.

Answer: C ?

28. Let θ be angle between the lines $\ell_1: y = 2x+3$ with slope $m_1 = 2$ and $\ell_2: y = 1-3x$ with slope $m_1 = -3$. Then, $0 \le \theta < \pi$ and

$$\tan\theta = \frac{m_2-m_1}{1+m_1m_2} = \frac{-3-2}{1+2(-3)} = 1 \text{ so that } \theta = \tan^{-1} 1 = \pi/4.$$

Answer: B ?

29. Given position vectors $\mathbf{u} = (-1, 6)$ and $\mathbf{v} = (2, 4)$. Then

$$4\mathbf{u}-3\mathbf{v} = 4(-1, 6)-3(2, 4) = (-4, 24) - (6, 12)$$
$$= (-4-6, 24-12) = (-10, 12).$$

Answer: B ?

30. Given a function $f(x) = 3\sin(4x)$ has period $p = \frac{2\pi}{4} = \frac{\pi}{2}$. And, the function $y = \sin x$ lies below x-axis on $(\pi, 2\pi)$; dividing bounds of this interval by 4, f lies below x-axis on $(\pi/4, \pi/2)$ so that A is **true**.

Period $p = \frac{\pi}{2}$ implies graph of f completes one cycle on interval $[0, \frac{\pi}{2}]$ so that B & D are false.

The function $y = \sin x$ rises up on the interval $[0, \pi/2]$; dividing bounds of this interval by 4, f rises up on $[0, \pi/8]$ but $[0, \pi/4] \not\subseteq [0, \pi/8]$ so that C is false.

Answer: A ?

31. Given a translation T that takes a point **(-1, 1) to point (3, 2).**

The translation vector is: $\vec{U}_T = (3, 2) - (-1, 1) = (4, 1)$.

Translating a circle means translating the center and use the same radius. $x^2+y^2-2x+4y+1=0$ has center $C = (-(-2)/2, -4/2) = (1, -2)$.

So, $T(1, -2) = \vec{U}_T + (1, -2) = (5, -1)$; Center of image circle

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And, for radius r, $r^2 = \frac{1}{4}((-2)^2+4^2-4(1)) = 4$.

Thus, the image circle is $(x-5)^2+(y+1)^2 = 4$.

Answer: D

32. Let A be a square matrix of order n and let B be a matrix obtained by multiplying one row of A by 2 and C be a matrix found by adding one row of A to another row of A.

If a row/a column is multiplied by a scalar r, then the new determinant equals r times the old determinant; and the determinant remains the same if a multiple of a row/column is added to another row/column.

Thus,

$\det(B) = 2\det(A) = 20$ and $\det(C) = \det(A) = 10$

Answer: B

33. You may check that $\Delta = 0$, $\Delta_x = 0$, $\Delta_y = 0$ and $\Delta_z = 0$ so that from Cramer's rule we conclude that the system has infinitely many solution. And it is better to apply row/column operation method.

The augmented matrix is $\left(\begin{array}{ccc|c} 2 & -1 & 3 & 1 \\ 1 & 1 & 3 & 5 \\ 1 & -1 & 1 & -1 \end{array}\right)$

Pivoting R_2 and R_3 using R_1 to make $a_{21} = 0 = a_{31}$ as:

$$\left(\begin{array}{ccc|c} 2 & -1 & 3 & 1 \\ 1 & 1 & 3 & 5 \\ 1 & -1 & 1 & -1 \end{array}\right) \xrightarrow[R_3 \to R_3 - \frac{1}{2}R_1]{R_2 \to R_2 - \frac{1}{2}R_1} \left(\begin{array}{ccc|c} 2 & -1 & 3 & 1 \\ 0 & 3/2 & 3/2 & 9/2 \\ 0 & -1/2 & -1/2 & -3/2 \end{array}\right)$$

After rescaling as $R_2 \longrightarrow 2R_2$ and $R_3 \longrightarrow -2R_3$: Pivoting R_3 using R_2 to make $a_{32} = 0$ as:

$$\left(\begin{array}{ccc|c} 2 & -1 & 3 & 1 \\ 0 & 3 & 3 & 9 \\ 0 & 1 & 1 & 3 \end{array}\right) \xrightarrow{R_3 \to R_3 + \frac{1}{3}R_2} \left(\begin{array}{ccc|c} 2 & -1 & 3 & 1 \\ 0 & 3 & 3 & 9 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Rescaling as $R_2 \longrightarrow \frac{1}{3}R_2$: $\left(\begin{array}{ccc|c} 2 & -1 & 3 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right)$.

Therefore, the system is **Dependent**.

From R_1 and R_2 we get two equations in three unknowns as

$$\begin{cases} 2x - y + 3z = 1 \ldots\ldots (1) \\ y + z = 3 \ldots\ldots (2) \end{cases}$$

Let $z = k$ for $k \in \mathbb{R}$. Then (2) gives $y = 3 - k$.

Then, substituting $z = k$ and $y = 3 - k$ in to (1) gives $x = 2-2k$

Therefore, the solution set is: $\{(2-2k, 3-k, k) \square k \in \mathbb{R}\}$, and we say the system has infinitely many solution.

Answer: D

34.
$$\begin{aligned} z_1 - (z_2 - z_3) &= z_1 + (-1)(z_2 - z_3) \\ &= z_1 + (-1)(z_2) - (-1)z_3 \\ &= z_1 - z_2 + z_3 \neq (z_1 - z_2) - z_3 \text{ so that A is False} \end{aligned}$$

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B, C, and, D are ,commutativity, associativity, and distributivity properties which hold in real number system so that they also hold in set of complex numbers.

Answer: A ☑

35. In the Argand plane, the complex numbers a + bi corresponds to the point (a, b) so that −3 − i corresponds to the point (-3, -1).

Answer: C ☑

36. A group of students, 5 from grade 10, 8 from grade 11 and 2 from grade 12 to be formed. If a student is chosen randomly from this group, the probability that the chosen student is either from grade 10 or from grade 12 is:

$$p(G10 \text{ or } G12) = \frac{n(G10 \text{ and } G12)}{n(\text{all students})} = \frac{7}{15}$$

Answer: B ☑

37. Use matrices multiplication rule and definition of equal matrices.

$$\begin{pmatrix} 2 & -1 \\ a & 4 \end{pmatrix}\begin{pmatrix} 1 & b \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ -1 & 18 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2(1)+(-1)(-1) & 2b+(-1)3 \\ a(1)+4(-1) & ab+4(3) \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ -1 & 18 \end{pmatrix}, \text{ multiplication}$$

$$\Rightarrow \begin{pmatrix} 3 & 2b-3 \\ a-4 & ab+12 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ -1 & 18 \end{pmatrix}$$

$$\Rightarrow \begin{cases} 3=3 \\ 2b-3=1 \\ a-4=-1 \\ ab+12=18 \end{cases}, \quad \text{by definition of equal matrices}$$

$$\Rightarrow \begin{cases} 3=3 \\ b=2 \\ a=3 \\ ab=6 \end{cases}, \quad \text{by definition of equal matrices}$$

Since ab = 6 is also satisfied, a = 3, and b = 2.

Answer: B ☑

38. A^T is the transpose of A that is found by making the i^{th} row in to i^{th} column, i.e., $A^T = (a_{ji})$ for $A = (a_{ij})$ so that

$$A = \begin{pmatrix} 1 & -1 \\ 0 & 2 \\ -3 & 6 \end{pmatrix} \Rightarrow A^T = \begin{pmatrix} 1 & 0 & -3 \\ -1 & 2 & 6 \end{pmatrix}.$$

Answer: B ☑

39. Cumulative frequency (commonly the less than cumulative frequency) of a class (except the first class whose cumulative frequency and frequency are equal) is the sum of frequencies of all the classes before it and its frequency.

Temperature(in °C)	10-14	15-19	20-24	25-29	30-34

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Frequency (Numberofdays)	2	9	11	5	3
Cumulative frequencies	2	11	22	27	30

Hence, as shown in bold the cumulative frequency of the fourth class is 27 (i.e., 2+9+11+5).

Answer: A ☑

40. Given a universal set $\mathbb{N}$, the proposition $(\exists_x)(\exists_y)(x-y = y-x)$ is True just by taking $x = y$ so that B is True.

To show the falseness of A, C, and D we use disprove by counter example. Take $x = 1$ and $y = 2$. Then,

$x \neq y \Rightarrow x > y \equiv T \Rightarrow F \equiv F$ so that A is False.

No value of $y \in \mathbb{N}$ such that $xy < x$ if $x = 1$ so that C is False.

No value of $y \in \mathbb{N}$ such that $x - y = 5$ if $x = 1$ so that D is False.

Answer: B ☑

41. The following is distribution of the weight of 40 students in a certain class.

Weight (in Kg)	40-48	49-57	58-66	67-75	76-84
Frequency, f_i	8	18	7	5	2

Then, to get the median weight in Kg, first calculate class boundaries for correction factor 0.5 & (Less than) cumulative frequencies as follows.

Class boundaries	39.5-48.5	48.5-57.5	57.5-66.5	66.5-75.5	75.5-84.5
Frequency, f_i	8	18	7	5	2
Cumulative frequencies, Cf_i	8	26	33	38	40

Then, $n = \sum_{i=1}^{5} f_i = 40 \Rightarrow Cf_i \geq \frac{n}{2} = 20$ so that $Cf_i \geq 20$ for first time in 2nd class, i.e., the median class is the 2nd class with class boundaries 48.5 - 57.5 and lower bound $B_L = 48.5$, cumulative frequency before the median class $Cf_b = 8$, median class frequency $f_c = 18$, and class size $i = 9$ so that the median is:

$$\tilde{x} = B_L + \left[\frac{\frac{n}{2} - Cf_b}{f_c}\right] i = 48.5 + \left[\frac{20-8}{18}\right] 9 = 54.5 \text{ Kg}$$

Answer: B ☑

42. Recall that: Two events are said to be ***mutually exclusive***, if both cannot occur simultaneously. If a fair die is rolled once, then the sample space is $S = \{1, 2, 3, 4, 5, 6\}$ and among events:

E_1 = an even number occurs = {2, 4, 6};

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E_2 = a prime number occurs = {2, 3, 5};
E_3 = a multiple of 3 occurs = {3, 6};
E_4 = a multiple of 5 occurs = {5}.

the pairs of mutually exclusive events are: E_1 and E_4; and E_3 and E_4, because their intersection is empty set.

Answer: A

43. This question must be specified for a polynomial function; otherwise the question is wrong. For instance: $f(x) = 2^x - 1$ has zero at 0, but x is not factor of f.

Answer: Wrong Item

44. Mean Value Theorem: If a function f is continuous on the interval [a, b] and differentiable on (a, b), then there exists (at least one) $c \in (a, b)$ such that $f'(c) = \frac{f(b)-f(a)}{b-a}$.

$f(x) = 3x^2 - 5x + 1$ is continuous on [0, 1] and differentiable on (0, 1) so that by Mean Value Theorem, there exists $c \in (0, 1)$ such that

$$f'(c) = \frac{f(1)-f(0)}{1-0} \Rightarrow 6c - 5 = \frac{-1-1}{1} \Rightarrow c = 1/2.$$

Answer: B

45. Volume of a cone with height h and base radius r is $V = \frac{1}{3}\pi r^2 h$. Given $h = 10$ cm and $r = 5$ cm implies

$$\frac{r}{h} = \frac{5}{10} \Rightarrow r = \frac{1}{2}h \text{ or } h = 2r.$$

Then, $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(\frac{1}{2}h)^2 h = \frac{1}{12}\pi h^3$ so that by chain rule

$$\frac{dV}{dt} = \frac{1}{4}\pi h^2 \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{4}{\pi h^2}\frac{dV}{dt}$$

By given at the moment $\frac{dV}{dt} = 6\pi cm^3/sec$, the rate at which the water level rising when the water reaches $h = 2$ cm deep is

$$\frac{dh}{dt} = \frac{4}{\pi h^2}\frac{dV}{dt} = \frac{4}{\pi(2cm)^2}(6\pi cm^3/sec) = 6 \text{ cm/sec}.$$

Answer: A

46. The graphs of $f(x) = \frac{1}{\sqrt{x}}$ and $g(x) = 2 - \frac{1}{6}x^3$ have tangent lines of the same slope at x where $f'(x) = g'(x)$. Thus,

$$f'(x) = g'(x) \Rightarrow -\frac{1}{2}x^{-\frac{3}{2}} = -\frac{1}{2}x^2 \Rightarrow x = 1 \text{ (WHY?)}$$

Answer: A

47. Mechanical (also called physical) meaning of derivative is its interpretation as an instantaneous rate of change.

Answer: D

48. Using sum rule of derivative:

$$f(x) = 5x^3 + 2x - \sin x + e^x \Rightarrow f'(x) = 15x^2 + 2 - \cos x + e^x$$

Thus, $f'(0) = 15(0)^2 + 2 - \cos 0 + e^0 = 2$.

Answer: B

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49. Given $h(x) = g(f(3x+1))$, $f'(1) = f(1) = 2$, and $g'(2) = 1$. Then, using chain rule

$$h'(x) = (g(f(3x+1)))' = g'(f(3x+1))\, f'(3x+1)(3x+1)'$$
$$= 3g'(f(3x+1))\, f'(3x+1)$$

Thus, $h'(0) = 3g'(f(3(0)+1))f'(3(0)+1)$
$= 3g'(f(1))f'(1)$
$= 3g'(2)(2)$, given $f'(1) = f(1) = 2$
$= 3(1)(2)$, given $g'(2) = 1$
$= 6$.

Answer: D

50. Validity of an argument can be checked using either of Formal proof, Truth table, Tautology or Rules of Inference.

A. [Using Formal proof]

Given argument: $p, p \Rightarrow q, r \vee \neg q \vdash p \wedge \neg r$

(1) $p \equiv T$ and $p \Rightarrow q \equiv T$ ...premise
(2) $q \equiv T$ ...(1) and rule of "$\Rightarrow$"
(3) $\neg q \equiv F$ ...(2) and rule of "$\neg$"
(4) $r \vee \neg q \equiv T$ ...premise
(5) $r \equiv T$ ...(3) and rule of "$\vee$" on (4)
(6) $\neg r \equiv F$ ...(5) and rule of "$\neg$"
(7) $p \wedge \neg r \equiv F$ ...(6) and rule of "$\wedge$"

Therefore, the argument form is invalid.

B. [Using Truth table]

Given argument: $p \vee q, p \Rightarrow q \vdash \neg q$

p	q	$\neg q$	$p \vee q$	$p \Rightarrow q$
T	T	F	T	T
T	F	T	T	F
F	T	F	T	T
F	F	T	F	T

In first row the premises are T but the conclusion is F so that the argument form is invalid.

C. [Using Formal proof]

Given argument: $p \Rightarrow q, \neg r \wedge q \vdash p \vee r$

(1) $\neg r \wedge q \equiv T$ ...premise
(2) $\neg r \equiv T$ and $q \equiv T$ ...(1) and rule of "$\wedge$"
(3) $r \equiv F$ ...(2) and rule of "$\neg$"
(4) $p \Rightarrow q \equiv T$ ...premise

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(5) $p \equiv T$ or $p \equiv F$ ...(2) and (4) with rule of "$\Rightarrow$"
(6) $p \vee r \equiv T$ or $p \vee r \equiv F$ ...by (3) and (5) with rule of "$\vee$"
Therefore, the argument form is invalid.

D. [Using Formal proof]
Given argument: $p \Rightarrow q, q, r \Rightarrow p \vdash r$
(1) $q \equiv T$ and $p \Rightarrow q$ ...premise
(2) $p \equiv T$ or $p \equiv F$ ...(1) and rule of "$\Rightarrow$"
(3) $r \Rightarrow p \equiv T$ ... premise
(4) If $p \equiv T$, then $r \equiv T$ or $r \equiv F$...(2) and rule of "$\Rightarrow$"
(5) If $p \equiv F$, then $r \equiv F$...(2) and rule of "$\Rightarrow$"
Therefore, the argument form is invalid.

Answer: No Answer is given

51. The mid-point of line segment joining (-1, -3, 3) and (-1, 5, -7) in space is $(\frac{-1+(-1)}{2}, \frac{-3+5}{2}, \frac{3+(-7)}{2}) = (-1, 1, -2)$.

Answer: D

52. Principle of Mathematical Induction (PMI) is applicable on a proposition with variables' value set of Natural Numbers.
A. $n = 0 \Rightarrow 0! \geq 2^0$ which is false so that no use of PMI.
B. $n = 0 \Rightarrow 2^{0+1} - 1 = 1$ (True). Hence, first note that
$1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$ for $n \geq 0$ is similar to
$1 + 2 + 2^2 + \cdots + 2^{n-1} = 2^n - 1$ for $n \geq 1$
$n = 1 \Rightarrow 2^1 - 1 = 1$ (True).
Assume it holds for $n = k > 1$ and $k \in \mathbb{N}$
, i.e., $1 + 2 + 2^2 + \cdots + 2^{k-1} = 2^k - 1$ is true. (IH)

Then, for $n = k + 1$,
$1 + 2 + 2^2 + \cdots + 2^{k-1} + 2^k = 2^k - 1 + 2^k$, by IH
$= 2 \times 2^k - 1$
$= 2^{k+1} - 1$

Thus, the formula holds also for $n = k + 1$ so that this assertion is true applying the Principle of Mathematical Induction.
C. $n = 1 \Rightarrow 1^2 \neq 3$ so that no use of PMI.
D. $n = 2 \Rightarrow \frac{2(2+1)}{2} + 1 = 4 \neq 2$ so that no use of PMI.

Answer: B

53. Let F be the anti-derivative of f. For constant of integration c
A. $F' = f(x)$ so that A is true.
B. $\frac{d}{dx}\int F(x)dx = F(x)$ for every function F so that B is **NOT** true.
C. $\int \frac{d}{dx} f(x)dx = f(x) + c$ so that C is true.
D. $\int f(x)dx = F(x) + c$ so that D is **NOT** true.

Answer: B and D

54. $\int 3\cos 3x\,dx = 3\int \cos 3x\,dx$

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Let $u = 3x$. Then, $dx = \frac{1}{3}du$ so that

$$3\int \cos 3x\,dx = 3\int \cos u \frac{1}{3}du$$
$$= \int \cos u\,du = \sin u + C$$
$$= \sin 3x + C \text{ by } u = 3x$$

Answer: A

55. $f(x) = x^2 - 4$ crosses x-axis at $x = \pm 2$ and $f \leq 0$ on $[-2, 2]$ so that the area of the region bounded by the graph of f and x-axis, using the evenness of f is

$$A = 2\int_0^2 -f(x)dx = 2\int_0^2 (4 - x^2)dx$$
$$= 2(4x - \frac{x^3}{3})_0^2$$
$$= 2[(4(2) - \frac{2^3}{3}) - (4(0) - \frac{0^3}{3})] = \frac{32}{3}$$

Answer: C

56. Let f be a continuous function on $[0, 1]$ and let F be an anti-derivative of f with $F(1) = -11$ and $F(0) = 11$. Then

$$\int_0^1 f(x)dx = F(x)]_0^1 = F(1) - F(0) = -11 - 11 = -22$$

Answer: B

57. On xz-plane $y = 0$ so that the set describing equation of the xz-plane is $\{(x, y, z) \mid x, y, z \in \mathbb{R} \text{ and } y = 0\}$.

Answer: A

58. If the sequence $\{a_n\}$ converges to 3 and the sequence $\{b_n\}$ converges to -1, then by convergence theorem, $\{a_n - b_n\}$ converges to $3 - (-1) = 4$.

Answer: A

59. $f(x) = \frac{x^2-1}{x^2+1}$ is defined and continuous on $\mathbb{R}$ so that

$$\lim_{x\to 1} \frac{x^2-1}{x^2+1} = \frac{1^2-1}{1^2+1} = 0.$$

Answer: C

60. $f(x) = \begin{cases} \alpha x + 1, & x \geq 1 \\ x^2 - 4, & x < 1 \end{cases}$ is continuous at 1 if

i) $1 \in$ domain of f. Of course assume $f(1) = \alpha + 1 \in \mathbb{R}$.

ii) $\lim_{x\to 1} f(x)$ must exist so that side limits exist and

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x) \Rightarrow \lim_{x\to 1^-} (x^2 - 4) = \lim_{x\to 1^+} (\alpha x + 1)$$
$$\Rightarrow 1^2 - 4 = \alpha(1) + 1$$
$$\Rightarrow \alpha = -4$$

Answer: A

61. First length of the interval $[\frac{\pi}{4}, \frac{11\pi}{4}]$ is $\ell = \frac{11\pi}{4} - (-\frac{\pi}{4}) = 3\pi$ which is a period of the function which completes one cycle.

Period of $f(x) = 2\sin(\frac{2}{3}x + \frac{\pi}{6})$ is: $p = \frac{2\pi}{2/3} = 3\pi$ (see solution of Q62).

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Thus, $\ell = p$ implies $f(x) = 2\sin(\frac{2}{3}x + \frac{\pi}{6})$ has a graph that completes one full cycle on $[-\frac{\pi}{4}, \frac{11\pi}{4}]$.

Answer: D

62. Graph of $f(x) = c + a\cos(kx + b)$ has:

Period: $p = \frac{2\pi}{k}$ and Phase shift: $-\frac{b}{k}$.

Thus, for $f(x) = -\frac{1}{3}\cos(\frac{3}{2}x - \frac{\pi}{3})$ we have $k = \frac{3}{2}$, $b = -\frac{\pi}{3}$ so that

Period: $p = \frac{2\pi}{3/2} = \frac{4\pi}{3}$ and Phase shift: $-\frac{b}{k} = -\frac{-\pi/3}{3/2} = \frac{2\pi}{9}$.

Answer: B

63. Since n starts from 0, the fifth term is at $n = 4$

, i.e., $2^4 + 4(4) - 5 = 27$.

Answer: B

64. By long division $\frac{2n+1}{n+1} = 2 - \frac{1}{n+1}$

Then, $n \geq 1 \Rightarrow n + 1 \geq 2 \Rightarrow 0 < \frac{1}{n+1} \leq \frac{1}{2}$, reciprocal

$\Rightarrow -\frac{1}{2} \leq -\frac{1}{n+1} < 0$, multiplying by -1

$\Rightarrow \frac{3}{2} \leq 2 - \frac{1}{n+1} < 2$, adding 2

$\Rightarrow 1 \leq \frac{2n+1}{n+1} < 2$

Thus, every real number greater than or equal to 2 is an upper bound.

Or simply by listing some elements of $\frac{2n+1}{n+1} = 2 - \frac{1}{n+1}$ as:

$2 - \frac{1}{2}, 2 - \frac{1}{3}, 2 - \frac{1}{4}, \ldots, 2 - 0$

, we can see it is going to 2 increasingly so that every real number greater than or equal to 2 is an upper bound.

Answer: C

65. Given arithmetic sequence $\{a_n\}$ with $a_6 = 35$, and $a_{11} = 75$. Then, the common difference is: $d = \frac{a_{11} - a_6}{11-6} = \frac{75-35}{5} = 8$.

And, using $a_n = a_1 + (n-1)d$ for $n = 6$:

$a_1 = 35 - (6-1)8 = -5$.

Thus, by $S_n = \frac{n}{2}(2a_1 + (n-1)d)$ the sum of the first 31 terms is

$S_{31} = \frac{31}{2}(2(-5) + (31-1)8) = 3565$

Answer: No Answer is given

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BOOKLET CODE; 097

1. Which one of the following statements is correct about transformation of mechanical energy into other forms and the law of conservation of total energy?
   A. Neglecting air resistance, an object moving vertically downward gains potential energy and loses an equal amount of kinetic energy.
   B. Neglecting air resistance, an object moving vertically upward gains kinetic energy and loses an equal amount of gravitational potential energy.
   C. Due to dissipative force, the mechanical energy of a wooden block sliding on a rough horizontal surface is not conserved.
   D. Mechanical energy of a wooden block sliding on a rough horizontal surface is not conserved.

2. A wheel rotating at angular speed of $10rev/s$ is brought to rest after $40rev$ under a constant angular acceleration. The angular acceleration of a point on the wheel is
   A. $-0.5rev/s^2$. B. $0.25rev/s^2$. C. $5rev/s^2$. D. $-5rev/s^2$.

3. The parallel axis theorem states that if I_{cm} is the moment of inertia of a body of mass M about an axis passing through its center of mass (cm), its moment of inertia about any axis
   A. perpendicular to and a displacement d from the axis through the cm is $I = I_{cm} + Md^2$.
   B. parallel to and a displacement d from the axis through the cm is $I = I_{cm} + \frac{1}{2}Md^2$.
   C. perpendicular to and a displacement d from the axis through the cm is $I = I_{cm} + \frac{1}{2}Md^2$.
   D. parallel to and a displacement d from the axis through the cm is $I = I_{cm} + Md^2$.

4. Three forces of equal magnitude are exerted at the center of a uniform beam
   of length L and negligible mass, as shown in the figure.

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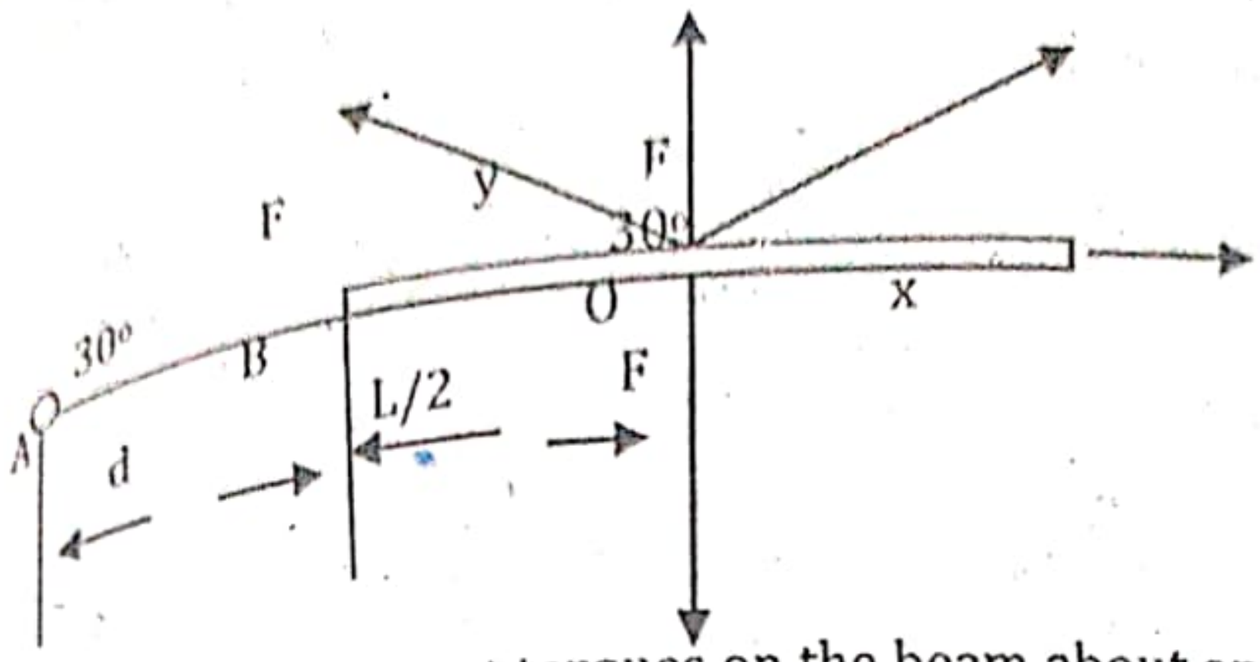

If $\tau_A, \tau_B, and \tau_O$ are net torques on the beam about an axis parallel to the z - axis and passing through point A, B, and O, respectively, which one of the following expression is correct?

A. $\tau_A < \tau_B < \tau_O$

B. $\tau_O = 0, \tau_A > \tau_B$.

C. $\tau_A = \tau_B = \tau_O = 0$

D. $\tau_A > \tau_B > \tau_O$.

5. Tensile stress is the
   A. perpendicular component of the applied force divided by the cross - sectional are of an object.
   B. ratio of original length to the change in length of the object.
   C. parallel component of the applied force divided by the cross - sectional are of an object.
   D. ratio of change length of an object to the original length.
6. A certain fluid has a density of $1080 kg/m^3$ and is observed to rise to a height of 2.1 cm in a 1.00 mm diameter tube. If the contact angle between the wall and the surface is zero, the surface tension of the fluids
   A. $0.1134 N/m$
   B. $0.0567 N/m$.
   C. $5670 N/m$.
   D. $0.02835 N/m$.
7. A slab of thermal insulator has a cross - section of 0.1 m^2, a length of 2 cm, and thermal conductivityof $0.1 J/(m.s^\circ C)$. If the temperature difference between the opposite faces of the insulator is 100°C, the amount of heat flowing through the slab in 24 h is
   A. $7.2\times10^4 J$.
   B. $5.0\times10^1 J$
   C. $1.2\times10^3 J$.
   D. $4.32\times10^6 J$.
8. A 100 J of work is done on the system and 418.6 J of heat is extracted from it. The change in the internal energy of the system is
   A. 318.6 J. B. 518.6 J. C. - 518.6 J. D. - 318.6 J.
9. Which of the following statements describe the difference between reversible and irreversible processes?

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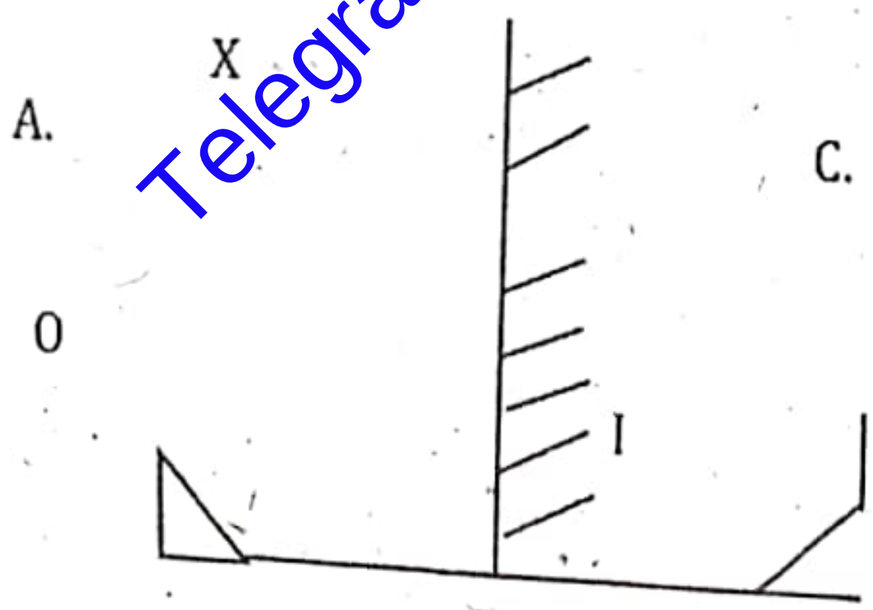
A. Ice at 0°C placed in a metallic can at 40°C undergoes a reversible process;
whereas ice at 0°C undergoes irreversible process.
B. A process is reversible if its entropy increases, but irreversible if its entropy is constant.
C. In an irreversible process, a system can move from one thermodynamic state to another and back to its initial state; whereas a system in a reversible process cannot move back to its initial state.
D. A system undergoing reversible process is always close to thermodynamic equilibrium, whereasa system undergoing irreversible process is not in thermal equilibrium.

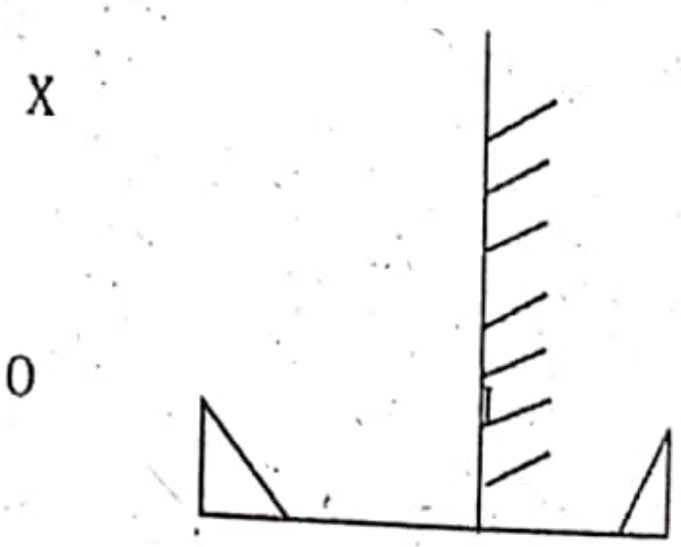
10. Which one of the following statements is correct about resonance in an oscillating object?
A. Resonance occurs when the frequency of a periodic driving force is greater than the natural frequency of the oscillator.
B. Resonance occurs when the oscillator is subject to a constant driving force.
C. At resonance, the driving frequency is equal to the natural frequency and the amplitude becomes maximum.
D. Dumping increases the amplitude and natural frequency of the oscillator.

11. Suppose a triangular object (O) is placed in front of a plane mirror and its image is observed at position (X). Which one of the following diagram represents the image (I) formed by the mirror when it is observed from position (X)?

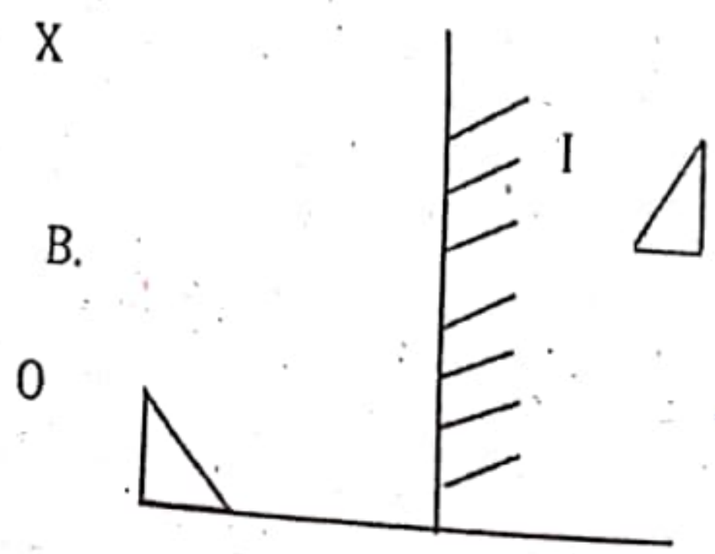
A.



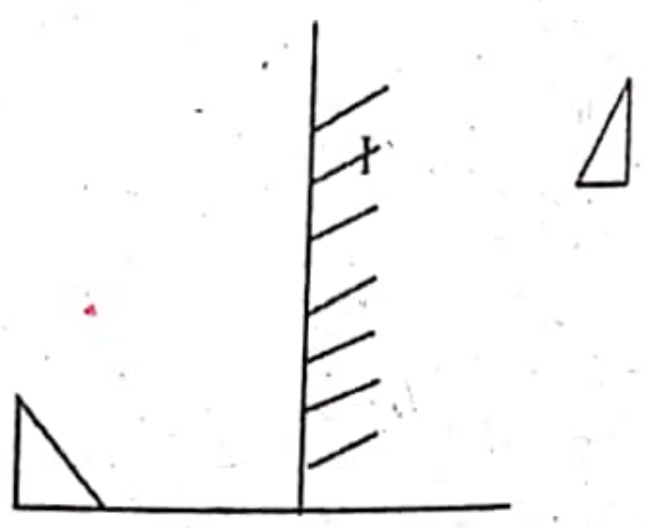
C.



B.



D.



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...n object is placed 35.0 cm in front of a concave mirror having focal ...gth of 10.0 cm. The magnification of the mirror is.....

A. 1.5 B. 2.5 C. 2.5 D. 16.7.

14. Which one of the following statements distinguishes between random and systematic errors?

A. Random error makes measured values to be always above or below the true value; whereas systemic error causes measured values to be sometimes above and sometimes below the true value.

B. Random error is biased, whereas systemic error does not show bias.

C. Systemic error can be minimized by taking several readings and averaging; whereas random error cannot be minimized.

D. Systemic error makes measured values to be always above or below the true value; whereas random error causes measured values to be sometimes above and sometimes below the true value.

15. The unit vector in the direction of $\vec{A} = 3\hat{i} + 4\hat{j}$ is.....

A. $\frac{3}{2}\hat{i} + \frac{4}{2}\hat{j}$ B. $\hat{i} + \hat{j}$ C. $\frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}$ D. $2\hat{i} - \hat{j}$

16. Which of the following pair of vectors are collinear?

A. $2\hat{i} + 3\hat{j}$ and $-4\hat{i} + 6\hat{j}$ C. $2\hat{i} - 3\hat{j}$ and $-4\hat{i} - 6\hat{j}$

B. $2\hat{i} + 3\hat{j}$ and $-4\hat{i} - 6\hat{j}$ D. $\hat{i} + \hat{j}$ and $-\hat{i} + \hat{j}$

17. For an object moving in straight line,

A. if the velocity increases by equal amount in equal time interval, the velocity - time graph is straight line with positive slope and the acceleration - time graph is horizontal.

B. the slope of displacement - time graph and the area of the region under acceleration - time graph are, respectively, the acceleration and velocity of the object.

C. if the displacement changes by the equal amount in equal time interval, the displacement - time graph is horizontal line.

D. the area of the region under the displacement - time graph and the slope of velocity - time graph are, respectively, the acceleration and displacement of the object.

18. Which of the following is true about satellite motion?

A. For equal orbital radius, a satellite moving around the Earth has smaller speed than a satellite moving about the Moon.

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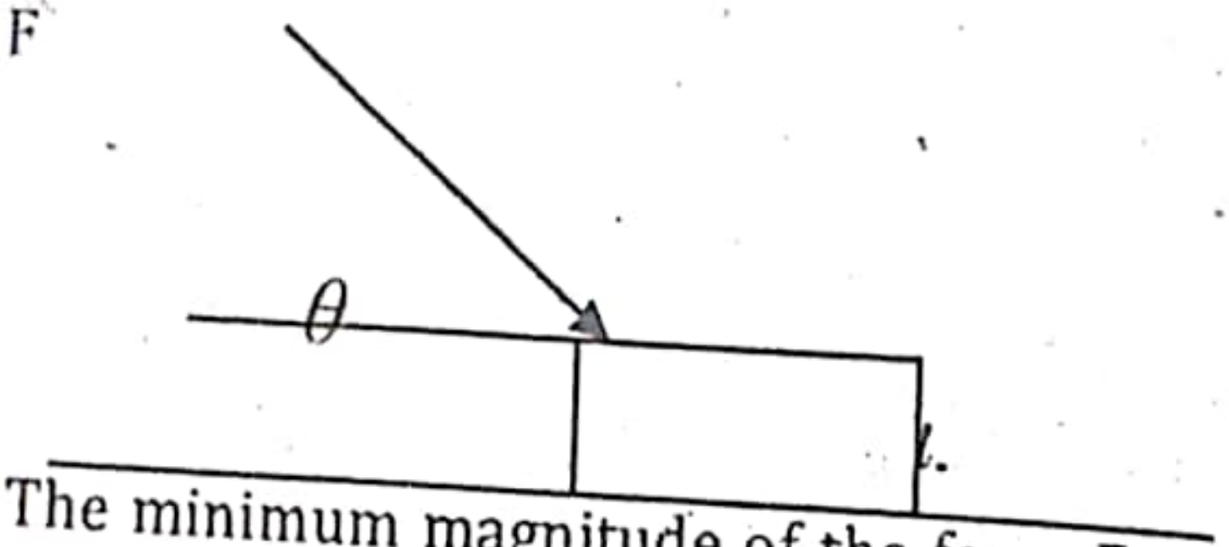
B. A satellite closer to the Earth is faster than a satellite farther from the Earth.

C. A satellite closer to the Earth takes longer time to complete one revolution than a satellite farther from the Earth.

D. For two satellites moving around the Earth at equal orbital radius, a heavier satellite has larger speed than a lighter one.

18. A block of mass m is placed on horizontal surface. The coefficient of static friction between the block and the surface is μ. A force of magnitude F acts on the block at an angle θ as shown in the figure below.



The minimum magnitude of the force F required to slide the block on the surface is.....

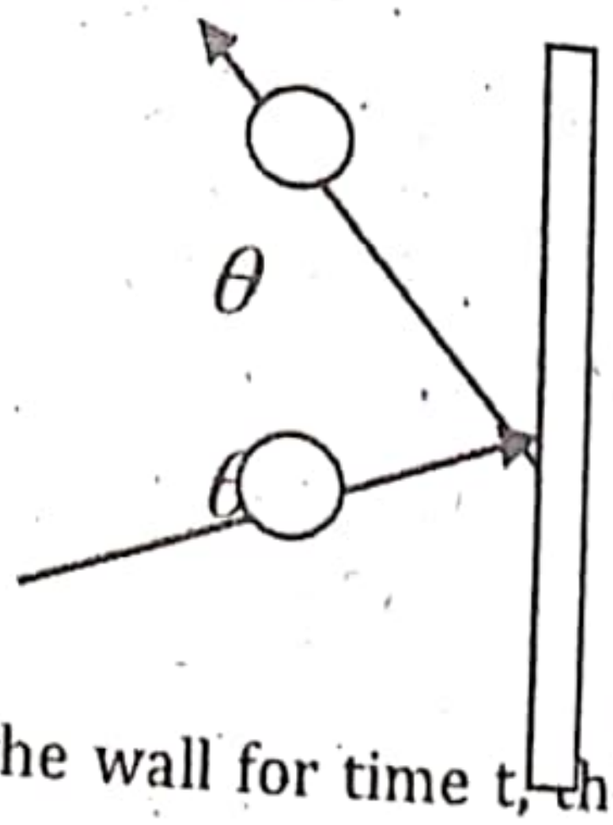
A. $\dfrac{\mu mg}{\cos\theta - \sin\theta}$

C. $\dfrac{\mu mg}{\cos\theta - \mu\sin\theta}$

B. $\dfrac{\mu mg}{\cos\theta + \sin\theta}$

D. $\dfrac{\mu mg}{\cos\theta + \mu\sin\theta}$

19. A ball of mass m moving with a speed v strikes a hard wall at an angle θ with the wall. It is reflected with the same speed at the same angle as shown in the figure below.



If the ball is in contact with the wall for time t, then the average force acting on the wall is given by.....

A. $\dfrac{2mv}{t}\cos\theta$

C. $\dfrac{mv}{t}\cos\theta$

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B. $\frac{2mv}{t}\sin\theta$ D. $\frac{mv}{t}\sin\theta$

20. According to the work - energy theorem,
A. work done by gravity on an object thrown upward and moving upward is negative.
B. work done by gravitational force on a satellite moving around the earth in a circular orbit is positive.
C. work done by tension on an object tied ti a string and moving in a horizontal circle is negative.
D. work done by gravity on an object thrown downward is negative.

21. Standing wave is formed when two identical waves travelling in opposite directions in the same medium interfere with each other. If y_1 and y_2 are the interfering waves and $y = y_1 + y_2$ is the resulting standing wave, which of the following is a correct combination?
A. $y_1 = A\sin(kx-\omega t), y_2 = A\sin(kx-\omega t), y = 2A\sin(kx)\cos(\omega t)$
B. $y_1 = A\sin(kx-\omega t), y_2 = A\cos(kx+\omega t), y = 2A\sin(kx-\omega t)$
C. $y_1 = A\sin(kx-\omega t), y_2 = A\sin(kx+\omega t), y = A\sin(kx)\cos(2\omega t)$
D. $y_1 = A\sin(kx-\omega t), y_2 = A\sin(kx+\omega t), y = 2A\sin(kx)\cos(\omega t)$,

22. A Young's double slit experiment consists of two narrow slits separated by 0.66 mm and 1.2 m away from a screen. If the slits are illuminated with a light of wavelength 563 mm, the location of the fourth bright fringe on the screen is
A. 3.94 cm. B. 4.59 cm. C. 0.394 cm. D. 0.45 cm.

23. An electric dipole consists of a negative charge $-q$ placed at a point with x and y coordinates (-2,-2) and a positive charge $+q$ at (2,2). The dipole is placed in a uniform electric field $\vec{E}$ directed in the positive x direction. If the dipole axis makes an angle θ with the field, which of the following is true about the dipole?
A. It is in both translational and rotational equilibrium.
B. It travels in the direction of the electric field.
C. The field rotates the dipole in the direction of decreasing the angle θ, until the dipole becomes parallel to the field.
D. The field rotates it in the direction of increasing the angle θ, until the dipole is perpendicular to the field.

24. The electric potential at a distance of 2 cm from a $-1.6x10^{-9}C$ point charge is

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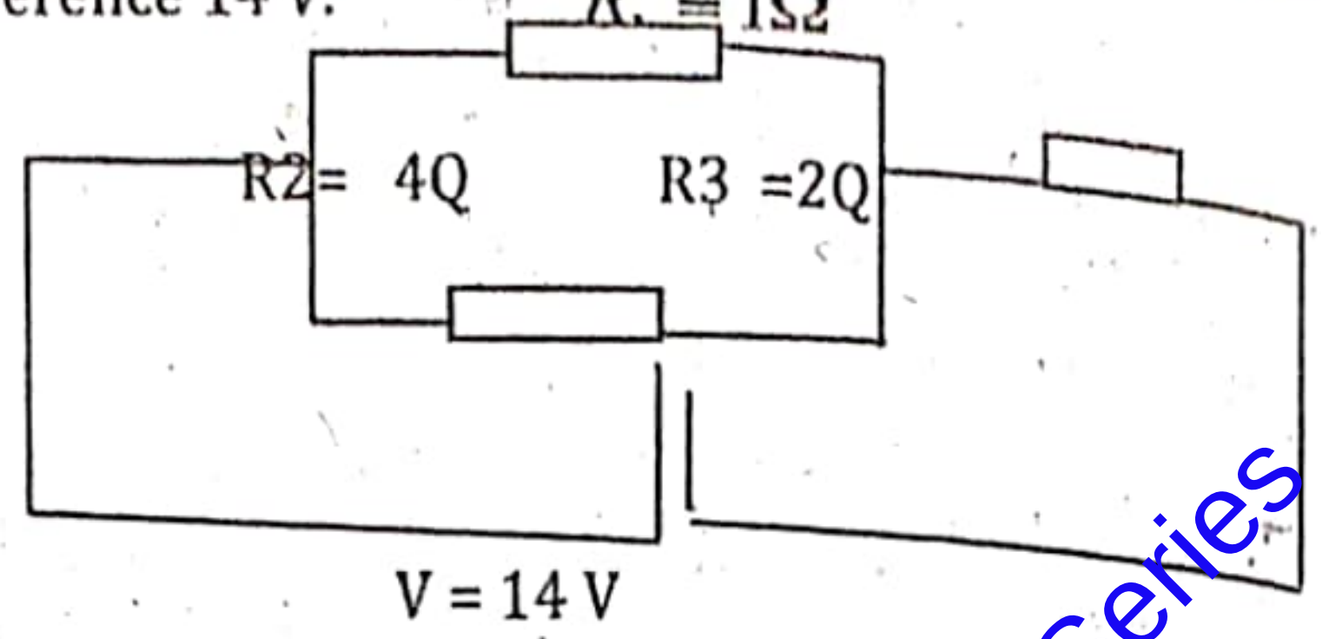
A. 4.4 V. B. - 720 V. C. -14.4 V. D. 720 V.

25. Two resistors $R_1 = 1\Omega$ and $R_2 = 4\Omega$ are connected in parallel to each other and their combination is connected in series with a third resistor $R_3 = 2\Omega$. The whole combination is connected to a voltage source of potential difference 14 V.



The potential difference across $R_1 = 1\Omega$ is

A. 5 V. B. 4 V. C. 10 V. D. 14 V.

26. Consider a long and tightly wound solenoid carrying current I, having length L and N turns of winding.If n is the number of turns per unit length, which of the following describes the magnetic field produced by the solenoid?

A. The magnetic field inside the solenoid is uniform and perpendicular to the axis of the solenoid.

B. The magnetic field outside the solenoid is as strong as the magnetic field inside it.

C. The magnetic field inside the solenoid is zero, whereas the field outside is perpendicular to the length of the solenoid.

D. The magnetic field inside the solenoid is constant and is parallel to the solenoid axis, whereas the field outside is negligible.

27. According to the law of electromagnetic induction,

A. a constant magnetic flux crossing the surface of a circuit induced an emf in the circuit.

B. an emf is induced in a circuit when a constant magnetic field crosses its surface.

C. the magnitude of emf induced in a circuit is directly proportional to the rate of change of the magnetic flux crossing any surface bounded by the circuit.

D. a changing magnetic field parallel to the plane of a circuit induces an emf in it.

28. A resistor R, a capacitor C, and an inductor L are connected in series to an AC voltage source that generates a sinusoidal voltage, $V(t) = V_P \sin(\omega t)$, with V_P and ω being the peak voltage and angular frequency of the source. If X_C, X_L, and Z are capacitive

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reactance, inductive reactance, and impedance, respectively; I_{rms} and V_{rms} are the root - mean - square (rms) values of the current and voltage, the average power dissipated in series the RLC circuit is given by

A. $I_{rms}V_{rms}\dfrac{R}{\sqrt{R^2+(X_L-X_C)^2}}$

C. $2I_pV_p\dfrac{R}{\sqrt{R^2+(X_L-X_C)^2}}$.

D. $I_{rms}V_{rms}$.

B. $I_{rms}{}^2Z$.

29. According to the Heisenberg's uncertainty principle,
A. the more precisely we measure the position of a particle; we become less precise of its momentum and vice versa.
B. we van measure both the position and momentum of a particle simultaneously with unlimited precision.
C. the larger the position uncertainty, the larger the momentum uncertainty.
D. if the uncertainty in position decreases, the momentum uncertainty decreases, too.

30. The graph of nuclear binding energy for nucleon (BE/A) versus mass number (A) has its peak near an atomic weight A = 56. Which of the following statements is true about the graph?
A. For heavier nuclei (A>56), the binding energy per nucleon decreases with the increase of mass number; implying that heavier atoms can increase their stability by increasing their protons number.
B. For smaller atoms (A<56), the binding energy per nucleon generally increases with the increase of mass number; indicating that lighter atoms can increase stability through nuclear fission
C. Uranium (A = 235), which has larger binding energy than iron (A = 56), is more stable than iron.
D. Nucleus with large number of neutron is more stable than nucleus of the same atom with smaller number of neutron.

31. Which of the following statements is NOT correct about reflection and refraction of sound waves and their applications?
A. Reflection and refraction of sound are used to detect flaws in metals and to help people park their cars.
B. Refraction of sound is the change in its direction as it travels through a medium having different densities.
C. In pre - natal scanning, reflection and refraction of sound allows doctors to monitor the progress of a a developing body.
D. Reflection of sound (echo) is formed when sound enters a medium of varying densities.

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32. A torque τ rotates an object of mass m and moment of inertia I through an angle θ. The work done on the object by the torque is.....

A. $\tau\theta$ B. τI C. $I\theta$ D. $m\theta$

33. A satellite revolves around the Earth in a circular orbit of radius $r = 4x10^{16} m$. The period of the satellite is

A. $3.7\times10^{26} s$. B. $2.5\times10^{18} s$. C. $2.5\times10^{19} s$. D. $3.7\times10^{27} s$.

34. One of the following statements is a correct description of the charging processes?

A. When a positively charged object is momentarily brought into contact with a neutral metallic sphere, the sphere acquires a net positive charge.

B. When a plastic rod is rubbed with wool, the two objects acquires a net charge of the same sign.

C. During charging a neutral object by induction using a negatively charged rod, charge flows from the rod to the object.

D. When a neutral metallic sphere is charged by induction using a negatively charged rod the sphere acquires a net negative charge.

35. A parallel plate capacitor is charged with a battery. The battery is then disconnected, and a dielectric material is inserted between the plates of the capacitor. Which of the following is correct about the effect of the dielectric?

A. The electric field between the plates and the energy stored decrease.

B. The capacitance remains the same and the electric field between the plates increases.

C. The potential difference between the plates remains the same and the charge increases.

D. The charge on the plates decreases and the energy stored increases.

36. Which one of the following statements is correct about the resistance of a conductor with a circular diameter?

A. The resistance of the conductor increases as its length and diameter increases.

B. The resistance of the conductor decreases as it is heated.

C. At a given temperature, the resistance of copper and silver wires of the same length and diameter is the same.

D. The resistance of the conductor increases as its length increases and its diameter decreases.

37. A student wants to investigate Ohm's law using an ammeter (A) and a voltmeter (V) to measure current through and potential difference across a resistor R. Moreover, she employs a rheostat to vary the potential difference across the resistor. Which of the following

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